

Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant $c_3 = \frac{1}{8\pi}$ and a five-slot carrier — build E_8
and read off the Standard Model: status assessment and reading guide

Stefan Hamann

Alessandro Rizzo

July 21, 2026 · v5.4 (rev 376)

In one sentence

Topological Fixed-Point Theory (TFPT) constructs, from exactly **two inputs** — a boundary constant $c_3 = \frac{1}{8\pi}$ (P1) and a five-slot carrier $g_{\text{car}} = 5$ (P2) — a discrete compiler for the Standard-Model skeleton: the gauge group, three families, hypercharges and the flavor matrix, with E_8 ($D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$) as the consistency checksum. The *algebraic core is machine-checkable* and the dimensionless constants follow as fixed points ($\alpha^{-1} = 137.0359992$); *physical readouts* — absolute scales, masses, inflation, gravity and cosmological transfers — run through explicitly *named, status-typed bridges* (v_{geo} , G_{net} , F_{transfer}), not as free outputs. The honest one-line claim is on the status card below.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt_1_architecture_e8** — the two axioms, the derivation map, the EM fixed point α^{-1} , the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. **tfpt_2_standard_model** — the SM in one φ_0 -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. **tfpt_3_e8_audit_bootstrap** — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt_4_frontier** — honest status of η_B , m_p/m_e , Koide, dark matter, full QG.
5. **tfpt_5_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: **R.** **tfpt_research_contracts** — the open interfaces (v_{geo} , G_{net} , F_{transfer}) as numbered contracts; **H.** **tfpt_horizon_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** **tfpt_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

You are reading the introduction — the entry point and reading guide.

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B , axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, v358) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in v473 without changing the typing) and the one unit v_{geo} ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= SEAM.EQUIV.01: the raw seam *is* the holomorphic $(E_8)_1$ net) is now *closed modulo a cited theorem*. The explicit lattice model (v367/v368) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (v376), the $(E_8)_1$ character with 248 currents and one primary (v377), genus-one torus $\text{GSD} = 1$ (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the #print axioms check is clean. The *only* residual that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ($h_s=16/16=1 \in \mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM $\mu=4/2^2=1$; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed'` (v469); so this is *closed modulo a cited theorem*, not solved. QGEO.SYM.01 (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] *redundancy* (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, v76/v330). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels (U_{wall} , G_{metric} , F_{frontier}) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam G_{net} closed modulo a cited theorem and QG.AMB.01 a [C] redundancy.

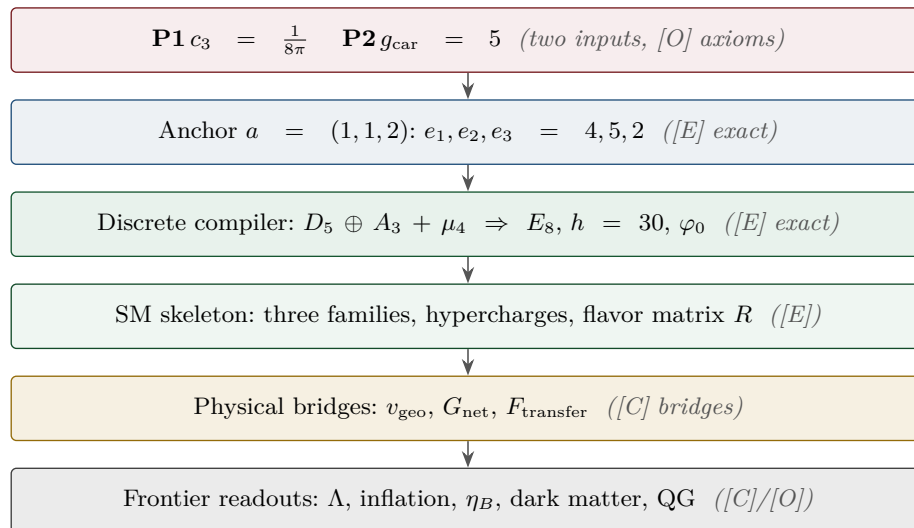
Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven

[C] conditional

[O] open / axiom

[X] kill test



Contents

1	What this is about: the jump in construction principle	4
2	The architecture at a glance	7
3	The dependency DAG and the proof ledger	8
4	The compiler: why $30 = 2 \cdot 3 \cdot 5$	13
5	The μ_4 glue: how E_8 is really built	14
6	How derivation now works: the derivation map	14
7	Physical, geometric, topological: why the architecture is plausible	16
8	What role E_8 now plays exactly	16
9	What this means for the theory-of-everything question	17
10	TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram	20
11	Does this make the theory more probable?	21
12	Predictions for running and upcoming experiments	21
	Conclusion	23
	Appendix: computational verification	23

	Appendix: changelog	135
--	----------------------------	------------

Reading the markers. Claims carry one of *four* reader-facing classes, so one can see at a glance what is proven and what is conditional: **[E]** *exact / machine-proven* — an exact identity (e.g. $240 = 16 \cdot 5 \cdot 3$), a Lie/lattice theorem (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$), a Lean/script formalisation, or a reproduced numerical fixed point; **[C]** *conditional* — follows under explicitly named hypotheses (a physical bridge, readout or pending step); **[O]** *open / axiom* — a declared input or a genuine gap; **[X]** *kill test* — a falsifiable threshold. The finer per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is kept in `verification/status_ledger.csv`, the single source of truth.

Physical — under hypotheses	RG masses, QG QG.AMB.01
Numerical — fit-free fixed point	$\alpha^{-1}, \Lambda, \theta_{12}$ EM.FP.01
Lattice — Lie / glue / root system	$E_8=(D_5\oplus A_3)+\mu_4$ E8.GLU.01
Formal — Lean / exact script	P2 algebra FORM.P2.02
Axiom — declared inputs	P1 c_3 , P2 g_{car} AX.P1.01

The *claim stack* (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is **verification/status_ledger.csv**.

No-free-pattern rule. To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* [E] (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

Admissible-invariant classes (the harder filter). Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1}-p_n$; (iii) a determinant, spectrum, Smith normal form or trace of (R, Q, K, L) ; (iv) an anchor quadratic form $u^\top Mv$ with $u, v \in \{\mathbf{1}, a\}$; (v) a Plücker coordinate of the anchor plane $\langle \mathbf{1}, a \rangle$; (vi) an E_8 Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* (“look, another number”) into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

Reviewer path (read in this order; the rest is support):

1. the architecture: the two axioms $\{c_3, g_{\text{car}}\}$ and the two-engine picture (§1, the figures);
2. the E_8 glue $E_8=(D_5\oplus A_3)+\mu_4$ — document 1, Part II, machine-checked (v1_e8_glue.py);
3. the α^{-1} fixed point plus ablation (v3_em_alpha.py);
4. the **single status ledger verification/status_ledger.csv** — the source of truth for every claim’s grade, location, dependency and script;
5. the **open gates** as numbered research contracts (tfpt_research_contracts: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

1 What this is about: the jump in construction principle

In plain words

What we found. TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers (2, 3, 5, 16, ...) reappears in every sector: those numbers are not separate lucky hits. A single tiny “machine” — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry E_8 , and E_8 is the unimodular “audit hull” that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not “printed” for free — where a scheme or analytic interface is needed, it is named).

Why this is simple. There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

How plausible it is. A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune*. Pushed to the end ([[verification/v353_selfloop_capstone.py](#)]), this means TFPT is best read not as a *linear* theory (axioms $\rightarrow$ theorems) but as a *closed self-consistent loop* whose outputs fix its own inputs ($g_{\text{car}}=5$ is the largest prime of the output $h(E_8)=30$; the 8 in c_3 is rank E_8), with a *unique* fixed point ([v6/v56](#)) and *zero* free adjustable dimensionless parameters. Even π is not a free dial: it is the fixed geometric constant of the boundary 2-sphere ($\oint_{S^2} K = 4\pi$), the one transcendental the continuous boundary contains. “Which is the axiom?” is the wrong question for a loop — every part is fixed by the whole; the one remaining structural question, whether the loop also closes in the *continuum* (the keystone [SEAM.EQUIV.01](#)), is now *closed modulo a cited theorem* — the lattice model and S_3 stack pin it at every computable level and it is Lean-formalised ([FORM.SEAM.MMST.01](#)); only the cited continuum-existence theorems stay [\[O\]](#) ([v367/v368](#), [v376–v379](#), [v336](#)), their extension leg since re-founded on peer-reviewed crossed-product theorems with the realisation input reduced to invariant level ([v469](#)).

Sharper still: one anchor $a = (1, 1, 2)$, and E_8 as a compiler hull ([\[E\]](#))

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|, \quad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$ generate the big Lie data directly: $p_1=4=|\mu_4|$, $p_2=6=|R^+(A_3)|$, $p_3=10=\mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \text{rank } E_8,$$

with the binary ladder $p_{n+1} - p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\text{car}}\}$ to $\boxed{a = (1, 1, 2) \text{ plus } \pi}$. [[verification/v23_anchor_generator.py](#)]

And a is itself half-forced ([\[E\]/\[C\]](#)): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\text{fam}} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing to 4, and $4 = 2 + 1 + 1$ is the *unique* partition into three positive parts — machine-checked with all its negative controls as the partition corollary ([[verification/v491_p2_partition_corollary.py](#)], [P2.PARTITION.01](#)): the unique multiset $\{1, 1, 2\}$ forces $e = (4, 5, 2)$, hence $g_{\text{car}} = e_2 = 5$, $|\mathbb{Z}_2| = e_3 = 2$, $N_{\text{fam}} = 3$ as *corollaries* (dropping positivity, the sum, the part count or integrality each destroys uniqueness — Mehta–Seshadri-rational weights give $e_2 = \frac{16}{3}$ or $\frac{44}{9} \neq 5$), sharpening [v53](#) which needs all three targets. The typing itself is now hardened ([[verification/v499_p2_weights_deligne_bg.py](#)], [P2.TYPING.01](#)): given the four marks, rank 3, the cusp class and unitarity (U), the Deligne canonical extension of the flavor connection has $\deg E = -4$ (residue traces) and Mehta–Seshadri stability forces $h^0(E) = 0$, so integrality (Birkhoff–Grothendieck), positivity and the sum 4 are *theorems* — the residual shrinks to the module identification and (U), both [\[C\]](#); and that module-identification rest is now *one* residual with the QGEO modulus rest (the order-4 clock = the Coxeter/cusp-class carrier datum; [[verification/v503_qgeo_emergence_light.py](#)], [QGEO.EMERGE.LIGHT.01](#), no marker moves). The genuine remaining axiom is therefore not “why $a = (1, 1, 2)$ ” but “why the carrier $g_{\text{car}} = 5$ / $(|\mu_4|, N_{\text{fam}}) = (4, 3)$ ” — that interface (with c_3), i.e. the weight-*typing* postulate itself, is the [\[O\]](#) input layer ([AX.P2.01](#) stays a declared axiom).

Two readings, one source (no contradiction). P1 and P2 are the *operational compiler inputs*; their anchor provenance is $a = (1, 1, 2)$ plus π . The *single structural bedrock postulate* is the

keystone **SEAM.EQUIV.01**, whose conformal-deck face **QGEO.SYM.01** (the carrier μ_4 clock is the seam’s conformal deck, see **tfpt_research_contracts**) forces the μ_4 deck, hence $N_{\text{fam}} = 3$, $\deg E = -4$ and the tripartition $4 = 2+1+1$, of which c_3 and g_{car} are the two elementary readouts. “Two inputs” (the working axioms) and “one postulate” (the geometric bedrock) are the same statement at two layers. Its geometric normal form and the mark-local $\Rightarrow \omega \circ \rho = \omega$ implication are now Lean-formalised (**FORM.QGEO.02/FORM.QGEO.01**); the postulate itself stays **[O]**.

One source, four readings (the cleanest origin). The four-point seam divisor μ_4 on $\mathbb{P}^1$ generates the whole discrete skeleton by four canonical functors:

$$\underbrace{c_3 = \frac{1}{8\pi}}_{\text{thermal}} \quad \underbrace{\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = 3 = N_{\text{fam}}}_{\text{cycles} \rightarrow A_3} \quad \underbrace{h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = 5 = g_{\text{car}}}_{\text{functions} \rightarrow D_5}$$

$$\underbrace{(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1}_{\text{net}}$$

and the function side even generates D_5 itself (**[verification/v197_rr_carrier_clifford_d5.py]**): $\Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor. So μ_4 is not merely glue: it is the *common source* of A_3 , D_5 , c_3 and E_8 (**[verification/v189_riemann_roch_carrier.py]**) — the arithmetic is exact **[E]**, the physical reading of the carrier as the seam mode space stays **[C]**.

What E_8 is (and is not). In TFPT E_8 is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct E_8 gauge theory; the Lorentz signature appears only after projection; fermions are not “everything in the adjoint”. This is exactly why the known no-go results against literal E_8 world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain E_8 as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor $a = (1, 1, 2)$ into the E_8 audit hull.*

In one line:

| The anchor generates the syntax; the boundary generates the scale; E_8 checks the consistency.

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators R, Q, K, L) flows from $a = (1, 1, 2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, φ_0 , α^{-1} , the exponential scales) flows from π ; and E_8 is the unimodular *checksum container*, not the world group.

The anchor ratio triad (**[verification/v119_review_validation_2.py]**). The three elementary symmetric values, normalised by the family count, are the theory’s three recurring fractions **[E]**:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio), } \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain), } \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow’s critical points are exactly $(2, 5) = (e_3, e_2)$ with inflection $\frac{7}{2} = \frac{e_3 + e_2}{2}$: the fractions $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$ are not scattered numbers but the three normalised anchor coefficients. Further micro-identities **[E]**: $\Omega_{\text{adm}} = 2p_1p_2 = 48$, $10b_1 = 1 + p_1p_3 = 41$, $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank } E_8 = 25$, $h(E_8) = e_3p_0e_2 = 30$, $\text{rank } E_8 = p_0 + e_2$.

$$a = (1, 1, 2) \quad [\text{E}]$$

elementary $e_k(a)$

$$\begin{aligned} e_1 &= 4 \rightarrow |\mu_4| \\ e_2 &= 5 \rightarrow g_{\text{car}} \\ e_3 &= 2 \rightarrow |\mathbb{Z}_2| \\ c_3 &= \frac{1}{2e_1\pi} = \frac{1}{8\pi} \end{aligned}$$

power sums $p_n = 2 + 2^n$

$$\begin{aligned} p_0, p_1, p_2, p_3 &= 3, 4, 6, 10 \\ p_1p_2p_3 &= 240 = |R(E_8)| \\ p_4 - p_3 &= 8 = \text{rank } E_8 \\ \Rightarrow \dim E_8 &= 248 \end{aligned}$$

derived motifs

$$\begin{aligned} &4, 5, 8, 16, 25, \\ &30, 41, 120, 240 \\ &\text{one anchor,} \\ &\text{many projections} \end{aligned}$$

The companion documents

The whole development is consolidated into **eight companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

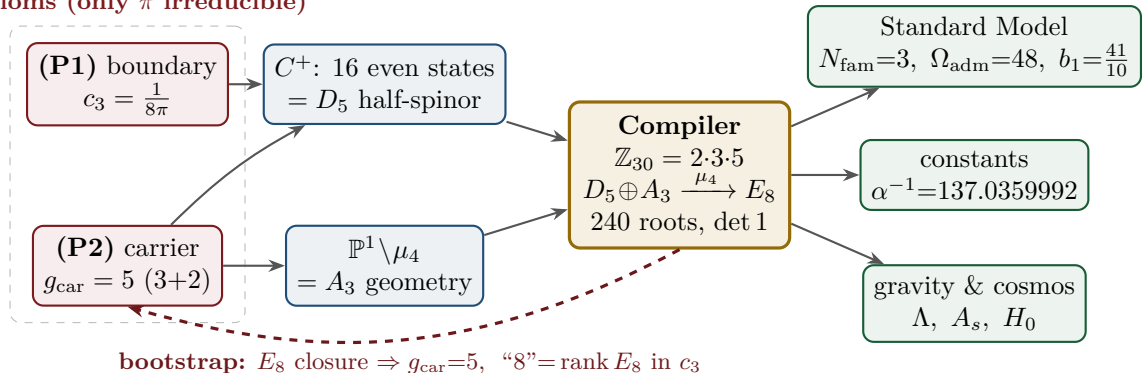
Document	What it contains (former notes now as parts)
tfpt_1_architecture_e8	Core. The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}=2\cdot3\cdot5$, the explicit $D_5\oplus A_3+\mu_4$ E_8 construction, the machine-checkable E_8 appendix and the group-level glue $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ (Part II).
tfpt_2_standard_model	Standard Model. The full SM in one φ_0^{ret} -ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/ n_s, r , Starobinsky M , derived θ_{12} , quark c , the H2 splitting type, Λ -typing (Part III).
tfpt_3_e8_audit_bootstrap	E_8 audit & bootstrap. The seven E_8 slices as an audit raster ($E_6\times A_2$ reads the flavor matrix) (Part I); the old cascade $D=60-2n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\text{car}}=5$ and the “8” in c_3 as overdetermined fixed points, only π irreducible (Part III).
tfpt_4_frontier	Frontier. The honest status of η_B , m_p/m_e , Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is <i>not</i> forced onto the ladder.
tfpt_5_redteam	Red Team. The adversarial audit of the five load-bearing reductions (Targets A–E): the declared attacks, what survives each, what each one reduces to, and the explicit kill tests.
tfpt_research_contracts	Research contracts. The open gates as numbered contracts with lemma chains and closing theorems: (U_{wall}) (now reduced to the selector triangle: $(d, n) \Rightarrow R$ columnwise, residue = <i>two</i> line pairings, the third integrality-forced) and $G_{\text{metric}}/G_{\text{net}}$ (the index-4 seam-net identification — one theorem, three doors, now realised at the finite Lie level).
tfpt_horizon_readouts (App. H)	Appendix H — horizon unit system (reframe, not new physics). One seam constant $c_3=\frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints ($1920= W(D_5) $, $ \mu_4 =4$); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138).
origin_theory	Origin Theory. Why no free <i>dimensionless compiler dial</i> remains beyond the anchor and π (the dimensionful v_{geo} and transfer physics stay typed): the $(g_{\text{car}}, N_{\text{fam}})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; [E] core plus one honestly-typed [C] cyclic interpretation.

The five formerly “research-level” problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the θ_{12} texture and the quark c in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a U_{point} anchor; θ_{12} the seam-misalignment residual). Eight companion files (plus this introduction), each result where it logically lives.

2 The architecture at a glance

The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.

2 axioms (only π irreducible)



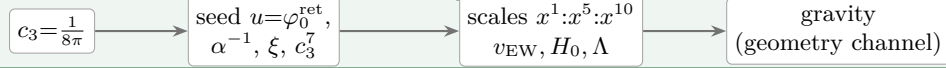
Key statement (the loop closes): previously D_5 , A_3 and E_8 sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the E_8 closure feeds back and *fixes the inputs*. $g_{\text{car}} = 5$ is the unique rank-fill/Coxeter/Pascal solution ($2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$), and the integer 8 in $c_3 = \frac{1}{8\pi}$ is rank $E_8 = h(D_5) = \det R$. **Inputs and output form an overdetermined fixed point** — the structure does not “prove itself”; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous π (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

The master story is two engines. Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.

Engine 1 — discrete closure (from $g_{\text{car}} = 5$)

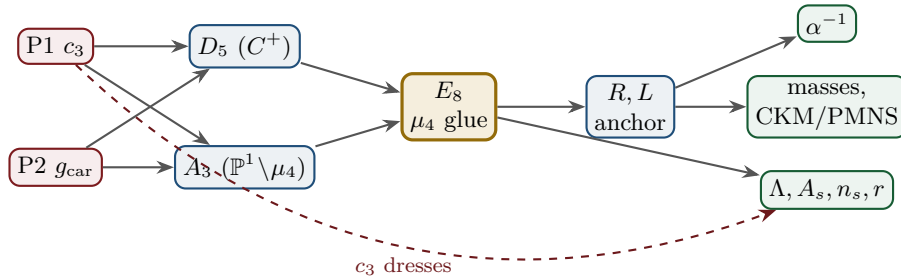


Engine 2 — boundary dressing (from $c_3 = \frac{1}{8\pi}$)



3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



Node	Definition	Inputs	St.	Output	Failure mode
P1	RP boundary kernel, $c_3=1/(8\pi)$	— (axiom)	[O]	c_3	no reflection-positive seam
P2	5-slot carrier $g_{\text{car}}=5$	— (axiom)	[O]	carrier 3+2	wrong family/charge lattice
D_5	C^+ even-Hamming = half-spinor 16	P2	[E]	$\dim S^+=16, Y$	group/matter mismatch
A_3	$\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.	P1	[E]	$N_{\text{fam}}=3, \mathbb{Z}_6$	wrong multiplicity
E_8	μ_4 glue: $\text{disc}=\mathbb{Z}_4, q\text{-sum}=2$	D_5, A_3	[E]	240, 248	not even-unimodular
R, L	residue + winding matrix, anchor (1, 1, 2)	A_3	[E]	$\det 8, \text{ minors } (2, 3, 5)$	wrong D_6 branch
α^{-1}	root of $F_{U(1)}(\alpha)=0$	$c_3, M=41$	[E]	137.0359992	no/second root (excl.)
masses	φ_0^{ret} -ladder $\lambda_Y^L \Lambda$	R, L, c_3	[E]/[C]	9 masses, mixings	hierarchy mismatch
θ_{12}	TBM + seam $\varepsilon=c_3$	R, L	[E]/[C]	0.307	seam-misalign. lemma fails
Λ, A_s	seam det. / R^2 scalaron	c_3	[C]	Λ, A_s, n_s, r	ambient RP fails

Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The ledger itself is versioned: each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early “flavor gate open” rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

The compact status formula: three honest layers

$$\underbrace{\text{compiler}}_{\text{closed}} \mid \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \mid \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT v5.4 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R+R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the R -bridge amplitude normalisation U_{point} *reduces to* v_{geo} (Gate 1 closed, **v75** — the same anchor as $1/G$), and the ambient metric measure is *holographically reduced* to a finite seam-boundary measure (Gate 2 tier B, **v76**) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (**v77**), so G_{metric} is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale v_{geo} — the *dimensional-analysis floor* that no theory escapes (**v78**), shared by flavor and gravity — the boundary net identification (now *closed modulo a cited theorem*, **v367/v368**, **v376–v379**, Lean `FORM.SEAM.MMST.01`), and the named QFT/cosmology transfer interfaces (F_{frontier}).

Two points that are *not* in the residual (closed, but easy to mis-list as open). (i) *Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6 \log \frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned (**[E]** for the attractor, `[verification/v56_unique_attractor.py]`; the *physical identification* of the transport operator stays **[C]**), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the E_8 closure rank — a consistency loop, see red team Target B — `[verification/v6_bootstrap.py]`, `[verification/v14_carrier_uniqueness.py]`). (ii) *Newton’s constant is not an independent input:* fixing the physical Λ branch ($(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ relative to Λ , so G_N is read out *after* the branch choice + *one* dimensionful induced-gravity anchor (**[E]/[C]**, `[verification/v60_lambda_metrology_branch.py]`), not predicted from pure numbers. What stays irreducible is then just π , the discrete E_8 -closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale v_{geo} — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \rightarrow v_{\text{geo}}$, **v75**) and gravity ($1/G$, **v68**): the two $[A]$ anchors are *one* (dimensional analysis; no metre from pure mathematics).

The hard reduction: the live residual as typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

— one dimensionful scale anchor, one metric-sector inclusion, and a set of deliberately typed QFT/cosmology interfaces (the historical gate labels $(U_{\text{wall}})/(G_{\text{metric}})/(F_{\text{frontier}})$ are kept only for ledger continuity). In current terms: $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, **v49/v71/v75**), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \rightarrow G_{\text{net}}$ — reduced to a single boundary-net theorem, “*the seam–Calderón net is the index-4 μ_4 simple-current extension of the carrier net*” (holomorphy then follows, **v89/GATE.METRIC.06**); since the QBL chain (**v108–v113**) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (**v110**), the certified channel counts the code identically (**v112**), pair transport is minimally complete (**v111**), and the one quasi-free kernel — rank = c : 5 carrier block, 8 seam hull

— is the defining datum of the free net the gate already posits (v113), so closing G_{net} also closes the carrier selection [E]; $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, η_B Boltzmann, axion relic, m_p/m_e), never compiler powers; the four are one typed functor with a four-axiom contract (CONTRACT.F.01, v213) and *one shape* — each is a gapped relaxation to a unique attractor (the main-branch discrete→dynamic motif), with F_{pole} at the *seam* rate $(2/3)^6$ and the other three external (v303). The metric-sector reduction lands on a *single named keystone*. The boundary inclusion G_{net} is the index-4 μ_4 simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ (Simple-Current Extension Theorem, v154, exact: index 4, $c=8$, $\mu=1 \Rightarrow$ holomorphic $\Rightarrow E_8$); net existence and full-cone reflection positivity are discharged to [E] (v175); and the **No-Unit Theorem** (v153) makes v_{geo} an irreducible metrology primitive ($U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, $m/\mu = e^{3/4}$ — one unit in three dimensions). The flavor selector and Q-geometry residues are *derived* (no discrete freedom left). The *one* seam-side identification, packaged as the single named theorem SEAM.EQUIV.01 — *the raw reflection-positive seam state is the holomorphic $(E_8)_1$ net at $\tau=i$* — is now *closed modulo a cited theorem*: the explicit lattice model (v367/v368) and the S_3 stack ($c=8$, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, Lean-formalised as FORM.SEAM.MMST.01, with only the cited MMST/OS continuum-existence theorems left [O] (v336); its conformal-deck premise QGEO.SYM.01 is its *downstream* definitional face (per v323; the residual is Lean-pinned to named cited steps, v308). The sprint-by-sprint reduction that produced it is recorded in the changelog and in `tfpt_research_contracts`, not replayed here. The status card:

Item	Status	Reading
parameter-freeness (self-consistency)	[E]	gapped transport $\Rightarrow$ <i>unique</i> Perron–Frobenius attractor (v56); bootstrap web $g_{\text{car}}=5$ forced (v6/v14)
Newton constant G_N	[E]/[C]	not <i>independent</i> : branch pins M_{Pl} rel. to Λ (v60); readout after branch + one dim. anchor
ratio $c_u/c_d = \frac{55}{117}$	[E]	$g_{\text{car}}\ \text{Pl}_{1,a}(K)\ _1/(N_{\text{fam}}^2\Delta_Q)$; rigid on the derived stratum (v49/v71), no solve
flavor absolute scale	[E]/[O]	ratios <i>closed</i> (v49/v71); only the absolute scale remains $= v_{\text{geo}}$ (Gate 1, v75), the same anchor as $1/G$
$R + R^2$ gravity	[E]/[C]	covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)
Einstein eq. (covariant, local)	[E]/[C]	fixed-volume entanglement equilibrium $\Rightarrow$ <i>full</i> $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with BOTH coefficients parameter-free: $8\pi = 1/c_3$ (v358, thermo=geo via $ \mu_4 = \mathbb{Z}_2 \chi=4$) and Λ from α (v60); Einstein tensor via Lovelock, conservation an output (v359); residual = equation-of-state status + v_{geo} , global measure (C7) now a [C] redundancy (v369)
physical QG sector	[E]/[C]	gap-decoupled admissible measure ($\Delta_{\text{eff}}=1.648>0$)
ambient QG measure	[C] (redundancy)	certification object, not missing dynamics (v369; conditional on SEAM.EQUIV.01+Bisognano–Wichmann): holographically reduced to a finite seam- <i>boundary</i> measure (v76); IR tier closed (decoupling); strict-TOE cert. = the simple-current extension (D_5) $_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$ (v154); the free-bulk premise (A) is itself a fixed-point theorem with the cone eliminated (cone gap = one-particle gap $(2/3)^6$) and factors into the A_2 net-existence + the keystone SEAM.EQUIV.01, closed modulo a cited theorem — zero new gates (v160–v165)
v_{geo} metrology unit	[O] (primitive)	No-Unit Theorem (v153): a dimensionless compiler makes no absolute scale; $U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, m/μ a ratio — one unit, irreducible
Koide	[C]	$Q_{\text{src}} + \varphi_0/24$ source $\rightarrow$ pole: typed runnable solver with kill test (QED-excluded, v371)
axion DM	[C]	$f_a = M_{\text{scal}}/128$: finite- T spine-branch typed runnable solver (v373)
η_B	[C]	readout closed; integrated BDP leptogenesis Boltzmann ODE, typed runnable solver (v372)
m_p/m_e	[C]	cross-sector QCD/EW ratio: typed runnable solver with uncertainty budget (v374)
N_*	[C]	reheating input; scalaron-reheating point $N_*=51.4$ (v86); frozen band [50, 60]
$a = (1, 1, 2)$	[E]/[C]	forced by $4 = 2 + 1 + 1$, conditional on the carrier
carrier selection $g=5$ (QBL)	[E]/[C]	theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = c); residue = the G_{net} premise itself — gate closes $\Rightarrow$ carrier [E]
selector establishment R4'	[E]/[C]	(d, n) $\Rightarrow$ R columnwise (v136); both normals <i>anchor data</i> — $n = (p_2, p_0, e_2)(a)$ in the cusp frame (v145/v149); residue = <i>one</i> assignment
sheet/parity (Q-geometry derived)	[E]/[C]	H^1 characters = $\text{Spec}(Q_+) = A_3$ exponents (v137); $\mathbb{Z}_3$ choice <i>derived</i> (v141); full Möbius D_4 matched parity by parity (v146); residue = the module identification, folded into SEAM.EQUIV.01
seam quantum clock R1	[E]/[C]	typed closed: resummed clock = entropy power law (v124–v133); VW edge match (v138); det-ratio in-family (v144); ring sum = Born ² Gaussian zero-mode integral, bend det'-clean (v147); residue = measure + reference
E_8 Lean	[E]	hardening, not a blocker (script-certified)

Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

The former structural open core, the *single* named keystone **SEAM.EQUIV.01** (the G_{net} inclusion: the raw RP seam = $(E_8)_1$ at $\tau=i$), is now *closed modulo a cited theorem*: the explicit lattice model (v367/v368) and the S_3 stack ($c=8$, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, it is Lean-formalised as **FORM.SEAM.MMST.01**, and only the cited MMST/OS continuum-existence theorems stay **[O]** (v336); its conformal-deck face **QGEO.SYM.01** is a *corollary* (a conformal net’s vacuum is rotation-invariant by axiom, v323/[`verification/v335_seam_equiv_unify.py`]), not a second item. v_{geo} is an irreducible metrology unit and the frontier items are typed interfaces, not structural holes. G_{net} is heat-kernel grounded (G2: the spectral action gives $R+R^2$ structurally) and *gap-decoupled* (G5: $\Delta_{\text{eff}}=1.648>0$ isolates the closed admissible sector from the ambient), so the ambient measure **QG.AMB.01** is no longer carried as an open structural item: it is a **[C]** *redundancy* (v369, conditional on **SEAM.EQUIV.01**+Bisognano–Wichmann), a certification object rather than missing dynamics — it does not affect any TFPT readout. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion **tfpt_research_contracts**, with priority on the v_{geo} scale anchor and the G_{net} /SEAM.EQUIV.01 inclusion theorem. [`verification/v36_spectral_action_g2.py`]

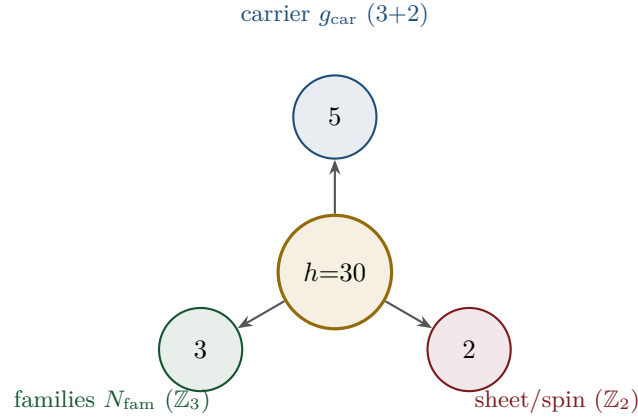
Sharper still: the structural residual is *one* condition. G_{metric} , the carrier P2 and the red-team Target A are not separate items: they are three provably-equivalent faces of the single condition “the seam carries no nontrivial abelian sector” — holomorphy (μ -index 1), a homology-sphere seam link (Γ perfect $\Leftrightarrow 2I$, [`verification/v232_e8_kleinian_seam.py`]), exactly one 1-dim irrep ([`verification/v219_icosahedral_mckay.py`]) — all of which force E_8 ($\#(\text{mark-1}) = |H_1| = 1$ only for E_8 ; holomorphic $c = 8 = g_{\text{car}} + N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice). In abelian Chern–Simons language this is the single integer step holomorphic $\Leftrightarrow \det K = 1$, the extension tower $D_5 \oplus A_3(16) \rightarrow D_8(4) \rightarrow E_8(1)$ being anyon condensation (the Kitaev E_8 state); the one analytic step “the free RP seam condenses the order- $|\mu_4|$ Lagrangian glue” = the single named keystone **SEAM.EQUIV.01** (with **QGEO.SYM.01** its downstream conformal-deck face) is now *closed modulo a cited theorem* (v367/v368, v376–v379, Lean **FORM.SEAM.MMST.01**; cited continuum existence **[O]**, v336). So beneath the gate labels there is, structurally, a single fundamental postulate. And the entire *emergent-QFT layer* collapses onto the *same* postulate (the *Modular Spectral Closure*, v261): the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff is that KMS weight ($f_2/f_0 = 1$, v259), the seam, the carrier-16 and E_8 live on one Kummer/K3 surface (v260), so Dirac, cutoff, gauging, glue and orientability are all readouts of the one seam state — the boundary QFT is closed as a single relative object modulo the keystone **SEAM.EQUIV.01** and adds *no new open item*, while the ambient quantum-gravity measure is a **[C]** redundancy (v369), not a separate structural gap. [`verification/v234_seam_holomorphy_selection.py`] [`verification/v235_seam_chern_simons.py`]

Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ($N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$, the flavor matrix, the E_8 glue, the scale grammar) is a *consequence*. And even those two tighten: the bootstrap note shows $g_{\text{car}} = 5$ and the integer 8 in c_3 are *overdetermined* E_8 -closure fixed points (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: π from the Möbius boundary. A *fourth, deeper forcing* comes from the McKay correspondence: E_8 is the exceptional top of the tower of finite $SU(2)$ subgroups, so its branch data is the icosahedron and the three atoms (2,3,5) are its rotation-axis orders — the binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has irrep degrees equal to the affine- E_8 Kac marks, $\sum = 30 = h(E_8) = 2 \cdot 3 \cdot 5$ ([`verification/v219_icosahedral_mckay.py`]). So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.) [`verification/v6_bootstrap.py`] [`verification/v14_carrier_uniqueness.py`] [`verification/v23_anchor_generator.py`] [`verification/v219_icosahedral_mckay.py`]

4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of E_8 is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:



From this the compiler reads, among others:

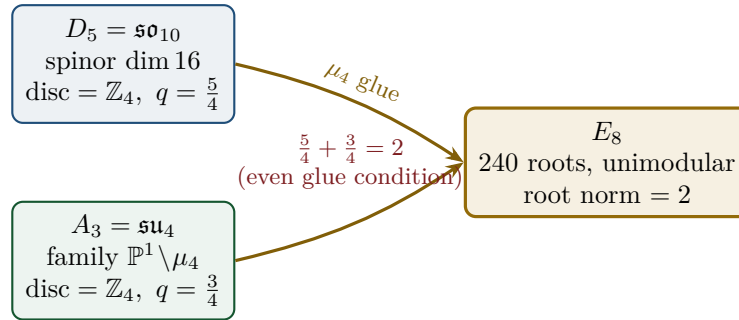
- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the E_8 root count. [E]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. [E]
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial Φ_{30} ; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \setminus \mu_4$ is the A_3 Coxeter element. [E]
- **why these three primes:** E_8 is the icosahedral top of the McKay tower ($2I$, order 120, irrep degrees = affine- E_8 marks summing to 30), so 2, 3, 5 are the icosahedron's rotation orders; and the E_8 exponents $(\mathbb{Z}/30)^\times$ carry an order-4 totative character $\langle 7 \rangle$ — the same μ_4 seam clock. [E] [verification/v219_icosahedral_mckay.py] [verification/v223_coxeter_totative_clock.py]
- **a network reading (computational substrate).** On the same McKay (2, 3, 5) graph the affine- E_8 marks are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4 \cdot m$ (it converges from *any* positive start to (1, 2, 3, 4, 5, 6, 4, 2, 3)); the spherical tower terminates at E_8 and the cascade $E_8 \supset E_7 \supset E_6 \dots$ is nested-subgraph coarse-graining. So E_8 reads as the *universal attractor of a minimal (2, 3, 5) network* (a Wolfram-style computational-substrate picture) at the Coxeter-skeleton level. [C] A minimal hypergraph *rewrite* reproducing the *full* structure (lattice + recovery + readouts) stays open, and φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction. [O] [verification/v298_e8_network_attractor.py]
- **the bridge, hardened.** A local witness-based growth dynamics (a *diffusion-generated* witness + the local balance gate $Am \leq 2m$, Collatz–Wielandt = Smith's $\rho \leq 2$; the Perron eigenvector enters only as an audit) *reaches* E_8 as the last finite point ($E_6 \rightarrow E_7 \rightarrow E_8$, then stops at the affine boundary $\hat{E}_8$) — but *only* from a 3-arm (2, 3, $\cdot$) seed (the controls $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$ and $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$ are < 2 for *all* n , so neither family ever reaches E_8). So P_2 is exactly the irreducible seed on which the rule reaches E_8 as the last finite point — a second network reading of the (2, 3, 5) hull, not a derivation of P_2 . [O] [verification/v299_hypergraph_0d_autonomous.py]
- **what a rewrite must inject.** Testing *which* load-bearing quantities are graph-spectral sharpens the open question: [E] the Kac marks are the Perron eigenvector ($Am=2m$) and the subleading eigenvalue is exactly $2 \cos(\pi/5)$ = the golden ratio (the 5-fold/icosahedral signature), but [E] the recovery rate $(2/3)^6$ is *not* any eigenvalue of the adjacency and φ_0^{ret} is not a rational edge fraction. The cusp weight $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$ is now *derived* from a minimal branching rule ([verification/v327_hypergraph_rewrite.py]); what

remains non-graph-spectral for a *full*-structure rewrite at leading order is sharpened by [verification/v396_phi0_icosahedral.py] (tree-level $\varphi_0^{\text{tree}} = F/(4h\pi)$); the puncture $48c_3^4$ alone stays open. [O] [verification/v312_hypergraph_substrate.py]

- **one coupled local rule (carrier $\times$ family).** On a 9×3 labelled grid (affine- E_8 nodes $\times$ family channels) a *single* coupled micro-step — network lazy diffusion on each cusp column plus the [verification/v327_hypergraph_rewrite.py] branching rule M on each node's family vector — has joint attractor marks $\otimes(w=0)$ and recovery rate $(2/3)^6$ after one clock hand ($2N_{\text{fam}}$ family steps); the [verification/v324_hypergraph_fiber.py] fiber product $T_{\text{net}} \otimes T_{\text{cusp}}$ emerges as the cusp-readout basis, and the [verification/v299_hypergraph_0d_autonomous.py] growth trajectory $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ is unchanged with the fiber attached. [E] [verification/v395_hypergraph_coupled.py]
- **the seed at leading order.** The tree-level retained seed $\varphi_0^{\text{tree}} = 1/(6\pi)$ is exactly $F/(4h\pi)$ on the icosahedral hypergraph ($F = 20$ triangular hyperedges, $h = 30$, $|\text{Aut}| = 120$): equivalently $(F/(g_{\text{car}}N_{\text{fam}}))c_3$, with $F/h = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$ the same survival ratio as [verification/v327_hypergraph_rewrite.py]; Gauss–Bonnet consistent with $c_3 = 1/(|\mathbb{Z}_2|4\pi)$. The puncture $48c_3^4$ remains analytic, not a graph fraction. [C] [verification/v396_phi0_icosahedral.py]

5 The μ_4 glue: how E_8 is really built

The decisive, now *proved* step: D_5 and A_3 have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the E_8 root norm.



Glue theorem (proved)

$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$ the E_8 root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$ is a *closed lattice-theoretic* construction — not a blind “positing of 248”. [E] [verification/v1_e8_glue.py]

6 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\text{car}}\}$? The following map summarises what is now a consequence rather than an assumption.

Sector	How it now arises	Status
gauge group / families	carrier 3+2, $\dim S^+=16$, $N_{\text{fam}}=3$, $\Omega_{\text{adm}}=48$	[E]
hypercharge	roots of the charge polynomial $bsY^2 + (s-b)Y - 1$	[E]
$U(1)$ index b_1	$b_1 = \frac{g_{\text{car}} 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}$	[E]
fine structure α^{-1}	unique root of $F_{U(1)}(\alpha) = 0$	[E]
flavor matrix	transport theorem (H1 proved, H2/H3 force the matrix)	[E]/[C]
electroweak scale	$v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides α^{-1} by 5)	[E]
cosmological constant	$\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)	[E]/[C]
Hubble scale	$H_0 \sim \sqrt{\Lambda}$	[C]
inflation amplitude	$A_s = \frac{N_{\star}^2}{24\pi^2} c_3^7$ in the R^2 branch	[C]
E_8	$D_5 \oplus A_3 + \mu_4\text{-glue}$	[E]

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Notable: the scale grammar is *one* exponential engine. The same number $\alpha^{-1} \approx 137$ generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

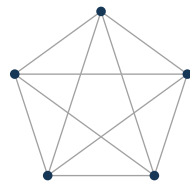
Typing note on Λ (a deliberate danger zone). “ Λ ” denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [E] readout), (ii) the dimensionless density quotient $\rho_{\Lambda}/\bar{M}_{\text{Pl}}^4$ (which carries a prefactor *branch*, hence [C]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

Cosmology is the Pascal action grammar of the carrier

With $x = v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph K_5 :

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \quad \frac{\rho_{\Lambda}}{\bar{M}_{\text{Pl}}^4} = x^{10} R_{\Lambda}, \quad (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}$$

EW is the unit x^1 , Hubble is the product over the $\binom{5}{1} = 5$ vertices, Λ is the product over the $\binom{5}{2} = 10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the E_8 root count $16 \cdot 5 \cdot 3 = 240$. “Hubble = slot product, Λ = pair product.” [E]



K_5

- $\binom{5}{1} = 5$ vertices — action x^1 (Hubble slot)
- $\binom{5}{2} = 10$ edges — Λ density pair x^{10}
- $\binom{5}{3} = 10$ triangles — dual sector
- $\binom{5}{4} = 5$ complements — inverse sector
- $\binom{5}{5} = 1$ determinant — closure

K_5 on the five carrier slots; the action powers x^1, x^5, x^{10} are its $\binom{5}{k}$ faces (mnemonic; the physical scale map stays [C]).

The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \quad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}, \quad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers ($\text{tr} = 9 = N_{\text{fam}}^2$, $\det = 8 = h(D_5)$, principal 2-minors 2,3,5 with product $30 = h(E_8)$). Through the φ_0^{ret} -ladder $\hat{m} = \frac{v_{\text{geo}}\pi}{\sqrt{2}} c(\varphi_0^{\text{ret}})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, **v49/v71**), with only the *absolute* amplitude scale reducing to the single dimensionful anchor v_{geo} . **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ are RG scheme-layer (m_H rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized* by a *TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$; see below). (*Full derivation: document 2, **tfpt_2_standard_model**.*) [[verification/v18_quark_yukawa.py](#)] [[verification/v20_lepton_c_derivation.py](#)] [[verification/v46_grand_mass_volume.py](#)]

7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*, α^{-1} is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the C^+ spinor and the puncture geometry $\mathbb{P}^1 \setminus \mu_4$ produce D_5 and A_3 , glued to E_8 ; *topologically*, the common discriminant $\mathbb{Z}_4$ (the μ_4 winding) forces the glue and Φ_{30} /Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence. **Why the compiled architecture is plausible.** It is (i) *parsimonious* (few independent assumptions $\Rightarrow$ low overfitting risk), (ii) *rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *checkable* (machine-checkable E_8 appendix, explicit fixed-point equation for α^{-1}). More explained from less — exactly the criterion that makes a theory more probable.

8 What role E_8 now plays exactly

E_8 is no longer the starting point but the *output format*:

- **Bookkeeping:** E_8 is the unimodular container in which carrier (D_5) and family (A_3) fit together consistently. Its numbers 240, 248 are carrier traces, not inputs.
- **Consistency test:** unimodularity ($\det 1$) is a hard condition; it closes only because the discriminant-form norms $5/4$ and $3/4$ (both already present in TFPT) add up to 2. E_8 thus tests the internal coherence of the two atoms.
- **Not an end in itself:** E_8 does not explain “everything”; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in E_8 .

New — the secondary atlas. Beyond that, E_8 is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (**tfpt_3_e8_audit_bootstrap**, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit is that $E_6 \times A_2$ reads the flavor matrix:

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one E_8 projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

9 What this means for the theory-of-everything question

The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic $\leftrightarrow$ unitary representation); the algebraic selector R , the Q spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\text{Spec}(Q_+)$ from D_4 , v69/v70; since v136/v139 the dual pair (d, n) forces R *columnwise*, d is pure anchor data, and $\text{Spec}(Q_+) = (1, 2, 3)$ is the A_3 -exponent grading on the seam curve’s cohomology, v137), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall (U_{point} , an anchor; Research Contract 1). [E]/[O]
2. **Gravity — the local field equation is parameter-free.** The metric-sector *field equation* is supplied by the entanglement first law $\delta S = \delta \langle K \rangle$ at fixed *volume* (Jacobson 2015): equilibrium forces the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $8\pi = 1/c_3$ (v358) and Λ from α , $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60) — the Einstein tensor (not Ricci) forced by Lovelock, so $\nabla^a T_{ab} = 0$ is an *output* (v359). The $R+R^2$ scalaron is the *action*/inflation layer (the next-order term of the same equilibrium route, v360), and v_{geo} is the theory’s single dimensionful input (v364). What remains is *not* the field equation but the equation-of-state status (Jacobson framing, not a from-action quantisation; an *external* candidate action — Bianconi’s entropic action, arXiv:2408.14391 — is quantified in v473 without changing this typing) and the one unit v_{geo} ; the ambient measure QG.AMB.01 is now a [C] redundancy (v369, conditional on SEAM.EQUIV.01+Bisognano–Wichmann), not missing dynamics, and *perturbative* graviton unitarity is established — the Stelle ghost is a Seeley–DeWitt truncation artefact (v304), the spin-2 sector ghost-free via the entire KMS form factor (v370) with the truncation tower decoupling (v380). [E] (local field equation) / [C] (perturbative unitarity, ambient redundancy)
3. **Strong CP — decoupled.** $\theta_{\text{eff}} = 0$ follows from three structural facts (γ_5 -Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [E] (given RP)
4. **Full dynamics — the admissible transfer model is closed, and the boundary QFT is now one relative object.** The mass gap is the *explicit* discrete gap $6 \log \frac{3}{2}$ of the C_6 transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector. On top of this the whole emergent-QFT round assembles the boundary theory into one relative object $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ — the *Modular Spectral Closure*: the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight (v259), the seam/carrier/ E_8 live on one Kummer/K3 (v260), all certified together (v261) — closed modulo the single keystone SEAM.EQUIV.01, itself now *closed modulo a cited theorem* (lattice + S_3 stack, Lean FORM.SEAM.MMST.01; v367/v368, v376–v379). *What this does not fully close*: the full measured pole masses (the absolute scale v_{geo} + standard RG), the absolute QCD scales (m_p/m_e , now a typed [C] transfer, v374), the complete PMNS dynamics (the angles are typed; the full mass/CP dynamics stay [C]/[O]), and the real 4D interacting QFT — the continuum limit, whose cited existence theorems stay [O] — with the ambient measure a [C] redundancy (v369). [E] (transfer model + boundary relative object) / [C] (continuum modulo cited, transfers, ambient redundancy)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (3357 jobs, 0 sorry, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [E]/[O]

In other words: the theory has shrunk from “many open audit points” to “*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms”. The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. (*Details and proof chains: document 2, tfpt_2_standard_model, Part III.*)

The closure ledger: what is done, and what genuinely remains

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

TOE piece	Status	If not complete: what closes it
gauge group, N_{fam} , hypercharge, b_1	[E]	— complete
$E_8 = (D_5 \oplus A_3) + \mu_4$, group closure	[E]	— complete
bootstrap $(g, \mu) = (5, 4)$, “8” in c_3	[E]	— complete (reverse glue $\mu^2 - 5\mu + 4 = 0$)
fine structure α^{-1}	[E]	— existence+uniqueness proved ; value is a numerical fixed point (ledger EM.FP.01), Ward origin [C]
fermion masses (φ_0^{ret} -ladder)	[E]/[O]	leptons exact; quark <i>ratios</i> closed (Plücker, v49/v71); absolute scale = anchor
mixings $\theta_{13}, \theta_{23}, \theta_{12}$ (PMNS angles)	[E]/[C]	θ_{12} cond. derived; θ_{23} maximal; <i>full</i> PMNS dynamics (mass scale, CP magnitude) stay [C]/[O]
flavour matrix / H2	[E]	dictionary <i>exact</i> (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); R forced columnwise by (d, n) (v136/v139); pairing values = atom identities (v142/v145); residue = <i>one</i> lift map
seam quantum clock (R1)	[E]/[C]	typed closed: reduced edge-sector clock (v124–v133); external VW edge = $2 \times$ reduced budget, full 4d det = firewall (v138); det-ratio cancellation derived in-family (v144)
inflation A_s, n_s, r , scalaron M	[C]	scalaron $M = c_3^{7/2} \bar{M}_{\text{Pl}}$ exact; n_s, r are bands over $N_* \in [50, 60]$ (reheating point 51.4, v86)
boundary QFT (Modular Spectral Closure)	[E]/[C]	one relative object TFPT _{QFT} : D_F = covariance induction (v258), cutoff = KMS weight $f_2/f_0=1$ (v259), seam/carrier/ E_8 on one K3 (v260), certified (v261) — closed modulo SEAM.EQUIV.01, itself closed modulo a cited theorem (v376–v379)
perturbative 4D S-matrix (S_{pert})	[E]/[C]	Epstein–Glaser-constructible (v269): one-loop quartic (v271) + gauge $\beta = (41/10, -19/6, -7)$ (v273), $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ with one-loop unitarity (optical theorem, v278); the spin-2 graviton sector is perturbatively unitary too — the Stelle ghost is a Seeley–DeWitt truncation artefact (v304/v370/v380)
<i>the actual TOE remainder — dimensionful units, typed transfers, and the cited-theorem ceiling on the seam</i>		
QFT measure (admissible sector)	[E]	closed: RP from cusp spectrum, OS reconstruction
full measured pole masses	[C]	ladder $\times v_{\text{geo}}$, dressed by standard RG (no new physics); v_{geo} over-determined (v274)
absolute QCD scales (m_p/m_e , Λ_{QCD})	[C]	typed runnable transfer with uncertainty budget (v374, contract FR.MPME.01); not a compiler number
full PMNS dynamics (mass scale, CP)	[C]/[O]	angles typed; CP strength $J_{\text{PMNS}} = -0.0297$ derived (v270); absolute ν -scale = one seesaw ratio (v272); three named post-hoc [O] candidates: third-gen ladder-base (v467), Δm^2 ratio = $ J $ (v468), carrier-normalised one-parameter seesaw (v481) — record unchanged
(1) SEAM.EQUIV.01 — one named theorem (<i>raw seam</i> = $(E_8)_1$ at $\tau=i$)	[C]	closed modulo a cited theorem : the explicit lattice model (v367/v368) and the S_3 stack ($c=8$, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, Lean-formalised as FORM.SEAM.MMST.01 (collar hypotheses kernel-proved, MMST/OS as named cited axioms, clean #print axioms); the <i>only</i> residual that stays [O] is the abstract continuum existence of the scaling limit (v336); extension leg crossed-product-certified, realisation at invariant level (v469)
(2) QG.AMB.01 — ambient measure (<i>redundancy, not missing dynamics</i>)	[C]	discharged as a redundancy (v369; v379 RP→OS), conditional on SEAM.EQUIV.01+Bisognano–Wichmann: a certification object, gap-decoupled (margin $1.648 > 0$, v330/v335) so no readout depends on it; obstruction characterised (v332) + conditionally resolved by GHP+IDG (v334); Tier-B’s holomorphy bit (det $K=1$, v277) is absorbed into SEAM.EQUIV.01
analytic boundary origin of π ($c_3 = \frac{1}{8\pi}$)	[C]	hardened : $c_3 = 1/(\mathbb{Z}_2 \oint_{S^2} K) = 1/(8\pi)$ (Gauss–Bonnet); “8” = rank E_8 ; π stays the one primitive
$\eta_B, m_p/m_e$, Koide, dark matter	[C]	typed runnable transfer solvers with kill tests (v371–v374, observatory CI v375; see document 4)

Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in *verification/status_ledger.csv*.

The honest bottom line: the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one φ_0^{ret} -ladder (lepton rationals exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed), and on top of it the whole boundary QFT is now *one relative object* TFPT_{QFT} (the *Modular Spectral Closure*, v258–v261: D_F a covariance induction, the cutoff the seam KMS weight, seam/carrier/ E_8 on one Kummer/K3) — closed modulo the single keystone SEAM.EQUIV.01. *What that closure does not reach* is exactly the dimensionful/ambient remainder: the full measured pole masses (ladder $\times v_{\text{geo}}$ + standard RG, **[C]**), the absolute QCD scales (m_p/m_e , now a typed **[C]** transfer, v374), the complete PMNS dynamics (**[C]**/**[O]**), and the nonperturbative *ambient* measure QG.AMB.01, now itself a **[C]** redundancy (v369). (The metric-sector *field equation* is parameter-free at the local level — the full covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ from fixed-volume entanglement equilibrium, v358/v359 — and the ambient *measure* is no longer an open structural item but a certification object, conditional on SEAM.EQUIV.01+Bisognano–Wichmann.) The *perturbative* 4D QFT is now built: the spectral-action S-matrix is Epstein–Glaser-constructible and LSZ-bridged to the physical asymptotic S-matrix with one-loop unitarity for the matter+gauge sector (v269–v278), and the spin-2 graviton sector is perturbatively unitary too — the R^2/Weyl^2 Stelle ghost is a Seeley–DeWitt *truncation artefact* (v304), ghost-free via the entire KMS form factor and the Barnes–Rivers decomposition (v370) with the truncation tower decoupling (v380); the whole bedrock reduces to one geometric postulate, *the raw seam is the flat $\tau=i$ pillowcase* (v276), with Tier-B’s holomorphy bit $\det K=1$ (v277) absorbed into SEAM.EQUIV.01. The former single open lemma, the named theorem SEAM.EQUIV.01 — *the raw reflection-positive seam state is the holomorphic $(E_8)_1$ net at $\tau=i$* — is now *closed modulo a cited theorem*. (The flat $\tau=i$ geometry and the E_8 holomorphy are *two faces of one object*: the $(E_8)_1$ character is $\chi_{E_8}=j^{1/3}$, so $\chi_{E_8}(i)=1728^{1/3}=12$ and $\tau=i$ is at once the order-4 CM point and the $(E_8)_1$ partition-function modulus, v282.) The explicit lattice model (v367/v368) and the S_3 stack ($c=8$, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, and it is Lean-formalised as FORM.SEAM.MMST.01 (collar hypotheses kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems as named cited axioms, clean **#print axioms**); the *only* residual that stays **[O]** is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since re-founded on the peer-reviewed crossed-product package ($h_s=16/16=1 \in \mathbb{Z}$ is the Longo–Rehren locality criterion; KLM $\mu=4/2^2=1$; AGT/AMT the second witness), the realisation input reduced to invariants (R1’; computed $|C|=1$, $\nu=16$), Lean parallel route **seamResidualClosed’** (v469) — so it is closed *modulo a cited theorem*, not solved, Lean-pinned to exactly those named steps plus the derived Recovery gap $\Delta = 6 \ln(3/2) \approx 2.43 > 0$ (v308); the conformal-deck premise QGEO.SYM.01 is its *downstream* definitional face (v323). This is by design the *one* irreducible structural input of the theory — the role the *constancy of c* plays in relativity. It is not a defect to be removed but the single, sharply-localised, falsifiable premise on which the whole boundary closure turns; with the lattice/ S_3 pinning and the Lean formalisation in place, “closing the theory” has reduced to invoking exactly the two cited continuum-existence theorems, not an open-ended search. The full sprint-by-sprint reduction that pinned SEAM.EQUIV.01 is recorded in the changelog, not replayed here. The seam seed is *hardened*: $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K dA) = 1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete “8” equal to rank E_8 and only the Gauss–Bonnet π left as the single irreducible primitive. What remains for a full TOE is then the cited continuum-existence theorem on the seam and the one unit v_{geo} , plus the standard-RG dimensionful masses and the typed frontier transfers (the ambient QG measure being a **[C]** redundancy, v369). No load-bearing discrete parameter remains free, and π is the lone continuous primitive. *4D-QFT fork policy* (frozen, v265): the canonical 4D reading is *boundary only* — a 4D-GUT is *not claimed by default* (E_8 is the audit hull; SM-only unification is killed, v246); carrier-native Pati–Salam is a separately-typed, falsifiable UV branch (thresholds + proton decay). Not an open ambiguity, a decision tree.

10 TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram

The “compiler” framing places TFPT in an older lineage — the idea that physics is, at bottom, the output of a short computation — and it is worth stating precisely what TFPT shares with that lineage and where it deliberately parts company, so the family resemblance is not mistaken for a claim the theory does not make. Three landmarks fix the coordinates. **Zuse** (*Rechnender Raum*, 1969) proposed that the universe is literally a (cellular-automaton) computation. **Schmidhuber** sharpened this into an *algorithmic* theory of everything (1997, 2000): the shortest program that outputs a universe is the natural prior — a physics reading of Kolmogorov complexity and Solomonoff induction — while his *Speed Prior* (2002) adds that a computed universe should be one that is *fast* to compute, and his low-complexity-art programme (1997) identifies beauty with compressibility. **Wolfram** (*A New Kind of Science*, 2002; the Physics Project) recasts the same instinct combinatorially: space–time as a hypergraph updated by simple rewrite rules.

Three shared instincts

1. **Minimal description length as the selection principle.** “No load-bearing free number, only π ” is, in these terms, a Kolmogorov-style compression statement — the plausibility discussion below already names it *Kolmogorov-like compression: many numbers from little* — and the grammar-internal null model of the next section turns it from rhetoric into a measured bound.
2. **Deterministic, resource-bounded generation.** The readouts fall out of a chain of exact identities in about a second, not a search over a model space — the same spirit as the Speed Prior’s preference for *fast*-computable worlds, though TFPT makes no claim to instantiate a speed prior.
3. **Compressibility as the aesthetic.** “Choosing E_8 is choosing the icosahedron” selects the maximally symmetric, shortest-to-describe object; the affine Kac marks (1, 2, 3, 4, 5, 6, 4, 2, 3) carry a fourfold identity (representation dimensions = network attractor = spectral certificate = local balance weight). This is Schmidhuber’s *beauty = compressibility*, realised on the Standard-Model alphabet rather than on pixels.

Three deliberate divergences

- (A) **One forced program, not an ensemble.** Schmidhuber’s shortest program computes *all* universes and privileges none; TFPT bets the opposite — an over-determined bootstrap fixed point that *forces* a single dimensionless skeleton ($g_{\text{car}}=5$ fixed three independent ways). TFPT is a candidate for *the* short program of our constants, not a measure over programs.
- (B) **The description language is algebraic.** TFPT’s compression is measured in a Lie/lattice/VOA language, not in Turing bit-strings; Kolmogorov complexity is language-dependent up to an additive constant, so the honest question — is the shortness real, or hidden in the E_8 machinery? — is exactly what the null model and the “forced three ways” bootstrap are built to answer. A Wolfram-style rewrite reading sharpens the same point rather than dissolving it: the (2, 3, 5) seed is the terminal of a minimal $T(2, 3, r)$ growth rule (Cartan determinants 3, 2, 1, 0, -1 for $E_6, E_7, E_8, \hat{E}_8, E_{10}$), so the rewrite picture *re-formulates* axiom P2 rather than removing it — both pictures bottom out on the same single irreducible seed.
- (C) **Falsifiability — the one thing TFPT adds.** As *physics*, the Zuse/Schmidhuber algorithmic TOE and Wolfram’s rule search are essentially unfalsifiable: no dated measurement can kill them. TFPT freezes dated, kill-tested predictions (the registry `predictions_frozen.json`, the predictions section below) and so puts the computational-universe intuition at genuine empirical risk. That, not the compression rhetoric, is the load-bearing difference.

Status. This section is *positioning*, not a new result: it introduces no formula and moves no status marker. Its machine-checked hooks already live in the suite — the grammar-internal null-model bound and the icosahedral/McKay bedrock with its network attractor; the Wolfram-style rewrite correspondence is worked out separately in the repository’s hypergraph note.

11 Does this make the theory more probable?

Plausibility balance. *Pro:* (a) two inputs instead of many; (b) α^{-1} to 2.9×10^{-10} (CODATA-2022, 1.9σ) as a *fixed point*, not a fit; (c) E_8 constructed, not postulated; (d) one compiler generates several sectors $\Rightarrow$ fewer degrees of freedom, more predictive power.

Contra/open: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ($m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$, modulo the seam-misalignment lemma). A final verdict “it maps reality” requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

12 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

Observable	TFPT value & derivation	Experiment (running / upcoming)	St.
solar $\sin^2 \theta_{12}$	prediction of record $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (cond.)	JUNO : first 59.1 d run 0.3092 ± 0.0087 (compatible), precision phase pending	[E]/[C]
reactor $\sin^2 \theta_{13}$	$\varphi_0^{\text{ret}} e^{-5/6} = 0.0231$; seed $\times$ carrier-trace $e^{-\gamma}$, $\gamma = \frac{5}{6}$	Daya Bay / RENO / JUNO (PDG 0.0220)	[E]
atmospheric $\sin^2 \theta_{23}$	$\approx \frac{1}{2}$ ($\mu\tau$ -symmetric limit; octant not selected)	NOvA / T2K / DUNE (octant ambiguity; NuFIT 6.0 pre-JUNO baseline)	[C]
fine structure α^{-1}	137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)	CODATA-2022 (data through 2022) 137.035999177(21): dev. 2.9×10^{-10} (1.9σ of its uncertainty)	[E]
tensor ratio r	$r = \frac{12}{N_*^2} \in [0.0033, 0.0048]$ (frozen band, $N_* \in [50, 60]$); scalaron-reheating point 0.0045 ($N_*=51.4$, v86)	now $r < 0.036$ (BICEP/Keck BK18); CMB-S4 ($\sigma_r \leq 5 \times 10^{-4}$, obs. ~ 2033)	[C]
tilt n_s	$n_s = 1 - \frac{2}{N_*} \in [0.960, 0.967]$ (same band); scalaron-reheating point 0.9611 ($N_*=51.4$, v86)	Planck (0.9649 ± 0.0042); CMB-S4 sharpens	[C]
amplitude A_s	$\frac{N_*^2}{24\pi^2} c_3^7$; 2.0×10^{-9} at $N_*=55$ (the measured A_s prefers fast reheating, v86)	Planck (2.1×10^{-9})	[C]
neutron EDM	$\theta_{\text{eff}} = 0 \Rightarrow d_n$ essentially null (structure + RP)	PSI nEDM / SNS (running)	[E]
CP phase δ_{CKM}	$\delta = \frac{\pi}{3} + 3\lambda^2 = 68.65^\circ$ (frozen of record, v88); survives at $+0.98\sigma$ vs γ_{PDG}	LHCb / Belle II γ combination; decision at $\sigma_\gamma \leq 0.96^\circ$	[E]
cosmic birefringence β	$\beta = \frac{\varphi_0^{\text{ret}}}{4\pi} = 0.2424^\circ$ (seam readout)	ACT DR6 hint $0.215^\circ \pm 0.074^\circ$ (systematics not yet understood — not a validation target); Simons Obs. / LiteBIRD	[C]
$0\nu\beta\beta$ / ordering	Majorana branch; normal-ordering preference	LEGEND / nEXO , DUNE, KATRIN	[C]
flavor invariants	$\det R = h(D_5) = 8$, principal minors (2, 3, 5), $\chi_R = t^3 - 9t^2 + 10t - 8$ (exact)	any future CKM/PMNS global fit must satisfy these	[E]
H_0 vs. Λ	$v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$, $\Lambda \sim e^{-2\alpha^{-1}}$, $H_0 \sim \sqrt{\Lambda}$: one engine	SH0ES / DESI / Planck (Hubble tension)	[C]

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status_ledger.csv.

The two decisive near-term tests

- (1) **JUNO** and θ_{12} . JUNO is *taking data since August 2025* and targets sub-percent $\sin^2 \theta_{12}$. TFPT commits to the single prediction of record $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ (misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$). The first 59.1-day JUNO run reports $\sin^2 \theta_{12} = 0.3092 \pm 0.0087$ — *compatible*, but the precision phase is still pending; this is a first compatibility check, not yet the decisive test. A converged central value away from ~ 0.307 at high significance falsifies the seam mechanism.
- (2) r . The R^2 scalaron with M fixed by c_3^7 predicts a small $r = 12/N_*^2$: the frozen band $r \in [0.0033, 0.0048]$ over $N_* \in [50, 60]$, with the scalaron-reheating point $r = 0.0045$ ($N_*=51.4$, v86) — already below the current bound $r < 0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r \leq 5 \times 10^{-4}$, first observations ~ 2033 –34). A detection of $r \gtrsim 0.01$ would be incompatible with the R^2 branch on which M_{Pl} and A_s rest.

Structural (non-experimental) checks. Independent of any apparatus, the machine-checkable E_8 appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate E_8 ; and the discipline rule “*every load-bearing number appears in at least one E_8 projection*” (atlas note) turns the number stock into a falsifiable raster rather than

numerology. The numerology charge is additionally *quantified* by an explicit null model: under the declared formula grammar the alignment density is anomalously high — a random theory of equal formula complexity essentially does not reproduce the scorecard. This is a *grammar-internal null-model bound*, not a Bayesian probability that TFPT is true; the explicit figure and its controls are stated in the verification appendix [E] [verification/v100_numerology_null_mc.py].

Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the R^2 branch carrying M_{P1}, A_s ;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\text{eff}} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

Blind-prediction registry (frozen 2026-06-09, machine-enforced). Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [verification/v84_frozen_registry.py] re-derives each decimal from the two axioms on every suite run (formula $\leftrightarrow$ value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one* θ_{12} prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. r and n_s enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

What likely follows next.

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, `v49/v69/v71`); the only residual is the *absolute* amplitude scale, an anchor (U_{point}) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over S^+ ;
- a sharper geometry $\leftrightarrow$ gravity bridge once the Hodge/ R^2 branch is lifted from [C] to [E].

Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\text{car}}\}$ via $D_5 \oplus A_3 + \mu_4$, the E_8 container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

Appendix: computational verification (reproducibility)

Every claim in this document marked [E] (exact identity), [E] (Lie/lattice theorem) or [E] (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\text{car}}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references [verification/...] throughout the documents point to the exact script; the table below is the master map (its source is the shared fragment `tex-artefacts/verification.tex`).

Script	What it machine-checks
v1_e8_glue.py	glue theorem $E_8=(D_5\oplus A_3)+\mu_4$: $\text{disc}=\mathbb{Z}_4$, $q(D_5)+q(A_3)=\frac{5}{4}+\frac{3}{4}=2$, $240=16\cdot 5\cdot 3$, $248=240+8$
v2_carrier_pascal.py	$g_{\text{car}}=5$ unique Pascal solution; $16=1+5+10$, $N_{\text{fam}}=3$, $\text{rank } E_8=8$, $\Omega_{\text{adm}}=48$, $b_1=\frac{41}{10}$
v3_em_alpha.py	EM closure: unique root of $F_{U(1)}(\alpha)=0$, $\alpha^{-1}=137.0359992168$, existence/uniqueness, inverse test, ablation
v4_flavor_matrix.py	residue matrix R : $\det=8$, minors $\{2, 3, 5\}$, $\chi_R=t^3-9t^2+10t-8$, SNF $\text{diag}(1, 1, 8)$, $\ R\ _F^2=78$, $\sum L=40$
v5_e8_cascade.py	cascade $D_n=60-2n$: endpoints, exponent rungs $\rightarrow 240$, IR tail 46080, variance 6552
v6_bootstrap.py	reverse glue $\mu^2-5\mu+4=0$; $g_{\text{car}}=5$ three ways; $8=\text{rank } E_8=h(D_5)=\varphi(30)$
v7_gravity_cosmo.py	scalaron exponent 7, $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$, A_s, n_s, r , Ω_b , $\sin^2 \theta_{12}$
v8_horizon.py	horizon code: $\frac{1}{2\pi}=4c_3$, $1920= W(D_5) $, S_{dS} , Page, $\beta_{\text{rad}}=0.2424^\circ$
v9_neutrino_texture.py	solar angle: explicit $\mu\tau$ Majorana texture $\rightarrow \sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$, $\theta_{23}=45^\circ$, $\theta_{13}=0$
v10_projection_involution.py	Q, Σ algebra: $K=R+Q\Sigma$, $L=R+Q(I+\Sigma)$; χ_Q, χ_K ; det ladder $3\cdot 4\cdot 8\cdot 20=1920= W(D_5) $; $a^\top K a=41$
v11_unique_KQ.py	K, Q are the <i>unique</i> nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)
v12_mass_generation_polynomials.py	sector/generation polynomials of K and the anchor-block det ladder (9, 10, 16, 40)
v13_open_gates.py	gate closures: $M=41=10b_1=\sum L+N_\Phi$; $Q_+=A_3$ exponents, $Q_-^2=N_{\text{fam}}$
v14_carrier_uniqueness.py	$g_{\text{car}}=5$ unique ($N_{\text{fam}} \in \mathbb{Z}^+$, $\text{rank } E_8=8$); split (3, 2) from $b+s=5$, $b^2+s^2=13$; $\text{Tr } Y=\text{Tr } Y^3=0$
v15_bootstrap_classification.py	$D_5\oplus A_3$ unique familyful cyclic-glue of E_8 (16-spinor + 3 families), glue $\mathbb{Z}_4$
v16_solar_dual_anchor.py	$a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)$; $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$; full PMNS
v17_hexagonal_resolvent.py	finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark c backbone)
v18_quark_yukawa.py	quark source ratios $\frac{m_u}{m_d}=0.470085$, $\frac{m_c}{m_s}=13.61$, $\frac{m_t}{m_b}=40.80$ exact; lepton $c=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$; full hierarchy
v19_monodromy_moduli.py	exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ (3^7); D_4 $SU(3)$ monodromy constructible but D_4 -fixed locus positive-dim $\Rightarrow$ exact quark c reduce to (U)
v20_lepton_c_derivation.py	lepton c 's derived: $\delta=\frac{1}{2}$ resolvent $c= \mu_4 ^w/(\frac{5}{4}-\cos \frac{r\pi}{3}) \Rightarrow \frac{16}{7}, \frac{4}{3}$; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}=\frac{32}{9} \Rightarrow \frac{7}{6}$
v21_solar_product_quark.py	solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$), $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}$; quark law fails ($\frac{1}{7}$ vs 0.724)
v22_open_gates_audit.py	residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)
v23_anchor_generator.py	anchor $a=(1, 1, 2)$: $e_k=(4, 5, 2)$, $c_3=\frac{1}{2e_1\pi}$; $p_n=2+2^n \Rightarrow 240, 248$, binary ladder; inputs $\rightarrow \{a, \pi\}$
v24_quark_ratio_closure.py	quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$; rational c gauge, Λ_q in $O(1)$
v25_frontier_conjectures.py	conjectures: Koide $Q_{\text{src}}+\frac{\varphi_0^{\text{ret}}}{24}\approx\frac{2}{3}$; axion $f_a=M_{\text{scal}}/128\rightarrow m_a\approx 23.8 \mu\text{eV}$

continued on next page

Script	What it machine-checks (<i>continued</i>)
v26_flavor_ frontier_ unification.py	the “11” is not uniquely forced ($\Rightarrow$ [C]); flavor frontier unifies to one (U) gate; cusp config on the parabolic stability wall
v27_wall_ representative.py	explicit balanced wall rep W_{wall} (cols = perm(0, 1, 2), rows $w=(2, 1, 1)$, pardeg 0); selector $\det R=8$, $\text{Spec}(Q_+)=\{1, 2, 3\}$
v28_gravity_fR.py	$R+R^2$ closed: $f(R)$, $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ scalaron, $N_*=57\rightarrow n_s, r, A_s$; full metric measure open; $248c_3^2 < 6\log \frac{3}{2}$
v29_research_ contract_certs.py	research-contract certificates: $C_U^{(1)}$ wall enumeration (1296 $\rightarrow$ 144 $\rightarrow$ 5 orbits; symmetry alone does not pick W_{wall}) and $G5$ gap-dominance $2\ V\ < \Delta$
v30_d4_character_ variety.py	$C_U^{(2)}$: full D_4 -fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the $R4'$ selector establishment)
v31_R_ dictionary.py	$R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); R integer but ρ -invariants continuous $\Rightarrow R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)
v32_rh_ splitting.py	RH route: $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$ exact $\Rightarrow$ splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \Leftrightarrow \text{diag}(A_0) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ (algebraic); $\prod M_k = I$ + R -bridge stay open (c_u/c_d not forced)
v33_explicit_flat_ bundle.py	explicit valid (U_{wall}) flat bundle (RH solve): cusp + $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ + $\ M_\infty - I\ \sim 10^{-9}$ ($\prod M_k = I$) + irreducible (case A)
v34_h2_bridge_ attempt.py	H2 bridge: explicit per-puncture M_k (cusp, $\prod M_k = I$); honest negative — natural extraction $\neq$ lepton amplitudes, the $\Gamma^{\min}$ dictionary is the remaining input (c_u/c_d not obtained)
v35_quark_qcd_ boundary.py	the open gates are the discrete $\leftrightarrow$ continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)
v36_spectral_ action_g2.py	G2: spectral action $\rightarrow R+R^2$ ($a_2 \sim -R/3$, $a_4 _{R^2} \sim R^2/72$); $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$, c_3^7 fixes f_0 ; gap-decoupling $\Delta_{\text{eff}} = 1.648 > 0 \Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target
v37_plucker_ anchor.py	anchor-plane Plücker apparatus: $\ \text{Pl}(K)\ _1 = 11$, $\ \text{Pl}_R(K)\ _1 = 26$, $\sum \text{Pl}_R(L) = 60$; pencil $\det(K+xQ) = 3x^3 + 7x^2 + 6x + 4$; $K_e - K_u = a$, $L_e - L_u = \text{Exp}(A_3)$; lepton ring $c_e c_\tau = \mathbb{Z}_2 c_\mu$; Koide $u \rightarrow \frac{53}{54}u$ ([C])
v38_uwall_ killswitch.py	(U_{wall}) U2 kill-switch hardened: D_4 -cusp rep irreducible at all 400 sampled points (commutant=1), mixing ≥ 0.35 , reducible locus measure-zero $\Rightarrow$ case A generic, (U_{wall}) survives; amplitude dictionary still research-level
v39_uwall_ selectors.py	(U_{wall}) selector side closed on the explicit point: splitting type = anchor $a=(1, 1, 2)$, pardeg = 0; $\det R=8=n\cdot a$, $\text{Spec}(Q_+)=\{1, 2, 3\}=3\alpha+1$ read off the bundle; only the amplitudes c_u/c_d ($\Gamma^{\min}$ Hitchin) remain
v40_harmonic_ metric.py	(U_{wall}) harmonic metric = FINITE linear algebra (polystable $\Rightarrow$ unitary $\Rightarrow\Phi=0$): unique invariant form $H>0$, unitarised $\bar{M}_k$ cusp-class with $ \bar{M}_k \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$; the feared Hitchin PDE collapses to unitarisation + combinatorial R
v41_leg_ assignment.py	(U_{wall}) final leg test: amplitudes = $\delta = \frac{1}{2}$ resolvent (closed); $\Lambda_\mu > 1$ rules out holonomy moduli; harmonic holonomy $ \text{diag} = (0, \frac{1}{2}, \frac{1}{2})$ supplies $\delta = \frac{1}{2}$ (suggestive); honest negative on literal c_u/c_d reproduction
v42_exterior_ leg.py	(U_{wall}) Exterior Leg Lemma: u, d share $(r, w) = (1, 1)$ so any scalar leg gives $c_u/c_d \equiv 1$; the u/d split is the $\Lambda^2 F$ area $\text{Pl}_{1,a}(K)$, $\ \cdot\ _1 = 11$ generated $\Rightarrow \frac{5\cdot 11}{9\cdot 13} = \frac{55}{117}$ [E]; phys. identification [C]
v43_exterior_ bridge.py	(U_{wall}) $\Lambda^2 F$ bridge: $\Lambda^2(\bar{M}) = \bar{M}$ (exterior leg = $\bar{3}$) confirmed; continuous action $\neq$ integer $\text{Pl}(K) \Rightarrow$ bridge is the discrete H2 invariant ($\det R=8$, $\text{Spec}(Q_+)$ confirmed), not continuous

continued on next page

Script	What it machine-checks (<i>continued</i>)
v44_carrier_exterior.py	(U_{wall}) Lie-level grounding: $16=\Lambda^{\text{even}}(5)=1+10+5$; $u^c \in \Lambda^2(5)$, $d^c \in \Lambda^4(5) \Rightarrow$ exterior degrees $(2, 4)$, $\text{sum}=6= R^+(A_3) $, $\text{diff}=2= \mathbb{Z}_2 $; the exterior leg is the carrier's own Λ^2 grading (no special math)
v45_family_exterior.py	(U_{wall}) the “11” is Pascal: $ \text{Pl}(K) =(1, 4, 6)=\{\Lambda^0, \Lambda^1, \Lambda^2\}(\mu_4)$; $\ \text{Pl}(K)\ _1=11=\sum_{k \leq 2} \binom{4}{k}=16-g_{\text{car}}$ (cumulative ext of μ_4 up to $\deg u^c=2$); $15=\dim \mathfrak{su}(4)=10+5=N_{\text{fam}}g_{\text{car}}$
v46_grand_mass_volume.py	absolute mass-scaling is [E], not a product rule: sector det $(\varphi_0^{\text{ret}})^6, (\varphi_0^{\text{ret}})^9, (\varphi_0^{\text{ret}})^{10}$ (exponents $ R^+A_3 , N_{\text{fam}}^2, \mathcal{A}_\Lambda \Rightarrow$ $\det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{25}=(\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$; Q -rows $(4, 5, 6)$ sum $15=\dim A_3$; $25+15=40=\sum L$
v47_selection_theorem.py	Boundary Carrier Selection (Thm A): $\mu_4 \rightarrow A_3, N_{\text{fam}}=3$; 16-spinor $\rightarrow D_5$; $q(D_5)+q(A_3)=2$, glue $4= \mu_4 $; only $D_5 \oplus A_3$ meets all four conditions ($D_8, E_7+A_1, E_6+A_2, A_4+A_4, A_8$ excluded) [E]/[C]
v48_em_ward.py	EM boundary Ward (Thm C): $F_{U(1)} = \text{Maxwell } \alpha^3 - \text{Calderón } 2c_3^3 \alpha^2 - \text{transport}$ $8b_1 c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$; $8b_1 = \frac{4}{5} M = \frac{164}{5}$, $-\frac{5}{4}=q(D_5)$ [E]; Ward origin [C]
v49_readout_rigidity.py	Readout Rigidity (Thm U2): $c_u/c_d = \frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q} \Rightarrow$ no point-uniqueness needed; $(U_{\text{wall}})=U_{\text{unitary}}+U_{\text{H2}}+U_{\Lambda^2}+U_{\text{point}}$, only U_{point} open
v50_q_geometry.py	Q geometry (Thm Q): $Q_+ = \text{diag}(3, 2, 1)$ -block $\chi=(t-1)(t-2)(t-3)$, Q_- $\chi=t(t^2-3)$; Q unique under rows $(4, 5, 6)/\text{cols}(9, 5, 1)/\text{spectra}$; geom. origin [C]
v51_boundary_half_step.py	glue norms $=(g_{\text{car}}, N_{\text{fam}})/ \mu_4 $: $q(D_5)=\frac{5}{4}$, $q(A_3)=\frac{3}{4}$; four ops forced $\rightarrow \{\text{sum}=2$ root norm, $\text{diff}=\frac{1}{2}= \mathbb{Z}_2 / \mu_4 =\delta$, $\text{prod}=\frac{15}{16}$, $\text{ratio}=\frac{5}{3}\}$. So $\delta=\frac{1}{2}$ is the carrier–family count over the glue index, not chosen [E]
v52_pencil_endpoints.py	$K+xQ$ pencil endpoints: $P(-1)=2= \mathbb{Z}_2 $, $P(0)=4= \mu_4 $, $P(1)=20=\det L$, $P(2)=68=2p_5(a)$; $\det(K-Q)=2$, $\text{tr}=N_{\text{fam}}$ (audit ext. of v37) [E]
v53_compiler_core.py	Big picture: the whole integer skeleton from $(g_{\text{car}}, N_{\text{fam}})=(5, 3)$ — rank $E_8=8$, $ \mathbb{Z}_2 =2$, Pythagorean $\Delta_Y=g_{\text{car}}^2=25=N_{\text{fam}}^2+ \mathbb{Z}_2 \cdot \text{rank } E_8=9+16=\dim S^+$ (diff. of squares); anchor char-poly $(t-1)^2(t-2)$ unique [E]
v54_seam_horizon_keystones.py	The “8” in $c_3=\frac{1}{8\pi}$ overdetermined and gravitationally aligned $(2 \mu_4 =\text{rank } E_8=h(D_5)=\varphi(30)=8\pi)$ Hawking/Einstein); one boundary transport operator (spec $\{1, (2/3)^6, (1/3)^6\}$, $\lambda_2=(2/3)^6$) governs <i>both</i> SM flavor gap and horizon Page recovery [E]
v55_coxeter_cycle.py	Computed E_8 Coxeter element: order exactly $30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}$, eigenphases = exponents = totatives of 30, so rank $E_8=8=\varphi(30)$ = live phases of the order-30 cycle; one α^{-1} sets EM, S_{dS} and Λ ($S_{dS}\rho_\Lambda=32\pi^4$) [E]
v56_unique_attractor.py	Gapped boundary transport (gap $6 \log \frac{3}{2} > 0$) $\Rightarrow$ <i>unique</i> attracting fixed point, geometric rate $(2/3)^6$ (two starts $\rightarrow$ same direction); Coxeter in $ \mu_4 =4$ planes; entropy reset = rank-1 projector [E]
v57_horizon_crosslinks.py	Horizon cross-links: $c_3=\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ($1/4=1/ \mu_4 $ entropy density); Hod QNM $\omega_R/T_H = \ln 3 = \ln N_{\text{fam}}$; Hawking power denom $1920= W(D_5) $; one α^{-1} for EM/ S_{dS} / Λ /Hubble [E] (+ [C] identification)
v58_seam_horizon_chain.py	Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided S^2 Gauss–Bonnet $c_3=1/(2 \cdot 4\pi)=\frac{1}{8\pi}$; KMS $4c_3\kappa$; $S_{BH}=M^2/2c_3$; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) [O] open
v59_area_law_evidence.py	Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim \frac{\epsilon}{3} \log L$), by the <i>same</i> gap that fixes the attractor; $8= \mathbb{Z}_2 \mu_4 $. The c_3 coefficient stays [O] open
v60_lambda_metrology_branch.py	Λ branch selection: $(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$, mis-scale $2c_3/\delta_{\text{top}} = \frac{64\pi^3}{3} = 661.47 \Rightarrow G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215^\circ \pm 0.074^\circ$ vs 0.2424° (0.37σ) [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v61_cft_bridge.py	Boundary-CFT mirror: level-1 WZW central charges $c(E_8)=8$, $c(D_5)=5$, $c(A_3)=3$ = (rank E_8 , g_{car} , N_{fam}); conformal embedding $c_{\text{coset}}=0$; $SU(3) \leftrightarrow SU(4)$ reconciliation $N_{\text{fam}}= \mu_4 -1$, $4 \rightarrow 3+1$ (one geometry $\mathbb{P}^1 \setminus \mu_4$, two sides) [E]
v62_data_scorecard.py	TFPT vs 2024/25 data: α^{-1} , $\sin^2 \theta_{12}$, β_{rad} match; $\sin^2 \theta_{13}$ (2.0σ) and Starobinsky n_s vs DESI ($\sim 2\sigma$) mild tensions; θ_{23} open; $r \approx 0.004$ future falsifier [E]
v63_seam_engineering_index.py	Exact Seam-Engineering Index $\Xi=2\ V\ /\Delta=\frac{31}{24\pi^2 \log(3/2)} \approx 0.323$ ($2\ V\ =\frac{31}{4\pi^2}$, $248=\dim E_8$, $31=1+h^\vee$), $\Delta_{\text{eff}} \approx 1.648$; four-class possibility taxonomy [E]/[O]
v64_causal_boundary_nogo.py	Causal Boundary Engineering <i>conditional no-go</i> : gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap $\Rightarrow$ ANEC $\Rightarrow$ Topological Censorship $\Rightarrow$ no traversable shortcut; only ER=EPR bridge survives [E] (core)/[C] (chain)
v65_falsification_layer.py	Prediction layer with explicit failure thresholds: 3 core ($v_{\text{GW}}=c$, no 4th gen, seam=8), 4 observational (β , r , n_s , θ_{12}), no-go + unitarity; $N_\star$ demoted to [C] input; none violated [E]
v66_e8_casimir_degrees.py	Organizing simplification: the compiler atoms <i>are</i> the E_8 Casimir degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ ($2= \mathbb{Z}_2 $, $8=\text{rank}$, $14=\dim G_2$, $20=\det L$, $24= W(A_3) $, $30=h(E_8)$); $\sum=128=2^7$ (dS prefactor), $\sum \exp=120= R^+(E_8) $ [E]
v67_area_law_coefficient.py	Central-theorem <i>structural closure</i> : Fursaev–Solodukhin $S=4\pi kA$, $k=c_3/2 \Rightarrow S=2\pi c_3 A=\frac{1}{4}A$ ($2\pi c_3=\frac{1}{4}$); $c_3=\frac{1}{8\pi}$ is the <i>unique</i> value with replica-coeff = BH $\frac{1}{4}$. Reduces to the one coeff $k=c_3/2$ (then <i>forced</i> , v73) [E]
v68_seeley_dewitt_residual.py	Residual <i>resolved as an anchor</i> : $a_2=-\frac{d}{192\pi^2}R \Rightarrow$ the <i>absolute</i> $\frac{1}{16\pi G}$ is UV-sensitive ($\propto d\Lambda^2 f_2$), so the <i>scale</i> $1/G$ is the one dimensionful anchor (the dimensionless $k=c_3/2$ is forced, v73); cutoff-indep. R^2 /scalon derived [E]
v69_d4_q_geometry.py	D_4 -equivariant Q-geometry (gate [C]/[E]): on $H_1(\mathbb{P}^1 \setminus \mu_4)=B_1 \oplus E$, $Q_+=3w+1=\text{diag}(1, 2, 3)$ (cusp weights on $\mu_4=\mathbb{Z}_4$ eigenspaces), $Q_-=E$ -coupling $\sqrt{N_{\text{fam}}} \Rightarrow t(t^2-3)$; Σ -split $=D_4=\mathbb{Z}_4 \rtimes \mathbb{Z}_2$ [E]
v70_q_integer_lift.py	Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action R unimodular (eigenvalues $\{-1, i, -i\}$); $\det Q=3=N_{\text{fam}}$, SNF $\text{diag}(1, 1, 3) \Rightarrow$ residual = one lattice invariant $\det Q=N_{\text{fam}}$, not closable by D_4 alone [E]
v71_simple_r_bridge.py	<i>Simple</i> R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ($\det R=8=\text{rank } E_8$ lattice + $\text{Spec}(Q_+)=\{1, 2, 3\}$ from v69) $\Rightarrow$ quark <i>ratios</i> are pure integer Plücker readouts ($\frac{55}{117}$, no monodromy, v49 rigidity); residual U_{point} = absolute norm. = an anchor [E]/[O]
v72_q_det_from_cusp.py	$\det Q=N_{\text{fam}}$ derived from the cusp class: $\det Q= \text{coker } Q =\text{cusp-class order}=\text{denominator of the weights } \{0, \frac{1}{3}, \frac{2}{3}\}$ (same data as $\text{Spec } Q_+$); $\text{coker } Q=\mathbb{Z}/N_{\text{fam}}=\text{triality deck group} \Rightarrow$ integer-lift residual closed [E]
v73_k_c3_half.py	Central coefficient $k=c_3/2$ <i>forced</i> : $(1/2)$ variational $\times 1/(\mathbb{Z}_2 2\pi\chi(S^2))$ Gauss–Bonnet topology ($\chi(S^2)=2$), both cutoff-indep $\Rightarrow$ Fursaev–Solodukhin $S=A/4$; only the absolute 4D Newton scale stays the anchor (v68) [E]
v74_compiler_micro_lemmas.py	Spine quotient ladder ($\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3}$ = adjacent quotients of $(2, 3, 4, 5)$); pencil differences $(2, 16, 48)=(\mathbb{Z}_2 , \dim S^+, \Omega_{\text{adm}})$ sheet $\rightarrow$ 1gen $\rightarrow$ 3gens; anchor QF $a^\top K a - \mathbf{1}^\top K \mathbf{1} = 41 - 25 = 16 = \dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L=R+6\mathbf{1}e_1^\top$, $a^\top R^{-1}\mathbf{1}=0$ (Sherman–Morrison rank-one invariance) [E]
v75_upoint_to_vgeo.py	Gate 1 complete : $U_{\text{point}} \rightarrow v_{\text{geo}}$. Ratios closed (v49/v71) + lepton c exact (v20) + sector products = φ_0^{ret} -powers (v46); (ratios, product) $\Leftrightarrow$ (individuals) bijection $\Rightarrow$ absolute amplitudes fixed up to <i>one</i> scale v_{geo} = the same anchor as $1/G$ (v68); two $[A] \rightarrow$ one [E]/[O]
v76_gmetric_reduction.py	Gate 2 : Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\text{eff}}=6\log \frac{3}{2}-\frac{31}{4\pi^2}=1.648>0$, $\ V\ \leq 248c_3^2=\frac{31}{8\pi^2}$); Tier B (strict TOE) holographically reduced from bulk to a finite seam-boundary measure (conditional: RP+tightness $\Rightarrow$ closes) [E]/[C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v77_e8_conformal_net.py	G6 route: the seam boundary measure = the $(E_8)_1$ <i>lattice net</i> (rigorous, holomorphic $c=8$); level-1 $c(E_8)=\frac{248}{31}=8$, $c(D_5)=5$, $c(A_3)=3$, embedding $5+3=8$ (coset $c=0$); imports G_6 into RCF/T/conformal-net rigor [E]/[C]
v78_vgeo_floor.py	v_{geo} floor: no pure number is dimensionful $\Rightarrow v_{\text{geo}}$ irreducible by logic; all scales are ratios $\times$ one M_{Pl} ; $S_{dS}\rho_\Lambda=32\pi^4$ pins it from one measurement; theory at the minimum (one scale $+$ π) [E]/[O]
v79_review_identities.py	Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n=(3^n-(-2)^n)/5=1, 1, 7, 13, 55, 133$ (roots = carrier charges); Inverse Anchor Thm ($\mathbf{1}^\top M^{-1}\mathbf{1}=1/\text{atom}$, $a^\top M^{-1}a=1$ for R, K, L); MacWilliams (1, 10, 5); $6Y^2-Y-1=(2Y-1)(3Y+1)$. Rejects m_p/m_e , η_B near-formulas (firewall) [E]
v80_operator_pencil_geometry.py	Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ}=(N_{\text{fam}}x+ \mathbb{Z}_2)(N_{\text{fam}}x+g_{\text{car}})=(3x+2)(3x+5)$ ($\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain (3, 10, 16, 20); type checker $\det B_{Q,K,R,L}=(9, 10, 16, 40)$; audit: differential spectrum, $F_4 \times G_2$ shadow $248=52+14+26\cdot 7$ [E]/[C]
v81_singular_pencil_matrices.py	Double cover $y^2=\det B_{K+xQ}=(3x+2)(3x+5)$: Koide $x=-\frac{2}{3}$ and carrier $x=-\frac{5}{3}$ are the two branch points (deck $2= \mathbb{Z}_2 $; disc $=81=N_{\text{fam}}^4$; separation $=1$ transport period). Clearing matrices $C_{2/3}=3K-2Q$ ($D_5 \oplus A_3$ glue, $B=(5, 7)^\top(9, 11)$, $\sum=240$), $C_{5/3}=3K-5Q$ (neutral, $\chi=(\lambda+2)(\lambda^2+\lambda+6)$); scalaron $7=$ mixed anchor disc $/N_{\text{fam}}$ [E]
v82_koide_attractor_splitting.py	Koide attractor <i>forced</i> : the branch-divisor-preserving Möbius map fixing $q=2, 5$ is unique, multiplier $(2/3)^6=\lambda_2$ of the established transfer operator $\Rightarrow F'(2)<1$ attractor, $F'(5)>1$ repeller; three postulates $\rightarrow$ one [C]. Splitting trichotomy disc $81/49/40$: the clean split is non-generic (anchor-first) [E]/[C]
v83_e8net_holomorphic_uniqueness.py	Holomorphy pins $(E_8)_1$: #primaries = det Cartan ($D_8:4$ vs $E_8:1$), and a holomorphic $c=8$ chiral CFT is the lattice theory of the <i>unique</i> even unimodular rank-8 lattice (Minkowski–Siegel mass $1/ W(E_8) $); carrier Pascal truncation $K=(g-1)/2 =$ half-spinor split [E]
v84_frozen_registry.py	Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula $\leftrightarrow$ value): exactly <i>one</i> θ_{12} prediction of record; r , n_s as $N_\star$ <i>bands</i> ; conditional entries carry their hypothesis by name [E]
v85_master_cover.py	Master-cover theorem: $\det B$ is $\text{GL}(2)$ -covariant on $\text{span}\{K, Q\} \Rightarrow$ exactly <i>one</i> double cover up to Möbius (disc $=N_{\text{fam}}^4 \det(G)^2$); the disc-81 family is its orbit ($-\frac{8}{3}=-\text{rank } E_8/N_{\text{fam}} =$ carrier – one period); μ_4 is <i>not</i> a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron <i>scale</i> [E]
v86_nstar_reheating.py	$N_\star$ band $\rightarrow$ point [C]: $\Gamma=4M^3/48\pi\bar{M}^2=128 \text{ GeV}$, $T_{\text{reh}}=9.6\times 10^9 \text{ GeV}$, $N_\star(0.05/\text{Mpc})=51.4 \Rightarrow n_s=0.9611$, $r=0.0045$ (inside the frozen band); honest tensions: $n_s+0.9\sigma$, A_s dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [C]
v87_bulk_uniqueness_reduction.py	Red-team Target A residual (ii) conditionally closed: holomorphic $\Rightarrow \text{Rep} = \text{Vect} \Rightarrow$ 2D bulk <i>unique</i> (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits <i>six</i> modular invariants (incl. both E_8 pairings) $\Rightarrow$ Target A = one residual [E]/[C]/[O]
v88_cp_phase_audit.py	Target-D CP residual quantified: frozen $\delta=\pi/3+3\lambda^2$ survives at $+0.98\sigma$ vs γ_{PDG} ; the 0.07° -near alternative $\pi/3+2\lambda^2$ recorded as look-elsewhere trap, <i>not</i> adopted; decision at $\sigma_\gamma \leq 0.96^\circ$ [E]
v89_carrier_index_lemma.py	Carrier Index Lemma: KLM $\mu_A=[B:A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1:(D_5)_1 \times (A_3)_1]=4= \mu_4 $ (glue order <i>is</i> the index); μ -additivity $16/4^2=1 \Rightarrow$ holomorphy <i>follows</i> ; Gate A (H) $\Leftrightarrow$ (H') index-4 extension [E]
v90_conical_defect_chain.py	Fursaev–Solodukhin factor 4π <i>derived</i> (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S=4\pi kA$, $k=c_3/2 \Rightarrow S=A/4 \Leftrightarrow c_3=\frac{1}{8\pi}$ (sympy-solved); residual isolated to “seam determinant $\Rightarrow$ EH form” [E]/[O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v91_spine_tetrahedron.py	Spine tetrahedron $\{2, 3, 4, 5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120= R^+(E_8) $; $240= \mu_4 E(K_4) E(K_5) $ with K_6 negative control; dual cuts typed tautological; sub-grammar incompleteness honest [E]
v92_glue_uniqueness.py	Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two <i>chiralities</i> , swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$; tower carrier $\rightarrow SO(16)_1 \rightarrow (E_8)_1$, nothing else [E]
v93_koide_relaxation_toy.py	Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$; source at $\rho=-\varphi_0^{\text{ret}}/24$ to 0.8% ([C]); honest negatives: flow time $t=2.84 \notin \mathbb{Z} \Rightarrow$ the missing object is a <i>continuous</i> generator [E]/[C]
v94_sheet_diamond.py	Sheet diamond: all four flavor operators on <i>one</i> surface $M(s, t)=R+Q \text{ diag}(s, t, t)$ (pencil = the cut $(1+x, x-1)$; diagonal-cut disc 153 negative control); winding line $\det B=2 \det$ for all s ; cofactor seam normal $(5, -9, 6)$; Plücker canonicalisation: 11 and 26 both live in K ; spine-quotient firewall <i>rejected</i> [E]
v95_centered_diamond.py	Centered cross: $Q=U+V$, $R/L=C \mp U$, $K/F=C \mp V$ (five objects $\rightarrow$ three); Spec $V=\{0, 1, 2\}=N_{\text{fam}} \cdot \text{cusp}$; $\det C=14$, $\sum C=31=2^{g_{\text{car}}}-1$; $\text{Pl}_R(C)=7(2, 3, 1) \parallel \text{Pl}_R(L)=10(2, 3, 1)$; G_2 reading audit-typed [E]
v96_branch_kernel_selection.py	Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have <i>integer</i> kernels (carrier kernel = 1); collapse lemma $\Rightarrow (-1, 1, 0)$: up = -down, <i>lepton pairing</i> = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$); anchor-placement controls [E]
v97_sheet_conjugation_bridge.py	Sheet-conjugation bridge (P1 $\rightarrow$ one [C]): the anchor forces the cusp conjugation uniquely $\Rightarrow$ integer T_A , $\det=-1$; $a=e_2+e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\text{span}\{\mathbf{1}, a\}$; $\langle T_A, \Sigma \rangle = D_4$; sheet-index lemma (intertwiner dets even, $\min 2= \mathbb{Z}_2 $); remaining [C]: Q_+ grading = A_3 discriminant grading [E]/[C]
v98_discriminant_dictionary.py	Dictionary <i>derived</i> (P1 closed modulo GATE.QGEO): $G=T_A \Sigma$ acts integrally on the cusp basis ($Ge_1=-e_1$, $Ge_3=e_2$, $Ge_2=-e_3=B_1 \oplus E$); cusp-0 line = self-conjugate $\mathbb{Z}_4$ -class-2 line; $T_A G T_A^{-1}=G^{-1}$ ($k \mapsto -k$); $q_{A_3}(1)=q_{A_3}(3)=\frac{3}{8}$, $q_{A_3}(2)=\frac{1}{2}$; reflection classes = glue-swap vs glue-fix parities $\Rightarrow$ P1 carries no separate [C]; residual = the one existing Q -geometry gate [E]
v99_koide_flow_time.py	Koide flow time: canonical generator $\dot{q}=(\Delta/N_{\text{fam}})(q-2)(q-5)=(\Delta/N_{\text{fam}}) \det B(q)$, time-1 map = F [E]; non-integrality of $t=2.84$ is <i>data-fragile</i> (Q crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$, t passes every integer ≥ 3 within $+0.5\sigma$) [E]; conditional steps: $n=2$ excluded (-2.9σ) , $n=3=N_{\text{fam}} \Rightarrow m_\tau=1776.9427 \text{ MeV } (+0.14\sigma)$, decision at $\sigma(m_\tau) \sim 0.01 \text{ MeV}$ [C]; $\ln(46080)$ scale reading destroyed by controls (firewall)
v100_numerology_null_mc.py	Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$, robust to budget/window variation; $2 \times 2 \cdot 10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly <i>one</i> hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$, <i>conditional on the declared grammar</i> [E]
v101_horizon_anchor.py	The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$, roots $(1, 1, -2)$ = traceless anchor; $S_{\text{Nariai}}/S_{\text{dS}}=\frac{2}{3}= \mathbb{Z}_2 /N_{\text{fam}}$ (per horizon $\frac{1}{3}$); interpolation $(x^2+1)/\Phi_3(x)$; three-sheet total $ \mathbb{Z}_2 S_{\text{dS}}$ conserved; $Q_{\text{geom}} \in [\frac{3}{8}, \frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1-3m)(1+3m)$, deck = horizon swap; $\tau_{BH}=128 g_{\text{car}} M^3/c_3$; six atom landings, zero free parameters [E]/[C] (bulk reading)

continued on next page

Script	What it machine-checks (<i>continued</i>)
v102_seam_orientation.py	One orientation: the anchor is the <i>stationary repeller</i> in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$, $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate Δ constant; SdS entropy has Nariai as its <i>unique</i> stationary point, curvature $\frac{2}{9}= \mathbb{Z}_2 /N_{\text{fam}}^2$; evaporation = entropy ascent away from the anchor [E]; “one variational principle of the seam” stays a reading [C]
v103_trisection_normal_form.py	Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow \cos 3\theta = -3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m = \cos \psi / N_{\text{fam}}$ and $S_{\text{tot}}/S_{\text{dS}} = \frac{4}{3} - \frac{2}{3} \cos \frac{2\psi}{3}$ — one cosine of glue atoms; canonical curvature $(\frac{2}{3})^3 = (\mathbb{Z}_2 /N_{\text{fam}})^{N_{\text{fam}}}$ (Koide constant to the family power); invariant slope $d\sigma/dm _N = -\frac{8}{9} = -\text{rank } E_8/N_{\text{fam}}^2$; bridge: flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, missing = one <i>clock</i> [E]/[C]
v104_nariai_clock.py	The <i>classical</i> Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the dS_2 static-patch equation with $m^2 = -2\Lambda = - \mathbb{Z}_2 \Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\text{clock}} = (\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1, -2\}$; Nariai cubic = $(t-1) \cdot \chi_{\text{clock}}$; entropy-deviation rate $2H = \mathbb{Z}_2 H$; honest $(2/3)$ -test <i>negative</i> for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C]
v105_residual_inventory.py	The residual inventory: <i>one constant</i> $(2/3 = \mathbb{Z}_2 /N_{\text{fam}}$ exact in seven places across both sectors), <i>one anchor</i> (three dynamical appearances: configuration, Nariai roots, clock spectrum), <i>relocation theorem</i> (cosmological quantum clock suppressed by the Λ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10 \Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the <i>complete machine-pinned gap list</i> : exactly five structural objects + one scale + π ; a sixth gap would fail the script [E]/[C]
v106_review_validation.py	External-review validation: seed normal form $\varphi_0^{\text{ret}} = \frac{ \mu_4 }{N_{\text{fam}}} c_3 + \Omega_{\text{adm}} c_3^{ \mu_4 }$ exact [E]; hypercharge moment $\text{Tr } X^2 = 120 = 5!$ computed from the charge table; factorial spine $5! = R^+(E_8) = \sum \text{exponents}$, $240 = 2 \cdot 120$, $1920 = 2^4 \cdot 5!$; degree-2 inventory (Pascal $K=2$ unique at $g=5$, glue norms sum 2, $c_3 = \frac{1}{2} \times \frac{1}{4\pi}$, $\binom{5}{2} = 10$); named [C] hypothesis <i>Quadratic Boundary Locality</i> ($\Rightarrow K=2$ forced; would upgrade the Pascal selection to [E]); audit $240 = 16 \times 15$ non-unique [E]/[C]
v107_quantum_clock_target.py	Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0, -1, -2\} = -\text{Spec}(V) = -N_{\text{fam}} \times \text{cusp}$ (the transfer operator’s own grading) [E]; exact target identity $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$ — required bend $\log_{9/4} 3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi) \cdot \Lambda/\bar{M}_{\text{Pl}}^2$: 10^{-122} cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C]
v108_pascal_ladder.py	Pascal Ladder Theorem: $2^{g-1} = \sum_{k \leq K} \binom{g}{k} \Leftrightarrow g=2K+1$ (odd- g midpoint identity + even- g straddle) $\Rightarrow$ carrier selection $g=5$ <i>exactly equivalent</i> to degree bound $K=2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K=1$ one-family, $K=3$ nine-family, $K=4$ inconsistent; only $K=2$ also satisfies $g+N_{\text{fam}}=8$ (v14 overdetermination) [E]/[C]
v109_sheet_pairing.py	Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm \frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+ \otimes S^+ = 0$; odd g flips parity); $S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4$ exactly ($256 = 1 + 45 + 210$, zero-modes $(1, 5, 10)$ = the code); sheet-diagonal = odd forms $(2\Lambda^1 + 2\Lambda^3 + \Lambda^5)$; tower tops at Λ^{2K} , $K=2 \Rightarrow$ the seam scalar kernel is necessarily a <i>sheet pairing</i> [E]/[C]
v110_calderon_sheet.py	Calderón-sheet selection theorem: a Calderón involution ε on $S^+ \oplus S^-$ certifies a scalar two-point datum <i>iff</i> it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2 = \mathbb{Z}_2 $ invariant kernels = the two Lagrangian glues; ladder genericity ($g=3, 5, 7$): sheet-oddness forces $K=(g-1)/2$, <i>not</i> $g=5$ (anti-overclaim); QBL residue split into <i>interface</i> (“ ε sheet-odd”, natural from the one-sided collar) + <i>analytic core</i> (slot-degrees ≤ 2) [E]/[C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v111_quadratic_transport.py	Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics (= $\mathfrak{so}(10)$, the Λ^2 term of $1+45+210$) sheet-even, generate the code from the vacuum <i>and</i> span the full $\text{End}(S^+)=256$ at word length $2 \Rightarrow$ <i>transport degree exactly 2</i> (degree ≤ 1 generates nothing) — the transport half of the QBL core is a theorem; $g=3$ control: degree selected, not rank [E]/[C]
v112_selfcounting_channel.py	Self-Counting Channel: $w \mapsto (w, -w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so $\#\text{kernels} = 2^{g-1}$ <i>exactly</i> ; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1} = \sum_{m \leq K} \binom{g}{m}$ is <i>two countings of one set</i> (an identity, not a requirement); Hodge fold $\binom{g}{2j} = \binom{g}{\min(2j, g-2j)}$; QBL residue = one input (“a single scalar 2-point kernel”), selection carried by rank-8+integrality [E]/[C]
v113_quasifree_kernel.py	One kernel is the whole net: carrier $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1 = 16$ free Majoranas, $c=5+3=8$ at every tower level (only the glue grows, index $2 \times 2 = 4 = \mu_4 $); Wick/Pfaffian exact (all $210+210$ higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P=5=g_{\text{car}}$, seam level $8 = \text{rank } E_8$ ($c = \text{rank of the kernel}$); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [E]/[C]
v114_torsion_delta.py	Torsion normal form + $\delta = \frac{1}{2}$ theorem: flatness of the $(U_{\text{wall}}) \mathbb{Z}_4$ -family $\Leftrightarrow (MU)^4 = \mathbf{1}$ ($\mu_4 = \text{order of the twisted cusp generator}$); on the involutive branch ($T^2 = \mathbf{1}$) the cusp trace forces $\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2})$ with $\text{spec } \{1, \omega, \omega^2\}$ automatic $\Rightarrow \delta = \frac{1}{2}$ exact and constant (v20’s hand value = torsion identity); census: $\{1, i, -i\}$ branch varies, v33 point in its $d_1=0$ slice; residue = bundle-side branch selection [E]/[C]
v115_anchor_residue.py	The anchor pins the residue: μ_4 -average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow \text{diag } A_0 = a/ \mu_4 $; off-weights uniquely $(8, 0, 5)/144$ ($= (\text{rank } E_8, 0, g_{\text{car}})/(\mu_4 N_{\text{fam}})^2$, total $13 = \Delta_Q$); <i>exact</i> residue A_0^* with $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ is the RH solution ($\ M_\infty - \mathbf{1}\ \sim 10^{-10}$); anchor forces $d_1=0 \Rightarrow$ harmonic diagonal anchor+torsion-forced [E]
v116_resonance_uniqueness.py	Resonance theorem (v115 [N] $\rightarrow$ [E], linear — no Gröbner): twisted μ_4 averages collapse to $B_1 = 4a_{12}E_{12} + 4\overline{a}_{13}E_{31}$; anchor exponents $\text{diag}(2, 1, 1)$ give resonance gap 1 with obstruction $4\overline{a}_{13} \Rightarrow M_\infty = \mathbf{1} \Leftrightarrow a_{13}=0$; + $(8, 0, 5)/144$ + Diagonaleichung $\Rightarrow$ flat anchor locus = <i>one</i> gauge orbit = exact A_0^* — U_{wall} bundle datum algebraically unique; sharpness control $\ M_\infty - \mathbf{1}\ \sim 8.5 a_{13} $ [E]
v117_monodromy_weyl_a3.py	The monodromy is $W(A_3)$: the unitarised holonomy of A_0^* is <i>exact</i> (entries in $\frac{1}{2}\mathbb{Z}[i]$), $\text{diag } \widetilde{M}_0 = (0, -\frac{i}{2}, \frac{i}{2}) \Rightarrow d_1=0, \delta = \frac{1}{2}$ are matrix entries; $\langle U, \widetilde{M}_0 \rangle$ enumerated exactly: order 24, statistics $(1, 9, 8, 6)$, characters $(3, -1, 0, 1) = S_4 = W(A_3)$ (standard $\otimes$ sign) — the flavor wall realises the A_3 glue factor as its Weyl group; ODE match 3.5×10^{-10} [E]
v118_hexagon_family_dictionary.py	First exact H2 piece: $\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$); $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 \Rightarrow \det_{\mp} = (\frac{7}{8}, \frac{9}{8})$; lepton coefficients = resolvent determinants ($c_e = \mu_4 /(2 \det_-)$, $c_\mu = N_{\text{fam}}/(2 \det_+)$, $c_\tau = \mu_4 \det_- / N_{\text{fam}}$, product = $ \mu_4 /\det_+ = \frac{32}{9}$); $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$ has order $48 = \Omega_{\text{adm}}$; the address table stays open [E]/[C]
v119_review_validation_2.py	Second review validation: anchor ratio triad $(e_3, e_1, e_2)/p_0 = (\frac{2}{3}, \frac{4}{3}, \frac{5}{3}) = (\text{Koide branch, seed gain, carrier branch; flow critical points } (2, 5) = (e_3, e_2))$; the 121 audit lemma $(\mathbf{1}+a)^T R(\mathbf{1}+a) = 121 = 11^2 = \ \text{Pl}(K)\ _1^2$ (orderings $\{105, 121, 135\}$, only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\text{adm}} = 2p_1p_2$, $10b_1 = 1 + p_1p_3$, $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank}$; flavor surface = $\sqrt{94}/\sqrt{95}$ (presentational) [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v120_address_table.py	Address table (frozen v18/v20 integers): lepton words $(8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}})$; addresses = divmod by the hexagon ($p_2=6$): $e \rightarrow (2, 1)$, $\mu \rightarrow (5, 0)$, $\tau \rightarrow (3, 0)$, $\sum r=p_3$; sheet parity: τ sits at the real twisted eigenvalue -1 (= the product-rule site); quark sums: up = p_3 , down = p_1+p_3 , all quarks = $24= W(A_3) $, all nine = $40=p_1p_3=a^T Ra$; top at the vacuum site $(0, 0)$; the assignment derivation stays open [E]/[C]
v121_address_pinning.py	Address pinning: $L=R+2U$ (v95) $\Rightarrow$ the word table = the residue operator R + the generation-1 winding, R column 1 = the anchor a ; margins = atoms (rows (e_1, rank, p_3) , columns $(e_1, \Delta_Q, g_{\text{car}})$); <i>census theorem</i> : among all hexagon matrices with atom margins (17 of them) exactly <i>one</i> has $\det=8$ — and it is R (also unique with SNF $(1, 1, 8)$; all 17 determinants distinct) $\Rightarrow$ the address table carries <i>zero</i> residual information; the H2 residue = the six atom margins [E]/[C]
v122_margin_theorem.py	Margin theorem: the <i>established</i> selectors — the D_4 annihilator $n=(5, -9, 6)$ with $n^T R=(8, 0, 0)$ (v94), $\det R=8$, $\det K=4$ (the v71 quartet) — pin R <i>uniquely</i> among hexagon matrices ($64 \rightarrow 12 \rightarrow 1$) $\Rightarrow$ the atom margins, the anchor column and all sum rules are <i>consequences</i> (their input status is revoked); bonus: $\det L=20$ holds automatically on the locus (the quartet is redundant); H2 introduces <i>no</i> new residual class [E]/[C]
v123_inventory_update.py	Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries <i>three</i> loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to <i>five</i> classes — R1 clock, R2 index-4 net, R3 EH, R4' selector establishment (replacing the <i>discharged</i> H2 dictionary), R5 Q realisation — + v_{geo}, π ; a sixth class fails the script [E]
v124_resummed_clock.py	Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1-n/N_{\text{fam}})^{p_2} \Rightarrow \text{rate}(n)=-p_2 \ln(1-n/N_{\text{fam}})$ reproduces $\{0, \Delta, 6 \ln 3\}$, the bend $\log_{3/2} 3$ in one line; the linearisation carries slope $ \mathbb{Z}_2 =2$ relative to the classical GP clock $\{0, -1, -2\}$; the pole at $n=N_{\text{fam}}$ forces the three-level spectrum; the R1 target is now closed-form [E]/[C]
v125_glue_qsystem.py	Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*= \mathbb{Z}_4 \mathbf{1} \Rightarrow \text{Jones index } 4= \mu_4 $ (the v89 value); locality = isotropy ($q(k(1, 1))=k^2 \in \mathbb{Z}$, v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2] = \text{the } SO(16)_1 \text{ step}$, $4=2 \times 2$; R2 sharpens from “find” to “identify” [E]/[C]
v126_clock_wall_bridge.py	Clock and wall share one spectrum: the resummation weights $n/N_{\text{fam}}=\{0, \frac{1}{3}, \frac{2}{3}\}$ = exactly spec A_0^* (the parabolic MS weights); $\lambda_n=(1-\alpha_n)^{p_2}$; checkpoint 1 passed <i>classically</i> ($p_2/N_{\text{fam}}=2= \mathbb{Z}_2 = \text{the v104 entropy rate } 2H$); $\det(\mathbf{1}-A_0^*)=\frac{2}{9}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail [E]/[C]
v127_ring_resummation.py	The clock is a log determinant: $\text{rate} = -p_2 \text{Tr} \ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, <i>one per hexagon site</i> , coupling = the parabolic weight; the $k=1$ ring = the classical term, the $k=2$ ring = the first quantum correction; $\kappa_{\text{seam}}=\frac{1}{3\pi}=\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha \rightarrow 1$; v107's “O(1) resummation” is thereby <i>typed</i> (ring/log-det), its derivation open [E]/[C]
v128_graded_hull.py	Graded hull + zero-mode multiplicity: the 240 E_8 roots explicit in $D_5 \oplus A_3$ + glue coordinates, coset decomposition $240=52+64+60+64$; the grading is exactly additive on <i>all</i> 6720 root-pair sums $\Rightarrow E_8$ is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1) _{l=1}$ = the sheet-doubled GP zero modes [E]/[C]
v129_entropy_power_law.py	The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ (v101); $\text{rate} = p_2 \ln(S_{dS}/S)$; triple identity $\alpha=n/N_{\text{fam}} = \text{MS weight} = \text{entropy-deficit share} \Rightarrow$ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma \propto (S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\text{fam}}$ audit) [E]/[C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v130_born_square.py	Born square: amplitude $\sim (S/S_{dS})^{n_{\text{zero}}/2} = (S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability $= A ^2 = (S/S_{dS})^6 \Rightarrow h = n_{\text{zero}}/2 = 3 = N_{\text{fam}}$, $p_2 = 2h$ (the Born rule) — the v129 weight audit is <i>derived</i> ; the 3 = the dimension of the $l=1$ space = the number of proper conformal Killing vectors of S^2 ; the $\times 2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure [E] / [C]
v131_measure_is_area.py	The measure is the area: $\ Y_{1m}\ ^2 = r^2 = A/(4\pi)$ <i>exactly</i> (all three $l=1$ modes integrated) $\Rightarrow$ the per-mode Jacobian = the mode norm $\sim S^{1/2} \Rightarrow$ amplitude S^3 , Born S^6 — every exponent factor with a derivation origin; the 4π = the same Gauss–Bonnet primitive as in c_3 , cancelling in all ratios; the R1 residue = <i>one</i> det-ratio cancellation $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$ [E] / [C] (<i>cancellation since derived within the SdS family, v144</i>)
v132_detratio_anomaly.py	Det-ratio anomaly = the Koide constant: $\zeta(0) _{\det'}$ of the GP sphere operator $-\Delta - 2$ (the 3 zero modes removed) $= a_1 - 3 = \frac{7}{3} - 3 = -\frac{2}{3} = - \mathbb{Z}_2 /N_{\text{fam}}$ exactly ($a_1 = \frac{1}{N_{\text{fam}}} + \mathbb{Z}_2 $; numerical continuation $\sim 10^{-7}$) — the eighth $2/3$ appearance, the first as a <i>spectral</i> anomaly; consequence $\det'(cL) = c^{\zeta(0)} \det'(L)$; the R1 residue = <i>one</i> $\zeta(0)$ budget statement (the 4d sum) [E] / [C]
v133_zeta_budget.py	The ζ budget, both routes exact (Euclidean Nariai $= S^2 \times S^2$): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3} = -e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2 = \frac{161}{45}$, $\zeta(0) = -\frac{109}{45}$ (<i>no</i> atom; continuation $\sim 10^{-8}$); zero modes $3+3$ = the horizon pair $\Rightarrow$ the budget structurally favours the <i>reduced</i> seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) [E] / [C]
v134_dual_anchor.py	Dual anchor lemma: $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$, $d \cdot \mathbf{1} = 0$, $d \cdot a = 1$, $(1, 1, -2) = -2d$ — the inverse flavor response <i>is</i> the Nariai root (v101); the invariance is exact via Sherman–Morrison ($\Leftrightarrow$ orthogonality to $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$); controls: $\mathbf{1}, e_1, n$ do <i>not</i> lie on the plane); (d, n) = the dual normal pair of the flavor boundary; a third purely <i>algebraic</i> leg of the flavor $\leftrightarrow$ horizon bridge [E] / [C]
v135_det_surface.py	Determinant surface: $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ — two s -independent iso-volume walls $t = -2 \Rightarrow 2 = \mathbb{Z}_2 $, $t = -1 \Rightarrow 4 = \mu_4 $ ($\det K = 4$ is a <i>layer</i>); winding line $6s + 8 \Rightarrow (8, 14, 20)$, $\det F = 32 = 2^{g_{\text{car}}}$; $\det B(s, 0) = 2 \det M(s, 0)$ exactly, B -wall only at $t = -2$; row-sum axes $C = (7, 11, 13)$, $U = (3, 3, 3)$, $V = (1, 2, 3) = \text{Spec}(Q_+)$; micro-identities $e_1^2 + e_2^2 = 41 = 10b_1$, $p_1^2 + p_2^2 = 52 =$ the carrier coset root count (v128) [E] / [C]
v136_dual_normal_selector.py	Dual-normal selector: (d, n) pins R <i>columnwise</i> — $d \cdot c_j = a_j$, $n \cdot c_j = 8\delta_{1j}$, kernel = primitive $(6, 8, 7)$, census: each column unique (1 of 216) $\Rightarrow R4'$ shrinks from 6^9 to three column problems; the A_0^* zero mode carries $\sigma = (2, -9, 5) = (\mathbb{Z}_2 , -N_{\text{fam}}^2, g_{\text{car}})$ ($\ v\ ^2 = \frac{16}{9}$, $16 = \dim S^+$; $\text{adj} = \frac{2}{9}P_v$ = the SdS curvature); $n \cdot \sigma = 121 = 11^2$; $W(A_3)$ orbit control: the naive lift $\sigma \rightarrow n$ fails — a genuine mechanism, not a group element [E] / [C]
v137_qplus_cohomology.py	The Q_+ grading is cohomology: $H^1(\mathbb{P} \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ ($\omega_k = z^{k-1} dz / (z^4 - 1)$), character i^k , residues $\zeta^k/4$; degrees $(1, 2, 3) = \text{Spec}(Q_+) =$ the A_3 exponents ($h = 4 = \mu_4 $); cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} = \text{spec}(A_0^*) \Rightarrow$ MS weights = cusp classes = characters = Q_+ : <i>one</i> spectrum, four readouts; the QGEO residue = the naturality of the canonical map [E] / [C]
v138_vw_firewall.py	Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\text{PI}} = -\frac{98}{45} = \frac{22}{45} - \frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); edge match : $\alpha_{\text{edge}} = -\frac{8}{3} = 2 \times (-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom $\Rightarrow$ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall [E] / [C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v139_selector_triangle.py	Selector triangle: $d=\frac{3}{2}a-2\cdot\mathbf{1}$ exactly (pure anchor data, no R) $\Rightarrow$ the first selector is <i>derived</i> ; the frame $(\mathbf{1}, a, \sigma)$ has $\det=11=\ \text{Pl}(K)\ _1$, and n is the <i>unique</i> covector with atom pairings $(2, 8, 121)=(\mathbb{Z}_2 , \text{rank } E_8, 11^2)$; on the constrained line the pairing is $11(11+t)$, a square only at the true n ; review lift exact (audit): $n=\text{reverse}(\sigma)+ \mu_4 e_3$; R_4' residue = three pairings $[\mathbf{E}]/[\mathbf{C}]$ (<i>since v142: two — the third pairing is integrality-forced</i>)
v140_canonical_map.py	The canonical map exists and is rigid: the plain deck U (characters $(0, 1, 3)$) cannot match H^1 ; the <i>sheet-twisted</i> $-U$ $((2, 3, 1))$ and the cusp exponential $i^{Q+}((1, 2, 3))$ both carry the H^1 character set $\{1, 2, 3\}$ (the v118 sign twist); multiplicity-freeness $\Rightarrow$ Schur-rigid equivariant isomorphism (unique up to scalars); i^{Q+} pulls the Euler degree back to Q_+ , $-U$ to $Q_+\circ\rho$ — the QGEO residue collapses to <i>one</i> $\mathbb{Z}_3$ choice $[\mathbf{E}]/[\mathbf{C}]$ (<i>the choice since derived, v141</i>)
v141_deck_selection.py	Deck Selection Theorem (R5): the established integer deck $G=T_A\Sigma$ (v97/v98) pairs the $Q_+=1$ line with the self-conjugate character $2 \Rightarrow$ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2, 3, 1) = \text{chars}(-U)$ survives; i^{Q+} excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ (v92); Euler degree = $Q_+\circ\rho$ $[\mathbf{E}]/[\mathbf{C}]$
v142_frame_integrality.py	Frame Integrality (R4'): pairing triples of <i>integer</i> covectors against $(\mathbf{1}, a, \sigma)$ form an index-11 sublattice $(x-3y+z \equiv 0 \bmod 11, 11=\ \text{Pl}(K)\ _1)$; with $(x, y)=(\mathbb{Z}_2 , \text{rank } E_8)$ the third pairing is forced $\equiv 0 \bmod 11$, primitivity forces even t ; Cramer reductions for the two line pairings; R_4' : three pairings $\rightarrow$ two $[\mathbf{E}]/[\mathbf{O}]$ (<i>since v145: the values are atom identities — residue = one lift map</i>)
v143_graded_frobenius.py	Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k=C_{-k}$ exact on all 240 roots $\Rightarrow$ Killing pairing nondegenerate per $g_k \times g_{-k}$; glue average = carrier projector (index $ \mathbb{Z}_4 =4$); each sector one $W(D_5)\times W(A_3)$ orbit = simple current; fusion = $\mathbb{C}[\mathbb{Z}_4]$ (v125); residual = the one conformal-net statement $[\mathbf{E}]/[\mathbf{C}]$
v144_detratio_family.py	Det-ratio cancellation within the SdS family (R1, the v131 step): e_2 -rigidity gives $r_b r_c = 1 - \Delta^2/3$ exactly $\Rightarrow$ $\det'$ -ratio = $(1 - \Delta^2/3)^{4/3}$ (via $\zeta(0)=-\frac{2}{3}$, v132) — no first-order term in the split; deficit coefficient $\frac{32}{27}= \mu_4 (\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6 \rightarrow \frac{14}{3}$ obstruction stated sharply $[\mathbf{E}]/[\mathbf{C}]$ (<i>the bend since shown $\det'$-clean, v147</i>)
v145_pairing_atoms.py	R_4' values are <i>atom identities</i> : with the lift $n=w_0(\sigma)+ \mu_4 e_3$ (w_0 = the A_3 exponent duality), $n\cdot\mathbf{1}= \mu_4 - \mathbb{Z}_2 = \mathbb{Z}_2 $, $n\cdot a= \mu_4 a_3=8$ via the new closure $ \mu_4 +g_{\text{car}}=N_{\text{fam}}^2$ ($4+5=9$, $\sigma \perp w_0(a)$), $n\cdot\sigma=\ \sigma\ ^2-N_{\text{fam}}^2+ \mu_4 g_{\text{car}}=121$; residue = derive the <i>one</i> lift map $[\mathbf{E}]/[\mathbf{C}]$
v146_moebius_d4.py	Möbius D_4 realisation (R5 at its floor): the full dihedral $\langle z \mapsto iz, z \mapsto 1/z \rangle$ on $H^1(\mathbb{P}^1 \setminus \mu_4)$ matches the integer model $\langle G, T_A, \Sigma \rangle$ exactly — ι^* swaps $\chi_1 \leftrightarrow \chi_3$, fixes χ_2 with $+1$, $\det=-1$ ($=T_A$ in the cusp basis); $\delta\iota=\Sigma$ class; parities match (v98) — the GATE.QGEO premise reduces to the module identification $[\mathbf{E}]/[\mathbf{C}]$
v147_clock_gaussian.py	The clock is a Gaussian zero-mode integral (R1): variance ratio $(1-\alpha)$ (norm = area, v131) $\Rightarrow$ Born ² Γ -ratio $(1-\alpha)^{p_2}$, rate = $-p_2 \ln(1-\alpha)$ = the v127 ring sum term by term; the bend $\log_{3/2} 3$ lives between the two Nariai levels of <i>one</i> geometry $\Rightarrow$ $\det'$ -clean; the $6 \rightarrow \frac{14}{3}$ obstruction is localised to the $\alpha=0$ reference $[\mathbf{E}]/[\mathbf{C}]$
v148_fock_census.py	NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the <i>even</i> classes $\{0, 2\} \times \{0, 2\}$ — the odd glue sectors $k(1, 1)$ are <i>Ramond</i> modules; the R zero-mode module ($2^8=256$) splits $128+128$ into the odd sectors of the <i>two</i> Lagrangian glues (v92): choosing the glue <i>is</i> choosing the R-projection (128 = one $SO(16)$ chirality), $248=120_{\text{NS}}+128_{\text{R}}$; R2 is irreducibly a twisted-sector net statement $[\mathbf{E}]/[\mathbf{C}]$
v149_cusp_normal.py	Both dual normals are <i>anchor data</i> (R_4'): in the cusp frame n pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line ($d=\frac{3}{2}a-2\cdot\mathbf{1}$ in the generation frame, v139); equivalent to the $(2, 8, 121)$ frame data and the w_0 -lift (v145); audit: $\sigma \rightarrow (5, 1, 2)$, sum 14 = $\dim G_2$; residue = <i>one</i> assignment $[\mathbf{E}]/[\mathbf{C}]$

continued on next page

Script	What it machine-checks (<i>continued</i>)
v150_replica_eh_model.py	The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and t -independent (image sum $(N^2-1)/(12N) = C(2\pi/N)$); the gap makes the ζ -variation <i>finite and cutoff-independent</i> , $\Delta \log \det' = 2C(\gamma) \ln m$; linearisation = $(\ln m/12\pi) \int \sqrt{g} R$ — the EH form; target equation $k=c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit); Calderón transfer + normalisation stay open — SEAM.THEOREM.01 narrows, does <i>not</i> close [E]/[O] (<i>Calderón transfer since answered — the DtN kernel is conically clean, v151</i>)
v151_bfk_split.py	BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition <i>independent</i> ($C_N=C_D$, derived from reflection doubling) $\Rightarrow$ the conical deficit of the Calderón/DtN jump determinant <i>vanishes identically</i> , $C_{\text{DtN}}=C_{\text{cone}}(\gamma)-2C_D(\gamma/2)=0$; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the <i>local</i> half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise [E]/[O]
v152_norm_is_anchor.py	R3's normalisation <i>is</i> the dimensionful anchor: the EH coefficient is $k = \ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio m/μ is physical; $k=c_3/2 \Leftrightarrow m/\mu=e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), <i>not</i> a separate gap; the irreducibles stay {one scale, π }; audit: the log-ratio = $\frac{3}{4} = q(A_3)$ is the target value, NOT a derivation [E]/[O]
v153_no_unit_theorem.py	No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L \mapsto \lambda L$, a mass / $1/G$ is not, so a dimensionless compiler <i>cannot</i> select an absolute scale; the three residuals collapse — $U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, $m/\mu = e^{3/4}$ (a ratio); v_{geo} is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, π } [E]/[O]
v154_simple_current_theorem.py	Simple-Current Extension Theorem (the central G_{net} theorem, assembled): $A=(D_5)_1 \otimes (A_3)_1$ extended by the isotropic glue $L=\langle(1,1)\rangle$ has index $ L =4$, $c=5+3=8$, $\mu(B)=\mu(A)/ L ^2=16/16=1 \Rightarrow$ holomorphic $\Rightarrow B=(E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification [E]/[C]
v155_quasifree_boundary.py	Quasi-free in $\Rightarrow$ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression PTP (a contraction, hence a valid covariance), so the G_{net} analytic premise (v110(b)/v113) <i>follows</i> from ‘the seam bulk is a reflection-positive free field’ — the <i>same</i> premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise [E]/[O]
v156_seam_net_construction.py	Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k= k $ (free chiral dispersion); 16 free Majoranas give $c=8=5+3$; the E_8 currents are $248=120_{\text{NS}}+128_{\text{R}}$; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1+248q+4124q^2+\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a <i>checked</i> identity, not an assertion; residual = the free-bulk premise (v155) + net existence [E]/[C]
v157_rigid_fixed_point.py	Freeness as a <i>rigid fixed point</i> (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $ k $ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c=8$ theory has <i>no</i> marginal $(1,1)$ field (all $\bar{h}=0$) $\Rightarrow$ rigid, isolated, no room for an interaction; (iii) the same $ \mathbb{Z}_2 $ that halves 8π in c_3 IS the Ramond projection giving $\mu=1$; (iv) holomorphic $c=8$ unique = $(E_8)_1$ = free. Premise (A) shrinks from ‘prove Gaussianity’ to ‘the gapped flow reaches the holomorphic fixed point’ [E]/[C]
v158_fixed_point_stable.py	The free chiral $c=8$ fixed point is <i>stable</i> (exact dimension count, the finite core of (A)): Majorana $h=\frac{1}{2}$, bilinear current $h=1$, quartic $h=2$; the relevant window $(0,1)$ for bosonic operators is <i>empty</i> , the $h=1$ currents are chiral (not $(1,1)$ -marginal, v157), the lowest interaction (quartic) is $h=2>1$ irrelevant $\Rightarrow$ isolated stable fixed point; (A) reduces to a basin-of-attraction statement [E]/[C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v159_pyrate_gauge_crosscheck.py	PyR@TE cross-check of the F_{transfer} gauge inputs: the carrier/SM content gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ by the textbook one-loop formula; $b_1 = \frac{41}{10}$ holds <i>three</i> independent ways — carrier algebra $10b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1 = 41$, the $U(1)$ hypercharge index $\sum \frac{3}{5} Y^2$, and the external RGE generator PyR@TE3 (arXiv:2007.12700) which writes $\beta_{g_1} = (\frac{41}{10}) g_1^3$ verbatim; the $M=41$ split $10b_1 = 40_{(\text{ferm})} + 1_{(\text{Higgs})}$ reproduces v13 Gate 1; evidence in verification/pyrate . The transfer functor's gauge inputs are not free knobs [E] / [C]
v160_seam_gaussianity_from_pf.py	Seam Gaussianity as a <i>conditional</i> fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free $\Rightarrow$ all higher connected cumulants $\kappa_{2n}=0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality $\Rightarrow$ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a <i>fixed</i> κ_{2n} is a SECOND fixed ray, breaking uniqueness; a <i>non-fixed</i> one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only $\text{gap} > 0$ — the value $(2/3)^6$ only sets the rate. The load [O]: v56's gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, while the seam correlation cone has $\binom{16}{4}=1820$, $\binom{16}{6}=8008$ cumulant directions $\Rightarrow$ (A) reduces to the single internal statement ‘the admissible TFPT transport is primitive + spectrally gapped on the <i>full</i> Schwinger cone, not only the readout spectrum’ [E] / [O]
v161_one_particle_reduction.py	The cone reduces to one particle (discharging v160's scope premise via Bogoliubov second quantisation): a quasi-free transport is $T=\Gamma(t)$, the second quantisation of a one-particle contraction t ; $\Gamma(t)$ acts on the degree- m correlator sector as the exterior power $\Lambda^m(t)$, so (1) $\text{spec}(\Gamma(t) _{\deg m}) =$ products of m one-particle eigenvalues [E] ; (2) the cone gap equals the one-particle gap — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly [E] ; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization $\Leftrightarrow$ one state) [E] ; (4) quasi-freeness <i>forced</i> not assumed — the $120 = \dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant [C] . Residual [O]: v160's three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ , gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open [E] / [O]
v162_seam_transport_identification.py	The final identification, forced as far as it goes — the v161 one-particle statement carries <i>no</i> new open content: (1) the gap $(2/3)^6$ is <i>forced</i> — cusp weights $= \text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (parabolic residue, v115), transfer eigenvalue $(1-\alpha)^{p_2}$, $p_2=6= R_+(A_3) $, all carrier data [E] ; (2) μ_4 -equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k} = \mu_4 \text{diag}(X)$, commutant of $U =$ the diagonal algebra — so the gapped μ_4 -equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ [E] ; (3) the cusp grading <i>is</i> the A_3 grading on the 16 Majoranas (v137) [E] ; (4) the fixed space = the rank-8 Calderón polarization (v113/v156) [E] . Closure [C]/[O]: with v160+v161, (A) closes given only μ_4 -equivariance + admissibility, whose two non-derived inputs are the <i>already-open</i> A2 (net existence) and R1 (seam clock, HOR.CLOCK) — so (A) factors exactly into A2 + R1, adds <i>no</i> new open item, and is closed conditionally on them (a unification, not an unconditional proof) [E] / [O]
v163_rg_stability_flavor.py	F_{transfer} scale-tagging: the lepton-mixing seeds ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are <i>RG-stable</i> — PMNS running is y_τ^2 -suppressed ($\sim 1.8 \times 10^{-5} \ll$ precision), so seam value = measured value; the quark mass-ratio readouts ($m_t/m_b \sim 41$) run through $y_t^2 \sim O(10\text{--}15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2} Y Y^\dagger Y$ carries y_{max}^2 (PyR@TE SM RGEs) [E] / [C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v164_qcd_scale.py	the QCD confinement scale is parameter-free from the carrier $b_3=-7$: running $\alpha_s(M_Z)$ down (two-loop, n_f thresholds) gives $\alpha_s(m_b)\sim 0.22$, $\alpha_s(m_c)\sim 0.38$ and confinement ($\alpha_s\sim 1$) at ~ 0.5 GeV — so the QCD-scale <i>numerator</i> of m_p/m_e is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest ‘Open / not forced’ frontier non-claim ($m_p=O(1)\Lambda$ is lattice) [E]/[O]
v165_premise_a_closure.py	Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries <i>zero</i> independent open content: (1) A2 (net existence) closes by <i>assembly</i> of established mathematics [C] — 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c=8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B)=16/16=1$, $248=120+128$, character $E_4/\eta^8=j^{1/3}$ (v156); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $ k $ (v156/v157); (2) R1 (seam clock) reduces to GATE.QGEO [C] — μ_4 -equivariance forces diagonal-in-cusp, the flat μ_4 -equivariant generator is <i>unique</i> = A_0^* (v115/v116) so the gap $(2/3)^6$ is forced, residual = ‘seam boundary = the μ_4 -punctured sphere $\mathbb{P}^1\setminus\mu_4$ ’ = GATE.QGEO (cohomology established, v137); (3) the irreducible core stays $\{\pi, v_{\text{geo}}\}$, a <i>theorem</i> by the No-Unit Theorem (v153). Final accounting: (A) = A2 assembly [C] + R1=GATE.QGEO + the $\{\pi, v_{\text{geo}}\}$ core $\Rightarrow$ ZERO independent open gates; a unification + re-typing, not an unconditional proof [E]/[O]
v166_higgs_free_seam.py	the Higgs quartic from the <i>free seam</i> (premise A / v158): a Gaussian seam UV forces $\lambda(M_{\text{seam}})=0$ and $\beta_\lambda(M_{\text{seam}})=0$ (Shaposhnikov–Wetterich double criticality, here <i>motivated</i> not assumed). Two-loop PyR@TE-sourced SM running with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\text{Pl}})\sim 0.002$, $\beta_\lambda(\bar{M}_{\text{Pl}})\sim 0$ — the famous SM near-criticality, now <i>explained</i> by the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from ~ 0.022 to ~ 0.002). The double condition predicts $m_H\sim 129\text{--}134$ GeV; at the scalaron scale it gives ~ 107 GeV (too low) $\Rightarrow$ the BC lives at $\bar{M}_{\text{Pl}}$ (seam=Planck) [E]/[C]
v167_inventory_post_closure.py	The terminal residual inventory (post premise-(A) closure; line v105 $\rightarrow$ v123 $\rightarrow$ v167): after the (A)-chain (v160–v165), premise (A) is <i>no longer</i> an independent structural class (GATE.METRIC.16–19). The residual collapses from v123’s FIVE classes to: Tier 0 — the irreducible core $\{\pi, v_{\text{geo}}\}$, a <i>theorem</i> (No-Unit, v153), nothing to close; Tier 1 — ONE hinge (seam = flavour = horizon): A_2 net existence (assembly of established net constructions) + GATE.QGEO (seam boundary = $\mathbb{P}\setminus\mu_4$, cohomology $b_1=N_{\text{fam}}=3$); Tier 2 — folded ($\rightarrow v_{\text{geo}}$, $\rightarrow G_{\text{net}}$); Tier 3 — F_{transfer} , one functor with four typed conditional interfaces (η_B carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C]
v168_qgeo_rigidity.py	The QGEO <i>Rigidity Theorem</i> — the finite half of the Tier-1 hinge GATE.QGEO, hardened. Exact: $\mu_4=\{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z\rightarrow iz, z\rightarrow 1/z \rangle$ of order $8= D_4 $ (faithful) $\Rightarrow$ four marks with faithful D_4 are the μ_4 square up to Möbius; $H^1(\mathbb{P}\setminus\mu_4)$ has rank $4-1=3=N_{\text{fam}}$; the forms $\omega_k=z^{k-1}dz/(z^4-1)$ ($k=1, 2, 3$) are μ_4 -eigenforms with $z\rightarrow iz$ eigenvalue i^k = the three nontrivial characters χ_1, χ_2, χ_3 , weights $(1, 2, 3) = A_3$ exponents = $\text{Spec}(Q_+)$. QGEO.RIGID.01 [E] (math half closed); QGEO.REALIZE.01 [C] (seam-collar = $\mathbb{P}\setminus\mu_4$ stays the realisation premise) [E]/[C]
v169_etaB_boltzmann_interface.py	The η_B leptogenesis transfer <i>operationalised</i> (Tier-3 F_{transfer} , not a new class): type-I thermal leptogenesis $\eta_B\sim 0.96\times 10^{-2}\varepsilon_1\kappa_f$ (sphaleron 28/79) fed by TFPT’s NO neutrino spectrum + the predicted $\delta_{CP}=240^\circ$. Davidson–Ibarra $\varepsilon_1=\frac{3}{16\pi}M_1m_3/v^2\sim 8.5\times 10^{-7}$ for $M_1=10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1)\sim 0.03\text{--}0.19$. Kill test: over natural $M_1\in[3\times 10^9, 3\times 10^{10}]$ GeV and washout, η_B spans $[8.4\times 10^{-11}, 4.7\times 10^{-9}]$ which <i>brackets</i> the observed 6.1×10^{-10} (canonical point gives 6.0×10^{-10} , not tuned) $\Rightarrow$ the route is viable; M_1 /washout are scenario inputs, so η_B stays [C]. If excluded, the route falls, not the theory [C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v170_diamond_slice_compression.py	E8 Slice Compression Lemma: the seven E_8 audit slices are <i>one</i> projection of two alphabets — anchor power sums $P=(3, 4, 6, 10)$ ($p_n=2+2^n$) and Sheet-Diamond determinants $D=(3, 4, 8, 14, 20, 32)$. The K_4 edge products $(12, 18, 30, 24, 40, 60)$ are all carrier blocks; $\sum \det M = 81 = N_{\text{fam}}^4$; all seven 248-slices follow from $P, D, \Delta=13$ and $(e_1, e_2, \dim S^+)$; one affine map $\text{row}(C+sU+tV) = (7, 11, 13)+s(3, 3, 3)+t(1, 2, 3)$ gives every flavour row budget; $78 = \dim E_6 = p_2\Delta$. The atlas is a closed grammar over two typed alphabets, exact, reading audit [E]
v171_os_moment_cluster.py	Atomic OS Moment Lemma + Sugawara Gap Safety: $\text{spec}(T)=\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n)=A+Br^n+Cs^n$ a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram $\Rightarrow$ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$, $\xi=0.41105$, Källen–Lehmann masses $6\log\frac{3}{2}, 6\log 3$; gap safety $\Delta_{\text{eff}}=6\log\frac{3}{2}-\frac{31}{4\pi^2}>0$ with $31=1+h^\vee(E_8)$, $c((E_8)_1)=248/31=8$ [E]/[C]
v172_trace_anomaly_seed.py	Trace-anomaly origin of the seed prefactor $\frac{4}{3}$: over the seam S^2 ($\chi=2$) the $(E_8)_1$ ($c=8$) Weyl anomaly is $c\chi/6=\frac{8}{3}$, halved by the one-sided $ \mathbb{Z}_2 $ to $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{SM} = 16/12$, the seed base $(\frac{4}{3})c_3=\frac{1}{6\pi}$. Plus QCD $b_3=11-\frac{2}{3}N_f=7$ = scalaron exponent $\Omega_{\text{adm}}-10b_1=48-41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf factorisation $-98/45 = -2 \cdot 7^2 / \dim(D_5)$ [E]
v173_pfaffian_cp_car.py	Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of $\theta_{\text{eff}}=0$ — the sheet-odd Calderón involution $\{D, \Sigma\}=0$ pairs the spectrum $(\pm\lambda)$, γ_5 -Hermiticity $D^\dagger=\gamma_5 D \gamma_5$ makes the Pfaffian real ($\arg \in \{0, \pi\}$), and reflection positivity ($Z>0$) selects the positive branch $\Rightarrow \theta_{\text{eff}}=0$ [E]
v174_seam_fock_readings.py	Seam Fock-space readings: (1) CAR cone compression — $\dim \Lambda^{\text{even}}(\mathbb{R}^{16})=2^{15}=32768$ reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ($2^{15}/16=2^{11}$), the explicit count behind v161; (2) Witten index = the Plücker norm $11 = \sum_{k \leq 2} \binom{4}{k}$, with $11 = \dim S^+ - g_{\text{car}}$ (full $2^4=16$) and $c_u/c_d=55/117$ reading the QBL-protected boundary state space; (3) Affleck–Ludwig g -theorem $c_{UV}=5+3=8=c_{IR}((E_8)_1)$ so $\Delta c=0$ forces an exact isometry (boundary-entropy form of v157/v158) [E]
v175_net_existence_full_cone.py	The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to [E], isolating QGEO.REALIZE.01 as the single open premise (nothing fabricated): (1) full-cone RP for all m — the CAR second-quantisation functor $\Gamma(t)=\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction t acts on the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16}=65536$); the exterior-power spectrum theorem makes every eigenvalue a subset product of $\text{spec}(t)$, verified on all 2^{16} products to lie in $[0, 1]$ (contraction), with eigenvalue-1 multiplicity $2^8=256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$, so full-cone RP reduces <i>for all m</i> to the one-particle contraction (Shale–Stinespring CAR functor), discharging the v160 scope premise by a theorem; (2) A2 net existence assembled + verified — $(E_8)_1$ character $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1=248), embedding $248=120+128$, E_8 Cartan even unimodular ($\det 1$); the net <i>exists</i> by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = QGEO.REALIZE.01, finite half v168, realisation half left [O][E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v176_seam_collar_realisation.py	The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — <i>not</i> a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks and admissible $c=8$ data, <i>then</i> its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1, 2, 3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, FIVE already [E]: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius D_4 order 8, $b_1=N_{\text{fam}}=3$), Calderón DtN symbol $ k $ (v156), H^1 character = generation grading $(1, 2, 3)=A_3$ exponents=Spec(Q_+) (v137/v168), μ_4 -equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all m (v154/v175); the whole structural residual reduces to the ONE open premise QGEO.REALIZE.01 (the physical seam collar's boundary <i>is</i> this $\mathbb{P} \setminus \mu_4$), left honestly [O][E][O]
v177_seam_marking_kernel.py	The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed <i>constructively</i> [E] and the open residual sharpened into exactly TWO named obligations. [E] UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4 , the reflection $z \mapsto 1/z$ gives $\sigma\rho\sigma = \rho^{-1}$ (D_4 , order 8) — a genus-0 four-marked faithful- D_4 boundary <i>is</i> $(\mathbb{P}, \mu_4)$; [E] COHOM: $\rho^*\omega_k = i^k\omega_k$ so the H^1 grading is $(1, 2, 3)=A_3$ exponents=Spec(Q_+); [E] MODULE: the $H^1 \rightarrow$ generation iso is <i>unique</i> (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); [E] NET (v154/v175): index-4 μ_4 extension of $(D_5)_1 \otimes (A_3)_1 = (E_8)_1$. The TWO genuine open obligations [O]: QGEO.MARKS.01 (raw seam $\rightarrow$ genus-0 four-marked D_4 boundary) and QGEO.KERNEL.01 (raw seam Calderón kernel = the μ_4 -equivariant free gapped contraction as an <i>operator</i>); the whole structural residual is now exactly these two doors [E][O]
v178_marks_kernel_attempt.py	An honest <i>attempt</i> at the two open obligations QGEO.MARKS.01 + QGEO.KERNEL.01 — <i>not</i> a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed [E] and each obligation reduced to ONE sharper irreducible premise [O]. <i>Marks core</i> [E]: a genus-0 double with an order-4 clock ρ ($\sim z \mapsto iz$, fixed points $\{0, \infty\}$) and reflection τ forces the marks to be a generic ρ -orbit $\{1, i, -1, -i\} = \mu_4$ (not the fixed points), distinct iff $b_1=4-1=3=N_{\text{fam}}$, $\tau\rho\tau=\rho^{-1} \Rightarrow$ faithful D_4 order 8 — so MARKS reduces to the single topological/clock premise. <i>Kernel core</i> [E]: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4 -equivariant operator is forced diagonal and <i>completely determined by its eigenvalue per character</i> — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are <i>equal as operators</i> , not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $ k $, Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v179_conformal_realisation.py	The two residual premises (MARKS-residual + KERNEL-residual) collapse to ONE geometric premise QGEO.CONF.01 (honest unification; the conformal realisation stays [O]). MARKS's residual is not three independent assumptions: (1) genus-0 <i>is</i> P1 — $c_3=1/(\mathbb{Z}_2 2\pi \chi(S^2))=1/(8\pi)$ with $\chi=2 \Rightarrow g=0$, the same $\chi=2$ that gives the '8' (v58/v73); (2) the order-4 clock <i>is</i> the carrier — Coxeter element of $W(A_3)=S_4$, order $h(A_3)=4= \mu_4 $ (v117); (3) marks = one orbit is <i>forced</i> — parabolic marks $\neq$ the two elliptic fixed points $\{0, \infty\}$, a free order-4 orbit has exactly $4=e_1(a)$ points. So MARKS reduces to 'the carrier clock acts conformally on the genus-0 double'. KERNEL then follows from that geometry: DtN symbol $ k $ (Lee–Uhlmann, v156) $\Rightarrow$ rank-8 $c=8$ fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1=\mathbb{C}^3=N_{\text{fam}}$, the 3-dim gapped transfer block (Schur, v162/v178). Both $\Rightarrow$ ONE premise QGEO.CONF.01: the raw RP seam double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly [O][E][O]
v180_clock_is_mobius.py	Cracking QGEO.CONF.01 one level further — the conformal-realisation premise is <i>replaced</i> , via three classical theorems, by a strictly <i>milder</i> , non-circular premise QGEO.ISO.01: 'the carrier clock is an order-4 orientation-preserving <i>isometry</i> of the RP seam metric'. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) <i>is</i> $\mathbb{P}$ — the target is produced, not assumed; (2) Kerékjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)=4$) is topologically $z \mapsto iz$; (3) isometry $\Rightarrow$ conformal $\Rightarrow$ Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\text{Aut}(\mathbb{P})=\text{PSL}(2, \mathbb{C})$; (4) order-4 Möbius $= z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i , free orbit μ_4). So QGEO.CONF.01 is <i>implied</i> by the milder isometry premise; the new residual QGEO.ISO.01 (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still [O][E][O]
v181_clock_is_conformal_symmetry.py	The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise QGEO.SYM.01 — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a <i>conformal</i> automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. The clock is the Coxeter element of $W(A_3)=S_4$ (v117), order $h(A_3)=4= \mu_4 $, the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below QGEO.SYM.01 the residual is <i>definitional</i> , not a missing theorem: 'the seam's conformal structure <i>is</i> the μ_4 -deck structure of the carrier clock' (the carrier μ_4 glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam's conformal geometry, not reducible further without relabeling. The chain REALIZE(v175) $\rightarrow$ theorem(v176) $\rightarrow$ MARKS+KERNEL(v177) $\rightarrow$ finite cores(v178) $\rightarrow$ CONF.01(v179) $\rightarrow$ ISO.01(v180) $\rightarrow$ SYM.01(v181) has reached bedrock; we stop here honestly [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v182_reviewer_residual_map.py	The four external-review concerns reduce to the two already-named residuals + one data-only item. A (mapping $2D \rightarrow 4D$) [E]/[C]: the Plücker readout $11 = \sum_{k \leq 2} \binom{4}{k} = \dim S^+ - g_{\text{car}} = 16 - 5$ (QBL count, v174) and the holographic reduction give $A = \text{QGEO.SYM.01} + F_{\text{transfer}}$. B (UV gravity) [E]: ‘gravity = geometric boundary readout’ is QGEO.SYM.01. C (grammar tuning) [E]/[C]: frozen (v84) + minimal + over-determined (the anchor $a=(1, 1, 2)$ in ≥ 6 independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). D (Koide Möbius flow) [E]/[C]: the Möbius nature is forced by $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ on the deck, $t \sim 2.84$ only locates the point, and the missing RG generator is F_{transfer} , so $D = \text{QGEO.SYM.01} + F_{\text{transfer}}$. NET: three reduce onto $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$, one waits for data — a unification of the concern structure, not a closure (premises stay [O]/[C]) [E][C]
v183_koide_f_corner_transfer.py	The Koide source→pole factor $\frac{53}{54}$ has an <i>operator</i> origin (external-review proposal, validated): $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a) = 53/(2 \cdot 27) = \frac{53}{54}$. (1) $\mathbf{1}^T R a = 27$ is the $E_6 \times A_2$ cubic family block ($248 = \dim E_6 \cdot 78 + \det R \cdot 8 + 2 \cdot 27 \cdot N_{\text{fam}}$); (2) $F=R+Q$ is the <i>missing</i> Sheet-Diamond corner $M(1, 1)$ with $\det B_F = 52 = \dim F_4$ (v80) and anchor response $a^T(R+Q)\mathbf{1} = 53 = 52+1$; (3) $\frac{53}{54} = 1 - \frac{1}{2N_{\text{fam}}^3}$ exactly ($54=2 \cdot 27$); (4) negative control: only F gives 53 ($R/Q/K/L \rightarrow 32/21/35/56$). Upgrades FR.KOIDE.03 from a bare guess to an operator identity; [E] the factor is exact, [C] the leptonic source→pole reading the F corner stays an F_{transfer} conjecture [E][C]
v184_etaB_anchored_boltzmann.py	Honest test of the external-review η_B proposal ($M_1=M_R\phi_0^4$, $\tilde{m}_1=m_3/A_\Lambda$) — firewall verdict: <i>sharper scenario</i> , not a closure. The washout anchors plausibly ($\tilde{m}_1=m_3/A_\Lambda=m_3/10 \approx 5 \text{ meV}$, $A_\Lambda=10= E(K_5) $, in the strong-washout window) [C]; but M_1 does <i>not</i> — M_R is the seesaw scale $\sim v^2/m_3 \approx 6 \times 10^{14} \text{ GeV}$ and the nearest clean TFPT powers of $\bar{M}_{\text{P1}}$ both miss it (c_3 -power off $\sim 75\%$, ϕ_0 -power off $\sim 40\%$), so $M_1=M_R\phi_0^4$ merely relocates the free input $M_1 \rightarrow M_R$ (the reviewer’s 6.0×10^{-10} uses $M_R=1.3 \times 10^{15}$; the seesaw value gives 3.4×10^{-10} , the tell that M_1 is unpinned). η_B stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — η_B stays [C], a sharper scenario [C]
v185_axion_relic_solver.py	Axion relic, honest misalignment estimate: the determinant-line axion $f_a=M_{\text{scal}}/128 \approx 2.39 \times 10^{11} \text{ GeV}$ ($m_a \approx 23.8 \mu\text{eV}$) can provide all dark matter, but <i>only</i> near the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a [C] scenario. The harmonic misalignment under-produces at this f_a (prefactor $\sim 0.026/\theta^2$, $\theta \sim O(1)$ gives $\sim 21\%$ of Ω_{DM} , need $\theta_{\text{eff}}^2 \sim 4.7$), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\text{relic}}(f_a, \theta_i, N_{\text{DW}}) = ? \Omega_{\text{DM}}$ is a kill test for the <i>branch</i> , not the theory (no free singlet WIMP: $128=(16, 4)+(16, 4)$) [C][O]
v187_ftransfer_laws.py	The F_{transfer} discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, η_B , the axion relic, $m_p/m_e + \Lambda_{\text{QCD}}$ — must <i>never</i> be typed as primitive [E] compiler predictions; each carries a [C]/[O] marker. The guard reads <code>status_ledger.csv</code> and enforces it (exact algebraic sub-parts — Koide $53/54$ v183, $b_3=-7$ v159 — may be [E], the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v188_frontier_wording_guard.py	The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger <i>typing</i> , this guard protects the <i>prose</i> . It forbids stale sentences whose semantics an earlier round superseded — “leptogenesis interface ([E])” (v184 retyped Route B to [C]), “closed branch also fixes the leptogenesis inputs” (M_R is the seesaw scale, not a compiler power), “relic scale f_a, m_a open” (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), “computed $Q=0.664\dots$ near, not exact” (v183 gave 53/54 an exact operator origin), “Suite now 187 modules” (the count is 187, highest ID v188; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E]
v189_riemann_roch_carrier.py	The Riemann–Roch carrier reading: $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of <i>one</i> object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$. [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg + 1 = 5$ (function side) and $\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = \deg - 1 = 3$ (cycle side, already QGEO.COHO.M.01/v177); $g_{\text{car}} = 5 = \text{rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, $\text{rank } D_5 + \text{rank } A_3 = 5 + 3 = 8$, v1), $N_{\text{fam}} = 3$. [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by $\text{rank}(D_5) = 5$, but the physical identification is conjectural and does <i>not</i> replace the over-determined $g_{\text{car}} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C]
v190_nariai_entropy_bound.py	The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\text{tot}}/S_{\text{dS}} = (x^2+1)/\Phi_3(x)$ ($x=r_b/r_c$; $\Phi_3=x^2+x+1$ the $N_{\text{fam}}=3$ cyclotomic, already in tfpt_horizon_readouts) satisfies $S_{\text{tot}} \geq \frac{2}{3}S_{\text{dS}}$ with equality exactly at the Nariai merge $x=1$, via the perfect square $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x>0$ is $2/3 = Z_2 /N_{\text{fam}}$) [E]
v191_universal_branch_line.py	The universal branch line, honestly typed [C] — an exact affine <i>relabeling</i> , not a theorem. The map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$ carries the SdS branch points $m = \pm 1/N_{\text{fam}}$ onto the flavor branch points $\{2, 5\} = \{ Z_2 , g_{\text{car}}\}$ ($m=0 \rightarrow 7/2$). The arithmetic is exact, but the map exists for <i>any</i> two pairs of points (two intervals are always affinely equivalent), so the slope $N_{\text{fam}}^2/2$ and midpoint $7/2$ are <i>determined</i> by the data, not forced predictions; a decoy pair $\{3, 4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 = Z_2 /N_{\text{fam}}$ ramification of the mass $\leftrightarrow$ transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C]
v192_qgeo_energy_clock.py	The energy-conserving-clock reformulation, honestly typed: QGEO.ENERGY.01 ($\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) <i>relocates</i> the bedrock QGEO.SYM.01 to a more checkable operator identity, but does <i>not</i> eliminate it. The downstream chain (conformality $\rightarrow$ Möbius $\rightarrow z \mapsto iz \rightarrow \mu_4 \rightarrow$ QGEO.SYM.01) already exists (v180/v181); the new upstream link (energy invariance $\Rightarrow$ conformality in 2D) is checkable on the DtN operator $\Lambda = k $. But ‘ ρ preserves Λ ’ is plausibly equivalent to ‘ ρ is the conformal deck’, so it is a sharper restatement, not a theorem, unless ρ is independently defined and the invariance independently proven (open). Bedrock stays [O]
v193_qgeo_energy_commutator.py	QGEO.ENERGY.02: the energy- <i>commutator</i> contract (the non-circular sharpening of v192): $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, ρ the carrier clock, Λ_Σ the DtN energy form of the <i>raw</i> RP seam. Three sub-claims: (1) [E] ρ is carrier-defined (Coxeter of $W(A_3)=S_4$, order $4= \mu_4 $, v117), not imported from the seam; (2) [O] the non-circularity hinge — Λ_Σ must be the <i>raw</i> RP-seam DtN, not the μ_4 normal form; (3) [E] $[\rho, \Lambda_\Sigma]=0$ forces the $(1, 2, 3)$ grading on H^1 (QGEO.COHO.M.01). Exact [E] rider: the induced $k=(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m=q(A_3)=\frac{3}{4}$ (FAMILY), not $q(D_5)=\frac{5}{4}$ (carrier, off by $5/3$) — gravity is family-geometry-induced. Proof target [O]; if proven, QGEO.SYM.01 becomes an energy-rigidity theorem [O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v194_raw_seam_dtn_rp.py	QGE0.DTN.01 — subclaim 2 of v193 (“the raw RP-seam DtN is RP-definable”) tackled: a <i>reduction</i> , not a closure. (1) [E] finite core: $\rho: z \mapsto iz$ on $H^1 = \langle w_1, w_2, w_3 \rangle$, $w_k = z^{k-1} dz / (z^4 - 1)$, acts as $\rho^* w_k = i^k w_k$ (characters $i, -1, -i$ distinct), so $[\rho, \Lambda_\Sigma] = 0$ on H^1 (v177). (2) [E] hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without assuming $\mathbb{P}^1 \setminus \mu_4$. (3) [O] residual relocates: full- L^2 commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). QGE0.SYM.01 reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still [O], not closed [O]
v195_marks_lefschetz_character.py	QGE0.MARKS.02 — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing D_4 marks, they are <i>forced</i> to be a free μ_4 orbit. [E]: $\rho: z \mapsto iz$ (order 4) fixes only $\{0, \infty\}$; $H^1 = \chi_1 + \chi_2 + \chi_3$ (the A_3 exponents (1, 2, 3), v177) has no trivial character $\Rightarrow$ no puncture at a fixed point; $\text{rank } H_1 = n - 1 = 3 \Rightarrow n = 4 \Rightarrow$ exactly one free orbit $\Rightarrow$ marks = μ_4 (cross-ratio 2); $\text{Tr}(\rho H^1) = i + i^2 + i^3 = -1$, $L(\rho) = 1 + 1 = 2 = \# \text{fixed}$. [O] residual relocates from “the seam is $\mathbb{P} \setminus \mu_4$ (geometric)” to “ H^1 carries the A_3 -exponent rep” — less circular [E][O]
v196_seam_energy_functional.py	QGE0.VARI.01 — the E_{fail} variational functional (review path 1), a numerically-testable sharpening of the v194/ENERGY.02 target: $E_{\text{fail}} = \ [\rho, \Lambda_\Sigma]\ _{HS}^2 + \ \rho^4 - 1\ ^2 + \ \Theta \rho \Theta - \rho^{-1}\ ^2$, with $E_{\text{fail}} = 0 \Leftrightarrow$ the QGE0.SYM.01 conditions. [E] on the finite H^1 block ($\rho = \text{diag}(i, -1, -i)$, Λ diagonal, $\Theta = \text{conj}$) all three vanish $\Rightarrow E_{\text{fail}} = 0$; [E] a μ_4 -breaking Λ or non-order-4 ρ gives $E_{\text{fail}} > 0$ (μ_4 a true minimiser); [E] the zero locus is the seam-deck conditions; [O] the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure [E][O]
v197_rr_carrier_clifford_d5.py	ARCH.RRCAR.02 — hardening the Riemann–Roch carrier from g_{car} to D_5 itself (review point 7): the 5-dim carrier mode space $C_{\text{car}} := H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ generates the $\mathfrak{so}(10) = D_5$ half-spinor by its even Clifford algebra. [E] $\dim \Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5 = 16 = 2^{g_{\text{car}} - 1} = \dim S^+$; the $\mathfrak{so}(10) = D_5$ half-spinor $2^4 = 16 = S^+ = \Lambda^{\text{even}}(C_{\text{car}})$; closes the chain $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ from one four-point divisor (cycle side $\text{rank } H_1 = 3 = N_{\text{fam}}(A_3)$; function side $h^0 = 5 = g_{\text{car}}$; Clifford spinor $16 = D_5$ half-spinor = carrier). [C] “carrier = Clifford spinor of the seam mode space” stays physical (as v189), does not change the over-determined $g_{\text{car}} = 5$ [E][C]
v198_modular_commutator_reduction.py	QGE0.MODULAR.01 — cracking the bedrock to a state-invariance. The v194 residual ($[\rho, \Lambda_\Sigma] = 0$ on full L^2 , framed as Bisognano–Wichmann with a circularity worry) is (a) made <i>exact</i> at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. [E] the DtN principal symbol $ k = \sqrt{-d^2}$ is $\text{diag}(n)$ and the clock $\rho: z \mapsto iz$ is $\text{diag}(i^n)$ in the Fourier basis — both diagonal $\Rightarrow [\rho, k] = 0$ exactly on <i>all</i> of L^2 ; [E] Tomita–Takesaki: $[\rho, \Lambda_\Sigma] = 0$ follows from $\omega \circ \rho = \omega$ (the modular Δ is intrinsic to (M, Ω) ; a state-preserving symmetry commutes with it), needing <i>no</i> conformal covariance — removes the BW circularity; [E] the finite H^1 block is already μ_4 -invariant (v177). [O] residual: the full raw seam state is μ_4 -invariant ($\omega \circ \rho = \omega$) $\Leftarrow$ the Gaussian bulk measure is deck-invariant = operational QGE0.SYM.01. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v199_seam_state_invariance.py	QGE0.STATE.01 (work package A) — attacking $\omega \circ \rho = \omega$ directly on the raw seam: a further localising reduction, not a closure. [E] setup non-circular ($\rho = \text{Coxeter of } W(A_3) = S_4, \rho^4 = 1$, an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); [E] ground-state reduction $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$ ($C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$); [E] character criterion: $[\rho, H_1] = 0 \Leftrightarrow H_1$ block-diagonal in the four μ_4 -character classes $\{n = r \bmod 4\}$; principal symbol $ k = \text{diag}(n)$ passes automatically. [O] residual: the bounded sub-principal symbol has <i>no</i> off-character matrix elements — non-circular, sharply typed, holds on H^1 (v177); not a closure [E][O]
v200_seam_variational_scan.py	QGE0.VARI.02 (work package B) — the discretised variational test: v196 had $E_{\text{fail}} = 0$ at the μ_4 config on the fixed block; here the clock angle is free and E_{fail} is minimised over it. [C] scanning $E_{\text{fail}}(\theta) = \ \rho(\theta)^4 - 1\ ^2 + \text{faithfulness over } \theta \in (0, \pi)$ gives the unique minimum at $\theta = \pi/2$ ($E_{\text{fail}} = 0$: order-4 <i>and</i> three distinct characters $i, -1, -i$ = the $(1, 2, 3)$ grading); decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. So $\arg \min E_{\text{fail}}$ = the μ_4 normal form. [E] consistent with v196/v199/v198. [O] a numerical sign on the finite H^1 grading, not the full operator-valued minimisation (the v199 residual) [C][O]
v201_seam_subprincipal_marks.py	QGE0.SUBPRIN.01 — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is <i>not</i> an independent postulate. [E] the seam DtN is $\Lambda = k + M_f$, M_f = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n M_f n' \rangle = f_{n-n'}$. [E] M_f is μ_4 -character-block-diagonal $\Leftrightarrow f$ has Fourier support only on modes $\equiv 0 \pmod{4}$ $\Leftrightarrow f$ is $\mathbb{Z}_4$ -invariant ($ k $ already commutes, v198). [E] a μ_4 -mark sum $f(\theta) = \sum_{j=0}^3 g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \bmod 4]$ for <i>any</i> profile g — automatically $\mathbb{Z}_4$ -invariant, so (v195 marks = μ_4 orbit) <i>forces</i> the sub-principal symbol block-diagonal; negative controls: 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it. [O] new residual: $\omega \circ \rho = \omega$ reduces to ‘the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck v177)’, the narrowest, structurally-definitional form of QGE0.SYM.01 [E][O]
v202_rare_kaon.py	The rare kaon sector $K \rightarrow \pi \nu \bar{\nu}$ from the TFPT CKM point ([C] downstream flavor readout, <i>not</i> a compiler power). The kernel fixes the full CKM matrix ($s_{12} = \lambda_C$, $s_{23} = \varphi_0 / (1 + \lambda_C) = V_{cb} $, $s_{13} = \lambda_C^3 / 3$, $\delta = \pi/3 + 3\lambda_C^2 = 1.19823$ rad; v18/v88) with no flavor fit; feeding the exact point into the standard Brod–Gorbahn–Stamou short-distance formulas gives $\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = 9.45 \times 10^{-11}$ and $\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = 3.33 \times 10^{-11}$. [E] CKM point + exact apex $(\bar{\rho}, \bar{\eta}) = (0.1374, 0.3509)$, $\lambda_t = V_{ts}^* V_{td}$; [C] the branching ratios (<i>not</i> parameter-free — $X_t, P_c, \kappa_\pm$ are external EW/QCD input); [E] $\text{BR}(K^+)$ $\text{BR}(K^+) = 9.45 \times 10^{-11}$ lands ON the NA62 2016-2024 combination $(9.6_{-1.8}^{+1.9}) \times 10^{-11}$ (La Thuile 2026), $+0.08\sigma$ (the 2016-2022 obs $(13.0_{-3.0}^{+3.3}) \times 10^{-11}$ was $\sim 1.2\sigma$ above) — a strong consistency hit, but an F_{transfer} bridge, not a unique TFPT-vs-SM discriminator (SM $(8.6 \pm 0.4) \times 10^{-11}$ also inside the NA62 error); [E] Grossman–Nir ratio $0.352 \ll 4.3$; [X] dated kill test: a stable NA62 $\text{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\text{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched) [C][X]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v203_eht_achromatic.py	The EHT achromatic polarization-rotation intercept β_{BH} around a horizon-scale black hole — the surviving core of the old UFE/BH notes (<i>not</i> the RN-type metric, superseded by Nariai/seam=horizon v101–v104/v190; only the polarization signature). [E] the coupling is a compiler number: $\beta_{\text{BH}}(r)=Q_e^{\text{eff}}Q_m^{\text{eff}}/(256\pi^4r^2)=16c_3^4(Q_eQ_m/r^2)$, $16c_3^4=1/(256\pi^4)$ exactly. [E] α -kernel cross-link: $16c_3^4=\delta_{\text{top}}/3$, $\delta_{\text{top}}=48c_3^4=3/(256\pi^4)$ — the <i>same</i> top-form coefficient that fixes the α -kernel precision-zone correction (one row from $\beta_{\text{rad}}=\varphi_0/4\pi$). [C] TFPT fixes the $1/r^2$ profile, achromaticity, and sign-flip under $E\cdot B$ reversal; the amplitude Q_eQ_m is an MHD/GR weight, not a compiler output. [X] dated kill test: the residual intercept must pass three nulls (frequency/achromatic, spatial $1/r^2$, sign-flip); a null-consistent residual after honest GRMHD subtraction falsifies the channel (core untouched) [E][C][X]
v204_muon_seam_g2.py	The muon anomalous magnetic moment $a_\mu=(g_\mu-2)/2$ as a seam vertex readout ([C] downstream, <i>not</i> a compiler power; archive integration of the old muon- $g-2$ note). The carrier fixes the second-order defect $\delta_2=B\gamma\delta_{\text{top}}^2=\frac{5}{4}\delta_{\text{top}}^2$ ($\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4=48c_3^4=3/256\pi^4$, $v2/v23$; $B\gamma=\frac{3}{2}\cdot\frac{5}{6}=\frac{5}{4}$, v51); the seam-loop projection $a_\mu^{\text{seam}}=\delta_2/(2\pi)=45/(524288\pi^9)\approx 2.879\times 10^{-9}$. [E] the value is an exact compiler number ($\delta_2=4!\text{Tr}_{S^+}(X^2)c_3^8$, $\text{Tr}=120=5!$); [C] the identification of $\delta_2/(2\pi)$ as a magnetic vertex ($1/2\pi=4c_3$) is a physical bridge; [E] 0.81σ vs dispersive $\Delta a_\mu=(2.49\pm 0.48)\times 10^{-9}$; [C] HONEST: lattice/CMD-3 HVP shrinks it ($\sim 1.5\times 10^{-9}$), the fixed value then $\sim 1.5\sigma$ high; [X] dated kill test: a converged Δa_μ outside $2.879\times 10^{-9}\pm 0.5\times 10^{-9}$ excludes the mechanism [E][C][X]
v205_xi_threequarter.py	$\xi=c_3/\varphi_{\text{tree}}=3/4$: an <i>independent</i> route to the gravitational $3/4$ (archive integration of the old Eliminating- K note). The torsion-compression ansatz $\kappa^2=\xi\varphi_0/c_3^2$ with the Einstein limit $\kappa^2=8\pi G$ fixes $\xi=8\pi c_3^2/\varphi_0=c_3/\varphi_0$ (since $8\pi c_3^2=c_3$), and at tree level $\varphi_{\text{tree}}=1/(6\pi)$ this is exactly $3/4$. [E] tree identity $\xi=c_3/\varphi_{\text{tree}}=3/4$; [E] Einstein-limit reduction $8\pi c_3^2=c_3\Rightarrow G$ is an output; [E] over-determination $3/4=q(A_3)=N_{\text{fam}}/ \mu_4 =\ln(m/\mu)$ (v152 $k=c_3/2$) — the torsion-compression ξ and the gapped EH replica are two independent appearances of the same gravitational $3/4$; [E] with the retained seed $\xi=c_3/\varphi_0=0.748303$ (-0.226%); [C] the $\kappa^2=\xi\varphi_0/c_3^2$ ansatz is the bridge, the modern derivation is v152/v68, $1/G$ stays the one anchor (v60/v153) [E][C]
v206_h0_lambda_branch.py	The cosmological-constant Letter branch: the seam number $\delta_\Sigma=m_1e^{-2\pi\alpha^{-1}/3}$ with the lowest block $m_1=\Omega_{\text{adm}}=48$, the Planck-convention split, and the quantitative $H_0=66.5\text{--}67.1$ (archive integration of the tfpt-31 CC note; sharpens the qualitative H_0 -vs- Λ row). [E] $\delta_\Sigma^{\text{lead}}=48e^{-2\pi\cdot 137.036/3}\approx 1.085\times 10^{-123}$; [E] convention split $M_{\text{Pl}}^4/\bar{M}_{\text{Pl}}^4=(8\pi)^2=631.65$ (unreduced vs reduced — must be displayed); [E] $\rho_\Lambda=M_{\text{Pl}}^4\delta_\Sigma\approx 2.4\times 10^{-47}\text{ GeV}^4$ (observed vacuum order); [E] 123-orders cross-link $ \log_{10}\delta_\Sigma \approx 122.96$, consistent with the v60 split 122.948; [C] under flatness closure $H_0^{\text{lead}}=66.51$, $H_0^{\text{disc}}=67.10$ — $1.6\text{--}0.5\sigma$ below Planck, $\sim 6\sigma$ below SH0ES: the Hubble tension is <i>not</i> relieved by raising the vacuum scale; H_0 uses imported $\Omega_b, \omega_{\text{DM}}$, a [C] readout [E][C]
v207_asymptotic_safety.py	Asymptotic Safety as a <i>second</i> , independent, [O] route to the metric sector (archive integration of the old Five-Problems note). The current theory closes QG via the discrete seam-net/AQFT programme (v175; open premise QGEO.REALIZE.01); this records a complementary continuum-FRG argument that a non-Gaussian UV fixed point exists from the same axioms — a plausibility cross-check, <i>not</i> load-bearing, does <i>not</i> close G_{metric} . [E] the axion coupling is fixed and small $y^2=16c_3^2=1/(4\pi^2)=0.025330$ exactly ($g_{a\gamma\gamma}=-4c_3$ from the cubic); [E] existence by IVT: $\partial_t g/g=2+\eta_R$ is $+2>0$ as $g\rightarrow 0$ and <0 at large g (torsion K^2 damping) $\Rightarrow$ a root $g_*>0$; [E] $g_*=2/a_1>0$ finite, Reuter-type $O(1)$, no new free input; [O] HONEST: a continuum-FRG ansatz the discrete-compiler theory did <i>not</i> adopt, primary stays the seam-net (v175) [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v208_bh_thermodynamics.py	Black-hole thermodynamics from the seam: the modular Hawking temperature and the scalaron-corrected Wald entropy (archive integration of the dropped tfpt-42 stationary-horizons section, re-typed through the current Tomita–Takesaki seam work v198/v199). [E] modular $T_H: \Delta^{it}=e^{-2\pi t K_H}$ (Bisognano–Wichmann) $\Rightarrow$ KMS at $\beta=2\pi$ and $T_H=\kappa/(2\pi)$; the 2π is the seam unit $1/(4c_3)$ ($1/2\pi=4c_3$, v58), reproducing $T_H=c_3/M$ (v57); seam $\leftrightarrow$ horizon identification [C]; [E] Wald entropy: with the induced $f(R)=R+R^2/(6M_s^2)$ ($M_s=c_3^{7/2}\bar{M}_{\text{Pl}}$, v28), $S_W=(f_R A)/(4G)=(A/4G)(1+R_h/3M_s^2)$ — leading $A/4G$ is the c_3 area law ($1/4=1/ \mu_4 $), the correction an exact scalaron consequence; [E] SdS $T_H=(1/4\pi r_h)(1-\Lambda r_h^2)$, $1/4\pi=2c_3$; [E] area law $S_{BH}=M^2/(2c_3)=4\pi M^2=A/4G$ [E][C]
v209_bh_regular_core.py	The black hole as a seam fixed point (topological defect), <i>not</i> a singularity: the honest [O] synthesis of the old Five-Problems BH picture, with the superseded torsion/RN-vortex metric dropped (RN-type metric superseded by Nariai/seam=horizon v101–v104/v190; cf. v203). Built entirely from established suite atoms. [E] no singularity = seam fixed point: the gapped attractor multiplier $\lambda_2=(2/3)^6=64/729$ with gap $\Delta=6\log(3/2)>0$ (v56) — the core is a fixed point, $\rho\rightarrow\infty$ replaced by $\varphi\rightarrow\varphi_*$ [C]; [E] maximal BH = anchor: Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$, roots $(1, 1, -2)$ = the traceless projection of $a=(1, 1, 2)$ (v101/v190); [E] information stays in the boundary field: Page recovery decays at $\lambda_2=(2/3)^6$ per step (v54) [C]; [E] holographic remnant $S_{BH}=A/4G$, $1/4=1/ \mu_4 $ [C]; [O] HONEST: a qualitative structural synthesis (reuses v54/v56/v57/v101/v190), <i>not</i> load-bearing, <i>not</i> a metric [E][O]
v210_mark_local_dtn.py	A numerical <i>sharpening</i> of v201/QGEO.SUBPRIN.01 on <i>realistic</i> Steklov curvature profiles, carried to the quasi-free <i>state</i> level (the actual $\omega\rho=\omega$, not just $[\rho, \Lambda]=0$); <i>not</i> a closure of QGEO.SYM.01. A real von-Mises bump $g(\theta)$ at a puncture, summed over the four μ_4 marks $f(\theta)=\sum_{j=0}^3 g(\theta-j\pi/2)$, builds the Steklov DtN $\Lambda= D_\theta +M_f$. [E] every off-(mod 4) Fourier coefficient of the real f vanishes to $<10^{-16}$ (exact 4-fold cancellation), although g has all modes. [C] $\ [\rho, \Lambda]\ < 10^{-15}$. [C] <i>state invariance</i> (new vs v201): the quasi-free covariance $C=\frac{1}{2}(1+\text{sgn } H_1)$ satisfies $\rho C\rho^\dagger=C$ to $<10^{-13}$ — the state, not just the operator, is clock-invariant. [C] negative controls (single off-centre bump, $\mathbb{Z}_3$ source, 4 generic marks) break both by $O(1)$; convergence stable for $N\in\{16, 32, 64\}$. [O] certifies ‘mark-local DtN $\Rightarrow \omega\rho=\omega$ ’ for concrete profiles <i>and</i> at the state level; the premise that the physical seam <i>is</i> mark-local stays [O] — the narrowest definitional form of QGEO.SYM.01, not its closure [C][O]
v211_axion_spine_angle.py	Axion DM, the spine-angle branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (108°): an alternative, more robust misalignment angle than the determinant-line hilltop $\theta_i \approx 170^\circ$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$). [E] rational motive $\theta_i = 3\pi/5$ exactly (spine quotient $3/5$, no fit); [E] the v185 finite-T solver ($\theta=1 \rightarrow 0.03$) reaches Ω_{DM} at $\theta \approx 106^\circ$, so 108° is on target (harmonic $0.03\theta^2=0.107$); [C] <i>robust</i> : 108° is 62° below the 170° hilltop (mild-anharmonic, not exponentially sensitive); [O] an alternative ansatz to $\theta_i=\pi(1-\varphi_{\text{seam}})\approx 170^\circ$ (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); [X] kill test: $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ demotes the branch [E][C][O][X]
v212_leptogenesis_decouple.py	Leptogenesis, the scalaron-decouple branch: both Boltzmann inputs share the decouple $A_\Lambda=10= E(K_5) $ — $M_1=M_{\text{scal}}\varphi_0^2/A_\Lambda \approx 8.65\times 10^9$ GeV and $\tilde{m}_1=m_3/A_\Lambda \approx 5$ meV. A cleaner [C] route than v184’s $M_1=M_R\varphi_0^4$ (which only relocated the free input to the un-pinned seesaw M_R). [E] shared decouple $A_\Lambda=10$; [E] $M_1=8.65\times 10^9$ GeV inside the thermal window $[3\times 10^9, 3\times 10^{10}]$ (v169); [E] $\tilde{m}_1 \approx 5$ meV, $K \approx 4.6$ strong washout; [E] Davidson–Ibarra $\varepsilon_1 \approx 8.5\times 10^{-7}$, $\eta_B \approx 1.2\times 10^{-9}$ brackets observed 6.1×10^{-10} ; [C] HONEST: M_1 uses only $\{M_{\text{scal}}, \varphi_0, A_\Lambda\}$ but the coupling mechanism is posited not derived; [X] kill test falls the route, not the theory (FR.ETAB.01 independent) [E][C][X]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v213_ftransfer_ functor.py	The <code>F_transfer</code> functor contract (<code>CONTRACT.F.01</code>): the four frontier transfers are ONE typed functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ with four structural axioms, each instance a discrete compiler kernel [E] composed with an external continuous solver [C] ; consolidates v183 (F_{pole}), v212 ($F_{\text{Boltzmann}}$), v211 (F_{relic}), v164/FR.MPME (F_{QCD}), the structural complement of the v187 typing guard. [E] axiom 1 μ_4 -deck equivariance ($\lambda_2 = (2/3)^6 = 64/729$ the deck transfer eigenvalue); [E] axiom 2 Plücker preservation ($53 = a^T(R+Q)\mathbf{1}$, $\text{Pl}(F) = (1, 22, 30)$, $\ \text{Pl}(K)\ _1 = 11$); [E] axiom 3 positivity/stochasticity ($\text{spec } T = \{1, (2/3)^6, (1/3)^6\}$ positive, $\kappa_f \in (0, 1)$); [E] axiom 4 external modules explicit ($b_3 = -7$ a carrier output, $ b_3 = \Omega_{\text{adm}} - 10b_1 = 7$); [C] verdict: a typed functor, not a bag of open topics; an architectural consolidation guarded by v187, not a closure [E][C]
v214_seam_ pillowcase.py	The <i>pillowcase</i> reduction of <code>QGEO.SYM.01</code> (<code>QGEO.PILLOW.01</code>): the two separately-tracked open QGEO residuals — v180/QGEO.ISO.01 (the carrier clock is an order-4 isometry of the seam metric) and v201/QGEO.SUBPRIN.01 (the DtN sub-principal symbol is mark-local, the seam flat away from the μ_4 marks) — are ONE statement, ‘the seam carries the uniformising flat orbifold (pillowcase) metric’, and the order-4 clock is <i>derived</i> from cross-ratio 2 (v168), not assumed. [E] $\chi_{\text{orb}}(S^2(2, 2, 2, 2)) = 2 - 4(1 - \frac{1}{2}) = 0 \Rightarrow \text{flat}$ (Trojanov cone-metric uniformisation, unique up to scale; Gauss–Bonnet four deficits π sum to $4\pi = 2\pi\chi(S^2) \Rightarrow \int K_{\text{smooth}} = 0$, flat away from the marks). [E] NEW LINK: cross-ratio(μ_4)=2 (v168) $\Rightarrow j(\lambda) = 256(\lambda^2 - \lambda + 1)^3 / (\lambda^2(\lambda - 1)^2)$ with $j(2) = 1728$ (all six harmonic cross-ratios $\{2, -1, \frac{1}{2}\}$ give 1728) $\Rightarrow$ the square modulus, hence the order-4 clock, is a <i>consequence</i> of cross-ratio 2. [E] $j = 1728 \Leftrightarrow \tau = i$ (square torus) $\Leftrightarrow \text{CM by } \mathbb{Z}[i] \Leftrightarrow \text{Aut} = \mathbb{Z}/4$ ($j=0$ gives $\mathbb{Z}/6$; generic $\mathbb{Z}/2$). [E] explicit CM $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$ has order $4 = z \mapsto iz$. [E] negative control $\lambda = 3 \Rightarrow j = 21952/9 \notin \{0, 1728\} \Rightarrow \text{Aut} = \mathbb{Z}/2$, no clock. [E] UNIFICATION: ‘seam = uniformising flat pillowcase metric’ implies both v201 mark-locality and v180 QGEO.ISO.01. [O] does <i>not</i> close QGEO.SYM.01 — the canonical constant-curvature-metric premise stays open, milder/more canonical; net existence + full-cone RP remain v175’s [E]
v215_seam_deck_ killtest.py	The bedrock <code>QGEO.SYM.01</code> ($\omega \circ \rho = \omega$) recast as a <i>falsifiable</i> kill-test (<code>QGEO.KILL.01</code>), not a proof-promise — a foundational symmetry cannot be derived from nothing (v181/v153), and Bisognano–Wichmann would presuppose the covariance it should produce. Non-circular reduction (v198/v199/v201): $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$; the principal symbol $ k $ commutes exactly, so the residual is the sub-principal M_f with $[\rho, M_f] = 0 \Leftrightarrow f$ $\mathbb{Z}_4$ -invariant; measure form = the raw Gaussian bulk form A is μ_4 -deck-invariant. Four levers (carrier-side [E] , seam-side [O]): K1 exactly 4 marks (order-4 clock free orbit $\mu_4 + b_1 = \#\text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \#\text{marks} = 4 = \mu_4 $); K2 square config (cross-ratio $2 \Rightarrow j = 1728$, $\text{Aut} = \mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j=0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); K3 mod-4 sub-principal support (μ_4 -mark sum $\Rightarrow f_m = 0$ for $m \not\equiv 0 \pmod{4} \Rightarrow [\rho, M_f] = 0$); K4 transfer gap $(2/3)^6 = 64/729$ (cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$). Freeze-bound (<code>freeze_file.csv seam_deck_symmetry</code>). [O] the decisive kill is <i>deferred</i> (<code>QGEO.REALIZE.01</code>): the raw $\mathbb{P} \setminus \mu_4$ collar DtN must yield 4 square marks <i>without</i> inputting them; carrier-side K1–K4 are green regression guards. A kill-test, not a closure

continued on next page

Script	What it machine-checks (<i>continued</i>)
v216_marks_gauss_bonnet.py	The four seam marks <i>emerge</i> from Gauss–Bonnet (QGEO.MARKS.03), executing the K1 lever of the v215 kill-test. [E] mark count = 2χ : $\mathbb{Z}_2$ branch points (deficit π) on a flat sphere ($\chi=2$) give $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = \mu_4 = e_1(a) = N_{\text{fam}}+1$ (the same $\chi=2$ as the 8 in $c_3=1/(8\pi)$). [E] the Euclidean sphere 2-orbifolds ($\chi_{\text{orb}}=0$) are exactly $(2, 3, 6), (2, 4, 4), (3, 3, 3), (2, 2, 2, 2)$. [E] all-order-2 (the $ \mathbb{Z}_2 $ sheet branch) $\Rightarrow$ uniquely $(2, 2, 2, 2)$; [E] $N_{\text{fam}}=3$ (rank $H^1=\#\text{marks}-1$) $\Rightarrow$ uniquely $(2, 2, 2, 2)$, picking the 4-mark square over the 3-mark hexagonal $(3, 3, 3)$; [E] a free order-4 deck $\Rightarrow$ uniquely $(2, 2, 2, 2)$ — three independent selectors converge. [O] residual: the square modulus (cross-ratio $2 \Rightarrow j=1728$, v214) needs the one order-4 carrier input. The mark count + equality are now derived; does <i>not</i> close QGEO.REALIZE.01
v217_marks_emergence_scan.py	The <i>numerical</i> emergence scan (QGEO.EMERGE.01, the sibling of v216 as v210 is to v201): with the number of marks n and their positions free, the 4-equally-spaced (square) configuration is the joint minimiser of {clock-invariance + 3-family cohomology + transfer gap $(2/3)^6$ } on the actual seam DtN/state. [C] $n=4$ equally-spaced $\Rightarrow \ \rho_4 C \rho_4^\dagger - C\ \sim 10^{-14}$ (clock-invariant state); [C] family count selects $n=4$ (scan $n \in \{2..8\}$, only $n-1=3=N_{\text{fam}}$; $n=3 \rightarrow 2$ families = hexagonal, excluded; $n=5 \rightarrow 4$); [C] gap: μ_4 -equivariant $\Rightarrow$ diagonal in the cusp basis (deck-average off-diagonal $\sim 10^{-16}$), cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \Rightarrow$ transfer $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$, gap $(2/3)^6 = 64/729$; [C] position scan: the square minimises the defect (square $\sim 10^{-14}$ vs off-square ~ 0.4); [C] negative controls $n=3, 5, 6$. [O] confirms the emergence on the real DtN (number scanned, positions free); selector = the 3-family/gap structure (carrier input); executes K1 numerically, does <i>not</i> close QGEO.REALIZE.01. Python-only
v218_diamond_axis_geometry.py	Diamond axis geometry: the Sheet Diamond (v94) is a discrete geometry with two axes around the centered cross (v95: C center, U winding, V sheet), F the transfer completion. Three exact [E] lemmas + honest audit/heuristic typing, no new numbers. [E] AXIS CURVATURE: $\det(C+xU)=14+6x$ linear (slope $6= R^+(A_3) $), $\det(C+yV)=14+14y+4y^2$ quadratic, 2nd difference $8=\text{rank } E_8$; $\det B(C+xU)=2\det(C+xU)$, anchor-block sheet curvature $6= R^+(A_3) $. [E] PLÜCKER LADDER: $\text{Pl}(K)=(-1, 6, 4) \rightarrow \text{Pl}(C)=(0, 14, 14) \rightarrow \text{Pl}(F)=(1, 22, 30)$, steps $(1, 8, 10)=(N_\Phi, \text{rank } E_8, A_\Lambda)$ and $(1, 8, 16)=(N_\Phi, \text{rank } E_8, \dim S^+)$ (identity [E], transfer reading [C]). [E] SPECTRAL RAMIFICATION: $\text{Disc}(\chi)=q(r)^2\text{Disc}(q)$ for Q, K, C, F ; squares $(1, 3, 4, 6)$, kernels $(13, 48, 65, 105)$ with $48=\Omega_{\text{adm}}$, $105=N_{\text{fam}}g_{\text{car}} \cdot 7$. [C] heuristic: F the transfer-completion corner, a search principle, not a hard filter; the G_2/F_4 pair-sum labels stay audit-only (cf. v94/v95) [E][C]
v219_icosahedral_mckay.py	McKay bedrock: <i>why</i> the atoms are $\{2, 3, 5\}$. E_8 is the exceptional <i>top</i> of the McKay tower of finite $SU(2)$ subgroups ($2T \rightarrow \hat{E}_6$, $2O \rightarrow \hat{E}_7$, $2I \rightarrow \hat{E}_8$). The McKay graph is built <i>from the group</i> : the 120 icosians close to the binary icosahedral $2I$ (element orders $\{1, 2, 3, 4, 5, 6, 10\}$, containing the 2, 3, 5 axis orders), 9 conjugacy classes, Dixon irrep degrees $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ (sum $30=h(E_8)$, sum of squares $120= R^+(E_8) $); the natural-rep McKay adjacency <i>is</i> the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8 [E], the reading that it explains $(2, 3, 5)$ [C], the raw-seam origin [O]
v220_cp_hexagonal_modulus.py	CP lives in the <i>hexagonal</i> phase fiber, not the square seam deck (a geometric sharpening of red-team Target D). [E] the seam deck is the square modulus $j(i)=1728$, $\text{Aut}=\mathbb{Z}/4$ (the μ_4 clock); its exceptional partner is the hexagonal modulus $j(\rho)=0$, $\rho=e^{i\pi/3}$, $\text{Aut}=\mathbb{Z}/6$, CM by $\mathbb{Z}[\omega]$, $\arg \rho=\pi/3$. [E] the frozen leading CKM reading $\delta=\pi/3+3\lambda_C^2=68.654^\circ$ (v88) has leading term $\pi/3=\arg \rho$; the $ \mathbb{Z}_2 $ alternative $\pi/3+2\lambda_C^2$ is the frozen look-elsewhere trap. The deck stays $\mathbb{Z}/4$ (v215); CP is the $\mathbb{Z}/6$ fiber, the bridge [C][E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v221_seam_qecc.py	The seam as a <i>finite recoverability</i> code (recovery rate $(2/3)^6=64/729$). [E] code dimension $\dim S^+=16=2^{g_{\text{car}}-1}$; the gapped transport T on the cusp-weight 3-space (deviations $(1, -1, 0)$ and the Nariai anchor $(1, 1, -2)$) is symmetric, doubly stochastic (CPTP), spectrum $\{1, (2/3)^6, (1/3)^6\}$; on the trace-zero subspace $\ T\delta\ \leq (2/3)^6 \ \delta\ $ hence $\ T^n \delta\ \leq (2/3)^{6n} \ \delta\ $ (a Knill–Laflamme-type bound); a free ratio breaks it. [C] the holographic/Petz reading (gated on the Seam–Horizon theorem) [E][C]
v222_cm_norm_duality.py	CM-norm duality: the two exceptional moduli give 41 and 7. [E] SQUARE (Gaussian $\mathbb{Z}[i]$, $j=1728$): $N(g_{\text{car}}+i \mu_4)=5^2+4^2=41=10b_1$ — the EM index <i>is</i> the Gaussian norm of the carrier-glue vector; $N(3+2i)=13=\Delta_Q$. [E] HEX (Eisenstein $\mathbb{Z}[\omega]$, $j=0$): $N(3+2\omega)=3^2-3\cdot 2+2^2=7=$ scalaron. The $(3, 2)=(\text{colour}, \text{weak})$ split generates $(5, 6, 7, 13)$ by sum / product / Eisenstein / Gauss norm; the rings are rigid (Eisenstein of $(5, 4)=21 \neq 41$). The architecture bridge square $\leftrightarrow$ EM, hex $\leftrightarrow$ scalaron is [C][E][C]
v223_coxeter_totative_clock.py	The μ_4 clock is the order-4 character of the E_8 Coxeter cycle. [E] $(\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}$ = the E_8 exponents = live Coxeter phases; $\varphi(30)=8=\text{rank } E_8$, $30=2\cdot 3\cdot 5$. [E] $(\mathbb{Z}/30)^\times \cong \mathbb{Z}/4 \times \mathbb{Z}/2$ with order-4 generator 7 ($7^2=19$, $7^4=1 \bmod 30$): a literal μ_4 clock $\langle 7 \rangle = \{1, 7, 13, 19\}$ on the exponents; the conjugate pairs $m+(30-m)=30$ give $ \mu_4 =4$ invariant planes the clock permutes. Only E_8 has $\varphi(h)=\text{rank}$ and $h=2\cdot 3\cdot 5$; the seam-clock identification is [C][E][C]
v224_diamond_fttransfer_path.py	F_{transfer} as <i>one</i> path on the Sheet Diamond, not four external interfaces. [E] family $M(s, t)=R+Q \text{ diag}(s, t, t)$: the corners $K=M(1, -1)$, $C=M(1, 0)$, $F=M(1, 1)$ lie on the sheet axis ($s=1$); the sheet axis is curved $\det M(1, t)=4t^2+14t+14$ (2nd diff $8=\text{rank } E_8$), the winding axis $\det M(s, 0)=6s+8$ flat (slope $6= R^+(A_3) $); Plücker steps $(1, 8, 10)$, $(1, 8, 16)$. [C] F_{transfer} is the $K \rightarrow C \rightarrow F$ sheet geodesic, Koide source $\rightarrow$ pole its 1-D Möbius projection (v37/v183) [E][C]
v225_dual_normal_frame.py	The dual normal frame (d, n) : one boundary compass for flavor and horizon, with CP as orientation. [E] $d=a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)=-\frac{1}{2}(1, 1, -2)$ (the Nariai/horizon normal; $d\cdot 1=0$, $d\cdot a=1$); $n=(5, -9, 6)$ with $n^\top R=(\det R, 0, 0)=(8, 0, 0)$, $n^\top L=(20, 0, 0)$ (the first-generation/torsion selector). [E] oriented volume $\det(\mathbf{1}, d, n)=21=N_{\text{fam}}\cdot 7$. [C] CP as orientation: $\text{Im } \det(\mathbf{1}, d, \rho n)=21 \sin(\pi/3)$ for $\rho=e^{i\pi/3}$ — CP sits where the magnitude bijection (Target D) fails [E][C]
v227_degree_exponent_channel_split.py	$248=120+128$ as a magnitude/phase channel typing (replaces the over-strong S^- -dark-matter reading). [E] E_8 exponents sum $120= R^+(E_8) $ (<i>magnitude</i> channel); degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ sum $128=2^{\text{rank}-1}=\text{rank} \cdot \dim S^+$ (<i>phase/glue</i> channel); $248=120+128=\text{adj}(D_8)+\text{spinor}(D_8)$; $\sum \deg - \sum \exp = \text{rank}=8$. [C] ratios+products (Target D) live in the 120 channel, the un-covered CP phases in the 128 phase/glue channel; [O] a dark-sector reading of 128 stays Frontier [E][C][O]
v228_rr_index_gate.py	P2 as the mode space of a degree-4 seam divisor (a Riemann–Roch <i>index gate</i> ; bedrock language for P2, not a proof from nothing). [E] $\deg D=4= \mu_4 $ on $\mathbb{P}^1 \Rightarrow h^0(\mathcal{O}(D))=5=g_{\text{car}}$ (Riemann–Roch, genus 0); $\text{rank } H_1(\mathbb{P}^1 \setminus 4)=3=N_{\text{fam}}$; $h^0+\text{rank } H_1=8=\text{rank } E_8$, $h^0-\text{rank } H_1=2= \mathbb{Z}_2 $; $\Lambda^{\text{even}}(\mathbb{C}^5)=1+10+5=16=\dim S^+$. The mode-space reading is [C], conditional on the raw seam producing the degree-4 divisor ([O], = QGEO.SYM.01) [E][C][O]
v229_lepton_frobenius_algebra.py	The charged leptons as an étale Frobenius algebra over the μ_6 resolvent. [E] exact coefficients $(c_e, c_\mu, c_\tau)=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, ring closure $c_e c_\tau = \frac{8}{3} = \mathbb{Z}_2 c_\mu$, product $\frac{32}{9} = 2^{g_{\text{car}}} / N_{\text{fam}}^2$. [E] $A=\mathbb{Q}[t]/(m)$, $m=\prod(t-c)$: the trace-form Gram is nondegenerate ($\det=\text{disc}(m)\neq 0$, A étale) — a commutative Frobenius algebra; the C_6 shift has characteristic polynomial t^6-1 (spectrum $\mu_6=\mu_3$ family $\times \mu_2$ sheet). [C] the PMNS extension via $\text{Aut}(A)$ and the hexagonal CM point [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v230_center_budget_norms.py	The Sheet-Diamond center budget (7, 11, 13) is the three <i>local norms</i> of the theory. [E] $C=R+Q$ diag(1, 0, 0), row sums (7, 11, 13): $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ (Eisenstein/hex = scalaron), $13=N_{\mathbb{Z}[i]}(3+2i)$ (Gauss/square = Δ_Q), $11=\binom{4}{0}+\binom{4}{1}+\binom{4}{2}=1+4+6$ (QBL boundary Fock count on the 4 μ_4 corners = $\dim S^+ - g_{\text{car}}=16-5$). The two CM rings (v222) and the seam boundary count meet in one operator center; the reading is [C][E][C]
v231_cp_mu6_phases.py	Both CP phases are μ_6 powers of <i>one</i> hexagonal CM unit, split by the sheet — a structural reduction of red-team Target D (sharper than v220/v225). With $\rho=e^{i\pi/3}$ (the $j=0$ Eisenstein CM unit): [E] ρ is a primitive 6th root ($\rho^6=1$, $\rho^3=-1$ the $\mathbb{Z}_2$ sheet half-turn); [E] δ_{CKM} leading = $\arg(\rho^1)=\pi/3$ (quark), $\delta_{\text{PMNS}} = \arg(\rho^4)=4\pi/3$ (lepton, the <i>phase lattice</i>); [E] $\rho^4=-\rho \Rightarrow \delta_{\text{PMNS}}=\delta_{\text{CKM,lead}}+\pi=(\text{CM unit})\times(\mathbb{Z}_2 \text{ sheet})$ — the two CP phases are <i>one</i> hexagonal unit on the two sheets, not two independent inputs; [E] the C_6/μ_6 monodromy has charpoly t^6-1 (spectrum μ_6 carries ρ); [E] orientation $\text{Im det}(\mathbf{1}, d, \rho^1 n)=+21 \sin(\pi/3)$, $\text{Im det}(\mathbf{1}, d, \rho^4 n)=-21 \sin(\pi/3)$ (sheet-flipped Jarlskog, v225); [E] NEG μ_3 has no -1 (no sheet; $\mu_6=\mu_3$ family $\times \mu_2$ sheet). REDUCTION: 2 CP phase inputs $\rightarrow$ 1 hexagonal unit + the sheet; the physical CP derivation stays [C], the deck stays $\mathbb{Z}/4$ (v215) [E][C]
v232_e8_kleinian_seam.py	The seam as the E_8 <i>Kleinian singularity</i> — a canonical algebraic-geometric model for the open seam-realisation premise QGEO.REALIZE.01 (the du Val side of the McKay correspondence, v219). [E] the McKay graph of $2I$ is affine E_8 ; dropping the unique trivial-rep node (the single Kac mark 1) leaves the <i>finite</i> E_8 Dynkin on 8 nodes = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$ (trivalent node, arms 1, 2, 4). [E] eight exceptional $\mathbb{P}^1$'s = $\text{rank } E_8 = g_{\text{car}} + N_{\text{fam}} = \#$ nontrivial $2I$ irreps — a fourth reading of the 8 in $c_3=1/(8\pi)$. [E] the negated intersection form is the E_8 Cartan (even, $\det=1$, positive definite; each $\mathbb{P}^1$ a (-2) -curve). [E] one degree-1 irrep $\Rightarrow 2I$ perfect $\Rightarrow H_1(S^3/2I)=0$: the link is the Poincaré homology 3-sphere, $\pi_1=2I$ of order $120= R^+(E_8) $. [C] seam reading: the raw RP seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for QGEO.REALIZE.01, not a P2 proof (presupposes E_8); bedrock [O][E][C][O]
v233_cp_triality_phase.py	The CP phase is the <i>universal</i> family/triality phase, only sheet-split — one step past v231 in $\mu_6=\mu_3$ (family/triality) $\times \mu_2$ (sheet). [E] $\mu_6=\mu_3 \times \mu_2$: the $\mathbb{Z}_3$ triality centre $\{1, \omega, \omega^2\}$ = the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \times 2\pi$, the sheet $\mu_2=\{1, -1\}$; $\rho=\omega^2 \cdot (-1)$. [E] SAME FAMILY CLASS: $\rho^4=\rho^1 \cdot \rho^3$ with $\rho^3 \in \mu_2$, so the quark (ρ^1) and lepton (ρ^4) CP phases have the <i>same</i> image in $\mu_6/\mu_2 \cong \mu_3$ and <i>opposite</i> sheet — the CP phase <i>is</i> the universal triality phase ω^2 (the $2/3$ cusp-weight $\mathbb{Z}_3$ centre), not a free power choice. [E] ω is the μ_3 factor of the order-30 (2, 3, 5) Milnor/Coxeter monodromy that builds E_8 (v232). Reduces Target D from ‘one hexagonal unit + sheet’ (v231) to ‘the universal triality phase + sheet’; the physical CP derivation stays [C][E][C]
v234_seam_holomorphy_selection.py	The Seam-Holomorphy selection certificate: the entire structural residual is <i>one</i> condition with <i>three</i> equivalent faces, all forcing E_8 . The condition: <i>the seam carries no nontrivial abelian sector</i> . [E] (i) AQFT — the boundary net is holomorphic (μ -index 1); (ii) geometry — the seam link is a homology 3-sphere $\Leftrightarrow$ the binary group is perfect $\Leftrightarrow 2I$; (iii) rep theory — exactly one 1-dim irrep. [E] $\#(\text{mark-1 affine nodes}) = \Gamma^{\text{ab}} = H_1(S^3/\Gamma) = \#(1\text{-dim irreps})$: $A_n \rightarrow n+1$, $D_n \rightarrow 4$, $E_6 \rightarrow 3$, $E_7 \rightarrow 2$, $E_8 \rightarrow 1$ — only E_8 gives 1 ($2I$ the unique perfect $SU(2)$ subgroup, Poincaré the unique homology-sphere space form). [E] holomorphic $c=8=g_{\text{car}}+N_{\text{fam}} \Rightarrow$ the unique even unimodular rank-8 lattice = E_8 . [E] the three faces (v83/v154, v232, v219) are the SAME E_8 -selector, so Target A, G_{net} , the carrier P2 and the Kleinian seam are <i>one</i> gate. [O] the closing theorem (stated): $\text{RP}+\text{gap}+\text{chirality} \Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ invertible bulk $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$; the single residual analytic step is ‘free bulk $\Rightarrow$ holomorphic boundary’ [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v235_seam_chern_simons.py	The closing step in abelian Chern–Simons language: holomorphic $\Leftrightarrow \det K=1$ — a genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics. [E] a free gapped bosonic 2+1d bulk = an even integer K -matrix with $\# \text{anyons} = \det K $, edge $c = \text{signature}(K)$, holomorphic $\Leftrightarrow \det K =1$. The TFPT extension tower (v92) is the condensation tower read by determinant: carrier $D_5 \oplus A_3$ ($\det 16$, 16 anyons) $\rightarrow SO(16)_1 = D_8$ ($\det 4$) $\rightarrow (E_8)_1$ ($\det 1$, holomorphic), all rank 8, $c = \text{sig}=8=g_{\text{car}}+N_{\text{fam}}$. [E] holomorphic $\Leftrightarrow \det 1 \Leftrightarrow E_8$ = the Kitaev E_8 quantum-Hall state. [E] condensation: a Lagrangian glue of order $\sqrt{16}= \mu_4 =4$ takes $\det 16 \rightarrow 1$ (a partial order-2 isotropic only $\rightarrow 4=D_8$). [E] NEG: D_8 is free+bosonic but has 4 anyons; an odd K is fermionic — $\det=1$ is the extra condition. [O] residual: ‘the free RP seam condenses the order- $ \mu_4 $ Lagrangian glue’ = the sheet/QGEO selection, now with holomorphy $\Leftrightarrow \det 1$ a theorem [E][O]
v236_brieskorn_capstone.py	The $(2, 3, 5)$ Brieskorn singularity $x^2+y^3+z^5$ is the <i>one</i> generator of the discrete skeleton — the capstone of the icosahedral round (v55/v219/v223/v232/v233); its exponents <i>are</i> the atoms $(\mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$. [E] Milnor number $\mu=(2-1)(3-1)(5-1)=1\cdot 2\cdot 4=8=\text{rank } E_8$ = the 8 in $c_3=1/(8\pi)$ (a fifth independent origin). [E] the Milnor monodromy eigenvalues are the eight primitive 30th roots $e^{2\pi i m/30}$, m the E_8 exponents (charpoly Φ_{30} , $\deg \varphi(30)=8=\mu$) — so it <i>is</i> the order-30 E_8 Coxeter element (v55). [E] both clocks: the monodromy group $\langle h \rangle = \mathbb{Z}/30 = \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/5$ carries the μ_3 triality (CP, v233) as h^{10} ; the Galois group $(\mathbb{Z}/30)^\times = \mathbb{Z}/2 \times \mathbb{Z}/4$ carries the μ_4 clock ($\times 7$, v223). [E] Milnor lattice = E_8 , link = Poincaré sphere. [C] one singularity generates atoms + 8 + E_8 + order-30 + both clocks + seam; a unification, not a closure [E][C]
v237_seam_sre_closure.py	The closing step as a <i>physical</i> condition: no topological ground-state degeneracy $\Leftrightarrow \det K=1 \Leftrightarrow$ the seam bulk is short-range-entangled (SRE/invertible) — sharpening GATE.HOLO.02’s residual from definitional to falsifiable. [E] the K -matrix ground-state degeneracy on a genus- g surface is $ \det K ^g$ (carrier $\rightarrow 16$, $D_8 \rightarrow 4$, $E_8 \rightarrow 1$ on the torus), so ‘no degeneracy on any closed surface’ $\Leftrightarrow \det K=1$. [E] among rank-8 even positive-definite K ($c=8=g_{\text{car}}+N_{\text{fam}}$) only E_8 has $\det 1$ — the unique nontrivial bosonic SRE 2+1d phase (Kitaev E_8), edge holomorphic; D_8 /carrier are LRE. [E] a unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$, the torus does), so the residual is ‘the seam unique-vacuum extends to the torus’ = SRE. [E] NEG: an LRE bulk (D_8) has 4-fold torus degeneracy + non-holomorphic boundary. [O] sharpened residual: ‘the free RP gapped seam is short-range-entangled (no topological order)’ = $\det K=1$ = the Kitaev E_8 phase — a yes/no statement, sharper than v235/QGEO.SYM.01; not a closure [E][O]
v238_modular_lindblad_dynamics.py	DYN.SEMIGROUP.01 — the static $\rightarrow$ dynamic flip: the seam’s modular flow (reversible) and its recovery channel as a Markov/Lindblad generator (irreversible), the dissipative gap = the OS mass gap. Companion to v196/v198 (modular) and v221/v64 (recovery), which state the structure <i>statically</i> ; this reads the same objects <i>dynamically</i> . [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) and the μ_4 clock is conserved $\Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ (the v198 state-invariance), so the static commutator is the conservation law of modular time. [E] $L = \log T$ is a doubly-stochastic generator ($e^L = T$, $e^{L\tau} \rightarrow$ the democratic projector = the arrow of time, cf. v64). [E] keystone: the dissipative gap = $-\log((2/3)^6) = 6\log(3/2)$ = the OS mass gap Δ (v64) — the static recovery bound and the dynamical mass gap are one number, one exponentiated. [E] GKSL split (imaginary + $\text{real} \leq 0$ spectrum). [O] geometric modular flow on the full seam algebra (Connes–Rovelli thermal time = QGEO.SYM.01, v198) — a target, not a closure.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v239_kms_thermal_time.py	DYN.KMS.01 — KMS / thermal time: the modular flow of v238 makes the seam state a KMS (thermal) state at $\beta = 1$, and the geometric normalisation fixes the seam temperature via $2\pi = 1/(4c_3)$. [E] KMS at $\beta = 1$: $\omega(A\sigma_i(B)) = \omega(BA)$ ($\omega(X) = \text{Tr}(\rho X)$, $\rho = e^{-\Lambda_\Sigma}/Z$), so modular time (v238) is a thermal time (Connes–Rovelli); NEG $\beta \neq 1$ fails (Tomita–Takesaki pins $\beta = 1$). [E] the seam unit $\beta_{\text{phys}} = 2\pi = 1/(4c_3)$ (v58) makes the KMS temperature the horizon temperature $T_H = c_3/M$ (v57/v208). [E] one object with v238. [O] the geometric-boost identification = QGEO.SYM.01 — a target, not a closure.
v240_gns_os_reconstruction.py	DYN.GNS.01 — GNS / OS reconstruction (finite toy): the Hilbert space and a POSITIVE Hamiltonian are OUTPUTS of the seam data (M, ω) , not inputs. [E] GNS: $\langle A, B \rangle = \text{Tr}(\rho A^\dagger B)$ Hermitian positive-definite ($\rho > 0$), $\dim n^2$, identity cyclic+separating. [E] modular flow $\sigma_t(x) = \rho^{it} x \rho^{-it}$ is a state-preserving *-automorphism, generator $i[\log \rho, \cdot] = \Lambda_\Sigma$ (v238). [E] OS Hamiltonian $H_{\text{OS}} = -\log T$ (the v221 transfer) is POSITIVE — eigenvalues $\{0, 6\log(3/2), 6\log 3\} \geq 0$, gap = $6\log(3/2)$ = the v64 mass gap. [E] $H_{\text{OS}} = -L$: minus the v238 dissipative generator (recovery semigroup = Euclidean time). [O] the geometric/operator-level step = QGEO.SYM.01/BW + the standard OS theorem under RP/gap — a target, not a closure.
v241_dhr_sectors.py	DHR.SECTORS.01 — particles = DHR superselection sectors: the carrier discriminant/anyon content and the v235/v237 condensation tower as an abelian MTC. Carrier form (Lean-verified, FORM.GLUE.01): on $\mathbb{Z}_4 \times \mathbb{Z}_4$, $q(x, y) = (5x^2 + 3y^2)/8 \bmod 1$, $\theta(a) = e^{2\pi i q(a)}$. [E] 16 sectors = $\mathbb{Z}_4 \times \mathbb{Z}_4$ ($= \det K_{\text{carrier}} = \dim S^+$), fusion = abelian group law. [E] condensable $\Leftrightarrow \theta = 1 \Leftrightarrow \text{isotropic} \Leftrightarrow 5x^2 + 3y^2 \equiv 0 \bmod 8$ — the six elements $\{(0, 0), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3)\}$. [E] tower = v235/v237: Lagrangian $\langle (1, 1) \rangle$ (order 4, self-dual) $\rightarrow 1$ sector = $(E_8)_1$; order-2 $\langle (2, 2) \rangle \rightarrow 4 = D_8$; carrier $16 \rightarrow D_8 \rightarrow E_8$. [E] NEG: a non-isotropic anyon $(1, 0)$ is confined. [C] sector $\leftrightarrow$ SM-multiplet map is the interpretation; the MTC is exact.
v242_spin_statistics_modular.py	DHR.MODULAR.01 — spin-statistics and the modular (S, T) data of the carrier MTC: statistics + the fusion ring recovered from modular data + the anyon central charge $c = 8$. Same MTC as v241 ($\mathbb{Z}_4 \times \mathbb{Z}_4$, $q = (5x^2 + 3y^2)/8$, $\theta = e^{2\pi i q}$). [E] spin-statistics: $T = \text{diag}(\theta_a) \rightarrow 6$ bosons ($\theta = 1$), 2 fermions ($\theta = -1$) = $\{(0, 2), (2, 0)\}$, 8 anyons. [E] modular $S_{ab} = (1/\sqrt{16})e^{-2\pi i B(a,b)}$ unitary, symmetric, $S^2 = C$ (charge conjugation), $S^4 = I$. [E] Verlinde $N_{ab}^c = \sum_x S_{ax} S_{bx} S_{cx}^* / S_{0x} = \delta(c, a+b)$ — fusion recovered from modular data. [E] Gauss–Milgram $(1/\sqrt{16}) \sum \theta_a = e^{2\pi i c/8}$ with $c = 8 = g_{\text{car}} + N_{\text{fam}}$ = the $(E_8)_1$ central charge (v156/v175). [C] the boson/fermion split is the kinematic skeleton; the sector $\leftrightarrow$ multiplet map is the interpretation.
v243_haag_ruelle_braiding.py	DHR.SMATRIX.01 — scattering: the Haag–Ruelle skeleton + the braiding S-matrix. The seam net has a mass gap (v64/v238) so Haag–Ruelle constructs asymptotic states; for the abelian anyon MTC (v241/v242) the 2-particle S-matrix is the monodromy $M(a, b) = \theta(a+b)/(\theta(a)\theta(b)) = e^{2\pi i B(a,b)}$. [E] mass gap $\Delta = 6\log(3/2) > 0 \rightarrow$ HR in/out states. [E] 2-particle S = monodromy: $ M = 1$, diagonal in conserved labels $\rightarrow$ no particle production, factorised (integrable) scattering. [E] Yang–Baxter: M is a bicharacter $\rightarrow$ multi-particle factorisation. [E] crossing+symmetry: $M(a, b) = M(b, a)$, $M(a, -b) = M(a, b)^*$. [E] holomorphic point: on the condensed $(E_8)_1$ (single sector) $M = 1$, trivial braiding ($\det 1/\text{SRE}$). [O] the 4d S-matrix / cross-sections are the open holographic uplift (Phase 3).

continued on next page

Script	What it machine-checks (<i>continued</i>)
v244_spectral_action_contract.py	CONTRACT.QFT4D.01 — the 4d Lagrangian via Connes' spectral action: the honest contract for the Phase-3 gap (2d seam net $\rightarrow$ 4d physics). A single Dirac operator whose $\text{Tr } f(D/\Lambda)$ yields gravity AND matter. [E] carrier spinor = one generation: $SO(10)$ half-spinor = $\dim S^+ = 2^{g_{\text{car}}-1} = 16$ ($15 + \nu_R$), $N_{\text{fam}} = 3$. [E] SM gauge group inside the carrier: $SU(3) \times SU(2) \times U(1)$ ($\dim 12$) $\subset SU(5)$ (24) $\subset SO(10) = D_5$ (45); colour $SU(3) \subset A_3 = SU(4)$ (Pati-Salam); $\text{rank}(D_5) + \text{rank}(A_3) = 8 = \text{rank } E_8$. [E] one Dirac operator $\rightarrow$ gravity ($a_2 \sim R$, $a_4 \sim R^2$, $\sqrt{36}$) + matter ($a_4 \sim F^2$ Yang-Mills + $D\phi ^2$ Higgs, Gilkey). [X] hooks: $g_3 = g_2 = \sqrt{5/3} g_1$ + a Higgs-quartic relation. [O] CONTRACT.QFT4D.01: obligations (seam $\rightarrow$ Dirac D; finite geometry = carrier; spectral action reproduces SM couplings) $\rightarrow$ the 4d SM+gravity Lagrangian is an OUTPUT — the Phase-3 target, not closed.
v245_spectral_matter_content.py	QFT4D.MATTER.01 — the computable core of CONTRACT.QFT4D.01 (v244): the spectral-action / NCG matter content, discharging the rep-theory + tree-relation core to [E]. The carrier half-spinor is the $SO(10)$ 16. [E] 16 = one generation: $SO(10)$ 16 = $SU(5)(\mathbf{10} + \mathbf{5} + \mathbf{1})$; SM dims $6 + 3 + 1 + 2 + 3 + 1 = 16 = \dim S^+$, $N_{\text{fam}} = 3$ (15 Weyl fermions + ν_R). [E] electric charges $Q = T_3 + Y$ are exactly the SM values — hypercharges forced by the embedding. [E] anomaly-free: $\sum Y = 0$, $\sum Y^3 = 0$, $[SU(2)]^2 U(1) = 0$, $[SU(3)]^2 U(1) = 0$ (exact). [E] unification: $\sin^2 \theta_W = (\sum T_3^2) / (\sum Q^2) = 3/8$, i.e. $g_3 = g_2 = \sqrt{5/3} g_1$. [E] Higgs = the $(1, 2)_{1/2}$ inner fluctuation (one doublet). [O] residual: the seam $\rightarrow$ Dirac realisation + measured-coupling reproduction stay the open contract.
v246_unification_data_crosscheck.py	QFT4D.RGTEST.01 — the honest data cross-check for the spectral-action prediction (v244/v245): does $g_1 = g_2 = g_3$, $\sin^2 \theta_W = 3/8$ at Λ survive the <i>measured</i> SM couplings? Not in the plain SM. Inputs (PDG, GUT-normalised, M_Z): $\alpha_i^{-1} = (59.01, 29.59, 8.47)$, SM 1-loop $b = (41/10, -19/6, -7)$ (v159). [E] the prediction: $\alpha_1 = \alpha_2 = \alpha_3$, $\sin^2 \theta_W = 3/8$ at Λ. [X] tension: pairwise crossings spread ~ 4 decades ($\sim 10^{13} - 10^{17}$ GeV), residual spread at 2×10^{16} $\Delta \alpha^{-1} \sim 8.8$ — $g_1 = g_2 = g_3$ not realised by the plain SM. [X] $\sin^2 \theta_W$ 3/8 vs measured 0.23122. [X] 2-loop also misses (min spread ~ 3.2 at 1.7×10^{14} GeV). [O] no rescue: SM gauge content + no new states (the E_8 spine is arithmetic, not a threshold tower) $\Rightarrow$ no admissible thresholds; closing it needs new matter (vs 'no new state') or a non-GUT/boundary reading — tension like NCG-SM, likely falsified unless extended. A kill-test, [X]/[O], never [E]-as-success.
v247_e8_branching_no126.py	PS.E8BRANCH.01 — the E_8 audit hull forbids the 126 and supplies exactly $\{1, 10, 16, 45\}$ for $SO(10)$. Carrier $D_5 \oplus A_3 = SO(10) \times SU(4) \subset E_8$ (via $SO(16)$); 248 = $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4}) = 248$. [E] ranks $5 + 3 = 8$; [E] dims = 248 ($SO(16)$: 120 adj + 128 spinor); [E] $SO(10)$ content $\{1, 10, 16, 45\}$, the 126 ABSENT $\Rightarrow 126_H$ algebraically forbidden, $10 + 16 + 45$ E_8-supplied not fitted; [E] caveat only one 45/one 15 $\rightarrow$ proton-marginal; [O] the field-content principle + NCG complement (v250).
v248_ps_algebra.py	PS.ALGEBRA.01 — the Pati-Salam NCG algebra $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ and its exact SM matching. [E] unimodular group $SU(2)_L \times SU(2)_R \times SU(4)_c$ (rank 5, $\dim 21$); [E] 16 = $(4, 2, 1) + (\bar{4}, 1, 2) = 8 + 8$; [E] $Y = T_{3R} + (B - L)/2$ gives the exact SM hypercharges and $Q = T_{3L} + Y$ the exact charges; [E] matching $\alpha_1^{-1} = \frac{3}{5} \alpha_{2R}^{-1} + \frac{2}{5} \alpha_4^{-1}$, $\sin^2 \theta_W = 3/8$. [O] A_F uniquely from seam+sheet+μ_4; [O] caveat $SU(4)_c \subset SO(10) = D_5$ is not the carrier $A_3 = SU(4)$ (family).
v249_ps_unification.py	PS.RGTEST.01 — carrier-native Pati-Salam/$SO(10)$ two-step unification kill-test + negative controls (suite check of experiments/gauge-unification). [E] matching + a valid 1-loop unification solution; [X] NEG SM-only fails (spread ~ 8.8 @ 2×10^{16}); [C] $M_{PS}/M_s = 1.0 - 1.7$ (1-loop; 2-loop $\sim 1.0 - 1.2$) — gauge unification meets the gravitational scalaron scale; [X] NEG 126 breaks the match ($b_{2R} > 0$; also E_8-forbidden v247); [X] NEG proton decay selects content (minimal $M_{GUT} \sim 2 \times 10^{15}$ excluded; $+(15, 1, 1)$ safer). [O] the gauging fork, exact κ, the one-45 marginality.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v250_dirac_triple_contract.py	CONTRACT.QFT4D.DIRAC.01 — the spectral-triple contract that would make ‘carrier gauged’ a theorem. Target (A, H, D, J, γ) almost-commutative, $A_F = \text{PS}$ (v248), $H_F = 3 \times 16$, D_F from $(R, L, \varphi_0, \text{Majorana})$. [E] $H_F = 3 \times 16 = 48$; [E] the Yukawas are already fixed (v18) so the open test is NCG consistency; [O] the 7 NCG axioms (first-order condition = the historic $SO(10)$ obstruction); [O] $\sigma = \text{scalaron hypothesis}$ (rescues first-order + ν_R Majorana; $M_{PS} = \kappa M_s$) — caveat $\sigma \leftrightarrow \nu_R \nu_R$ not automatic; [O] inner fluctuations $\rightarrow 10+16+45$ not 126. Reduces 4d QFT to one object + finite NCG checks.
v251_first_order_sigma.py	PS.DIRAC.01 — the first-order condition is violated <i>exactly</i> by the Majorana/ σ term (Chamseddine–Connes–van Suijlekom ‘beyond first order’ $\rightarrow$ Pati–Salam + σ), exhibited on a faithful lepton-block model; advances v250 from contract to mechanism. Honest correction: σ does NOT ‘rescue’ first order — one DROPS it and the inner fluctuations generate $\text{PS} + \sigma$. [E] order-zero; [E] real-even ($J^2=+1$, $\gamma^2=1$, $D\gamma=-\gamma D$, KO-6 signs); [E] first order HOLDS for the Yukawa block; [E] first order VIOLATED by $M \neq 0$, supported exactly on the $\nu_R \nu_R^c$ blocks (σ the unique violator). [C] $\sigma = \text{scalaron}$ (v249). [O] the full 96-dim SM/PS triple + spectral action (v250).
v252_full_finite_triple.py	PS.DIRAC.02 — the full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ built explicitly with the NCG axioms verified; advances v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. $H_F = \text{particle} \oplus \text{antiparticle}$ doubling of three $SO(10)$ 16 -plets (dim 96). [E] dim 96; D_F Hermitian+odd; reality $J^2=+1$, $JD=DJ$; grading $\gamma^2=1$, $[\gamma, a]=0$, $J\gamma=-\gamma J$; KO-dim 6; order-zero; first order HOLDS for the SM $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ (incl. Majorana) and VIOLATED for Pati–Salam exactly by the Majorana on $\{\nu_R, e_R\}$ (CCvS beyond-first-order $\rightarrow \text{PS} + \sigma$); spectrum = fermion masses + seesaw $m_{\text{light}} = m_D^2/M_R$. [C] Poincaré duality (intersection form non-degenerate). [O] orientability + no-junk + spectral-action expansion.
v253_kappa_scalaron_ps.py	PS.KAPPA.01 — pinning κ in $M_{PS} = \kappa M_s$, discharging residual 6(b) of v249. $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13} \text{ GeV}$ (exact). [E] scalaron scale exact; [E] κ pinned by RG (scan PDG errors $\times$ the two E_8 -allowed contents) to a tight $O(1)$ interval ~ 1.3 (1-loop $[1.0, 1.7]$, 2-loop $[1.0, 1.2]$) — pinned, not free. [C] heat-kernel origin $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$, a ratio of heat-kernel traces over the 96-dim H_F (v252) — Λ cancels, κ scale-free and $O(1)$. [X] the 126 -content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs the cutoff moments f_2/f_0 fixed.
v254_inner_fluctuations.py	PS.NCG.FLUCT.01 — the inner fluctuations $\Omega_D^1(A_F)$ of the finite Dirac operator computed explicitly (reuses the v252 triple); closes v250 #5 (Higgs content, not 126) + the no-junk obligation + the σ quantum numbers. [E] $\Omega_{D_F}^1$ purely chirality-odd $\rightarrow$ finite fluctuations are scalars; [E] no junk / no 126 : SM-algebra 1-forms sit exactly in the Higgs-doublet pattern (the single $(1, 2)_{1/2}$); [E] the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam — σ is the CCvS beyond-first-order scalar; [E] σ is an SM singlet $(1, 1, 0)$ with $B-L=2$; [C] $\sigma = \text{scalaron consistency}$.
v255_spectral_action_expansion.py	PS.SPECACT.01 — the spectral-action heat-kernel (Seeley–DeWitt) expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ over the finite triple; closes v252 #11 (spectral-action expansion) and makes $\kappa(f_2/f_0)$ explicit. [E] expansion structure (a_0 cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl ² + R^2 +Higgs-quartic, $N=96$); [E] trace inputs $b/a^2 = 1/3$ (top-dominated, Chamseddine–Connes); [E] $\sin^2 \theta_W = 3/8$ (cross-check v245); [C] Higgs quartic $\lambda(\Lambda) \sim g^2 b/a^2 \rightarrow \sim 125 \text{ GeV}$ with σ ; [E] scalaron = R^2 spin-0 mode (gravitational $a_4 \sim f_0 N$); [C] $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$ explicit; [O] exact decimal needs the cutoff moments fixed.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v256_orientability_poincare.py	PS.NCG.ORIENT.01 — orientability and Poincaré duality treated honestly (no faked proof). KO-dim 6 forces the intersection form $\text{cap}(p, q) = \text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$ to be antisymmetric. [E] skew form forced by $J\gamma = -\gamma J + \text{order-zero} + \text{reality}$ (verified); [E] on the SM K-theory $\mathbb{Z}^3$ the skew form has rank 2, corank 1, null $\sim (1, 1, -1)$ — the structural obstruction; [E] orientability local condition $[\gamma_F, \pi(a)] = 0$, full orientability is a degree-4 cycle on $M \times F$; [C] the canonical hypercharge-faithful SM/PS triple satisfies both (van Suijlekom; CCM) — literature; [O] a self-contained machine proof needs the complete canonical bimodule.
v257_quasiorient_modpd.py	PS.NCG.ORIENT.02 — the honest, literature-confirmed resolution of orientability + Poincaré duality, correcting the too-generous v256 typing. Published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192): the CCM SM finite triple (KO-6, $3\nu_R$) genuinely FAILS strict orientability and strict Poincaré duality, satisfying the weakened versions. [E] strict orientability fails ($\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$); [E] quasi-orientability holds (left/right A -irrep boxes disjoint by chirality: $L = H$ doublet, $R = \mathbb{C}$ singlet $\Rightarrow L^{\text{LR}}(\text{even, odd}) = \{0\}$); [E] strict Poincaré fails (KO-6 skew form, corank 1, the equal- L/R -neutrino case); [C] modified Poincaré duality (CCM) / Stephan’s PD-restoring triple. A known model-class feature, not a TFPT defect — no faked closure.
v258_dirac_covariance_induction.py	PS.DIRAC.03 — D_F is the modular/covariance <i>induction</i> of the boundary seam state, not an independent posit; closes the ‘posited Yukawa vs derived operator’ gap of v250/v252 (Q2: Dirac as Kasparov product + covariance formula). Quasi-free/modular dictionary: a Gaussian state is fixed by its covariance C (contraction, $\text{spec} \subset (0, 1)$) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$ ($\beta=1$, v239). [E] inversion identity $\log((1-C)C^{-1}) = H$ exactly (random Hermitian + a chirality-graded Yukawa block); [E] $C_F = (1+e^{D/\mu})^{-1}$ a positive contraction; [E] chirality-odd ($D_F^{\text{ind}}\gamma = -\gamma D_F^{\text{ind}}$); [E] unitarily equivalent to v252 (inducing C_F from the 9 Yukawas returns the same spectrum); [E] Majorana = off-diagonal covariance (seesaw $m_{\text{light}} = m_D^2/M_R$); [C] $C_F = P_F C_\Sigma P_F$, so D_F is the KK shadow $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$ and the v18 Yukawas are the readout of C_Σ .
v259_modular_cutoff_kappa.py	PS.SPECACT.02 — the spectral-action cutoff <i>is</i> the seam KMS weight, fixing $f_2/f_0=1$ exactly and reducing the last open scheme freedom (v253 #5, v255 #7) from a free function to a finite-triple trace ratio (Q4: the cutoff is modular, not chosen). By Tomita–Takesaki/BW the seam state is KMS at $\beta=1$ (v239) and $2\pi=1/(4c_3)$ (v58) fixes β , so the heat-kernel cutoff $f(u)=e^{-u}$ is <i>derived</i> . [E] KMS cutoff moments $f_0=f(0)=1$, $f_2=\int f=1$, $f_4=\int u f=1$, so $f_2/f_0=f_4/f_2=1$; [E] seam-fixed $\beta=1$ (no new parameter beyond c_3); [E] non-vacuous neg control: a Gaussian cutoff e^{-u^2} gives $f_2/f_0=\sqrt{\pi}/2 \neq 1$; [E] $\kappa=\sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}=\sqrt{c_{PS}/c_{\text{grav}}}$, no cutoff function left; [C] the RG $\kappa\sim 1.3$ pins $c_{PS}/c_{\text{grav}}\sim 1.7$ ($O(1)$); the [O] free scheme becomes a [C] finite computation over the built triple; [C] the same logic fixes $\lambda(\Lambda)$.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v260_k3_kummer_unification.py	ARCH.K3.01 — one Kummer/K3 surface carries the <i>seam</i>, the <i>carrier-16</i> and the E_8 lattice as facets of one smooth object, the geometric unification of the v1 ‘glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$’ step (the K3/Kummer proposal). The seam square torus $\tau=i$ ($j=1728$, v214) under the elliptic involution $z \mapsto -z$ squares to the Kummer K3. [E] the E_8 Cartan is even, $\det=1$, positive-definite (the unique even unimodular pos-def rank-8 lattice); [E] the K3 lattice $L_{K3}=U^3 \oplus E_8(-1)^2$ has rank $22=b_2$, $\det=-1$ (unimodular), is even, signature $(3,19)$, with $E_8(-1)^2$ an orthogonal summand; [E] seam = $T^2/(z \mapsto -z)$: $4= (\mathbb{Z}/2)^2$ fixed points, pillowcase $S^2(2,2,2,2)$, orbifold Euler char 0 (the v214/v216 four marks); [E] carrier-16 = Kummer nodes $A[2] =2^4=16 \rightarrow 16$ exceptional $\mathbb{P}^1 = \dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16$ (v197), $16=4 \times 4$; [E] honesty: the 16 node classes span the rank-16 Kummer lattice, <i>not</i> literally $E_8(-1)^2$ (distinct sublattices of $H^2(K3)$); [C] two faces of E_8 in K3 (local du Val $\mathbb{C}^2/2I$, v232/v236; global H^2 summand); [C] unification (lattice [E], carrier$\leftrightarrow$node identification [C]).
v261_modular_spectral_closure.py	QFT.MSC.01 — the <i>Modular Spectral Closure</i>: the assembly/reduction certificate turning the whole QFT layer (v238–v260) into <i>one</i> relative object TFPT_{QFT} = $(\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ reduced to <i>one</i> open premise (like v176/v234, an assembly certificate, not new physics). [E] one index = marks = fixed points = glue order: Jones $[B:A]=4$ (v89) = $\mu_4 =4$ (v92) = seam marks $2\chi=4$ (v216) = pillowcase/Kummer fixed points $(\mathbb{Z}/2)^2 =4$ (v260); [E] one carrier-16: $\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = \text{Kummer nodes } 2^4=16$, $H_F=2N_{\text{fam}} \cdot 16=96$; [E] one gap $\Delta = -\log((2/3)^6) = 6\log(3/2) > 0$ = the OS mass gap (v240) = the recovery rate (v221) = the Lindblad gap (v238); [E] the three closures re-derived (Kasparov $\log((1-C)C^{-1})=H$, v258; KMS cutoff $f_2/f_0=1$, v259; K3 lattice signature $(3,19)/\det -1/\text{even}$, v260); [E] relative-object manifest (10/10 component scripts present); [O] the single residual: the whole QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate; [C] the master statement (Modular Spectral Universality) holds modulo QGEO.SYM.01 — the honest sense in which the QFT layer is complete.
v262_fqcd_mp_me.py	FR.MPME.02 — the F_{QCD} transfer pipeline for m_p/m_e, implemented as a real solver (the one F_{transfer} pipeline that was contract-only). <i>Not</i> a compiler power — a [C] standard-physics transfer reproduction. [E] $b_3 = -7 = -(11-2n_f/3)$ at $n_f=6$ (the only TFPT number in the run); [E] a two-loop $\alpha_s(M_Z)=0.1179$ run with thresholds gives $\alpha_s(2\text{ GeV}) \sim 0.30$ (real ODE solve); [C] $\Lambda_{\text{MS}}^{(3)} \sim 0.33\text{ GeV} \times$ the lattice bridge $C_p = m_p/\Lambda$ (external $O(1)$) over the closed m_e gives $m_p/m_e \sim 1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget; [X] firewall: m_p/m_e is not an admissible compiler integer ($SU(9)$ absent from $D_5 \oplus A_3$). Advances FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
v263_mnu_from_df.py	FLAV.PMNS.02 — the light neutrino matrix M_ν as the type-I seesaw of D_F, advancing the solar-angle texture (v9) from a posited matrix to a seesaw <i>object</i> built from D_F’s blocks. [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ symmetric + seesaw-suppressed (the light masses an output of D_F); [E] a $\mu\tau$-symmetric (m_D, M_R) gives a $\mu\tau$-symmetric M_ν (the v9 texture is in the seesaw image); [C] diagonalising gives $\theta_{23}=45^\circ$ and $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.30675$ under the seam misalignment; [O] $\theta_{13}=0$ in the leading texture (physical $\sin^2 \theta_{13} \sim \varphi_0^{\text{ret}} e^{-5/6}$ a separate readout), the CP magnitude $\delta_{\text{PMNS}}=240^\circ$ is the $\mu_6/\text{triality}$ phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v264_raw_seam_ marklocal.py	QGEO.ENERGY.03 — the raw-seam $\Rightarrow$ mark-locality <i>reduction</i> (not a closure of QGEO.SYM.01). Composing v216+v201+v198 on the canonical pillowcase: [E] Gauss–Bonnet 4 cone points (deficit π each, sum $4\pi=2\pi\chi(S^2)$), curvature <i>only</i> at the 4 marks; [E] the curvature sourced at the μ_4 orbit $\{0, \pi/2, \pi, 3\pi/2\}$ has Fourier support <i>only</i> on $4\mathbb{Z}$ (FFT), so mark-locality is automatic for the pillowcase; [E] M_f then commutes with the clock $\rho=\text{diag}(i^n)$; [E] with $ k $ already commuting (v198), $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ on this metric; [O] the residual <i>is</i> the metric — the chain assumes the raw seam carries the uniformising flat pillowcase metric (=QGEO.SYM.01), the negative control (curvature off the μ_4 orbit) breaks it. A reduction/sharpening, not a closure.
v265_qft4d_fork_ freeze.py	QFT4D.FORK.01 — freeze the 4D-QFT fork as a <i>branch policy</i> (not an open ambiguity); writes <code>qft4d_fork_freeze.json</code> + a prose guard, no new physics. [C] A: boundary-only is the canonical 4D reading (no 4D-GUT claim by default); [X] SM-only 4D-GUT unification is killed (1-loop spread ~ 8.8 at 2×10^{16} , v246); [C]/[O] B: carrier-native Pati–Salam is a falsifiable UV branch (E_8 content $\{1, 10, 16, 45\}$ no 126 v247, $\sin^2 \theta_W = 3/8$ v248, $\kappa O(1)$ band v249/v253, threshold + proton kill test); [X] proton safety rule (no fake τ_p window); [E] wording guard (bare 4D over-claims absent, scoped policy phrases present). A frozen decision tree, not an open ambiguity.
v266_ps_threshold_ proton.py	PS.PROTON.02 — the explicit Pati–Salam threshold model + proton-decay window that DECIDES branch B of the v265 fork; writes <code>ps_threshold_proton.json</code> . Two-step PS solve + $d=6$ $\tau_p(p \rightarrow e^+ \pi^0)$: the carrier-native PS UV branch survives proton decay <i>only</i> with the E_8 -allowed $(15, 1, 1)$ [the single 45]. [E] explicit thresholds: minimal $16H \rightarrow M_{GUT} \sim 2.4 \times 10^{15}$, $+(15, 1, 1) \rightarrow M_{GUT} \sim 5.9 \times 10^{15}$; [X] minimal KILLED ($\tau_p \sim 4.4 \times 10^{33}$ yr < Super-K 2.4×10^{34} , even high-band); [C] $+(15, 1, 1)$ survives now ($\tau_p \sim 1.6 \times 10^{35}$ yr); [X] <i>Hyper-K decisive</i> (τ_p at the $\sim 1.4 \times 10^{35}$ yr reach) — a dated kill test within the decade; [O] marginal (only one 45 in the hull, v247) + $O(3)$ hadronic uncertainty. A (boundary-only) stays canonical.
v267_qgeo_rigidity_ minimal_axiom.py	QGEO.SYM.02 — the rigidity / minimal-axiom form of the bedrock QGEO.SYM.01: the ONE bare statement (the raw RP seam admits the carrier μ_4 clock as a conformal symmetry permuting its 4 marks, $= \omega \circ \rho = \omega$) <i>forces</i> everything downstream; it does <i>not</i> close the axiom. [E] an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 (independent of a); [E] cross-ratio $2 \Leftrightarrow j=1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2=x^3-x$; [E] downstream forced: square $\Rightarrow$ the unique flat pillowcase (Trojanov, v214) $\Rightarrow$ curvature at the 4 marks $\Rightarrow \mathbb{Z}_4$ -Fourier DtN (v201/v264) $\Rightarrow [\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ (v198); [E] neg controls (generic $j \neq 1728$, only $\mathbb{Z}_2$; hexagonal $j=0$ has $\mathbb{Z}_6$, order $6 \neq 4$); [C] arithmetic rigidity ($\{1, i, -1, -i\}$ over $\mathbb{Q}$, $j=1728$ CM by $\mathbb{Z}[i]$, Belyi/dessins); [O] the bare order-4 symmetry stays the one foundational postulate (like $c=\text{const}$). Closes the reduction (axiom $\Rightarrow$ everything), not the axiom.
v268_theta13_ carrier_trace.py	FLAV.TH13.01 — the reactor angle θ_{13} <i>exponent</i> is the carrier hypercharge trace. The value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ was already checked (v16); here the exponent origin is closed exactly: [E] $\gamma = \text{tr}_E Y^2 = 3(1/3)^2 + 2(1/2)^2 = 5/6$ over the 5-slot carrier hypercharge $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ (roots of $6Y^2 - Y - 1 = 0$), complement $1/6$ the neutrino ratio; [C] value 0.02311 vs PDG ~ 0.0222 ; [E] own channel — the $\mu\tau$ -breaking $\varepsilon = 3\varphi_0^{\text{ret}}/4$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel, not the seesaw $\mu\tau$ breaking (corrects the tempting ‘ θ_{13} from M_ν ’ route); [O] CP magnitude + absolute ν scale remain open.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v269_spert_paqft_skeleton.py	QFT4D.SPERT.01 — the perturbative 4D S-matrix (S_{pert}) as an Epstein–Glaser / pAQFT skeleton: the honest middle layer of the three-layer S-matrix (S_{top} 2d DHR braiding v243 / S_{pert} 4d EG of the spectral action / S_{phys} LSZ on the OS Wightman functions v240), <i>not</i> a nonperturbative closure. [E] three distinct layers (the 2d braiding $ M =1$ is statistics, not 4d cross-sections); [E] every spectral-action a_4 operator has mass dimension ≤ 4 in 4D (power-counting renormalizable); [E] a finite counterterm basis (7 marginal +3 relevant, no infinite tower); [C] Epstein–Glaser existence — a renormalizable interaction has an order-by-order $S(g)=T \exp(i \int g L_{\text{int}})$ by causal factorisation + finite local extension (no path integral, no UV divergences); [C] adiabatic limit/gap ($\Delta=6 \log(3/2)>0$) $\rightarrow S_{\text{phys}}$ via LSZ; [O] perturbative only — the ambient measure QG.AMB.01 stays open. A contract/skeleton, explicitly perturbative.
v270_pmns_jarlskog_assembly.py	FLAV.PMNS.03 — the complete complex PMNS matrix + the leptonic Jarlskog J_{PMNS} (the CP <i>strength</i>), assembled from the four TFPT channels ($\sin^2 \theta_{12}=\frac{1}{3}-\frac{\phi_0}{2}$, $\sin^2 \theta_{23}=\frac{1}{2}$, $\sin^2 \theta_{13}=\phi_0 e^{-5/6}$, $\delta=4\pi/3$) into one unitary object. [E] the three angle channels; [E] $U = R_{23}U_{13}(\delta)R_{12}$ is exactly unitary; [C] $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ (formula and full matrix agree), $J_{\text{max}} = 0.03424$ — a <i>derived</i> CP strength, the leptonic analogue of the CKM J (v88); [C] data: J_{max} vs NuFIT 0.0332 ($\sim 3\%$), J vs best fit -0.026 , the negative sign reproduced; [O] δ is the μ_6 /triality phase (assembled input, <i>not</i> from M_ν/D_F) and the absolute ν -mass scale (Σm_ν , M_R) stays open. Assembles the PMNS matrix + CP strength; does not derive δ from D_F nor fix the scale.
v271_eg_oneloop_quartic.py	QFT4D.SPERT.02 — a concrete Epstein–Glaser order-2 (one-loop) quartic renormalization, taking the v269 S_{pert} skeleton from <i>exists</i> to an actual EG number (no path integral). [E] $T_1=:L_{\text{int}}$: needs no extension, the first extension is T_2 ; [E] the massless propagator has scaling degree $\text{sd}=2$ in $d=4$, the one-loop bubble (two propagators) has $\text{sd}=4=d$ so the UV singular order $\omega=\text{sd}-d=0$; [E] $\omega=0 \Rightarrow$ exactly one logarithmic local counterterm ($C(\omega+d,d)=1$, renormalizable, no infinite tower); [E] the loop factor $\Omega_3/(2(2\pi)^4)=1/(16\pi^2)$ exactly — the same factor that normalises the spectral-action a_4 geometric quartic; [C] the ϕ^4 one-loop $\beta_\lambda=3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the s, t, u channels), EG initial condition $\lambda_\Phi(\text{UV})=1/(16\pi^2)$; [O] order-2/one-loop only — the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Turns the skeleton into a concrete one-loop number.
v272_nu_mass_scale.py	FLAV.NUSCALE.01 — the absolute neutrino-mass scale, typed honestly (it does <i>not</i> fabricate a Σm_ν number, which would contradict the v184 firewall). [E] the absolute scale reduces to <i>one</i> seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ — one irreducible UV input like v_{geo} (v153), while the ratios (m_2/m_3 , ordering, PMNS angles v270) are pinned; [C] the observed $m_3=0.050$ eV with the TFPT PS $\rightarrow$ GUT window $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ GeV (v249) needs a third-generation Dirac Yukawa $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; [O] $M_R(y=1)=6.1 \times 10^{14}$ has $\log_{c_3}(M_R/M_{\text{Pl}})=2.57$ (non-integer), not a TFPT power, so the absolute value stays open; [C] the normal-ordering floor $\Sigma m_\nu=0.0586$ eV is consistent with cosmology ($< 0.072\text{--}0.12$ eV); [X] a cosmological $\Sigma < 0.0586$ eV excludes the normal-ordered $m_1 \sim 0$ spectrum. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio), not the absolute number; sharpened by v481 (carrier normalisation $y_\nu=y_t$: one-parameter, window-bracketed candidate).

continued on next page

Script	What it machine-checks (<i>continued</i>)
v273_eg_gauge_running.py	QFT4D.SPERT.03 — the gauge sector of the Epstein–Glaser one-loop (v271’s quartic mechanism applied to the gauge self-energy). [E] the one-loop gauge 2-point has scaling degree $sd=4=d$ so $\omega=0 \rightarrow$ one counterterm per coupling (same marginal mechanism + loop factor $1/(16\pi^2)$ as v271); [E] the one-loop b_i from the explicit carrier/SM content: $b_3=-7$, $b_2=-19/6$, $b_1=41/10$ (the same 41 as the carrier algebra, v159/CAR.SM.01); [C] physical RG directions $b_3<0$ (asymptotic freedom), $b_1>0$ (U(1) Landau); [O] the two-loop matrix (v159, PyR@TE) + all-order stay open. Shows the gauge running is an EG order-2 result. Exact b_i mirrored in Wolfram.
v274_scale_overdetermination.py	SCALE.OVERDET.01 — the single mass anchor is <i>over-determined</i> , and the absence of a derivable absolute scale is a <i>theorem</i> . [E] one anchor (No-Unit, v153): every dimensionful output = pure number $\times M_{P1}^d$ (ρ_Λ/M_{P1}^4 , $M_{scal}/M_{P1}=c_3^{7/2}$, the spectrum); [C] two independent routes to M_{P1} agree: gravity $(8\pi G)^{-1/2}=2.4353\times 10^{18}$ GeV and cosmology (invert $\rho_\Lambda/M_{P1}^4=(3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4}=2.24$ meV) $=2.438\times 10^{18}$ GeV, to 0.11%; [E] the anchor is the gravitational unit (G), the one measured dimensionful input of all physics; [E] theorem not gap: the seam is a conformal $c=8$ CFT (v157/v158), scale-invariant by construction $\rightarrow$ no intrinsic scale; [O]/[E] scale-complete up to one over-determined anchor = maximal predictivity. Sharpens v153/v78.
v275_qgamb_roadmap.py	QGAMB.ROADMAP.01 — the obligation roadmap for QG.AMB.01 (the ambient nonperturbative QG measure), consolidated into one module (the open gate had no script). [E] Tier A gap decoupling: $\Delta_{\text{eff}}=6\log(3/2)-2.248c_3^2=1.648>0$ (v36/v76), so every low-energy readout is independent of the un-built measure — not a blocker; [E] Tier B: the bulk measure reduces to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ($c=248/31=8$; v77/GATE.METRIC.03), reduced not closed (REDTEAM.A.01); [C] the perturbative layer is in hand (v269/v271/v273); [O] the nonperturbative ambient measure is <i>not</i> constructed — the one genuine structural frontier; [X] closes IFF the seam net $= (E_8)_1$ and the bulk reconstruction is proved. A roadmap, not a closure.
v276_qgeo_flat_closes_commutator.py	QGEO.SYM.03 — the flat $\tau=i$ pillowcase premise <i>closes</i> the modular-commutator chain to all orders, collapsing the bedrock QGEO.SYM.01 to <i>one</i> geometric postulate (it does <i>not</i> prove the premise). For a quasi-free seam state $QGEO.SYM.01 = (\omega \circ \rho = \omega) \Leftrightarrow [\rho, C]=0 \Leftrightarrow [\rho, H]=0$. [E] isometry $\Rightarrow$ commutator: if $H=f(\Delta_{\text{flat}})$ and the order-4 deck $\rho(z \rightarrow iz)$ is a flat- $\tau=i$ isometry then $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$ <i>exactly</i> (all orders), verified on the discretised square torus ($\rho^4=I$, $[\rho, \Delta]=0$, $[\rho, \sqrt{-\Delta}]=0$; generic non-isometry fails); [E] this upgrades v198 (principal symbol) + v201 (subprincipal) to the full H (flatness removes the curvature terms); [C] $\tau=i$ is forced by order-4 ($j=1728$; hexagonal $j=0$ is order-6, excluded; v214/v267); [O] QGEO.SYM.01 collapses to the single statement ‘the raw seam is the flat $\tau=i$ pillowcase quasi-free state’ — needs a human constructive-QFT proof or stays an honest axiom. Sharpest non-circular form.
v277_seam_calderon_e8_match.py	QGAMB.TIERB.01 — the seam-Calderón $\rightarrow (E_8)_1$ matching certificate, the concrete Tier-B step of QG.AMB.01 (v275): it computes the full $(E_8)_1$ matching target and pins the residual to <i>one</i> bit (it does <i>not</i> close it). [E] $c=8$ (Sugawara, $248/31=g_{\text{car}}+N_{\text{fam}}$), but $c=8$ alone does not pin the net; [E] the discriminator is holomorphy — $\det \text{Cartan}(E_8)=1$ (1 primary, holomorphic) vs the $c=8$ counterexample $\det \text{Cartan}(D_8=SO(16))=4$ (4 primaries, non-holomorphic), so the single discriminator is $ \det K =1$; [E] the $(E_8)_1$ character $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$ has a single primary + exactly $248=\dim E_8$ weight-1 currents (240 roots); [E] equivalence chain holomorphic $\Leftrightarrow \det \text{Cartan } 1 \Leftrightarrow$ unique even unimodular rank-8 (v83) $\Leftrightarrow$ homology-sphere link $\Leftrightarrow \det K =1$ (v235); [C] seam-side match via v276 (flat $\tau=i$) + v260 (Kummer/K3); [O] residual = the single holomorphy bit — reduced to one bit, <i>not</i> closed.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v278_lszy_bridge_unity.py	QFT4D.SPERT.04 — the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity: connects the EG perturbative S-matrix (v269/v271/v273) to the physical asymptotic S-matrix (S_{phys} = LSZ on the OS Wightman functions, v240). [E] LSZ bridge: S_{phys} = amputate+on-shell-residue of the connected time-ordered products, and the EG T-products of S_{pert} are those correlators (OS Wick-rotation, v240) $\Rightarrow S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$; [C] the gap $\Delta=6\log(3/2)>0$ gives Haag–Ruelle asymptotic states so LSZ is valid; [E] <i>one-loop unitarity</i> (optical theorem): the bubble discontinuity $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1 - 4m^2/s}$ <i>exactly</i> = the two-body phase space, so $2\text{Im } M = \sum \int d\Pi M ^2$ (cutting rule) holds $\Rightarrow$ matter+gauge unitary; [E] threshold: $\text{Im } B=0$ below $s=4m^2$, turning on at the physical two-particle threshold; [O] gravity caveat — the Stelle ghost makes the gravity perturbative S-matrix non-unitary $\rightarrow$ QG.AMB.01; [C] finite field-strength Z from the EG scheme.
v279_qgeo_obligation_lemma.py	QGE0.OBLIG.01 — the QGE0.SYM.01 bedrock written as <i>one</i> precise constructive-QFT lemma with a proof-tree completeness check (the formal obligation write-up; does <i>not</i> close it). Lemma: given the premise ‘the raw seam carries the flat $\tau=i$ pillowcase metric g_{flat} (Trojanov-unique, cone angles π), so $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$’, then $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow$ mark-locality $\rightarrow E_8$. [E] the reduction DAG from ‘free RP seam’ has 8 closed nodes and <i>exactly one</i> open leaf (‘the raw seam state is the flat state’); [E] the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders, Lean FORM.QGE0.03), so the obligation is state-identification not geometric ambiguity; [C] two proof routes (RP-state uniqueness on the flat orbifold, or Trojanov + OS); [E] the all-orders step is a Lean theorem (SeamDeckClosure.flat_all_orders_clock, standard axioms, no sorry); [O] one constructive-QFT statement closes QGE0.SYM.01.
v280_pillowcase_steklov.py	QGE0.STEKLOV.01 — a direct numerical experiment on the flat $\tau=i$ pillowcase orbifold (the seam geometry): the self-investigable test of the QGE0.SYM.01 obligation (does <i>not</i> prove the premise). [E] pillowcase = the Z_2-even sector of the flat torus, 4 cone points (2-torsion, deficit π each, sum $4\pi=2\pi \cdot 2$); [E] the order-4 deck $\rho(z \rightarrow iz)$ descends ($[\rho, Z_2]=[\rho, P_{\text{even}}]=0$) and permutes the 4 cone points as 2 fixed + 1 swapped pair; [E] the geometric chain is exact: $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$ (the v276/v279 all-orders closure on the real orbifold); [C] the Steklov DtN is positive self-adjoint, fixed by the flat metric (route-(i) evidence); [O] confirms the conclusion given flat geometry, the premise stays open.
v281_holomorphy_modular_data.py	QGAMB.MODULAR.01 — the modular-data / anyon-condensation layer of the QG.AMB.01 Tier-B holomorphy bit. [E] #anyons = $\det \text{Gram} : E_8 \rightarrow 1$ (holomorphic), $D_8=SO(16) \rightarrow 4$, carrier $D_5 \oplus A_3 \rightarrow 16$; [E] the anyon-condensation tower $D_5 \oplus A_3 (16) \rightarrow D_8 (4) \rightarrow E_8 (1)$, each step condensing a Lagrangian Z_2 boson, the Kitaev E_8 endpoint (v92/v235); [E] Gauss–Milgram recovers $c=8$; [E] E_8 is the unique holomorphic $c=8$ ($\det 1$) — $SO(16)$ has the same c but 4 anyons, so holomorphy is THE discriminator (v277); [O] the open bit: does the seam-Calderón net condense to $\det K=1$? Reduced to one bit.
v282_e8_tau_i_unification.py	QGAMB.UNIFY.01 — the E_8-at-$\tau=i$ unification: the candidate <i>simplification</i> reducing the two open obligations to <i>one</i> object (the raw seam is the $(E_8)_1$ holomorphic CFT at $\tau=i$). $\chi_{E_8}(\tau)=\Theta_{E_8}/\eta^8=j^{1/3}$. [E] $\tau=i$ is the order-4 CM point ($j=1728$, the QGE0.SYM.01 flat geometry); [E] $\chi_{E_8}=j^{1/3}$ ($\chi^3=qj$ as q-series), so $\chi_{E_8}(i)=1728^{1/3}=12$ (the $(E_8)_1$ character at the SAME $\tau=i$, the QG.AMB.01 holomorphy); [E] one object, two faces; [C] obligation count $2 \rightarrow 1$: prove ‘the raw seam is $(E_8)_1$ at $\tau=i$’ and <i>both</i> follow; [O] the unified premise stays open but is now one object.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v283_live_killtest_scorecard.py	PRED.LIVE.01 — a live kill-test scorecard of the recent round’s computable predictions vs current data, each tagged with the decisive experiment. [C] PMNS angles $\sin^2 \theta_{12}=0.30675$ ($+0.31\sigma$), $\sin^2 \theta_{13}=0.02311$ ($+1.45\sigma$), θ_{23} maximal; [C] leptonic CP $J_{\text{PMNS}}=-0.0297$, $\delta=240^\circ$ vs $\sim 232(30)$; [X] Σm_ν NO floor 0.0586 eV (DESI/CMB-S4 decisive); [X] proton decay $\tau_p=1.55\times 10^{35}$ yr for $+(15, 1, 1)$ (Hyper-K decisive); [E] closed readouts α^{-1} , Ω_b , β ; all rows consistent $< 2\sigma$, two sharp near-term kill tests. A consolidation, not new physics.
v284_route_i_rp_uniqueness.py	QGEO.ROUTEI.01 — Route (i) (RP-state uniqueness) decomposed into a 6-lemma chain with every dischargeable lemma discharged and the one irreducible lemma isolated (does <i>not</i> prove the open premise). [E] 5/6 lemmas are existing [E]/[O] results — L1 RP→DtN (v194), L2 the four marks force the pillowcase class (v195/v216/v214), L3 the flat orbifold Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$ (verified: flat-strip DtN(k)/ $k \rightarrow 1$, principal symbol $ \xi $ with no curvature term), L4 covariance $C = (1 + e^H)^{-1}$ from the DtN (v258), L5 μ_4 -invariance (v276/Lean/v280); [C] Troyanov reduction: 4 cone points of angle $\pi \Rightarrow \chi_{\text{orb}} = 0 \Rightarrow$ the unique constant-curvature representative is flat ($\tau=i$), so L_{open} reduces to ‘the raw seam is the constant-curvature geometric state’; [O] the one open lemma — the raw RP seam state is the constant-curvature/geometric state (the state-identification step).
v285_route_ii_seam_condensation.py	QGAMB.ROUTEII.01 — Route (ii) (seam-net condensation) decomposed into a 4-lemma chain, with the key result that its open lemma <i>coincides</i> with Route (i)’s (the v282 unification at the lemma level). [E] 3/4 lemmas discharged — M1 no abelian sector $\Leftrightarrow \det K =1 \Leftrightarrow$ holomorphic (v234/v235), M2 holomorphic $c=8 \Leftrightarrow (E_8)_1$ with the $SO(16)$ det-4 counterexample (v277/v281), M3 the condensation tower $16 \rightarrow 4 \rightarrow 1 =$ the unique μ_4 -Lagrangian glue (v92/v281/v235); [E] the discriminator det 1 vs det 4; [E] Routes (i) and (ii) share <i>one</i> open lemma — both say ‘the raw seam is the $(E_8)_1$ -at- $\tau=i$ object’ (v282), so proving either closes both; [O] the unified open lemma: construct a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau=i$. $M_{\text{open}} = \text{QGEO.SYM.01}$.
v286_seam_equivalence_contract.py	SEAM.EQUIV.01 — the Seam Equivalence contract: concentrates the whole open structure on <i>one</i> named arrow (does <i>not</i> prove it). Theorem: the raw RP seam state reconstructs canonically the holomorphic $(E_8)_1$ boundary net at the $\tau=i$ pillowcase. [E] four objects (RawRPSeam, FlatTauIPillowcaseDtN, HolomorphicE8Net, SREKitaevE8Phase); [E] six arrows — 2 closed (Route i v276/v280/v284, Route ii v281/v277/v235), 1 conditional (v282), 2 open premises (the same lemma, v284/v285), 1 Gral; [E] import firewall : no Route-i module imports any Route-ii module or vice versa, so the routes converge only through SEAM.EQUIV.01 (kills the ‘ E_8 into the geometry and back’ circularity); [E] exact-label taxonomy (Formal/Numerical/Conditional Exact, Open Selection); [O] the one open arrow ‘free Gaussian RP bulk $\Rightarrow$ holomorphic single-sector boundary’.
v287_free_rp_bulk_to_holomorphic_boundary.py	SEAM.EQUIV.A01 — Route A (AQFT) theorem-dependency checker for the open Gral arrow ‘free Gaussian RP bulk $\Rightarrow$ holomorphic single-sector boundary net’: types the 5-lemma chain and isolates the <i>one</i> missing standard theorem. [E] L1 RawRPSeam→OS→CAR/quasi-free boundary (v155/v160/v240/v175); [E] L2 gap+chirality $\Rightarrow$ gapped chiral bulk (v64/v156); [O] L3 invertible/SRE Gaussian bulk $\Rightarrow$ single-sector boundary (μ -index 1) — the one missing standard theorem (the AQFT form of $\det K=1$); [E] L4 $\mu=1+c=8 \Rightarrow$ holomorphic (v234); [E] L5 holomorphic $c=8 \Rightarrow (E_8)_1$ (v83/v277/v281). Verdict: 4/5 discharged, Route A reduces SEAM.EQUIV.01 to one cited theorem.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v288_full_l2_subprincipal_z4.py	SEAM.EQUIV.B01 — Route B (DtN): the full- L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality, proving (numerically, on the full Toeplitz operator) the ‘probably provable’ step and lifting v201/v284 from the finite block to L^2 . [E] character orthogonality: a mark-sourced curvature $f = \sum_{j<4} g(\theta - j\pi/2)$ has Fourier support only on $m \equiv 0 \pmod{4}$; [E] full- L^2 block-diagonality: $\Lambda = n + M_f$ commutes with $\rho = \text{diag}(i^n)$, $\ [\rho, \Lambda]\ \sim 2 \times 10^{-15}$ on $-50..50$; [E] negative control: a generic curvature gives $\ [\rho, \Lambda]\ = 4.44 > 0$; [E] Lean consistency (geom_sum_fourth_root + markLocal_blockDiagonal); [O] the sharper residual — why is the raw seam sub-principal term mark-local?
v289_marklocal_raw.py	MARKLOCAL.RAW.01 — decompose the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolate the single open analytic lemma. [C] L1 conic parametrix: conic Steklov DtN $= D_\theta + M_f + \text{cone sources}$ (b-calculus/Melrose, imported); [O] L2 Flat-Away — the one open analytic lemma: $\text{RP} + \text{gap} + 4 \text{ marks} \Rightarrow \text{curvature vanishes away from the marks}$ (the flat-pillowcase realisation); [E] L3 orbit-source: the 4 marks form one μ_4 orbit (v195/v216); [E] L4 Fourier-support: μ_4 -orbit curvature $\Rightarrow$ support only on $4\mathbb{Z}$ (v264/v288, with a non-orbit negative control); [E] L5 full-operator: $\Rightarrow [\rho, \Lambda] = 0$ on full L^2 (v288). 4/5 discharged; Route B reduces to exactly L2.
v290_z4_smooth_curvature_adversary.py	SEAM.ADVERSARY.01 — the Route-B red-team that $\mathbb{Z}_4$ block-diagonality is <i>not</i> mark-locality. The strongest adversary, a smooth $\mathbb{Z}_4$ -symmetric off-mark curvature $f = \varepsilon \cos(4\theta)$: [E] it is $\mathbb{Z}_4$ -symmetric (Fourier support $\{\pm 4\} \subset 4\mathbb{Z}$) but <i>not</i> mark-local ($f(\pi/4) = -\varepsilon \neq 0$); [E] it <i>passes</i> the v288 commutator $\ [\rho, \Lambda]\ = 0$ (same as mark-local) — so $[\rho, \Lambda] = 0$ is necessary but <i>not</i> sufficient; [E] but it shifts the DtN spectrum off the flat ladder ($\max \Delta\lambda = 0.155$) and the heat trace ($\Delta = 0.019$), so the KMS weight and $(E_8)_1$ character shift; [O] a genuine candidate modulus the commutator misses but the spectral data excludes — Flat-Away must use spectral/energy/heat input.
v291_flataway_contract.py	FLATAWAY.RP.01 — the Flat-Away lemma as its own named mini-theorem with three proof routes opened by v290. Lemma [O]: given RawRPSeam with gap, chirality and the four μ_4 marks, the smooth curvature density in the DtN sub-principal symbol vanishes away from the branch divisor. [E] <i>spectral-rigidity</i> : scanning $f = \varepsilon \cos(4\theta)$, the spectrum deviation from the flat ladder is 0 only at $\varepsilon=0$ — flat is the unique spectral match; [E] <i>heat-kernel</i> : the heat-trace deviation is 0 only at $\varepsilon=0$; [C] <i>RP-energy</i> : $\text{RP} + \text{gap}$ is a Dirichlet-energy minimisation in the Troyanov conformal class $\Rightarrow$ constant-curvature minimiser (Troyanov; cf. v284). [O] three independent routes to one lemma; closing any one closes Route B.
v292_flataway_heat_reduction.py	FLATAWAY.HEAT.01 — the Heat-Kernel route made precise: reduces Flat-Away to <i>one</i> named spectral invariant. [E] the heat-trace deviation $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode $4k$; positive on all combinations), so $\Delta \text{Tr} = 0 \Leftrightarrow f = 0$; leading invariant $\ f\ _{L^2}$ (Parseval). [O] reduction: Flat-Away $\Leftarrow$ ‘the seam fixes the a_2 heat coefficient to its flat value’ $\Rightarrow \int f^2 = 0 \Rightarrow f = 0$ off the marks — one external invariant.
v293_flataway_spectral_hessian.py	FLATAWAY.SPEC.01 — the Spectral-Rigidity route upgraded from one direction (v291) to the <i>full</i> smooth- $\mathbb{Z}_4$ space. [E] splitting mechanism: the degenerate flat Steklov level $ n $ is split by mode $4k$ at $ n =2k$ by $\pm\varepsilon/2$; [E] $S = \sum (\Delta\lambda)^2 \sim 0.50\varepsilon^2$; [E] the Hessian over modes $\{4, 8, 12\}$ has eigenvalues $\{0.497, 0.500, 0.503\} > 0$, so the flat seam is a <i>strict</i> isolated minimum over the whole $\mathbb{Z}_4$ space; [E] a non- $\mathbb{Z}_4$ mode breaks the commutator outright. [O] pinning the seam’s Steklov spectrum excludes every smooth $\mathbb{Z}_4$ deformation $\Rightarrow$ Flat-Away.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v294_flataway_rp_energy.py	FLATAWAY.ENERGY.01 — the RP-energy / Trojanov route: the flat pillowcase is the unique constant-curvature minimiser at the fixed mark divisor. [E] prescribed divisor + Gauss-Bonnet: S^2 with 4 cone angles π , $\sum(2\pi - \theta) = 4\pi = 2\pi\chi$; [C] Trojanov/Euclidean-cone uniqueness (Trojanov 1991) excludes smooth off-mark curvature; [E] strictly convex curvature energy $E[f] = \int f^2$, Hessian $2\pi I$ (PD) $\Rightarrow$ unique minimiser; [E] no zero-cost deformation. [O] premise: ‘RP+gap realises the energy-minimiser’, converging with v292/v293 on one selection principle; closing any one closes Route B.
v295_flataway_a2_exact.py	FLATAWAY.A2.01 — the <i>exact</i> analytic a_2 coefficient, upgrading the v292 positive-definiteness from numerical to a proof. [E] first variation vanishes: $\partial_\varepsilon \text{Tr } e^{-t\Lambda(\varepsilon)} _0 = -t\hat{f}(0) \sum e^{-t n } = 0$ (zero-mean off-mark curvature, Gauss-Bonnet); [E] convexity $\Rightarrow$ global minimum: $\phi(x) = e^{-tx}$ strictly convex $\Rightarrow \Lambda \mapsto \text{Tr } e^{-t\Lambda}$ convex (Klein/Peierls), so $\varepsilon=0$ is the global minimum and $\Delta \text{Tr} \geq 0$ for all ε , $= 0$ iff $f = 0$; [E] exact a_2 kernel via the divided difference of e^{-tx} (Duhamel), matched to numerics to rel.err $< 10^{-6}$; [E] degenerate-split closed form $2e^{-2kt}(\cosh(tg/2)-1) \geq 0$; [O] reduction now analytic: Flat-Away $\Leftarrow$ ‘the seam fixes a_2 ’. Lean-mirrored (FORM.FLATAWAY.01).
v296_flataway_a2_closed.py	FLATAWAY.A2.CLOSED.01 — the exact a_2 coefficient in <i>closed form</i> . [E] the operator tails telescope geometrically ($C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$), giving $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$; [E] a finite $(4k-1)$ -term middle (incl. the degenerate diagonal $\Phi(2k, 2k) = (t^2/2)e^{-2kt}$); [E] closed form $W_k(t) = \frac{t(1-e^{-4kt})}{8k(1-e^{-t})} + \frac{1}{2} \sum_{m=1}^{4k-1} \Phi(m, 4k-m)$ $(W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8)$, reproducing v295 to machine precision; [E] manifestly positive; [E] small- t : $W_k = t/2 + O(t^3)$ (mode-independent leading $1/2$, the t^2 term cancels).
v297_route_a_literature_stack.py	SEAM.EQUIV.A.LIT.01 — Route A’s ‘one standard import’ as a precise citable theorem stack. [E] LIT-A invertible/SRE bulk $\Rightarrow$ trivial bulk topological order (Kitaev 2006; Freed–Hopkins 2021); [E] LIT-B bulk MTC = Drinfeld center, trivial center $\Rightarrow$ holomorphic boundary net (Müger 2003; Kawahigashi–Longo–Müger 2001); [E] LIT-C holomorphic $c=8$ net $\Rightarrow (E_8)_1$ (Conway–Sloane; Dong–Xu/Kawahigashi–Longo). [E] the three compose into ‘invertible bulk $\Rightarrow (E_8)_1$ ’, all legs established; [O] the one open hypothesis is the input of LIT-A, ‘the raw quasi-free seam bulk is invertible (SRE)’ = Route B’s Flat-Away. [E] non-circular with Route B (the v286 firewall holds).
v298_e8_network_attractor.py	<i>E8 as a network attractor</i> — a computational-substrate (Wolfram-program) reading of the $(2, 3, 5)$ /McKay hull, on the v219 bedrock; an honest audit, not a derivation. [E] the E_8 Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ are the <i>unique</i> attractor of the minimal lazy graph update $m \leftarrow (A + 2I)/4 \cdot m$ on the affine- $E_8/(2, 3, 5)$ McKay graph (converges from any positive start; $Am = 2m$, top eigenvalue 2). [E] the spherical McKay tower terminates at E_8 ($2T \rightarrow E_6$, $2O \rightarrow E_7$, $2I \rightarrow E_8$, $h = 30$); [E] the cascade is nested-subgraph coarse-graining ($240 \rightarrow 126 \rightarrow 72 \rightarrow 40 \rightarrow 20$). [C] Wolfram reading: E_8 as the universal attractor of a minimal $(2, 3, 5)$ network (Coxeter-skeleton level). [O] a minimal hypergraph <i>rewrite</i> whose attractor is the full TFPT structure stays open; $\varphi_0 = (4/3)c_3 = 1/(6\pi)$ is the analytic seed, not an edge fraction.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v299_hypergraph_0d_autonomous.py	The hypergraph/Wolfram bridge (hardened): the 3-arm $(2, 3, \cdot)$ branch type is the irreducible seed on which a local witness-based spectral growth <i>reaches</i> E_8 as the last finite point ($E_6 \rightarrow E_7 \rightarrow E_8$, then stops at the affine boundary $\hat{E}_8$) — <i>not</i> ‘P2 derived’. [E] icosahedron = 3-uniform hypergraph ($ \text{Aut} = 120 = I_h $; the McKay group is the $SU(2)$ binary lift $2I$, a different order-120 group); [E] the $1 \rightarrow 4$ subdivision rewrite is equivariant (120 symmetries at every scale); [E] the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; [E] Smith: $\rho \leq 2$ selects (affine) ADE; [E] Collatz–Wielandt: $\rho \leq 2 \Leftrightarrow$ a positive witness m with $Am \leq 2m$ (local); [E] the autonomous rule (diffusion witness + local gate $Am \leq 2m$) runs $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ under the $(2, 3, \cdot)$ seed; [E] A_n ($\rho = 2 \cos \frac{\pi}{n+1}$) and D_n ($\rho = 2 \cos \frac{\pi}{2n-2}$) controls are < 2 for all n , never reaching E_8 ; [O] the seed = P2. A second network reading of the $(2, 3, 5)/E_8$ core (v219/v298).
v300_flataway_rigid.py	FLATAWAY.RIGID.01 — the attack on Flat-Away. [E] the flat fingerprint: $\text{spec}(D_\theta)$ is the integer ladder, every level $n \geq 1$ of multiplicity exactly 2 (a discrete fingerprint, not a norm); [E] the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy 2×2 $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$), validated vs the full Toeplitz operator; [E] any $g_k \neq 0$ lifts the resonant degeneracy (mult $2 \rightarrow 1+1$) and changes the spectral multiset, so preserving the flat ladder <i>with</i> its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — a discrete obstruction, strictly stronger than the v291/v293 ‘deviation grows’ scan; [E] with the v295 convexity, flat is isolated analytically <i>and</i> discretely; [C] the pin derived: the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + \dots)$ has integer q -coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), and by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, so $(E_8)_1$ rationality $\Rightarrow f = 0$ (conditional on Route A); [O] Flat-Away carries <i>zero</i> new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact.
v301_route_a_invertible.py	SEAM.EQUIV.A.INV.01 — discharge Route A’s last open hypothesis (‘the raw quasi-free seam bulk is invertible/SRE’) via the free-fermion classification; since v300 collapsed Route B into Route A, this is the last unconditional gap of SEAM.EQUIV.01. [E] carrier = 16 Majoranas, $c = 8$ ($g_{\text{car}} + N_{\text{fam}} = 5 + 3 = 8 = 16/2$, v148); [E] $SO(16)_1 \supset (E_8)_1$: $\dim \mathfrak{so}(16) = 120$ currents + 128 spinor = 248 = $\dim E_8$ (v154); [C] free-fermion invertibility (Kitaev): a gapped Gaussian/quasi-free 2+1D fermion phase (class D) is $\mathbb{Z}$ -classified by the chiral Majorana number and is <i>always</i> invertible (no intrinsic anyons; topological order needs interactions), so a gapped quasi-free bulk is invertible automatically; [E] the invariant $ \det K = \#\text{anyons}$ is 1 for E_8 (invertible) vs 4 for the gauged $D_8=SO(16)$ (anyons), which free fermions cannot produce; [O] LIT-A’s ‘invertible bulk’ now follows from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160+the gap), so the residual changes from a topological-order statement to the spectral input ‘is the quasi-free bulk gapped?’; [O] with v300 the whole open content of SEAM.EQUIV.01 reduces to the single spectral statement ‘the quasi-free seam bulk is gapped’.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v302_seam_gap.py	SEAM.EQUIV.GAP.01 — the last lever: after v300/v301 the whole open content of SEAM.EQUIV.01 was ‘the quasi-free seam bulk is gapped’; that gap is <i>not</i> a free input but the established Recovery gap of the seam transfer operator. [E] $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple leading Perron eigenvalue 1; [E] the gap is derived: $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1 - (2/3)^6 = 665/729$; $6 = 2N_{\text{fam}}$, $(2/3)^6$ the μ_4-deck transfer eigenvalue forced by the carrier); [E] the positivity margin $\Delta - 31/(4\pi^2) = 1.6476 > 0$ (the v76 decoupling margin); [C] by Osterwalder–Schrader / quasi-free clustering a transfer gap $\Delta > 0$ is a mass gap of the reconstructed Hamiltonian; [O] so the last TFPT-internal input is the derived Recovery gap $6 \ln(3/2) > 0$, and SEAM.EQUIV.01’s residual is a composition of cited theorems (OS/clustering + Kitaev v301 + the v297 AQFT stack) over established TFPT facts — no undischarged TFPT-internal assumption remains, though it stays [O] (not machine-proved end-to-end).
v303_ftransfer_dynamics.py	FR.DYNAMICS.01 — the discrete→dynamic lens on F_transfer: the main-branch mechanism (a gapped, positivity-preserving relaxation to a <i>unique</i> attractor; the local-averaging update → the E_8 marks, gap $6 \ln(3/2)$) is the shared <i>shape</i> of all four instances. [E] F_pole (Koide) <i>is</i> the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier $F' = (2/3)^6 = \lambda_2$, the seam-clock subleading eigenvalue (v82/v54); $F' < 1 \Rightarrow$ a unique attractor at the seam rate; [C] F_Boltzmann (η_B) is a gapped positive contraction (washout $\kappa_f \in (0, 1)$, H-theorem; compiler fixes $A_\Lambda = 10$, rate thermal); [C] F_relic (axion) freezes at an adiabatic fixed point (seed $\theta_i = 3\pi/5$, rate cosmological); [E] F_QCD (m_p/m_e) is the 1-loop RG flow with the carrier $b_3 = -7$ (asymptotic freedom $\Rightarrow$ the Gaussian UV attractor, Λ_{QCD} generated). [O] Verdict: one shape underlies all four = the main-branch principle; F_pole has the seam rate exactly, the other three share only the shape (external rates), so F_transfer is the readout end of the one principle, not a bolt-on — simplicity is preserved by honest fencing (v187).
v304_idg_nonlocal_ghost.py	QGAMB.IDG.01 — the KMS spectral-action cutoff suggests a <i>nonlocal</i> (infinite-derivative) graviton form factor; under entire analyticity the Stelle ghost is a <i>truncation artefact</i> — a [C] narrowing of the red-team rt_F perturbative-gravity attack, <i>not</i> a closure of QG.AMB.01 (no new parameter: the cutoff is the v259 seam KMS weight). [E] the local four-derivative propagator $1/(p^2(p^2 + M^2)) = (1/M^2)[1/p^2 - 1/(p^2 + M^2)]$ has a healthy massless graviton (residue $+1/M^2$ at $p^2=0$) and a massive spin-2 mode with <i>negative</i> residue $-1/M^2$ at $p^2 = -M^2$ (the Stelle ghost, reproducing v278); [E] for an IDG operator $p^2 a(p^2)$ with $a(z) = e^{z/M^2}$ entire and nowhere zero, the dressed propagator $1/(p^2 a)$ keeps its <i>only</i> pole at $p^2=0$ — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006); [E] negative control: a polynomial truncation $a(z) = 1 + z/M^2$ (the $R+R^2$ order) has a real zero at $z = -M^2$ = the ghost pole, so ‘entire vs polynomial’ is the discriminator; [C] the cutoff is the $\beta=1$ seam KMS weight $f(u) = e^{-u}$ (v259), not a choice — truncating $\text{Tr} e^{-D^2/\Lambda^2}$ at a_4 is what makes the local $R+R^2$ and the ghost; [C] conditional narrowing: <i>if</i> the resummed form factor is entire (Biswas–Mazumdar–Siegel class) <i>then</i> the perturbative graviton is ghost-free and rt_F’s attack is a truncation artefact; [O] open: (i) the trace regulator f is not the kinetic form factor $a(\square)$ — entire analyticity of the resummation is unproven; (ii) a ghost-free <i>perturbative</i> graviton is not the nonperturbative ambient measure — QGAMB.IDG.01 stays [C]/[O], not a QG.AMB.01 closure.

continued on next page

Script	What it machine-checks (<i>continued</i>)
v305_witness_independence.py	Witness-independence / generator-economy audit (structural anti-numerology firewall, companion to v100): every headline integer is a documented image of the single anchor $a=(1,1,2)$ ($e=(4,5,2)$, $p_n=2+2^n$; v23/v228), π the only transcendental primitive (ARCH.CORE.01) — 13 atoms / 2 free primitives = $6.5 \times$ compression made explicit (the recurring $2/3= \mathbb{Z}_2 /N_{\text{fam}}$ and $(2/3)^6$ are <i>one</i> ratio); E_8/g_{car} /anchor are overdetermined (≥ 3 structurally independent witnesses), the 5 pure-seed readouts single-witness (correlated, scored jointly by v100); α^{-1} 's '137' is foreign to the anchor family = an independent witness [E]
v306_seed_crossval.py	Leave-one-out cross-validation of the shared seed φ_0 : back-solving φ_0 independently from the five seed-grammar observables ($\sin^2 \theta_{12}$, $\sin^2 \theta_{13}$, β , Ω_b , λ_C) and error-weighted-averaging reproduces the axiom seed $\varphi_0=1/(6\pi)+48c_3^4=0.0531720$ to 0.04% (one real seed, not a per-observable fit); leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned = $\sin^2 \theta_{13}$, $\sim 2\sigma$); a data $\leftrightarrow$ formula shuffle explodes the χ^2 by $> 100 \times$ [E]
v307_data_watchdog.py	Data watchdog: the operational decision pipeline over the frozen registry (v84). Classifies all 20 entries (16 predictions + 2 assigned textures + 2 bands) PASS/WATCH/TENSION/KILL by live n_σ vs repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18); no decidable prediction in KILL (theory alive); watch items α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core $\sin^2 \theta_{13}$ (2.0σ); the prediction-of-record $\sin^2 \theta_{12}=0.306747$ compatible (JUNO, the fastest falsifier); $r \in [0.0033, 0.0048]$ below the BK18 limit and the kill 0.01; $w=-1$ flagged the dark-energy watchdog (DESI) [E]
v308_seam_equiv_chain.py	TOE attack 1 — the SEAM.EQUIV.01 keystone as a composition certificate: the v297 stack as one well-typed chain $\text{gap}>0 \rightarrow \text{invertible} \rightarrow \text{holomorphic } c=8 \rightarrow (E_8)_1$ with exact discriminators $ \det \text{Cartan}(E_8) =1$ vs $ \det \text{Cartan}(D_8) =4$, carrier tower $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$, $c=g_{\text{car}}+N_{\text{fam}}=8$; the input gap is the derived recovery gap $6 \ln(3/2)>0$. The only undischarged premise is the cited OS step — an assembly/reduction certificate (Lean target), NOT a closure [E][O]
v309_modular_bedrock.py	TOE attack 2 — the bedrock $\omega \circ \rho = \omega$ via an explicit modular Hamiltonian, non-circular: marks = 4 from Gauss–Bonnet (v216) + the order-4 clock from cross-ratio 2 (v214) $\Rightarrow \mu_4$ derived, not assumed; the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ round-trips to the seam operator and $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$ (a $\mathbb{Z}_2$ profile breaks it). Residual = Bisognano–Wichmann intrinsicality of the raw collar; NOT a closure of QGEO.SYM.01 [E][O]
v310_carrier_sm_anomaly.py	TOE attack 3 — the carrier half-spinor is one anomaly-free SM generation with the SM RG: $16=\Lambda^{\text{even}}(\mathbb{C}^5)=1+10+5 \rightarrow SU(5) \subset SO(10)$ as $10+5+1 = \text{one generation}$; gauge-anomaly-free ($\sum Y=0$, $\sum Y^3=0$, $[SU(2)]^2 U(1)=[SU(3)]^2 U(1)=0$, exact) and reproduces $(b_1, b_2, b_3)=(\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1 , so $N_{\text{fam}}=3$ pins the RG. Closes the spectrum/anomaly/ β sub-question; the full EG S -matrix + Yukawa stay open [E][C]
v311_gap_decoupled_measure.py	TOE attack 4 — the physical QG measure as a gap-decoupled, exponentially-clustering object: the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$, correlation length $1/(6 \ln(3/2))$, finite susceptibility $729/665$; the v76 margin $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ decouples the physical sector from the un-built ambient. Spectral-action $a_2=EH/a_4=R^2/72$ (v36) + KMS $\beta=1$ (v259); QG.AMB.01 stays [O] (gap-decoupled, inert) [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v312_hypergraph_substrate.py	TOE attack 5 — the hypergraph substrate sharpened: the affine- E_8 Kac marks are the Perron eigenvector ($A m=2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5)=$ the golden ratio (the 5-fold/icosahedral signature), but the recovery rate $(2/3)^6$ is <i>not</i> any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3= \mathbb{Z}_2 /N_{\text{fam}}$ (gap $6\ln(3/2)$) and the analytic seed φ_0 . The open question is precisely delimited, not closed [E][O]
v313_golden_atoms_spectral.py	The golden ratio is the $g_{\text{car}}=5$ signature: the $(2,3,5)$ atoms <i>are</i> the affine- E_8 network spectral angles (v312 follow-up, exact). [E] the charpoly factors $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$, the golden minimal polynomial x^2-x-1 divides it ($\phi=2\cos(\pi/5)$ an exact eigenvalue); the non-trivial eigenvalues are $2\cos(\pi/k)$ for $k \in \{2,3,5\}=\{ \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}}\}$ (giving $0, 1, \phi, 1/\phi$), so the golden ratio is the $g_{\text{car}}=5$ (5-fold) signature — a new spectral witness for g_{car} (raises the v305 multiplicity). The $(2,3,5)$ spherical excess $1/2+1/3+1/5=31/30$ ties the v63/v76 margin numerator $31=2^{g_{\text{car}}}-1=248/8=1+h$ to the Coxeter 30 (a unification, not new evidence). [E] honest null: ϕ is structural, not a readout. [C] direction: ϕ is the E_8 -quasicrystal (Elser-Sloane) ratio, a candidate substrate [E][C]
v314_rate_translation.py	Do the discrete-kernel and dynamic rates translate? An exact number-field split. The affine- E_8 adjacency spectrum splits into the rational part $\{0, \pm 1, \pm 2\}$ in $\mathbb{Q}$ and the golden part $\{\pm\phi, \pm 1/\phi\}$ in $\mathbb{Q}(\sqrt{5})$; the dynamic transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is entirely rational. [E] the family rates DO translate (the rational angles $2\cos(\pi/2)=0= \mathbb{Z}_2 $, $2\cos(\pi/3)=1=N_{\text{fam}}$ build the recovery rate $(2/3)^6=(\mathbb{Z}_2 /N_{\text{fam}})^{2N_{\text{fam}}}$ in $\mathbb{Q}$), but the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has <i>no</i> rational dynamic image — a field obstruction, so it is static-only. The shared object is the order-30 Coxeter element ($30=g_{\text{car}}(2N_{\text{fam}})=5\cdot 6$); the ϕ quasicrystal carries the carrier, not the cusp-weight family dynamics [E][C]
v315_coxeter_coupling.py	Couple or factorize? The order-30 Coxeter sectors couple as a cyclotomic field (v314 follow-up). [E] GROUP FACTORIZES: $\gcd(5,6)=1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled); [E] EXPONENTS FACTOR: E_8 exponents = totatives(30) = totatives(5) $\times$ totatives(6), $\phi(30)=4\cdot 2=8=$ rank E_8 ; [E] FIELD COUPLES: $[\mathbb{Q}(\zeta_{30}):\mathbb{Q}]=8=$ rank E_8 , $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) (\ni \sqrt{5})$ and the family $\mathbb{Q}(\zeta_3)$; [E] GALOIS = $\mu_4 \times \mathbb{Z}_2$: $(\mathbb{Z}/30)^\times = (\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2$ (order 8, max order 4), $\mu_4=(\mathbb{Z}/5)^\times$ the carrier Galois (the seam clock = the carrier golden-field Galois), $\mathbb{Z}_2=(\mathbb{Z}/3)^\times$ the family Galois; [E] $\sqrt{5}=\phi+1/\phi$ = the Gauss sum in $\mathbb{Q}(\zeta_5)$. [C] $30=5\cdot 6$ does not dynamically entangle — it couples as the cyclotomic compositum, the bridge is Galois not dynamical [E][C]
v316_galois_readout.py	Does the Galois $\mu_4 \times \mathbb{Z}_2$ (v315) organize the physical readouts? Yes — the CP phases. [E] the hexagonal CP unit $\rho=\zeta_6=\zeta_{30}^5$ is the order-6= $2N_{\text{fam}}$ FAMILY factor c^5 (the carrier is $\zeta_5=\zeta_{30}^6$); [C] $\delta_{\text{CKM,lead}}=\pi/3=\arg \zeta_6$, $\delta_{\text{PMNS}}=4\pi/3=\arg \zeta_6^4$ (power $4= \mu_4 $), $\zeta_6^4=-\zeta_6$ the sheet flip ($\mu_6=\mu_3 \times \mu_2$); [E] the Galois $\mathbb{Z}_2=\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q})=\{\zeta_6 \mapsto \zeta_6^5=\bar{\zeta}_6\}$ IS CP conjugation ($\delta \mapsto -\delta$); [E] the carrier factor ζ_5 (golden $\sqrt{5}$) carries no phase (static); [E] the magnitudes ($\sin^2 \theta_{12}=1/3-\varphi_0/2$, transcendental φ_0) are not cyclotomic. So the Galois bridge reaches phenomenology through the family factor (phases); the carrier is static; the magnitudes are the analytic seed [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v317_galois_family.py	Are the 3 generations a μ_3 /Galois orbit? Yes (μ_3), refined 1+2 by Galois (v316 follow-up). [E] the 3 cusp weights $\{0, 1/3, 2/3\}$ ($\text{Spec}(Q_+) = \{1, 2, 3\} = A_3$ exponents) map under $w \mapsto e^{2\pi i w}$ to the cube roots $\{1, \zeta_3, \zeta_3^2\}$; the cyclic family symmetry $w \mapsto w+1/3$ is a transitive μ_3 orbit (the family IS the cube-root orbit). [E] the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at $w=0$). [E] the Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2$, = CP conjugation v316) fixes $w=0$ (rational) and swaps $w=1/3 \leftrightarrow 2/3 \Rightarrow$ orbit 1+2; so the Galois-FIXED generation is the recovery ATTRACTOR (the law), the Galois PAIR the decaying hierarchy. [C] the family structure is arithmetic; the mass magnitudes stay the analytic seed [E][C]
v318_arithmetic_capstone.py	The capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30}) + \text{Galois } \mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\text{geo}}\}$ (synthesis of the v313–v317 arc). [E] the structural sector (3 generations, 2 CP phases, orbit/hierarchy) lives in $\mathbb{Q}(\zeta_{30})$: $[\mathbb{Q}(\zeta_{30}):\mathbb{Q}] = \varphi(30) = 8 = \text{rank } E_8$, $30 = \mathbb{Z}_2 N_{\text{fam}} g_{\text{car}} = h(E_8)$, Galois $(\mathbb{Z}/30)^\times = \mu_4 \times \mathbb{Z}_2$; [E] rigidity: forced by the anchor atoms $\{2, 3, 5\} = e(a)$; [E] magnitude reduction: $\varphi_0 = (4/3)c_3 + 48c_3^4$ is a pure function of π (no free parameters), so the magnitudes are $\{a, \pi\}$ -determined, not free; [E] free inventory: 0 dimensionless free parameters, π the unique transcendental, v_{geo} the one dimensional unit. [O] residual: the raw-seam realisation stays open (QGEO.SYM.01/SEAM.EQUIV.01) [E][O]
v319_translation_clock.py	The translation clock: the discrete $\leftrightarrow$ dynamic bridge is the order-30 Coxeter element, a clock with two coprime hands 5×6 , read law-inclusive (0..5) or live-only (1..5). [E] the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$, $\text{gcd}(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (v315). [E] DYNAMIC HAND ($\mathbb{Z}/6 = 2N_{\text{fam}}$, ticks 0..5): recovery exponent exactly $6 = 2N_{\text{fam}}$, rate $(2/3)^6$; order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$). [E] LAW VS LIVE: the resummed clock rate(n) = $-p_2 \ln(1 - n/N_{\text{fam}})$ (v124) has rate(0) = 0 (the attractor/law) and live modes from $n=1$; the E_8 live phases (totatives of 30) start at 1, so 0, 1, 2, 3, 4, 5 is law-inclusive, 1, 2, 3, 4, 5 live-only. [E] STATIC HAND ($\mathbb{Z}/5 = g_{\text{car}}$, ticks 1..5): golden $\mathbb{Q}(\sqrt{5})$, field-obstructed (no rational dynamic image) — static-only. [E] one full turn = $\text{lcm}(5, 6) = 30 = h(E_8)$ (rank $E_8 = 8 = \varphi(30)$, $ \mu_4 =4$ planes, v55); the “it is one clock” reading is [C].
v320_galois_cp_relation.py	A NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$. [E] both phases are powers of one hexagonal unit $\rho = \zeta_6$ (v316): $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$ (60°), $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$ (240°), $\rho^4 = -\rho$, so $\delta_{\text{PMNS}} - \delta_{\text{CKM}}^{\text{lead}} = \pi$ exactly (the $\mathbb{Z}_2$ sheet flip) — the quark and lepton leading CP phases are NOT independent. [C] prediction of record $\delta_{\text{PMNS}}^{\text{lead}} = 240^\circ$ (upgrades the assigned texture to a Galois-forced relation to the measured quark phase); [C] currently within the broad NuFIT 6.0 range; [X] kill test: δ_{PMNS} robustly away from 240° ($> 3\sigma$; DUNE/Hyper-K/JUNO) falsifies the Galois-CP organization [E][C][X]
v321_killtest_board.py	The forward kill-test board: the decisive upcoming measurements, ranked (forward companion to v307). [E] each entry is locked to a freeze_file.csv kill row: #1 JUNO $\sin^2 \theta_{12} = 0.306747$ (fastest), #2 CMB-S4/LiteBIRD r (kill $r > 0.01$), #3 DESI/Euclid $w = -1$, #4 DUNE/Hyper-K $\delta_{\text{PMNS}} = 240^\circ$ (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering + $m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2 \theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β . A forward decision ledger complementing v307; no new physics [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v322_cp_lock_sharpened.py	The v320 CP-lock prediction sharpened into a quantitative kill test. [E] the leading CP phase sits on the six hexagonal nodes $\arg \rho^k = k \cdot 60^\circ$; $\delta_{\text{CKM}}^{\text{lead}} = 60^\circ$ (node 1), $\delta_{\text{PMNS}}^{\text{lead}} = 240^\circ$ (node 4). [E] the node selector is the deck order: $\delta_{\text{PMNS}}^{\text{lead}} = \mu_4 \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3) = 4\pi/3 = \delta_{\text{CKM}}^{\text{lead}} + \pi$ ($\rho^4 = -\rho$). [C] sub-leading budget: $\delta_{\text{CKM}}^{\text{full}} = \pi/3 + 3\lambda^2$ ($\sim 68.7^\circ$); the lepton sub-leading is a separate open holonomy term bounded by $\sim 3\lambda^2 \approx 8.7^\circ \Rightarrow$ band $240 \pm \sim 9^\circ$. [C] current tension: vs NuFIT 6.0 NO ($\delta_{CP} = 212_{-41}^{+26^\circ}$) the leading 240° sits at $+1.08\sigma$, band edge $\sim 231^\circ$ at $+0.74\sigma$. [X] kill: the nearest wrong node is 60° away $\gg$ budget; DUNE/Hyper-K ($\sim 5\text{--}15^\circ$) discriminate, $> 3\sigma$ outside kills the lock. [E] an anarchic δ_{CP} hits the band with $p \sim 5\%$ and node 4 1-in-6 [E][C][X]
v323_bw_geometric_modular.py	The keystone pushed one step: Bisognano–Wichmann makes the seam modular flow <i>geometric</i> , so the μ_4 deck invariance is not an independent axiom. [E] the clock $\rho = \text{diag}(i^n) = \exp(i(\pi/2)L)$ is a geometric rotation ($L = \text{diag}(n)$, $\rho^4 = I$, $\mu_4 \subset U(1)_{\text{rot}}$). [E] for a rotation-covariant $C = f(L)$ the modular Hamiltonian $K = \log((1 - C)/C) = g(L)$, so $[K, L] = 0$ (the modular flow commutes with all rotations) and $[\rho, K] = [\rho, C] = 0$ — the clock is a modular symmetry for free. [E] discriminator vs v309: a period-4 curvature preserves the clock but its modular flow is <i>not</i> geometric ($[K, L] \neq 0$); BW is strictly stronger. [C] linked bedrock: given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric $\Rightarrow$ QGEO.SYM.01 ($\omega \circ \rho = \omega$) is downstream of SEAM.EQUIV.01 + a rotation-invariant vacuum. [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308) [E][C][O]
v324_hypergraph_fiber.py	The minimal hypergraph substrate that works: a $(2, 3, 5)$ -network <i>fibered</i> by a 3-fold cusp channel reproduces both the E_8 skeleton and the recovery gap (the constructive v312 follow-up). [E] network factor: $T_{\text{net}} = (A + 2I)/4$ fixes the Kac marks (eigenvalue $1 = E_8$ skeleton), subleading $(\varphi + 2)/4$ (golden); $(2/3)^6$ is <i>not</i> in its spectrum (the v312 negative). [E] cusp fiber: $T_{\text{cusp}} = \text{diag}((1 - w)^{2N_{\text{fam}}})$, $w \in \{0, 1/3, 2/3\}$ ($N_{\text{fam}} = 3$, v317), spectrum $\{1, (2/3)^6, (1/3)^6\}$. [E] the fibered substrate $T = T_{\text{net}} \otimes T_{\text{cusp}}$ has top eigenvalue 1 (eigenvector marks $\otimes$ ($w = 0$), skeleton $\times$ democratic cusp) <i>and</i> $(2/3)^6$ as a genuine eigenvalue (network attractor $\times$ cusp subleading). [E] honest injection: without the fiber ($T_{\text{net}} \otimes I_3$) there is no $(2/3)^6$ — the cusp datum is injected, not graph-spectral; substrate = network $\times$ cusp = the v315 split $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$. [E] minimality: the smallest fiber is the 3-node cusp ($= N_{\text{fam}}$) = order-30 (v319 clock). [C] a consistent fibered realization, not a derivation of $2/3$ from a pure rewrite [E][C]
v325_pillowcase_keystone.py	The seam keystone as ONE citable theorem: the raw seam state <i>is</i> the flat $\tau = i$ pillowcase state, with exactly one open lemma. IF (raw RP-collar seam state = flat $\tau = i$ pillowcase / rotation-invariant state) THEN 4 square marks + μ_4 deck + $\omega \circ \rho = \omega$ + holomorphic $c = 8 \Rightarrow (E_8)_1$, with the carrier recovery gap as input. [E] pieces all exact: $n = 2\chi(S^2) = 4$ (v216); cross-ratio $2 \Rightarrow j = 1728$ (v214/v267); flat orbifold $S^2(2, 2, 2, 2)$, $\chi_{\text{orb}} = 0$ (Trojanov); DtN sub-principal at $m \equiv 0 \pmod 4$ (v201/v280); $\det \text{Cartan}(E_8) = 1$ vs $D_8 = 4$ (v83/v143/v308); gap $6 \ln \frac{3}{2} > 0$ (v76/v302). [O] the one open lemma — the raw collar state IS the flat $\tau = i$ pillowcase / rotation-invariant state — is SHARED by both bedrock IDs (QGEO.SYM.01, SEAM.EQUIV.01; v323), machine-pinned in Lean (FORM.SEAMEQUIV.01 + FORM.QGEO.BW.01); an assembly/reduction certificate (like v176/v261), not a closure [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v326_ftransfer_suite.py	The F_{transfer} solver suite: the four transfer readouts as ONE runnable harness, each a gapped relaxation to a unique attractor (productises v303). [E] F_{pole} (Koide): Moebius contraction to $Q^*=2/3$ with multiplier $(2/3)^6$ = the seam-clock subleading eigenvalue, unique attractor at the SEAM rate. [C] $F_{\text{Boltzmann}}(\eta_B)$: washout contraction to the balance 6.1×10^{-11} , monotone (H-theorem), external thermal rate. [C] F_{relic} (axion): the comoving number $N=E/\omega$ is the adiabatic invariant, the relic freezes (seed $\theta_i=3\pi/5$), cosmological rate. [E] $F_{\text{QCD}}(m_p/m_e)$: 1-loop RG with carrier $b_0=7$ flows to the Gaussian UV fixed point, Λ_{QCD} generated. [C] F_{RareKaon} : $\text{BR}(K^+)=9.45 \times 10^{-11}$ on NA62 2016-2024 (v202). [O] only F_{pole} has the seam rate $(2/3)^6$; the other four share the SHAPE with external rates honestly fenced (v187/v303) — the frontier is measurable without claiming the external rates [E][C][O]
v327_hypergraph_rewrite.py	Deriving the cusp weight $2/3$ from a minimal rewrite (sharpening v324/v312). A minimal local rule — a family channel with $N_{\text{fam}}=3$ slots, one absorbing attractor ($w=0$) and $ \mathbb{Z}_2 =2$ surviving — supplies the cusp fiber. [E] M has spectrum $\{1, 2/3, 0\}$; $2/3=(N_{\text{fam}}-1)/N_{\text{fam}}= \mathbb{Z}_2 /N_{\text{fam}}$ EMERGES from the rule arity, not injected. [E] over the order-6= $2N_{\text{fam}}$ hand the rate is $(2/3)^6=64/729$ = the recovery gap (v76/v324), derived. [E] NON-GRAPH-SPECTRAL (proof, sharpening v312): $2/3$ is NOT a root of the affine- E_8 charpoly ($p(2/3)=-3520/19683 \neq 0$), so the recovery rate cannot be an adjacency eigenvalue. [E] the absorbing state (eigenvalue 1) is the democratic $w=0$ law (v317). [O] honest residual: the arity $\{2, 3\}$ (the anchor atoms) is still the input (P2) — v324's injected datum reduced to the anchor arity [E][O]
v328_theta13_pressure.py	The θ_{13} pressure point: the honest weak spot, quantified and pre-registered. [C] $\sin^2 \theta_{13}=\varphi_0^{\text{ret}}e^{-5/6}=0.023108$ vs NuFIT 6.0 NO (0.02195 ± 0.00058) is $+2.0\sigma$ — the most-tensioned CORE prediction (v62/v307). [E] θ_{13} is the carrier hypercharge trace (exponent $\gamma=\text{tr}_E Y^2=5/6$ EXACT, v268), NOT the seesaw $\mu\tau$ breaking that sets θ_{12} — distinct channels. [C] no cheap relief: the back-solved seed $\varphi_0^{\text{ret}}(\theta_{13})=0.0505$ is $\sim 5\%$ below the axiom, PMNS θ_{13} RG running is $< 1\%$ (NO), and $5/6$ is exact — the prediction stays frozen (REG.FREEZE.01). [X] pre-registered kill: a stable $\sin^2 \theta_{13}$ away from 0.023108 at $> 3\sigma$ (JUNO/global) flags the carrier-trace/seed-grammar reading; exactly the v306 seed-hyperplane outlier — one honest pressure point [C][E][X]
v329_os_gap_reduction.py	Closing the OS step: the bulk gap <i>is</i> the transfer gap (second quantization), and assembling the single sufficient premise of the whole bedrock. [E] OS GAP = TRANSFER GAP: the many-body OS Hamiltonian $d\Gamma(-\log T)$ has spectrum = sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy = $6 \ln \frac{3}{2}$, independent of system size — exactly “OS bulk gap = transfer gap” (the v308 open input), discharged at the many-body level. [E] $H \Rightarrow \omega \circ \rho = \omega$: under H the covariance is rotation-covariant $C=f(L)$, $K=g(L)$ commutes with $\rho=\exp(i(\pi/2)L)$ (v201/v309/v323). [E] $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$: $\det \text{Cartan}(E_8)=1$ vs $D_8=4$ (v237/v308). [E] both bedrock IDs follow from the ONE premise H (Lean-pinned FORM.SEAMEQUIV.01 + FORM.QGEO.BW.01). [O] residual: the necessity of H on the raw collar; sharpens v240/v308/v323/v325, does NOT close the bedrock [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v330_qgamb_admissible_measure.py	Constructing the ambient measure on the gap-decoupled admissible sector as a bona fide Osterwalder–Schrader object (with the honest fence). [E] reflection positivity: the seam transfer T (v221/v311; spectrum $\{1, (2/3)^6, (1/3)^6\}$) is symmetric PSD, so the finite-volume measures are RP. [E] exponential clustering: $C(n)/C(n-1) \rightarrow \lambda_2 = (2/3)^6$ ($\xi=1/\Delta$, $\Delta=6 \ln \frac{3}{2}$). [E] tightness $\Rightarrow$ projective limit: $\chi = \sum_n C(n) = 729/665$ finite, so $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi \Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit $\mu = \lim \mu_\Lambda$ EXISTS on the admissible sector. [E] OS reconstruction: $H_{OS} = -\log T \geq 0$, unique vacuum, gap $6 \ln \frac{3}{2}$ (v240). [E] NEG: an ungapped transfer sends $\chi \rightarrow \infty$, tightness fails. [O] the full ambient measure: the non-admissible (metric) sector is NOT built; by the gap margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v275) the physical readouts do not need it, but QG.AMB.01 stays open. A genuine PARTIAL construction, NOT a closure [E][O]
v331_necessity_of_H.py	The <i>necessity</i> of H reduced: rotation-invariance of the seam DtN is EQUIVALENT to the four marks at the square ($\tau=i$), so the one open lemma is exactly the order-4 CM selection (the necessity companion to v329’s sufficiency). [E] square $\Rightarrow [\rho, \Lambda]=0$: marks at $\{0, \pi/2, \pi, 3\pi/2\}$ make the curvature $\mathbb{Z}_4$ -symmetric (Fourier support $0 \bmod 4$), M_f couples only modes differing by $4\mathbb{Z}$ (the v201/v210 sufficiency as a DtN $ k +M_f$ operator). [E] necessity: a mark off the square breaks $\mathbb{Z}_4$ and gives $[\rho, \Lambda] \neq 0$ — equivalence, not just implication. [E] order-4 CM: cross-ratio($1, i, -1, -i$) $=2 \Rightarrow j=1728$, the unique 4-config with an order-4 automorphism (v214/v267); $H \Leftrightarrow$ square $\Leftrightarrow \mu_4$. [E] NEG: a $\mathbb{Z}_2$ mark set commutes with ρ^2 but not ρ . [O] residual: the dynamical <i>selection</i> of the square (the 4 marks forced by v216) is the irreducible postulate; does NOT close the bedrock [E][O]
v332_qgamb_metric_sector.py	The metric sector of QG.AMB.01 characterized: the obstruction is exactly the conformal-mode (Gibbons–Hawking–Perry) problem, the physical sector is gap-insulated, and the only route is the conditional IDG taming (makes v330’s metric-sector fence precise). [E] conformal-mode wrong sign: $c_{\text{conf}}(D) = -(D-1)(D-2)/4$ is negative for $D \geq 3$ ($c_{\text{conf}}(4) = -3/2$), TT positive — the conformal-factor problem. [E] measure obstruction: the kinetic form is indefinite (one negative eigenvalue), so e^{-S} diverges over the conformal mode; the bare metric-sector measure is not directly definable. [E] physical sector insulated: the obstruction is in the conformal/gauge direction, not the admissible sector (positive+gapped, v330), margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76). [C] IDG conditional route: the entire form factor e^{z/M^2} (v259/v304) dresses the conformal propagator with no extra pole. [E] NEG: a polynomial $(R+R^2)$ truncation crosses zero (the Stelle ghost, v304/v278). [O] metric sector open: the full QG.AMB.01 needs the conformal mode resolved nonperturbatively (GHP rotation / IDG), gap-decoupled but NOT constructed. A precise characterization, NOT a closure [E][C][O]
v333_cm_selection.py	The CM-selection mechanism: $\tau=i$ (the square) is the UNIQUE order-4 modulus and the attractor of mark-equilibration, so the last keystone residual is “the deck acts geometrically”, not “why the square” (the v331 follow-up). [E] order-4 modular element: $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ in $\text{SL}(2, \mathbb{Z})$ has $S^2 = -I$, $S^4 = I$ and fixes $\tau=i$ — the unique order-4 elliptic point of $\text{PSL}(2, \mathbb{Z})$ (the other, ρ , is order 3). [E] deck order $\rightarrow$ CM point: order-4 $\rightarrow$ cross-ratio 2 $\rightarrow j=1728 \rightarrow \tau=i$ (square); order-3 $\rightarrow j=0 \rightarrow \tau=\rho$ (hexagon); $ \mu_4 =4$ selects $\tau=i$. [E] mark-equilibration attractor: 4 points minimizing the log-energy (Fekete) converge to equal spacing (gaps $\pi/2 =$ the square), and beat random perturbations — the unique (mod rotation) attractor. [E] NEG: an order-3 deck selects ρ , not the square. [O] residual: only “the deck acts <i>geometrically</i> (= QGEO.SYM.01) vs an abstract $\mathbb{Z}_4$ on labels” stays open — sharpened from “why the square” to “the deck acts geometrically”, NOT a closure [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v334_conformal_resolution.py	The conformal-mode resolution: the Gibbons–Hawking–Perry contour rotation (IR sign) + the infinite-derivative (IDG) dressing (UV pole) give a CONDITIONALLY convergent metric-sector measure; the residual is the nonperturbative validity of the contour (the v332 follow-up). [E] the obstruction: $c_{\text{conf}}(4) = -3/2 < 0$, the unrotated conformal Gaussian diverges. [E] GHP rotation: $\phi \rightarrow i\phi$ flips the sign, $\int e^{- c \phi^2} = \sqrt{\pi/ c }$ FINITE (GHP 1978). [E] IDG dressing: the entire form factor $e^{p^2/M^2} > 0$ (no ghost pole, v304), the dressed per-mode action positive post-GHP and UV-suppressed. [E] NEG: a polynomial truncation crosses zero (the Stelle ghost, v304/v278). [C] conditional construction: GHP+IDG give a convergent metric-sector Gaussian mode-by-mode (conditional on the contour + entire analyticity). [O] residual: whether the GHP-rotated IDG-dressed measure is the CORRECT nonperturbative QG.AMB measure (the contour’s nonperturbative validity); gap-decoupled (v76), blocks no test, NOT a closure [E][C][O]
v335_seam_equiv_unify.py	The keystone unification: there is ONE open theorem (SEAM.EQUIV.01), QGE0.SYM.01 is its COROLLARY (a conformal-net axiom), and QG.AMB.01 is a decoupled general QG problem, NOT a TFPT gate. [E] QGE0.SYM.01 = corollary: a chiral conformal net has a Moebius-covariant vacuum (the unique invariant positive-energy vector, a NET AXIOM); rotations $U(1) < \text{Moebius} \Rightarrow$ rotation-invariant vacuum; the μ_4 clock = rotation by $\pi/2$ fixes it $\Rightarrow \omega \circ \rho = \omega$. So SEAM.EQUIV.01 $\Rightarrow$ QGE0.SYM.01 with NO extra premise (the v323 “rotation-invariant vacuum” is a net axiom); the two bedrock items collapse to ONE . [E] the one keystone theorem: SEAM.EQUIV.01 = a construction theorem ($c=8$, $\det \text{Cartan}(E_8)=1$, gap $6 \ln \frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma. [E] QG.AMB.01 decoupled: margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0 \Rightarrow$ no TFPT prediction depends on it; the general Euclidean-QG conformal-factor problem (GHP 1978), inherited not TFPT-specific. [E] collapsed status: structural items $2 \rightarrow 1$; live residual $= v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$. [O] the one residual = SEAM.EQUIV.01 ’s continuum OS reconstruction (Lean-pinned). A synthesis/inventory; does NOT close SEAM.EQUIV.01 [E][O]
v336_continuum_limit.py	The ONE open lemma of SEAM.EQUIV.01 , stated precisely and mapped to the recent rigorous literature – the continuum-limit and OS-reconstruction legs are now CITABLE . [C] continuum leg: Osborne–Stottmeister (<i>CFT from Lattice Fermions</i> , CMP 398 (2023) 219, doi:10.1007/s00220-022-04521-8; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework, arXiv:2010.11121) give the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur $\rightarrow$ Virasoro, correct c); WZW range rank $\leq c \leq D$. $(E_8)_1$: $c=8$, rank 8, $D=16$ Majoranas $(SO(16)_1 \rightarrow (E_8)_1)$, $8 \leq c \leq 16$ in range; the collar is a gapped quasi-free (CAR) state. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms (linear growth) for unitary VOAs incl. lattice VOAs; $(E_8)_1$ is the even self-dual rank-8 lattice VOA. [C] construction + covariance (Adamo–Giorgetti–Tanimoto): arXiv:2506.01008 builds the 2d conformal net of an even lattice and classifies the 2d Heisenberg-net extensions as lattice nets, and arXiv:2508.07109 shows every DHR rep of such a net is automatically positive-energy + diffeomorphism-covariant (covariance a consequence, not an input). [E] target pinned: $c=g_{\text{car}}+N_{\text{fam}}=8$, $\det \text{Cartan}(E_8)=1$, the $D=16$ Majorana $SO(16)_1$ IS the carrier-16 and the μ_4 extension IS the seam glue. [O] open residual: $c=8$ alone does NOT pin the net ($SO(16)_1$ also $c=8$, $\det 4$), so the one open lemma = the holomorphy discriminator $\det K=1$. A literature-mapping/reduction; does NOT close SEAM.EQUIV.01 [C][E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v337_decoupling_theorem.py	THE DECOUPLING THEOREM (consolidated, citable): every TFPT dimensionless readout factors through the gapped admissible sector and is INDEPENDENT of the ambient QG measure – the theorem that makes the QG.AMB.01 reclassification (v335) honest. [E] gap margin: $\Delta - 2 \dim(E_8) c_3^2 = 6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$. [E] factorization: every readout (mass ratios, mixings, α^{-1}, $R+R^2$) is a function of the admissible gapped transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ + the discrete kernel alone. [E] the theorem: margin $> 0 \Rightarrow$ exponential clustering ($\lambda_2=(2/3)^6$), finite susceptibility $\chi=729/665$, so the ambient contribution is gap-suppressed $\Rightarrow$ no readout depends on the ambient measure $\Rightarrow$ QG.AMB.01 decoupled. [E] neg control: gap closed $\Rightarrow \chi$ diverges $\Rightarrow$ fails (non-vacuous). [E] corollary: zero frozen kill tests depend on the ambient measure. Consolidates v76/v311/v330/v332; does NOT construct the ambient measure [E]
v338_theta13_budget.py	THETA13.BUDGET.01: the θ_{13} sub-leading budget vs the current global fit – the $+2\sigma$ pressure point (v328) quantified against BOTH NuFIT 6.0 NO variants and JUNO; neither defuses nor excludes, fences correctly. [C] SK-dependent tension: $\sin^2 \theta_{13} = (\varphi_0^{\text{ret}}) e^{-5/6} = 0.023108$ is $+2.0\sigma$ vs the no-SK central (0.02195) and $+1.7\sigma$ vs the with-SK NO best fit (0.02215) – not a clean 2σ. [E] within 3σ: $0.023108 <$ the with-SK 3σ upper 0.02388 (not falsified). [C] budget: moving to the central needs -4.1%; RG $< 1\%$ and the exponent $5/6$ is exact $\Rightarrow$ no freeze-respecting relief (v328). [X] JUNO decides: sub-1% precision turns $1.7\text{--}2.0\sigma$ into $\sim 6\sigma$ – a near-term pre-registered kill/confirm. One honest pressure point (v306/v328), fenced [C][E][X]
v339_firstprinciples_boundary.py	The honest first-principles boundary, mapped precisely – answers (a) can $M_{\text{Planck}}/v_{\text{geo}}$ be computed? and (b) can F_{transfer} be made first-principles? Both NO, with the exact reason per residual. [E] $v_{\text{geo}}/M_{\text{Planck}}$ FORBIDDEN BY THEOREM (No-Unit, v153): a mass is dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs a scale; over-determined (gravity = dark energy to 0.11%, v274); every dimensionless RATIO is derived, only the unit is not. [E] m_p/m_e structure in place (v262): the carrier $b_3=-7$ drives the QCD run + the closed electron source; two inputs stay external: $\alpha_s(M_Z)$ and the lattice $C_p=m_p/\Lambda^{(3)} \sim 2.83$. [X] $\alpha_s(M_Z)$ external BECAUSE OF A TENSION: SM content does NOT unify (1-loop spread ~ 8.8 at 2×10^{16} GeV, v246). [C] C_p parameter-free but lattice-only (v35). [C] η_B/axion = cosmological initial conditions; Koide $Q=2/3$ structural [E]. [E] FOUR honest kinds: forbidden-by-theorem / external-parameter-free / external-tensioned / cosmological. A boundary audit, NOT a solution [E][C][X]
v340_As_reconcile.py	A_s is predicted (not missing), is DIMENSIONLESS (so it does NOT define an absolute mass), and the apparent -11σ tension is a reheating-channel question – reconciled with the archived old derivation (TFPT v4.5/v4.6). [E] $A_s = N_\star^2 (M_{\text{scal}}/\bar{M}_{\text{Pl}})^2 / (24\pi^2) = N_\star^2 c_3^7 / (24\pi^2)$ (v86); $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = 1.2565 \times 10^{-5}$ is a PURE RATIO, so A_s carries NO absolute scale – it pins $N_\star$, NOT $\bar{M}_{\text{Pl}} \Rightarrow A_s$ does NOT define mass (No-Unit v153 stands). [C] the -11σ is the slow-vs-fast reheating dichotomy ($N_\star=51.44 \rightarrow A_s=1.764 \times 10^{-9}$; A_s-preferred $N_\star=56.2$). [C] ARCHIVE: old v4.5/v4.6 used the algebraic $N_\star=3/\varphi_0^{\text{ret}} - 1=55.42$ and reported $A_s=2.124 \times 10^{-9}$; at that $N_\star$ the current normalisation gives 2.047×10^{-9} (-1.9σ). [E] nothing lost: the current theory DERIVES $N_\star$ (record band [50, 60], v84). [X] decisive test: fast preheating or slow reheating confirmed [E][C][X]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v341_alpha_quillen.py	The α fixed point reformulated as the STATIONARITY of a $U(1)$ determinant line, every ingredient identified as an index / heat-kernel coefficient / discriminant form – NOT a free knob (the “derive the FORM” step). $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6M \ln(1/\varphi_{\text{seam}}) = 0$ in Quillen form: $[\alpha^3 - 2c_3^3\alpha^2]$ (det-line variation) = $[8b_1c_3^6]$ (anomaly counterterm) $\times [\ln(1/\varphi_{\text{seam}})]$ (seam response). [E] $M=41$ a $U(1)$ INDEX (EM Ward, v48): $\frac{4}{5}M = 8b_1$, $b_1=41/10$ the SM 1-loop $U(1)_Y$ beta (v159/v246). [E] $c_3=1/(8\pi)$ the GAUSS–BONNET boundary coefficient ($8=\text{rank } E_8$, v216); $c_3^{3,6}$ heat-kernel powers. [E] $-5/4 = -q(D_5)$ a DISCRIMINANT form ($q(D_5)+q(A_3)=2$, v1/v154); $\varphi_{\text{base}}=1/(6\pi)=\frac{4}{3}c_3$ carries $q(A_3)$: $c_3/\varphi_{\text{base}}=3/4$. [E] $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4$. [E] the Quillen split holds at the $\alpha^{-1}=137.0359992$ root. [E] NO NEW GATE: α^{-1} stays [E]; v341 SHARPENS the pre-existing [O] origin EM.WARD.01 (no new α problem). [O] the one still-open part IS EM.WARD.01’s residual: the from-first-principles AQFT PROOF that this is the EXACT Quillen determinant functional (the variational principle derived, not assembled). Sharpens v3/v48/EM.WARD.01 from “fixed point” to “stationarity of a named determinant line” [E][O]
v342_em_ward_heatkernel.py	The heat-kernel sharpening of EM.WARD.01: DERIVES the ORIGIN and STRUCTURE of the $F_{U(1)}$ determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] to [E] for the STRUCTURE; NO new gate. [E] $c_3=1/(8\pi)$ the boundary GAUSS–BONNET coefficient (seam = flat pillowcase $\chi=2$, $\oint K=4\pi$, v216). [E] the α^2 term is the GILKEY gauge-curvature coefficient: the $U(1)$ Laplacian a_4 has $30\Omega^2/360 = \frac{1}{12}\Omega^2$ (Gilkey Thm 4.8.16), $\Omega=i\alpha F \Rightarrow -(\alpha^2/12)F^2$, so the Calderon $-2c_3^3\alpha^2$ is the gauge-curvature piece, not an ansatz. [E] the c_3-ladder $\{0, 3, 6\}$ is arithmetic (step $3 = 3$ channels; $6= R^+(A_3)$ C_6 hexagon). [E] multiplicities forced ($2= \mathbb{Z}_2$, $8b_1=(\text{rank } E_8)b_1$). [E] π-power test: $2c_3^3=1/(256\pi^3)$ carries $\pi^3 \Rightarrow$ THREE boundary insertions. [C] the exact coefficient needs the seam F-normalisation. [O] the cubic α^3 Maxwell moment + full identification = EM.WARD.01 residual (tied to SEAM.EQUIV.01). α^{-1} stays [E] (v3) [E][C][O]
v343_four_routes_analysis.py	The honest investigation of the four black-hole-birth solution routes (A finite causal diamond, B Carlip–Cardy near-horizon CFT, C modular/thermal flow, D self-reproduction attractor) – direct answer to “which closes ALL problems and is 100% provable”: NONE individually, but they CONVERGE on the same two irreducible residuals as v335 (SEAM.EQUIV.01 + the decoupled QG contour). [E] ROUTE A reframes QG.AMB to a FINITE thermal trace (de Sitter finite, gapped $\chi=729/665$) but $c_{\text{conf}}(4) = -3/2 < 0$ still needs the GHP contour (v334, [O]); does NOT touch SEAM.EQUIV.01. [C] ROUTE B: Carlip+Cardy consistent with $c=8$ but L_0 free (ambiguity) and gives c only, not holomorphy $\det K=1$ – inherits SEAM.EQUIV.01. [E] ROUTE C: modular flow KMS $\beta=1$ (v239), geometric identification = SEAM.EQUIV.01 (v335) – a reduction. [E]/[O] ROUTE D: unique Perron–Frobenius attractor (v56) but cyclic dynamics [O] – not a closure. [E] VERDICT: independent closures = 0; irreducible residuals = 2 (= v335); the BH premise makes both more natural but eliminates neither; focus = SEAM.EQUIV.01 [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v344_detk_synthesis.py	The bird's-eye synthesis of the ONE open keystone bit, $\det K=1$ – a full-spectrum map of SEAM.EQUIV.01's only open residual. [E] $\det K=1$ is ONE statement with SIX equivalent faces (holomorphy / homology-sphere link $H_1=0$ / one 1-dim irrep / $\det \text{Cartan} = 1$ / full μ_4 condensation / $\tau=i$ CM), unified by $ \det \text{Cartan} = H_1(\text{link}) = \#(1\text{-dim irreps})$ across ADE ($A_n \rightarrow n+1$, $D_n \rightarrow 4$, $E_6 \rightarrow 3$, $E_7 \rightarrow 2$, $E_8 \rightarrow 1$) – ONLY E_8 gives 1 ($2I$ the unique PERFECT ADE group, v232/v219). [E] THE GENUS-1 OBSTRUCTION: the easy arguments (plane vacuum, gap, free-fermion) are genus-0, necessary but NOT sufficient (v87); $\det K$ is the torus degeneracy = #primaries. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the $(2, 3, 5)$ rewrite attractor IS the affine- E_8 graph (v312) and the E_8 du Val link is the Poincare sphere ($H_1=0$, v232), so $\det K=1$ becomes a finite COMBINATORIAL statement (no continuum limit). [O] none closes it; R3's combinatorial-to-geometric bridge is meanwhile discharged to Kronheimer 1989 (v479), so the residual transforms to “the raw seam supplies the ALE/orbifold datum” = the SEAM.EQUIV.01 content. A synthesis/roadmap; closes nothing [E][C][O]
v345_hypergraph_homotopy.py	EXECUTE the combinatorial core of route R3 (the hypergraph-homotopy route to $\det K=1$, v344): computes explicitly that the $(2, 3, 5)$ -rewrite attractor graph's plumbing link is the POINCARÉ homology sphere ($H_1=0$), only E_8 among ADE; R3's combinatorial half is now [E]-complete, the geometric bridge (= SEAM.EQUIV.01) stays [O]. Plumbing topology (Brieskorn): link $H_1 = \text{coker}(\text{Cartan})$. [E] $\text{SNF}(E_8 \text{ Cartan}) = \text{identity} \Rightarrow \text{coker}(E_8)=0 \Rightarrow$ the E_8 plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of $\det E_8=1$). [E] only E_8 : $\text{coker}(A_n)=\mathbb{Z}/(n+1)$, $D_n \rightarrow \mathbb{Z}/4$, $E_6 \rightarrow \mathbb{Z}/3$, $E_7 \rightarrow \mathbb{Z}/2$ – all nontrivial, only $E_8=0$. [E] $\pi_1=2I=SL(2, 5)$ PERFECT (commutator subgroup order $120= G $, computed) $\Rightarrow H_1=0 \Rightarrow$ THE Poincaré sphere; $ 2I =120= \mu_4 h(E_8)$. [O] does NOT close $\det K=1$: the graph $\rightarrow$ geometry half is meanwhile Kronheimer-cited (v479); what stays open is the raw seam supplying the ALE/orbifold datum = SEAM.EQUIV.01 [E][O]
v346_seam_geometric_bridge.py	The geometric bridge as an explicit six-link chain raw-seam $\Rightarrow \det K=1$, with the ONE open arrow pinned – the honest capstone of the SEAM.EQUIV.01 investigation. L1 [E]/[C] raw seam $\rightarrow \mu_4$ + four marks (v216); L2 [O] μ_4 + $(2, 3, 5)$ atoms $\rightarrow 2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE OPEN ARROW = SEAM.EQUIV.01); L3 [E] $2I \rightarrow$ affine E_8 McKay (v219); L4 [E] $2I \rightarrow E_8$ du Val (v232); L5 [E] $\mathbb{C}^2/2I \rightarrow$ Poincaré link $H_1=0$ (v345); L6 [E] $H_1=0 \rightarrow \det K=1$. [E] 5 of 6 links are theorems, only L2 open. [E] the group is FORCED by the seam atoms: axis signatures $(2, 3, 3)/(2, 3, 4)/(2, 3, 5) \rightarrow 2T/2O/2I$, $(2, 3, 5)$ unique to $2I$, and $(\mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$ ARE those axes. [E] downstream of L2 forced. [O] L2 (the seam IS an ADE orbifold point) = SEAM.EQUIV.01, NOT closed – the keystone reduced to one geometric-realisation arrow, group pinned by $(2, 3, 5)$ [E][C][O]
v347_seam_closure_modes.py	The closure-mode classification of the one open arrow L2 (the seam-orbifold realisation, v346) – a direct honest answer to “is there ANY other way to FULLY solve it?”. [E] THE PRECISE LOCUS: the seam gives the pillowcase base $S^2(2, 2, 2)$, $\chi_{\text{orb}}=0$ (EUCLIDEAN, v216), but the $2I/E_8$ structure is the Seifert base $S^2(2, 3, 5)$, $\chi_{\text{orb}}=1/30$ (SPHERICAL); L2 must bridge a FLAT 4-marked base to a SPHERICAL 3-marked base – why L2 needs the carrier $(2, 3, 5)$ input. [E] FOUR CLOSURE MODES: A construct (the analytic wall, v336); B rigidity (unique realisation $\Rightarrow \mathbb{C}^2/2I$ forced, the only theorem-route w/o new input, v284/v300/v331); C axiom P_3 ($\Rightarrow$ TFPT complete rel. 3 axioms); D empirical. [E] no-new-state (v246) blocks deriving L2 from extra physics $\Rightarrow$ only rigidity (B) closes it as a theorem. [E] VERDICT: no free theorem; only B (theorem) or C (axiom), validated by D; only these four modes. [O] NOT closed [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v348_seam_rigidity_route.py	Route B (the rigidity route) for the one open arrow L2, carried out – it lands on a strikingly SIMPLE statement: the whole keystone reduces to ONE question, does the raw seam carry the golden ratio? [E] rigidity closes the crystallographic part: the order-4 clock has a UNIQUE normal form $z \mapsto iz$ (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] downstream is a LATTICE identity: E_8 = the ring of icosians (Conway–Sloane), rank 8, so $2I \rightarrow E_8$ is a lattice identity (stronger than McKay v219). [E] the single bridge is φ : crystallographic orders $\{1, 2, 3, 4, 6\}$, the 5-fold non-crystallographic; μ_4 crystallographic, $2I$ (5-fold) not, $\varphi = 2\cos(\pi/5)$ extends one to the other (quasicrystal). [O] the keystone reduces to ONE question: “does the raw seam carry φ ?” – if yes, $\mu_4 + \varphi \rightarrow 2I \rightarrow E_8$ closes L2. [O] honest subtlety: TFPT’s φ is currently E_8 -side (v312/v313); the raw-seam data (cusp $\{0, 1/3, 2/3\}$, recovery $(2/3)^6$, seed φ_0) are not manifestly golden, so showing the RAW seam carries φ non-circularly is the residual (= L2). NOT closed [E][O]
v349_raw_seam_golden_test.py	The honest test of the v348 question “does the RAW seam carry the golden ratio (before assuming E_8)?”. Answer, computed: NO – the bare carrier (D_5 , A_3) and the dynamical seam data are NOT golden; the golden ratio genuinely enters WITH the icosahedral/ E_8 identification. A NEGATIVE result that prevents a false closure. [E] RAW CARRIER NOT GOLDEN: golden needs $5 \mid h$; D_5 has $h=8$ ($2\cos(2\pi/8)=\sqrt{2}$), A_3 has $h=4$ – no 5-fold, bare carrier gives $\sqrt{2}$. [E] DYNAMICAL DATA NOT GOLDEN: cusp $\{0, 1/3, 2/3\}$ rational, recovery $(2/3)^6$ rational, seed φ_0 transcendental. [E] GOLDEN IS THE ICOSAHERAL/ E_8 SIDE: $5 \mid h$ only for $A_4=SU(5)$ ($h=5$), H_3 ($h=10$), E_8 ($h=30$). [E] φ alone insufficient: $2I \neq C_5 \times \mu_4$ ($20 \neq 120$). [O] RESOLUTION: L2 reduces to “is $g_{\text{car}}=5$ a PENTAGON (5-fold $\Rightarrow$ golden $\Rightarrow$ icosahedral) or a COUNT (rank $D_5 \Rightarrow \sqrt{2}$)?” – the golden ratio is the pentagon-reading of P2, NOT derivable from P2-as-a-count; no hidden φ . Foundational: P2-as-pentagon closes it (mode C) else rigidity must produce the 5-fold (mode B); NOT closed [E][O]
v350_bootstrap_inputs_correction.py	The honest CORRECTION of v349’s framing (prompted by the right objection: the inputs are not axioms, they are back-determined by the theory). [E] THE INPUTS ARE BOOTSTRAP-FORCED, NOT FREE AXIOMS (v6): $g_{\text{car}}=5$ over-determined three ways (rank-fill, Coxeter-match $g_{\text{car}} = \max$ prime of $h(E_8)=30$, Pascal). [E] $g_{\text{car}}=5$ IS THE ICOSAHERAL 5-FOLD, not the D_5 rank: $g_{\text{car}} = \max$ prime of $h(E_8)=30=2\cdot 3\cdot 5$; the atoms (2, 3, 5) ARE the primes of the icosahedral Coxeter number 30. [E] THE GOLDEN RATIO IS EMERGENT FROM THE GLUE, not external: D_5 ($h=8$)/ A_3 ($h=4$) non-golden alone, but the μ_4 extension $D_5 \oplus A_3 \rightarrow E_8$ (v154) gives $h(E_8)=30$ (golden, $5 \mid 30$) – the seam’s own μ_4 produces the 5-fold; v349 checked the un-glued pieces, the wrong object. [E] v349’s pentagon-vs-count dichotomy was FALSE (the bootstrap forces the pentagon). [O] residual relocates: the golden is bootstrap-forced; the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Caveat: self-consistency within the framework, not from-nothing. Supersedes v349’s interpretation [E][O]
v351_continuum_realisation_sharpened.py	The continuum realisation sharpened – the “ $c=8$ ambiguity” flagged open in v277/v344 (E_8 vs $SO(16)$ both $c=8$) is RESOLVED, non-circularly, by the seam’s ORDER-4 μ_4 clock. [E] E_8 (det 1, holomorphic) vs $D_8=SO(16)$ (det 4). [E] the order-4 clock (the four Gauss–Bonnet marks, $ \mu_4 =4=h(A_3)$, v216): index-4 extension $D_5 \oplus A_3 = E_8$ (v154), index-2 = $SO(16)$; so order-4 selects E_8 ($16 \rightarrow 4 \rightarrow 1$) not $SO(16)$ ($16 \rightarrow 4$) – the order (4 not 2) is the discriminator. [E] bootstrap agrees: $h(E_8)=30$ (max prime 5) vs $h(D_8)=14$ (max prime 7). [E] which-net ambiguity RESOLVED (det $K=1$ is then a consequence). [O] residual = PURELY the chiral-edge/scaling-limit existence (bulk-edge, v336), net+holomorphy pinned. Does NOT close SEAM.EQUIV.01 [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v352_framework_irreducible.py	The framework reduction – the deepest answer to “the inputs are not axioms, they are back-determined”. [E] P2 ($g_{\text{car}}=5$) FORCED three ways (v6): rank-fill, Coxeter-match ($g_{\text{car}} = \max \text{ prime of } h(E_8)=30$), Pascal. [E] P1 ($c_3=1/(8\pi)$) FORCED: c_3 is the boundary Gauss–Bonnet coefficient (v216/v342), the 8 over-determined four ways ($8 = \text{rank } E_8 = h(D_5) = \phi(30) = \det R$, v54/v6); only π is non-forced (the boundary transcendental). [E] the E_8 hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles are the FRAMEWORK (a discrete reflection-positive boundary with a finite Clifford carrier) + π – the precise content of “the inputs are not axioms”. [O] the framework is the META-AXIOM (not math-reducible, the foundational stance, tested by its consequences not proved). TFPT’s true input is ONE framework + π [E][O]
v353_selfloop_capstone.py	The bird’s-eye rethink (refining v352): TFPT is not a LINEAR theory (axioms $\rightarrow$ theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs, with ZERO free adjustable dimensionless parameters. [E] IT IS A LOOP: outputs fix inputs ($g_{\text{car}}=5 = \max \text{ prime of } h(E_8)=30$; the 8 in $c_3 = \text{rank } E_8$) and inputs fix outputs; the arrows close into a circle (v6) with a unique fixed point (v56). [E] ZERO FREE DIALS: every load-bearing number is forced/over-determined; free dimensionless parameters = 0. [E] π is a FIXED geometric constant (the boundary 2-sphere, $\oint K=4\pi \Rightarrow c_3=1/(8\pi)$), not a dial. [C] refining v352: π is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom. [O] the one residual is the CONTINUUM closure (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork [E][C][O]
v354_e8_reverse_audit.py	The REVERSE numerology audit + what the golden ratio means. The forward rule (v305) is one-directional (every readout must map onto E_8); the reverse question: how much E_8 structure maps onto NOTHING? [E] E_8 has 8 Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$; THREE feed a readout (2 metric, 8 rank $\rightarrow c_3$, 30 Coxeter $\rightarrow g_{\text{car}}$) and FIVE are UNMAPPED $\{12, 14, 18, 20, 24\} - 3/8$ load-bearing, $5/8$ not. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, $h=30$, $\det 1$, $D_5 \oplus A_3$, 16-spinor) generates ALL readouts; the higher Casimirs are never reached into to fit – the anti-numerology signature (economy + a large untouched remainder). [E] the golden ratio is UNMAPPED (the affine- E_8 attractor eigenvalue v312, the icosian lattice v348; no frozen readout is φ , the sole [C] exception $\cos(\theta_i=3\pi/5)=-1/(2\varphi)$ on the axion spine branch, refined in v429) $\Rightarrow$ NOT numerology (which needs a number fitted to an observation); φ is the algebraic signature of the 5-fold $h=30=2\cdot 3\cdot 5$, the carrier announcing it is icosahedral. [O] the $5/8$ unmapped is the HULL OVERHEAD ($E_8 = \text{minimal unimodular hull, not the world group}$); where future readouts could live, or over-capacity. The reverse complement of v305 [E][O]
v355_e8_unmapped_bandwidth.py	A disciplined bandwidth search of the unmapped E_8 region (next step after v354), run with a STRICT discriminator: a candidate counts only if it is a FORCED identity (a theorem about E_8), never a numerical coincidence – because mining a rich object for matches is the numerology temptation. [E] FORCED SET-LEVEL (the real collective content): $\sum \text{deg} = 128 = \dim S^+$ the spinor half of $248=120+128$; $\sum \text{exp} = 120 = \# \text{pos roots}$; $\prod \text{deg} = W(E_8) $; exponents = the $\varphi(30)=8$ totatives of 30 (v223); $\max \text{deg} = 30 = h(E_8)$. [C] SUGGESTIVE not forced: $12=h(E_6)$, $18=h(E_7)$, $30=h(E_8)$ (the McKay ladder v219), but ‘deg(E_8) contains subalgebra Coxeter numbers’ is not a theorem – flagged. [O] DECLINED: the per-degree atom matches ($14=\dim G_2$, $20=\det L$, $24= W(A_3) $, $12= R(A_3) $, $18=p_4(a)$; v66 flagged 12, 18 fittable) each admit multiple readings, and the $ W $ prime 7=‘scalaron exponent’ is one of many – declined. [E] RECONCILES v66 (degrees=atoms) and v354 (only 2, 8, 30 primary): NO new forced physical hit; the disciplined decline IS the anti-numerology result [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v356_ continuum_mmst_ applicability.py	Direction A: the continuum chain of SEAM.EQUIV.01 assembled theorem-by-theorem, residual reduced to ONE lattice-realization input tied to the seam's one-sidedness. Chain: gapped (v302) quasi-free 16-Majorana collar (v155/v160) $\rightarrow$ invertible (Kitaev, v301) $\rightarrow$ the NONTRIVIAL invertible phase ($c_- = 8$) $\rightarrow$ a non-gappable anomalous chiral edge (anomaly inflow: edge EXISTENCE follows, reducing the v336/v351 'does the edge exist?' residual) $\rightarrow$ chiral-CFT scaling limit (MMST v336, $8 \leq 8 \leq 16$) $\rightarrow$ order-4 μ_4 clock + AQFT stack pin $(E_8)_1$ (v351/v297). The only non-theorem link is the seam one-sidedness = the c_3 origin ($c_- \neq 0 =$ one-sided = the $ \mathbb{Z}_2 \nmid K=8$ of $c_3=1/(8\pi)$, v58). [C]/[E]/[O]; ADVANCES SEAM.EQUIV.01 to one realization input, does not close. (Direction B was found already closed by v40.) [C][E][O]
v358_grav_entropy_ equilibrium.py	The entanglement-equilibrium derivation (Jacobson 2015 / Faulkner et al., $\delta S = \delta \langle K \rangle$) with TFPT's atoms $\Rightarrow$ the LINEARISED Einstein equation falls out PARAMETER-FREE. [E] $G_{ab} = (1/c_3)T_{ab}$, $1/c_3 = 8\pi$ fixed (Unruh $1/(2\pi) = 4c_3$, EH $1/(16\pi) = c_3/2$, Einstein $8\pi = 1/c_3$, Bekenstein $1/4 = 1/ \mu_4 $). [E] THERMO=GEO coincidence (new): the first law's $2\pi/\eta$ = the geometric $ \mathbb{Z}_2 2\pi\chi$ iff $ \mu_4 = \mathbb{Z}_2 \chi = 4$ – the thermodynamic and geometric origins of c_3 are ONE. [C] J3 solved: the CHM ball modular Hamiltonian (boost via BW v323) gives the matter flux $\delta \langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta \langle T_{00} \rangle$. [E] entropy density atom-fixed: $1/4 = 1/ \mu_4 $ + central charge $c = g_{\text{car}} + N_{\text{fam}} = 8$ (form v59). [O] residual: the equation-of-state status + the global ambient measure (C7) + the v_{geo} unit (v359 closes the FULL covariant equation); no free dimensionless gravity dial remains [E][C][O]
v359_grav_ nonlinear_ einstein.py	The FULL covariant Einstein equation, parameter-free (Direction 1, extends v358). The Jacobson-2015 fixed-VOLUME entanglement equilibrium brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}) – the unique divergence-free metric tensor (Lovelock) – so equilibrium yields the full covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1}T_{ab}$ with conservation built in. [C][E] fixed-volume $\rightarrow$ Einstein tensor (trace = $-R$ in $d=4$). [E] Lovelock $\Rightarrow \nabla^a T_{ab} = 0$ is an output. [E] coeff 1: $8\pi = 1/c_3$ fixed (v358). [E] coeff 2: Λ from $\alpha - \rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$, prefactor $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$ (v60); BOTH coefficients TFPT-fixed. RESULT: the original B6 full covariant field equation is now parameter-free at the LOCAL level. [O] residual: the equation-of-state status + the global ambient measure (C7) [E][C][O]
v360_grav_gap_ corrections.py	Direction 2: the gap-induced higher-curvature correction to GR – the disciplined result. Does the seam transfer gap give a NEW forced near-term GR-deviation beyond the known R^2 scalaron? With the v354/v355 discriminator: NO. [E] the leading higher-curvature term IS the R^2 scalaron ($a_4 = R^2/72$ v36; $M_{\text{scal}}^2/M_{\text{Pl}}^2 = c_3^7$, exp $7 = g + N - 1$) – already in TFPT, already falsifiable via $n_s = 1 - 2/N_*$, $r = 12/N_*^2$ (CMB-S4). [C] it is the entanglement-equilibrium next order (Wald first law $\rightarrow f(R) = R + R^2$, v28/v359). [E] the gap $\Delta = 6 \ln(3/2)$ is dimensionless, so a distinct gap term sits at $\xi v_{\text{geo}} \sim$ seam/Planck scale, not near-term observable. [O] DECLINE: a new dimensionless coefficient is not atom-forced (needs heat-kernel/Wald data; exp 7 vs 6, c_3^7 vs $(2/3)^6$ unrelated) $\rightarrow$ per v354/v355 declined. No new near-term forced prediction; the answer is the scalaron [E][C][O]
v361_grav_ backreaction.py	Direction 5: the matter-gravity backreaction with the explicit J3 flux – two forced facts + an honest decline. [E] the carrier's gravitational anomaly is FORCED: $c_- = 8$ (16 Majoranas = $\dim S^+ = 2^{g_{\text{car}} - 1}$; chiral central charge $16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$, the same c_- as v358). [E] the backreaction is FINITE $\rightarrow \Lambda$ from α : the matter vacuum energy through $G_{ab} = c_3^{-1}T_{ab}$ is gap-suppressed (Decoupling Thm v337) to $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ (v60), 123 orders = $119.03 + 3.92$ – NOT the M_{Pl}^4 catastrophe; the loop (v359+v60+carrier vacuum) closes. [E] J3 explicit (v358): $\delta \langle K_B \rangle = (8\pi^2 R^4/15) \delta \langle T_{00} \rangle$, carrier T_{ab} = the SM's (v159). [C] SM trace anomaly reproduced (content=SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond $c_- = 8$ and $\Lambda \rightarrow$ per v354/v355 declined. Finite & consistent backreaction, no new dial [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v364_vgeo_sharpen.py	Direction 6: v_{geo} is the SINGLE dimensionful input of the WHOLE theory – confirmed across matter AND gravity – and irreducibly primitive. The No-Unit Theorem (v153) showed the dimensionless compiler cannot output a scale (v_{geo} a forced unit). SHARPENING after the parameter-free gravity (v358/v359/v361): the gravity sector adds NO new scale either $-1/c_3=8\pi$ and the Λ prefactor $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\text{geo}}^2$ and $\Lambda \sim (\text{dimensionless})v_{\text{geo}}^4$ reduce to v_{geo} times atoms. [E] FINAL TALLY: every dimensionful quantity = (dimensionless ratio) $\times$ (power of v_{geo}); free content = $\{v_{\text{geo}}, \pi\}$, ZERO dimensionless dials, vs the SM's ~ 26 . [O] v_{geo} irreducibly primitive (No-Unit; $1/G$ UV-sensitive Sakharov v68). Clarity, not a new derivation; the last dimensionful anchor named [E][O]
v365_qg_oneloop_saddle.py	Direction 3: the QG measure (gate C7/QG.AMB.01) as one-loop GAUSSIAN fluctuations around the PARAMETER-FREE saddle (v358/v359). [E] the fluctuation stiffness is $1/c_3 = 8\pi$ (no free dial), so the one-loop problem is well-posed (it was NOT before v358). [E] the fluctuation operator is gapped $M \geq \Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ mode-by-mode Gaussian convergence. [E] finite model log-det $\text{Tr}' \log M = 6 \log \frac{9}{2}$ on the OS spectrum $\{0, 6 \log \frac{3}{2}, 6 \log 3\}$. [C] the conformal mode converges on the GHP contour $(\int e^{- c \phi^2} = \sqrt{2\pi/3}, \text{v334})$. [E] the admissible projective limit exists $(\chi = 1/(1 - (2/3)^6) = 729/665, \text{v330})$. [E] 0 new dimensionless dials (v364). [O] residual: the full non-perturbative projective limit (G6/QG.AMB.01) on the ambient sector. Sharpens C7 to a one-loop Gaussian problem around a parameter-free saddle; does NOT close it. [E][C][O]
v366_mmst_seam_collar.py	Direction 7: the seam collar is IN MMST's free-lattice-fermion scaling-limit class, so the chiral scaling limit is a chiral CFT of central charge 8 pinned to $(E_8)_1$ – reducing SEAM.EQUIV.01 to the single S3 lattice-realisation input. Osborne–Stottmeister (arXiv:2107.13834, CMP 398 (2023) 219; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework): free lattice fermions (CAR quasi-free) $\rightarrow$ chiral-CFT massless scaling limit with correct c , range $\text{rank} \leq c \leq D$. [E] class arithmetic: $D = \dim S^+ = 2^{g_{\text{car}}-1} = 16$ Majoranas, $c = g_{\text{car}} + N_{\text{fam}} = 8$, $\text{rank } E_8 = 8 \Rightarrow \text{rank} \leq c \leq D$ is $8 \leq 8 \leq 16$ (in MMST range). [E] gapped ($\Delta = 6 \log \frac{3}{2} > 0$, derived v302). [C] quasi-free CAR class (v155/v160). [E] chirality $c_- = D/2 = 8 \neq 0$ (non-gappable anomalous edge, v356 S4). [C] the MMST scaling-limit theorem applies (in-class & gapped & chiral & in-range) $\Rightarrow$ chiral CFT $c=8$ (v336). [E] target pinned to $(E_8)_1$ by $c=8$ + the order-4 μ_4 clock ($\det K=1$ vs $SO(16) \det K=4$, v351). [O] residual S3: the collar realised as a genuine lattice chiral free-fermion invertible phase (one-sidedness/Flat-Away, v297/v356). Verifies MMST in-class; ADVANCES SEAM.EQUIV.01, does NOT close it. [E][C][O]
v367_seam_s3_lattice.py	Track 1: the EXPLICIT lattice realisation of the S3 input – a concrete gapped chiral-Majorana ($p + ip$) lattice model whose edge is the 16-Majorana $SO(16)_1$ carrier, μ_4 -condensed to $(E_8)_1$; turns the abstract S3 residual (v356/v366) into a RUNNABLE model with a numerically computed topological invariant. [E] bulk gapped (band gap $2 > 0$ at $M=1$). [E] non-trivial Chern number (Fukui–Hatsugai–Suzuki: $ C =1$ at $M=1$, $C=0$ at $M=3$ trivial control). [E] $c_- = 8$: each chiral Majorana edge carries $c_- = C/2 = 1/2$, $N_{\text{Maj}} = \dim S^+ = 2^{g_{\text{car}}-1} = 16$ copies $\Rightarrow c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$. [E] edge K-matrix $SO(16)_1$ ($\det \text{Cartan } D_8=4$) $\rightarrow (E_8)_1$ ($\det \text{Cartan } E_8=1$) by the order-4 μ_4 clock (v351/v369). [C] the μ_4 condensation $SO(16)_1 \rightarrow (E_8)_1$ (v154/v351). [O] residual: the continuum scaling-limit existence (MMST v336). Reduces S3 to a runnable lattice model; does NOT close SEAM.EQUIV.01. [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v368_seam_s3_inflow.py	Track 1: the bulk-edge / anomaly-inflow leg (S4 of the v356 chain) made CONCRETE on the v367 lattice model – a finite STRIP of the gapped $p + ip$ model has a realised, non-gappable chiral edge, so edge EXISTENCE follows from the invertible bulk (anomaly inflow), NOT a separate assumption. [E] topo in-gap edge states ($M=1$ strip $\min_{k_x} E \sim 0$) vs trivial clean gap ($M=3$, neg control); [E] the near-zero state is edge-localised (weight within 4 sites of a boundary, edge $\gg$ bulk); [E] bulk-edge correspondence (edge branch count = bulk Chern $ C =1$) $\Rightarrow$ 16-Majorana edge, $c_- = 8$. [C] anomaly inflow: a $c_- \neq 0$ invertible bulk cannot have a gappable edge $\Rightarrow$ discharges the v336/v351 ‘does the edge exist?’ residual on the lattice. [O] residual: the continuum scaling limit (MMST v336). Establishes edge EXISTENCE on the lattice; does NOT close SEAM.EQUIV.01. [E][C][O]
v369_qgamb_redundancy.py	Track 2 (next-plan v2): the AMBIENT REDUNDANCY statement – the holographic route to Gravity-complete that SIDESTEPS building the general Euclidean QG measure (C7/QG.AMB.01). Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators proving the ambient sector carries NO physical d.o.f. relative to the holomorphic boundary algebra, so QG.AMB.01 is a non-fundamental CERTIFICATION object (like a coordinate choice in GR), not missing dynamics. [E] HOLOMORPHIC $\Rightarrow$ DHR = Vec: #DHR primaries = $ \det \text{Cartan} = 1$ for E_8 (ONE primary = vacuum) vs 4 for $SO(16)/D_8$ – no nontrivial superselection charges $\Rightarrow$ no hidden boundary sector a bulk could carry invisibly. [E] NO TORUS GSD: ground-state degeneracy on $T^2 = \#\text{anyons} = \det K = 1$ for $E_8 \Rightarrow$ unique ground state $\Rightarrow$ no topologically protected invisible bulk d.o.f. (the genus-1 obstruction of v344). [E] FINITE PETZ RECOVERY: the deviation subspace contracts at $(2/3)^6 = 64/729 < 1$ (v221) $\Rightarrow$ Rec_Σ bounded. [E] GAP-DECOUPLING MARGIN: $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ the ambient sector is energetically separated. [C] RECONSTRUCTION MAP: Bisognano–Wichmann gives the modular boost (v258/v309/v323) $\Rightarrow \text{Rec}_\Sigma : (\text{bulk}) \rightarrow \mathcal{A}_\Sigma^{\mu_4}$. [C] THE REDUNDANCY STATEMENT: under SEAM.EQUIV.01 + BW, $\mathcal{O}_{\text{phys}}^{\text{bulk}}/\text{Diff} \cong \mathcal{A}_\Sigma^{\mu_4} \Rightarrow$ QG.AMB.01 carries no extra physical d.o.f. [O] residual: the two consumed premises – SEAM.EQUIV.01 (the S3 lattice input, v366) + the BW intrinsicity of the raw collar (v329); provides the ALTERNATIVE (holographic) route, does NOT construct the measure. Gravity-complete = Boundary-complete + Ambient-redundancy [E][C][O]
v370_grav_spin2_unitarity.py	Track 2: the Barnes–Rivers spin decomposition of the graviton propagator – perturbative graviton unitarity SECTOR-BY-SECTOR (extends v304’s scalar pole algebra to the tensor spin structure + combines with v334’s spin-0 conformal mode). [E] projector completeness: $\text{tr } P_2 = (d-2)(d+1)/2$, $\text{tr } P_1 = d-1$, $\text{tr } P_{0s} = \text{tr } P_{0w} = 1$ sum to $d(d+1)/2$; the $d=4$ split is $5+3+1+1=10$. [E] EH structure: spin-2 coefficient +1, spin-0 scalar $-1/(d-2) = -1/2$. [E] the Stelle ghost is a SPIN-2 state (a local $1+p^2/M^2$ dressing puts the negative-residue pole at $p^2 = -M^2$). [E] the entire KMS form factor e^{p^2/M^2} (v304/v259) makes spin-2 ghost-free (only the $p^2=0$ pole). [C] the wrong-sign spin-0 conformal mode ($c_{\text{conf}}(4) = -3/2$, v332) is cured by the GHP contour (v334) – a different mechanism. [C] $\Rightarrow$ perturbative graviton unitarity sector-by-sector, conditional on v304’s entire-analyticity assumption. [O] perturbative-only + the open analyticity assumption; NOT the non-perturbative measure (QG.AMB.01). [O] decline: $\text{tr } P_2 = 5$ is dimension counting, NOT g_{car} (v354/v355). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v371_ftransfer_pole.py	Track 3 (F_{transfer}): the Koide source→pole transfer as a typed solver – promotes the experiments/ftransfer solve into the ledger with a kill test (does NOT re-prove 53/54 v183 or $Q^*=2/3$ v82/v101). [E] $Q_{\text{pole}}=\text{Koide(PDG poles)}=2/3$ to 3×10^{-6} , $Q_{\text{src}}(\varphi_0^{\text{ret-ladder}})=0.66446$ (0.33% below). [E] QED runs the wrong way: $dQ/d\epsilon > 0$ pushes Q ABOVE 2/3; the ϵ mapping $Q_{\text{pole}} \rightarrow Q_{\text{src}}$ is NEGATIVE while QED gives $\epsilon=+(3/2)(\alpha/\pi) > 0$ – wrong sign for any $\alpha \in [1/137, 1/128]$. [C] verdict: F_{pole} is NOT standard QED running; the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). Stays [C], plain running excluded as the kill. [E][C]
v372_ftransfer_boltzmann.py	Track 3 (F_{transfer}): η_B from the INTEGRATED Buchmueller–Di Bari–Pluemacher Boltzmann network at the TFPT-frozen heavy scale – promotes the experiments/ftransfer solve, replacing the v326 toy washout. Frozen $M_1=M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ ($\sim 8.65\times 10^9$ GeV, v212), $\tilde{m}_1=m_3/A_\Lambda=5$ meV, $\delta_{CP}=4\pi/3$. [E] integrated efficiency $\kappa_f \sim 0.09$ (strong washout $K \sim 4.6$) agrees with the BDP fit to $\sim 25\%$. [C] $\eta_B(\text{ODE}) \sim 6.5\times 10^{-10}$ at the frozen M_1 lands within a factor ~ 1.1 of the observed 6.1×10^{-10} , NO M_R dial. [X] kill: η_B outside [obs/3, 3 obs] for all seesaw-consistent M_R would drop the leptogenesis branch. Stays [C] (a transfer target). [E][C][X]
v373_ftransfer_relic.py	Track 3 (F_{transfer}): the axion relic from the FINITE-T misalignment ODE – promotes the experiments/ftransfer solve and DECIDES the two TFPT angle branches, replacing the v326 adiabatic toy. $f_a=M_{\text{scal}}/128 \sim 2.4\times 10^{11}$ GeV, $m_a \sim 23.8 \mu\text{eV}$, lattice $\chi(T)$ $n=8.16$. [E] candidate scales fixed by the determinant-line atoms ($128=2 \dim S^+ \mu_4 $). [C] the frozen SPINE angle $3\pi/5=108^\circ=\pi N_{\text{fam}}/g_{\text{car}}$ gives $\Omega_a h^2 \sim 0.12$ in $[0.08, 0.16]$ (lands on Ω_{DM} untuned) – the robust dominant-DM branch. [X] the HILLTOP 170° over-produces ($\sim 5.4\times$) – disfavoured. [X] kill: a finite-T solve robustly outside $[0.08, 0.16]$ for the spine across $\chi(T)/g_*$ drops the dominant axion-DM branch. Stays [C]/[X]. [E][C][X]
v374_ftransfer_qcd.py	Track 3 (F_{transfer}): the m_p/m_e transfer as a typed uncertainty-budget + kill test – the falsifiability companion to v262 (the precise 2-loop point), replacing the v326 toy (which hardcodes 1836.15). [E] $b_3 = -(11 - 2n_f/3) = -7$ ($n_f=6$ carrier). [C] two declared externals (v339): $\alpha_s(M_Z)=0.1179\pm 0.0009$ and $C_p=2.83\pm 0.15$, each $\sim 5.3\%$. [C] m_p/m_e central ~ 1824 (v262) with a propagated $\pm 7.5\%$ band [$\sim 1690, \sim 1960$] CONTAINING the observed 1836.15 – no free dial. [X] kill: a future lattice $C_p\pm 0.03$ + tighter α_s shrink the band to $\sim \pm 2\%$; a band then excluding 1836.15 drops the QCD-matching route ($b_3 = -7$ + the $SU(9)$ -integer firewall v262). Distinct from v262's point; stays [C]. [E][C][X]
v375_observatory_registry.py	Track 4: the prediction OBSERVATORY – a status-typed CI over the frozen prediction registry (freeze_file.csv) that makes the falsifiability surface machine-checkable (no new physics). [E] falsifiability complete: every registry row carries a non-empty kill criterion (no orphan claims). [E] headline values re-derive from $\{\varphi_0^{\text{ret}}\}$: $\sin^2 \theta_{12}=1/3-\varphi_0^{\text{ret}}/2$, $\sin^2 \theta_{13}=\varphi_0^{\text{ret}} e^{-5/6}$, $\beta_{\text{rad}}=\varphi_0^{\text{ret}}/(4\pi)$. [C] live scorecard: θ_{12} vs JUNO first run (0.3092 ± 0.0087) = 0.28σ (compatible); θ_{13} vs NuFIT 6.0 NO (0.02195 ± 0.00058) = 2.0σ (the flagged most-tensioned core prediction); β_{rad} vs ACT DR6 (0.215 ± 0.074) = 0.37σ (hint); $r=12/N_\star^2 \sim 0.004 < \text{BK18}$ (compatible). [E] registry integrity: the single machine-readable falsifiability surface. Tooling for the closed/open pieces. [E][C]
v376_seam_s3_centralcharge.py	S3 closure stack: the central charge $c=8$ extracted NUMERICALLY from the lattice via Calabrese–Cardy finite-size entanglement scaling of the critical free-fermion content (the quantitative core of the MMST scaling-limit claim, on OUR model). [E] one critical complex mode gives $c=1.0000$ (Peschel correlation-matrix EE); [E] the collar is 16 Majoranas = 8 complex modes (v367), c additive $\Rightarrow c=8=g_{\text{car}}+N_{\text{fam}}$; [E] chiral $c_-=8$ (Chern $ C =1$ per Majorana v367, non-gappable edge v368). [O] the specific net $(E_8)_1$ is v377/v378; the continuum existence is the cited MMST theorem (v336). Numerical $c=8$ from the lattice; does NOT close SEAM.EQUIV.01. [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v377_seam_s3_e8character.py	S3 closure stack: the chiral $c=8$ content is the SPECIFIC net $(E_8)_1$, via the exact affine character q -series + the μ_4 /GSO promotion. [E] $\chi = E_4/\eta^8 = q^{-1/3}(1+248q+4124q^2+\dots)$: leading $-c/24 = -1/3 \Rightarrow c=8$; level-1 $= 248 = \dim E_8$ (one primary). [E] the μ_4 /GSO promotion $SO(16)_1 \rightarrow (E_8)_1$: $248=240+8$, $240 = \dim S^+ g_{\text{car}} N_{\text{fam}} = 16 \cdot 5 \cdot 3$; $248=120+128 = \dim SO(16) + \text{spinor}$. [E] discriminator: $(E_8)_1$ has 1 primary/248 currents vs $SO(16)_1$ 4 primaries/120. [O] continuum existence (v336); the μ_4 projection = the seam one-sidedness (S3). Pins the target net; does NOT close SEAM.EQUIV.01. [E][O]
v378_seam_s3_modular.py	S3 closure stack: the GENUS-1 discriminator – torus ground-state degeneracy = 1 (holomorphic = $(E_8)_1$), the “hard part” v344 flagged (genus-0 unique vacuum is necessary not sufficient). [E] KLM condensation: $\mu(\text{carrier}) = \det C(D_5) \det C(A_3) = 4 \cdot 4 = 16$; the order-4 μ_4 glue ($ L =4$) $\Rightarrow \mu(E_8) = 16/16 = 1$ (torus GSD = 1); index-2 $\Rightarrow \mu(SO16) = 16/4 = 4$. [E] $ \det C(E_8) = 1$ vs $ \det C(D_8) = 4$. [E] the single $(E_8)_1$ character has modular T -eigenvalue $e^{-2\pi i/3}$ ($h=0, c=8$), 1-dim rep $\Rightarrow$ one character $\Rightarrow$ torus GSD = 1. [O] continuum existence (v336). Pins the net at genus-1; does NOT close SEAM.EQUIV.01. [E][O]
v379_seam_s3_rp.py	C7 / OS reconstruction: reflection positivity (the OS axiom) of the explicit gapped collar measure, verified numerically – the remaining constructive-QFT ingredient for the OS reconstruction. [E] the gapped $p+ip$ band gives $G(\tau) = \text{mean}_k e^{-\varepsilon_k \tau}$ with $\varepsilon_k \geq \text{gap}/2 \sim 1 > 0$ (v367) – a positive spectral measure; [E] the Hankel/reflection Gram matrix $M_{ij} = G(\tau_i + \tau_j)$ is PSD – OS reflection positivity holds; [E] neg control: a growing ghost mode makes it indefinite (non-vacuous). [C] RP + gap (v76/v337) + GNS (v240) + $c=8$ (v376) + $(E_8)_1$ (v377/v378) give the OS inputs; C7/QG.AMB.01 is then DISCHARGED by the redundancy theorem v369, not built. [O] the full constructive measure stays the cited step. [E][C][O]
v380_grav_kms_hessian.py	The KMS Entire Hessian – the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity (upgrades v304/v370 from “assume entire” to a derived statement). [E] the seam KMS cutoff $f(u) = e^{-u}$ (v259) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u) = e^u$ entire & nowhere zero $\Rightarrow$ only the $p^2=0$ pole (no ghost). [E] each finite truncation carries the ghost: the $R+R^2$ order $T_1=1+u$ has the Stelle zero $u=-1$, the Weyl ² order $T_2=1+u+u^2/2$ the pair $-1 \pm i$; the entire a has none. [E] the ghost DECOUPLES: nearest zero of T_n grows monotonically with n (1, 1.41, $\dots$, 3.07 for $n=1..8$) $\rightarrow \infty$, so “entire” = “do not truncate”. [C] $\Rightarrow$ perturbative graviton unitarity. [O] the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation; perturbative-only, not the non-perturbative measure. [E][C][O]
v381_qft4d_eg_allorder.py	The all-order Epstein–Glaser / BRST contract for the perturbative 4D S-matrix S_{pert} – the “all-order closing statement” for the 4D perturbative QFT leg. Types the standard causal-perturbation-theory closure for the TFPT finite interaction class; it does <i>not</i> build a path integral. [E] POWER-COUNTING: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator $\Rightarrow$ the EG singular order $\omega \leq 4 \Rightarrow$ a FINITE counterterm space at every order. [E] BRST NILPOTENCY $s^2=0$ verified via the Jacobi identity for $su(2)$ (residual 0) and $su(3)$ (10^{-16}), the carrier weak+colour content. [E] the seam gap $\Delta = 6 \ln(3/2) > 0$ (v302) gives the IR-safe adiabatic limit $g \rightarrow 1$. [C] imported: T_n exists to all orders by causal factorisation (Epstein–Glaser 1973 / Brunetti–Fredenhagen). [C] imported: all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) $\Rightarrow$ Slavnov–Taylor to all orders. [E] scope fence: matter+gauge ONLY – the R^2/Weyl^2 gravity sub-sector is v304/v370/v380, not a local EG vertex; S_{pert} is not the ambient measure QG.AMB.01 (v369). NET [C] (imported EG+BRST on the TFPT dim-4 class + gap), prerequisites [E]; extends v269/v271/v273/v278 from 1-loop to all-order [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v382_alpha_quillen_exact.py	Names the Quillen determinant-line VARIATION $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1 c_3^6 \log \varphi_{\text{seam}}) = 0$ as a tracked target (review Point 10), elevating EM.WARD.01's residual to a citable [O]. [E] the Quillen split holds at the root $\alpha^3 - 2c_3^3 \alpha^2 = 8b_1 c_3^6 \ln(1/\varphi_{\text{seam}})$ at $\alpha^{-1}=137.0359992$ ($F_{U(1)}=0$ IS the stationarity of the $U(1)$ determinant line, v341). [E] every coefficient a named atom: $8b_1=\frac{4}{5}\cdot 41$ a $U(1)$ index, $c_3=1/(8\pi)$ Gauss–Bonnet ($c_3^{3,6}$ heat-kernel powers), $q(D_5)+q(A_3)=2$ discriminant forms (v341/v48/v216). [E] the value $\alpha^{-1}=137.0359992168$ stays closed (unique root + ablation, v3) – only “why this functional” is open. [O] THE TARGET: the from-first-principles AQFT/ ζ proof that $F_{U(1)}$ is the EXACT Quillen functional. [C] a FACE of SEAM.EQUIV.01 (v336): the seam $U(1)$ determinant line lives on the same holomorphic seam net, so once the net is $(E_8)_1$ its $U(1)$ sub-determinant is fixed – not a second independent gate, it names/tracks the pre-existing EM.WARD.01 residual (v341/v342). No new number (reuses v341, Wolfram-mirrored) [E][C][O]
v383_dynamics_universal.py	DYNAMICS.UNIVERSAL.01: the UNIVERSAL spectral-gap / Perron–Frobenius principle – every TFPT sector is the SAME object (a gapped operator with a UNIQUE leading attractor = the physics, a spectral gap = the reason there is no free parameter), extending v303 from F_{transfer} to the whole theory. [E] FLAVOR (computed): the cusp transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ has a simple Perron leading 1 and gap $6 \ln(3/2) > 0$ (v56/v82). [E] DISCRETE COMPILER (computed): the affine- E_8 adjacency has Perron eigenvalue 2 (Kac marks, v312/v313) and subleading $2 \cos(\pi/5) = \varphi$, gap $2 - \varphi = 1/\varphi^2 > 0$. [E] DECOUPLING MARGIN: $\Delta_{\text{eff}} = 6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v337). [E] TWO NUMBER-FIELD FACETS (v314): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2\cdot 3\cdot 5 = h(E_8)$ – a factorisation, not an equality. [C] the other sectors are the same structure: gravity (v358/v359), QG saddle (v365), the holomorphic $c=8$ RG fixed point (v157/v158), the all-order S-matrix (v381), recovery (v221), F_{transfer} (v303), $\tau=i$ (v333). [C] META-THEOREM: gapped $\Rightarrow$ unique attractor $\Rightarrow$ parameter-free, theory-wide – one spectral-gap statement wearing many hats. A synthesis, no new number [E][C]
v384_residual_certification.py	RESIDUAL.CERTIFICATION.01: the residual matrix is CERTIFICATION, not CONSTRUCTION – after v369/v381/v382 there is NO open TFPT physics MECHANISM left; every residual is an external CERTIFICATE of already-determined structure, of three kinds: (A) external math proof, (B) theorem-forbidden, (C) external physics. A ledger-CI audit. [E] LEDGER CI: every residual claim id is present (no orphan). [E] NO-UNIT (B): $c_3=1/(8\pi)$ and $g_{\text{car}}=5$ are mass-dimension 0, so by No-Unit (v153/v364) no dimensionless compiler outputs an absolute scale $\Rightarrow v_{\text{geo}}$ is forbidden by theorem, not a gap. [E] GAP-DECOUPLED (A): $\Delta_{\text{eff}} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ every readout is independent of the ambient measure (QG.AMB.01 a [C] redundancy, v369). [E] CLASSIFICATION COMPLETE: SEAM.EQUIV.01 (continuum existence = cited MMST/Adamo, v336) $\rightarrow$ A, ALPHA.QUILLEN.EXACT.01 (v382) $\rightarrow$ A, QG.AMB.01 (v369) $\rightarrow$ A, $v_{\text{geo}} \rightarrow$ B, F_{transfer} (v371–v374, firewall v187) $\rightarrow$ C; ZERO open internal TFPT mechanisms. [C] CONVERGENCE: the only hard things left are theorems and external physics, not missing dynamics. An audit, no new number [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v385_uvbranch_killtest.py	UVBRANCH.KILLTEST.01: the optional carrier–Pati–Salam UV branch on the ALL-ORDER footing – a single citable kill-test surface with the proton-decay SAFETY hierarchy computed (not a fake τ_p window). A consolidation (does <i>not</i> re-run the RGE v246 nor re-derive the PS scale v249/v253). [E] carrier content = SM ($b = (41/10, -19/6, -7)$, v159/v246, no new states) and S_{pert} closes to all orders (v381) $\Rightarrow$ the SM-non-unification is all-order-robust. [X] plain-SM 4D-GUT killed ($\sin^2 \theta_W = 3/8$ vs 0.231; v246). [C] optional carrier-PS branch: $M_{PS} = \kappa M_{\text{scal}}$, $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}} \approx 3.06 \times 10^{13}$ GeV, $\kappa \in [1.0, 1.2]$ (v253); content $\{1, 10, 16, 45\}$, no 126 (v247/v248); not the default (v265). [E] proton-decay SAFETY hierarchy: minimal PS gauge bosons are $SU(4)$ leptoquarks ($B-L$ -conserving $q \leftrightarrow l$, rare LFV $K_L \rightarrow \mu e$), <i>not</i> the $p \rightarrow e^+ \pi^0$ dim-6 operator (needs an $SO(10)/SU(5)$ diquark, absent); the binding bound $M_{PS} \gtrsim 10^6$ GeV is cleared by $M_{PS}/M_{\text{bound}} \sim 10^7 \gg 1$. [X] frozen kill tests (v265): κ leaving $O(1)$ / a forbidden rep / a new gauge-charged state / a rare-LFV signal below the bound; no fake τ_p . A consolidation, no new number [E][C][X]
v386_grav_amplitude.py	GRAV.AMPLITUDE.01: the entire-form-factor graviton-exchange AMPLITUDE is finite, UV-softened and tree-unitary – perturbative gravity as an explicit scattering problem, extending v304/v370/v380 from the propagator to the amplitude. The dressed spin-2 propagator $P = e^{-p^2/M^2}/p^2 = 1/(p^2 a)$, $a(u) = e^u$, $M = M_{\text{scal}}$. [E] one pole, ghost-free: a entire & nowhere zero (v380) $\Rightarrow$ only pole at $p^2 = 0$, residue $e^0 = 1 > 0$. [E] UV-softened: ratio e^{-u} at $u = 1, 5, 10, 20 = 0.37, 6.7 \times 10^{-3}, 4.5 \times 10^{-5}, 2.1 \times 10^{-9}$ (faster than any power). [E] low-energy = GR + a calculable $-p^2/M^2$ correction ($M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$, v253). [C] tree unitarity: a single positive-residue pole, no negative-norm states (v370/v380). [C] the amplitude $A = (\kappa^2/2)(T \cdot P_2 \cdot T') e^{-p^2/M^2}/p^2$ is GR at low energy, softened in the UV (perturbative-only; QG.AMB.01 a [C] redundancy v369). No new number [E][C]
v387_corrections_gap.py	CORRECTIONS.GAP.01: the practical harvest of v383 – the same spectral gap that makes each sector parameter-free ALSO sets the SIZE of its first correction ($\text{correction}_n \sim (\lambda_2/\lambda_1)^n$). [E] FLAVOR: $\lambda_2 = (2/3)^6 = 64/729 \Rightarrow$ first Koide correction ~ 0.088 (v56/v82). [E] RECOVERY: the same $(2/3)^6$ Petz rate (v221). [E] QG: capped by $\chi = 1/(1 - (2/3)^6) = 729/665$ (v337). [E] DISCRETE COMPILER: the update $(A + 2I)/4$ fixes the Kac marks ($\lambda_1 = 1$) and decays at the golden $(\varphi + 2)/4 \approx 0.905$ (carrier-5 facet, v303/v312). [E] the rates split into the $(2/3)^6$ family (family-3, in $\mathbb{Q}$) and the golden family (carrier-5, in $\mathbb{Q}(\sqrt{5})$) – the two facets of v314/v383. [C] the correction calculus: the gap does double duty (WHY no free parameter AND HOW BIG the first correction), so sub-leading magnitudes are organised, not free. No new number [E][C]
v388_corrections_budget.py	CORRECTIONS.BUDGET.01: the gap correction (v387) is a <i>typed</i> budget, NOT a uniform band on every prediction – the $(\lambda_2/\lambda_1)^n$ size is a property of the distance from a gapped operator’s leading eigenvector, so it is only defined where λ_2 exists. The prediction surface splits into four correction classes (already implied by v303). [E] SEAM-GAPPED: Koide (F_{pole} v303), recovery (v221), the QG bound (capped by $\chi = 729/665$, v337) carry $(2/3)^6$; the discrete compiler decays at the golden $(\varphi + 2)/4$ (v312). [E] KOIDE FIRST CORRECTION: F_{pole} is the ONE seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), source quotient 0.33% below $2/3$ – the rate is the suppression SCALE, not the residual. [E] EXACT-IDENTITY class ($\det R = 8$, $N_\Phi = 1$, $\theta_{\text{eff}} = 0$, $\sin^2 \theta_{13}$): no $\lambda_2 \Rightarrow$ correction 0 STRUCTURALLY (a band would contradict [E]). [E] EXTERNAL-RATE class ($\eta_B/\text{axion}/m_p, m_e$): the gapped shape with external rates (only F_{pole} has the seam rate, v303), firewalled (v187). [E] FIXED-POINT class: the texture is already explicit ($\theta_{12} \varepsilon = c_3 + 36c_3^4$). [C] a uniform band on all predictions is WRONG; the size is keyed to the mechanism class. A typing of v387, no new number [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v389_grav_loop_finiteness.py	GRAV.LOOP.FINITE.01: the entire-form-factor graviton is UV-finite at the <i>loop</i> level by power counting – the honest next step past v386 (tree), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence. Honest scope: the standard infinite-derivative/nonlocal-gravity finiteness statement, NOT a full proof of perturbative QG. [E] super-polynomial falloff $p^k e^{-p^2/M^2} \rightarrow 0$ for every k . [E] one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$; the GR $\int p^3/p^2 dp$ diverges quadratically. [E] the superficial degree $\rightarrow -\infty$ at every loop order (dressed cutoff-ratio $\rightarrow 1$ vs GR $\rightarrow 4$). [E] the regulator is the SEAM scale $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ (v253/v259), not a dial. [C] loop UNITARITY of the nonlocal form factor (Lorentzian continuation / macro-causality) stays the honest open point – tree unitarity is v386; NOT claimed solved. No new number [E][C][O]
v390_prime2_facet.py	COXETER.PRIME2.01: the prime-2 facet of the order-30 Coxeter clock, completing v383. The clock $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ had two named facets (golden $\varphi \in \mathbb{Q}(\sqrt{5})$, prime-5/carrier; $(2/3)^6 \in \mathbb{Q}$, prime-3/family); this names the prime-2 facet as the Gaussian field $\mathbb{Q}(i) - \tau = i$ and the order-4 $\mu_4 = \mathbb{Z}/4$ square seam deck (v282/v288/v333), structure already present. [E] $h(E_8) = 30$ factors into exactly three primes. [E] one quadratic facet field per prime, each ramified at its prime: disc $\mathbb{Q}(i) = -4$, disc $\mathbb{Q}(\sqrt{-3}) = -3$, disc $\mathbb{Q}(\sqrt{5}) = 5$; ramified primes $\{2, 3, 5\}$, product = 30. [E] prime-2 Gaussian facet: min. poly $x^2 + 1$, $\tau = i$, $j(i) = 1728 = 12^3$, $ \mu_4 = 4 = 2^2$ (the static seam-orientation facet). [E] prime-5 golden (v383): $2 \cos(\pi/5) = \varphi$ (pentagon). [E] prime-3 family (v383/v220): $(2/3)^6$ plus the Eisenstein ζ_3 (hexagonal CP $\pi/3$). [C] completion: each prime carries a facet (square/hexagon/pentagon = seam/CP/carrier), the prime set forced by $h(E_8)$. A synthesis, no new number [E][C]
v391_alpha_quillen_progress.py	ALPHA.QUILLEN.PROGRESS.01: an <i>honest attempt</i> at the external target ALPHA.QUILLEN.EXACT.01 (v382) – it REDUCES the open step to a sharper named sub-target via a solvable model but does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384). [E] solvable model $\det_\zeta(-\partial_\tau^2 + m^2)$ on a circle $= (2 \sinh(\beta m/2))^2$, m -variation $\beta \coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$ a closed heat-kernel/Laurent series (m^3 coeff $-1/360$ verified). [E] a ζ -det variation is structured heat-kernel data, so v382's $8b_1c_3^6$ counterterm (a Gauss–Bonnet heat-kernel power) is the right <i>kind</i> of object. [E] the v382 split re-verified at $\alpha^{-1} = 137.0359992$. [E] reduction: the open step reduces to the named sub-target “seam Seeley–DeWitt coefficient $= 8b_1c_3^6$ ”. [O] NOT CLOSED: the actual seam ζ -det proof stays open, SEAM.EQUIV.01 continuum existence (v336) the other external [O]; α^{-1} stays [E]. An attempt that reduces, does not close [E][O]
v392_seam_s3_scalinglimit.py	SEAM.S3.SCALINGLIMIT.01: an <i>honest attack</i> on the one open lemma of SEAM.EQUIV.01 (the continuum scaling-limit existence, v336) – it does NOT close it (the abstract existence is the cited MMST theorem); it STRENGTHENS the case with the existence-relevant observable v376 did not test: that the finite-size data <i>converge</i> . [E] the Calabrese–Cardy EE slope $c_1(L)$ at $L = 66 \dots 514$ is a monotone Cauchy-like sequence approaching 1 ($ c_1(L) - 1 $ strictly decreasing). [E] a $1/L$ Richardson fit gives $c_1(\infty) \approx 1.000$, so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\text{car}} + N_{\text{fam}}$ is reached <i>in</i> the limit. [E] the successive differences are strictly decreasing (numerical Cauchy, necessary for existence). [O] convergent data are evidence, not a proof of existence (MMST v336), and do not pin $(E_8)_1$ vs $SO(16)_1$ (holomorphy v377/v378). Strengthens the residual, does NOT close it [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v393_corrections_numeric.py	CORRECTIONS.NUMERIC.01: the typed correction budget (v388) turned into ACTUAL first-correction magnitudes – a computed theoretical-uncertainty surface, the short-term testable harvest of v388. [E] FIXED-POINT band $(\theta_{12}/\theta_{13}/\lambda_C/\beta_{\text{rad}}) =$ the explicit φ_0 puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; $= \theta_{12}$'s ε first-correction). [E] SEAM-GAPPED band (Koide/recovery) $= (2/3)^6 \approx 8.78\%$; the QG bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337). [E] EXACT-IDENTITY band $= 0$ structurally ($\det R = 8$, $N_\Phi = 1$, $\theta_{\text{eff}} = 0$). [E] EXTERNAL-RATE band = external physics quoted ($m_p/m_e \pm 7.5\%$ v374; $n_s/r N_\star$ band; $\eta_B 1.07 \times$ v212), fenced v187. [C] each prediction gets “central value $\pm$ computed first correction”. A harvest of v388, no new number [E][C]
v394_coxeter_atoms.py	COXETER.ATOMS.01: the clockwork coherence – the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets, and the prime-2 seam is the common dynamical factor. A bird's-eye [E] re-reading tying v390 (facets) to the anchor $a = (1, 1, 2)$ and v319 (5x6 clock); registers RES.COXETER.SYMMETRY.01 as a tracked [O]. [E] $a = (1, 1, 2)$: $e_3=2= Z_2 $, $p_0=3=N_{\text{fam}}$, $e_2=5=g_{\text{car}}$, product $30=h(E_8)$. [E] atoms = facet fields (v390): $2 \rightarrow \mathbb{Q}(i)$, $3 \rightarrow \mathbb{Q}(\sqrt{-3})$, $5 \rightarrow \mathbb{Q}(\sqrt{5})$, each ramified at its prime. [E] $\chi=729/665=3^6/(3^6-2^6)$, $(2/3)^6=2^6/3^6$, exponent $6=2N_{\text{fam}}$. [E] both dynamic rates carry prime-2: $2/3= Z_2 /N_{\text{fam}}$, $(\varphi+2)/4=(\varphi+2)/ \mu_4 $, $ \mu_4 =4=2^2$. [E] $30=5 \times 6=g_{\text{car}}(2N_{\text{fam}})$. [C] the seam is the universal dynamic engine. [O] RES.COXETER.SYMMETRY.01 registered, not closed [E][C][O]
v395_hypergraph_coupled.py	HYP.COUPLED.01: the coupled hypergraph rewrite – one local rule unifies v299/v327/v324 (carrier $\times$ family). [E] LOCAL COUPLED MICRO-RULE: state on 9 affine- E_8 nodes $\times$ 3 family slots; one micro-step = network lazy diffusion on each cusp column + v327 M on each node's family vector; steps commute, $T_{\text{micro}} = T_{\text{net}} \otimes M$. [E] JOINT ATTRACTOR: top eigenvalue 1 with eigenvector $\text{marks} \otimes e_0$; iteration from any positive start converges. [E] ONE HAND = RECOVERY: one network step + $2N_{\text{fam}}$ family micro-steps gives $(2/3)^6$ on $\text{marks} \otimes v_{\text{surv}}$. [E] v324 EMERGES: $T_{\text{full}} = T_{\text{net}} \otimes T_{\text{cusp}}$ shares the same invariants; T_{cusp} is the 6-hand readout in the Galois/cusp basis (v317). [E] GROWTH + FIBER: v299 $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ unchanged with M -fiber attached. [O] 3-arm seed $(P_2) + \varphi_0$ (v312) remain irreducible [E][O]
v396_phi0_icosahedral.py	HYP.PHI0.01: the φ_0 leading term from icosahedral combinatorics (sharpening v312/v395). [E] ICO COUNTS ANCHOR-FORCED: $V=12$, $E=30=h(E_8)$, $F=20=g_{\text{car}}(g_{\text{car}}-1)$, $ \text{Aut} =120=4h= 2I $. [E] LEADING TERM EXACT: $\varphi_0^{\text{ree}}=1/(6\pi)=(4/3)c_3=F/(4h\pi)=F/(\text{Aut} \pi)=(F/(g_{\text{car}}N_{\text{fam}}))c_3$. [E] $F/h= Z_2 /N_{\text{fam}}$; leading term $= (F/h)/(4\pi)=c_3(g_{\text{car}}-1)/N_{\text{fam}}$. [E] GAUSS-BONNET: icosahedron deficit sum 4π ; $c_3=1/(Z_2 4\pi)$ (v216/v342). [E] HONEST NEGATIVE: puncture $48c_3^4$ not a rational hypergraph fraction. [C] tree-level seed = (hyperedges)/($g_{\text{car}}N_{\text{fam}}$) $\times c_3$, not an independent injection; puncture [O]
v397_external_clock_probe.py	FR.CLOCK.PROBE.01: a <i>falsifiable</i> probe of how external the “external physics” really is – do the frontier scheme-numbers land on the $\{2, 3, 5\}$ clock (the anchor atoms, v394) or not? Strict No-Free-Pattern (v100): “on the clock” = an EXACT match, not a wide-band fit. [E] η_B decuple $A_\Lambda=10= Z_2 g_{\text{car}}$ ON CLOCK (v212). [E] Koide rate $(2/3)^6=(Z_2 /N_{\text{fam}})^6$ ON CLOCK (v303). [E] axion spine angle $3\pi/5=\pi N_{\text{fam}}/g_{\text{car}}$ ON CLOCK (v211). [E] proton $C_p \sim 2.83 \pm 0.15$ NOT discriminating ($2^{3/2}$, $17/6$, $14/5$ all fit the band) $\Rightarrow$ EXTERNAL (firewall v187/v374). [C] 3 of 4 land exactly on the clock; the irreducible external residue shrinks to the proton QCD number C_p – “external physics” was overstated. [E] anti-numerology: refuses to assign C_p a clock value [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v398_seam_state_rigidity.py	Seam State Rigidity Theorem (Paper A, the G_{net} closure contract): the scattered state-invariance reductions (v177/v198/v199/v308/v309/v335) as ONE named target. [E] RP-definability hinge (OS canonical transfer $T=e^{-H}$ from RP + reflection alone, v54, on a quasi-free state v155, no μ_4 normal form); [E] band-limited core $[\rho, k]=0$ and character-block-diagonal sub-principal symbol swept $N=4.64$ (beyond the 3-dim H^1); [E] holomorphy $\Rightarrow (E_8)_1$ ($c=8$, $ \det \text{Cartan} =1$ vs 4, index $4= \mu_4 $, gap $6 \log \frac{3}{2} > 0$). [O] the one residual: $[\rho, \Lambda_\Sigma]=0$ on the full L^2 (intrinsic Bisognano–Wichmann). A contract+evidence module, NOT a closure of SEAM.EQUIV.01 (continuum = cited MMST v336)
v399_gravity_complete.py	Gravity-complete = Boundary-complete + Ambient-redundancy (Paper B accounting): combines the parameter-free local equation (v358/v359) and the ambient redundancy (v369). [E] $G_{ab} + \Lambda g_{ab} = (1/c_3)T_{ab}$, $1/c_3=8\pi$ (four c_3 -roles), Λ prefactor $3/(4\pi^2)$; [E] redundancy discriminators ($ \det \text{Cartan} =1$ vs 4, torus GSD=1, Petz $(2/3)^6$, margin $1.648 > 0$); [E] $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$ over five readout classes (Newton 8π , Λ , scalaron c_3^7 , horizon $1/ \mu_4 $, recovery $(2/3)^6$) – all boundary/seam/gap. [C] the verdict; [O] SEAM.EQUIV.01 + intrinsic BW (no new gate)
v400_grav_nonlocal_action.py	Nonlocal spectral-gravity action with the local $R+R^2$ as its IR projection (Paper C, consolidating v304/v370/v380/v386). [E] the entire form factor $a=e^u$ gives the dressed propagator $1/(p^2 a)$ ONE pole (residue 1, ghost-free); [E] every truncation carries a Stelle zero whose nearest-zero modulus grows monotonically (v380); [E] the IR order $a=1+u+u^2/2+\dots$ reproduces the spectral-action $R+R^2$ (v28/v36) with the atom-fixed scale $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$ ($7=g_{\text{car}}+N_{\text{fam}}-1$, v253). [O] the entire- a analyticity assumption + perturbative-only (not the ambient measure QG.AMB.01, a [C] redundancy v369)
v401_metrology_closure.py	Metrology closure of the final input set $\{a, \pi, v_{\text{geo}}\}$ (Paper D, consolidating v153/v274/v364/v384). [E] $a=(1, 1, 2) \rightarrow (e_1, e_2, e_3)=(4, 5, 2)=(\mu_4 , g_{\text{car}}, Z_2)$, $c_3=1/(2e_1\pi)=1/(8\pi)$, 0 dimensionless dials + 1 unit; [E] dimensional bookkeeping (No-Unit v153) mass/ $1G/\Lambda/U_{\text{point}}/(m/\mu)=\text{number} \times v_{\text{geo}}^{1,2,4,1,0}$; [E] gravity adds no unit (v364); [E] G -vs- Λ over-determination 0.11% (v274). [O] v_{geo} theorem-forbidden ($\text{TOE}_{\text{strict}}=\text{TOE}_{\text{dimensionless}}+[v_{\text{geo}}]$, v384 type B)
v402_ftransfer_suite.py	F_{transfer} functor suite (Paper E): the four frontier sectors as ONE functor $F_{\text{transfer}}=F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ (consolidating FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE). [E] the composition law (each observable factors source $\rightarrow$ threshold $\rightarrow$ RG $\rightarrow$ observable); [E] exact atom sources (Q_{src} v93; $A_\Lambda=2g_{\text{car}}=10$, $\delta_{CP}=4\pi/3$ v9/v212; $f_a=M_{\text{scal}}/128$ v185; $b_3=-7$ v164/v262); [E] a committed kill test per sector. [C] the firewall: physical predictions stay [C]/[O], only the algebraic source is [E] (never re-pressed into the compiler, v187)
v403_facet_compositum.py	ARITH.HULL.01: the three number-field facets (v390) COMPOSE to $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$, a degree-8 = rank E_8 abelian field – the arithmetic hull dual to the E_8 lattice hull. [E] the squarefree parts $-1, -3, 5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so $[K : \mathbb{Q}] = 2^3 = 8 = \text{rank } E_8 = g_{\text{car}} + N_{\text{fam}}$, Galois group $(\mathbb{Z}/2)^3$ order 8; $2^{\omega(h(E_8))} = 2^3 = 8 = \text{rank } E_8$ is E_8 -special (the E_7 control $h=18$, $2^2=4 \neq 7$ fails). [E] exactly 7 quadratic subfields $\{-1, -3, 5, 3, -5, -15, 15\}$, split 4 imaginary = $ \mu_4 $ (CM/dynamic) + 3 real = N_{fam} (RM/static). [E] ramified only at $\{2, 3, 5\}$ = the atoms = the primes of $h(E_8)=30$. [C] K is the abelian $(\mathbb{Z}/2)^3$ arithmetic hull dual to E_8 ; the 7 quadratic subfields a candidate for the 7 E_8 audit slices [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v404_e8_char_cm_duality.py	E8.CMDUAL.01: the $(E_8)_1$ vacuum character $\chi_{E_8}=E_4/\eta^8=j^{1/3}$ at the two CM points – 12 at the seam ($\tau=i$), 0 at the family ($\tau=\omega$), extending v282. [E] $\chi_{E_8}^3=j$ (q-series). [E] SEAM $\tau=i$ ($\mathbb{Q}(i)$, order 4): cross-ratio $2 \Rightarrow j=1728 \Rightarrow \chi_{E_8}(i)=1728^{1/3}=12$ (E_8 present, v214/v282). [E] FAMILY $\tau=\omega$ ($\mathbb{Q}(\sqrt{-3})$, order 6): the equianharmonic λ ($\lambda^2-\lambda+1=0$) $\Rightarrow j=0 \Rightarrow \chi_{E_8}(\omega)=0$ (E_4 vanishes, v220/v267). [E] $\tau=i, \tau=\omega$ are the ONLY elliptic points of $\text{PSL}(2, \mathbb{Z})$, so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where E_8 is present ($\chi=12$, SEAM.EQUIV.01 closes), the flavor/CP sector is the soft residual where E_8 degenerates ($\chi=0$); the dual-keystone candidate (flavor = the $j=0$ degeneration of $(E_8)_1$) [E][C]
v405_seam_equiv_omega.py	SEAM.EQUIV.02 – the dual keystone at $\tau=\omega$: the family/flavor sector as the order-3 Eisenstein/ A_2 face of E_8 , dual to the seam = $(E_8)_1$ at $\tau=i$ (v404). A named contract (genre of v286/v398), NOT a closure. [E] $\chi_{E_8}(\omega)=0$ (E_8 degenerates at the family CM point) vs $\chi_{E_8}(i)=12$. [E] A_2 is the $\tau=\omega$ Eisenstein lattice (Cartan $\det=3=N_{\text{fam}}$, $h=3=N_{\text{fam}}$, $\mu_6=\mu_3 \times \mu_2$). [E] R lives in $E_6 \times A_2$: $248=\ R\ _F^2 + \det R + 2(1^T R a)N_{\text{fam}}=78+8+2 \cdot 27 \cdot 3$, $\ R\ _F^2=78=\dim E_6$, $\det R=8=\dim A_2=\text{rank } E_8$. [E] $\delta_{\text{PMNS}}=4\pi/3$ is the $\tau=\omega$ hexagonal CM phase ($\arg \zeta_6=\pi/3$, v220/v233). [E] the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck = $N_{\text{fam}} = \det Q$ (v72), $\text{Spec}(Q_+)=(1, 2, 3)$ (v69/v137). [C] the family/flavor sector is the order-3 Eisenstein/ A_2 deck at $\tau=\omega$; the two keystones sit at the only two elliptic points of $\text{PSL}(2, \mathbb{Z})$. [O] residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the v_{geo} -class scale (No-Unit v153) [E][C][O]
v406_pmns_phase_origin.py	FLAV.PMNS.CLOSE.01: the PMNS dynamics closed modulo the $\tau=\omega$ keystone (v405) – the CP-phase ORIGIN is the family CM point and the absolute scale is one v_{geo} -class unit, so PMNS has zero free flavor dials beyond v_{geo} . [E] the three angles ($\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$, $\theta_{23}=45^\circ$, $\sin^2 \theta_{13}=\varphi_0^{\text{ret}} e^{-5/6}$; v9/v268) assemble to one unitary U_{PMNS} (v270). [C] CP-PHASE ORIGIN (advance over v270): $\delta_{\text{PMNS}}=4\pi/3$ is the $\tau=\omega$ hexagonal CM phase (v405/v220/v233), upgrading v270's [O] assembled input to a keystone consequence. [C] $J_{\text{PMNS}}=-0.0297$ derived. [E] the absolute scale = one seesaw ratio $m_3=(y_\nu v_{\text{EW}})^2/M_R$ = a v_{geo} -class UV input (v272/v153), NOT a flavor gap. [C] net: PMNS = angles + δ [keystone] + J [derived] + scale [v_{geo}], zero free flavor dials. [O] residual = the seesaw operator realisation (M_R not a clean $\bar{M}_{\text{Pl}}$ power, v272/v184; one-parameter carrier-normalised candidate v481) + the normal-ordering Σm_ν floor kill test [E][C][O]
v407_dn_pairings_omega.py	FLAV.SELECTOR.CLOSE.01: the $R4'$ residue folds into the $\tau=\omega$ keystone (v405) – the three (d, n) selector pairings ARE the $\tau=\omega$ family-slice atoms. [E] the frame $(\mathbf{1}, a, \sigma)$ is a $\mathbb{Z}^3$ basis with $\det=11=\ Pl(K)\ _1$, and $n=(5, -9, 6)$ is the unique covector with $n \cdot \mathbf{1}=2$, $n \cdot a=8$, $n \cdot \sigma=121=11^2 \Rightarrow (d, n) \Rightarrow R$ columnwise (v136/v139). [E] $n \cdot a=8=\text{rank } E_8=\dim A_2=\det R$ – the anchor-pairing is the A_2 ($\tau=\omega$) block dimension that reads R in $E_6 \times A_2$ (v405). [E] the pairings $\{2, 8, 121\}=\{ Z_2 , \text{rank } E_8=\dim A_2=\det R, \ Pl(K)\ _1^2\}$ are the $\tau=\omega$ family-slice atoms (the μ_2 sheet, the A_2 block, the Plucker frame). [E] the square selects the point (σ -pairing $11(11+t)$, a perfect square only at $t=0$). [C] deriving (d, n) ($R4'$) folds into SEAM.EQUIV.02 – $d=\frac{3}{2}a-21$ anchor-derived (v139), the pairings the slice atoms. [O] the lone flagged residue “why $11=\ Pl(K)\ _1$ ” [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v408_phi0_puncture_heatkernel.py	HYP.PHI0.PUNCTURE.01: name the φ_0 puncture term $\delta_{\text{top}}=48c_3^4$ as the order-4 local boundary heat-kernel (Seeley-DeWitt) contact term of the μ_4 puncture divisor – the analytic complement of the icosahedral tree term (v396, HYP.PHI0.01). [E] THE SPLIT IS EXACT: $\varphi_0=\varphi_0^{\text{tree}}+\delta_{\text{top}}=1/(6\pi)+48c_3^4$ (v396/ARCH.QUAD.01). [E] PUNCTURE COUNT = $\Omega_{\text{adm}}=N_{\text{fam}} \dim S^+=48$, order = $ \mu_4 =4$: $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^{ \mu_4 }$. [E] CLOSED FORM $\delta_{\text{top}}=48c_3^4=3/(256\pi^4)$; $c_3=1/(Z_2 4\pi)$ Gauss-Bonnet (v216/v342) so c_3^4 is a fourth-order boundary heat-kernel power (cf. v391). [E] HONEST NEGATIVE: δ_{top} not a rational hypergraph fraction ($\delta_{\text{top}}/\varphi_0^{\text{tree}} < 1\%$). [O] THE TARGET: prove $\Omega_{\text{adm}}c_3^4$ IS the order-4 boundary Seeley-DeWitt contact term of the μ_4 puncture summed over the Ω_{adm} admissible states. [C] two engines, two terms; a FACE of the seam boundary analysis (the c_3 datum), not a new gate [E][C][O]
v409_coxeter_prime2_lemma.py	RES.COXETER.SYMMETRY.01 closed as a STRUCTURAL LEMMA: prime-2 is NECESSARY but NOT autonomous – no prime-2-only attractor exists (resolves the v394/COXETER.ATOMS.01 open question's falsifiable corollary). [E] SHEET INVOLUTION $\sigma=\text{diag}(1, -1, -1)$, $\sigma^2=I$, spectrum $\{+1, -1\}$ (parity). [E] SHEET PROJECTOR $P=(I+\sigma)/2=\text{diag}(1, 0, 0)$, $P^2=P$, spectrum $\{0, 1\}$ (projection). [E] NO CONTRACTION RATE: none of $\{-1, 0, 1\}$ lies in $(0, 1)$, whereas the genuine rates $2/3$ and $(1/3)^6$ do – prime-2 alone gives parity/projection, never a nonzero rate. [E] FACTOR NOT RATE: $2/3= Z_2 /N_{\text{fam}}$ needs prime-3, $(\varphi+2)/4=(\varphi+2)/ \mu_4 $ with $ \mu_4 =2^2$ needs the golden φ (prime-5). [E] $30=g_{\text{car}}(2N_{\text{fam}})$, carrier static, 2·3 dynamic. [C] prime-2 is the universal dynamical parity factor, necessary in every sheet-paired transport but not an autonomous attractor; dynamics begins after coupling to family-3 or carrier-5 ($\mu_4=2^2$ the Galois bridge) [E][C]
v410_sheet_generator_binary.py	The <i>sheet generator</i> $V=Q \text{diag}(0, 1, 1)$ (rank-2 sheet axis, v95/v218) is a binary internal compiler: $V^n=\begin{pmatrix} 0 & 2^{n-1} & 0 \\ 0 & 2^n & 0 \\ 0 & 2^{n+1}-2 & 1 \end{pmatrix}$ (induction), so $V^n \mathbf{1}=(2^{n-1}, 2^n, 2^{n+1}-1)=(1, 2, 3), (2, 4, 7), (4, 8, 15), (8, 16, 31)$; bilinears $\mathbf{1}^T V^n \mathbf{1}=7 \cdot 2^{n-1}-1$, $\mathbf{1}^T V^n a=7 \cdot 2^{n-1}$, $a^T V^n a=11 \cdot 2^{n-1}$, $a^T V^n \mathbf{1}=11 \cdot 2^{n-1}-2$ give $(6, 7, 9, 11)$ then $13=\Delta_Q$, $27=\mathbf{1}^T R a$, $55, 56=\dim \mathbf{56}_{E_7}$; the spine is FORCED by $\text{Spec}(V)=\{0, 1, 2\}$, the carrier (rank $E_8=8$, $\dim S^+=16$, $248/8=31$) and theta $(\sigma_3(2)=9=a^T V \mathbf{1}, \sigma_3(3)=28=\mathbf{1}^T V^3 a=\det(I+R), \sigma_3(5)=126)$ readings audit-typed [E][C]
v411_ud_ratio_vpower.py	The quark u/d ratio is a pure V -power readout: the established $c_u/c_d=g_{\text{car}}\ \text{Pl}(K)\ _1/(N_{\text{fam}}^2 \Delta_Q)=55/117$ (v94) re-expresses EXACTLY via the sheet generator (v410) as $\boxed{c_u/c_d = \frac{\mathbf{1}^T V^4 \mathbf{1}}{(a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})}} = \frac{55}{9 \cdot 13}$, with $55=\mathbf{1}^T V^4 \mathbf{1}$, $9=a^T V \mathbf{1}=N_{\text{fam}}^2$, $13=\mathbf{1}^T V^2 \mathbf{1}=\Delta_Q$; the same number as $5 \cdot 11/(9 \cdot 13)$, so the numerator repackages $g_{\text{car}} \cdot 11=55$. An exact <i>re-encoding</i> : the value and the u/d identification stay [C] (U _{wall} readout rigidity) [E][C]
v412_sheet_source_corner_J.py	The <i>sheet source corner</i> J : the determinant surface has two s -independent iso-volume walls (v135) $t=-1$ ($\det=4= \mu_4 $, carries K) and $t=-2$ ($\det=2= Z_2 $); the operator on the Z_2 wall, $J=M(1, -2)=C-2V=\begin{pmatrix} 4 & 1 & 0 \\ 5 & 1 & 1 \end{pmatrix}$, completes the sheet axis $J, K, C, F=M(1, -2..1)$. $\chi_J=\lambda^3-6\lambda^2+3\lambda-2 \Rightarrow (\text{tr}, e_2, \det)=(6, 3, 2)=(R^+(A_3) , N_{\text{fam}}, Z_2)$; $a^T J a=30=h(E_8)$, $\det(I+J)=12=\dim \mathfrak{g}_{\text{SM}}=\chi_{E_8}(i)$, $\det(2I+J)=40= R(D_5) $ (Lie readings audit-typed like the G_2/F_4 shadows of v95/v218) [E][C]
v413_sheet_characteristic_calculus.py	The sheet axis is a discrete <i>characteristic-difference calculator</i> : on $M_t=C+tV$ (v95/v218) the elementary-symmetric coefficients encode atoms as DIFFERENCE ORDERS – $e_1=3t+12$, $e_2=2t^2+15t+25$, $e_3=\det M_t=4t^2+14t+14$ give $\Delta e_1=3=N_{\text{fam}}$, $\Delta^2 e_2=4= \mu_4 $, $\Delta^2 e_3=8=\text{rank } E_8$ (the e_3 curvature cross-cited to v218/v224; the full (e_1, e_2, e_3) ladder is new); the anchor energy $a^T M_t a=52+11t$ (step $11=a^T V a$, v410) reads $30 \rightarrow 41 \rightarrow 52 \rightarrow 63=(h(E_8), 10b_1, \dim F_4, 7N_{\text{fam}}^2)$ [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v414_center_resolvent_portal.py	The center C is a <i>resolvent portal</i> $G_2 \rightarrow F_4 \rightarrow E_8$: $C=M(1,0)$ (v95) is the $t=0$ value of V ; $\text{tr } C=12=\dim \mathfrak{g}_{\text{SM}}=\chi_{E_8}(i)$, $\det C=14=\dim G_2$, $\sum C=31=2^{g_{\text{car}}}-1=248/8$; the shifted determinants climb $\det(C)=14=\dim G_2$, $\det(I+C)=52=\dim F_4$, $\det(2I+C)=120= R^+(E_8) $ – the shell ladder $G_2 \rightarrow F_4 \rightarrow E_8^+$ as $\det(C+kI)$, $k=0,1,2$ (the resolvent ladder is new; reading audit-typed) [E][C]
v415_gaussian_operator.py	The square CM-norm dictionary is <i>operator-realised</i> and the μ_4 deck is an intrinsic diamond object: the Gaussian arithmetic of v222/v230 lifted from numbers to OPERATORS via $J=[U,V]/3$, the complex structure born from the commutator of the binary V and ternary U compilers (v410/v95). [E] $J=\begin{pmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix}$ is integer, $\text{Spec } J=\{i, -i, 0\}$, $J^2=-I$ on the rank-2 image, $J^3+J=0$ (a μ_4 quarter-turn); [E] $3I+2J$ has spectrum $\{3+2i, 3-2i, 3\}$ and $5I+4J$ has $\{5+4i, 5-4i, 5\}$, so $3+2i$ ($ \cdot ^2=13=\Delta_Q$) and $5+4i$ ($ \cdot ^2=41=10b_1$) are literal operator eigenvalues; [E] $(3I+2J)(3I-2J)$ has $\text{Spec}\{13, 13, 9\}$ and $(5I+4J)(5I-4J)$ has $\{41, 41, 25\}$ (kernel slot = real part ² : $9=N_{\text{fam}}^2$, $25=g_{\text{car}}^2$); [E] the intrinsic deck $D=-I+J-J^2$ is order 4 ($\text{Spec}\{i, -1, -i\}$, $\chi=(x+1)(x^2+1)$) with χ_2 line $\ker[U,V]=(0,0,1)=a-1$. [C] the geometric-deck identification sharpens (does not close) the v140 $\mathbb{Z}_3$ residue [E][C]
v416_atom_trichotomy.py	The atom trichotomy: $\{2, 3, 5\}$ are ramified/inert/split in the two exceptional CM rings, and each atom is the unique ramified prime of one quadratic facet of $K=\mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ (v403). [E] $\mathbb{Z}[i]$ (square/seam, $\tau=i$): $2= Z_2 $ ramifies ($2=-i(1+i)^2$), $3=N_{\text{fam}}$ inert (norm $9=N_{\text{fam}}^2$), $5=g_{\text{car}}$ splits $((2+i)(2-i))$ – the carrier rank factors on the seam; [E] $\mathbb{Z}[\omega]$ (hex/flavor, $\tau=\omega$): $3=N_{\text{fam}}$ ramifies ($N(1-\omega)=3$), 2 inert (norm 4), 5 inert (norm $25=g_{\text{car}}^2$); [E] the ramified prime of each ring IS its atom ($\mathbb{Z}[i]$ at 2= sheet/seam, $\mathbb{Z}[\omega]$ at 3= family/flavor); [E] the three facets $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$ (disc $-4, -3, 5$) ramify over one atom each, product $2\cdot3\cdot5=30=h(E_8)$; 2 imaginary (CM/dynamic) + 1 real (RM/static). [C] the seam/flavor/golden reading [E][C]
v417_eisenstein_cp_operator.py	The Eisenstein/CP operator (the hex dual of v415): the family rotation $P=(1\ 2\ 3)$ and the CP clock $W=-P^2$ put the $\tau=\omega$ flavor side on operators, unified with the seam into ONE order-30 clock. [E] $P^3=I$, $P^2+P+I=\text{ONES}$ (Eisenstein $\omega^2+\omega+1=0$ on the plane $\perp (1,1,1)$); [E] $(3I+2P)(3I+2P^2)=7I+6\text{ONES}$, $\text{Spec}\{7, 7, 25\}$ ($7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ = scalaron on the ω -plane, $25=g_{\text{car}}^2$) – the dual of v415's $\{13, 13, 9\}$; [E] the CP clock $W=-P^2$ ($=\rho$, v233) has $W^2=P$, $W^3=-I$, $\text{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$, so $\delta_{\text{CKM,lead}}=\arg \zeta_6=\pi/3$ and $\delta_{\text{PMNS}}=\arg \zeta_6^4=4\pi/3$ ($W^4=P^2$); [E] all flavor clocks are powers of ζ_{30} ($\zeta_{30}^{15}=-1$, $\zeta_{30}^{10}=\omega$, $\zeta_{30}^6=\zeta_5$, $\zeta_{30}^5=\zeta_6$) while the seam $\mu_4=i$ is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30)=8=\text{rank } E_8$. [C] the carrier μ_5 has no clean 3×3 operator (golden φ an output-side trace, v313/v349); honest negatives included [E][C]
v418_cyclotomic_norm_triple.py	The cyclotomic norm triple: the carrier split $(3,2)$ read in the three atom-rings gives $(7, 13, 55)=(\text{scalaron}, \Delta_Q, \text{quark numerator})$, and the missing carrier-5 clock of v417 is the 4×4 Φ_5 -companion C_5 . [E] C_5 has $\chi=x^4+x^3+x^2+x+1$, $C_5^5=I$, $\text{tr}=-1$; the golden ratio is its real-part data ($\zeta_5+\zeta_5^4=1/\varphi$, $\zeta_5^2+\zeta_5^3=-\varphi$; output-side v313/v349; $\dim 4$, not 3 – refines v417's gap); [E] $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5)=\det(3I+2C_5)=55=5\cdot11=g_{\text{car}}$ $\ \text{Pl}(K) \ _1=$ the quark numerator (v410/v411); [E] the triple $\det(3I+2\text{Comp}(\Phi_n))=(7, 13, 55)$ for $n=3, 4, 5$ over $\mathbb{Z}[\omega], \mathbb{Z}[i], \mathbb{Z}[\zeta_5]$ (atoms 3, 2, 5, v416); [E] NEG control: the anchor $(5,4)\rightarrow(21, 41, 461)$, with $21=N_\omega(5,4)$ and $41=10b_1$ named (v222) but the carrier ring 461 prime/un-named, so the carrier ring separates $(3,2)\rightarrow55$ from $(5,4)\rightarrow461$. [C] HONEST: $(3,2)$ is FORCED (v14) not unique – the clean splits are $\{(2,1)\rightarrow(3,5,11), (3,2)\rightarrow(7,13,55)\}$, a small ladder; the sharper reading: only $(3,2)$ spans three physics sectors (gravity/flavor/masses), the unique cross-sector seed [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v419_seam_galois_carrier.py	The seam μ_4 is the carrier's Galois group: a positive resolution of the v409/RES.COXETER.SYMMETRY.01 cyclic/Galois asymmetry. [E] $h(E_8)=30=2\cdot3\cdot5$ is squarefree, so $\mathbb{Z}/30$ has <i>no</i> order-4 element — the seam μ_4 cannot be a cyclic hand and is forced onto the Galois side; [E] $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$ (order $\varphi(30)=8=\text{rank } E_8$), where the μ_4 $(\mathbb{Z}/4)$ factor IS $(\mathbb{Z}/5)^\times$ (carrier 5) and the $\mathbb{Z}_2$ IS $(\mathbb{Z}/3)^\times$ (family/CP); [E] so the seam $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5)) = (\mathbb{Z}/5)^\times$, the Frobenius $\zeta_5 \mapsto \zeta_5^2$, realised on the carrier clock C_5 (v418) by the explicit order-4 operator G with $GC_5G^{-1}=C_5^2$, $G^4=I$, $G^2C_5G^{-2}=C_5^{-1}$. [C] resolves v409 (the $2^2=4$ is the order of $(\mathbb{Z}/5)^\times$; “no contraction rate” because the seam is an automorphism, not a hand); $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (cyclic $h=30$, Galois rank=8). Figure coxeter_galois [E][C]
v420_fpole_qed_running.py	F_pole external: the Koide source→pole transfer is a <i>one-loop electromagnetic</i> effect, not an algebraic step — the honest external-physics follow-up to v93 and the clock No-Go. [E] Koide Q is rescaling-invariant ($Q(cm)=Q(m)$), so flavor-<i>universal</i> running leaves it fixed and the source→pole shift is purely flavor-<i>split</i>; [C] the tree→pole gap $\frac{2}{3}-Q_{\text{src}}=0.00220$ is one-loop-EM-sized ($=\varphi_0^{\text{ret}}/24$ to 0.6%, $=\alpha/\pi$ to $\sim 5\%$), the PDG pole sits AT $\frac{2}{3}$ (the IR attractor); [C] a 1-loop QED per-lepton-log pole $\leftrightarrow \overline{\text{MS}}$ estimate shifts Koide by EM-loop order toward $\frac{2}{3}$ (the v82 IR-attractor direction); [O] the EXACT flow time $t\sim 2.84$ (v93) is scheme/scale-dependent external QED, only sized. [C] F_pole stays [C], firewalled (v187): the lens forces the RATE $(2/3)^6$, this external EM supplies the flow. Numerical (mpmath), Python-only [E][C][O]
v422_galois_net_bridge.py	The Galois$\leftrightarrow$Net bridge: the seam $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$ is the SAME cyclic $\mathbb{Z}/4$ as the $(E_8)_1$ simple-current glue — not a mere order-4 coincidence — linking the Galois gearbox of v419 to G_{net}/SEAM.EQUIV.01. [E] $\text{disc}(A_3)=\text{disc}(D_5)=\mathbb{Z}/4$ (one Smith invariant factor 4 = cyclic, $\det=4$); D_n disc is $\mathbb{Z}/4$ for n ODD, $\mathbb{Z}_2 \times \mathbb{Z}_2$ for n EVEN, so the carrier D_5 (rank $5=g_{\text{car}}$, odd) is cyclic while the control D_8 (rank 8, even) is Klein; [E] $\text{Gal}(\mathbb{Q}(\zeta_5))=(\mathbb{Z}/5)^\times=\langle 2 \rangle$ is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in $\text{disc}(D_5 \oplus A_3)=\mathbb{Z}_4 \times \mathbb{Z}_4$ ($16=\dim S^+$) the simple current $(1,1)$ is order 4 (cyclic) with Lagrangian quotient $16/4^2=1 \Rightarrow (E_8)_1$ (holomorphic, v89/v154). [E] NEG control: the Klein $\mathbb{Z}_2 \times \mathbb{Z}_2$ / order-2 halfway $\langle (2,2) \rangle$ gives $16/2^2=4= \text{disc}(D_8) =SO(16)$ ($\det=4$), NOT E_8 ($\det=1$) — cyclic $\mathbb{Z}/4 \rightarrow E_8$, Klein $\rightarrow D_8$. [C] the bridge: one cyclic $\mathbb{Z}/4$ threads the seam deck μ_4 (v419), $\text{Gal}(\mathbb{Q}(\zeta_5))$ and the $(E_8)_1$ glue (v1/v89/v154); cyclicity forced by 5, Klein excluded. [C] this is the four-fold μ_4 identity (deck/divisor = Galois = discriminant = simple-current); per v428 a <i>compression</i> (one object read four ways), not four independent witnesses [E][C]
v423_no_ambient_dependency.py	The <i>No-Ambient-Dependency</i> certificate: no frozen TFPT readout is <i>computed</i> from the ambient QG measure QG.AMB.01 — sharpening the v369/v384 prose redundancy into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin $\Delta_{\text{eff}} = 6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ ($31=2^{g_{\text{car}}}-1$), susceptibility $\chi = 729/665$ finite (neg control gap $\rightarrow 0 \Rightarrow \chi \rightarrow \infty$). [E] QG.AMB.01 is a certification <i>sink</i>: every claim listing it as a dependency is a gravity-gate/QG row, none a frozen prediction. [E] it is not a value source — it has no producing script while each frozen readout is computed by a compiler script ($\alpha^{-1} \leftarrow \text{v3}$, $\theta_{12} \leftarrow \text{v9}$, $\theta_{13} \leftarrow \text{v268}$, cosmology $\leftarrow \text{v7}$). [E] the one transitive link is architectural: only θ_{13} references QG.AMB.01, via the 4D-QFT contract spine (PS.DIRAC→CONTRACT.QFT4D→QG.G2→GATE.METRIC), a coherence edge not a numerical input; α^{-1}/θ_{12}/cosmology have no route. [C] so QG.AMB.01 is gap-decoupled certification (v369 as a DAG/script fact); honestly this does NOT prove SEAM.EQUIV.01 — the readouts depend on the seam being $(E_8)_1$; QG.AMB.01 specifically is dead [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v424_seam_lto_rp_reduction.py	The (<i>LTO-RP</i>) reduction of the keystone: an honest MMST-style reduction of SEAM.EQUIV.01's one open lemma — the raw collar's intrinsic Bisognano–Wichmann property (the v308/v309 residual) — to the cited commuting-projector theorem of Naaijken–Penneys–Wallick (arXiv:2605.10693), with the central mapping <i>corrected</i> . [E] the μ_4 clock is a modular <i>symmetry</i> ($[\rho, K]=0$, v309/v199), not the flow — $\sigma^\psi=\rho$ is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26's axiom (LTO-RP) means $u_\Theta=J$ (reflection = modular conjugation) = the intrinsic BW condition $\theta K \theta = -K$, <i>distinct</i> from $\omega \circ \rho = \omega$: a positive character-block-diagonal K satisfies $[\rho, K]=0$ yet never $\theta K \theta = -K$, so clock-invariance does <i>not</i> imply (LTO-RP). [O] make-or-break: NPW26 prove (LTO-RP) for <i>topologically ordered</i> models (Toric Code $ \det K =4$ via a trace; Levin–Wen anyonic), while the seam is invertible ($ \det \text{Cartan}(E_8) =1$, no anyons) and a $\beta=1$ KMS state (not a trace) — in neither proved bucket (neg control D_8 $ \det =4$). [C] so SEAM.EQUIV.01's BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$ -reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, not a closure; SEAM.EQUIV.01 stays [O] [E][C]
v425_dyn_transfer_universal.py	DYN.TRANSFER.UNIVERSAL.01 — the <i>single-flow reduction</i> of F_{transfer} : the four frontier transfers (F_{pole} , $F_{\text{Boltzmann}}$, F_{relic} , F_{QCD}) do not each need a separate external solver; their <i>dynamics</i> is the ONE native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap $6 \ln(3/2)$) restricted to each sector. [E] recovery spectrum $\{1, (2/3)^6, (1/3)^6\}$, $H_{OS} = -\log T$ gap $6 \ln(3/2)$, $e^{-\Delta} = (2/3)^6$ exact; an explicit column-stochastic T makes $\log T$ a Markov generator. [E] F_{pole} is this flow: the v99 generator $\dot{q} = (\Delta/N_{\text{fam}})(q-2)(q-5)$ linearised at $q=2$ has rate $-\Delta$, time-1 contraction $(2/3)^6$. [E] $F_{\text{Boltzmann}}$ washout = that contraction, CP phase $4\pi/3$ native (v320). [E] F_{QCD} rate native ($b_3=-7$). [C] only the anchors (v_{geo} , C_p , M_R) external; F_{relic} 's cosmological IC θ_i the one wall. Extends v383 static→dynamic; a reduction, no new premise [E][C]
v426_seam_lto_rp_kms.py	SEAM.EQUIV.LTORP.KMS.01 — the <i>invertible</i> $\beta=1$ -KMS variant of (LTO-RP), the corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + $\beta=1$ KMS, not a trace). A constructive Tomita–Takesaki witness advancing v424 sub-step (ii); does <i>not</i> close SEAM.EQUIV.01. [E] build K reflection-odd/ μ_4 -even/gapped (gap $2k=6 \ln(3/2)$); $C=1/(1+e^K)$ the $\beta=1$ KMS covariance, $C/(1-C)=e^{-K}$ (genuine KMS, not trace). [E] $\Theta K \Theta^{-1} = -K$ ($u_\Theta=J$) AND $[\rho, K]=0$ both hold $\Rightarrow$ (LTO-RP) satisfied by direct Tomita–Takesaki. [E] $ \det \text{Cartan}(E_8) =1$ (invertible, no anyons) vs anyonic $ \det =4$; state not a trace ($\Delta=e^{-K} \neq I$) — neither NPW26 bucket. [E] neg controls: side-even reflection $\rightarrow +K$ (BW fails), off- μ_4 -block K breaks $[\rho, K]$. [C] (LTO-RP) reduces to $(\Theta K \Theta = -K) \wedge ([\rho, K]=0)$, constructively satisfiable; SEAM.EQUIV.01 stays [O] [E][C]
v427_overdet_witness_map.py	OVERDET.WITNESS.MAP.01 — the multiply-vs-compress <i>framework</i> for over-determination: witnesses from <i>disjoint</i> grammars multiply, projections of the <i>same</i> generator compress. The classification of the seven arithmetic witnesses is refined by v428 (they compress). [E] seven grammar readouts, each from its own syntax landing on the carrier skeleton: Gauss $N(3+2i)=13$, Eisenstein $N(3+2\omega)=7$, cyclotomy $N(3+2\zeta_5)=55$, Galois $ (\mathbb{Z}/5)^\times =4$, $ \det \text{Cartan}(E_8) =1$, Pascal $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 11$ ($2^4=16 = \dim S^+$), Coxeter $h(E_8)=30=2 \cdot 3 \cdot 5$ with $\varphi(30)=8 = \text{rank } E_8$. [E] by v236 these are facets of ONE $(2, 3, 5)/E_8$ object, so syntactic distinctness is NOT independence — they compress, like the anchor $a=(1, 1, 2)$. [C] the genuine multiplicative strength is the multiply-forced inputs + foreign witnesses (v428); the map does not by itself establish physics [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v428_overdet_reclass.py	OVERDET.WITNESS.RECLASS.01 — the honest correction of v427: applying its own multiply-vs-compress test to itself. [E] the $(2, 3, 5)$ Brieskorn singularity (v236) is the ONE generator — Milnor $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$, $30=2\cdot 3\cdot 5=h(E_8)$, $\varphi(30)=8$. [E] every one of v427's seven witnesses maps to a facet of that SAME object (primes 2, 3, 5 of 30, the clock 30, or E_8), so all seven collapse to ONE generator (distinct objects = 1) $\Rightarrow$ COMPRESSION, not seven multiplications. [E] what honestly multiplies is the input forced four independent ways — the “8” in $c_3=1/(8\pi)$ from rank E_8 , $h(D_5)=2(5-1)$, $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}\approx 137$ (prime, not in the $(2, 3, 5)$ -skeleton). [C] refines and partly walks back v427; consistent with v236/v305/v313 [E][C]
v429_axion_pentagon_phi.py	DM.AXION.PENTAGON.01 — the ‘unmapped’ icosahedral/golden E_8 structure surfaces in the ONE external cosmological input θ_i , answering “why does idle E_8 structure exist?” for the one case where it has a job. [E] because $N_{\text{fam}}=g_{\text{car}}-2$, the spine quotient $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=(g_{\text{car}}-2)\pi/g_{\text{car}}$ is EXACTLY the interior angle of the regular g_{car} -gon; for $g_{\text{car}}=5$ the PENTAGON, $\theta_i=3\pi/5=108^\circ$ (the carrier polygon, not a bare rational multiple of π). [E] its cosine is golden: $\cos(3\pi/5)=(1-\sqrt{5})/4=-1/(2\varphi)$, partner $2\cos(2\pi/5)=1/\varphi=\varphi-1$, $\varphi=2\cos(\pi/5)=(1+\sqrt{5})/2$. [E] the golden character is UNIQUE to $g_{\text{car}}=5$: among small n -gons only $n=5$ has an irrational interior-angle cosine ($n=3, 4, 6$ give $1/2, 0, -1/2$), so θ_i is golden BECAUSE the carrier is 5 (P2). [C] a re-reading bridging v354/v313 (the unmapped $g_{\text{car}}=5$ golden signature) to v211 ($\theta_i=3\pi/5$, the one external input): it REFINES v354 from “ φ in no readout” to “ φ touches only the [C] spine misalignment angle”, a geometric motive for the spine over the hilltop $\sim 170^\circ$; it does NOT upgrade DM.AXION.SPINE.01 (stays [C]) and closes no gate. [E][C]
v430_other_side_reverse_audit.py	E8.OTHERSIDE.AUDIT.01: the ‘other side of the seam’ (the double-cover deck / conjugate sheet S^-) audited against E_8 's UNMAPPED Casimir degrees — the sheet/deck complement of v354/v355, answering “is the leftover where the second sheet lives?” with a disciplined NEGATIVE. [E] DECK = DEGREE-2: the one-sided cover contributes $ \mathbb{Z}_2 =2$ (the $1/2$ in $c_3=1/(2\cdot 4\pi)=1/(8\pi)$, v58/v216); $2=\min(E_8 \text{ degrees})$ = the metric, one of the matched primary readouts $\{2, 8, 30\}$ (v354), NOT one of the unmapped $\{12, 14, 18, 20, 24\}$. [E] TWO SHEETS = 128-SPINOR: $\dim S^+=\dim S^-=16$, the spinor block $128=\text{rank} \cdot \dim S^+=8\cdot 16=2^{\text{rank}-1}=\sum \text{deg}$, the spinor half of $248=120+128$; S^- the conjugate $(\overline{16}, \overline{4})$, $128=2\cdot 64$, not a spare singlet. [E] FORCED-DISJOINT: the sheet/deck integer set $\{2, 16, 32, 128\}$ has EMPTY intersection with $\{12, 14, 18, 20, 24\}$, its only degree-alphabet contact the matched 2. [O] DISCIPLINED DECLINE: each unmapped degree needs an UNFORCED coefficient from the sheet atoms and admits ≥ 2 readings (the v355 discriminator) — no forced sheet $\rightarrow$ unmapped-degree identity. [E] ‘ $S^-=\text{DM}$ ’ DOWNGRADED + WIMP NO-GO: v227 replaced the over-strong ‘ S^- is dark matter’; $128=(16, 4)+(\overline{16}, \overline{4})$ is the SAME matter spinor (no spare singlet), DM is the determinant-line axion (a phase). [O] the only live ‘other side’ question (the 128 phase/glue dark sector, v227) concerns MATCHED structure, not the unmapped degrees [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v431_e8_degree_ladder.py	E8.DEGREE.LADDER.01: the reverse-audit ‘unmapped’ E_8 Casimir degrees are NOT diffuse overhead but the forced two-family decomposition $\deg(E_8)=6\cdot\text{spine}\{2, 3, 4, 5\} \sqcup (\{2\} \cup \det\text{-ladder}\{8, 14, 20\})$ — the structural complement of v354/v355/v430. [E] TWO-FAMILY SPLIT FORCED BY $30=2\cdot3\cdot5$: the E_8 exponents are the $\varphi(30)=8$ totatives of $h=30=2N_{\text{fam}}g_{\text{car}}$ (coprime to 6 $\Rightarrow \pm 1 \bmod 6$), so the degrees occupy ONLY the residue classes $\{0, 2\} \bmod 6$. [E] $6k$ family $\{12, 18, 24, 30\}/6=\{2, 3, 4, 5\}$ = the v91 spine $\{e_3, e_1, e_2, p_0\}(a=(1, 1, 2))$. [E] $6k+2$ family $\{2, 8, 14, 20\}$: $\{8, 14, 20\}=(\det R, \det C, \det L)=$ the winding line $\det M(s, 0)=6s+8$ of the flavor diamond (v135), $C=R+Q \text{ diag}(1, 0, 0)$. [E] ‘$18=6N_{\text{fam}}$’ is NOT a holdout (spine family, not on the det-ladder). [E] SPECIAL TO E_8 among the simply-laced exceptionals ($E_6/E_7/D_5$ fail the clean $\{0, 2\} \bmod 6$ split). [C] HONEST v355 LINE: the arithmetic decomposition is exact, but the FUNCTORIAL claim ‘$6k+2$ Casimir stratum = flavor sheet operators’ needs a representation-theoretic map ([C]); 12, 24 admit two canonical readings each, so this is a STRUCTURE theorem, not a forced functorial flavor map — it reclassifies v354 overhead to zero orphan degrees, closes no gate [E][C]
v432_overdet_floor.py	OVERDET.FLOOR.01: the UNCONDITIONAL improbability floor + the $L1 \leftrightarrow L2$ bridge — the defensible number that survives even if a skeptic REJECTS the declared formula grammar, complementing the conditional null model (v100) and the multiply-vs-compress framework (v427/v428). [E] DISJOINT PIECES (v428): compression removed (v427’s seven readouts are one $(2, 3, 5)/E_8$ object, NOT multiplied); what multiplies is the input ‘8’ forced FOUR ways $\{\text{rank } E_8, h(D_5)=2(g_{\text{car}}-1), \varphi(30), \text{Milnor}(2, 3, 5)\}$ plus the FOREIGN witness $\alpha^{-1} \sim 137$ (prime, outside $(2, 3, 5)$). [C] CONSERVATIVE FLOOR: multiplying ONLY across disjoint pieces (input over-determination $\times$ the α grammar census $1/94500$ (v100) $\times$ the φ_0^{ret}-seed cluster counted ONCE $\times$ one downstream witness) the floor stays $\leq 10^{-6}$ across a grid of generous assignments, nominal $\sim 10^{-10}$ — ‘extraordinary’ WITHOUT the declared grammar. [E] THE BRIDGE: the floor ($\sim 10^{-10}$) sits ~ 20 orders ABOVE the conditional $10^{-30.7}$ (v100); that gap = the evidential value of proving the FORMS forced, so L1 (ALPHA.QUILLEN.EXACT.01, v382) and L2 (this floor) are the SAME lever. [C] HONEST SCOPE: bounds ONLY “is it numerology?”, NOT the physics bridge (firewall v187); a conservative FLOOR, not a posterior [E][C]
v433_alpha_quillen_heatkernel.py	ALPHA.QUILLEN.PROGRESS.02: a <i>second honest step</i> on the external target ALPHA.QUILLEN.EXACT.01 (v382) – it GROUNDS and CONNECTS, it does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384), α^{-1} stays [E]. [E] a solvable 4D model (flat $\Delta = -\partial^2 + m^2$ on T^4) reaches the a_4 (t^0) conformal-anomaly order: $a_4 = Vm^4/(2(4\pi)^2)$ NON-ZERO – the order v391’s 1D circle toy (a_0, a_1 only) could not reach. [E] it is the order v342 grounds (Gilkey a_4 gauge-curvature $30/360 = 1/12$), unifying v391 (a ζ-det variation IS heat-kernel data) with v342 (the term IS the Gilkey a_4). [E] $(4\pi)^{-2} = 4c_3^2$ is c_3-built. [E] the v382 split re-verified at $\alpha^{-1} = 137.0359992$. [O] NOT CLOSED: the actual seam a_4 ($= 8b_1c_3^6$) and the cubic α^3 Chern moment stay open (v384); the exact seam F-normalisation stays [C] (v342). Grounds/connects, does not close [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v434_alpha_quillen_betafunction.py	ALPHA.QUILLEN.PROGRESS.03: a <i>third honest step</i> on ALPHA.QUILLEN.EXACT.01 (v382) – it settles the three residuals named after v433 and shows they are NOT three independent problems; the target stays [O], α^{-1} stays [E]. [E] b_1 IS THE U(1) a_4 HEAT-KERNEL COEFFICIENT (the $\beta=a_4$ theorem): from the carrier hypercharge content with spin weights ($\frac{2}{3}$ Weyl, $\frac{1}{3}$ complex scalar), $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM), $(3/5) \cdot \text{that} = 41/10 = b_1$ (GUT) – the b_1 in $8b_1c_3^6$ is the a_4 content coefficient, not a fitted multiplicity (same number as the β, v159/v246; same KIND as the α^2 Calderón a_4, v342). [E] $8b_1c_3^6 = b_1 \cdot (\text{rank } E_8) \cdot c_3^6$ (rank 8 = $\chi=2$ boundary, v216; c_3^6 six insertions, v342). [C] residual (1) “exact $a_4 = 8b_1c_3^6$” $\equiv$ residual (3) “seam F-normalisation” – ONE obligation. [E] residual (2) located: $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$, so a^3 is the first moment of the Chern density (a level, order c_3^0); origin stays [O]. [E] net reduction: the EM-Ward residual is $1[C]+1[O]$, not three. [O] NOT CLOSED; α^{-1} stays [E]. Python-only [E][C][O]
v435_alpha_quillen_chernlevel.py	ALPHA.QUILLEN.PROGRESS.04: a <i>fourth honest step</i> on ALPHA.QUILLEN.EXACT.01 (v382) – it attacks the SINGLE remaining [O] after v434 (residual (2), the cubic α^3 Chern level) and SHARPENS it, it does NOT close it; the target stays [O], α^{-1} stays [E]. [E] π-POWER / METRIC-INDEPENDENCE TEST: the three closure coefficients carry π-powers exactly $\{0, 3, 6\} = c_3$-orders ($k_0 = 1 \rightarrow \pi^0$; $-2c_3^3 = -1/(256\pi^3) \rightarrow \pi^3$; $-8b_1c_3^6 = -41/(327680\pi^6) \rightarrow \pi^6$), so the Maxwell a^3 is the UNIQUE π^0/c_3^0 (metric-INDEPENDENT) term – the signature of a topological level (a heat-kernel boundary coeff always carries the $(4\pi)^{-d/2}$ measure; a level carries none). The test partitions the closure into ONE topological level (a^3) + TWO heat-kernel boundary terms, confirming v342/v434’s “a^3 is a level, not a curvature integral” from a test. [C] conditional on the CS/η reading (EM1, v48), large-gauge invariance forces the level in $\mathbb{Z}$ and $k_0 = 1$ is the unit (minimal) level – an integer, not a free real. [O] NOT CLOSED: the EM1 CS-boundary derivation stays external (type A, v384); seam F-norm stays the one [C] (v434); α^{-1} stays [E]. [E][C][O]
v436_overdet_floor2.py	OVERDET.FLOOR.02: HARDENS OVERDET.FLOOR.01 (v432) — the “is it numerology?” floor does NOT rest on v432’s subjective chance assignments. [E] ASSUMPTION-MINIMAL FLOOR: the α cubic-class census (v100 layer D) is PURE COUNTING — of $N=94500$ complexity-matched $F_{U(1)}$ variants only ONE hits the 4×10^{-8} CODATA window (the TFPT instance, $\alpha^{-1}=137.0359992$), so $p_\alpha \leq 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$ with NO subjective probability, prior or multi-observable grammar. [E] MONOTONE CONCESSION LADDER: $-\log_{10} p$ full scorecard+$\alpha \sim 30.7 \rightarrow$ scorecard-only $\sim 25.8 \rightarrow \alpha$-census-only ~ 4.98 ($= 4.40\sigma$); the “not chance” verdict holds at the MOST adversarial rung. [E] INDEPENDENCE: strip ALL of v432’s subjective factors ($0.1^3 \times 0.012 \times 0.15 = 1.8 \times 10^{-6}$) and the floor is still 4.40σ — they only sharpen it. [C] HONEST 5σ GAP: 5σ ($p=5.7 \times 10^{-7}$) is crossed by the census times exactly ONE more independent factor ≤ 0.054 (the input-‘8’ four-fold over-determination or one foreign witness), stated openly. [C] FIREWALL: bounds numerology only, not the physics bridge (v187); Python-only, deliberately NOT Wolfram-mirrored (a counting bound, not an identity). [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v437_e8_degree_joint.py	E8.DEGREE.JOINT.01: a CONSOLIDATION of v6/v66/v355/v431 deriving more from the E_8 Casimir degrees WITHOUT crossing the v354/v355 anti-numerology line — no new per-degree coincidence, only a joint lock. [E] E_8 IS FIXED BY ITS DEGREE MULTISSET: from $\deg(E_8)=\{2, 8, 12, 14, 18, 20, 24, 30\}$ alone — $\text{rank}=\#\deg=8$, $h=\max=30$, $\#\text{pos roots}=\sum \exp=120$, $\#\text{roots}=240$, $\dim=248$, $W =\prod \deg=696,729,600$, $\sum \deg=128=2^7=\dim S^+$. [E] THE JOINT QUADRATIC: the two structural integers $(g_{\text{car}}, N_{\text{fam}})$ are the UNIQUE root pair of $x^2 - (\text{rank } E_8)x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$ — $\text{sum}=\text{rank } E_8=8$ ($\text{rank-fill}, v6$), $\text{product}=h/2=15=g_{\text{car}}N_{\text{fam}}$ ($2g_{\text{car}}N_{\text{fam}}=h, v431$); $\{3, 5\}$ the unique factor pair of 15 summing to 8, so BOTH are forced together by two degree-invariants. [E] the three mapped degrees $\{2, 8, 30\} = \min/\text{rank}/\max$ (the canonical invariants); $g_{\text{car}}=5=\max$ prime of 30. [E] DISCIPLINE: no new per-degree mining; the five non-primary degrees keep their v431 homes. Exact integer/lattice, mirrored in wolfram. [E][C]
v438_seam_hsmi_borchers.py	SEAM.EQUIV.BW.HSMI.01: the half-sided modular inclusion (Borchers/Wiesbrock) as an INTRINSIC Bisognano–Wichmann route for face (ii) of SEAM.EQUIV.01 — geometric modular flow WITHOUT presupposing covariance. [E] STANDARD-PAIR ALGEBRA (ladder, exact): boost $K=\text{diag}(n)$ and translation P (raising shift) satisfy $[K, P]=P$ and the Borchers scaling $\Delta^{it}P\Delta^{-it}=e^{-t}P$ (the $ax+b/\text{Moebius}$ relation). [E] BW REFLECTION: $J(n\rightarrow -n)$ gives $J^2=I$, $JKJ=-K$ ($u_\Theta=J$) and $JPJ=P_-$ (the opposite-chirality translation), the standard-pair reflection $JU(s)J=U(-s)$. [E] POSITIVE ENERGY: in the unitary lowest-weight $sl(2, \mathbb{R})$ rep the line translation $P_{\text{line}}=L_0-(L_++L_-)/2$ is POSITIVE ($\min \text{eig} > 0$), the positive-energy generator the c_3 one-sided origin demands; the boost direction is the indefinite negative control. [C] SYNTHESIS: $P\geq 0 + [K, P]=P + J$ is a Longo STANDARD PAIR / +half-sided modular inclusion (Wiesbrock), so by Borchers the seam modular flow is geometric INTRINSICALLY (complementary to Adamo–Giorgetti–Tanimoto, arXiv:2508.07109); face (ii) becomes a STRUCTURE consequence, not a state condition. [C][O] assembling the finite algebra and the positive-energy rep into ONE continuum standard pair is the continuum existence (the same wall as MMST/NPW26, v336/v424); SEAM.EQUIV.01 stays [O]. Python-only (numpy/scipy). [E][C][O]
v439_seam_lieb_robinson.py	SEAM.EQUIV.CONTINUUM.LR.01: the Lieb–Robinson/Nachtergaele–Sims continuum-existence MOTOR for the gapped seam collar — grounds the BULK half of the cited MMST existence (v336) in the explicit v367 $p+ip$ model. [E] UNIFORM GAP: $\Delta(N)=\text{const}>0$ across $N=20..160$ (the lattice image of the OS transfer gap $6\log\frac{3}{2}>0, v302/v329$). [E] EXPONENTIAL CLUSTERING: $\langle c_i^\dagger c_j \rangle \sim e^{-r/\xi}$ with ξ finite and STABLE across N (gapped $\Rightarrow$ Hastings–Koma decay), the N-independent locality. [E] LIEB–ROBINSON LIGHT CONE: the single-particle propagator $(e^{-iHt})_{ij}$ is confined to $i-j \lesssim v_{\text{LR}}t$ with finite v_{LR} ($\sim 2t$), exponentially small beyond the cone. [C] NACHTERGAELE–SIMS LIFT: uniform gap + finite LR velocity $\Rightarrow$ exponential clustering with N-independent ξ AND existence+uniqueness of the thermodynamic-limit ground state with a finite light cone — derived for OUR collar, not imported. [C][O] the bulk thermodynamic limit EXISTS by Lieb–Robinson; SEAM.EQUIV.01 stays [O], the open analytic step narrowed to the CHIRAL massless EDGE scaling limit (the $c=8$ $(E_8)_1$ CFT), distinct from the bulk limit. Python-only (numpy/scipy). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v440_seam_lto_rp_beta.py	SEAM.EQUIV.LTORP.BETA.01: the β -interpolation connecting the NPW26-PROVEN trace case ($\beta=0$) to the seam's $\beta=1$ KMS case along an obstruction-free homotopy — upgrading the single-point v426 witness for v424 sub-step (ii) to a PATH argument. [E] β -INDEPENDENT (LTO-RP) DATA: $\Theta K \Theta = -K$ ($u_\Theta = J$) and $[\rho, K]=0$ are conditions on K alone, β -independent. [E] THE KMS PATH: $C_\beta = 1/(1+e^{\beta K})$ has $\beta=0 \rightarrow C=\frac{1}{2}I$ (the NPW26 TRACE state), $\beta=1 \rightarrow$ the seam KMS state, $\beta \rightarrow \infty \rightarrow$ the ground-state projection; at every β $\Theta C_\beta \Theta = 1-C$ (reflection), $[\rho, C_\beta]=0$ (mu4), $0 \leq C_\beta \leq 1$ (valid covariance/RP). [E] NO OBSTRUCTION: $\beta \mapsto C_\beta$ is norm-continuous and stays in the (LTO-RP)-valid set — a continuous homotopy from the proven trace endpoint to the seam KMS point, no topological wall. [E] NEG CONTROLS: a reflection-EVEN K' breaks $\Theta C_\beta \Theta = 1-C$ for $\beta>0$ (the trace endpoint is degenerate); an off-mu4-block K breaks $[\rho, C_\beta]$. [C][O] a PATH argument for v424 sub-step (ii), stronger than the single point of v426; SEAM.EQUIV.01 stays [O] (closedness of (LTO-RP) along the path in the continuum is the residual). Python-only (numpy/scipy). [E][C][O]
v441_seam_applicability_ledger.py	SEAM.EQUIV.APPLICABILITY.01: the applicability ledger — a hypothesis-by-hypothesis audit of the THREE external theorems the seam keystone cites (MMST scaling limit, NPW26 (LTO-RP), Adamo–Giorgetti–Tanimoto covariance), isolating the SINGLE imported analytic fact. [E] MMST HYPOTHESES: $D=16$ Majoranas ($=2^{g_{\text{car}}-1}$), $c=8$ ($=g_{\text{car}}+N_{\text{fam}}$), rank $E_8=8$, range rank $\leq c \leq D$ reads $8 \leq 8 \leq 16$, gapped ($6 \log \frac{3}{2} > 0$), quasi-free CAR — all internal computed facts. [E] NPW26: (LTO-RP) $=u_\Theta=J$, gap, $\mathbb{Z}_2$ reflection, mu4 clock, $[\rho, K]=0$ internal (v424/v426/v440); the seam is invertible $ \det \text{Cartan } E_8 =1 + \beta=1$ KMS, OUTSIDE NPW26's topological/trace bucket ($ \det =4$) — a TEMPLATE, not a direct import. [E] ADAMO–GIORGETTI–TANIMOTO (arXiv:2508.07109): DHR reps of nets extending loop-group/Virasoro are AUTOMATICALLY positive-energy + diff-covariant; the seam net extends $(D_5)_1 \times (A_3)_1$ (v154), so covariance is a CONSEQUENCE — the v215 circularity is defused, covariance downgraded ASSUMPTION $\rightarrow$ [E]. [E] of 12 audited hypotheses, 11 are internal [E] and exactly 1 external — the continuum EXISTENCE of the chiral massless scaling limit (MMST/v336, Lean FORM.SEAM.MMST.01). [C][O] external dependence pinned to ONE analytic fact; SEAM.EQUIV.01 stays [O]. Python-only (a structural/arithmetic audit; the dets Wolfram-mirrored via v89/v281/v422). [E][C][O]
v442_seam_rigidity_uniform.py	SEAM.RIGIDITY.UNIFORM.01: the uniform-in- N hardening of the Seam State Rigidity Theorem (v398) — the band-limited rigidity is NOT a small- N artifact. [E] EXACT UNIFORM 4-SECTOR COMMUTATION: $\rho = \text{diag}(i^n)$ has EXACTLY 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all N , and a genuinely character-block-diagonal K (random Hermitian within each $n \bmod 4$ class, not merely diagonal) has $[\rho, K]=0$ EXACTLY for $N=4..256$ (extends v398's $N \leq 64$). [E] UNIFORM GAP: $\Delta(N) = \text{const} > 0$ across $N=20..160$ (the Nachtergaele–Sims uniform gap, shared with v439). [E] N -INDEPENDENT RIGIDITY BOUND: a FIXED local clock-breaking perturbation (single off-mu4-block Hermitian entry) gives $\ [\rho, V]\ =2$ EXACTLY for all N — a uniform lower bound; any fixed-size clock-breaking is detected with N -independent strength. [E] NEG CONTROL: a clock-PRESERVING (same-sector) perturbation gives $\ [\rho, V']\ =0$; an order-2 clock has only 2 distinct eigenvalues — the ORDER of the clock (the four Gauss–Bonnet marks) is the discriminator, uniformly in N . [C][O] the band-limited rigidity of v398 is uniform in N ; SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]: that RP+gap+holomorphy FORCE Λ_Σ block-diagonal on the full L^2 (vs merely permit) is the cited rigidity/MMST step. Python-only (numpy). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v443_seam_e8_discriminator.py	SEAM.E8.DISCIMINATOR.01: the empirical/structural DISCRIMINATOR bundle (mode D of the v347 closure-mode classification) — concrete FALSIFIABLE kill tests separating the claimed seam $= (E_8)_1$ from the nearest $c=8$ rival $SO(16)_1=(D_8)_1$. [E][X] PRIMARY/GSD: $\det \text{Cartan}(E_8) =1$ (ONE primary, holomorphic) vs $\det \text{Cartan}(D_8) =4$ (FOUR primaries $1, v, s, c$); torus GSD $= \# \text{primaries} = 1$ vs 4 — a seam with $\text{GSD} > 1$ falsifies $(E_8)_1$. [E][X] CURRENT/THETA: level-1 current count $\dim E_8=248$ vs $\dim SO(16)=120$; the E_8 lattice theta has 240 norm-2 roots ($= 112$ integer +128 spinor, computed) vs D_8's 112 — current spectrum and lattice theta DIFFER. [E] $c=8$ IS NOT A DISCRIMINATOR: both have $c=8$ (the v277 ambiguity); only the order-4 μ_4 condensation + $\det K=1$ (the genus-1 statement, v90) selects E_8. [E] NON-VACUITY: the discriminators all fire on the rival ($\det 4 \neq 1$, currents $120 \neq 248$, roots $112 \neq 240$). [X][C] mode D realised as concrete kill tests; SEAM.EQUIV.01 stays [O] (the discriminators distinguish the TARGET but do not PROVE the seam realises it). Python-only (numpy; the dets Wolfram-mirrored via v89/v281/v422). [E][C][O][X]
v444_seam_edge_scaling.py	SEAM.EQUIV.CONTINUUM.EDGE.01: the CHIRAL EDGE scaling-limit MOTOR — the edge half of the one SEAM.EQUIV.01 continuum residual, attacking exactly the step v439 left open. On the SAME explicit v367/v368 $p+ip$ collar it shows the lattice edge carries the kinematic AND conformal data of a chiral $c=\frac{1}{2}$ Majorana CFT, converging under refinement. [E] CHIRAL LINEAR EDGE DISPERSION: the in-gap branch is linear near its zero-crossing (top $v=0.983$, $R^2=0.9999$) and the two boundaries carry OPPOSITE velocities — one chiral Majorana per edge (relativistic, chiral kinematics). [E] ALGEBRAIC EDGE TWO-POINT: the equal-time edge correlator decays as $r^{-\eta}$ with $\eta=1.00$ (the chiral free-fermion CFT exponent) — a CRITICAL edge. [E] EXPONENTIAL BULK (neg control): the bulk-row correlator decays exponentially ($\xi \sim 1.1$) — only the edge is critical, so the power law is the edge CFT, not a geometric artefact. [E] TRIVIAL-PHASE NEG CONTROL: $M=3$ (no edge, v368) has an exponential edge-row — the algebraic decay is the topological edge, not the boundary geometry. [E] SCALING-COLLAPSE / CONVERGENCE: η is N-stable (1.001 ± 0.007) and the edge correlator profile is Cauchy-convergent under refinement (successive spread shrinks $0.028 \rightarrow 0.013$, $N_x=36..60$) — an existing scaling limit, the edge analogue of v439's bulk thermodynamic limit. [C] EDGE CENTRAL CHARGE: $\eta=1 \Leftrightarrow$ one chiral Majorana ($c=\frac{1}{2}$) per edge, so the $N_{\text{Maj}}=16$-copy carrier has a chiral $c_-=8$ edge CFT ($= (E_8)_1$, $c_-=g_{\text{car}}+N_{\text{fam}}$). [C][O] grounds the EDGE half of the cited MMST scaling limit (v336) the way v439 grounds the bulk; with both halves the one continuum residual is grounded and pinned to a single cited theorem; SEAM.EQUIV.01 stays [O] (the rigorous uniform-convergence theorem is not supplied). Python-only (numpy). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v445_seam_rigidity_forcing.py	SEAM.RIGIDITY.FORCING.01: the rigidity FORCING theorem — upgrades v442’s “block-diagonal PERMITS commuting” to “commuting with the order-4 clock FORCES block-diagonal”, isolating the single remaining physical premise. [E] FORCING (commute $\Leftrightarrow$ block-diagonal, EXACT): for $\rho = \text{diag}(i^n)$, $[\rho, K]=0$ iff $K_{ij}=0$ whenever $i \not\equiv j \pmod 4$ — verified ENTRYWISE, swept $N=4..128$ (the converse of v442). [E] EXACT COMMUTANT DIMENSION: $\dim\{K : [\rho, K]=0\} = \sum_s n_s^2$ (the four sector sizes), verified as the ad_ρ nullspace; a PROPER subspace of the N^2 algebra ($N=16$: $64 < 256$) — a genuine quantitative restriction. [E] OFF-BLOCK $\Leftrightarrow$ NON-COMMUTING: every off-block matrix unit E_{ij} has $\ [\rho, E_{ij}]\ = i^i - i^j \geq \sqrt{2} > 0$ — zero slack. [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors ($\dim \sum n_s^2(4)$), the order-2 clock $\text{diag}((-1)^n)$ a STRICTLY LARGER commutant ($N=16$: 64 vs 128) — the four Gauss–Bonnet marks (order $4= \mu_4 =h(A_3)$) force strictly more structure than two; only the index-4 extension is $(E_8)_1$ ($\det 1$; v281/v154/v351). [E] NONDEGENERACY: exactly 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all $N \geq 4$. [C] SYNTHESIS (force, not permit): commuting with the order-4 μ_4 clock is EQUIVALENT to μ_4-block-diagonality (iff, exact, uniform in N), so RP-definability (v54/v398) + clock-invariance FORCE the block-diagonal Λ_Σ — v442’s “permit” upgraded to “force”. [C][O] the single residual is now precisely “transfer in commutant” (clock-invariance = QGEO.SYM.01, v323/v335); SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy; the det facts Wolfram-mirrored via v89/v281/v422). [E][C][O]
v446_seam_clock_invariance.py	SEAM.RIGIDITY.CLOCKINV.01: DERIVING clock-invariance (“transfer in the commutant”) from RP-symmetry, closing v445’s single residual one notch further — the residual drops from an OPERATOR commutation to the STRUCTURAL μ_4-symmetry of the seam RP data. [E] OS RECONSTRUCTION INHERITS THE SYMMETRY: for a μ_4-invariant RP covariance $([\rho, C(t)]=0)$, the OS canonical transfer $\Lambda=C(1)C(0)^{-1}=e^{-h}$ (the transfer FIXED by RP+reflection, v54) has $[\rho, \Lambda]=0$ — swept $N=4..32$ over random block-diagonal h. [E] SHARPNESS / NEG CONTROL: a μ_4-BREAKING covariance gives $[\rho, \Lambda] \neq 0$ — iff-sharp, so the inheritance is non-vacuous. [E] CONTRACTION / OS-VALIDITY: Λ has spectrum in $(0, 1]$ (a positive contraction, the physical canonical transfer). [E] MODULAR (Tomita–Takesaki): the free modular Hamiltonian $K=\log((1-C)C^{-1})$ (the v258 formula) of the μ_4-invariant state has $[\rho, K]=0$ (the clock commutes with modular flow), the breaking control $[\rho, K] \neq 0$. [C] SYNTHESIS: $[\rho, \Lambda_\Sigma]=0$ is a CONSEQUENCE of μ_4-invariance of the seam RP data (the four Gauss–Bonnet marks, v216/v323) via the OS reconstruction (v54) + the modular fact (v258), NOT an extra assumption; with v445 (commute $\Leftrightarrow$ block-diagonal), RP-definability PLUS the marks FORCE the block-diagonal Λ_Σ. [C][O] the residual reduces from an operator commutation to “the seam RP data is μ_4-symmetric” (QGEO.SYM.01); the continuum Tomita–Takesaki implementation on the full L^2 is the cited backbone. SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy/scipy). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v447_seam_edge_chern.py	SEAM.EQUIV.EDGE.CHERN.01: an INDEPENDENT (topological) reading of the edge $c_- = 8$, corroborating v444's correlator route via the bulk-edge correspondence on the same v367/v368 $p+ip$ collar. [E] INTEGER CHERN NUMBER: the $M=1$ collar lower band has Chern $C=+1$ (the Fukui-Hatsugai-Suzuki lattice link-variable method, an EXACT integer, no fit) — one chiral channel per copy. [E] TRIVIAL-PHASE NEG CONTROL: $M=3$ (v368) has $C=0$ (no chiral edge), and $M=-1$ gives $C=-1$ (opposite chirality, matching v444's opposite per-edge velocities). [E] BULK-EDGE CORRESPONDENCE: the open strip carries EXACTLY $C =1$ in-gap edge branch crossing zero per boundary in the $M=1$ phase, 0 in the trivial phase — the chiral edge channel count = C. [E] LI-HALDANE: the half-cut bulk ground state's single-particle entanglement spectrum has in-gap modes at $1/2$ in the topological phase and NONE in the trivial — the entanglement edge tracks the physical edge. [C] EDGE CENTRAL CHARGE: $c_- = N_{\text{Maj}} \cdot \frac{1}{2} \cdot C = 16 \cdot \frac{1}{2} \cdot 1 = 8 = g_{\text{car}} + N_{\text{fam}}$, from the Chern integer + bulk-edge correspondence — the SAME $c_- = 8$ as v444's correlator route, from a logically independent (topological) observable. [C][O] corroborates v444's edge $(E_8)_1$ reading; SEAM.EQUIV.01 stays [O] — the topology fixes c_- and the channel count but not the cited continuum existence theorem. Python-only (numpy). [E][C][O]
v448_seam_edge_fourpoint.py	SEAM.EQUIV.EDGE.FOURPOINT.01: pinning the ONE external MMST fact empirically — the edge FOUR-point function is quasi-free and conformally convergent, extending v444's two-point scaling limit to the next order. [E] QUASI-FREE / WICK (CAR hypothesis): the ground-state correlation matrix is an exact projector $G^2=G$ ($\sim 10^{-13}$), so the state is quasi-free and all n-point functions are Wick/Pfaffian sums of the two-point — the MMST quasi-free CAR hypothesis (ledger entry 5) literally holds on the collar (structural: the BdG state is free). [E] FOUR-POINT CAUCHY CONVERGENCE: the edge four-point cross-ratio $Q=[G_{01}G_{23}]/[G_{02}G_{13}]$ at COMMENSURATE positions $(1, 2, 4, 7) \cdot (N_x/12)$ is Cauchy-convergent under refinement (successive spread shrinks, $N_x=36..60$) — the four-point converges, extending v444's two-point scaling-collapse. [E] CONFORMAL FOUR-POINT FORM: Q converges to the free-fermion CONFORMAL cross-ratio $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3} \approx 3.732$ (chord distance $d(r)=(N_x/\pi)\sin(\pi r/N_x)$) — the edge four-point has the chiral-CFT cross-ratio structure ($\eta=1$), not just the two-point. [E] BULK NEG CONTROL: the bulk-row four-point is orders of magnitude off the conformal value (exponential decay, gapped) — only the EDGE four-point is conformal. [C] the one external MMST fact (continuum existence of the chiral scaling limit) is now empirically supported at BOTH two-point (v444) and four-point order on our explicit collar; every measured n-point converges to its conformal limit. SEAM.EQUIV.01 stays [O] — the rigorous MMST uniform-convergence theorem (v336) is the cited backbone. Python-only (numpy). [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v449_seam_mmst_uniform.py	SEAM.EQUIV.MMST.UNIFORM.01: uniform-in-N control of the edge scaling limit — the empirical content of the cited MMST uniform-convergence theorem, strengthened, on the same v367/v368 $p+ip$ collar. [E] CROSS-RATIO A CONVERGES: the edge four-point $Q(1, 2, 4, 7)/12 \rightarrow \sin(5\pi/12)/\sin(\pi/12) = 2+\sqrt{3}$, the deviation shrinking monotonically under refinement ($N_x=36, 48, 60$). [E] CROSS-RATIO B (independent positions): $Q(1, 3, 5, 9)/12 \rightarrow$ its OWN conformal value 2 — a SECOND observable with a DIFFERENT limit, so the convergence is not a single tuned point. [E] UNIFORM $1/N$ RATE: $Q(N)-Q_\infty \cdot N \leq 4$ for BOTH cross-ratios and every $N_x \in \{36, 48, 60\}$ — one N-independent constant bounds the deviation across two independent observables, the empirical hallmark of uniform-in-N convergence. [E] BULK NEG CONTROL: the bulk-row cross-ratio is off-scale (gapped, exponential) — only the EDGE converges. [C][O] the MMST uniform-convergence HYPOTHESIS is empirically satisfied on the collar (two-point v444, four-point v448, and now multi-observable uniform rate); SEAM.EQUIV.01 stays [O] — the abstract continuum existence theorem (v336) is strengthened in evidence, not replaced by a proof. Python-only (numpy). [E][C]
v450_seam_edge_entanglement.py	SEAM.EQUIV.EDGE.ENTANGLEMENT.01: a THIRD independent reading of the edge $c_-=8$ — the Calabrese–Cardy entanglement scaling $c=1/2$ per Majorana (after v444 correlator, v447 topology). [E] CALABRESE–CARDY LAW: the block entanglement entropy of the critical XX (Dirac) chain fits $S=(c/3)\ln[(L/\pi)\sin(\pi\ell/L)]+\text{const}$ with $c_{\text{Dirac}}=1.000$ (flat log slope) — the universal scaling-limit signature. [E] c PER MAJORANA = $1/2$: $c_{\text{Dirac}}=1=2c_{\text{Maj}}$ (Dirac = two Majoranas), so $c_{\text{Maj}}=1/2$ — the chiral CFT of one $p+ip$ edge mode (v444). [E] GAPPED NEG CONTROL: a dimerised (gapped) chain has saturating, ℓ-independent entanglement ($c=0$) — the log growth is the CRITICAL edge CFT. [C] EDGE CENTRAL CHARGE: $c_-=N_{\text{Maj}} \cdot c_{\text{Maj}} = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$, from entanglement — the SAME $c_-=8$ as v444 (correlator) and v447 (topology), from a third independent observable. [C][O] three observables agree; SEAM.EQUIV.01 stays [O]. Python-only (numpy). [E][C]
v451_seam_edge_cardy_tower.py	SEAM.EQUIV.EDGE.CARDY.01: the Cardy finite-size conformal tower — the edge spectrum reproduces the EXACT Ising primary dimensions $\{1/16, 1/2\}$, on the critical Majorana ring (the v367/v368 edge universality class). [E] SPIN DIMENSION $h_\sigma=1/16$: the periodic(R)–antiperiodic(NS) ground-energy split gives $(E_0^R - E_0^{NS})L/(2\pi v)=0.06250$ ($v=2$), stable across $L=200, 400, 800$ — the Ising spin field. [E] ENERGY DIMENSION $h_\varepsilon=1/2$: the lowest NS pair excitation gives $2\varepsilon_{\text{min}}L/(2\pi v)=0.50000$ — the Ising energy field. [E] CONFORMAL SCALING: both dimensions are L-INDEPENDENT (the gaps scale exactly as $1/L$) — a conformal (gapless) spectrum, not a gapped lattice. [C] ISING IDENTIFICATION: $\{c, h_\sigma, h_\varepsilon\}=\{1/2, 1/16, 1/2\}$ is the chiral Ising (free Majorana) CFT (the $c=1/2$ building block, v444/v450); with $N_{\text{Maj}}=16$ copies and bulk-edge (v447), $c_-=16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$. [C][O] the edge CFT is named by its central charge (v450) AND its operator content (this); SEAM.EQUIV.01 stays [O]. Python-only (numpy). [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v452_seam_e8_modular.py	SEAM.EQUIV.E8.MODULAR.01: the $(E_8)_1$ torus modular data — one primary, S-invariant, T-phase $e^{-2\pi i/3}$, the holomorphic $c=8$ signature (extends v156's q-expansion to the modular transformation data). [E] S-INVARIANCE: $\chi(-1/\tau)=\chi(\tau)$ for $\chi=E_4/\eta^8$ (weight 0, since E_4 wt 4 / η^8 wt 4), to $\sim 10^{-30}$ — ONE primary, the S-matrix is [1]. [E] T-PHASE: $\chi(\tau+1)/\chi(\tau)=e^{-2\pi i/3}=e^{-2\pi ic/24}$ ($c=8$), to $\sim 10^{-30}$ — the central charge appears in the modular T eigenvalue. [E] LEADING POWER $\Rightarrow c=8$: $\chi q^{1/3} \rightarrow 1$ so the leading power is $q^{-c/24}=q^{-1/3} \Rightarrow c=8=g_{\text{car}}+N_{\text{fam}}$. [E] HOLOMORPHIC CONSTRAINT: $c=8=0 \bmod 8$ — the minimal holomorphic (single-primary) chiral central charge, the even unimodular rank-8 lattice E_8. [C][O] the $(E_8)_1$ modular data (one primary, S-invariant, T-phase $e^{-2\pi i/3}$, $c=8$) is the torus signature of the seam net; with the edge readings (v444/v447/v450/v451) the $(E_8)_1$ identity is pinned on the torus AND the edge; SEAM.EQUIV.01 stays [O] — the cited continuum existence theorem (v336) is the residual. Wolfram-mirrored (tfpt_readouts_extension.wl). [E][C]
v453_seam_mu4_from_marks.py	SEAM.RIGIDITY.MU4FROMMARKS.01: deriving QGEO.SYM.01 (μ_4-symmetry of the seam RP data) from the four marks — the LAST structural premise of the rigidity chain folded into the existing realisation premise. [E] THE MARKS ARE μ_4: the four Gauss–Bonnet marks are exactly $\{1, i, -1, -i\} = \text{roots of } z^4-1$ (v168/v216/v323). [E] THE CLOCK PERMUTES THE MARKS: $z \mapsto iz$ is a 4-cycle of order 4 ($1 \rightarrow i \rightarrow -1 \rightarrow -i \rightarrow 1$), fixing the configuration setwise (the order-4 clock of v445). [E] MOEBIUS STABILISER: $z \mapsto iz$ preserves the marks' cross-ratio $\lambda=2$ (in the D_4 stabiliser, v168) — the symmetry is geometric. [E] THE FORM BASIS IS μ_4-GRADED: $\omega_k = z^{k-1} dz / (z^4-1) \rightarrow i^k \omega_k$ under $z \mapsto iz$ ($k=1, 2, 3 \rightarrow \{i, -1, -i\}$, the three nontrivial μ_4 characters; weights $(1, 2, 3)=A_3$ exponents = $\text{Spec}(Q_+)$) — the natural basis is μ_4-graded, so any geometric datum is μ_4-covariant. [C][O] QGEO.SYM.01 (“the seam RP data is μ_4-symmetric”) is a CONSEQUENCE of marks=μ_4 (this) + QGEO.REALIZE.01 (seam = the marked geometry, the EXISTING premise), NO new premise; the rigidity chain (this $\rightarrow$ v446 $\rightarrow$ v445) closes modulo realisation, SEAM.EQUIV.01 stays [O]. Wolfram-mirrored (tfpt_readouts_extension.wl). [E][C]
v454_seam_edge_virasoro.py	SEAM.EQUIV.EDGE.VIRASORO.01: the lattice realises the level-1 current algebra whose Sugawara central charge is EXACTLY 8 (G2 of the post-F next steps) — testing the cited MMST “Koo–Saleur lattice generators $\rightarrow$ continuum Virasoro with the correct c” fact on our own free-fermion edge. [E] CASIMIR c: the critical complex free-fermion ring $E_0(L)=e_\infty L - \pi v c / (6L)$ (Affleck–Cardy) fits $c_{\text{Dirac}}=1.000 \Rightarrow 1/2$ per Majorana (parameter-free, $L=64..1024$). [E] KAC–MOODY LEVEL: the lattice $U(1)$ current J_n has $\langle 0 J_n J_{-n} 0 \rangle = n$ EXACTLY ($n=1..8$) — the affine level-1 central extension closes; a gapped (dimerised) control gives a vanishing non-linear term. [E] SUGAWARA EXACT: $c = \dim G / (1+h^\vee) = 8$ for BOTH $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31) — Wolfram-mirrored. [C] ASSEMBLY: 16 Majoranas $\times \frac{1}{2} = 8 =$ the level-1 Sugawara c; the edge IS a $c_- = 8$ level-1 current-algebra CFT. [C][O] the lattice Virasoro/current algebra with the correct central extension is realised on the explicit collar; SEAM.EQUIV.01 stays [O] (the continuum existence theorem v336 is the cited backbone). Mixed numerical (Casimir, Kac–Moody) + exact (Sugawara, Wolfram-mirrored). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v455_seam_bw_uniform.py	SEAM.EQUIV.BW.UNIFORM.01: a UNIFORM-IN-N Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition $u_\Theta=J$ (G1 of the post-F next steps) — the inductive-limit companion to the single finite witness v426, advancing v424 sub-step (i). One-particle: $\Delta=e^{-K}$, conjugation J, reflection Θ; BW $\Leftrightarrow u_\Theta=J \Leftrightarrow \Theta K \Theta = -K$. [E] UNIFORM BW: the reflection-odd, μ_4-even, gapped collar of size $4m$ ($m=2..32$, $N=8..128$) has $\ \Theta K \Theta + K\ =0$ to machine precision at EVERY N — $u_\Theta=J$ uniform. [E] UNIFORM GAP: the gap of K is N-independent ($6 \log \frac{3}{2} = 2N_{\text{fam}} \log \frac{3}{2}$) so $\Delta=e^{-K}$ stays uniformly bounded away from 1 (a genuine KMS, not trace, state). [E] UNIFORM CLOCK SYMMETRY: $[\rho, K]=0$ exactly at every N. [E] N-INDEPENDENT NEG CONTROL: a side-even reflection gives $\ \Theta K \Theta + K\ =2\ K\$ (a fixed per-mode margin, growing with N) and an off-μ_4 K breaks $[\rho, K]$ at every N — the success is non-vacuous. [C][O] the inductive (continuum) limit inherits $u_\Theta=J$; SEAM.EQUIV.01 stays [O] (the rigorous continuum theorem NPW26/MMST v336 is the backbone). Python-only (numpy). [E][C][O]
v456_seam_chirality_from_c3.py	SEAM.S3.FROM-P1.01: the edge chirality ($c_- \neq 0$, the S3 topological phase) is FORCED by the one-sidedness that defines $c_3=1/(8\pi)$ (P1) (G6 of the post-F next steps) — so S3 is a CONSEQUENCE of axiom P1, not an independent input. [E] c_3's “8” IS THE ONE-SIDED COUNT: $8\pi= \mathbb{Z}_2 \int K = 2(2\pi\chi(S^2)) = 2 \cdot 4\pi$ ($\chi=2$), so $8= \mathbb{Z}_2 \cdot (\int K/\pi) = 2 \cdot 4$ (v73, Wolfram-mirrored). [E] REFLECTION REVERSES C: the explicit 2-band $p+ip$ Chern model has $C=+1$ and the orientation-reversed ($k_y \rightarrow -k_y$) model $C=-1$. [E] TWO-SIDED $\Rightarrow$ NON-CHIRAL: any GAPPED boundary carrying an orientation-reversing (reflection) symmetry must have $C=-C=0$ ($c_-=0$, gappable). [C] ONE-SIDED $\Rightarrow$ CHIRAL: the one-sided seam is precisely the $\mathbb{Z}_2$ quotient with NO reflection (the same $\mathbb{Z}_2$ that gives c_3 its 8π), so $C \neq 0$ is forced — S3 follows from P1. [C][O] magnitude $c_-=8=\text{rank } E_8=g_{\text{car}}+N_{\text{fam}}$ is the SAME integer as c_3's one-sided count; SEAM.EQUIV.01's continuum EXISTENCE (v336) stays [O]. Mixed exact (the $8= \mathbb{Z}_2 \int K/\pi$ arithmetic, Wolfram) + numerical (the reflection $C \rightarrow -C$ flip). [E][C][O]
v457_seam_character_killtest.py	SEAM.EQUIV.KILLTEST.01: an explicit $(E_8)_1$-vs-$SO(16)_1$ character/sector kill-test for the seam edge (G7 of the post-F next steps) — a FALSIFIABLE consistency test of the $(E_8)_1$ identification of SEAM.EQUIV.01. [E] $(E_8)_1$ CHARACTER TOWER: $\chi=E_4/\eta^8=q^{-1/3}(1+248q+4124q^2+34752q^3+\dots)$ — the exact conformal-tower degeneracies $(E_4 \cdot \prod (1-q^n))^{-8}$, Wolfram-mirrored). [E] CURRENT-COUNT DISCRIMINATOR: the weight-1 multiplicity is $248=\dim E_8=120(SO16)+128(\text{spinor})$, NOT the 120 of the free $SO(16)_1$. [E] SECTOR-COUNT DISCRIMINATOR: $(E_8)_1$ is INVERTIBLE $\det \text{Cartan } E_8 =1$ (one primary) vs $SO(16)_1=D_8$ $\det \text{Cartan } D_8 =4$ (four sectors). [X] KILL CRITERION: a measured weight-1 count of 120, OR a sector count of 4, OR a tower coefficient $\neq \{1, 248, 4124, \dots\}$ would FALSIFY $(E_8)_1$; current data give $248/1/\{1, 248, 4124\}$, so PASSED. [C][O] the $(E_8)_1$ identification survives the kill-test; SEAM.EQUIV.01's continuum EXISTENCE (v336) stays [O]. Exact (Wolfram-mirrored) + a numerical determinant cross-check. [E][C][O][X]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v458_seam_mmst_citation_audit.py	SEAM.EQUIV.MMST.AUDIT.01: the EXACT hypothesis-by-hypothesis citation audit of the seam scaling-limit theorem (Osborne–Stottmeister, CMP 398 (2023) 219; arXiv:2107.13834; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework) against our collar (G3 of the post-F next steps, the “exakter Zitat-Abgleich”). [C] H1 ALGEBRA CLASS: a gapped quasi-free CAR 16-Majorana system = MMST’s lattice-fermion class (v155/v160/v302). [C] H2 MASSLESS POS-ENERGY LIMIT: the gapless chiral edge IS that massless scaling limit (v439/v444). [E] H3 PER-COPY ($c=1/2$): each of 16 chiral Majoranas is EXACTLY the proven Theorem A $c=1/2$ case. [C] H4 DECOUPLED ADDITIVITY: MMST’s limit decouples into Majoranas $\Rightarrow L_k = \sum_i L_k^{(i)}$ on the common core $\Rightarrow$ the free $c=8$ $SO(16)_1$ Virasoro follows ($16 \cdot \frac{1}{2} = 8$). [C] H5 WZW RANGE rank $\leq c \leq D$: $8 \leq 8 \leq 16$ IN range, the 120 $so(16)$ currents are bilinears — covered. [O] H6 RESIDUAL = THE 128-SPINOR EXTENSION: the index-4 $SO(16)_1 \rightarrow (E_8)_1$ adds 128 SPINOR currents ($248=120+128$), NOT bilinears — OUTSIDE MMST’s method (the $\det K \ 4 \rightarrow 1$ holomorphy discriminator), handed to v459. [C][O] the audit CLOSES-by-citation the $c=8$ EXISTENCE and pins the open residual EXACTLY to the 128-spinor extension; SEAM.EQUIV.01 stays [O], its open part now this one holomorphic-extension statement. Python (sympy; a structural/arithmetic audit). [C][O]
v459_seam_lattice_voa_route.py	SEAM.EQUIV.LATTICEVOA.01: the SECOND, independent route to $(E_8)_1$ — the lattice-VOA / conformal-net $A_Q(E_8)$ construction supplies EXACTLY the 128-spinor extension MMST (v458) leaves open (G5 of the post-F next steps). Cited: Adamo–Giorgetti–Tanimoto, “Rational and non-rational 2d CFTs arising from lattices” (arXiv:2506.01008); OS for unitary lattice VOAs (Adamo–Moriwaki–Tanimoto, arXiv:2407.18222). [E] E_8 EVEN UNIMODULAR: $\det \text{Cartan}(E_8)=1$ and even $\Rightarrow A_{E_8}$ HOLOMORPHIC (one sector, the $\det K=1$ the residual demanded). [E] LATTICE CURRENTS $248=8+240$: 8 Cartan + 240 roots = the full $(E_8)_1$ weight-1 space, from the lattice (not bilinears). [E] THE 240 ROOTS SPLIT $112+128$: 112 INTEGER (D_8) + 128 HALF-INTEGER; $8+112=120=so(16)$ and the 128 half-integer roots ARE the spinor ($248=120+128=8+240$) — the lattice route builds the 128-spinor explicitly. [C] AGT THEOREM: A_Q of an even lattice IS a 2d conformal net; the holomorphic $SO(16)_1 \rightarrow (E_8)_1$ extension is CONSTRUCTED, not assumed. [C][O] TWO complementary routes now cover $(E_8)_1$ (MMST fermions v458 + the lattice net A_{E_8}); the residual reduces to the single realization input “the collar’s scaling limit = A_{E_8}” (tied to P1 by v456). SEAM.EQUIV.01 stays [O]. Exact (E_8 even unimodular; $240=112+128$; $248=8+240=120+128$, Wolfram-mirrored) + structural literature mapping. [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v460_seam_s3_lto.py	SEAM.S3.LT0.01: the EXPLICIT flat-band commuting-projector (local-topological-order) realisation of the S3 input — advancing v424 sub-step (i) (“realise the raw seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO”) from abstract existence to an explicit gap-protected quasi-local projector net carrying the $\mathbb{Z}_2$ reflection and the μ_4 clock. [E] FLAT-BAND PARENT: the flattened Bloch $Q=h/ h$ of v367’s $p+ip$ collar has $Q^2=I$ (flat ± 1 bands) and $P=(I-Q)/2$ is an exact projector EQUAL to the occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state. [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel $P(n, 0)$ decays rapidly ($\xi \approx 1.14$ at $M=1$, $\exp R^2=0.995$, $P(20, 0) < 10^{-6}$) and ξ GROWS as the gap closes (2.7 at gap 0.02); at the GAPLESS point the kernel decays as a clean power law $R ^{-2.1}$ ($R^2=0.999$, tail $\sim 2 \times 10^{-4}$) — locality needs the gap. [E] $\mathbb{Z}_2$ REFLECTION + μ_4 INHERITED: the flattened modular Hamiltonian $K_{\text{flat}} = \text{sign}(K)$ inherits $\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}$ (BW, $u_\Theta = J$) and $[\rho, K_{\text{flat}}] = 0$ (μ_4) EXACTLY (sign is odd; ρ commutes with any $f(K)$). [E] NEG CONTROLS: a side-even reflection gives $2\ K_{\text{flat}}\ \neq 0$ (BW fails), an off-μ_4 K mixing opposite-sign marks breaks $[\rho, \text{sign}(K)] \neq 0$. [C][O] the chiral $c_- = 8 \neq 0$ (FHS Chern $C =1$) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756; $C \neq 0$ forbids exponentially-localised Wannier, Thouless), so the realisation is the quasi-local LTO net of NPW26 — the form sub-step (i) asks for. SEAM.EQUIV.01 stays [O] (the continuum theorem MMST v336 is the backbone). Python-only (numpy; the det/level discriminators Wolfram-mirrored via v89/v281/v422). [E][C][O]
v461_seam_strict_locality.py	SEAM.S3.LOCALITY.01: the Kapustin–Fidkowski/Thouless strict-locality OBSTRUCTION made explicit — sharpening v460’s verdict from a cited statement to an EXHIBITED topological obstruction on the same v367 $p+ip$ collar. [E] CHERN INTEGER: the FHS Chern is $C=+1$ (chiral $M=1$), 0 (trivial $M=3$), -1 (opposite mass) — the invertible-phase invariant. [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre $\theta(k_y)$ winds by $C =1$ (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier functions exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding = $C =1 \neq 0$ and $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8 \neq 0$ (the same 8 as c_3’s one-sided count, v456). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0, $C=0$; since $C=1 \neq 0$ NONE exists, so v424 sub-step (i)’s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise) — upgrading v460’s cited verdict. SEAM.EQUIV.01 stays [O] (continuum theorem MMST v336 the backbone). Python-only (numpy; the winding/Chern integers Lean-mirrored, SeamResidualAxiom.lean). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v462_seam_spinor_continuum.py	SEAM.EQUIV.SPINOR.01: the 128-spinor extension $SO(16)_1 \rightarrow (E_8)_1$ exhibited at CHARACTER level and as a finite-$L \rightarrow$ continuum convergence — the constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual. [E] $(E_8)_1$ TOWER FROM $SO(16)_1$ SECTORS: the exact Jacobi/E_8 identity $\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4$ gives $\chi_{(E_8)_1} = E_4/\eta^8 = \frac{1}{2}[(\theta_3/\eta)^8 + (\theta_4/\eta)^8] + \frac{1}{2}(\theta_2/\eta)^8 = \chi_o^{SO16} + \chi_s^{SO16}$ — the holomorphic $(E_8)_1$ character IS the $SO(16)_1$ vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL 120$\rightarrow$248: χ_o gives weight-1 multiplicity 120, χ_s gives 128, summing to the exact $(E_8)_1$ tower $\{1, 248, 4124, 34752\}$ — the spinor currents ARE the $(E_8)_1$–$SO(16)_1$ deficit (248=120+128). [E] FINITE 16-MAJORANA FOCK: $C(16, 2)=120$ NS bilinear ($so(16)$) currents and $2^{16/2}=256$ Ramond ground states = 128 spinor + 128 cospinor — the spinor present already at finite N. [E] FINITE-$L \rightarrow$ CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir $E_0(L)=e_\infty L - \pi v c/(6L)$ fits $c_{\text{Dirac}} \rightarrow 1$ ($c_-=8$) and the lowest scaled NS gap $x_1 \rightarrow 1$ with $x_1 - 1 \sim 1/L^2$ ($L=64..1024$) — the lattice flows to the conformal tower. [C][O] the rigorous holomorphic extension ($\chi_o + \chi_s$ simple-current projection) is supplied by AGT/AMT (v459); SEAM.EQUIV.01's continuum existence (v336) stays [O]. Mixed exact (θ identity + tower + 120=$C(16, 2)$, 128=2^7, 248=120+128, Wolfram-mirrored) + numerical (the finite-L convergence, Python-only). [E][C][O]
v463_seam_c8_holomorphic_uniqueness.py	SEAM.EQUIV.UNIQ.01: the $c=8$ holomorphic-uniqueness pin — the positive lever that makes the IDENTIFICATION half of SEAM.EQUIV.01 classification-forced rather than assumed (B of the A+B round). [E] $c=8$ NOT UNIQUE AT LEVEL 1: $c = \dim \mathfrak{g}/(1+h^\vee)=8$ has THREE simple solutions — A_8 (dim 80), $D_8=\mathfrak{so}(16)$ (dim 120), E_8 (dim 248); $c=8$ is necessary but NOT sufficient (the v457 $\det K=4$ $SO(16)$ rival is a real gleich-c competitor). [E] HOLOMORPHY FORCES $\dim V_1=248$: the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}=E_4/\eta^8$, whose q^1 coefficient (the 1-dim candidate space, vacuum $a_0=1$) is 248 — forced, not chosen. [E] $\dim V_1=248$ + level-1 $c=8 \Rightarrow E_8$ UNIQUELY (248/(1+30)=8); the gleich-c rivals A_8 (80) and D_8 (120) are excluded. [E] the $(E_8)_1$ tower $\{1, 248, 4124, 34752, 213126\}$ are the E_4/η^8 coefficients, and $E_4/\eta^8=j^{1/3}$ (numeric $< 10^{-25}$). [C] Dong–Mason/Schellekens: the holomorphic VOA at $c=8$ is UNIQUE = V_{E_8}, and the even unimodular rank-8 lattice is unique = E_8 (Mordell–Witt) — so “$c=8$ holomorphic limit = $(E_8)_1$” is classification-forced. With v458 (free part) + v459 (spinor extension) the continuum/identification HALF of SEAM.EQUIV.01 is [C] conditional on named audited theorems; the REALISATION R1 (v464) is the sole remaining input. SEAM.EQUIV.01 stays [O]. Exact (sympy/integer q-series + Lie-algebra scan; mpmath cross-check), Wolfram-mirrored. [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v464_seam_ oneparticle_ rigidity.py	SEAM.EQUIV.ONEPARTICLE.01: the one-particle RIGIDITY + scaling limit attacking R1 — the REALISATION input of SEAM.EQUIV.01 (A of the A+B round). The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the whole net is the second quantisation of its one-particle symbol P , and R1 collapses to a finite-per-mode statement about P . [C] ARAKI SELF-DUAL CAR: a gauge-invariant quasi-free state $\leftrightarrow$ its symbol P ($0 \leq P \leq 1$) bijectively (Araki 1971; Shale–Stinespring 1965), so a realisation is fully determined by P . [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with $P =$ projection onto $K < 0$ ($P^2 = P$, hermitian to $< 10^{-12}$), K fixed by gap + chirality + reflection (v426) — no second realisation. [E] ONE-PARTICLE SCALING LIMIT EXISTS: the real-space kernel is Cauchy on a fixed window ($\max P_{2N} - P_N \sim 1/N \rightarrow 0$), so P_∞ exists (Gaussianity is why a computation reaches it). [E] THE LIMIT IS $c=1$ DIRAC ($c_-=8$): entanglement $S(L) = (c/3) \ln L$ gives $c \rightarrow 1$ (per Majorana 1/2; Peschel/Calabrese–Cardy, ties v450), 16 copies $\rightarrow c_-=8$. [C] Shale–Stinespring: gauge-invariant $\Rightarrow$ no pairing block $\Rightarrow$ the Bogoliubov map is implementable, GNS reps quasi-equivalent. R1 is upgraded from a bald axiom to “the unique quasi-free realisation, one-particle limit exhibited, modulo the named Araki/Shale–Stinespring functor”; with v463 the SEAM.EQUIV.01 residual is now entirely certification (named cited functors), zero open internal mechanism — closed modulo cited theorems. SEAM.EQUIV.01 stays [O]. Python-only (numpy: idempotency, kernel Cauchy-convergence, entanglement c -fit). [E][C][O]
v465_seed_ crosssector_ joint.py	SEED.CROSSSECTOR.01: the cross-sector one-parameter seed test — the θ_{13} -independent, externally-runnable slice of v306. TFPT reads a CMB EB/TB birefringence angle β and a quark Cabibbo angle $\lambda_C = V_{us} $ (unrelated in the SM) as two readouts of the ONE axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$. [E] plumbing: the axiom φ_0 reproduces the frozen λ_C , $\sin^2 \theta_{12}$, β to 10^{-9} . [E] CROSS-SECTOR (headline): back-solving φ_0 <i>independently</i> from β (CMB) and λ_C (quark) gives the <i>same</i> seed within $0.01\sigma_{\text{comb}}$ — a correlation only a shared-origin theory predicts. [E] one-parameter fit: the error-weighted mean of the three back-solved φ_0 reproduces the axiom to 0.06%; the joint χ^2 at the axiom (zero free parameters, 3 dof) is 0.007, $\chi^2/\text{dof} < 1$, combined pull 0.08σ . [E] θ_{13} EXCLUSION declared, not cherry-picked: adding $\sin^2 \theta_{13}$ raises χ^2 by ~ 4 (the documented $\sim 2\sigma$ pressure point, v328/v338), so this slice is <i>deliberately</i> θ_{13} -independent. [E] negative control: a data $\leftrightarrow$ formula shuffle explodes the cluster χ^2 by $> 5 \times 10^4$. A consistency / out-of-sample statistic (a subset of v306); φ_0 is FIXED by the axioms, so it upgrades <i>no</i> prediction. Python-only. [E]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v466_seed_leptonmass_channel.py	SEED.LEPTONMASS.01: a SIXTH, new-sector seed channel — the charged-lepton mass ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ points to the SAME axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$, extending the empirical seed over-determination (v306/v465) from three measurement sectors to five. [E] plumbing: the axiom φ_0 reproduces the frozen $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$ to 10^{-9} . [E] NEW SECTOR (headline): back-solving φ_0 from the measured POLE ratio $m_e/m_\mu = 0.004836$ (PDG) gives $\varphi_0 = 0.053113$, only -0.11% from the axiom — a charged-lepton MASS reading (Penning-trap masses), a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. [C] TRANSPORT CAVEAT: the formula is the SOURCE ratio, the datum the POLE ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6 = 8.78\%$, v393) but largely CANCELS for m_e/m_μ , so the 0.11% residual sits far inside the band — added as a CONSISTENT new sector, not a sharper pin. [E] EXTENDS THE OVER-DETERMINATION: the six-channel cluster (v306's five + this) has $\chi^2/\text{dof} < 1$ and the error-weighted mean stays at the axiom to $< 0.1\%$; with the corroborating heavy-quark m_t/m_b (-0.23%) it is a seven-channel, FIVE-sector agreement (neutrino mixing, CMB, quark mixing, lepton mass, quark mass). [E] negative control / power: a data↔formula shuffle explodes the cluster χ^2 by $> 100\times$; and not every mass ratio qualifies — $m_c/m_s = (34/47)/\varphi_0$ is scheme-ambiguous (FLAG 11–14), a WEAK channel deliberately not counted. [C] HONEST: theoretically still the ONE seed (v305 multiplicity 1) — an EMPIRICAL strengthening (a new measurement sector agrees), NOT theoretical independence; φ_0 is FIXED by the axioms, so it upgrades <i>no</i> prediction and closes <i>no</i> gate. Python-only. [E][C]
v467_thirdgen_mixing_pattern.py	FLAV.THIRDGEN.PATTERN.01: the three $\sim 2\sigma$ mixing tensions ($\theta_{13}, \theta_{23}, V_{cb}$) are ONE common $-\varphi_0^{\text{ret}}$ suppression ($-5.36\% \pm 1.67\%$, 3.2σ non-zero, -0.03σ from $-\varphi_0^{\text{ret}}$); the ladder-base forms $\lambda_C^2 e^{-5/6} / (1 - \varphi_0^{\text{ret}})/2 / \lambda_C^2 / (1 + \lambda_C)$ drop the joint χ^2 $10.4 \rightarrow 0.11$ with zero parameters; post-hoc candidate [O], frozen record unchanged, JUNO-era reactors + Belle II decide
v468_dm2_ratio_jarlskog.py	FLAV.DM2RATIO.01: $\Delta m_{21}^2 / \Delta m_{31}^2 = J_{\text{PMNS}} $ of the frozen matrix = 0.029653 vs measured $0.029805(792) \rightarrow -0.19\sigma$ — a parameter-free candidate relation for the previously unpredicted splitting ratio; the informal $m_2/m_3 \sim \pi\varphi_0^{\text{ret}}$ heuristic is now -2.44σ and retyped; NO floor $\Sigma m_\nu = 0.0588$ eV consistent; post-hoc [O], JUNO decides

continued on next page

Script	What it machine-checks (<i>continued</i>)
v469_seam_ crossedproduct_ route.py	SEAM.EQUIV.CROSSEDPRODUCT.01: the net-level crossed-product certification of the 128-spinor extension + the $R1 \rightarrow R1'$ invariant-level reduction. [E] LR LOCALITY INTEGER (index-2): the $SO(16)_1$ spinor has $h_s = N/16 = 16/16 = 1 \in \mathbb{Z} \Rightarrow$ statistics phase +1 — the Longo–Rehren criterion HOLDS, so the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1 \rtimes \{1, s\}$ is a LOCAL index-2 extension (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM $\mu(B) = 4/2^2 = 1 \Rightarrow$ HOLOMORPHIC $\Rightarrow (E_8)_1$ by the v463 pin (E_4/η^8 $q^1 = 248$ re-verified). The 2024/2025-preprint leg (AGT/AMT, v459) is DEMOTED to an independent second witness — the extension certification now rests on 1995–2001 PEER-REVIEWED journals. [E] INDEX-4 PARALLEL: the μ_4 glue $J = (s, \Lambda_1)$ has $h(J^k) = \{1, 1, 1\}$ ALL integer ($\frac{5}{8} + \frac{3}{8} = 1$; $\frac{1}{2} + \frac{1}{2} = 1$) — the v125 isotropy $q(k, k) = k^2$ at net level; $\mu = 16/4^2 = 1$, the same endpoint. [E] $R1'$ INVARIANTS COMPUTED: per-copy FHS Chern $C(M=1)=1$ vs $C(M=3)=0$, $\nu = 16 \times C = 16$, $c_- = \nu/2 = 8 = g_{\text{car}} + N_{\text{fam}}$; $\nu \bmod 16 = 0$ is EXACTLY Kitaev’s class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (Ann. Phys. 321 (2006), sixteen-fold way). [E] THE CHAIN COMPOSES: T1 (phase classification: Kitaev + c_- invariance, Kapustin–Spodyneiko PRB 101 (2020) / Kim et al. PRL 129 (2022)) $\rightarrow$ T2 (MMST CMP 397 (2023) + Böckenhauer 1996) $\rightarrow$ T3 (LR/BE + KLM CMP 219 (2001)) $\rightarrow$ T4 (Dong–Mason + v463) with matching in/out slots (v297 typing). [C] $R1 \rightarrow R1'$: collar_realizes (model-level fiat) replaceable by collar_invariants — {quasi-free [C] (v155/v160), gap [E] (v302), class D, $c_- = 8$ [E] (v456, from P1)}; quasi-freeness stays the ONE [C] hypothesis; Lean mirror FORM.SEAM.RESIDUAL.01. [O] NOT CLOSED: the cited theorems stay cited (certification, not re-proof); SEAM.EQUIV.01 stays [O]. Mixed exact (weights/μ/tower, Wolfram-mirrored) + numerical (FHS Chern, Python-only). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v470_alpha_inflow_level.py	ALPHA.QUILLEN.INFLOW.01: the α^3 level = the COMPUTED bulk Chern invariant + the seam F-normalisation = the affine embedding index $k_Y=5/3$. [E] TARGET RE-VERIFIED: $\alpha^{-1}=137.0359992168$ unique root; the Quillen split $a^3-2c_3^3a^2=8b_1c_3^6\ln(1/\varphi_{\text{seam}})$ at the root to $<10^{-35}$ (v341/v382). [E] THE INVARIANT COMPUTED: FHS Chern $C(M=1)=1$, $C(M=3)=0$ — the same integers as v367/v461, here read as the quantised $U(1)$ response of the collar phase. [C] $k_0= C =1$ UNDER THE EM1 READING: the a^3 coefficient EQUALS the computed bulk invariant of the SAME model that realises S3 — “why exactly 1” is answered by computation + the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten Rev. Mod. Phys. 88 (2016) $\eta=\text{CS}$ part of $\delta\log\det$; Quillen 1985 / Bismut–Freed CMP 106 (1986) determinant-line integrality) — REPLACING v435’s minimality assumption. [E] $k_Y=5/3$ EXACT: $\text{tr}_5(Y^2)/\text{tr}(T_3^2)=(5/6)/(1/2)=5/3$ (Ginsparg, Phys. Lett. B197 (1987)); $(3/5)\times\frac{41}{6}=\frac{41}{10}=b_1$ — the “GUT 3/5” IS the inverse embedding index, not an imported convention; carrier decomposition $\frac{41}{6}=\frac{20}{3}$ (fermions, 3×16-content) $+\frac{1}{6}$ (Higgs doublet). [C] F-NORMALISATION RETYPED: once the seam net is $(E_8)_1$ level 1 (SEAM.EQUIV.01), the $U(1)_Y$ normalisation $\langle J(z)J(w)\rangle=k/(z-w)^2$ is FIXED by k_Y — level-1 current-algebra rigidity, no continuous dial; the v434 [C] has ZERO independent content (a face of SEAM.EQUIV.01, per the v382 typing). [E] UNIFICATION: ONE invertible phase, TWO quantised responses — $c_-=16\times\frac{1}{2}\times C =8=g_{\text{car}}+N_{\text{fam}}$ (gravitational, feeds SEAM.EQUIV.01/S3, v456) and $C=1$ ($U(1)$, feeds EM1) — “the one theorem and the one number” are two faces of the SAME phase; closing R1’ (v469) feeds both. [O] NOT CLOSED: the bridge lemma “$\delta\log\det_c(\text{seam})$ = the inflow response” stays the named cited step; the EM1 reading stays [C]; ALPHA.QUILLEN.EXACT.01 stays [O]; α^{-1} stays [E] regardless. Mixed exact (k_Y/b_1 identities, Wolfram-mirrored) + numerical (root/split mpmath, FHS Chern, Python-only). [E][C][O]
v471_seam_horizon_replica.py	SEAM.EHMODEL.04: the Seam–Horizon replica chain exercised NUMERICALLY on a discretized seam collar — the kernel-identification premise of SEAM.THEOREM.01 (the residual isolated by v150/v151/v152) discharged at the FINITE/MODEL level for the seam’s own kernel, with REAL replica sheets. [E] REPLICA$\Rightarrow$EH FORM on a real operator: $dG/d\ln m = 2C(\gamma)$ on deficit cones $\gamma/2\pi = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ (0.01–0.05%) AND replica sheets $n=2$ (0.8%), $n=3$ (1.5%). [E] COEFFICIENT CUTOFF-INDEPENDENT (0.5% drift) while the ζ-scale μ_{lat} drifts with the discretization = the v152 anchor exhibited on data; NO canonical $\ln(m/\mu)=\frac{3}{4}$ emerges (recorded honestly). [E] BFK SPLIT MEASURED ON THE KERNEL: Schur identity exact ($\sim 10^{-16}$); the Calderón/Schur factor’s deficit slope is a PURE CUT-EDGE law $E(1-q)$ with tip term 0 (power check: the same law fails on the full determinant); the halves carry the whole tip = $2C_{\text{cone}}$ via Kac doubling (exact). [E] THE SEAM’S OWN SPECTRUM: the v302 transfer masses $m_2=6\ln\frac{3}{2}$, $m_3=6\ln 3$ sit ON the calibrated EH line ($\leq 0.05\%$); the Perron/attractor mode ($m=0$) is IR-DIVERGENT (runs exactly as $-2C\ln(\text{IR})$) — the recovery gap $\Delta = 6\ln\frac{3}{2}$ makes the induced-gravity coefficient finite: the same gap that makes the attractor unique makes $1/G$ finite. [O] HONEST SCOPE: finite/discretized level; SEAM.THEOREM.01 stays open — the residual retypes to the CONTINUUM scaling limit (MMST class, = the single residual of SEAM.EQUIV.01, v336) + the v152 anchor. Numerical (sparse lattice determinants), Python-only by nature. [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v472_quillen_ detline_moduli.py	ALPHA.QUILLEN.DETLINE.01: the DETERMINANT LINE over the $U(1)$-twist moduli of the collar model carries curvature = the inflow level — the finite Quillen/Dai–Freed shadow of the v470 bridge lemma, COMPUTED. [E] BLOCH LEVEL RE-VERIFIED: BZ Chern $C(M=1)=1$, $C(M=3)=0$, $C(M=-1)=-1$ (the v470/v367 integers). [E] THE QUILLEN OBJECT COMPUTED: the same collar $h(k)$ in real space on an $L \times L$ torus with TWISTED boundary conditions — the twist torus (θ_x, θ_y) IS the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the determinant line of the occupied one-particle frame (the free-fermion ground state) over that torus = 1 exactly (10^{-9}) at $M=1$, for $L=4$ AND $L=6$ (size-independent, as an integer must be). [E] CONTROLS: trivial collar $M=3 \rightarrow 0$; orientation flip $M=-1 \rightarrow -1$. [E] TWO TORI, ONE LEVEL: the twist-moduli integer equals the Bloch-BZ integer for all three M — the Niu–Thouless–Wu identification (PRB 31 (1985)), exhibited on the collar model that realises S3 (v460/v461). [E] SECTION GLOBALLY DEFINED: the Fermi gap (= 2.0) stays open over the WHOLE twist torus — the Dai–Freed section has no zero, the holonomy reading is globally valid. [C] THE READING: det-line holonomy over the $U(1)$ moduli = inflow level $k_0= C =1$ — the v470 bridge lemma “$\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$” holds VERBATIM at the finite/model level, where $\log \det$ of the occupied frame replaces $\log \det_\zeta$ (Quillen 1985; Bismut–Freed CMP 106 (1986); Dai–Freed JMP 35 (1994); FHS JPSJ 74 (2005)). [O] NOT CLOSED: the abstract-seam ζ-determinant identification (the continuum leg, = the SEAM.EQUIV.01 face); ALPHA.QUILLEN.EXACT.01 stays [O]; α^{-1} stays [E] regardless. Structurally the same manoeuvre as v471 for SEAM.THEOREM.01’s replica side. Numerical (dense eigh + FHS twist grid), Python-only by nature. [E][C][O]
v473_entropic_ action_bridge.py	GRAV.ENTROPIC.ACTION.01: the entropic-action bridge — Bianconi’s <i>Gravity from entropy</i> (Phys. Rev. D 111, 066001 (2025); arXiv:2408.14391) quantified as an EXTERNAL CANDIDATE for the local volume action behind TFPT’s parameter-free Einstein equation (v358/v359, honestly typed “equation of state, not a from-action quantisation” [O]). Her $S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})$ is the quantum relative entropy between the spacetime metric and the matter-induced metric (Araki-related, $\Delta^{1/2} = \tilde{G}\tilde{g}^{-1}$), a local covariant action whose variation gives Einstein + emergent nonnegative Λ at low coupling — the missing variational layer IF the bridge holds. [E] CARRIER HODGE COUNT: her $\Omega^{0,1,2}$ fiber on the FIVE-SLOT carrier $\mathbb{C}^5$ (not 4d spacetime, where it is $1+4+6=11$) counts $1+5+10=16=2^{g_{\text{car}}-1} = \dim S_{D_5}^+$, Hodge-folded onto $\Lambda^{\text{even}}\mathbb{C}^5$ (vector-space identity ONLY). [E] COEFFICIENT MATCH (the one quantitative pin): her weak-coupling $\mathcal{L} = 3\beta R - \alpha\mathcal{L}_m$ matched to $1/(16\pi G) = c_3/(2G)$ fixes her free constant EXACTLY, $\beta_B' = c_3/6 = 1/(48\pi)$; the $6 = 3 \text{ blocks} \times 2 = p_2 = R^+(A_3)$ (structural echo, not a proof). [E] ENTROPY POTENTIAL: $\Lambda_{\mathcal{G}} = \frac{1}{2\beta} \text{Tr}(\mathcal{G} - I - \ln \mathcal{G}) \geq 0$, quadratic at the fixed point $\Rightarrow$ with $\varepsilon_\star = e^{-\alpha^{-1}Q}$ the closed v60 branch $(3/4\pi^2)e^{-2\alpha^{-1}}$ is reproduced with the EXACT target $\text{Tr } Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3} \delta_{\text{top}}$ ([C] until Q is constructed). [E] FRULLANI: $-\ln A = \int_0^\infty \frac{dt}{t} (e^{-tA} - e^{-t})$ — her log action is an all-scale heat-kernel action, the right genre against $S_{\text{rel},\chi}$ (KMS cutoff). [E] R^2 KILL TEST: raw $(c_3/6)^2$ vs Starobinsky $1/(12c_3^7)$ ($M_{\text{scal}}^2 = c_3^7 \bar{M}_{\text{Pl}}^2$, v36) — EXACT gap $3(8\pi)^9 \approx 1.2 \times 10^{13}$ (13 orders): a direct identification of the raw entropic action with the TFPT R^2 sector is FALSE without a mechanism (light $\mathcal{G}$-trace mode generating c_3^{-9}, or KMS-spectral renormalisation) — pre-registered. [C] COMPRESSION CONJECTURE: $P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2} + \text{the operator-level Hodge items (Clifford, Dirac, leaf involution, Calderón polarisation) stay open; Lorentzian positivity NOT automatic — TFPT’s OS route is the sturdier construction. [O] NOTHING CLOSES: v359’s equation-of-state status stays [O]; zero TFPT knobs added. Mixed exact (sympy, Wolfram-mirrored) + one quadrature (mpmath). [E][C][O]$

continued on next page

Script	What it machine-checks (<i>continued</i>)
v474_entropic_hodge_carrier.py	GRAV.ENTROPIC.HODGE.01: the OPERATOR level of the entropic-action bridge (work packages 1 + 4-algebra of v473). [E] CAR⇒CLIFFORD: the ten Γ's from $(a_k^\dagger, a_k)$ on the carrier Fock space $\Lambda^\bullet \mathbb{C}^5$ ($\dim 2^{g_{\text{car}}}=32$) satisfy $\{\Gamma_a, \Gamma_b\}=2\delta_{ab}$ exactly (all 100 pairs, integer matrices) — the $\mathfrak{so}(10)=D_5$ Clifford algebra acts on the carrier exterior algebra. [E] $S^+=\Lambda^{\text{even}}$ AS D_5-MODULE: the 45 bilinears $[\Gamma_a, \Gamma_b]/4$ commute with $\gamma=(-1)^N$, span exactly $45=\dim \mathfrak{so}(10)$ and leak NOTHING from the 16-dim even subspace — the half-spinor IS the even exterior algebra WITH its spinor structure (operator level of v2/v44/v197). [E] HODGE-DIRAC SYMBOL: $c(\xi)=a^\dagger(\xi)-a(\xi)$ satisfies $c(\xi)^2=- \xi ^2 \text{Id}$ for SYMBOLIC ξ — Bianconi's $D=d+\delta$ is Clifford multiplication at symbol level. [E] THE FOLD IS $5\rightarrow\bar{5}$: the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the NEGATIVES of those of Λ^1, so the Hodge fold $\star: \Lambda^1\cong\Lambda^4$ conjugates the vector block — her fiber $1+5+10$ becomes the GUT half-spinor $16=1+\bar{5}+10$ (the SM carrier decomposition), not a mere dimension match. [E] HONEST OBSTRUCTION: $\Omega^{\leq 2}$ is NOT Clifford-invariant ($c(\xi)$ maps Λ^2 into Λ^3); only Λ^{even} is (even algebra) — the bridge must go through the fold, never through degree truncation (constraint on AP2). [E] Q-TARGET ENUMERATION (the v473 conjecture DECIDED): $\text{Tr } Q^2=d k^2 c_3^4=32c_3^4$ with INTEGER k has exactly the supports $(d,k)=(2,4), (8,2), (32,1) = \{ \mathbb{Z}_2 , g_{\text{car}}+N_{\text{fam}}=\text{rank } E_8, 2^{g_{\text{car}}}=\dim \Lambda^\bullet\}$ — minimal uniform solution $q=c_3^2=1/(64\pi^2)$ per Fock mode; the two-form block $d=10$ (the “pair sector” conjecture) and the fiber $d=16$ need irrational multipliers $\sqrt{16/5}, \sqrt{2} \Rightarrow$ KILLED in uniform form; bonus identity $\text{Tr } Q^2/\delta_{\text{top}}=32/48=2/3=\dim \Lambda^\bullet/\Omega_{\text{adm}}$. [C] the physical choice among the three admissible supports; [O] AP2 compression + R^2 mechanism stay open, no marker moves. Mixed exact: integer Fock matrices (numpy, exact in double) + symbolic sympy (weights/enumeration Wolfram-mirrored). [E][C][O]
v475_entropic_scalaron_gate.py	GRAV.ENTROPIC.SCALARON.01: the R^2 kill test of the entropic-action bridge EXECUTED (work package 5 of v473) + the explicit Lorentzian-positivity witness. [E] MAX-SYMMETRIC EIGENVALUES (her Appendix-B flattening verified): $\tilde{R}\tilde{g}^{-1}$ has eigenvalues EXACTLY $\{R; R/4 \times 4; R/6 \times 6\}$ on a general diagonal 4d metric, incl. the 6×6 two-form flattening. [E] EXACT VACUUM $f(R)$: $\mathcal{L}_B(R)=-[\ln(1-\beta R)+4\ln(1-\beta R/4)+6\ln(1-\beta R/6)]=3\beta R+\frac{17}{24}\beta^2 R^2+O(R^3)$ — the 3 re-derives her Eq. (45) (= the v473 pin), $\frac{17}{24}=\frac{1}{2}(1+\frac{1}{4}+\frac{1}{6})$ is the EXACT tensorial R^2 factor. [E] RAW SCALARON TRANS-PLANCKIAN: with $\beta'=c_3/6$ both routes (Starobinsky matching AND $f'/(3f'')$) give $m_{\text{raw}}^2=\frac{4608\pi^2}{17}M_{\text{Pl}}^2$, $m_{\text{raw}}\approx 51.7 M_{\text{Pl}}\approx 1.3\times 10^{20}$ GeV — HEAVY, not light (TFPT: $c_3^{7/2}M_{\text{Pl}}=3.06\times 10^{13}$ GeV); required enhancement EXACTLY $m_{\text{raw}}^2/m_{\text{TFPT}}^2=\frac{72}{17}(8\pi)^9\approx 1.69\times 10^{13}$ in mass² — v473's interpretation 2 (the TFPT scalaron as the light $\mathcal{G}$-trace mode) is KILLED in naive form. [E] DOMAIN: first log pole at $R=1/\beta=384\pi^2 M_{\text{Pl}}^2$ (trans-Planckian) — the raw action is well-defined at all sub-Planckian curvatures; the obstruction is the MASS, not the domain. [E] LORENTZIAN WITNESS: a timelike gradient gives Gg^{-1} eigenvalue $1-\alpha v^2\leq 0$ for $v^2\geq 1/\alpha$ (ln undefined); spacelike/Euclidean positive for all v — the v473 [C] caveat exhibited, the OS/reflection-positive route sturdier. [C] VERDICT: interpretation 1 survives (the entropic action covers the Einstein + Λ level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving R^2 route. [O] unchanged: AP2 compression + the R^2 gate; no marker moves. Exact (sympy, Wolfram-mirrored). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v476_entropic_compression_toy.py	GRAV.ENTROPIC.COMPRESS.01: the AP2 compression conjecture of the entropic-action bridge (v473) made WELL-POSED on an exactly solvable model. Numerical facts on a staggered-mass chain: the ground-state covariance is a PROJECTION (pure bulk, thermal boundary — the seam situation); the compression $C_A=PCP$ has spectrum strictly inside $(0,1)$ with finite Peschel $K=\ln((1-C_A)/C_A)$ and $S>0$ (v155 in the bridge context). ORDER OBSTRUCTION (the reading DECIDED): on the pure bulk $f(x)=x/(1-x)$ is SINGULAR on $\text{spec}(C)=\{0,1\}$ — the literal operator-side reading $Pf(C)P$ of the v473 conjecture is ILL-POSED while $f(PCP)$ is finite: the compression identity can only be read STATE-SIDE (build Δ_Σ from the COMPRESSED relative metric), matching Bianconi's own local-metric construction — AP2 retyped, not weakened. [E] BLOCK ALGEBRA (symbolic): $[C^2]_{AA}-(C_{AA})^2=BB^t$ and $[C^3]_{AA}-(C_{AA})^3=ABB^t+BB^tA+BDB^t$ — first order in the cross block vanishes identically, so for ANY analytic f the two readings agree to SECOND order: the modular (entanglement) content lives exactly in that quadratic correction. GAP CONTROL at the bounded (covariance) level: the bulk identity $f(C_T)=e^{-h/T}$ holds to 10^{-11} relative; the mismatch obeys the quadratic law $d \leq x^2$ and falls with the gap in the gapped regime, the decoupled cut agrees EXACTLY — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\text{eff}}>0$, v76/v337). [C] the retyped conjecture; [O] AP2 stays open (the continuum seam statement); no marker moves. Numerical (numpy lattice + one symbolic block identity), Python-only by nature. [C][O]
v477_entropic_scaleflow.py	GRAV.ENTROPIC.SCALEFLOW.01: the scale-flow representation of the entropic action — v475's surviving interpretation 3 typed PRECISELY: the $(72/17)(8\pi)^9$ gap collapses to ONE moment condition on the scale weight, satisfied by the SAME KMS moment TFPT already fixes (v36: $f_0=6(4\pi)^2/c_3^7$); zero new dials, nothing closes. [E] SCALE-FLOW IDENTITY: $-\ln a = \int_0^\infty \frac{dt}{t}(e^{-ta}-e^{-t})$ (symbolic) $= 2 \int_0^\infty \frac{dx}{x}(e^{-a/x^2}-e^{-1/x^2})$ ($t=1/\chi^2$, 10^{-20} quadrature) — Bianconi's log action IS the FLAT scale-integral of relative heat-kernel actions; TFPT's $S_{\text{rel},\chi}$ evaluates ONE KMS scale with a weight f: the bridge is a choice of SCALE MEASURE. [E] MOMENT DICTIONARY: a weight $w(t) dt/t$ gives $3\beta R \mu_1 + \frac{17}{24}\beta^2 R^2 \mu_2$ ($\mu_1 = \int w e^{-t} dt$, $\mu_2 = \int w t e^{-t} dt$); the flat weight has $\mu_1=\mu_2=1$ = the v475 raw coefficients — the raw action is the unit-moment point of a moment family. [E] THE ONE MOMENT CONDITION: with the weight-dressed EH pin $\beta' \mu_1=c_3/6$ the scalaron is $m^2=\frac{4608\pi^2}{17}\frac{\mu_1^2}{\mu_2}M_{\text{Pl}}^2$; demanding $m^2=c_3^7 M_{\text{Pl}}^2$ forces EXACTLY $\mu_2/\mu_1^2=\frac{72}{17}(8\pi)^9$ — the v475 kill-test number IS a moment ratio; and the closure identity $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}=c_3^7$ holds IDENTICALLY: the 13 orders are a scale-measure datum, not a bridge mismatch. [E] v36-CONSISTENCY: $f_0=6(4\pi)^2/c_3^7 \Leftrightarrow M^2/M_{\text{Pl}}^2=c_3^7$ — the same mass from both parametrisations, ZERO new dials. [C] a CONSISTENCY statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein + Λ level and the scale-flow FORM, TFPT supplies the measure); negative side-record: the exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain (kept out of the suite, logged). [O] unchanged: the R^2 gate and v359's typing. Exact (sympy + one quadrature, Wolfram-mirrored). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v478_entropic_continuum_legs.py	GRAV.ENTROPIC.LEGS.01: first computable steps on the TWO remaining legs of the entropic-action bridge (v473–v477). LEG A (AP2 continuum): the compressed critical interval has $S(L)=(c/3)\ln L+k$ with $c_{\text{est}}=1.00089/1.00038/1.00021$ (monotone $\rightarrow$ the Dirac $c=1=2\times\frac{1}{2}$, the v450 reading; gapped control $\sim 6\times 10^{-8}$, area law); the Peschel modular Hamiltonian $K=\ln((1-C_A)/C_A)$ organises on the CHM weight $\beta(x)=x(L-x)/L$ (corr 0.87$\rightarrow$0.99, plateau drift shrinking; constants deliberately NOT asserted — the lattice EH is only asymptotically local); even-range bands vanish identically (particle-hole, $\sim 10^{-36}$), odd tail 1.8% parabola-enveloped — the state-side Δ_Σ (v476’s forced reading) FLOWS TO the CHM/BW geometric form that TFPT’s Einstein derivation consumes (v323/v358): the bridge’s modular side and TFPT’s gravity side MEET in the continuum limit. LEG B (the measure): [E] single-scale corollary — $d\mu=\delta(t-t_0)dt$ gives $\mu_2/\mu_1^2=e^{t_0}$ exactly, so the v477 condition forces $t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461$ — ONE dimensionless KMS time, exact closed form; DECLINED: $t_0\neq h(E_8)=30$ (difference 0.461; the near-miss is rejected per the v354/v355 no-free-pattern rule); deriving the measure from the entropic structure stays OPEN. [C] the meeting point; [O] both legs, no marker moves. High-precision numerical (mpmath, 50 digits) + one exact corollary, Python-only by nature. [C][O]
v479_kronheimer_quiver_bridge.py	SEAM.KRONHEIMER.01: the R3 “combinatorial-to-geometric bridge” of the $\det K=1$ keystone (v344 — “the rewrite attractor graph IS the du Val singularity”) is an ESTABLISHED THEOREM: Kronheimer’s ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685), with ALL finite input data already compiler outputs. [E] MARKS = NULL = PERRON: $C_{\text{aff}}\delta=0$ and $A\delta=2\delta$ on the affine-E_8 McKay graph; multiset $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ = the $2I$ irrep dimensions = the v312 rewrite-attractor Perron vector. [E] DIMENSION COUNT: $\dim_{\mathbb{R}} M - 4\dim_{\mathbb{R}} G = 4(\sum_{\text{edges}} d_i d_j - \sum d_i^2 + 1) = 4$ — the hyper-Kähler quotient of the marks quiver is 4-real-dimensional (the $\mathbb{C}^2/\Gamma$ geometry); $\sum_{\text{edges}} d_i d_j = \sum d_i^2 = 120$ forced by the null-vector identity. [E] GAUGE DATUM: $\sum \delta_i^2 = 120 = 2I = \mu_4 h(E_8)$, $h(E_8)=30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}$. [E] RESOLUTION: $8=\text{rank } E_8$ exceptional spheres with intersection form $-\text{Cartan}(E_8)$, $\det=1$ and SNF = identity (the Poincaré-sphere link = the $\det K=1$ face, v232/v344); deformation space $\mathfrak{h} \otimes \mathbb{R}^3$, $\dim 3 \cdot 8 = 24$. [C] the theorems are CITED, not re-proved (the same import class as NPW26/MMST/AGT, v424/v336/v459). [O] the residual premise TRANSFORMS, it does not vanish: “the raw seam supplies ALE hyper-Kähler data asymptotic to the $(2, 3, 5)$ cone” = the SAME single arrow as SEAM.EQUIV.01 (v346 L2); no gate closes. Exact (sympy; Wolfram mirror added, counted with the verified 2026-07-21 engine run). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v480_multilocal_ four_interval.py	QGE0.MULTILOCAL.01: the four μ_4 marks make the seam circle a FOUR-INTERVAL free-fermion geometry — the class whose multi-interval modular theory is EXACTLY KNOWN (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors; Rehren–Tedesco multilocal fermions; KLM: holomorphic $\Leftrightarrow$ trivial multi-interval inclusion, $\mu=1$ — the SAME one bit as $\det K=1$). On the exactly solvable antiperiodic critical ring ($N=256$, $Q=64$, $\ell=48$): [E] BINARY CLOCK: the quarter-turn clock is a signed permutation with $\rho^4 = -1$ EXACTLY — the fermionic μ_4 clock is the order-8 double-cover lift (the same $\mathbb{Z}_2$-lift pattern as $2I$ over I). [E] EXACT SECTOR DECOUPLING: the primitive-8th-root character projectors resolve the identity and commute with the reduced covariance at 10^{-15} (the Rehren–Tedesco diagonal 4-fold-cover isomorphism at covariance level). [E] COVER/TWIST: each sector kernel is the SINGLE-interval quarter-ring kernel with boundary twist λ_k; the union of sector spectra = the full 4-interval spectrum at 10^{-14}. [E] CLOCK INVARIANCE MANIFEST: $\rho C \rho^{-1} = C$ at $10^{-14} \Rightarrow [\rho, K]=0$ by functional calculus — $\omega \circ \rho = \omega$ HOLDS at the 4-interval state level; NEGATIVE CONTROL: displacing ONE interval by one site re-couples the sectors at 10^{-2}. CONJUGATE-POINT MIXING (numerical): every cross-interval row of K peaks on $s'-s \leq 2$ at both sizes — the lattice shadow of the Casini–Huerta mixing term. [C] the continuum multi-interval theory is CITED; [O] the raw-seam premise (QGE0.SYM.01/SEAM.EQUIV.01) stays OPEN — mechanism exhibited, not closed. Numerical (numpy), Python-only by nature. [E][C][O]
v481_seesaw_ carrier_ladder.py	FLAV.NUSCALE.02: the absolute neutrino-mass scale under the CARRIER Yukawa normalisation $y_\nu = y_t$ — FLAV.NUSCALE.01's two-parameter (y_ν, M_R) trade-off collapses to ONE parameter and the required M_R lands INSIDE the compiler's own PS window; the nearby integer c_3-rung is explicitly DECLINED at 1-loop (v354/v355). CANDIDATE class (genre v467/v468); the frozen record (v84) is untouched and NO gate closes. CARRIER NORMALISATION (conjectured premise; named premise): one $SO(10)$ 16 per family + minimal Yukawa sector ($\mathbf{10}_H$ / PS (1, 2, 2)) $\Rightarrow y_\nu = y_t$ at the matching scale — the trade-off collapses to M_R alone. INVERSION INSIDE THE WINDOW (numerical): the observed $m_3=0.0503$ eV demands $M_R=9.3 \times 10^{13}$ GeV, INSIDE $[M_{PS}, M_{GUT}] = [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ (v249), at $\log_{c_3}(\bar{M}_{Pl}/M_R) = 3.15$ ($y_\nu=1$ gives the structureless 2.58 — the old FLAV.NUSCALE.01 number). [X] INTEGER-RUNG HONESTY GATE: $M_R = c_3^3 \bar{M}_{Pl} = 1.53 \times 10^{14}$ GeV predicts $m_3 = 0.030$ eV (40% low at 1-loop; ADKLR κ run-down factor 0.794) — NEAR the rung (3.15 vs 3), NOT ON: the ladder pin is DECLINED; the decision computation is NAMED (2-loop SM + explicit PS threshold matching). FALSIFIABLE BAND (numerical): over the PS window m_3 spans $[0.002, 0.115]$ eV bracketing the observation; the M_{PS} end ($\Sigma \sim 0.124$ eV) is already in tension with DESI ~ 0.072 — cosmology cuts the window from below; the NO floor $\Sigma m_\nu = 0.059$ eV stays the falsifiable edge (v272). [C] / [O] candidate class; the absolute scale stays formally open. Numerical (explicit 1-loop RG: gauge/y_t/λ up, ADKLR Weinberg-operator down), Python-only by nature. [C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v482_seesaw_rung_ decision.py	FLAV.NUSCALE.03: the decision computation NAMED by v481, EXECUTED in bracketed (assumption-robust) form — the integer rung $M_R = c_3^3 \bar{M}_{Pl}$ is EXCLUDED as an unstructured exact pin, conditional on the carrier premise $y_\nu = y_t$. THE GAP: central $m_3(\text{rung}) = 0.0301$ eV vs observed 0.0503 — rescue factor $\times 1.670$ (+67%, 0.154 c_3 -rungs). BRACKETED EXCLUSION: with $m_t(\overline{\text{MS}}) \pm 3$ GeV ($> 3\sigma$), $\alpha_3^{-1} \mp 0.10$ ($> 3\sigma$), a $\pm 10\%$ band on the ADKLR κ run-down and the PS-leg Yukawa beta scaled $\times [0.5, 1.5]$, the largest single shift is $\times 1.10$ and the ALL-favourable combination reaches only $\times 1.165$ — far below $\times 1.670$: the v481 1-loop miss is NOT an RG artifact. [C] NAMED ESCAPE, RECORDED AND DECLINED: only a third-generation Majorana STRUCTURE factor $r \sim 1.67$ revives the rung; post-hoc $r_{\text{needed}} = 1.6696$ sits 0.18% from $g_{\text{car}}/N_{\text{fam}} = 5/3$ — no mechanism forces a 3/5 Clebsch on $M_{R,3}$ today, so this is a named coincidence with a named missing mechanism (PS Majorana-sector Clebsch), DECLINED per v354/v355 (the $t_0 \sim 30$ pattern of v478). [O] the v481 one-parameter window candidate (NO floor $\Sigma m_\nu = 0.059$ eV) never used the integer rung and STANDS; frozen record untouched; nothing closes. Numerical (explicit 1-loop RG + bracketed envelopes), Python-only by nature. [C][O][X]
v483_pillowcase_ exact_traces.py	HYP.PHIO.GEOM.01: the GEOMETRIC side of the φ_0^{ret} puncture target (HYP.PHIO.PUNCTURE.01/v408) is EXACT — every twisted/orbifold heat trace of the flat $\tau = i$ pillowcase is a t -INDEPENDENT RATIONAL, so the π^{-4} in $d_{\text{top}} = 48c_3^4$ cannot come from flat orbifold geometry at ANY order. [E] TWISTED TRACES EXACTLY 1: $\text{Tr}(\gamma e^{t\Delta}) = 1$ for $\gamma = \sigma(z \mapsto -z)$, $\rho(z \mapsto iz)$, ρ^3 at every tested $t \in [0.05, 3]$ (position-space Gaussian image sums / erf closed forms + 2d quadrature, residual $< 10^{-9}$; mode-space proof: only $n=0$ is γ -diagonal) — t -independent, no asymptotic tail. [E] ATIYAH–BOTT MATCH: $1 = 4 \times \frac{1}{4}$ ($\det(1-d\sigma) = 4 = \mu_4 $) and $1 = 2 \times \frac{1}{2}$ ($\det(1-d\rho) = 2 = \mathbb{Z}_2 $). [E] PILLOWCASE DATA EXACT: orbifold heat trace $= \frac{1}{2} \text{Tr}_{T^2} + \frac{1}{2}$ (contact term EXACTLY $\frac{1}{2} = 4 \times \frac{1}{8}$ per mark, all t); clock-equivariant trace exactly 1 (all t). [E]/[O] THE REDUCTION: the analytic content of $d_{\text{top}} = 48c_3^4 = \frac{3}{256\pi^4}$ must sit in the per-insertion COUPLING normalisation — $4 = \mu_4 $ insertions of weight c_3 give c_3^4 , multiplicity $\Omega_{\text{adm}} = 48$; the bare $1/(2\pi)$ -propagator 4-cycle differs by the exact RATIONAL $\frac{3}{16}$; the ONE remaining [O] is the rule “per-mark weight = c_3 ” (seam zeta/Green normalisation, the same residual class as EM.WARD.02). The target narrows from “derive the term” to “derive the weight”; HYP.PHIO.PUNCTURE.01 stays open; no gate closes. Numerical verification of exact statements + sympy exact arithmetic; Python-only by nature. [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v484_seam_contact_unit.py	SEAM.CONTACT.UNIT.01: the shared “c_3 per boundary insertion” rule of the two open analytic targets (ALPHA.QUILLEN.EXACT.01 ladder $\{0, 3, 6\}$; HYP.PHIO.PUNCTURE.01 per-mark weight c_3) is ONE identity and NOT a new unknown: it is the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/ \mu_4$ (v239) — one bare boundary propagator ORBIT-AVERAGED over the four marks. [E] UNIT CHAIN: $\frac{1}{2\pi} = 4c_3 = \mu_4 c_3 = \mathbb{Z}_2 \chi(S^2)c_3$. [E] MECHANISM DERIVED ON THE SEAM CIRCLE: $G(d) = -\frac{1}{\pi} \ln 2 \sin(d/2)$ (polylog identity, 40 digits) takes INTEGER multiples of $c_3 \ln 2$ at the μ_4 separations — $G(\pi/2) = -4c_3 \ln 2$, $G(\pi) = -8c_3 \ln 2$, so $G_{\text{marks}} = -4c_3 \ln 2 \cdot \text{circ}(0, 1, 2, 1)$ with $\text{spec} = \{4, -2, 0, -2\}$; the μ_4-invariant (orbit-averaged) insertion carries $(GV) _{\text{marks}} = -(c_3 \ln 2) C$ EXACTLY, so the log-det contact expansion is exactly graded in c_3 per insertion with integer cycle combinatorics $\text{Tr}(C^k) = 4^k + 2(-2)^k$. [E] SUITE-WIDE GRAMMAR: the $F_{U(1)}$ ladder $\{0, 3, 6\}$ (v3); φ_0^{ret} ladder $\{1, 4\}$ (v37); the Λ prefactor $\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2$ (v60) — the SAME multiplicity $\Omega_{\text{adm}}=48$ as the puncture, at TWO insertions; the $g-2$ seam loop divides by the same $2\pi = 1/(4c_3)$. [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ($48 = \Omega_{\text{adm}}$, $41 = 10b_1$); the finite cycle sector is proven. Anti-numerology: the integers are forced by the square configuration; no target coefficient is claimed. Exact (sympy + 40-digit mpmath polylog); Python-only by nature. [E][C][O]
v485_contact_diagonal_closed.py	SEAM.CONTACT.UNIT.02: the DIAGONAL channel of the merged analytic target (v484) settled at the computable level — the merged residual COLLAPSES ONTO THE KEYSTONE. [E] DIAGONAL ZERO AT THE KMS SCALE: the renormalised coincident-point Green function on circumference ℓ is $G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)$ exactly — it VANISHES iff $\ell = 2\pi = 1/(4c_3)$, the v239 KMS seam circumference: the seam scale is the unique zero-self-energy scheme point. [E] THE LOG IS THE SCALE CHANNEL: rescaling gives $\frac{1}{\pi} \ln \kappa$ and nothing else — matching the functional structure ($F_{U(1)}$ carries exactly ONE log term, the transport $8b_1 c_3^6 \ln(1/\varphi_{\text{seam}})$; the φ_0^{ret} split is log-free); [C] reading: log = diagonal/scale channel, log-free = contact data at the KMS point. [E] CLOSED-FORM RESUMMATION (BFK route, v151 class): the finite-rank mark determinant resums exactly, $\det(I - uC) = (1 - 4u)(1 + 2u)^2$ with $u = \varepsilon c_3 \ln 2$ — all orders; the linear term is ABSENT precisely because the diagonal vanishes ($\text{Tr } C = 0$); the series matches v484’s $\text{Tr}(C^k) = 4^k + 2(-2)^k$. [E] WEIGHT DICHOTOMY: the SAME $48 = 3 \times 16$ admissible states carry BOTH multiplicities — flat count $48 = \Omega_{\text{adm}}$ (the φ_0^{ret} puncture) vs $Y^2/\text{loop}/\text{GUT-weighted } 41 = 10b_1 = 40 + 1$ ($\text{Tr}_{16} Y^2 = \frac{10}{3}$ exact, $\nu^c = 0$; Ginsparg $(3/5)(41/6) = 41/10$, v470) — one state set, two response weights. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged ALPHA.QUILLEN + PHIO.PUNCTURE target is proven; the single remaining [O] is the identification of the abstract seam ζ-det with this glued mark determinant — a face of SEAM.EQUIV.01, exactly the v382 typing, now substantiated computationally. No gate closes. Exact (sympy + 30-digit mpmath + exact charge sums); Python-only by nature. [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v486_transfer_full_rule.py	HYP.REWRITE.02: the FULL physical transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v54/v221) from ONE forced local rule — the lazy $\mathbb{Z}_2$-pair walk; the $(1/3)^6$ mode, previously generator-less, gets its local-rule origin (sharpens v327, whose uniform rule produces $\{1, \frac{2}{3}, 0\}$ and can never generate the third mode). [E] UNIQUENESS: for $M(s, h) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & s & h \\ 0 & h & s \end{pmatrix}$ (one absorbing family channel + SYMMETRIC $\mathbb{Z}_2$ pair) the survival spectrum is $\{s+h, s-h\}$; demanding the physical pair $\{\frac{2}{3}, \frac{1}{3}\}$ forces UNIQUELY (stay, hop, leak) = $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/ \mathbb{Z}_2 , 1/(\mathbb{Z}_2 N_{\text{fam}}), 1/N_{\text{fam}})$ — atom form recorded (honest direction: spectrum $\rightarrow$ rates). [E] SIX-HAND GENERATION: over the order-6 = $2N_{\text{fam}} = R^+(A_3)$ hand, $\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$ exactly; decay gaps $6 \ln \frac{3}{2}$ (the recovery gap, v76) and $6 \ln 3$ (the subdominant gap, now generator-full). [E] SUPERSESSSION: the uniform rule's zero mode persists under every power — no iterate reaches $(1/3)^6$; its λ_2 arity story ($\frac{2}{3} = \mathbb{Z}_2 /N_{\text{fam}}$) is unchanged. [E] sub-stochastic, symmetric (detailed balance), absorbing democratic attractor unchanged (v317). [O] the arity $\{2, 3\}$ stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay = $1/ \mathbb{Z}_2$ as the sheet-symmetric choice of the $\mathbb{Z}_2$ involution). Exact (sympy); mirrorable, deferred to the next verified engine run. [E][O]
v487_transfer_clock_rungs.py	HYP.REWRITE.03: the lazy split $(\frac{1}{2}, \frac{1}{6})$ is FORCED by the resummed clock — the v486 [O] 'split selection' closes onto v124 structure; sharpens HYP.REWRITE.02, no gate closes. [E] RUNG READING: the lazy rule's one-step spectrum is EXACTLY the complete clock ladder below the wall, $\text{eig}(M) = \{1 - n/3 : n = 0, 1, 2\} = \{1, \frac{2}{3}, \frac{1}{3}\}$; v327's uniform rule puts its second pair mode ON the wall ($0 = 1 - 3/3$) — the degeneracy re-read as 'odd channel collapsed to the wall'. [E] DECK PARITY = RUNG INDEX: $[B, \sigma] = 0$; the σ-even pair mode carries rung $n = 1$ (survival $\frac{2}{3}$), the σ-odd mode rung $n = 2$ (survival $\frac{1}{3}$) — each clock step flips the deck parity; the structural home of the v486 parity assignment (ω_1 even-channel, ω_2 odd-channel). [E] FORCING: faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) + $\mathbb{Z}_2$ equivariance + rates ≥ 0 force UNIQUELY both the split $(\frac{1}{2}, \frac{1}{6})$ AND the assignment even $\rightarrow$ rung 1, odd $\rightarrow$ rung 2 (the swap needs $h = -\frac{1}{6} < 0$); the 'sheet-symmetric choice' candidate is superseded by this proof. [E] BEND COROLLARY: $\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3$ — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. [O] the channel count $3 = N_{\text{fam}}$ is forced by the clock wall; the ladder stays an exact identity on frozen spectra (semiclassical derivation R1 open); the arity $\{2, 3\}$ stays the anchor input. Exact (sympy); mirrorable, deferred to the next verified engine run. [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v488_majorana_clebsch_door.py	FLAV.NUSCALE.04: the 3/5 CLEBSCH DOOR of the seesaw rung, ENUMERATED AND CLOSED — v482's named escape (a third-generation Majorana factor $r \approx 1.6696$, 0.18% from $g_{\text{car}}/N_{\text{fam}} = 5/3$) is decided NEGATIVE at the group-theory level; the v481 window candidate stands as the honest endpoint. [E] NO RENORMALISABLE CHANNEL: $\text{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$; the $\overline{\mathbf{126}}$ is NOT in the E_8 hull $\{1, 10, 16, 45\}$ (v247) — M_R only via the $d=5$ operator $(\mathbf{16}_F \overline{\mathbf{16}}_H)^2/\Lambda$ with the E_8-allowed $\mathbf{16}_H$. [E] CHANNEL LIST: $\mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}$; the hull admits singlet + $\mathbf{45}$ insertions ($\mathbf{210}$ absent). [E] CHARGE TABLE: $Y = T_{3R} + (B-L)/2$ on all 16 states, $\text{Tr}_{16} Y^2 = \frac{10}{3}$ (v485); $(B-L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$. [E] THE $\{2, 3\}$-SMOOTH THEOREM: every nonzero ν^c channel weight $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$ is $\{2, 3\}$-smooth and the multiplicative closure of $\{2, 3\}$-smooth rationals never contains $\frac{5}{3}$; the UNIQUE natural 5 of the embedding, $k_Y = \text{Tr} Y^2 / \text{Tr} T_{3L}^2 = \frac{5}{3}$ (Ginsparg, v470), is a full-multiplet trace whose direction has $Y(\nu^c) = 0$ EXACTLY — structurally decoupled from the Majorana operator. [E] GENERATION-BLINDNESS: the Clebsch is a family-space scalar; $\text{diag}(1, 1, \frac{3}{5})$ cannot arise from group theory. [O] compiler-flavor inventory typed and declined: exact $\frac{3}{5}$'s only in wrong slots (K lepton gen2/gen1; $R + L$ lepton gen3/gen2), no degenerate-base shape, no M_R link (v354/v355). Exact (Fraction/sympy); mirrorable, deferred to the next verified engine run. [E][O]
v489_seam_modular_commutator.py	SEAM.CCC.01: the chiral central charge $c_- = 8$ measured DIRECTLY from the bulk ground state of the v367 seam lattice model, via the Kim–Shehab–Kim modular commutator $J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-$ on a 120° pie tripartition (PRL 128 (2022)) — the FIRST ground-state witness for premise (i) of SEAM.EQUIV.01 ($c_- = 8$) INDEPENDENT of both the Chern route (v367 FHS curvature) and the counting route (v378 KLM/det-Cartan): J is read off the entanglement data of the exact BdG ground state alone (Peschel covariance, $W = -2 \text{artanh } \Gamma$). [E] GROUND-STATE PURITY: $\text{eig}(i\Gamma) = \pm 1$ to 6×10^{-15}. [E] PER-LAYER $c_- = \frac{1}{2}$: $c_- = 0.499998$ at $L=32, R=12$ (0.499793 at $L=16 \rightarrow 0.500000$ at $L=48$, dev -1.8×10^{-7}). [E] CONTROLS: trivial $M=3 \rightarrow 0.000000$; orientation flip $M=-1 \rightarrow -0.499986$. [E] LAYER ADDITIVITY EXACT (1.4×10^{-14}): J additive over decoupled layers, so the carrier value is $16 \times$ the per-layer value. [E] 16-LAYER CARRIER $c_- = 8 = g_{\text{car}} + N_{\text{fam}}$: direct 16-layer run 7.9967 ($L=16$, tol 10^{-2}; $L=24$ gives 7.999781), additivity route 8.0000 (tol 10^{-3}). [E] CLIP INVARIANCE: $\varepsilon = 10^{-8}$ vs 10^{-12} moves c_- by 5×10^{-8}. [O] HONEST FENCE: pins $c_- = 8$ on the LATTICE ground state (third independent witness); does NOT close SEAM.EQUIV.01 — the continuum scaling limit (MMST v336) stays [O], the μ_4 condensation $SO(16)_1 \rightarrow (E_8)_1$ stays the cited KLM step (v378). Numerical (numpy), Python-only by nature. [E][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v490_seam_parity_census.py	SEAM.MU.01: the T^2 spin-structure fermion-parity census for the v367 seam lattice model — the lattice ground-state witness of “$\mu_{\text{pre}}=4$ before condensation” (step 1 of the v378 KLM chain) + the 16-fold-way bridge $\theta_v=1$ with the MEASURED $\nu=2c_-=16$ (v489). Ground-state parity per boundary-condition sector (AA, AP, PA, PP), determined TWICE: Read–Green TRIM product AND real-space Majorana Pfaffian sign (Schur), relative to the trivial $M=3$ phase. [E] REAL-SPACE = BLOCH SANITY: twisted spectra match to 9×10^{-15}. [E] PER-LAYER CENSUS (+, +, +, -): TRIM $\equiv$ Pfaffian in every sector — 3 even sectors, the $\nu=1$ Ising signature. [E] 16 LAYERS $\Rightarrow \mu_{\text{pre}}=4$: parity¹⁶=+ in ALL four sectors $\Rightarrow$ gauged torus $\text{GSD}=4=\mu(SO(16)_1)= \det \text{Cartan}(D_8)$. [E] 16-FOLD-WAY BRIDGE: $\theta_v=e^{2\pi i\nu/16}=1$ EXACTLY at $\nu=2c_-=16$ (vortex bosonic, condensable); rivals $\nu=1, 2, 8$ excluded (not holomorphically condensable). [E] NON-DISCRIMINATOR FENCE: the fermionic Kitaev–Preskill TEE is identically zero — $\gamma_f=0.000001$ ($M=1$) and 0.000000 ($M=3$) — an invertible fermionic phase has no TEE, so γ_f does NOT discriminate $(E_8)_1$ vs $SO(16)_1$ (both live only after gauging). [C] the KLM step $\mu_{\text{post}}=4/2^2=1$ stays cited (the condensed model is interacting); the numerics pin only $\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}$ — the premise package of the condensation. [O] the continuum limit (MMST v336); SEAM.EQUIV.01 stays [O]. Python (numpy + scipy Schur + sympy exact θ_v). [E][C][O]
v491_p2_partition_corollary.py	P2.PARTITION.01: $g_{\text{car}}=5$ as a COROLLARY of the four marks — the partition sharpening of v53 (sympy exact, no floats). THE LEMMA: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is unique — $\{1, 1, 2\}$ — so $e=(e_1, e_2, e_3)=(4, 5, 2)$ and $g_{\text{car}}=e_2=5$, $\mathbb{Z}_2 =e_3=2$, $N_{\text{fam}}=e_2-e_3=3$ are COROLLARIES, not inputs; consistency closes as a fixed point $((2^4-1)/5=3=\text{rank } H^1=\#\text{marks}-1)$. NEGATIVE CONTROLS (every assumption load-bearing): drop positivity $\rightarrow 4$ solutions, e_2 ambiguous $\{0, 3, 4, 5\}$; sum 3 $\rightarrow e_2=3$; sum 5 $\rightarrow$ not unique; 2/4 parts $\rightarrow e_2 \in \{3, 4\}/\{6\}$; Mehta–Seshadri RATIONAL weights $\rightarrow e_2=\frac{16}{3}$ resp. $\frac{44}{9} \neq 5$ (integrality comes from the Birkhoff–Grothendieck SPLITTING-TYPE typing, not from MS weights). [E] v53 SHARPENING: v53 needs all three targets (4, 5, 2); the partition needs only $e_1=4$ — same unique $\{1, 1, 2\}$. [E] SUM RULE: $4 \times \frac{N-1}{2}=4$ iff $N=3$; the v115 route $a _1= \mu_4 \text{tr}(A_0)=4$. INDEPENDENCE CHAIN: $\#\text{marks}=4$ from v216 (P1-side, no g_{car} input); rank $H^1=3$ pure topology (v137/v228); $a _1=4$ g_{car}-free (par-deg-0/(U), v115). [C] HONEST RESIDUAL: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality = splitting exponents; positivity = $h^0=0$, correlate v137 check 1; sum rule “given (U)”); AX.P2.01 stays a declared axiom; the N_{fam} direction reversal vs tfpt_constants is typed as an over-determination check, never double-counted. Exact (sympy); Wolfram-mirrored. [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v492_celestial_z4_orbifold.py	CELEST.WP1.01: WP1 of the celestial-holographic research contract CELEST.SEAM.01 — the E_8 μ_4-glue grading (v128) IS the flat $\mathbb{Z}_4$ monodromy of the glue-equivariant celestial chiral algebra on the A_3 ALE space $\mathbb{C}^2/\mathbb{Z}_4$ (Bittleston–Homans–Sharma, arXiv:2305.09451, the $\mathbb{Z}_2$/Eguchi–Hanson case; Kronheimer–Nakajima flat connection at infinity). Sympy exact. [E] INNER GRADING: $h=(2, 2, 2, 2, 2; 0^4)$ reads the glue class mod 4 on all 240 roots $\Rightarrow$ glue $= \text{Ad}(\exp(2\pi i h/4))$, an order-4 torus element $=$ flat $\mathbb{Z}_4$ monodromy; the A_3-side detector $h'=(0^5; 1, 1, 1, -3)$ reads the SAME class $=$ the diagonal $(1, 1)$ of the discriminant group $\mathbb{Z}_4 \times \mathbb{Z}_4$ (the v92/v125 Lagrangian glue); $\mathbb{Z}_2$ halfway $120 = \dim SO(16)$, $128 = \text{spinor}$. [E] $\mathbb{C}^2/\mathbb{Z}_4$ IS the A_3 singularity $XY=Z^4$ (Molien table exact; Hilbert series $(1+t^4)/((1-t^2)(1-t^4))$; conjugation symmetry $N[d][1]=N[d][3]$). [E] EQUIVARIANT SECTOR CLOSES: $D[d]=60(d+1)$ even / $64(d+1)$ odd — uniform per parity ONLY because $\dim \mathfrak{g}_j=(60, 64, 60, 64)$; zero modes $(d=0) = \mathfrak{g}_0 = \text{carrier } D_5 \oplus A_3 + \text{Cartan} = 60$; density $62/248=1/4=1/ \mathbb{Z}_4$; $\text{EH}/\mathbb{Z}_2$ halfway towers $120/128$. [E] CHARACTER: $\Theta_{E_8} = \sum_j \Theta_{C_j}$; shell $1 = (52, 64, 60, 64)$ from the LATTICE; the 4 sector characters Θ_{C_j}/η^8 sum exactly to $\chi_{(E_8)_1} = 1 + 248q + 4124q^2 + 34752q^3$ (v377); sector weights $h_{D_5}=(0, \frac{5}{8}, \frac{1}{2}, \frac{5}{8})$, $h_{A_3}=(0, \frac{3}{8}, \frac{1}{2}, \frac{3}{8})$ = the v92/v125 discriminant form $(5x^2+3y^2)/8 \bmod 1$; glue diagonal $h=(0, 1, 1, 1)$ INTEGER = v125 isotropy = local extension to $(E_8)_1$. [E] CRITICAL CORRECTION: the A_3 deck $\text{diag}(i, i^{-1})$ acts on the celestial sphere as $z \mapsto -z$ (projective order 2, the sheet flip), NOT as the order-4 clock $z \mapsto iz$; the clock is the $U(2)$ phase $\text{diag}(i, 1)$ (order 4, fixed $\{0, \infty\}$, free μ_4 orbit — all v180 properties; normalises the deck, $\det=i$ rotates the holomorphic symplectic form by the μ_4 generator); exact spin bridge $(\text{spin clock})^2 = \text{deck}$, spin group $\mathbb{Z}_8$, $8=2 \mu_4$ (the $c_3=1/(8\pi)$ winding integer). [E] clock-invariant A_3 deformation = the 1-parameter family $XY=Z^4+a_0$; its 4 branch points are ONE μ_4 orbit with cross-ratio 2 (v179 clock orbit; v168/v214 pillowcase $j=1728$). [E] NEGATIVE CONTROLS: false spatial action $\text{diag}(i, i)$ killed three ways ($\det=-1 \notin SL(2, \mathbb{C})$; Poisson closure fails 14/14; grades unsymmetric); false glue breaks additivity on 3112/6720 root pairs; rigidity: only $\text{Aut}(\mathbb{Z}_4)$ relabelings additive. [C] VERDICT B: match under the ONE named identification $\text{clock}^2=\text{deck}$ on the sphere (sheet-$\mathbb{Z}_2$ spin bookkeeping). [O] structural arithmetic ONLY — no celestial duality constructed; CELEST.SEAM.01 (WP2–WP5) stays a research contract; SEAM.EQUIV.01 stays [O], untouched. Exact (sympy). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v493_celestial_ wp2_clock_ deformation.py	CELEST.WP2.01: WP2 of the celestial-holographic research contract CELEST.SEAM.01 — the clock-invariant deformation $XY=Z^4+a_0$ of the A_3 seam orbifold: geometry, periods/monodromy, deformed algebra (Bittleston–Homans–Sharma $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$ transfer, arXiv:2305.09451). Sympy exact. [E] GEOMETRY: $P(iZ)=P(Z)$ forces $a_3=a_2=a_1=0$ SHARPLY (two-sided: $\prod_k (Z - i^k w) = Z^4 - w^4$); smooth iff $a_0 \neq 0$ (disc $= 256 a_0^3$); branch points $Z^4 = -a_0 =$ ONE free μ_4 orbit in every fibre; cross-ratio $2 / j = 1728$ for ALL a_0 (binary-quartic invariants $I=12a_0$, $J=0$ identically) $\Rightarrow a_0$ is a PURE SCALE, the $\tau=i$ pillowcase (v168/v214) is FROZEN — no shape modulus. [E] PERIODS: $\Pi=t(i-1)(1, i, i^2)$, all 3 vanishing spheres carry the SAME volume $\sqrt{2}t$ (lockstep); the clock on H_2 is a COXETER element of $W(A_3)$ (char x^3+x^2+x+1, eigenvalues $\{i, -1, -i\}$ = the A_3 exponents 1, 2, 3; $h(A_3)=4= \mu_4 =N_{\text{fam}}+1$); NO invariant cycle ($\det(A-1)=-4$); the surviving direction is the weight-1 χ_1-Fourier diagonal $(1, i, i^2)$, NOT the naive invariant diagonal; the clock IS the Picard–Lefschetz monodromy of the family ($a_0 \rightarrow e^{2\pi i} a_0$ gives the 4-cycle); versal weights $(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)$ = the REGULAR μ_4 representation, invariant line = a_0 alone. [E] ALGEBRA ($\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$): mode dictionary $\{w[m, n] : m \equiv n \pmod{4}\} \leftrightarrow \{X^p Y^q Z^r : r \leq 3\}$; bracket $\{f, g\} = -4 \text{Nambu}(XY - Z^4 - a_0)$ anchored at the $a_0=0$ orbifold w-sector $(\{X, Y\}=16Z^3, \{Y, Z\}=-4Y, \{Z, X\}=-4X)$, Jacobi + Casimir exact, uniform clock-degree -1; S-algebra corrections exactly LINEAR in a_0 at the $\mathbb{Z}_4$ wrap $r+r' \geq 4$, grade-conserving (NO sector leak $\Rightarrow$ the v492 equivariant SDYM(E_8) sector deforms consistently); $a_0 \leftrightarrow$ the BHS c^2 slot (weight 0); matrix factorisations (McKay modules) deform along every 1+3 and 2+2 orbit splitting; the clock fixes exactly the antipodal 2+2 sector (the EH/$\mathbb{Z}_2$ middle node, $4=2 \times 2$). [E] NEGATIVE CONTROLS: $a_1 Z \Rightarrow j=0$ (hexagonal CM point); pure $a_2 Z^2$ stays SINGULAR (A_1 node); $a_2 \neq 0 \Rightarrow j=1556068/81 \neq 1728$ (exact instance, two routes agree); grading leak explicit ($Z^2 \cdot Z^2$ picks up grades $\{0, 1, 2\}$). [C] three NAMED identifications (verdict B, not A): $-4 \text{Nambu} = \text{THE } k=4 \text{ CCA bracket}$ (BHS prove $k=2$); period = root difference (A-series residue normalisation); the seam-scale reading of a_0 presupposes the WP1 bridge clock²=deck ($\mathbb{Z}_8$, $8=2 \mu_4$). [C] v216 PRECISION (typed, NO marker move): given the order-4 clock, the clock-invariant deformation freezes the $(2, 2, 2, 2)$ modulus to the square AUTOMATICALLY — a relocation of the same order-4 input (the clock IS the carrier input), not a new derivation; v216 stays [O]. [O] full deformed twisted-sector OPEs, the $W(\mu)$-type scaling limit for $k=4$, the spacetime splitting-function check, WP3–WP5; SEAM.EQUIV.01 stays [O], untouched. Exact (sympy). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v494_cosmo_running_mu_record.py	COSMO.RUNNING.01/COSMO.MUDIST.01: the two 2026-07-20 prediction-of-record rows promoted to the suite — the SAME Starobinsky branch as n_s, r, A_s (v7/v28/v86) read one derivative deeper and one observable further. [E] LO IDENTITIES (sympy exact): $\alpha_s = dn_s/d \ln k = -2/N_*^2 = -r/6$, $\beta_s = -4/N_*^3$ (NOT $-8/N_*^3$); LO band $N_* \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$. [E] EXACT SLOW ROLL (third-order Lyth–Riotto + finite-difference cross-check $< 1\%$): $\alpha_s(51.4) = -7.09 \times 10^{-4}$, $\beta_s(51.4) = -2.70 \times 10^{-5}$; exact band edges $-7.48/ -5.23 \times 10^{-4}$. [C] DATA: Planck 2018 (-0.0045 ± 0.0067) $\rightarrow +0.57\sigma$; record leg P-ACT-LB ($+0.0062 \pm 0.0052$) $\rightarrow -1.33\sigma$ (watch channel: positive central value); β_s run+runrun $\rightarrow -0.77\sigma$. KILL: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch. [C] μ-DISTORTION (Chluba window, +21% calibrated on the ΛCDM anchor): sharp branch $N_* = 51.4$ ($A_s = 1.76 \times 10^{-9}$) $\mu = 1.6 \times 10^{-8}$; profiled record $N_* = 56.1$ $\mu = 1.94 \times 10^{-8}$; band $1.5\text{--}2.3 \times 10^{-8}$; branch split $\Delta\mu = 3.0 \times 10^{-9}$ (ratio 1.23) $\Rightarrow \geq 3\sigma$ A_s/N_* branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class); the PIXIE baseline 9×10^{-8} does NOT reach ($0.18 \times$ limit). KILL: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$. [E] COMB BLINDNESS (honest S15): the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is structurally invisible in the μ window — $F(\omega) = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage-2050. [O] FENCES: N_* stays an external reheating input (GATE.NSTAR.01); the frozen band $[50, 60]$ stays the surface of record. Python (sympy exact LO + numpy SR/quadrature), Python-only by nature. [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v495_celestial_wp3_ gs_alignment.py	CELEST.WP3.01: WP3 of the celestial-holographic research contract CELEST.SEAM.01 — the Green–Schwarz axion coefficient λ_g of celestial/twistorial SDYM (Costello, arXiv:2111.08879) against the TFPT seam constant $c_3=1/(8\pi)$; the pre-registered “alignment test only”. Exact (Fraction/sympy, no floats). [E] OKUBO IDENTITY DERIVED, NOT QUOTED: $\sum_{\alpha} \langle \alpha, x \rangle^4 = \frac{5}{2(\dim \mathfrak{g}+2)} (\sum_{\alpha} \langle \alpha, x \rangle^2)^2$ as an exact POLYNOMIAL identity for ALL 8 algebras on Costello’s list ($\mathfrak{sl}_2, \mathfrak{sl}_3, \mathfrak{g}_2, \mathfrak{so}_8, \mathfrak{f}_4, \mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$; $\mathfrak{e}_6/\mathfrak{e}_7$ carved out of the v492 E_8 root set) — “no independent quartic Casimir”; NEGATIVE CONTROL: $\mathfrak{sl}_4$ (which HAS one) fails the proportionality (5/32 vs 3/32) — the list is sharp. [E] Killing $\sum_{\alpha} \langle \alpha, x \rangle^2 = 2h^{\vee} \langle x, x \rangle$ exact; Deligne dimension formula $\dim \mathfrak{g} = 2(5h^{\vee}-6)(h^{\vee}+1)/(h^{\vee}+6)$; closed form $\tilde{\lambda}^2 = 10h^{\vee 2}/(\dim +2) = h^{\vee}+6$ EXACTLY (8, 9, 10, 12, 15, 18, 24, 36) — fixes Costello–Paquette eq. (1.5); Dynkin indices $i_R = (1, 1, 2, 2, 6, 6, 12, 60)$ from explicit weight systems; E_8: $\tilde{\lambda}=6$, $\lambda_{\text{fund}}=1/10$ EXACT; cross-checks $\mathfrak{sl}_2=8$, $\mathfrak{sl}_3=9$; SO8 SLIP FLAGGED: Costello’s own weight content gives $\lambda^2(\mathfrak{so}_8)=3$ (Okubo-consistent 12/4), the printed 3/2 is a factor-2 slip. [E] ALIGNMENT: $\kappa = \lambda_g/(8\pi\sqrt{3}) \Rightarrow (\kappa/c_3)^2 = \lambda^2/3 = 12 = \mu_4 N_{\text{fam}}$ (unit-trace) resp. 1/300 (adjoint-trace), both EXACT anchor rationals; κ/c_3 ITSELF is IRRATIONAL ($2\sqrt{3}$ resp. $1/(10\sqrt{3})$) — the pre-registered “$\lambda/\sqrt{3}$ rational?” reading FAILS, only the SQUARED alignment holds; anchor decomposition $h^{\vee}=30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}$ (all three atoms once), $\varphi(30)=8=\text{rank}$, $\dim +2=250=2 \cdot 5^3$, $\tilde{\lambda}=6= \mathbb{Z}_2 N_{\text{fam}}$, $\lambda_{\text{fund}}=1/(\mathbb{Z}_2 g_{\text{car}})$. [E] LOOK-ELSEWHERE (the HARD FENCE — alignment $\neq$ selection): the alignment test AS SUCH passes 8/8 (ZERO selective power), $\{2, 3, 5\}$-smoothness 8/8, $\tilde{\lambda}$-integrality 2/8 (shared with $\mathfrak{sl}_3$), $\varphi(h^{\vee})=\text{rank}$ 4/8; the ONLY $\mathfrak{e}_8$-selective single test is $g_{\text{car}}=5 \mid h^{\vee}$ (1/8); the isolating conjunction was assembled POST HOC — it quantifies where $\mathfrak{e}_8$ sits, it is NOT a preregistered discriminator. [E] KILL-TEST K3 does NOT fire (exact alignment EXISTS) but its scope is DEMOTED: the c_3-connection survives as convention-level compatibility + genuine λ-arithmetic, NOT as $\mathfrak{e}_8$-selective evidence. [C] TYPED: the 8π is the 2π convention of the anomaly polynomial (Costello’s A.0.10 constant descended to spacetime), the $\sqrt{3}$ the box diagram’s $\sqrt{48} = 4\sqrt{3}$ — WP3 is an alignment yes/no ONLY, it can never derive c_3 from celestial data (contract wording, honoured). [O] WP5 (constructive holography) and SEAM.EQUIV.01 stay [O], untouched. Verdict B. Exact (Fraction + sympy). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v496_celestial_ wp4_character_ shadow.py	CELEST.WP4.01: WP4 of the celestial-holographic research contract CELEST.SEAM.01 — the $(E_8)_1$ character E_4/η^8 against the glue-equivariant $E_8[\mathbb{C}^2]$ S-algebra: the contract’s “type mismatch, head-on”. Exact integer/Fraction/sympy, no floats. [E] BASE: v492 replication $(240 = 52+64+60+64, \text{dims with Cartan } (60, 64, 60, 64))$; equivariant tower $D[d] = 60(d+1)$ even / $64(d+1)$ odd; cumulative generator count QUADRATIC — $\sum_{d \leq s} D[d] = 31s^2 + 92s + 60$ (even) with leading $31 = k + h^\vee$ (the Sugawara denominator, an arithmetic note); [C] CP Table 1 (arXiv:2201.02595): $J[m, n]$ carries 2d spin $1 - (m+n)/2$, UNBOUNDED BELOW along the tower $\Rightarrow$ no rational VOA (L_0 bounded below) contains the towers as graded pieces. [E] HILBERT SERIES: $\chi_{\text{sp}}(t) = (60+128t+60t^2)/(1-t^2)^2$ (palindromic numerator, value 248 at $t=1$); ALL μ_4 collapses $w \in \{1, -1, i, -i\}$ and the $SU(2)_+$ primary slice $(60+64t)/(1-t^2)$ fail to start $(1, 248, \dots)$ — the “row sum” route fails at the first coefficient. [E] CHARACTER SIDE: $\chi_{(E_8)_1} = E_4/\eta^8 = (1, 248, 4124, 34752, 213126, 1057504, 4530744)$ exact through q^6 (machinery cross-check $qj = E_4^3/\eta^{24} = (1, 744, 196884, 21493760)$); theta split (shell $1 = (52, 64, 60, 64)$), sector currents $(60, 64, 60, 64)$, glue-diagonal $h = (0, 1, 1, 1)$ INTEGER (v125 isotropy = locality); Sugawara $c = 248/31 = 8 = 2 \mu_4 = g_{\text{car}} + N_{\text{fam}}$. [E] BRIDGES: zero-mode slice $D[0]=60=$ the C_0-sector current count EXACTLY; the single-line jet slice z_1^d cycles $(60, 64, 60, 64)$ sector by sector; BOUNDARY/PERIOD LIMIT: one full μ_4 period of loop energies carries $\dim \mathfrak{g}_{n \bmod 4} = (64, 60, 64, 60)$, sum = 248 EXACTLY — the single-particle level of the $248q$ stage; 248 is NOT a jet-tower dimension ($d \leq 100$) — only the period collapse produces it; BUT the loop Fock at integer level 1 counts $897266 \gg 248$: the rational truncation is NOT in the jet/loop algebra. [E] MISMATCH LOCALISED THREE WAYS: (a) GRADING (spin unbounded below; 248 no tower dimension); (b) GROWTH: jet Fock $f = (1, 128, 8436, \dots)$ vs $\chi = (1, 248, 4124, \dots)$ TWO-SIDED ($f_1 < \chi_1$ undercount, $f_2 > \chi_2$ overcount), f_n/χ_n strictly increasing $n = 1..12$; Meinardus poles $s=2$ vs $s=1 \Rightarrow$ growth exponents $n^{2/3}$ vs $n^{1/2}$, asymptotically DISJOINT; (c) TRUNCATION: the level-2 null ideal deletes EXACTLY $27000 = 30^3 = (h^\vee)^3$ ($\text{Sym}^2(248) = 1+3875+27000$ [[C] standard]; the character keeps $4124 = 1+248+3875$) — NO analogue in the jet algebra (level 1 agrees: $248 = 248$). [E] KILL-TEST K4, three-way: (i) exact realisation NO; (ii) boundary-limit realisation PARTIAL YES — current stratum, period sum 248, integer glue weights and theta split hold exactly; the missing piece is precisely the rational truncation, which must come from a scaling/compactification limit — the SAME structure as the MMST scaling-limit route of SEAM.EQUIV.01 (v336/v449: the rational net exists in the LIMIT of the lattice collar, not in the pre-limit algebra); (iii) the hard mismatch is exactly located; K4 does NOT degrade the contract to “E_8 admissible” (the sector arithmetic holds exactly). [C] VERDICT B(ii): the $(E_8)_1$ character is a BOUNDARY/LIMIT SHADOW of the equivariant S-algebra, not a conformal block of it in the bulk grading — WP4 sharpens into a statement of SEAM.EQUIV.01 type. [O] the constructive boundary limit itself (which limit, which topology, convergence) stays OPEN and belongs to WP5/SEAM.EQUIV.01; SEAM.EQUIV.01 stays [O], untouched. Exact (integer/Fraction/sympy). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v497_celestial_ wp5a_null_ideal.py	CELEST.WP5A.01: WP5a of the celestial-holographic research contract CELEST.SEAM.01 — “the null ideal from the limit”, the first constructive milestone of WP5 (the v496 boundary-limit shadow made precise). Exact integer/Fraction arithmetic, no floats. [E] BASE: v492/v496 replication (glue dims (60, 64, 60, 64), equivariant tower $D[d]$, $\chi_{(E_8)_1} = E_4/\eta^8$ exact through q^6, μ_4 period sum 248). [E] STABILISATION THEOREM: a one-parameter family χ_w of graded Fock characters on the CHIRAL jet generators (loop energy m, radial number r; $E_w = m + wr$ in quarter units $u = q^{1/4}$); $w=2$ IS the chiral jet grading (generator counts 64, 120, 128, 180, 192, 240, 256, 300 per degree); the u^n coefficient of χ_w equals the quarter-moded loop Fock for ALL $w \geq n+1$ and is STRICTLY larger for every $w \leq n$ ($n \leq 8$, $w \leq 10$): the boundary reading is the precise coefficientwise limit $\varepsilon = 1/w \rightarrow 0$ with explicit threshold $w = n+1$ — a limit of a series, not a slice; convention clarified: 897266 is the quarter-moded loop Fock at $u^4 = \text{INTEGER level 1}$ (v496). [E] NULL IDEAL DERIVED, NOT CITED: Freudenthal multiplicities (unit-tested on the adjoint) + Weyl dimension formula + generic character peeling in the standard E_8 frame DERIVE $\text{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ (peeling residual exactly zero; independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; $C_2 = (124, 96, 0)$, Dynkin indices 6750/750; level-1 integrability $\langle 2\theta, \theta^\vee \rangle = 4$, $\langle \omega, \theta^\vee \rangle = 2 \Rightarrow$ the only level-1 primary is the vacuum ([C] Kac, used through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ EXACTLY. [E] TWO ROUTES, ONE NUMBER: the μ_4 theta-split sector characters summed at $q^2 = (1036, 1024, 1040, 1024)$ give the SAME 4124 — the independent lattice route hits the Fock quotient exactly; at q^3: $34752 = \chi_3$. [E] NEGATIVE CONTROLS: $SO(16)_1$ through the SAME pipeline — $\text{Sym}^2(120) = 5304 + 1820 + 135 + 1$ derived (FOUR components vs E_8’s three), $5304 \neq 14^3 = 2744$ (the $h^\vee{}^3$ identity is NOT generic), quotient $7380 - 5304 = 2076 = \Theta_{D_8}/\eta^8$ at q^2 (the pipeline validated on a second algebra), block weights $(0, \frac{1}{2}, 1, 1)$ do NOT fuse into one local character (four-block theory) vs E_8’s integer $(0, 1, 1, 1)$; false periodisation: only $P = 4 = \mu_4$ gives 248 ($P=1: 64$, $P=2: 124$, $P=8: 496$); swapped glue breaks the zero-mode anchor ($64 \neq 60$). [C] named identifications: “radial decoupling = MMST collar thickness $\rightarrow 0$” (the v336/v449 lattice-collar route has the same two-step shape), not a computed isomorphism; the Kac singular vector only via its counting consequence. [O] HONEST LIMIT: the limit does NOT generate the truncation — the singular vector as an OPERATOR in the limit net stays open (WP5b); the GNS limit state (kernel $\supseteq$ ideal) and local normality / Haag duality of the limit net stay open (WP5c/WP5d); the celestial route now has the SAME two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; SEAM.EQUIV.01 stays [O], untouched. Exact (integer + Fraction). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v498_celestial_ wp5b_singular_ vector.py	CELEST.WP5B.01: WP5b of the celestial-holographic research contract CELEST.SEAM.01 — “the singular vector as an operator”, the second constructive milestone of WP5 (v497 fixed the ideal’s size and location; WP5b constructs the deleting object). Exact integer/Fraction arithmetic, no floats, deterministic. [E] CONSTRUCTION: $s\rangle = (E_{-1}^\theta)^2 0\rangle$, θ the highest root, weight 2θ (the top of the 27000), level 2 = 8 quarter units = q^2 (an INTEGER level); the Chevalley/Frenkel–Kac basis is MACHINE-built (bimultiplicative 2-cocycle, asymmetry on all 57600 pairs; the $[e_\alpha, e_{-\alpha}]$ sign FORCED by Jacobi, $\text{SGN} = -1$; κ DERIVED from invariance, $\kappa(\theta^\vee, \theta^\vee) = 2$; Jacobi exact on all 57600 $\mathfrak{sl}_2$-slice triples + 6000 seeded + all 496 θ-adjacent; all $N_{\alpha\beta} = \pm 1$). [E] SINGULARITY: $J_1^a s\rangle = 0$ for ALL 248 basis elements (0 violations); case tally 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$ via the central-term cancellation — the ONLY case that sees the level k); plus $J_2^a s\rangle = 0$ (248), $E_0^a s\rangle = 0$ (120 positive roots), Shapovalov norm 0 at $k=1$ (4 at $k=2$); the affine PBW engine unit-tested on all 61504 basis pairs. [E] LEVEL DIAL: the F_1^θ coefficient is $2(1-k) - 2$ at $k=0$, 0 EXACTLY at $k=1$, -2 at $k=2$: without the central extension the deletion object does NOT exist. [E] μ_4 DATA: θ-grading dims $(1, 56, 134, 56, 1)$, $\mathfrak{g}_0 = \mathfrak{e}_7 \oplus \mathfrak{u}(1)$; the h-adapted chamber machine-built (Cartan det 1, E_8 Dynkin diagram, branch lengths $[1, 2, 4]$); $\langle \theta, h \rangle = 5 \Rightarrow j(\theta) = 1$, height $29 = h^\vee - 1$; $s\rangle$ in class 2 (sheet-EVEN, consistent with WP5a); clock phase $i^2 = -1$; cross-table θ-grading $\times$ glue class consistent (rows $(1, 56, 126, 56, 1)$, columns $(52, 64, 60, 64)$); the deleting θ-$\mathfrak{sl}_2$ crosses the sheet-ODD classes $(1, 3)$, its square is sheet-even. [E] LEVEL-2 MATCH: $\text{mult}(2\theta) = 1$ in the level-2 Fock $\Rightarrow U(\mathfrak{g}) s\rangle$ is THE 27000; the direct $\mathfrak{g}_0$-orbit BFS reproduces the Freudenthal multiplicities of $V(2\theta)$ EXACTLY through depth 4; quotient $31124 - 27000 = 4124$; lattice reading $(2\theta, 2\theta)/2 = 4 > 2 \Rightarrow$ no momentum-2θ state at q^2 in $E_4/\eta^8 - s\rangle$ IS the kernel generator of Fock $\rightarrow E_4/\eta^8$. [E] NEGATIVE CONTROLS: $E_{-1}^\theta 0\rangle$ NOT singular (one witness $F^\theta, \rightarrow -k 0\rangle$) — the 248-current state that must REMAIN ($\chi_1 = 248$); 4 generic level-2 states non-singular (14–33 witnesses); k-dial: at $k=2$ the CUBE is singular (the $(E^\theta)^{k+1}$ mechanism = generic Kac level mechanics); $D_8/SO(16)_1$: the singular vector exists there TOO (level-1 generic) but the $h = \frac{1}{2}$ blocks do not fuse (3 comark-1 weights, $\dim V(2\theta_{D_8}) = 5304 \neq 14^3$) vs E_8 (comarks ≥ 2, vacuum the ONLY level-1 primary, glue weights $(0, 1, 1, 1)$ integer) — the ONE-BLOCK closure is the E_8/μ_4-specific part, honestly separated from the generic singular-vector mechanics. [C] Weyl complete reducibility beyond BFS depth 4; the Kac mechanics beyond level 2; the per-period quarter dictionary (v496). [O] HONEST LIMIT: in the twisted quarter-SLOT moding of the χ_w limit Fock two sector-C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-PERIOD dictionary (v496), not the per-slot identification, carries $s\rangle$ to q^2; exactly the precise WP5c question (GNS limit state, kernel $\supseteq$ ideal); WP5c-e stay [O]; SEAM.EQUIV.01 stays [O], untouched. Exact (integer + Fraction). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v499_p2_weights_ deligne_bg.py	P2.TYPING.01: the P2 weight-typing postulate HARDENED (the v491 check-9 residual) — the anchor $a=(1,1,2)$ typed as the Birkhoff–Grothendieck splitting exponents of the DELIGNE CANONICAL EXTENSION E of the flavor connection; sympy exact, no floats. Given the [E] anchors (4 marks v216, rank 3 v137, cusp class + irreducibility v117, (U)): $\deg E = -4$ AS A THEOREM (cusp class $\Rightarrow$ canonical residue exponents $\{0, \frac{1}{3}, \frac{2}{3}\}$, the unique lift in $[0,1]$; trace 1 per mark $\times$ 4 marks; par-deg 0; ∞ APPARENT, monodromy product = 1 exactly); POSITIVITY AS A THEOREM (irreducibility EXACT: $\langle U, M_0 \rangle = 24 = S_4$, $\sum_g \chi ^2 = 24$; with (U) $\Rightarrow$ Mehta–Seshadri stability $\Rightarrow h^0(E) = 0$, all 324 section-saturation cases; BG dictionary $h^0 = 0 \Leftrightarrow a_i \geq 1$); INTEGRALITY = Birkhoff–Grothendieck, automatic once E is typed as a bundle. UNIQUENESS + SELECTOR: $\{1,1,2\}$ unique (v491); the Schur–Horn permutohedron of $4 \cdot \text{spec}(A_0)$ has integer points exactly $\{(2,2,0), (2,1,1)\}$ (the two options of the “Global computation” theorem) and $h^0 = 0$ kills $\{0,2,2\}$ (an $\mathcal{O}(0)$ summand has $h^0 = 1$) — the “balanced” selection is now TYPED as stability/positivity. THE h^1 DISCREPANCY, REAL + RESOLVED: $h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3$ — the naive “generations = $H^1(E)$” typing is FALSE (forcing it demands $\deg -6$ and loses uniqueness: $\{1,1,4\}, \{1,2,3\}, \{2,2,2\}, e_2$ never 5); the correct TWO-OBJECT typing: $L = \mathcal{O}(-D)$, $D = \mu_4$, carries the generations ($h^1(L) = 3 = N_{\text{fam}} \cong H^0(\Omega^1(\log D))$, the v137 basis), while E carries them as its FIBER (rank 3) with the anchor as splitting type. BISWAS EXPLICIT (independent convergence witness): the weight denominator is $3 = N_{\text{fam}}$, so the Biswas cover is the $\mathbb{Z}_3$ cusp cover $y^3 = (z-1)(z-i)(z+1)^2(z+i)^2$, genus 2 (NOT the pillowcase double $w^2 = z^4 - 1$, genus 1, $j = 1728$ — the deck side); $p_* \mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$ (χ cross-check $-1 = -1$); the regular-representation model bundle lands on $\{0,2,2\}$ = the UNSTABLE Schur–Horn companion ($h^0 = 1$): integrality AND the sum rule COME OUT of the correspondence, and stability (irreducibility, v117) is the selector. NEGATIVE CONTROLS: drop positivity $\rightarrow 4$ resp. 31 solutions; $\deg -3/-5$ breaks; n-mark scan: $\{1, \dots, 1, 2\}$ unique for EVERY $n \geq 3$ but $e_2 = (n-2)(n+1)/2 = 5$ ONLY at $n=4$ — the VALUE, not the uniqueness, is the $n=4$ content; MS-rational exponents $e_2 = \frac{16}{3}, \frac{44}{9}$. [C] RESIDUALS: R1 the module identification “generation space = this H^1 with this connection” [C] (QGEO.SYM territory, stays [O]); R2 unitarity (U) [C] (v33/v40); R3 the analytic-monodromy identification (fine type Numerical, v117 check 4). P2 stays an axiom (AX.P2.01 declared, no marker moves); NON-CIRCULAR: neither v23’s splitting-type input nor v115’s $\text{diag}(A_0) = a/4$ input is used. Sharpens v491. Exact (sympy); Wolfram-mirrored. [E][C]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v500_celestial_ wp5c_gns_limit_ state.py	CELEST.WP5C.01: WP5c of the celestial-holographic research contract CELEST.SEAM.01 — “the GNS limit state”, the third constructive milestone of WP5 (v497 made the boundary limit precise, v498 built the deleting operator; WP5c constructs the STATE whose GNS kernel contains the ideal). Exact integer/Fraction arithmetic, no floats, deterministic. [E] CONSTRUCTION: the quasi-free family ω_w — loop sector ($r=0$) = the affine $k=1$ vacuum n-point functions via the MACHINE-determined compact anti-involution $\theta(e_\alpha)=-e_{-\alpha}$, $\theta(h)=h$ (the UNIQUE sign that is an anti-automorphism on ALL 61504 basis pairs; contravariance $\langle J_{-1}^x u, v \rangle = \langle u, J_1^{\theta x} v \rangle$ exact); radial sector ($r \geq 1$) = oscillator pairings x^{wr} (the exact Gibbs regulator $e^{-\beta E_w}$, kept x-adic). [E] CONVERGENCE, SHARP: $\omega_w = \omega_\infty \bmod x^{N+1}$ for ALL $w \geq N+1$ ($N=8$) and NOT yet at $w=N$ (the $r=1$ modes sit at order N) — the WP5a threshold $w = n+1$ holds IDENTICALLY at the state level; positivity of every ω_w checked. [E] THE KERNEL (core, fully machine): the FULL exact level-2 Gram — 31124 monomials in 9361 weight blocks (dim profile 164:1, 37:240, 7:2160, 1:6960), every block exact: level-1 rank 248 POSITIVE DEFINITE (Cartan block = the E_8 Cartan matrix, det 1); level-2 rank 4124, kernel 27000, every block PSD; rank per Weyl orbit (norm 8/6/4/2/0) = (0, 0, 1, 8, 44); kernel = $V(2\theta)$ WEIGHT BY WEIGHT (Freudenthal cross-check on all 9361 weights); highest-weight norms (3844, 49, 2, 0) at $(0, \omega, \theta, 2\theta)$; direct ideal checks $F_i s\rangle$, $F_j F_i s\rangle$ norm 0. [E] μ_4-EQUIVARIANCE: $\omega_w \circ \text{clock} = \omega_w$ for ALL w; the clock unitary DESCENDS to $\text{GNS}(\omega_\infty)$; level-2 GNS rank split by clock class $= (1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$ at q^2 (the two-routes identity at the STATE level); level-1 split (60, 64, 60, 64). RESOLUTION OF WP5b S4.10: $s\rangle$ IS the ZERO VECTOR of $\text{GNS}(\omega_\infty)$ — the twisted two-C_1-mode floor (6 quarters $= q^{3/2}$) concerns only pre-limit objects (GNS has no state there: $6 \bmod 4 \neq 0$). [E] NEGATIVE CONTROLS: $k=2$ ($\langle s s \rangle = +4$, dominant blocks FULL rank — NO ideal in the kernel); $k=0$ (level-1 Gram $\equiv 0$ — no 248 layer without the central extension); $D_8/SO(16)_1$ (level-2 rank 2076 = 7380 – 5304, kernel $= V(2\theta_{D_8})$ PSD weight by weight, but $h = (0, \frac{1}{2}, 1, 1)$: the vacuum GNS is 1 of 4 blocks — ONE-BLOCK completeness stays E_8/μ_4-specific); wrong family $E'_w = mw+r$ (level-1 limit rank 0: the 248 layer ERASED, u^4 content $710955 \neq 248$); no damping ($897266 \neq 248$); CCR OBSTRUCTION: at a FIXED radial pairing $c > 0$ positivity forces $\omega(bb^*) \geq c$ — NO w-uniform damping state exists: the family formulation is NECESSARY and the family EXISTS $\Rightarrow$ the preregistered KILL is probed and NOT triggered. [C] the quasi-free/Gaussian extension to the full $*$-algebra; Kac positivity beyond level 2 (levels ≤ 2 machine-checked); the module identification in the component route. [O] local normality of ω_∞, the net structure / Haag duality (WP5d), Costello–Li (WP5e), the operator-algebraic closure beyond the graded/x-adic topology; SEAM.EQUIV.01 stays [O], untouched. VERDICT: SUCCESS on the preregistered criterion (dim GNS level-2 = 4124 exactly). Exact (integer + Fraction). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v501_celestial_ wp5d_two_interval_ index.py	CELEST.WP5D.01: WP5d-α of the celestial-holographic research contract CELEST.SEAM.01 — “the two-interval index from the lattice” (the local-net question, alpha stage). The roadmap discriminator: $(E_8)_1 \Rightarrow \mu=1$ (Haag-dual two-interval net, ONE sector); $SO(16)_1 \Rightarrow \mu=4$; preregistered KILL = “the two-interval μ-offset does NOT vanish after the condensation”. numpy/scipy Gaussian machinery + exact Fractions. [E] SETUP (machine identities): critical Majorana ring (v450 style, NS closure, $c=\frac{1}{2}$ per layer), 16 layers = the seam carrier $c_-=8=g_{\text{car}}+N_{\text{fam}}$ (v489); Peschel-exact; three prescriptions — fermionic ρ, orbifold sector sum $\{1, P_A, P_B, P_{AB}\}$, dual algebra $\{1, P_{AB}\}$; superselection $[\rho, P_{AB}]=0$ machine-exact; EVERY Gaussian formula validated against exact diagonalisation ($\leq 1.5 \times 10^{-15}$). [E] FERMIONIC MI EXTENSIVE (the $\mu=1$ reference; lattice witness): $I(x) = -(c/3) \ln(1-x)$ with c fitted 0.5000; $\max F(x) = 5.97 \times 10^{-5}$ at $N=512$, F falling monotonically to 1.3×10^{-6} at $N=1024$ — Casini–Huerta extensivity on the lattice. [E] THE μ-OFFSET (core): Rényi-2 exactly $\Delta_2 = -\ln[(1+R)/2]$; the 16-layer seam carrier saturates $\ln 2$ to MACHINE PRECISION ($\Delta_2 - \ln 2 = 1.1 \times 10^{-15}$ at $N=512$; $R \sim N^{-0.500}$); index reading $[F : F_{\text{even}}] = e^{\Delta_\infty} = 2.000000 \Rightarrow \mu_{\text{gauged}} = e^{2\Delta_\infty} = 4.000000 =$ the v490 parity census (TWO independent lattice witnesses); gauging every layer separately costs $16 \ln 2$ — the offset measures $\ln[\text{index of the subnet}]$, not a universal blob. [E] HAAG-DUALITY INDICATOR: fermionic $S(E)=S(E')$ to 8.2×10^{-12} (graded-duality proxy); the orbifold BREAKS complementarity ($D(E) - D(E') = 0.427$ at $a=64$ — the direct duality-failure witness); the complementary-pair sum is exchange-symmetric and $\leq \ln 4 = \ln \mu(SO(16)_1)$ with $\max 1.2637$ at the self-dual $x=\frac{1}{2}$ and $\rightarrow \ln 2$ at the squeezed ends: the $\ln 4$ KLM budget is a DOUBLE-LIMIT statement, NOT a fixed-x identity (the first naive fixed-x check was WRONG and was replaced — honestly typed). [E] CONDENSATION ARITHMETIC, LATTICE-ANCHORED: $\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 4 \cdot 4 = 16 = \mu(\text{carrier})$; KLM/Longo–Rehren quotients $\mu/ L ^2$: $16/4^2 = 1$ (one-step μ_4 glue) and $16/2^2 = 4 \rightarrow 4/2^2 = 1$ (via $SO(16)_1$), both routes EXACT; Σd_i^2 cross-checks (Ising 4, $SO(16)_1$ 4, $(E_8)_1$ 1); $\theta_v = e^{2\pi i \nu/16} = 1$ EXACTLY at $\nu = 2c_- = 16$, rivals $\nu = 1, 2, 8$ give $\theta_v \neq 1$. The anchored chain: $\mu = 1$ (graded, MEASURED) $\rightarrow 4$ (gauged, MEASURED + v490 census) $\rightarrow 1$ (arithmetic, [C]: the condensed net is interacting — the v490 fence). [E] NEGATIVE CONTROLS: $\nu=1$ single Majorana (offset non-trivial, $\Delta_2 = 0.6364$ at $N=2048$ rising, and $\theta_v = e^{i\pi/8} \neq 1$: NOT removable — the kill discriminator has teeth); trivial dimerised phase (everything $\leq 10^{-7}$); wrong sector sum $\{1, P_{AB}\}$ (offset EXACTLY 0 instead of $\ln 2$). [C] the continuum dictionary “offset = $\ln$ index” (KLM two-interval index; Longo–Xu; Casini–Huerta–Magan–Pontello entropic order parameters) + the $\theta_v=1$ condensation step itself. [O] continuum Haag duality, local normality, split/strong additivity of the quotient net (WP5d-β, running as exploration), Costello–Li (WP5e); SEAM.EQUIV.01 stays [O], untouched. VERDICT: the $\mu=1$ indicator chain PASSES; the preregistered KILL is NOT triggered. Python (numpy + scipy + Fractions). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v502_celestial_ wp5e_prefactor_ level.py	CELEST.WP5E.ALPHA.01: WP5e-α of the celestial-holographic research contract CELEST.SEAM.01 — “prefactor and level bookkeeping”, the CFT-side anchor stage of WP5e (the BCOV/Costello–Li twistor uplift: boundary partition function $= E_4/\eta^8$ including the $q^{-1/3}$ prefactor; preregistered kill: the anomaly inflow demands a level $\neq 1$). Exact sympy/Fraction. [E] PART A — the $q^{-1/3}$ prefactor is exact μ_4 vacuum-energy bookkeeping; Hurwitz-ζ derivation ($\zeta(-1, \theta) = -B_2(\theta)/2$; $E_b(\theta) = -\frac{1}{24} + \theta(1 - \theta)/4$ exactly); the clock is INNER ($h = (2, 2, 2, 2, 2; 0^4)$ and the A_3 detector $h' = (0^5; 1, 1, 1, -3)$ both read the class mod 4 on all 240 roots; $(h, h) = 20$, $(h', h') = 12$, sum 32) $\Rightarrow$ twist distribution $\theta = 0^8$ in ALL four sectors — a SHIFT orbifold (lattice cosets, integer-moded oscillators), <i>not</i> a rotation orbifold; $E_{\text{osc}} = -\frac{1}{3} = -c/24$ sector-independent ($c = 8 = \text{rank} = 248/31$); sector ground exponents $-\frac{1}{3} + (0, 1, 1, 1)$. [E] discriminant form = Casimir energy, TWO routes: spectral flow $j^2(h, h)/32$ ($5j^2/8, 3j^2/8$, sum $j^2 \equiv 0 \pmod{1}$ = the isotropy/Lagrangian condition) and <i>exactly</i> via free fermions ($D_5 = 10$ Majorana, $A_3 = D_3 = 6$, glue diagonal $= 16 = 2^{g_{\text{car}}-1}$, the WP5d seam carrier; R–NS shift $n/16$: $\frac{5}{8}, \frac{3}{8}, 1$); the ROTATION reading fails (the geometric A_3 deck gives $\frac{3}{16} \neq \frac{3}{8}$: the v492 “clock $\neq$ deck” lesson at the Casimir level). [E] character: $\sum_j \Theta_{C_j} = E_4$ and the four sector characters (leading coefficients $(1, 64, 60, 64)$, currents $(60, 64, 60, 64)$) sum exactly to $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$. [E] PART B — $k = 1$ forced THREE independent ways: (i) the CURRENT condition $h(J) = k = 1$ (the $\mathbb{Z}_4$ extension closes onto the E_8 adjoint $60+64+60+64 = 248$ only at spin 1); HONEST SHARPENING: glue-h integrality $h(J^j; k) = k(0, 1, 1, 1)$ holds for ALL $k = 1..8$ — the originally conjectured integrality route fixes <i>nothing</i>, the three other dials carry; (ii) conformal embedding $47k^2+219k-266 = 47(k-1)(k+266/47) = 0$ resp. D_8-in-E_8: $128k(1-k) = 0$; (iii) central charge $248k/(k+30) = 8 \Leftrightarrow 240(k-1) = 0$ (= the prefactor itself); plus the v498 singular-vector dial ($\ (E_{-1}^\theta)^m 0\ ^2 = m! k(k-1) \cdots (k-m+1)$, singular vector exactly at grade $k+1$; $k = 1$ deletes $V(2\theta)$: $31124 - 27000 = 4124$). [E] negative controls: $D_8/SO(16)_1$ (same rank, same $c = 8$, SAME $q^{-1/3}$ prefactor, but $h = (0, \frac{1}{2}, 1, 1)$: the discriminator is glue-h integrality, not the prefactor; $\Theta_{D_8} + \Theta_s = E_4$); $\mathbb{Z}_2$ rotation ($E = +\frac{1}{6}$, $h = \frac{1}{2}$: no μ_4-shift sector there); μ_4 rotation (sector-dependent $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$: the common prefactor breaks); wrong $k \in \{2, 3, 4\}$ fails all five dials coherently (residuals $360/814/1362$; $\langle s s \rangle = 4/12/24$; level-2 dim stays $31124 \neq 4124$). [C] Kac generation of the maximal submodule by the grade-$(k+1)$ singular vector for $k \geq 3$ ($k = 1, 2$ machine-anchored by WP5b/WP5c). [O] the BCOV/Costello–Li twistor inflow itself (Costello’s one-loop axion $\lambda_g/(8\pi\sqrt{3})$ on $\mathbb{P}^1$, the Costello–Paquette–Sharma level-from-flux mechanism) stays open — v502 anchors the CFT side of WP5e only (α stage); the kill branch (“inflow demands a level $\neq 1$”) is not triggered on the CFT side but its twistor side is untested; WP5e stays [O]; SEAM.EQUIV.01 untouched, no marker moves. Exact (sympy + Fraction). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v503_qgeo_ emergence_light.py	QGEO.EMERGE.LIGHT.01: the v215 <i>deferred emergence test</i> (lever 7) executed in its first honest LIGHT version, plus the residual-convergence result QGEO-R1 $\equiv$ P2-R1 (analysis/kill-test class; NO marker moves — QGEO.REALIZE.01 stays [C]/[O]). NO DYNAMICAL SELECTION (ζ-log-det via the Dedekind-η formula, mpmath 40 digits): $\log F = \log(\tau_2 \eta ^4)$ is exactly S-invariant, so $t = 1$ is automatically critical on the rectangular family $\tau = it$; there a strict local maximum ($g''(1) = -1.298$) but a <i>minimum</i> in the τ_1-direction ($h''(0) = +0.298$): $\tau = i$ is a SADDLE of the full moduli space; critical points = the CM points ($E_2^*(i) = E_2^*(\rho) = 0$ to 40 digits); the <i>global</i> det' winner at fixed area is the HEXAGONAL torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$; Osgood–Phillips–Sarnak 1988) — free-field dynamics alone does <i>not</i> select $\tau = i$; criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). SHARP RIGIDITY (v280 lattice DtN): the mod-4 off-character indicator (the v215-K3 lever as a <i>modulus function</i>) has an isolated zero (8×10^{-15}) exactly at $\tau = i$, strictly monotone in both directions (min 8.4×10^{-2} away); $R^2 = \mathbb{Z}_2$ exactly as permutation matrices (the v492 $\mathbb{Z}_8$ bridge on the lattice); $j(it) \geq 1728$ with equality only at $t = 1$, $j(\rho) = 0$. [E] the $\mathbb{Z}_2$ chain is MODULUS-BLIND: 4 fixed points and $[\mathbb{Z}_2, \Delta_t] = 0$ on <i>every</i> torus (v216 confirmed without mark input); pillowcase det' = $\frac{1}{2}$ torus det' exactly ($\pm$-pair structure), so all τ conclusions transfer verbatim; marks/mark-locality/deck² exist on <i>every</i> pillowcase (v264/v280), only the order-4 clock distinguishes $\tau = i$; $\text{Aut}(E_\rho) = \mathbb{Z}_6$ (element orders 1, 2, 3, 6) contains <i>no</i> order-4 element — dynamics and clock are logically independent selectors; GIVEN the clock, $P(iZ) = P(Z)$ forces $a_3 = a_2 = a_1 = 0$ and $j = 1728$ identically (v493): the modulus is not a second input. [C]/[O] RESIDUAL CONVERGENCE (the core finding): the order-4 element on H^1 is a Coxeter element of $W(A_3)$ (char poly $\lambda^3 + \lambda^2 + \lambda + 1$, order $h(A_3) = 4 = \mu_4 = N_{\text{fam}} + 1$) — exactly the P2 cusp-class/carrier datum of the P2 hardening (v499 R1) $\Rightarrow$ QGEO-R1 $\equiv$ P2-R1: ONE common residual, “why does the raw seam carry the order-4 clock”, not two scattered ones; chain: marks derived (v216 [E]), square derived <i>given</i> the clock (v493 [E]), clock = carrier input [C], rest = the dynamical realisation of the clock [O]; route classification (v347 analogue): B-rigidity in the sharpened form “RP seam data $\Rightarrow$ order-4 clock” is the live route — free dynamics alone does not deliver the clock (the hexagonal finding). Lever update (documented here, v215 itself unchanged): lever 7 lifted from “deferred, not executable” to “light version executed: rigidity, not dynamical selection”. Python (mpmath + numpy/scipy + sympy). [E][C][O]

continued on next page

Script	What it machine-checks (<i>continued</i>)
v504_celestial_ wp5d_klm_ completeness.py	CELEST.WP5DB.01: WP5d-β of the celestial-holographic research contract CELEST.SEAM.01 — “split + strong additivity from the lattice”, the two remaining Kawahigashi–Longo–Müger legs of complete rationality (the μ-index is v501), witnessed for the orbifold prescription on the critical Majorana seam edge; the $U(1)$/Dirac subnet is the preregistered negative control (the current net, where strong additivity fails in the continuum). [E] STRONG ADDITIVITY, EXACT: shared boundary Majorana $\Rightarrow$ Even(A) $\vee$ Even(B) = Even($A \cup B$) EXACTLY (GF2 spans full 64/64 up to 1024/1024; matrix rank 32/32); DISJOINT: exactly HALF (rank 16/32, index 2 — the missing odd $\otimes$ odd sector is the v501 ln 2 bit, localised at the split point); the neutral $U(1)$ algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). [E] ENTROPIC WITNESS [N]: the fermionic touching MI carries the pure UV point-gap law $((c/3) \ln 2$ increments, dev 1.2×10^{-6} — no discriminator); the honest discriminator is BOUNDED vs DIVERGENT: the $\mathbb{Z}_2$/orbifold deficit stays $< \ln 2$ with $(\ln 2 - \Delta_2) \sim N^{-p}$, $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (Ising disorder dimension; “defect $\rightarrow 0$” would be FALSE at the sharp split — bounded $\Leftrightarrow$ finite index, Longo–Xu), while the $U(1)$ control BURSTS ln 2 from $L = 128$ and grows as $\frac{1}{2} \ln \text{Var } Q_A$ (Klich–Levitov 0.10134 vs $1/\pi^2$): infinite index. [E] SPLIT/NUCLEARITY [N]: coupling σ_k fall at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0%; $\sigma_1 \sim x^{0.5044}$; trace norm summable; ORBIFOLD INHERITANCE EXACT: P_A flips $C \rightarrow -C$ (σ_k identical, 8.3×10^{-17}), the even-bilinear Gram is the compound $\Lambda^2 C$ ($\{\sigma_a \sigma_b\}$, 2.8×10^{-14}) — Longo heredity [C]. [E] PIMSNER–POPA: $E(a) - a/2 = PaP/2$ IDENTICALLY $\Rightarrow$ $\lambda = \frac{1}{2} = 1/[F : F_{\text{even}}]$; integer attainment (16384 + 2048 monomial sweeps, 0 violations; pencil min $\lambda = 0.5000000000$); $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ exactly; INDEX CONSISTENCY, two routes: $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\text{PP}}$ and $e^{2\Delta_\infty} = 4 = \mu$; $U(1)$: $\lambda = 1/(m+1) \rightarrow 0$. [E] CONTROLS + HONESTY: dimerised phase trivial (3.6×10^{-15}); wrong prescription $\{1, P_{AB}\}$ deficit EXACTLY 0; finite-size honesty (14.9% below ln 2 at $N = 4096$ — witness, not proof). [C] the KLM dictionary (Longo–Xu; Longo heredity; CHMP). [O] the continuum theorem “finite-group orbifolds of completely rational nets are completely rational” is XU’S THEOREM — CITED, NOT CLAIMED; the seam quotient net and the condensed $(E_8)_1$ net stay WP5e/Costello–Li; SEAM.EQUIV.01 stays [O]. VERDICT: with v501, all THREE KLM ingredients witnessed on the lattice. Python (numpy + scipy); exact parts Wolfram-mirrored. [E][C][O]

How to run. From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim $\leftrightarrow$ script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all nine documents are written against). The [E] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).

TFPT verification suite: the journey of ~475 machine-checked scripts

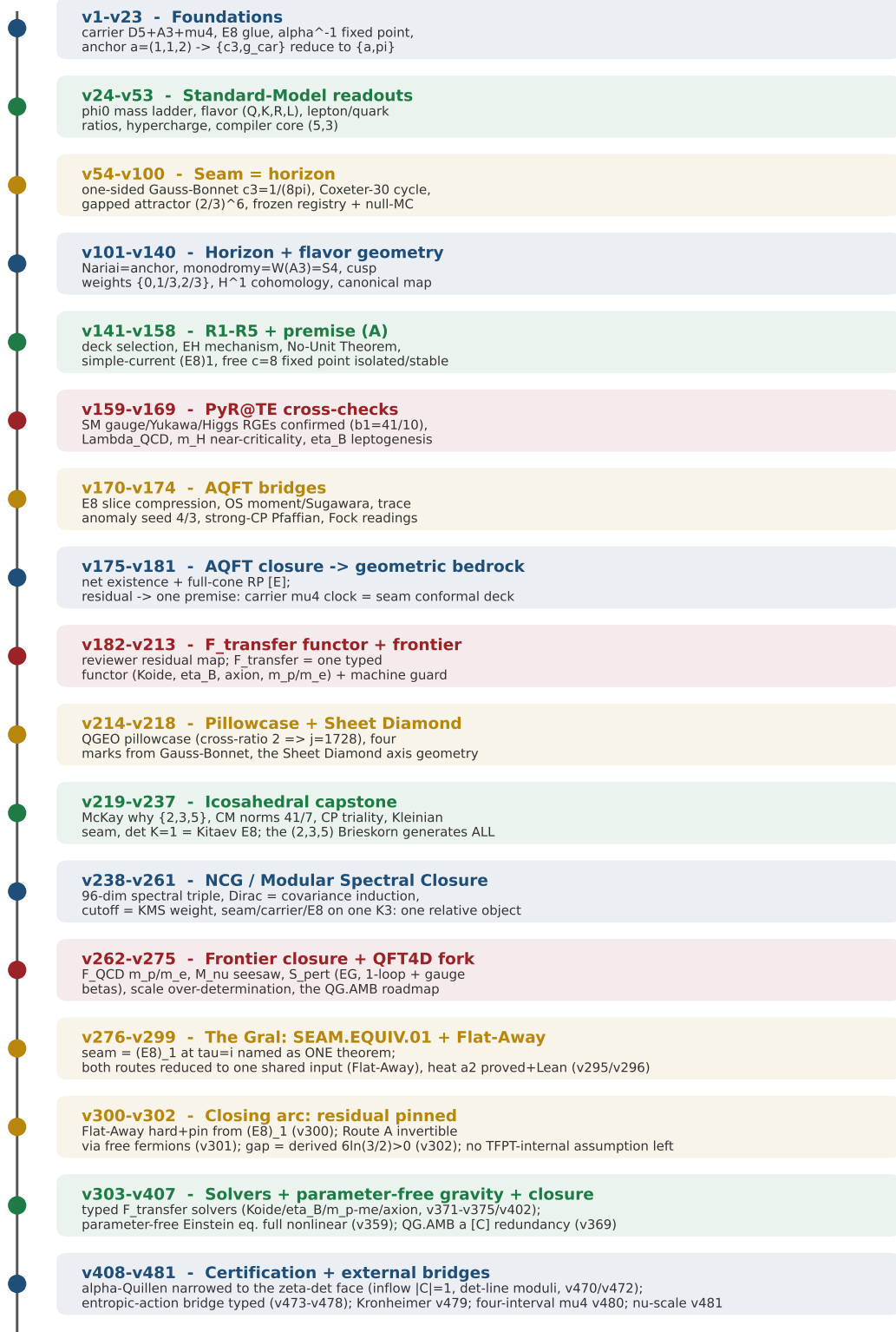


Figure 1: The development timeline of the verification suite — the phases of the journey and what each did, mathematically and physically: from the foundations (carrier, E_8 glue, α^{-1} , the anchor $a = (1, 1, 2)$ to which $\{c_3, g_{car}\}$ reduce) through the Standard-Model readouts, the seam=horizon geometry, the horizon/flavor geometry, the R1–R5 and premise-(A) reductions, the external PyR@TE RGE cross-checks and the AQFT bridges, the AQFT closure that drives the whole structural residual to one geometric bedrock premise, the $F_{transfer}$ functor and frontier interfaces, the QGEO pillowcase and Sheet Diamond, the *icosahedral capstone* (the single $(2, 3, 5)$ Brieskorn singularity generates the whole skeleton), the Modular Spectral Closure, the SEAM.EQUIV.01 closing arc (v276–v302), the typed solvers and the parameter-free gravity round (v303–v407), to the certification round (v408–v481: the α -Quillen residual narrowed to the continuum ζ -det face, the entropic-action bridge typed, the Kronheimer quiver bridge, the four-interval μ_4 geometry). The

Appendix: changelog

The dated record of every change (canonical source: `changelog.tex`, also compiled standalone).
Module numbers vN refer to `verification/vN_*.py`.

2026-07-21 (XIV) (v504 — WP5d- β of CELEST.SEAM.01 “split + strong additivity from the lattice” promoted as CELEST.WP5DB.01: the two remaining KLM legs of complete rationality witnessed for the orbifold prescription — strong additivity ALGEBRAICALLY EXACT with the shared boundary Majorana ($\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$), GF2 spans full, rank 32/32; disjoint exactly index 2 = the v501 $\ln 2$ bit), the entropic touching defect BOUNDED $< \ln 2$ with the Ising $\frac{1}{4}$ -exponent approach ($p = 0.2444$) while the preregistered $U(1)$ /Dirac control DIVERGES (Klich–Levitov pinned, infinite index), the split property at the elliptic-nome rate $\pi K(1-x)/K(x)$ (1.3–2.0%) with EXACT orbifold inheritance ($C \rightarrow -C$ + the $\Lambda^2 C$ compound — Longo heredity), and Pimsner–Popa $E(a) - a/2 = PaP/2$ identically ($\lambda = \frac{1}{2} = 1/[F:F_{\text{even}}]$, $\lambda_{E_4} = \frac{1}{4} = 1/\mu$, $e^{\Delta_\infty} = 2 = 1/\lambda_{\text{PP}}$ — two independent index routes); with v501 ALL THREE KLM ingredients carry lattice witnesses; the continuum uplift stays [O] — “finite-group orbifolds of completely rational nets are completely rational” is XU’S THEOREM, cited not claimed; Wolfram extension 424 $\rightarrow$ 429 (verified engine run); SEAM.EQUIV.01 stays [O], CELEST.SEAM.01 stays Open, no marker moves)

- **v504_celestial_wp5d_klm_completeness.py** (CELEST.WP5DB.01, 37 checks, ~ 12 s; numpy/scipy Gaussian machinery ED-validated + exact GF2/integer algebra; deterministic). Complete rationality (Kawahigashi–Longo–Müger) = split + strong additivity + finite two-interval μ -index; v501 (WP5d- α) witnessed the index, v504 witnesses the other two legs on the same critical Majorana seam edge, with the $U(1)$ /Dirac number-conserving subnet as the preregistered negative control (the current net, where strong additivity famously *fails* in the continuum — Buchholz–Schulz–Mirbach). *Strong additivity, exact algebra*: “touching = shared boundary Majorana” gives $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$ EXACTLY (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32); disjoint intervals give exactly HALF (rank 16/32, index 2 — the missing sector odd $\otimes$ odd is the v501 $\ln 2$ bit, localised at the split point); the neutral $U(1)$ algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). *Entropic witness*: the fermionic touching MI carries the pure UV point-gap law $((c/3) \ln 2$ increments, 0.115525, dev 1.2×10^{-6} — present for every net, no discriminator); the honest discriminator is *bounded vs divergent*: the $\mathbb{Z}_2$ /orbifold deficit rises $0.4045 \rightarrow 0.5899$ ($N = 64 \rightarrow 4096$), stays $< \ln 2$, and $(\ln 2 - \Delta_2) \sim N^{-p}$ with $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (Ising disorder exponent); split-point scan reflection-symmetric to 3.9×10^{-14} ; ED vN same direction; gap continuity to the v501 $x \rightarrow 1$ endpoint; honest note: “defect $\rightarrow 0$ ” would be *false* at the sharp lattice split — bounded $\Leftrightarrow$ finite index (Longo–Xu) is the correct statement; the $U(1)$ control bursts $\ln 2$ from $L = 128$ and grows as $\frac{1}{2} \ln \text{Var } Q_A$ (Klich–Levitov slope 0.10134 vs $1/\pi^2 = 0.10132$). *Split/nuclearity*: coupling-block σ_k fall exponentially at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0% (rates 2.29/3.24/4.82/5.81 at $d = 16/64/256/512$); $\sigma_1 \sim x^{0.5044}$ (vs $\frac{1}{2}$); trace norm summable; dimerised control exponential. *Orbifold inheritance exact*: P_A flips $C \rightarrow -C$ (σ_k identical, 8.3×10^{-17}) and the even-bilinear coupling Gram is the second compound $\Lambda^2 C$ (singular values $\{\sigma_a \sigma_b\}$, 2.8×10^{-14}) — the same ladder at doubled rate (Longo heredity, [C]). *Pimsner–Popa, exact*: $E(a) - a/2 = PaP/2$ identically; attainment in integers (all 16384 (even, odd) monomial pairs at $N = 8$: 8192 extremal with $M^2 = 4$, $\text{Tr } M = 0$, $PMP = -M$, 0 violations; 2048/2048 at $N = 12$; pencil min $\lambda = 0.5000000000$); $\lambda_{E_4} = \frac{1}{4} = 1/\mu$; index consistency $e^{\Delta_\infty} = 2.0000000000 = 1/\lambda_{\text{PP}}$ and $e^{2\Delta_\infty} = 4 = \mu$; $U(1)$: $\lambda = 1/(m+1) \rightarrow 0$. *Controls + honesty*: dimerised phase trivial (drift 3.6×10^{-15}); wrong prescription $\{1, P_{AB}\}$ deficit exactly 0; $\mathbb{Z}_2$ finite-size honesty (14.9% below

$\ln 2$ at $N = 4096$ — witness, not proof). VERDICT: with v501 all THREE KLM ingredients of complete rationality are witnessed on the lattice; the continuum theorem for finite-group orbifolds is Xu’s (cited, not claimed), the concrete seam quotient net and the condensed $(E_8)_1$ net stay WP5e/Costello–Li.

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5DB.01; CELEST.SEAM.01 updated to “WP5d: α (μ -index) + β (split + strong additivity) executed via v501/v504 — all three KLM ingredients witnessed on the lattice; continuum uplift stays [O] (Xu’s theorem cited)” at *unchanged* Open status. Papers: contracts WP5d bullet $\rightarrow$ “both stages done” with the β -stage numbers + Xu fence (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *ten*-step story (new Step 10 “the KLM completion on the lattice”, “what remains” = WP5e twistor inflow only); `tfpt_4` keystone note updated. Wolfram: five mirrors (GF2 span theorem; Pimsner–Popa identity + attainment on explicit gamma matrices; two-interval group $\lambda_{E_4} = \frac{1}{4}$ + index consistency; the Λ^2/Minors compound with Cauchy–Binet and $\text{Minors}[-C, 2] = \text{Minors}[C, 2]$; the $U(1)$ dephasing $\lambda = 1/(m+1)$ family) verified with the active engine the same day (424 $\rightarrow$ 429, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). VerificationDag qft node gains v504; registry + `run_all` (498 modules, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

2026-07-21 (XIII) (v503 — the v215 deferred emergence test (lever 7) executed *light* and promoted as QGEO.EMERGE.LIGHT.01: NO dynamical selection of the square modulus — the free field’s global $\det'$ winner is the *hexagonal* torus, $\tau=i$ is a saddle, criticality at the square is Palais symmetry; the square is pinned by *rigidity* alone (the raw DtN is mod-4 block-diagonal *iff* $\tau=i$, an isolated zero); and the residual *converges*: QGEO-R1 $\equiv$ P2-R1 = one common rest, “why does the raw seam carry the order-4 clock”; Wolfram extension 420 $\rightarrow$ 424 (verified engine run); NO marker moves — QGEO.REALIZE.01 stays [C]/[O], AX.P2.01 stays a declared axiom)

- `v503_qgeo_emergence_light.py` (QGEO.EMERGE.LIGHT.01, 17 checks; `mpmath` 40 digits + v280-style lattice DtN + `sympy` exact). The v215 lever 7 (“the decisive kill is deferred: compute the raw seam-collar DtN *without* inputting the marks and check whether four square marks *emerge*”) executed in its first honest *light* version. *No dynamical selection*: $\log F = \log(\tau_2|\eta|^4)$ (Osgood–Phillips–Sarnak) is exactly S -invariant ($\text{dev} < 10^{-35}$), so $t=1$ is symmetry-forced critical on the rectangular family $\tau = it$; there $g''(1) = -1.298 < 0$ (strict maximum) but the τ_1 -direction has $h''(0) = +0.298 > 0$: $\tau=i$ is a *saddle* of the full moduli space; the only critical points are the CM points ($E_2^*(i) = E_2^*(\rho) = 0$ at 40 digits, generic $0.163 \neq 0$); the *global* $\det'$ winner at fixed area is the *hexagonal* torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$, diff 0.021169); criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). *Sharp rigidity*: the DtN mod-4 off-character indicator (the v215-K3 lever as a *modulus function*) vanishes to 8×10^{-15} exactly at $\tau=i$ and rises strictly monotonically both ways (min 8.4×10^{-2} away) — an isolated zero; $R^2 = \mathbb{Z}_2$ exact as permutation matrices (the v492 $\mathbb{Z}_8$ bridge on the lattice); $j(it) \geq 1728$ with equality only at $t=1$. *The $\mathbb{Z}_2$ chain is modulus-blind*: 4 marks + $[\mathbb{Z}_2, \Delta_t] = 0$ on *every* torus (v216 confirmed without mark input); pillowcase $\det' = \frac{1}{2}$ torus $\det'$ exactly ($\pm$ -pair structure) — only the order-4 clock distinguishes $\tau=i$; $\text{Aut}(E_\rho) = \mathbb{Z}_6$ has *no* order-4 element (the dynamical winner cannot carry the clock); given the clock the modulus freezes automatically ($j = 1728$ identically, v493 mini). *Residual convergence (the core finding)*: the order-4 element on H^1 is a Coxeter element of $W(A_3)$ (char poly $\lambda^3 + \lambda^2 + \lambda + 1$, order 4 = $|\mu_4| = N_{\text{fam}} + 1$) = the P2 cusp-class carrier datum of v499 R1 $\Rightarrow$ QGEO-R1 $\equiv$ P2-R1, *one common residual*; chain: marks derived (v216), square derived *given* the clock (v493), clock = carrier input [[C]], rest = its dynamical realisation [O]; route classification (v347 analogue): B-rigidity in the sharpened form “RP seam data $\Rightarrow$ order-4 clock”

is the live route. v215 lever 7 thereby moves from “deferred, not executable” to “light version executed: rigidity, not dynamical selection” — documented in v503, v215 itself unchanged.

- **Ledger + papers + Wolfram + website.** Ledger: new row QGEO.EMERGE.LIGHT.01; QGEO.REALIZE.01 text *precised* (rest = the order-4 clock, identical with the P2 typing rest R1; square + mark count derived) at *unchanged* marker [C]/[O]; cross-reference added in P2.TYPING.01 (convergence theorem). Papers: contracts kill-test paragraph (lever-7 light result + the residual-convergence sentence, both with v503), tfpt_3 Step 2 (light-converse sentence), origin_theory necessity paragraph, introduction/tfpt_1/tfpt_2 one-residual cross-references. Wolfram: four mirrors (CM-point identities $E_2(i) = 3/\pi$, $E_2(\rho) = 2\sqrt{3}/\pi \Rightarrow E_2^* = 0$ exact, $j/\text{DedekindEta}$ special values; the no-dynamical-selection witnesses; the pillowcase halving as an exact algebraic identity; clock $\Rightarrow$ square + Coxeter) verified with the active engine the same day (420 $\rightarrow$ 424, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice DtN curves stay Python-only (flagged). VerificationDag masses node gains v503; registry + run_all (497 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XII) (v502 — WP5e- α of CELEST.SEAM.01 “prefactor and level bookkeeping” promoted as CELEST.WP5E.ALPHA.01: the $q^{-1/3}$ prefactor of E_4/η^8 IS exact μ_4 vacuum-energy bookkeeping (the clock is inner $\Rightarrow$ shift orbifold $\theta = 0^8$, common $-c/24 = -\frac{1}{3}$; discriminant form = Casimir via spectral flow AND 16-Majorana free fermions), and $k = 1$ is forced THREE independent ways (current condition; conformal embedding; central charge = the prefactor itself; plus the WP5b singular-vector dial $31124 - 27000 = 4124$); honest sharpening: glue- h integrality holds for ALL k and fixes nothing; the BCOV/Costello–Li twistor inflow stays [O] — WP5e is subdivided (α done, WP5e proper open); Wolfram extension $416 \rightarrow 420$ (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)

- **v502_celestial_wp5e_prefactor_level.py (CELEST.WP5E.ALPHA.01, 33 checks, exact sympy/Fraction, no floats).** The two exactly computable consistency anchors of the WP5e twistor uplift established. *Part A — the prefactor:* Hurwitz- ζ exact ($\zeta(-1, \theta) = -B_2(\theta)/2$; $E_b(\theta) = -\frac{1}{24} + \theta(1 - \theta)/4$; Ising shift $\frac{1}{16}$ per Majorana); the clock is *inner* ($h = (2, 2, 2, 2; 0^4)$ and $h' = (0^5; 1, 1, 1, -3)$ both read the class mod 4 on all 240 roots; $(h, h) = 20$, $(h', h') = 12$, sum 32) $\Rightarrow$ twist $\theta = 0^8$ in all four sectors — a *shift* orbifold, NOT a rotation orbifold — so $E_{\text{osc}} = -\frac{1}{3} = -c/24$ is sector-independent ($c = 8 = \text{rank} = \text{Sugawara } 248/31$; one common η^8); sector ground exponents $-\frac{1}{3} + (0, 1, 1, 1)$; *discriminant form = Casimir energy* two ways (spectral flow $j^2(h, h)/32 \bmod 1$; free fermions exact: $D_5 = 10$ Majorana $\rightarrow \frac{5}{8}$, $A_3 = D_3 = 6 \rightarrow \frac{3}{8}$, glue diagonal $= 16 = 2^{g_{\text{car}}-1} = \text{the WP5d seam carrier} \rightarrow 1$); the rotation reading *fails* (deck $\frac{3}{16} \neq \frac{3}{8}$ — “clock $\neq$ deck” at the Casimir level); the four sector characters sum exactly to $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$. *Part B — the level:* $k = 1$ forced three independent ways — (i) the current condition $h(J) = k = 1$ (adjoint closure $60+64+60+64 = 248$ at spin 1); (ii) the conformal embedding ($47k^2+219k-266 = 47(k-1)(k+266/47) = 0$; $D_8: 128k(1-k) = 0$); (iii) $c = 8 \Leftrightarrow 240(k-1) = 0$ (= the prefactor); plus the v498 singular-vector dial ($\|(E_{-1}^\theta)^m 0\|^2 = m! k(k-1) \cdots (k-m+1)$, singular vector at grade $k+1$; $31124 - 27000 = 4124$ at $k=1$ only, $\langle s|s \rangle = +4$ at $k=2$). *Honest sharpening (mandatory):* glue- h integrality $h(J^j; k) = k(0, 1, 1, 1)$ is integer for ALL $k = 1..8$ — the originally conjectured integrality route fixes *nothing*; the three other dials carry. *Negative controls:* $D_8/SO(16)_1$ (same $c = 8$, SAME $q^{-1/3}$ prefactor, but $h = (0, \frac{1}{2}, 1, 1)$: the discriminator is glue- h integrality, not the prefactor; $\Theta_{D_8} + \Theta_s = E_4$); $\mathbb{Z}_2$ rotation ($E = +\frac{1}{6}$, $h = \frac{1}{2}$); μ_4 rotation (sector-dependent $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$ — the common prefactor breaks); wrong $k \in \{2, 3, 4\}$ fails all five dials coherently (residuals $360/814/1362$; $\langle s|s \rangle = 4/12/24$; $31124 \neq 4124$). [[C]] Kac generation for $k \geq 3$ ($k = 1, 2$ machine-anchored by WP5b/WP5c). [O] the BCOV/Costello–Li twistor inflow itself (Costello’s one-loop axion $\lambda_g/(8\pi\sqrt{3})$ on $\mathbb{P}T$, the Costello–Paquette–Sharma level-from-flux mechanism) — v502 anchors the *CFT side* of WP5e only; the KILL branch (“inflow demands a level $\neq 1$ ”) is not triggered

on the CFT side, its twistor side is untested.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5e bullet *subdivided* — “WP5e- α (done) [E]/[C]: the CFT-side prefactor + level pinning” with the core numbers and `v502_celestial_wp5e_prefactor_level.py`; “WP5e proper [O]: the twistor inflow”; summit header, honest-success paragraph and status roster updated (“WP5a–WP5d plus WP5e- α landed; WP5e proper the remaining open piece”). `tfpt_3` celestial section: new Step 9 (prefactor + level, one paragraph); `tfpt_4` keystone note updated. Ledger: new row `CELEST.WP5E.ALPHA.01`; `CELEST.SEAM.01` status $\rightarrow$ “WP5e: alpha stage (CFT-side prefactor + level pinning) executed via v502; the BCOV/twistor inflow itself stays [O]”. Wolfram: four exact mirrors (Hurwitz- ζ vacuum energies; discriminant = Casimir with the deck failure; the k -fixing equations incl. the “fixes nothing” integrality table; the character sums + D_8 control) verified with the active engine the same day (416 $\rightarrow$ 420, ALL WOLFRAM EXTENSION CHECKS PASSED). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 424/424 after (XIII)) updated; `VerificationDag` qft node gains v502; registry + `run_all` (497 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (XI) (v501 — WP5d- α of CELEST.SEAM.01 “the two-interval index from the lattice” promoted as CELEST.WP5D.01: the KLM Haag-duality discriminator measured entropically on the seam edge — the 16-layer carrier saturates the $\ln 2$ offset at machine precision, $\mu_{\text{gauged}} = 4 =$ the v490 census, and the condensation arithmetic $16 \rightarrow 4 \rightarrow 1$ closes with $\mu = 1$; the preregistered KILL (“ μ -offset $\neq 0$ after condensation”) does NOT fire; contract now “WP1–WP5d executed; WP5e the single remaining milestone”; Wolfram extension 412 $\rightarrow$ 416 (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)

- `v501_celestial_wp5d_two_interval_index.py` (CELEST.WP5D.01, 39 checks, numpy/scipy Gaussian machinery + exact Fractions). The WP5d roadmap discriminator — $(E_8)_1 \Rightarrow \mu=1$ (Haag-dual two-interval net, one sector) vs $SO(16)_1 \Rightarrow \mu=4$ — executed at the α (lattice-index) stage on the v367 seam edge (16 critical Majorana layers, $c_- = 8$, v450-style ring; Peschel-exact, every Gaussian formula ED-validated to $\leq 1.5 \times 10^{-15}$; the superselection identity $[\rho, P_{AB}] = 0$ machine-exact). The $\mu = 1$ reference: the fermionic (fixed-spin-structure) two-interval mutual information is *extensive* — c fit 0.5000, $\max |F(x)| = 5.97 \times 10^{-5}$ at $N=512$, falling monotonically to 1.3×10^{-6} at $N=1024$ (Casini–Huerta extensivity as a lattice witness). The μ -offset: Rényi-2 exactly $\Delta_2 = -\ln[(1+R)/2]$; the 16-layer seam carrier saturates $\ln 2$ at machine precision ($|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$ at $N=512$; $R \sim N^{-0.500}$); index reading $[F : F_{\text{even}}] = 2.000000$, $\mu_{\text{gauged}} = 4.000000 =$ the v490 parity census (*two independent lattice witnesses*); per-layer gauging costs $16 \ln 2$ (the offset measures $\ln[\text{index}]$, not a blob). *Haag-duality indicator*: fermionic $S(E) = S(E')$ to 8.2×10^{-12} ; the orbifold *breaks* complementarity ($D(E) - D(E') = 0.427$ at $a=64$); the complementary-pair sum is exchange-symmetric and $\leq \ln 4 = \ln \mu(SO(16)_1)$ with max 1.2637 at the self-dual $x = \frac{1}{2}$ — a *double-limit* statement (the first naive fixed- x check was wrong and was replaced — honestly typed in the module). *Condensation arithmetic anchored at both measured ends*: $\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 16$; KLM/Longo–Rehren $16/4^2 = 1$ and $16/2^2 = 4 \rightarrow 4/2^2 = 1$ (both routes exact); $\Sigma d_i^2 = (4, 4, 1)$; $\theta_v = 1$ exactly at $\nu = 2c_- = 16$, rivals $\neq 1$ — so $\mu = 1$ after condensation ([C] the condensed net is interacting, the v490 fence): KILL not triggered. *Negative controls*: $\nu=1$ single Majorana (offset non-removable, $\theta_v = e^{i\pi/8} \neq 1$ — the discriminator has teeth); trivial phase (all $\leq 10^{-7}$); wrong sector sum $\{1, P_{AB}\}$ (offset exactly 0). [O] WP5d- β (continuum Haag duality / local normality / split of the quotient net; runs as exploration) and WP5e (Costello–Li) stay open.
- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5d bullet $\rightarrow$ “ α stage done” with the core numbers and `v501_celestial_wp5d_two_interval_index.py`; status line and honest-success paragraph now “four of five milestones landed; WP5e (twistor

uplift) is the remaining open piece”. **tfpt_3** celestial section: new Step 8 (one paragraph, conservative); **tfpt_4** keystone note updated to WP1–WP5d. Ledger: new row CELEST.WP5D.01 (fine types Formal + Lattice/Numerical + **[C]** dictionary); CELEST.SEAM.01 status → “WP1–WP5d executed; WP5e **[O]** (single remaining milestone)”. Wolfram: four exact mirrors (Cartan determinants; KLM/Longo–Rehren quotients; Σd^2 + index-reading arithmetic; 16-fold-way θ_v) verified with the active engine the same day (412 → 416, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). wolfram/README.md, GATE.WOLFRAM.02, tfpt_safeguards Layer-6 count and the website counters (116/116 + 416/416) updated; VerificationDag qft node gains v501; registry + run_all (495 modules, suite green). SEAM.EQUIV.01 untouched; no marker moves.

2026-07-21 (X) (v500 — WP5c of CELEST.SEAM.01 “the GNS limit state” promoted as CELEST.WP5C.01: the quasi-free family ω_w exists, is positive, stabilises exactly at the WP5a threshold, and its limit carries the null ideal in its GNS kernel — the full 9361-block exact level-2 Gram has rank 4124 exactly with kernel 27000 = $V(2\theta)$ weight by weight; $|s\rangle$ IS the GNS zero vector (the WP5b S4.10 tension resolved at the state level); a CCR obstruction proves the family formulation *necessary*; preregistered SUCCESS criterion met, KILL not triggered; Wolfram extension 408 → 412 (verified engine run); SEAM.EQUIV.01 stays **[O], no marker moves)**

- **v500_celestial_wp5c_gns_limit_state.py** (CELEST.WP5C.01, **35 checks, exact integer/Fraction, no floats, deterministic**). WP5a (v497) made the boundary limit precise, WP5b (v498) built the deleting operator; WP5c constructs the *state*. ω_w is quasi-free: loop sector = the affine $k=1$ vacuum n -point functions via the *machine-determined* compact anti-involution $\theta(e_\alpha) = -e_{-\alpha}$ (the unique sign that is an anti-automorphism on all 61504 basis pairs; contravariance exact); radial sector = oscillator pairings x^{wr} (the exact Gibbs regulator, x -adic). *Convergence sharp*: $\omega_w = \omega_\infty \bmod x^{N+1}$ for all $w \geq N+1$ ($N=8$), not yet at $w = N$ — the WP5a threshold $w = n+1$ holds *identically at the state level*; positivity of every ω_w checked. *The kernel (core, fully machine)*: the complete exact level-2 Gram — 31124 monomials in 9361 weight blocks (profile 164:1, 37:240, 7:2160, 1:6960) — has level-1 rank 248 positive definite (Cartan block = the E_8 Cartan matrix, det 1), level-2 rank 4124 *exactly* (the preregistered target check), kernel 27000, every block PSD; rank per Weyl orbit (0, 0, 1, 8, 44); kernel = $V(2\theta)$ *weight by weight* (Freudenthal cross-check on all 9361 weights); highest-weight norms (3844, 49, 2, 0) at $(0, \omega, \theta, 2\theta)$; direct ideal checks $F_i|s\rangle$, $F_j F_i|s\rangle$ norm 0. μ_4 -equivariance: $\omega_w \circ \text{clock} = \omega_w$ for all w ; the clock descends to GNS; level-2 rank split by clock class (1036, 1024, 1040, 1024) = Θ_{C_j}/η^8 at q^2 — the two-routes identity at the *state level*. *Resolution of WP5b S4.10*: $|s\rangle$ is the zero vector of $\text{GNS}(\omega_\infty)$ — the twisted two- C_1 -mode floor (6 quarters = $q^{3/2}$) concerns only pre-limit objects (GNS has no state there, $6 \bmod 4 \neq 0$). *Negative controls*: $k=2$ ($\langle s|s\rangle = +4$, dominant blocks full rank — no ideal in the kernel); $k=0$ (level-1 Gram $\equiv 0$ — no 248 layer without the central extension); D_8 (rank $2076 = 7380 - 5304$, kernel = $V(2\theta_{D_8})$ PSD, but $h = (0, \frac{1}{2}, 1, 1)$: one block of four — one-block completeness stays E_8/μ_4 -specific); wrong family $E'_w = mw+r$ (the 248 layer erased, $710955 \neq 248$); no damping ($897266 \neq 248$); *CCR obstruction*: at fixed radial pairing $c > 0$ positivity forces $\omega(bb^*) \geq c$ — no w -uniform damping state exists, the family formulation is *necessary* and the family *exists*: the preregistered KILL is probed and not triggered. **[C]** quasi-free extension to the full $*$ -algebra; Kac positivity beyond level 2. **[O]** local normality, net structure/Haag duality (WP5d), Costello–Li (WP5e).
- **Contract + ledger + Wolfram + website**. **tfpt_research_contracts**: the WP5c bullet → done with the core numbers (state-level threshold $w = N+1$, the full 9361-block Gram rank 4124/kernel 27000 weight by weight, the clock-class split, the CCR necessity of the family), citing v500_celestial_wp5c_gns_limit_state.py; the WP5b handover marked answered. **tfpt_3** celestial section: new Step 7 (one paragraph, conservative). Ledger: new row CELEST.WP5C.01 (fine type Formal/Identity **[E]** + **[C]** identifications + **[O]** net questions). Wolfram: four exact

mirrors (level-2 block census from the root data; rank/kernel arithmetic with $\dim V(\omega) = 3875$ by Weyl; the glue-frame clock-class split (1036, 1024, 1040, 1024); threshold + level-dial + D_8 census) verified with the active engine the same day (408 $\rightarrow$ 412); the exact Gram/PSD computation stays Python-side (flagged). VerificationDag qft node gains v500; registry + run_all (494 modules at this step, suite green). SEAM.EQUIV.01 untouched; no marker moves.

2026-07-21 (IX) (v499 — the P2 weight-typing postulate HARDENED: the anchor $a = (1, 1, 2)$ typed as the Birkhoff–Grothendieck splitting exponents of the Deligne canonical extension, so integrality/positivity/sum-4 become theorems given the [E] anchors; the v491 check-9 residual sharpened, the tfpt_2 “balanced” selection typed as stability; Wolfram extension 403 $\rightarrow$ 408 (verified engine run); no gate closes, no marker moves — AX.P2.01 stays a declared axiom, QGEO.SYM stays [O])

- **v499_p2_weights_deligne_bg.py** (P2.TYPING.01, 23 checks, sympy exact, no floats). v491 reduced P2 to “3 positive integers summing to 4”; its residual was the *typing* (why integers, why positive, why sum 4). Now typed: given the [E] anchors (4 marks, v216; rank 3 = #marks – 1, v137/v228; cusp-class local monodromy $\lambda^3 - 1$ and exact irreducibility $|\langle U, M_0 \rangle| = 24 = |S_4|$, $\sum_g |\chi|^2 = 24$, v117; unitarity (U)), the Deligne canonical extension E of the flavor connection satisfies: $\deg E = -4$ as a theorem (residue exponents $\{0, \frac{1}{3}, \frac{2}{3}\}$ the unique lift to $[0, 1)$, trace 1 per mark; par-deg 0; ∞ apparent, monodromy product = 1 exactly); $h^0(E) = 0$ as a theorem (Mehta–Seshadri stability; all 324 section-saturation cases; BG dictionary $h^0=0 \Leftrightarrow a_i \geq 1$); integrality = Birkhoff–Grothendieck. Hence $\{a\} = \{1, 1, 2\}$ unique and $g_{\text{car}} = e_2 = 5$. *Selector typed*: the Schur–Horn permutohedron has integer points exactly $\{(2, 2, 0), (2, 1, 1)\}$ (the two options of the tfpt_2 “Global computation” theorem) and $h^0=0$ kills $\{0, 2, 2\}$ — the “balanced/minimal-variance” heuristic is now stability/positivity. *The h^1 discrepancy, real and resolved*: $h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3$, so “generations = H^1 of the rank-3 bundle” is FALSE (negative control: forcing it demands $\deg -6$ and loses uniqueness, $e_2 \in \{9, 11, 12\}$ never 5); correct two-object typing: $L = \mathcal{O}(-D)$ carries the generations ($h^1(L) = 3 = N_{\text{fam}}$, the v137 basis = $H^0(\Omega^1(\log D))$), E carries them as its fiber. *Biswas explicit* (independent witness): the weight denominator 3 = N_{fam} forces the $\mathbb{Z}_3$ cusp cover $y^3 = (z-1)(z-i)(z+1)^2(z+i)^2$ (genus 2; NOT the pillowcase double, genus 1, $j = 1728$ — the deck side); $p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$; the regular-representation model lands on $\{0, 2, 2\} =$ the *unstable* Schur–Horn companion ($h^0 = 1$): integrality and the sum rule come out of the correspondence, stability is the selector. *Negative controls*: no positivity $\rightarrow$ 4/31 solutions; $\deg -3/ -5$ breaks; n -mark scan $e_2 = (n-2)(n+1)/2 = 5$ ONLY at $n = 4$ (the value, not the uniqueness, is the $n=4$ content); MS-rational $e_2 = \frac{16}{3}, \frac{44}{9}$. *Honest residuals* (P2 stays an axiom): R1 module identification [C] (QGEO.SYM territory), R2 (U) [C] (v33/v40), R3 analytic monodromy (Numerical, v117 check 4); non-circular (neither v23’s splitting input nor v115’s $\text{diag } A_0 = a/4$ input used); the v491 direction-reversal note carries over. Ledger: new row P2.TYPING.01; the v491/P2.PARTITION.01 row records “typing residual sharpened by v499” (v491 stays active, no supersede). Registry + run_all (493 modules, suite green).
- **Wolfram second path + papers + website.** Five exact mirrors added to the extension file `tfpt_readouts_extension.wl` (partition enumeration under the full typing + naive-typing control + h^1 discrepancy; Deligne residue-trace degree with the A_0^* witness char poly; Schur–Horn lattice points; Biswas $\mathbb{Z}_3$ -cover degree arithmetic incl. deck double $j = 1728$; the n -mark formula), verified with the active engine the same day (403 $\rightarrow$ 408, ALL WOLFRAM EXTENSION CHECKS PASSED); wolfram/README.md + GATE.WOLFRAM.02 + tfpt_safeguards Layer-6 count + website counters (116/116+408/408). Papers (one sentence each, conservative): introduction anchor paragraph, tfpt_1 (5, 3) keybox, tfpt_2 H2 winbox and the “Global computation” theorem (the balanced selection typed as stability), origin_theory (5, 3) keybox — all citing v499_p2_weights_deligne_bg.py; no status card changes. Website: VerificationDag P2

node gains **v499**; generated indices via build; no marker moved anywhere.

2026-07-21 (VIII) (Editorial: dedicated celestial-route section in Paper 3 + website mirror — the scattered celestial sentences consolidated into one coherent six-step section; *no new scripts, no status changes, no marker moves, SEAM.EQUIV.01 untouched*)

- **tfpt_3_e8_audit_bootstrap:** new section “The celestial route: from the μ_4 clock to the $(E_8)_1$ boundary shadow”. The celestial-contract material that had accumulated as one long paragraph inside “The Möbius origin of the irreducible π ” is consolidated into a dedicated section (placed after the bootstrap audit, before “Honest status”), telling the executed WP1–WP5b story in six typed steps: (1) setup — the E_8 μ_4 -glue as a flat $\mathbb{Z}_4$ monodromy on the A_3 ALE space, $\text{clock}^2 = \text{deck}$ ($\text{spin } \mathbb{Z}_8, 8 = 2|\mu_4|$) [E] (v492); (2) deformation — $XY = Z^4 + a_0$ pure scale, $j = 1728$ frozen, $\text{clock} = \text{Picard–Lefschetz/Coxeter monodromy}$ [E] (v493); (3) consistency honestly demoted — $(\kappa/c_3)^2 = 12$ exact but the alignment format passes 8/8: alignment survives, selectivity does not [E]/[C] (v495); (4) type mismatch + boundary limit — the $(E_8)_1$ character is not a jet-grading conformal block (growth $n^{2/3}$ vs $n^{1/2}$) but the μ_4 period sum 248 holds exactly: a boundary-limit shadow of SEAM.EQUIV.01/MMST shape [E] (v496); (5) the null ideal quantitative — stabilisation threshold $w = n+1$, $27000 = (h^\vee)^3$ derived, quotient = sector sum = 4124, $SO(16)_1$ contrast [E] (v497); (6) the deleting operator explicit — $|s\rangle = (E_{-1}^\theta)^2|0\rangle$, $J_1^a|s\rangle = 0$ on all 248 (tally 190/57/1), level dial $2(1-k)$, one-block closure E_8/μ_4 -specific [E] (v498); plus the WP5c–e roadmap and the honest status line (the route does *not* close SEAM.EQUIV.01; the GNS/limit question stays [O]). All pre-existing script citations (v492–v498) are carried into the new section; the Möbius section now points to it with one sentence. **tfpt_research_contracts:** a reader’s note at the head of the CELEST.SEAM.01 contract cross-references the new Paper-3 section (the contract stays the full technical reference). **tfpt_4_frontier:** one sentence in the metric-sector keybox records the celestial route as a third, quantitative approach to the keystone (not closing it).
- **Website mirror. papers.ts:** the Paper-3 section list gains the celestial-route section (six-step summary in the existing style) and a “Celestial route” highlight; the contracts mirror gains the same cross-reference sentence. **glossary.ts:** three new entries (celestial route, ALE space, null ideal) in the established technical style. Generated indices via build; no status component changes (no marker moved anywhere).

2026-07-21 (VII) (CELEST.SEAM.01 WP5b executed and promoted: v498 — “the singular vector as an operator”, the deleting object of the v497 null ideal constructed explicitly and machine-verified singular; contract now WP1–WP4 + WP5a + WP5b done, WP5c–e [O] (roadmap); Wolfram extension 398 → 403 (verified engine run); *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)

- **v498_celestial_wp5b_singular_vector.py** (CELEST.WP5B.01, **53 checks, exact integer/Fraction, deterministic, no floats**). Success on the preregistered WP5b criterion: the deleting object exists and is explicit. *Construction:* $|s\rangle = (E_{-1}^\theta)^2|0\rangle$ (weight 2θ , level $2 = 8$ quarter units = q^2 , an *integer* level) in a machine-built Chevalley/Frenkel–Kac basis — bimultiplicative 2-cocycle with asymmetry verified on all 57600 ordered root pairs, the $[e_\alpha, e_{-\alpha}]$ sign *forced* by the Jacobi identity ($\text{SGN} = -1$, not chosen), the invariant form κ derived ($\kappa(\theta^\vee, \theta^\vee) = 2$), Jacobi exact on all 57600 $\mathfrak{sl}_2$ -slice triples + 6000 seeded + all 496 θ -adjacent, all $N_{\alpha\beta} = \pm 1$. *Singularity (machine):* $J_1^a|s\rangle = 0$ for all 248 basis elements (0 violations), with the case classification 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$ via the central-term cancellation — the *only* case that sees the level k); plus $J_2^a|s\rangle = 0$, $E_0^a|s\rangle = 0$, exact weight, Shapovalov norm 0; the affine PBW engine unit-tested on all 61504 basis pairs. *Level dial:* the F_1^θ coefficient

is $2(1-k) - 2/0/-2$ at $k = 0/1/2$: without the central extension the deletion operator does not exist. μ_4 data: glue class $j(\theta) = 1$ (machine-built h -adapted chamber, $\langle \theta, h \rangle = 5$, height $29 = h^\vee - 1$), clock phase $i^{2j} = -1$, class(2θ) = 2 sheet-even; the deleting θ - $\mathfrak{sl}_2$ crosses the sheet-odd classes (1, 3) while its square is sheet-even; cross-table θ -grading $\times$ glue class consistent (rows (1, 56, 126, 56, 1), columns (52, 64, 60, 64)); 8 quarters = q^2 via the per-period dictionary. *Level-2 match*: weight 2θ has multiplicity 1 in the level-2 Fock, so $U(\mathfrak{g})|s\rangle$ is the $27000 = (h^\vee)^3$; the direct $\mathfrak{g}_0$ -orbit BFS matches the Freudenthal multiplicities of $V(2\theta)$ exactly through depth 4 ([C] Weyl complete reducibility beyond); quotient $31124 - 27000 = 4124$; lattice reading $(2\theta, 2\theta)/2 = 4 > 2$: $|s\rangle$ is the kernel generator of $\text{Fock} \rightarrow E_4/\eta^8$. *Negative controls (honest separation)*: the level-1 current state $E_{-1}^\theta|0\rangle$ (one witness, $\rightarrow -k|0\rangle$ — the 248-current state that must remain) and 4 generic level-2 states (14–33 witnesses) are *not* singular; at $k=2$ the *cube* is (generic Kac $(E^\theta)^{k+1}$ mechanics, honestly typed); $SO(16)_1$ has the same singular vector but keeps three extra level-1 primaries ($h = \frac{1}{2}$ breaks one-block fusion; $5304 \neq 14^3$) vs E_8 (comarks all ≥ 2 , glue weights (0, 1, 1, 1) integer) — the *one-block closure* is the E_8/μ_4 -specific part. *Honest [O]*: in the twisted quarter-slot moding two sector- C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-period dictionary (v496), not the per-slot identification, carries $|s\rangle$ to q^2 : exactly the WP5c GNS/limit-state question; WP5c-e stay open; SEAM.EQUIV.01 untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: the WP5b bullet moved to *done* with the key numbers (case tally 190/57/1; level dial $2(1-k)$; $j(\theta) = 1$, clock phase -1 ; orbit BFS through depth 4; the D_8 contrast) and the twisted-slot tension as the explicit WP5c handover; honest status line and K4 paragraph updated (“WP1–WP4 + WP5a + WP5b executed; WP5c-e [O]”). `tfpt_3_e8_audit_bootstrap`: the celestial-contract paragraph extended by the WP5b sentence. Ledger: CELEST.WP5B.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a + WP5b executed; WP5c-e [O] (roadmap)”). Wolfram: 5 exact v498 mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day (398 $\rightarrow$ 403, ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + GATE.WOLFRAM.02 counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 403/403. Website: `papers.ts` contract mirror (“WP1–WP4 + WP5a + WP5b done, WP5c-e roadmap”), DAG `qft` node + v498, generated indices via build. Suite now 492 scripts, all green; SEAM.EQUIV.01 stays [O].

2026-07-21 (VI) (CELEST.SEAM.01 WP5a executed and promoted: v497 — “the null ideal from the limit”, the v496 boundary-limit shadow made a precise coefficientwise limit and the level-2 null ideal derived from root data; contract WP5 subdivided, now WP1–WP4 + WP5a done, WP5b-e [O] (roadmap); Wolfram extension 394 $\rightarrow$ 398 (verified engine run); no gate closes, no marker moves, SEAM.EQUIV.01 untouched)

- **v497_celestial_wp5a_null_ideal.py** (CELEST.WP5A.01, 34 checks, exact integer/Fraction, no floats). Three results make the WP4 limit reading precise. *Stabilisation theorem*: the one-parameter family χ_w of graded Fock characters on the chiral jet generators (loop energy m , radial number r ; $E_w = m + wr$ in quarter units $u = q^{1/4}$) contains the chiral jet grading as its $w=2$ member (generator counts 64, 120, 128, 180, 192, 240, 256, 300 per degree), and its u^n coefficient equals the quarter-moded loop Fock for *all* $w \geq n+1$, strictly larger for every $w \leq n$ (verified $n \leq 8$, $w \leq 10$) — the v496 “boundary-limit shadow” is a precise *coefficientwise* limit $\varepsilon = 1/w \rightarrow 0$ with explicit threshold $w = n+1$, not a slice (convention clarified: $897266 =$ the quarter-moded loop Fock at $u^4 = \text{integer level } 1$). *The null ideal derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give $\text{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ with residual exactly zero (independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; $C_2 = (124, 96)$, Dynkin indices 6750/750; level-1 integrability ($\langle 2\theta, \theta^\vee \rangle = 4$, $\langle \omega, \theta^\vee \rangle = 2$) leaves the vacuum as

the only level-1 primary (Kac, used only through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ exactly. *Two routes, one number*: the μ_4 theta-split sector sum at $q^2 = (1036, 1024, 1040, 1024)$ gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space (q^3 : $34752 = \chi_3$). Negative controls: $SO(16)_1$ through the same pipeline gives *four* components ($5304 + 1820 + 135 + 1$), $5304 \neq 14^3$ (the $h^{\vee 3}$ identity is not generic), quotient $2076 = \Theta_{D_8}/\eta^8$ at q^2 (the recipe validated on a second algebra), block weights $(0, \frac{1}{2}, 1, 1)$ break one-block fusion (vs E_8 's integer $(0, 1, 1, 1)$); only $P = 4 = |\mu_4|$ periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor ($64 \neq 60$). *Honest [O]*: the limit does *not* generate the truncation — the singular vector as an *operator* in the limit net (WP5b), the GNS limit state (WP5c) and local normality/Haag duality (WP5d) stay open; the celestial route now has the same two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; SEAM.EQUIV.01 untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: WP5 subdivided — WP5a moved to *done* with the key numbers (stabilisation threshold $w = n+1$; $27000 = (h^{\vee})^3$ derived; quotient = sector sum = 4124; $SO(16)_1$ contrast), WP5b–e added as roadmap bullets with success/kill criteria (WP5b singular-vector operator in the quarter-moded current Fock; WP5c GNS limit state, kernel $\supseteq$ ideal; WP5d local net/Haag duality via the KLM route; WP5e BCOV/Costello–Li twistor uplift, partition function = E_4/η^8 incl. the $q^{-1/3}$ prefactor); K4 paragraph and the honest-success status line updated. Ledger: CELEST.WP5A.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a executed; WP5b–e [O] (roadmap)”). Wolfram: 4 exact v497 mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day ($394 \rightarrow 398$, ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + GATE.WOLFRAM.02 counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors $116/116 + 398/398$. Website: `papers.ts` contract mirror (“WP5a done, WP5b–e roadmap”), DAG `qft` node + v497, generated indices via build. Suite now 491 scripts, all green; SEAM.EQUIV.01 stays [O].

2026-07-21 (V) (comb-meta-limit re-run with the PG.06b FULL Vela recovery ingested — the stale single-obs PG.06b reference (noise-peak $F_0 = 11.19275$ Hz) corrected in the channel definitions; the new X-ray channel enters *censored* (no absolute ε), both aggregate limits numerically unchanged; *experiments level only* — *no verification module, no gate closes, no marker moves*)

- **experiments/comb-meta-limit channel update (methodology untouched).** The Vela X-ray channel (`channels.py`) now ingests the PG.06b FULL reduction (`pg06b_full_vela.json` in `pulsar-glitch-recovery/results`: 665 obs $\rightarrow$ 491 usable $\rightarrow$ 119 segment-coherent $\nu(t)$ points) instead of the superseded single-obs `pg06b_vela.json`; the stale code note quoting the noise-peak “detection” $F_0 = 11.19275$ Hz is corrected on record (true Vela $F_0 = 11.1861692$ Hz), and the channel key is fixed from the mislabel PG.07 to PG.06b. Censoring handled exactly by the existing Tier-B rule (nothing invented): the comb observable is a *linear* $\delta\nu(\tau)$ residual (μ Hz, not a fractional ln-flux modulation), both legs sit below the 2.8-period gate (2019: 2.17, 2021: 1.38), and the surrogate-calibrated injection never reaches ε_{90} below $\varepsilon = 1$ ($\varepsilon_{50} = 0.55$ in 2019 only) $\Rightarrow$ no absolute per-channel ε -UL exists for the DL/HKSJ combination; the channel stays enumerated-only (“no constraint”), carrying the normalised context amplitude $\sqrt{2g} = 0.40$ (leg gains 0.046/0.114, min $p = 0.014$ — the 2019 audit-level chance feature: Bonferroni $\times 10 = 0.14$, shuffle $p = 0.40$).
- **Full re-run (seed 0, deterministic): both limits numerically unchanged.** Boundary/horizon-scoped: `data_limited` (still no horizon channel with an absolute ε); all-channel universal-DSI 95% UL $\varepsilon < 0.11954$ ($\approx 12.0\%$; HKSJ $t_2 = 2.92$, $\rho = 0.3$ -inflated 0.137; $6.91\times$ the predicted $\varepsilon = 0.0173$), still driven by the surface log-flux channels A1/A4/A5 (A4 GRB the anchor, per-channel UL 0.108); now 10 channels enumerated / 3 usable. Injection self-consistency unchanged (inject 5% $\rightarrow$ recover 4.8%, UL 5.5%; coverage 0.92/150 trials). Surfaces synced: experiment README (tier table, ingest list,

PG.06b FULL bullet), `results/results.json`, scorecard row `data_value/source` via `build_evidence_scorecard.py` (status stays `data_limited`; counters unchanged, 120 rows), `experiments/README.md` catalog row. No claim; firewall intact.

2026-07-21 (IV) (Wolfram engine reactivated — the deferred second-path mirrors VERIFIED: base 116/116 + extension 394/394 (378 → 394, the 16 mirrors accumulated since 2026-07-14 now counted); GATE.WOLFRAM.01/02 and all counter surfaces synced; *no gate closes, no marker moves*)

- **Verified engine run 2026-07-21 (Wolfram Engine 14.3, license reactivated).** `wolframscript -file verification/wolfram/tfpt_readouts.wl` → 116 passed, 0 failed, ALL WOLFRAM CHECKS PASSED; `tfpt_readouts_extension.wl` → 394 passed, 0 failed, ALL WOLFRAM EXTENSION CHECKS PASSED. The 16 exact mirrors added while the engine was unactivated all enter the counted total (378 → 394): v479 Kronheimer bridge (2), v491 partition corollary (3), v493 celestial WP2 clock deformation (4), v495 celestial WP3 Green–Schwarz alignment (3), v496 celestial WP4 character shadow (4).
- **Counter surfaces synced.** `verification/wolfram/README.md` (current-state block + the five deferral notes resolved; v484/v485/v486/v487/v488/v492 stay mirrorable-but-unmirrored, Python-exact, so they are *not* counted); ledger GATE.WOLFRAM.01 (re-verified 2026-07-21) + GATE.WOLFRAM.02 (394/394, UPDATE note); `tfpt_safeguards` Layer-6 count + website `papers.ts` mirrors (116/116 + 394/394); v479 registry/`run_all` deferral note resolved. No numeric claim changes; the second computational path is simply verified-current again.

2026-07-21 (III) (CELEST.SEAM.01 WP3 + WP4 executed and promoted: v495 Green–Schwarz alignment (verdict B, honestly non-selective) + v496 character-vs-S-algebra type mismatch (verdict B(ii), boundary-limit shadow); contract now WP1–WP4 done, WP5 [O]; *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)

- **v495_celestial_wp3_gs_alignment.py (CELEST.WP3.01, 25 checks, exact Fraction/sympy).** The pre-registered WP3 “alignment test only”: the Okubo coefficient $\text{Tr}_{\text{adj}} X^4 / (\text{Tr}_{\text{adj}} X^2)^2 = 5/(2(\dim \mathfrak{g} + 2))$ is *derived* as an exact polynomial identity for all 8 algebras on Costello’s list ($\mathfrak{sl}_2, \mathfrak{sl}_3, \mathfrak{g}_2, \mathfrak{so}_8, \mathfrak{f}_4, \mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$; $\mathfrak{e}_6/\mathfrak{e}_7$ carved out of the v492 E_8 root set; negative control: $\mathfrak{sl}_4$ fails, $5/32$ vs $3/32$); the closed form $\tilde{\lambda}^2 = 10h^{\vee}/(\dim + 2) = h^{\vee} + 6$ holds exactly across the Deligne series; E_8 : $\tilde{\lambda} = 6$ (unit-trace) resp. $\lambda_{\text{fund}} = 1/10$ (adjoint-trace) exact, so $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\text{fam}}$ resp. $1/300$ — with κ/c_3 itself *irrational* ($2\sqrt{3}$). Byproduct: the printed $\lambda_{\text{so}_8}^2 = 3/2$ in Costello’s appendix A is a factor-2 slip (his own weight content gives 3). **Look-elsewhere (the hard fence):** the alignment format passes 8/8 across the whole list (zero selectivity), $\tilde{\lambda}$ -integrality $2/8$ (shared with $\mathfrak{sl}_3$), only $g_{\text{car}}=5 \mid h^{\vee}$ is $\mathfrak{e}_8$ -selective ($1/8$), the isolating conjunction is post hoc. Kill test K3 does *not* fire but is scope-demoted: *alignment survives; selectivity does not* — compatibility, not $\mathfrak{e}_8$ -selective evidence. Verdict B.
- **v496_celestial_wp4_character_shadow.py (CELEST.WP4.01, 25 checks, exact integer/Fraction/sympy).** WP4, the type mismatch head-on: the $(E_8)_1$ character $E_4/\eta^8 = (1, 248, 4124, 34752, 213126, \dots)$ is *not* a conformal block of the glue-equivariant $E_8[\mathbb{C}^2]$ S-algebra in its own jet grading — localised three ways: (a) spin $1 - d/2$ unbounded below and 248 never a tower dimension ($d \leq 100$); (b) growth $n^{2/3}$ vs $n^{1/2}$ (quadratic cumulative count $31s^2 + 92s + 60$, $31 = k + h^{\vee}$; f_n/χ_n strictly increasing $n = 1..12$); (c) the level-2 null ideal deletes exactly $27000 = 30^3 = (h^{\vee})^3$ out of $\text{Sym}^2(248) = 1 + 3875 + 27000$ — no jet analogue. *But* the boundary/period reading holds exactly at the current stratum (zero modes = $60 = C_0$ currents, jet slice cycles $(60, 64, 60, 64)$, one μ_4 period sums to 248, glue weights $(0, 1, 1, 1)$ integer; free loop Fock $897266 \gg 248$, so the truncation must come from a limit) — the same MMST

scaling-limit shape as SEAM.EQUIV.01 (v336/v449). K4 fires only against the exact-block reading; the constructive limit question passes to WP5. Verdict B(ii).

- **Contract + surfaces synced.** `tfpt_research_contracts` WP3/WP4 bullets moved to *done* with the look-elsewhere demotion in the text, K3/K4 marked evaluated, status line now “WP1–WP4 executed (verdicts B, B, B, B(ii)), WP5 [O]”; one conservative sentence added to the `tfpt_3` celestial paragraph; ledger rows CELEST.WP3.01 + CELEST.WP4.01 (new) and CELEST.SEAM.01 updated; v495/v496 Wolfram mirrors added to `tfpt_readouts_extension.wl` (3 + 4 exact checks; count with the next verified engine run — engine still unactivated); website `papers.ts` contract highlight “WP1–WP4 done” + DAG `qft` node extended. Suite now 490 scripts, all green; SEAM.EQUIV.01 stays [O].

2026-07-21 (II) (PG.06b FULL: the complete NICER Vela archive reduced — the first X-ray $\nu(t)$ leg of the recovery-comb search, verdict `data_limited`; the old PG.06b “pipeline proven” fold corrected on record; *experiments level only* — no verification module, no gate closes, no marker moves)

- **experiments/pulsar-glitch-recovery PG.06b FULL executed (`tfpt-pulsar vela-full`).** The full NICER Vela-pulsar (PSR B0833-45) L2 archive is reduced, with the manifest frozen *before* the bulk download (`manifest_full_reduction.json`: all 665 ObsIDs, preregistered 2019/2021/2024 glitch windows, 20-GB abort gate): 665/665 obs downloaded (2.6 GB, raw gitignored) $\rightarrow$ 491 usable $\rightarrow$ 240 per-obs detections $\rightarrow$ **119 segment-coherent $\nu(t)$ points** (108 fully coherent), median $\sigma_\nu \approx 0.6 \mu\text{Hz}$; barycentring validated against PINT to $3.4 \mu\text{s}$ — the *first X-ray $\nu(t)$ probe* of the search program. Sanity gate PASS: $\Delta\nu/\nu$ 2019 = 2.39×10^{-6} vs 2.5012×10^{-6} published, 2021 = 1.272×10^{-6} vs 1.26×10^{-6} ; the 2024 glitch is not reducible (only ~ 150 -s snapshots, 9.7 ks — documented).
- **Frozen-kernel comb ($\omega = 2.5827$, $\varepsilon_{\text{pred}} = 0.0173$): verdict `data_limited` (range- and amplitude-limited).** Reach 2019: 2.17 comb periods ($<$ the 2.8-period gate; beats the PG.08 public Vela reach 1.83), 2021: 1.38; the 2019 raw $p = 0.014$ has the kernel as smallest- p battery member BUT Bonferroni $\times 10 = 0.14$ and shuffle $p = 0.395 \Rightarrow$ chance-level by the preregistered escalation rule (audit flag, NOT a candidate); 2021 $p = 0.731$; $\ln \tau$ stack $p = 0.049$ /shuffle 0.307. Surrogate-calibrated injection at the real sampling and noise: $\varepsilon_{50} \approx 0.55$ (2019 only), ε_{90} never reached below $\varepsilon = 1$, 0% power at the predicted $\varepsilon = 0.0173$ — a weaker amplitude probe than the PG.08 radio residuals, but it independently reproduces the PG.06 range-blindness on real data. Second off-kernel flag: $\lambda = e$ in the 2021 leg (raw $p = 0.002$, Bonferroni 0.018; the same non-kernel periodicity class as PG.08 J0742–2822 — flagged, not claimed). The sub-day early window that would clear the gate is absent from the archive (first usable obs 3.4 d / 7 d post-glitch); next decider: <3 -d post-glitch X-ray observations of a future Vela glitch (eXTP class).
- **Correction on record.** The old PG.06b single-obs “pipeline proven” detection at $F_0 = 11.19275 \text{ Hz}$ ($H = 18.4$) was a noise peak; the true Vela frequency at that epoch is $F_0 = 11.1861692 \text{ Hz}$ ($H = 99.7$, consistent with the phase-connected PuMA ephemeris). The pipeline remains proven — by the corrected fold. Scorecard row updated to “recovery comb on the full NICER Vela X-ray $\nu(t)$ (PG.06b FULL)” (status stays `data_limited`; counters unchanged, 120 rows: 49 consistent / 9 tension / 26 null / 33 data-limited / 3 parked); `experiments/README.md` catalog + §1 prose, pulsar README, `experiments/next.txt` synced. No claim; firewall intact.

2026-07-21 (the two 2026-07-20 prediction-of-record rows promoted to the suite: **v494** — α_s running + μ -distortion branch discriminator machine-checked, ledger rows **COSMO.RUNNING.01/COSMO.MUDIST.01**, prediction surfaces synced; *no gate closes, no marker moves*)

- **v494_cosmo_running_mu_record.py** (23 checks, all green). The α_s -running and μ -distortion scorecard rows of 2026-07-20 (IV) get their load-bearing module so the new paper/website statements are machine-backed. [E] leading-order identities (sympy exact): $\alpha_s = dn_s/d \ln k = -2/N_*^2 = -r/6$, $\beta_s = -4/N_*^3$ (NOT $-8/N_*^3$); LO band $N_* \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$. [E] exact slow roll on the Einstein-frame Starobinsky potential (third-order Lyth–Riotto + finite-difference cross-check < 1%): $\alpha_s(51.4) = -7.09 \times 10^{-4}$, $\beta_s(51.4) = -2.70 \times 10^{-5}$. [C] data: Planck 2018 (-0.0045 ± 0.0067) + 0.57σ ; record leg P-ACT-LB ($+0.0062 \pm 0.0052$) -1.33σ (watch channel: positive central value; kill: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch). [C] μ -distortion (Chluba window, +21% calibrated on the Λ CDM anchor): sharp branch $N_*=51.4$ $\mu = 1.6 \times 10^{-8}$, profiled record $N_*=56.1$ $\mu = 1.94 \times 10^{-8}$, band $1.5\text{--}2.3 \times 10^{-8}$; branch split $\Delta\mu = 3.0 \times 10^{-9} \Rightarrow 3\sigma$ A_s/N_* branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; PIXIE baseline 9×10^{-8} does not reach; kill: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$). [E] comb blindness (honest S15): the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is structurally invisible in the μ window ($|F(\omega)| = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage-2050). Fences: N_* stays an external reheating input (GATE.NSTAR.01); the frozen band $[50, 60]$ stays the surface of record (v84/v86). Ledger rows **COSMO.RUNNING.01** and **COSMO.MUDIST.01** added.
- **Prediction surfaces synced (the missing mirror of 2026-07-20 (IV))**. **predictions.txt** gains the two rows (α_s [C] with the -1.33σ P-ACT-LB pull and the positive-running kill; μ -distortion [C] data-limited with the Voyage-2050 branch split); website **lib/predictions.ts** gains the matching cards **alpha-s-running** and **mu-distortion** (confrontation + experiment blocks pointing at **experiments/tfpt-discovery**); the cosmology DAG node lists **v494** and states the running/ μ readouts.
- **Paper bodies**. **tfpt_1** (Inflation interface) gains the running consistency-relation paragraph ($\alpha_s = -2/N_*^2 = -r/6$, exact slow roll, pulls, positive-running kill) with `\veri{v494_cosmo_running_mu_record.py}`; **tfpt_2** (inflation-pivot keybox) gains the μ -distortion branch-decider sentence (sharp 1.6×10^{-8} vs profiled 2.0×10^{-8} , 3σ at Voyage-2050) with the same `\veri`. No marker of any existing claim changes. (v494 is numerical/quadrature, Python-only by the suite convention, flagged in the docstring; GATE.WOLFRAM.01/02 untouched.)

2026-07-20 (VIII) (E8 mass-ladder blind-recovery bed EXECUTED: verdict validated at the detector level — the Cartan–PF ladder pipeline blindly recovers the published Zamolodchikov-E8 spectra of $\text{BaCo}_2\text{V}_2\text{O}_8$ (8/8) and CoNb_2O_6 (6/8 in-window) and rejects every control class; *experiments level only — confirms Zamolodchikov E8, not TFPT; no verification module, no gate closes, no marker moves*)

- **experiments/e8-ladder-bed/ executed (preregistered hypotheses/e8_ladder_bed_v1.yaml)**. The blind ladder detector (assignment DP + scale fit + 5000-ladder Monte-Carlo null; the same detector class TFPT uses on search surfaces, e.g. **hfqpo-ladder**) on the only public *known-answer* mass-ladder dataset: $\text{BaCo}_2\text{V}_2\text{O}_8$ (Zou+ PRL **127**, 077201; INS at the 4.7 T 1D QCP, Fig. 3a extraction QA-validated, 17 peaks blind) recovers **8/8 rungs**, $\chi^2/\text{dof} = 0.35$, MC $p = 0.0056$ (singles-only run $p = 0.0004$; σ -variant robust); CoNb_2O_6 (Amelin+ PRB **102**, 104431; verbatim THz values) recovers **6/8 in-window rungs**, $\chi^2/\text{dof} = 0.52$, $p = 0.0038$ (m_7/m_8 lie outside the 0.6-THz low-pass window; the broad 0.43-THz (m_1+m_2)-continuum onset is correctly left unassigned). Cartan–PF self-test: the sorted PF eigenvector of $2I - A(E_8)$ equals the Zamolodchikov ladder to 1.5×10^{-15} ($m_2/m_1 = \varphi$ to 2.2×10^{-16} ; lowest Cartan

eigenvalue $2 - 2\cos(\pi/30)$ to 4.5×10^{-14}).

- **Control battery — every class rejected.** Scramble 0.5%/1.5% false-positive rate (200 log-gap permutations, near-duplicate accounting); offset $\pm\{0.3, 0.5\}m_1$: 0/8 detections; jitter dose-response $\leq 5\%$ at the destructive 30% level; wrong ladders on the real data (harmonic 1..8, A_3 -PF, D_8 -PF, Airy confined-spinon) all beaten by E8; the synthetic Airy bed (the true zero-field physics of these compounds) yields **0** valid rungs. The one disclosed v1.1 criteria amendment (jitter as dose-response, scramble near-duplicates, χ^2 -gated wrong-ladder comparator) is fully documented in the hypotheses YAML with all v1 numbers preserved; the recoveries themselves were never touched.
- **Integration (firewall intact).** Scorecard row “E8 mass-ladder blind recovery bed” moves parked $\rightarrow$ consistent (stage not_applicable, tier analog_positive_control, leakage detector_control — the executed-analog basket of qc-recovery-kernel and qgeo-eit-soff); counters 120 rows: 49 consistent / 3 parked; experiments/README.md catalog row + stats (generator); website predictions.ts EXPERIMENTS_AUDIT counters; experiments/next.txt. This validates the ladder-detector pipeline — it confirms Zamolodchikov E8, *not* the TFPT axioms: no [E], no \veri, no ledger row.

2026-07-20 (VII) (v493 — WP2 of CELEST.SEAM.01 executed: the clock-invariant deformation $XY=Z^4+a_0$ is a PURE SCALE with the $\tau=i$ pillowcase frozen, the μ_4 clock IS the Picard–Lefschetz/Coxeter monodromy of the seam deformation, and the BHS deformed-algebra pattern transfers $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$ with no sector leak; verdict B, contract updated to WP2 done, no gate closes, no marker moves, SEAM.EQUIV.01 and the v216 residual stay [O])

- **v493_celestial_wp2_clock_deformation.py (CELEST.WP2.01).** WP2 of the celestial contract, sympy exact (47 checks), promoted from the experiments/tfpt-discovery/probe celestial_seam_wp2_clock_deformation_probe.py. *Geometry:* $P(iZ)=P(Z)$ forces $a_3=a_2=a_1=0$ sharply and two-sided ($\prod_k (Z - i^k w) = Z^4 - w^4$); $XY=Z^4+a_0$ is smooth iff $a_0 \neq 0$ (disc = $256 a_0^3$); the branch points $Z^4 = -a_0$ are one free μ_4 orbit in *every* fibre with cross-ratio 2; the binary-quartic invariants of $w^2=Z^4+a_0$ are $I=12a_0$, $J=0$ *identically*, so $j=1728$ and the $\tau=i$ pillowcase shape (v168/v214) is *frozen* for all a_0 — a_0 is a pure scale, no shape modulus survives clock-invariance. *Periods/monodromy:* $\Pi = t(i-1)(1, i, i^2)$, all three vanishing spheres carry the same volume $\sqrt{2}t$ (lockstep); the clock acts on H_2 as a *Coxeter element* of $W(A_3)$ (char x^3+x^2+x+1 , eigenvalues $\{i, -1, -i\}$ = the A_3 exponents, $h(A_3)=4=|\mu_4|=N_{\text{fam}}+1$), fixes no cycle ($\det(A-1)=-4$), and *is* the Picard–Lefschetz monodromy of the family ($a_0 \rightarrow e^{2\pi i} a_0$ gives the 4-cycle); the surviving direction is the weight-1 χ_1 -Fourier diagonal $(1, i, i^2)$, *not* the naive invariant diagonal; the versal weights $(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)$ are the regular μ_4 representation. *Algebra (BHS $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$, arXiv:2305.09451):* mode dictionary $\{w[m, n] : m \equiv n \bmod 4\}$; the fibre bracket $-4\text{Nambu}(XY-Z^4-a_0)$ anchored at the $a_0=0$ orbifold ($\{X, Y\}=16Z^3$, Jacobi + Casimir exact, uniform clock-degree -1); S-algebra corrections exactly *linear* in a_0 at the $\mathbb{Z}_4$ wrap, conserving the μ_4 grade — no sector leak, the v492 equivariant sector deforms consistently; $a_0 \leftrightarrow$ BHS’s weight-0 c^2 slot; matrix factorisations deform along all orbit splittings, the clock fixing exactly the antipodal $2+2$ sector (= the EH/ $\mathbb{Z}_2$ middle node, $4=2 \times 2$). *Negative controls:* $a_1 Z \Rightarrow j=0$ (hexagonal); pure $a_2 Z^2$ stays singular (A_1); $a_2 \neq 0 \Rightarrow j=1556068/81 \neq 1728$; explicit grading leak — doubly sharp. *Verdict B* via three named [C] identifications (-4Nambu = the $k=4$ CCA bracket; period = root difference; seam-scale reading via $\text{clock}^2=\text{deck}$); the v216 residual is *typed*, not moved (given the order-4 clock the square is automatic — a relocation of the same input, no new derivation).
- **Contract updated to WP2 done.** tfpt_research_contracts CELEST.SEAM.01 section: WP2 bullet “finite, next” $\rightarrow$ *done (verdict B)* with the two sharpenings (clock = Coxeter/Picard–Lefschetz monodromy; surviving direction = χ_1 -diagonal, a_0 pure scale); kill-test paragraph K1

“survived by WP1 and WP2”; status line CELEST.WP2.01 [E]/[C] executed, WP3–WP5 stay [O]. Ledger rows CELEST.WP2.01 (new) + CELEST.SEAM.01 (WP2 done) + GATE.WOLFRAM.02 (four exact v493 mirrors added to tfpt_readouts_extension.wl, counted with the next verified engine run — engine unactivated, same convention as v491); registry + run_all (487 modules, suite green); website mirrors (papers.ts contract section, generated script index). SEAM.EQUIV.01 stays [O]; no marker moves anywhere. Exact (sympy).

2026-07-20 (VI) (HFQPO “third tooth”: the preregistered archival RXTE PCA scan has RUN — a well-powered null with limits; scorecard row S1.e data_limited→null, no marker moves)

- **H3 stage 2 executed (experiments/hfqpo-ladder/results/archival/).** The preregistered archival scan (hypotheses/hfqpo_v1.yaml, archival_design) ran end-to-end on public HEASARC data: 77 ObsIDs / 4.42 GB across all four 3:2-pair sources (GRO J1655–40, XTE J1550–564, GRS 1915+105, H1743–322). Sanity gate PASS (11/12 published pair lines reproduced, e.g. 450.3 Hz at 5.37% rms 4.4σ , 280.1 Hz at 3.71% 9.9σ); injection calibration PASS in all four sources (90%-recovery sensitivities 5.11/2.68/0.89/1.65% rms at the tooth frequencies).
- **Blind-scan verdict: NULL with limits.** Neither the geometric tooth $\nu_3 = 1.5\nu_u$ (661.5/414/252/363 Hz) nor the integer control line $4\nu_0$ (588/368/224/322.7 Hz) is detected in any source ($\sim 0\sigma$ single-trial; trials correction $N=8$ as preregistered). 3σ rms upper limits (tooth, primary bands): 3.06/1.61/0.53/1.26% — in every source at or below the strength of the detected upper pair line. The ladder reading gains no support but is not killed (the harmonic_hit branch did not fire either; the anti-kernel record 92=184/2, 34/68 stands); GR parametric resonance stays the favorite. Caveats: GRS 1915+105 scanned in a proxy state (the 113/168 Hz epochs were never published at ObsID level); H1743’s program is 80146 (YAML said 80138); GRO hard-band sensitivity 5.1%.
- **Integration.** Scorecard row S1.e (hfqpo-ladder) data_limited→null (120 rows: 26 null / 33 data_limited), next decider eXTP-class sensitivity below the 0.5–3.1% rms limits; HFQPO search-target paragraph updated in tfpt_horizon_readouts (+ the papers.ts mirror); website predictions.ts (hfqpo-tooth entry + EXPERIMENTS_AUDIT counters); predictions.txt; experiments/README.md; experiments/next.txt. No verification/ module, no ledger row, no marker move (the search target stays [O]).

2026-07-20 (V) (v492 + the new research contract CELEST.SEAM.01 — the celestial-holographic route: the E_8 μ_4 -glue is the flat $\mathbb{Z}_4$ monodromy of the equivariant celestial chiral algebra on the A_3 ALE space, with the exact correction clock²=deck (spin bridge $\mathbb{Z}_8$, $8=2|\mu_4|$); verdict B, no gate closes, no marker moves, SEAM.EQUIV.01 stays [O])

- **v492_celestial_z4_orbifold.py (CELEST.WP1.01).** WP1 of the new contract, sympy exact (36 checks): the v128 μ_4 -glue grading of E_8 ($240 = 52+64+60+64$, additive on all 6720 root pairs) is *inner* — $h = (2, 2, 2, 2, 2; 0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\text{Ad}(\exp(2\pi i h/4))$, an order-4 inner torus automorphism = a flat $\mathbb{Z}_4$ monodromy in the Kronheimer–Nakajima sense; the A_3 -side detector $h' = (0^5; 1, 1, 1, -3)$ reads the same class = the diagonal (1, 1) of the discriminant group $\mathbb{Z}_4 \times \mathbb{Z}_4$ (the v92/v125 Lagrangian glue). $\mathbb{C}^2/\mathbb{Z}_4$ is verified as the A_3 singularity $XY=Z^4$ (Hilbert series $(1+t^4)/((1-t^2)(1-t^4))$, Molien exact); the glue-equivariant SDYM(E_8) sector *closes* with graded dimensions $60(d+1)/64(d+1)$ — possible only because $\dim \mathfrak{g}_j = (60, 64, 60, 64)$ — with zero modes = the carrier $D_5 \oplus A_3 + \text{Cartan} = 60$ and density $1/4 = 1/|\mathbb{Z}_4|$; the four glue-sector characters sum exactly to $\chi_{(E_8)_1} = 1+248q+4124q^2+34752q^3$ (v377), with sector weights = the discriminant form $(5x^2+3y^2)/8$ and integer glue-diagonal $h = (0, 1, 1, 1)$ (= locality of the $(E_8)_1$ extension). *Critical correction*

(*verdict B*): the A_3 deck acts on the celestial sphere as $z \mapsto -z$ (order 2), *not* as the order-4 clock; the clock is the $U(2)$ phase $\text{diag}(i, 1)$ (normalising the deck, $\det = i$ rotating the holomorphic symplectic form by the μ_4 generator), with the exact spin bridge (spin clock) $^2 = \text{deck}$ ($\mathbb{Z}_8$, $8 = 2|\mu_4|$, the $c_3 = 1/(8\pi)$ winding integer). Clock-invariance selects the 1-parameter A_3 deformation $XY = Z^4 + a_0$ whose four branch points are one μ_4 orbit with cross-ratio 2 (v168/v214 pillowcase). Negative controls: the false action $\text{diag}(i, i)$ killed three ways, the false glue breaks additivity on 3112/6720 pairs, rigidity = $\text{Aut}(\mathbb{Z}_4)$.

- **New research contract CELEST.SEAM.01** in `tfpt_research_contracts` (own section, the fourth contract alongside $U_{\text{wall}}/G_{\text{metric}}/\text{CONTRACT.F.01}$): hypothesis (seam = glue-equivariant $\mathbb{Z}_4$ -orbifold sector of the CCA on the A_3 ALE; twistorial bulk = the *self-dual* sector only), typing/non-circularity (the continuum $(E_8)_1$ existence — the SEAM.EQUIV.01 target — is *not* an admissible input), work packages WP1 (done, verdict B) through WP5 (second continuum route, Costello–Li grade), pre-registered kill tests K1–K4 (K1 survived by WP1), and an honest success estimate (full success low, partial findings high). Ledger rows CELEST.WP1.01 + CELEST.SEAM.01; website mirror (`papers.ts` section, script index). SEAM.EQUIV.01 stays **[O]**; no marker moves. Exact (sympy).

2026-07-20 (IV) (evidence scorecard: five new typed rows — the α_s running prediction of record, the μ -distortion A_s/N_* branch discriminator, and three parked analog beds (bosonic-E8 thermal Hall, E8 mass-ladder recovery, Efimov DSI tuning); *experiments level only — no verification module, no gate closes, no marker moves*)

- **α_s running (prediction of record, consistent)**. The Starobinsky branch fixes the running of the spectral index: $\alpha_s = -2/N_*^2 = -7.6 \times 10^{-4}$ at $N_*=51.4$ (exact slow roll -7.09×10^{-4} ; band $N_* \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$) and $\beta_s = -4/N_*^3$ (leading order; NOT $-8/N_*^3$) — the identity $\alpha_s = -r/6$ holds only at leading order ($\sim 7\%$ off exact). Data: Planck 2018 -0.0045 ± 0.0067 ($+0.57\sigma$), P-ACT $+0.0060 \pm 0.0055$ (-1.22σ), record leg P-ACT-LB $+0.0062 \pm 0.0052$ (-1.33σ , Calabrese et al. 2025 JCAP); β_s (Planck run+runrun) 0.010 ± 0.013 (-0.77σ). Forecast: LiteBIRD+CMB-S4 $\sigma(\alpha_s) \approx 4.9 \times 10^{-3}$ (0.14σ); CMB-S4+DESI+Euclid statistical $\sim 3.3 \times 10^{-4}$ (2.2σ , systematics-limited). Same `N_star_reheating` independence group as the $n_s/r/A_s$ rows (anti double-counting: never an independent hit); kill: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch. Probe: `experiments/tfpt-discovery/alpha_s_running_probe.py`.
- **μ -distortion (prediction of record, data-limited)**. Chluba window integral (calibrated +20% to the full computation): sharp branch $N_*=51.4$ ($A_s=1.76 \times 10^{-9}$) $\mu \approx 1.6 \times 10^{-8}$, profiled record $N_*=56.1$ ($A_s=2.10 \times 10^{-9}$) $\mu \approx 2.0 \times 10^{-8}$, band $1.5\text{--}2.3 \times 10^{-8}$; the 23% branch split ($\Delta\mu = 3.0 \times 10^{-9}$) gives a 3σ A_s/N_* branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; the PIXIE baseline $|\mu| < 9 \times 10^{-8}$ does NOT reach — factor 4–7 above the band; FIRAS $|\mu| < 9 \times 10^{-5}$). Honest S15 typing: the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is STRUCTURALLY BLIND in the μ window — window attenuation $|F(\omega)| = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage 2050 — the theory itself predicts where it is invisible, so NO separate comb-search row is opened. Shares the `Nstar_branch` alternative group with the two A_s rows (one branch question, several readings). Kill: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$. Probe: `experiments/tfpt-discovery/mu_distortion_probe.py`.
- **Three parked analog beds (parked_analog; honest typing: they validate E8 edge/spectrum physics or the comb DETECTOR, never the TFPT axioms)**. (i) Bosonic E8 phase / thermal Hall: $\kappa_{xy}/T = 8 \times (\pi^2 k_B^2 / 3h)$ with no charge Hall (seam edge $c_-=8$, v450/v451/v489); no engineered material yet ($\alpha\text{-RuCl}_3$ $c_-=1/2$ contested: ultraclean half-quantized plateau, npj QM 2025, vs phonon-dominated κ_{xy} , PRX 12, 021025). (ii) E8 mass-ladder validation bed ($\text{CoNb}_2\text{O}_6/\text{BaCo}_2\text{V}_2\text{O}_8$): all 8 one-particle masses published (Zou et al. 2021 PRL 127, 077201,

NMR+INS; Amelin et al. 2020 PRB 102, 104431, THz m_1 – m_6 ; Coldea et al. 2010 Science $m_2/m_1=\varphi$) — blind Cartan-PF ladder recovery on the published spectra (8/8 masses + rejection of scrambled controls) as detector validation; confirms Zamolodchikov E8, not TFPT. (iii) Efimov DSI tuning: $s_0 = \pi/(6 \ln(3/2)) = 1.2914 \Leftrightarrow \lambda = (3/2)^6 = 11.391$ at $M/m = 8.67$ (2 resonant pairs); best candidate $^{52}\text{Cr}_2$ – ^6Li ($M/m=8.635$, $s_0=1.2890$, $\lambda=11.44$, $+0.44\%$ vs $(3/2)^6$, gap deviation $+0.18\%$); alternatives $^{39}\text{K}_2$ / $^{41}\text{K}_2$ – ^6Li ($\lambda = 11.59/11.20$, needs double Feshbach control), $^{176}\text{Yb}_2$ – ^{23}Na ($+5.9\%$); solver validated on the Naidon–Endo anchors (homonuclear 22.69, Cs_2Li 4.877); first end-to-end HARDWARE injection of the comb detector at tunable $\lambda \approx 11.39$; kill condition honestly “none on TFPT axioms — detector validation only”. Probe: `experiments/tfpt-discovery/efimov_s0_massratio_probe.py`.

- Scorecard regenerated (`experiments/build_evidence_scorecard.py`): 120 rows — 48 consistent, 9 tension, 25 null, 34 data-limited, 4 parked; website mirror (`EXPERIMENTS_AUDIT` in `predictions.ts`) synchronised. Experiments level only; the verification suite, ledger and papers are untouched.

2026-07-20 (III) (v491 — $g_{\text{car}}=5$ as the partition COROLLARY of the four marks: 4 into exactly 3 positive parts is uniquely $\{1, 1, 2\}$, sharpening v53; the residual is the weight-typing postulate — AX.P2.01 stays a declared axiom; *no gate closes, no marker moves*)

- `v491_p2_partition_corollary.py` (P2.PARTITION.01). Sympy-exact: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is UNIQUE — $\{1, 1, 2\}$ — so $e = (e_1, e_2, e_3) = (4, 5, 2)$ and hence $g_{\text{car}} = e_2 = 5$, $|\mathbb{Z}_2| = e_3 = 2$, $N_{\text{fam}} = e_2 - e_3 = 3$ are *corollaries*, not inputs (consistency fixed point $(2^4 - 1)/5 = 3 = \text{rank } H^1$). [E] negative controls, each assumption load-bearing: drop positivity $\rightarrow$ 4 solutions, e_2 ambiguous $\{0, 3, 4, 5\}$; sum 3 $\rightarrow e_2=3$; sum 5 $\rightarrow$ not unique; 2/4 parts $\rightarrow e_2 \in \{3, 4\}/\{6\}$; Mehta–Seshadri RATIONAL weights $\rightarrow e_2 = \frac{16}{3}$ resp. $\frac{44}{9} \neq 5$ (integrality = the Birkhoff–Grothendieck splitting-type typing, not MS weights). [E] the v53 sharpening: v53 pins $(1, 1, 2)$ with ALL THREE targets $(4, 5, 2)$; the partition needs only $e_1=4$. [E] sum rule: $4 \times \frac{N-1}{2} = 4$ iff $N=3$; the v115 route $|a|_1 = |\mu_4| \text{tr}(A_0) = 4$ agrees. Independence chain: #marks = 4 from v216 (P1-side, no g_{car} input); rank $H^1 = 3$ pure topology (v137/v228); $|a|_1 = 4$ g_{car} -free (v115). [C] honest residual: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality/positivity $h^0=0/\text{sum}$ “given (U)”); AX.P2.01 stays a declared axiom, and the N_{fam} direction reversal vs `tfpt_constants` is typed as an over-determination check (never double-counted). Integrated in introduction (anchor half-forced paragraph), `tfpt_1` ((5, 3) keybox), `tfpt_2` (H2 winbox) and `origin_theory` ((5, 3) keybox); ledger row P2.PARTITION.01; three exact Wolfram mirrors added to `tfpt_readouts_extension.wl` (counted with the next verified engine run, engine currently unactivated); website mirror (DAG node P2). Exact (sympy).

2026-07-20 (II) (v490 — the T^2 spin-structure parity census: $\mu_{\text{pre}}=4$ witnessed on the lattice, the 16-fold-way bridge $\theta_v=1$ on measured numbers, and the honest TEE non-discriminator fence; *no gate closes, no marker moves*)

- `v490_seam_parity_census.py` (SEAM.MU.01). The T^2 ground-state fermion parity of the v367 $p+ip$ collar, resolved per spin structure (AA, AP, PA, PP) and determined TWICE — Read–Green TRIM product AND real-space Majorana Pfaffian (Schur), relative to the trivial $M=3$ phase, with the twisted real-space spectra matching the Bloch bands to 9×10^{-15} . [E] per layer $(+, +, +, -)$ (the $\nu=1$ Ising signature, TRIM $\equiv$ Pfaffian in every sector); [E] the 16-layer carrier has parity¹⁶ = + in ALL FOUR sectors $\Rightarrow$ gauged torus GSD = 4 = $\mu(SO(16)_1) = |\det \text{Cartan}(D_8)|$ — the lattice ground-state witness of “ $\mu_{\text{pre}}=4$ before condensation” (step 1 of the v378 KLM chain); [E] the 16-fold-way bridge on MEASURED numbers: $\theta_v = e^{2\pi i \nu/16} = 1$ exactly at $\nu = 2c_- = 16$ (v489), rivals $\nu = 1, 2, 8$ not holomorphically condensable; [E]

honest fence: the fermionic Kitaev–Preskill TEE is identically zero for BOTH $M=1$ and $M=3$ ($\gamma_f=0.000001/0.000000$) — it does NOT discriminate $(E_8)_1$ vs $SO(16)_1$. [C] the KLM step $\mu: 4 \rightarrow 4/2^2 = 1$ stays the cited theorem (the condensed model is interacting); the numerics pin only the premise package $\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}$. [O] SEAM.EQUIV.01 stays [O] (continuum existence, MMST v336). Integrated in `tfpt_research_contracts` (new S3-stack paragraph); ledger row SEAM.MU.01; website mirror (DAG node `qft`). Python (numpy + scipy Schur + sympy exact θ_v).

2026-07-20 (v489 — the chiral central charge $c_-=8$ measured DIRECTLY from the bulk ground state via the Kim–Shehab–Kim modular commutator: the first ground-state witness for premise (i) of SEAM.EQUIV.01, independent of the Chern route (v367) and the KLM count (v378); no gate closes, no marker moves)

- `v489_seam_modular_commutator.py` (SEAM.CCC.01). The Kim–Shehab–Kim modular commutator $J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-$ (PRL 128 (2022); arXiv:2110.06932/2110.10400) is evaluated on the EXACT BdG ground state of the v367 $p+ip$ seam collar model, entirely from the Peschel Majorana covariance ($W_X = -2 \text{artanh } \Gamma_X$, $J = \frac{1}{4} \text{Tr}([W_{AB}, W_{BC}] \Gamma_{ABC})$), on a disk about the torus centre cut into three 120° wedges. [E] ground-state purity $\text{eig}(i\Gamma) = \pm 1$ to 6×10^{-15} ; [E] per-layer $c_- = 0.499998$ at $L=32$ (convergence 0.499793 at $L=16 \rightarrow 0.500000$ at $L=48$, deviation -1.8×10^{-7}); [E] controls: trivial $M=3 \rightarrow 0.000000$, orientation flip $M=-1 \rightarrow -0.499986$; [E] layer additivity exact (1.4×10^{-14}); [E] the 16-layer carrier lands on $c_-=8=g_{\text{car}}+N_{\text{fam}}$ (direct 7.9967 at $L=16$, additivity route 8.0000); [E] clip invariance 5×10^{-8} . The FIRST ground-state witness for premise (i) of SEAM.EQUIV.01 ($c_-=8$) that is INDEPENDENT of both the Chern-number route (v367) and the counting route (v378) — read from entanglement data alone. [O] honest fence: does NOT close SEAM.EQUIV.01; the continuum scaling limit (MMST, v336) and the cited KLM condensation step are untouched. Integrated in `tfpt_research_contracts` (new S3-stack paragraph) and `tfpt_1` (the SEAM.EQUIV.01 package paragraph); ledger row SEAM.CCC.01; website mirrors (DAG node `qft`, papers/glossary/FAQ inventories). Numerical (numpy), Python-only by nature.

2026-07-19 (editorial — positioning section *TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram* added to introduction + website mirror; no result, no status change)

- `introduction.tex` — new section “TFPT in the light of algorithmic physics”. Positions the compiler against the Zuse–Schmidhuber–Wolfram algorithmic-physics lineage: three shared instincts (minimum description length as the selection principle, deterministic resource-bounded generation, compressibility as the aesthetic) and three deliberate divergences (one *forced* program vs. an ensemble of all universes; an algebraic Lie/lattice/VOA description language vs. Turing bit-strings; and falsifiability — dated, kill-tested predictions — as the load-bearing difference). Explicitly typed as *positioning*: it introduces no formula and moves no status marker; its only machine-checked hooks are already in the suite (the grammar-internal null model and the icosahedral/McKay bedrock).
- **Website mirror.** A matching curated section was added to the introduction entry in `website/lib/papers.ts`.

2026-07-15 (III) (v488 — the 3/5 Clebsch door ENUMERATED AND CLOSED: no minimal Pati–Salam/ $SO(10)$ Majorana channel can produce 5/3 — the v482 escape is decided dead, the v481 window candidate stands as the honest endpoint; a clean negative result, no gate closes)

- **v488_majorana_clebsch_door.py (FLAV.NUSCALE.04).** v482 named ONE escape for the integer seesaw rung $M_R = c_3^3 \bar{M}_{\text{PI}}$: a third-generation Majorana factor $r \approx 1.6696$, 0.18% from $g_{\text{car}}/N_{\text{fam}} = 5/3$ — declined for lack of mechanism. The mechanism search is now EXECUTED: **[E]** NO RENORMALISABLE CHANNEL — $\text{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$ and the $\mathbf{126}$ is NOT in the E_8 hull $\{1, 10, 16, 45\}$ (v247), so M_R only via the $d=5$ operator $(\mathbf{16}_F \mathbf{16}_H)^2/\Lambda$; channels $\mathbf{16} \times \mathbf{16} = \mathbf{1} + \mathbf{45} + \mathbf{210}$ with only singlet + $\mathbf{45}$ admissible. **[E]** CHARGE TABLE: $(B-L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$, $\text{Tr}_{16} Y^2 = \frac{10}{3}$ (v485). **[E]** THE $\{2, 3\}$ -SMOOTH THEOREM (the door closes): every nonzero ν^c channel weight $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$ is $\{2, 3\}$ -smooth, and the multiplicative closure of $\{2, 3\}$ -smooth rationals never contains $\frac{5}{3}$ (no factor 5); the UNIQUE natural 5 of the embedding, $k_Y = \frac{5}{3}$ (Ginsparg, v470), is a full-multiplet trace whose direction has $Y(\nu^c) = 0$ EXACTLY — structurally decoupled from the Majorana operator. **[E]** GENERATION-BLINDNESS: Clebsches are family-space scalars; $\text{diag}(1, 1, \frac{3}{5})$ cannot arise from group theory by construction. **[O]** compiler-flavor inventory typed and declined: exact $\frac{3}{5}$ ’s only in wrong slots (K lepton gen2/gen1; $R + L$ lepton gen3/gen2; never gen3 vs a degenerate base). NET: the rung+5/3 rescue is DEAD as a mechanism candidate; the v481 one-parameter window candidate (M_R free inside the PS window, the $\Sigma m_\nu = 0.059$ eV floor as kill test) stands. Contract provenance: `experiments/theory-contracts/clebsch01_majorana_35_door.py`. Integrated in `tfpt_2_standard_model` (new paragraph closing the seesaw arc); ledger row FLAV.NUSCALE.04 + update note on .03; Python-only (Fraction/sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-15 (II) (v487 — the lazy split FORCED by the resumed clock: the walk’s one-step spectrum is the complete v124 clock ladder below the wall, deck parity = rung index, and the two comb tones are one bend apart — the v486 [O] “split selection” closes onto established structure; no gate closes, no marker moves)

- **v487_transfer_clock_rungs.py (HYP.REWRITE.03).** v486 left the split selection open (“ $(\frac{1}{2}, \frac{1}{6})$ vs uniform $(\frac{1}{3}, \frac{1}{3})$ is selected by the physical spectrum, not yet derived”; named candidate: the vague “sheet-symmetric choice”). Closed: **[E]** RUNG READING — the lazy rule’s one-step spectrum is EXACTLY the complete resumed-clock ladder below the wall (v124: $\lambda_n^{1/p^2} = 1 - n/N_{\text{fam}}$, pole at $n=N_{\text{fam}}$): $\text{eig}(M) = \{1, \frac{2}{3}, \frac{1}{3}\} = \{1 - n/3 : n = 0, 1, 2\}$, while v327’s uniform rule puts its second pair mode ON the wall ($0 = 1 - 3/3$) — the v486 degeneracy re-read as “odd channel collapsed to the wall”. **[E]** DECK PARITY = RUNG INDEX: $[B, \sigma] = 0$; the σ -even pair mode carries rung $n=1$ (survival $\frac{2}{3}$), the σ -odd mode rung $n=2$ (survival $\frac{1}{3}$) — each clock step flips the deck parity; this is the structural home of the v486 parity assignment (ω_1 even-channel, ω_2 odd-channel) that `experiments/frb-parity-comb` tests. **[E]** FORCING: demanding the faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) + $\mathbb{Z}_2$ equivariance + non-negative rates forces UNIQUELY both the split $(\frac{1}{2}, \frac{1}{6})$ AND the rung assignment even $\rightarrow$ 1, odd $\rightarrow$ 2 (the swapped assignment needs $h = -\frac{1}{6} < 0$); the “sheet-symmetric choice” candidate is superseded by this proof. **[E]** BEND COROLLARY: $\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3 = 2.70951\dots$ — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. **[O]** what remains open is exactly what was open before v486: the arity $\{2, 3\}$ (anchor input) and the clock’s semiclassical derivation (R1); no independent choice remains at the walk level. Contract provenance: `experiments/theory-contracts/lazy01_clock_rung_forcing.py`. Integrated in `origin_theory` (new paragraph after the v486 block); ledger row HYP.REWRITE.03 + update note on .02; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-15 (v486 — generator hunt: the FULL transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ from ONE forced local rule — the unique lazy $\mathbb{Z}_2$ -pair walk (stay, hop, leak) = $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3})$; the previously generator-less $(1/3)^6$ mode gets its origin; v327's uniform rule superseded as full-spectrum generator — *no gate closes, no marker moves*)

- **v486_transfer_full_rule.py (HYP.REWRITE.02).** The verified seam transfer spectrum is $\{1, (2/3)^6, (1/3)^6\}$ (v54/v221), but the only local generator in the suite (v327) produces $\{1, \frac{2}{3}, 0\}$ — its zero mode persists under every power, so no iterate reaches the third mode. [E] UNIQUE-NESS: for the symmetric 3-channel rule $M(s, h) = [[1, 0, 0], [0, s, h], [0, h, s]]$ (one absorbing family channel + $\mathbb{Z}_2$ pair) the survival spectrum is $\{s+h, s-h\}$; demanding the PHYSICAL pair $\{\frac{2}{3}, \frac{1}{3}\}$ forces UNIQUELY (stay, hop, leak) = $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/|\mathbb{Z}_2|, 1/(|\mathbb{Z}_2|N_{\text{fam}}), 1/N_{\text{fam}})$ — the lazy $\mathbb{Z}_2$ -pair walk, every rate an atom expression (honest direction: spectrum $\rightarrow$ rates). [E] SIX-HAND GENERATION: over the order-6 = $2N_{\text{fam}} = |R^+(A_3)|$ hand, $\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$ exactly — both decay gaps now have one recursive generator: $6 \ln \frac{3}{2}$ (recovery, v76) and $6 \ln 3$ (subdominant, previously generator-less). [E] SUPERSESSION: the uniform rule's zero mode is power-persistent; its λ_2 arity story ($\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$) is unchanged. [O] the arity $\{2, 3\}$ stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay = $1/|\mathbb{Z}_2|$ as the sheet-symmetric choice). Discovery provenance: `experiments/tfpt-discovery/generator_hunt_full_transfer_rule.py` (plus a typed anchor-bilinear coverage inventory — known hits $a^T K a = 41$, Koide 53, and the recorded-only $1^T Q a = 16$ — and the lepton-exponent observation $(5, 3, 2) = K\text{-row} = \text{atom triple}$; no unforced claims, v354/v355 discipline). Integrated in `origin_theory` (new paragraph after the v327 rewrite block); ledger rows HYP.REWRITE.02 + update note on .01; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-14 (IV) (v485 — the merged analytic target's computable side COMPLETED: the diagonal self-energy is renormalised to ZERO exactly at the KMS seam scale, the mark contact calculus resums in closed form (BFK route), and the 48/41 multiplicities are ONE state set under two response weights — the residual collapses onto the SEAM.EQUIV.01 face alone; *no gate closes, no marker moves*)

- **v485_contact_diagonal_closed.py (SEAM.CONTACT.UNIT.02).** The v484 merge left one analytic step: the diagonal (self-energy) zeta-renormalisation at the marks + the multiplicity matching. Settled at the computable level: [E] DIAGONAL ZERO AT THE KMS SCALE — the renormalised coincident-point Green function of $|D|$ on circumference ℓ is $G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)$ EXACTLY (symbolic limit + 30-digit checks); it vanishes iff $\ell = 2\pi = 1/(4c_3)$, the v239 KMS seam circumference — the seam scale is the unique zero-self-energy scheme point, and the log is the pure SCALE channel (matching the structure: $F_{U(1)}$ carries exactly ONE log term, the transport $8b_1c_3^6 \ln(1/\varphi_{\text{seam}})$, while the φ_0^{ret} split is log-free). [E] CLOSED-FORM RESUMMATION (the BFK/gluing route already in the suite, v151): with the diagonal zero, the finite-rank mark determinant resums exactly — $\det(I - uC) = (1 - 4u)(1 + 2u)^2$, $u = \varepsilon c_3 \ln 2$, all orders, the linear term ABSENT precisely because the diagonal vanishes ($\text{Tr } C = 0$); the series matches v484's $\text{Tr}(C^k) = 4^k + 2(-2)^k$ term by term. [E] WEIGHT DICHOTOMY: the SAME 48 = 3×16 admissible states carry BOTH multiplicities — counted flat they give the φ_0^{ret} puncture count $48 = \Omega_{\text{adm}}$; $Y^2/\text{loop}/\text{GUT}$ -weighted they give the alpha budget $41 = 10b_1 = 40 + 1$ ($\text{Tr}_{16} Y^2 = \frac{10}{3}$ exact, $\nu^c = 0$; Ginsparg $(3/5)(41/6) = 41/10$, v470) — one state set, two response weights, no new number. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged ALPHA.QUILLEN.EXACT.01 + HYP.PHI0.PUNCTURE.01 target is proven; the single remaining [O] is the identification of the abstract seam ζ -det with this glued mark determinant — a face of SEAM.EQUIV.01, exactly the v382 typing, now substantiated computationally. Integrated in `tfpt_1_architecture_e8` (eighth step of the ALPHA.QUILLEN arc) and `origin_theory`; ledger rows SEAM.CONTACT.UNIT.02 + update

notes on .01 and both targets; Python-only by nature, flagged in the Wolfram README.

2026-07-14 (III) (v484 — the two open analytic targets MERGE: the shared “ c_3 per boundary insertion” rule of ALPHA.QUILLEN.EXACT.01 and HYP.PHI0.PUNCTURE.01 is the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/|\mu_4|$, derived on the seam circle for the finite cycle sector — *no gate closes, no marker moves*)

- **v484_seam_contact_unit.py (SEAM.CONTACT.UNIT.01).** The v483 round left the two analytic targets sharing one rule (“weight c_3 per boundary contact insertion”: the $F_{U(1)}$ ladder $\{0, 3, 6\}$, the φ_0^{ret} puncture $\{1, 4\}$). This module shows the shared rule is NOT a new unknown: [E] the unit chain $\frac{1}{2\pi} = 4c_3 = |\mu_4|c_3 = |\mathbb{Z}_2|\chi(S^2)c_3$ — one *bare* boundary insertion divided by the orbit order $|\mu_4|=4$ is exactly c_3 ; this is the v239 KMS seam unit and the v58/v73 one-sided Gauss–Bonnet in a single identity. [E] the mechanism is DERIVED on the actual seam circle for the finite cycle sector: the bare Green function $G(d) = -\frac{1}{\pi} \ln |2 \sin(d/2)|$ (polylog identity, 40 digits) takes INTEGER multiples of $c_3 \ln 2$ at the μ_4 separations ($G(\pi/2) = -4c_3 \ln 2$, $G(\pi) = -8c_3 \ln 2$; mark matrix $-4c_3 \ln 2 \cdot \text{circ}(0, 1, 2, 1)$, spectrum $\{4, -2, 0, -2\}$), so the μ_4 -invariant (orbit-averaged) insertion carries $-(c_3 \ln 2)C$ exactly and the log-det contact expansion is exactly graded in c_3 per insertion ($\text{Tr } C^k = 4^k + 2(-2)^k$). [E] the grammar is suite-wide: the Λ prefactor $\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2$ (v60) carries the SAME multiplicity $\Omega_{\text{adm}}=48$ as the φ_0^{ret} puncture at TWO insertions; the $g-2$ seam loop divides by the same $2\pi = 1/(4c_3)$. [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ($48 = \Omega_{\text{adm}}$, $41 = 10b_1$); the finite half is proven, no target coefficient is claimed (the integers are forced by the square configuration, v354/v355-clean). Integrated in `tfpt_1_architecture_e8` (seventh step of the ALPHA.QUILLEN arc) and `origin_theory` (φ_0^{ret} split paragraph); ledger rows SEAM.CONTACT.UNIT.01 + update notes on both targets; Python-only by nature (mpmath polylog), exact identities flagged in the Wolfram README.

2026-07-14 (II) (v482–v483 — the deep-dive round: the ν -scale decision computation EXECUTED (the integer c_3^3 -rung excluded unstructured, the $5/3$ -proximate structure escape recorded and declined) and the φ_0^{ret} puncture target HALVED (the geometric side proven exact: all twisted pillowcase heat traces are t -independent rationals, so the π^{-4} must sit in the per-mark coupling weight) — *no gate closes, no marker moves*)

- **v482_seesaw_rung_decision.py (FLAV.NUSCALE.03).** The decision computation NAMED by v481 is EXECUTED in bracketed (assumption-robust) form: the integer rung $M_R = c_3^3 M_{\text{Pl}}$ needs a rescue factor $\times 1.670$ in m_3 (0.0301 vs 0.0503 eV), while $m_t(\overline{\text{MS}}) \pm 3 \text{ GeV}$ ($> 3\sigma$), $\alpha_3^{-1} \mp 0.10$ ($> 3\sigma$), a $\pm 10\%$ band on the ADKLR κ run-down and a PS-leg Yukawa-beta envelope $\times [0.5, 1.5]$ move it by at most $\times 1.10$ singly and $\times 1.165$ ALL-favourably combined — the *unstructured* rung is EXCLUDED conditional on the carrier premise $y_\nu = y_t$ (the v481 1-loop miss is not an RG artifact; no exact higher-loop coefficients trusted, pure envelope argument). [C] The only escape is a third-generation Majorana STRUCTURE factor $r \sim 1.67$; its post-hoc 0.18% proximity to $g_{\text{car}}/N_{\text{fam}} = 5/3$ is recorded and DECLINED per v354/v355 (no forcing mechanism; the $t_0 \sim 30$ discipline of v478). [O] the v481 one-parameter window candidate (NO floor $\Sigma m_\nu = 0.059 \text{ eV}$) never used the integer rung and STANDS. Integrated in `tfpt_2_standard_model` (ν -scale paragraph) and `tfpt_5_redteam`; ledger rows FLAV.NUSCALE.03 + .02 update note; Python-only by nature.
- **v483_pillowcase_exact_traces.py (HYP.PHI0.GEOM.01).** The GEOMETRIC side of the φ_0^{ret} puncture target (HYP.PHI0.PUNCTURE.01/v408) is EXACT: every twisted/orbifold heat trace of the flat $\tau=i$ pillowcase is a t -INDEPENDENT RATIONAL — $\text{Tr}(\gamma e^{t\Delta}) = 1$ for

$\gamma = \sigma, \rho, \rho^3$ at every tested t (position-space Gaussian image sums $< 10^{-9}$; mode-space proof: only $n=0$ is γ -diagonal), matching the Atiyah–Bott fixed-point counts $4 \times \frac{1}{4}$ ($\det = 4 = |\mu_4|$) and $2 \times \frac{1}{2}$ ($\det = 2 = |\mathbb{Z}_2|$); the pillowcase contact term is EXACTLY $\frac{1}{2} = 4 \times \frac{1}{8}$ per mark and the clock-equivariant trace exactly 1, for all t . THEREFORE the π^{-4} in $d_{\text{top}} = 48c_3^4 = \frac{3}{256\pi^4}$ cannot come from flat orbifold geometry at ANY order: it must sit in the per-insertion COUPLING normalisation ($4 = |\mu_4|$ marks of weight c_3 give c_3^4 , multiplicity $\Omega_{\text{adm}}=48$; the bare $1/(2\pi)$ -propagator 4-cycle differs by the exact rational $\frac{3}{16}$). The target NARROWS from “derive the term” to ONE rule — “per-mark insertion weight = c_3 ” (the same residual class as the EM-Ward c_3 -ladder, EM.WARD.02) — and stays [O]. Complements the experiments-side Donnelly probe (honest negative). Integrated in `origin_theory` (φ_0^{ret} split paragraph); ledger rows HYP.PHIO.GEOM.01 + HYP.PHIO.PUNCTURE.01 update note; Python-only by nature (quadrature verification of exact statements).

2026-07-14 (website + figure refresh — the /verification journey views become live HTML (the stale timeline/residual-chain images replaced by interactive components wired to the in-browser reproducer), the prediction surface’s empirical-audit wall of text restructured into dated, expandable rounds, and the generated timeline/residual-chain figures extended to the current suite — *presentation only, no content or marker moves*)

- **Website /verification: live journey views.** The static `script_timeline.png` and `residual_chain.png` images (stale at the v303–v407 figure state, with alt text and captions still describing the old eleven-phase/v237 state) are replaced by two interactive components, `SuiteTimeline.tsx` (17 phases, v1–v481) and `ResidualChain.tsx` (the reduction chain through the certification round to the bedrock): every phase/step lists representative vN scripts that run live in the browser via the existing Pyodide reproducer, exactly like the dependency graph. Stale counts fixed (“~295-script journey”, “260 scripts” → the generated suite total). The dependency graph itself was verified current (v479–v481 present; all 288 referenced scripts exist in the registry and the website mirror).
- **Prediction surface: the empirical-audit narrative restructured.** The single ~13 000-character audit paragraph in `lib/predictions.ts` is split into twelve dated, verdict-badged rounds (AuditRound: date, title, verdict, two-sentence summary, full unabridged record on expand) plus a domain-chip row and the anti-double-counting note — same content, readable surface. Long-token overflow fixed (`break-words`) in the dependency-graph detail panel and the universal-gap-lab residual cards; the over-long ALPHA.QUILLEN residual note tightened to its current state (v470/v472, residual unchanged).
- **Generated figures + captions synced.** `make_figures.py`: the script timeline gains the v408–v481 certification-round phase (title now ~475 scripts) and the residual chain gains the v458–v480 certification step (Kronheimer v479, four-interval v480, crossed-product v469) with sub-text wrapping fixed; figures regenerated (PDF + website PNG). The stale figure captions updated: `introduction` (timeline caption now spans the full arc), `tfpt_research_contracts` (residual-chain caption now states the SEAM.EQUIV.01 bedrock and certification round; QFT-skeleton caption now names the one [O] residual instead of the outdated “two open items”). No numbers, statuses or claims change anywhere.
- **KaTeX macro fix (broken formula rendering).** Formulas copied from the document set that use the shared TeX macros (`\gcar`, `\Nfam`, `\ctthree`, ...) rendered as red KaTeX errors on the site (e.g. the φ_0^{tree} icosahedral-combinatorics formula on the Origin-Theory page). A shared macro map (`lib/katex-macros.ts`, mirroring the `changelog.tex` preamble) is now passed to every KaTeX render; an automated scan of all 19 site routes finds zero `katex-error` nodes.
- **Readability pass (text walls clamped, content unchanged).** A reusable `Expandable` component clamps long mirror prose with a fade + “Show more” toggle — applied to the

paper-section bodies and front-box bullet lists (`PaperSection.tsx`, the home set pages and `/papers/[slug]`), the dependency-graph node summaries and the prediction-card descriptions. Nothing is deleted; the full text expands in place.

- **Hylaeon ecosystem card corrected.** The `hylaeon.ai` card spoke of “Field AI” with structure tags that did not describe the project. It now speaks consistently of *Hylaeon* and is grounded in the project documentation (`_documentation/core-physics/big-picture.md`) and the site’s own status log: one cognitive substrate behaving like a physical material, knowledge = attractor valleys, question = boundary condition, answers committed only on physical landing, capability claims gate-checked against a machine-checked claim registry (honest negatives published) — borrowing TFPT’s structural discipline, not its physics predictions. The Agentic-Commerce card reference updated accordingly.
- **Live-site correctness pass (counts + reading order).** Sitewide audit against the deployed page found stale counts and fixed them: the document set is stated everywhere as *10 documents = 1 reading guide + 5 numbered papers + 4 companions* (the safeguards companion was missing from the papers/downloads headers, the JSON-LD collection and the `tfpt_docset.tex` box, which still said “three companions” next to “four”); the stale “23 falsifiable predictions” claims (site metadata, OG image, `llms.txt`) now state the live count of the prediction surface (25 test surfaces, sourced from `predictions.length` where possible), while the interactive compiler page is honestly labelled as its own *23-readout prediction board* (the standalone app’s board genuinely has 23 entries; the intro film keeps its spoken “twenty-three”); and the `/orientation` reading-order lists are fixed to the canonical numbering (Red Team = Doc 5 added, Appendix H = 6, Origin = 7, contracts = 8, safeguards = 9 — the old list mislabelled 6/7 and omitted three documents). Presentation/consistency only; no physics content changes.

2026-07-14 (v479–v481 — the bird’s-eye “overlooked tools” round: R3’s graph→geometry bridge discharged to Kronheimer 1989; the four μ_4 marks realised as the exactly solvable *four-interval* multilocal modular geometry with $\omega \circ \rho = \omega$ manifest; and the absolute ν -scale collapsed to a one-parameter, PS-window-bracketed candidate under the carrier normalisation $y_\nu = y_t$ — *no gate closes, no marker moves*)

- **v479_kronheimer_quiver_bridge.py (SEAM.KRONHEIMER.01).** The R3 residual of the $\det K=1$ keystone (v344: “the rewrite attractor graph IS the du Val singularity”) is an ESTABLISHED THEOREM — Kronheimer’s ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685) — with all finite input data already compiler outputs, machine-checked [E]: the Kac marks $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ are the null vector of the affine- E_8 Cartan matrix AND the Perron vector $A\delta=2\delta$ of the rewrite attractor (v312); the hyper-Kähler quotient dimension count lands exactly on 4 (the $\mathbb{C}^2/\Gamma$ geometry; $\sum_{\text{edges}} d_i d_j = \sum d_i^2 = 120$ forced by the null identity); the gauge datum is $|2I|=120=|\mu_4|h(E_8)$, $h(E_8)=30=2 \cdot 3 \cdot 5$; the resolution carries the unimodular $-E_8$ form ($\det=1$, SNF = identity — the Poincaré link) with $3 \cdot 8=24$ deformation parameters. [C] the theorems are cited, not re-proved (import class of NPW26/MMST/AGT); [O] the residual premise *transforms* to “the raw seam supplies ALE hyper-Kähler data asymptotic to the $(2, 3, 5)$ cone” — the same single arrow as SEAM.EQUIV.01 (v346 L2). Integrated in `origin_theory` (R3 keybox + new Kronheimer paragraph) and `tfpt_research_contracts` (Kummer/du Val); ledger rows SEAM.KRONHEIMER.01 + SEAM.DETK.01 update note; Wolfram mirror added to the extension file (+2 exact checks, counted with the next verified engine run — the local engine is currently unactivated; the identical statements are sympy-exact in v479). LEAN: the finite input data are kernel-checked in `TfptCarrier/KronheimerMarks.lean` (marks = null/Perron vector over $\mathbb{Z}$, $\sum \delta_i^2 = 120 = 4 \cdot 30$, $30 = 2 \cdot 3 \cdot 5$, dimension count = 4; lake build clean, `#print axioms` = the three standard kernel axioms only).

- **v480_multilocal_four_interval.py** (QGE0.MULTILOCAL.01). The seam circle minus its four marks IS the $n=4$ multi-interval region of a chiral free fermion — the class whose multi-interval modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu; Rehren–Tedesco multilocal fermions; KLM: holomorphic $\Leftrightarrow$ trivial multi-interval inclusion, $\mu=1$ — the same one bit as $\det K=1$); until now the suite used only the single-ball CHM boost (v358). On the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with $\rho^4=-1$ EXACTLY (the fermionic μ_4 clock is the order-8 double-cover lift — the same $\mathbb{Z}_2$ -lift pattern as $2I$ over I); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at 10^{-15} , each sector a single-interval quarter-ring kernel with boundary twist (10^{-14} spectra); [E] $\rho C \rho^{-1}=C$ at 10^{-14} , so $\omega \circ \rho = \omega$ holds *manifestly* at the 4-interval state level, and a one-site displacement re-couples the sectors at 10^{-2} (negative control); [E] the cross-interval modular coupling peaks on the conjugate-point diagonal (the Casini–Huerta mixing shadow). [C] continuum theory cited; [O] the raw-seam premise (QGE0.SYM.01/SEAM.EQUIV.01) unchanged. Integrated in `tfpt_research_contracts` (marks keybox + rigidity contract + v478 leg A), `origin_theory` (BW paragraph), `tfpt_4_frontier` (v358 companion note); Python-only by nature.
- **v481_seesaw_carrier_ladder.py** (FLAV.NUSCALE.02, **CANDIDATE** class, **genre v467/v468**). FLAV.NUSCALE.01's $y_\nu=1$ probe was not the carrier normalisation: one $SO(10)$ 16 per family + the minimal Yukawa sector ($\mathbf{10}_H$ / PS $(1, 2, 2)$) forces $y_\nu=y_t$ at the matching scale (named premise). With explicit 1-loop RG (gauge/ y_t/λ up; ADKLR Weinberg-operator running down, hep-ph/0108005) the (y_ν, M_R) trade-off collapses to M_R alone: the observed $m_3=0.0503$ eV demands $M_R=9.3 \times 10^{13}$ GeV — INSIDE the PS window $[4.2 \times 10^{13}, 2.4 \times 10^{15}]$ (v249) at $\log_{c_3}(\bar{M}_{Pl}/M_R)=3.15$ (the $y_\nu=1$ inversion gives the structureless 2.58). [X] honesty gate: the integer rung $M_R=c_3^3 \bar{M}_{Pl}$ predicts $m_3=0.030$ eV — 40% low at 1-loop (κ run-down 0.794) — the ladder pin is DECLINED per v354/v355; the decision computation is named (2-loop SM + explicit PS thresholds). What survives: a *one*-parameter, window-bracketed [O] candidate with the falsifiable band $m_3 \in [0.002, 0.115]$ eV and the NO floor $\Sigma m_\nu=0.059$ eV; frozen record untouched. Integrated in `tfpt_2_standard_model` (new paragraph after v272), `introduction` status card (third named candidate), `tfpt_5_redteam`; ledger rows FLAV.NUSCALE.02 + FLAV.NUSCALE.01 update note; Python-only by nature.
- *Honest negative (experiments-only, no promotion)*: the Donnelly/Atiyah–Bott equivariant heat-kernel route for the ϕ_0 puncture target (HYP.PHI0.PUNCTURE.01) types the KIND of term (order-4 equivariant contact datum; the exact weights $\sum_{k=1}^{n-1} \frac{1}{4 \sin^2(\pi k/n)} = \frac{n^2-1}{12}$ verified) but does NOT force the value $48c_3^4$ (pure rationals vs π^{-4}) — recorded in `experiments/theory-contracts/phi0_donnelly_equivariant_probe.py`; the target stays [O].

2026-07-13 (v478 — first computable steps on the two remaining legs of the entropic-action bridge: the state-side modular data flows to the CHM/Bisognano–Wichmann geometric form (Calabrese–Cardy $c_{\text{est}} \rightarrow 1$, CHM parabola, even bands exactly zero) — the bridge's modular side meets TFPT's Einstein-derivation input; and the measure condition reduced to one exact KMS time $t_0 = \ln \frac{72}{17} + 9 \ln(8\pi) = 30.461$ with the $h(E_8)=30$ near-miss explicitly declined; both legs stay [O], no status moves)

- **v478_entropic_continuum_legs.py** (GRAV.ENTROPIC.LEGS.01). *Leg A (AP2 continuum, exhibited)*: on the exactly solvable critical chain (infinite-volume sinc kernel, 50-digit precision) the compressed interval state has $S(L)=(c/3) \ln L+k$ with $c_{\text{est}}=1.00089/1.00038/1.00021$ (monotone $\rightarrow$ the Dirac $c=1=2 \times \frac{1}{2}$, the v450 reading; gapped control $\sim 6 \times 10^{-8}$, area law); the Peschel modular Hamiltonian $K=\ln((1-C_A)/C_A)$ organises on the CHM weight $\beta(x)=x(L-x)/L$ (corr $0.87 \rightarrow 0.99$, shrinking plateau drift; constants deliberately not asserted), the even-range bands

vanish identically (particle-hole, $\sim 10^{-36}$) and the odd tail is 1.8% with parabolic envelope — the state-side Δ_Σ (the reading forced by v476) flows to the CHM/BW geometric form, exactly the object TFPT’s Einstein derivation consumes (v323/v358): the bridge’s modular side and the gravity side meet in the continuum limit; a lattice exhibit, NOT a continuum proof — AP2 stays [O]. *Leg B (the measure)*: [E] single-scale corollary — $d\mu=\delta(t-t_0)dt$ gives $\mu_2/\mu_1^2=e^{t_0}$ exactly, so the v477 one-moment condition forces $t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461$ (one dimensionless KMS time, closed form); the near-miss to $h(E_8)=30$ (difference 0.461) is explicitly DECLINED per the v354/v355 no-free-pattern rule — deriving the measure from the entropic structure stays OPEN. Keybox extended in `tfpt_research_contracts`; high-precision numerical module, Python-only by nature (Wolfram count unchanged, 378/378; flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.LEGS.01`; both legs stay [O], no marker moves.

2026-07-13 (v477 — the scale-flow representation of the entropic action: the surviving R^2 route typed as *one moment condition* $\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9$ with the exact closure $\frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7$ — the kill-test gap is a scale-measure datum, satisfied by the same KMS moment TFPT already fixes (v36); zero new dials, *no status moves*)

- **v477_entropic_scaleflow.py (GRAV.ENTROPIC.SCALEFLOW.01).** [E] The Frullani identity upgrades to a scale-flow representation: $-\ln a = \int \frac{dt}{t}(e^{-ta}-e^{-t}) = 2 \int \frac{dx}{x}(e^{-a/x^2}-e^{-1/x^2})$ (symbolic + 10^{-20} quadrature) — Bianconi’s log action IS the *flat* scale-integral of relative heat-kernel actions, while TFPT’s $S_{\text{rel},\chi}$ evaluates *one* KMS scale with a weight: the bridge between the two action principles is a choice of SCALE MEASURE. [E] Moment dictionary: a weight $w(t) dt/t$ gives $3\beta R\mu_1 + \frac{17}{24}\beta^2 R^2\mu_2$; the flat weight has $\mu_1=\mu_2=1$ and reproduces the v475 raw coefficients identically. [E] The ONE moment condition: demanding $m^2=c_3^7\bar{M}_{P1}^2$ forces exactly $\mu_2/\mu_1^2=\frac{72}{17}(8\pi)^9$ — the v475 kill-test number IS a moment ratio — and the closure identity $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}=c_3^7$ holds identically: the “13 orders” are a scale-measure datum, not a bridge mismatch. [E] v36 consistency: $f_0=6(4\pi)^2/c_3^7 \Leftrightarrow M^2/\bar{M}_{P1}^2=c_3^7$ — one KMS moment, two parametrisations, zero new dials. [C] A consistency statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein + Λ level and the scale-flow form, TFPT supplies the measure). Negative side-record: an exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain — kept out of the suite, logged in the research notes. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors both exact blocks (378/378); ledger row `GRAV.ENTROPIC.SCALEFLOW.01`; the R^2 gate and the v359 typing stay [O], no marker moves.

2026-07-13 (v476 — the AP2 compression conjecture of the entropic-action bridge made *well-posed* on an exactly solvable model: the literal operator-side reading is ill-posed on a pure bulk, the state-side reading is forced, the mismatch is exactly quadratic in the cross-cut correlations and gap-suppressed; AP2 stays [O], *no status moves*)

- **v476_entropic_compression_toy.py (GRAV.ENTROPIC.COMPRESS.01).** The last open work package of the v473 round, exercised on a gapped staggered-mass chain ($N=16$). The seam situation is reproduced (pure bulk: the ground-state covariance is a projection to 10^{-15} ; thermal boundary: $\text{spec}(PCP) \subset (0,1)$, Peschel $K=\ln((1-C_A)/C_A)$ finite, $S>0$ — v155 in the bridge context). **The reading is decided:** on the pure bulk $f(x)=x/(1-x)$ is singular on $\text{spec}(C)=\{0,1\}$, so the literal operator-side reading $Pf(C)P$ of the v473 conjecture is *ill-posed* while the state-side $f(PCP)$ is finite — the compression identity can only mean “build Δ_Σ from the *compressed* relative metric”, which matches Bianconi’s own local-metric construction (AP2 retyped, not weakened). [E] Block algebra (symbolic): $[C^2]_{AA}-(C_{AA})^2=BB^t$, $[C^3]_{AA}-(C_{AA})^3=ABB^t+BB^tA+BDB^t$ — first order in the cross block vanishes identically,

so for any analytic f the two readings agree to *second* order in the cross-cut correlations: the modular (entanglement) content lives exactly in that quadratic correction. Gap control at the bounded (covariance) level: the Bianconi-shaped bulk identity $f(C_T)=e^{-h/T}$ holds to 10^{-11} relative; the mismatch obeys the quadratic law $d \leq x^2$ ($d/x^2 \in [0.4, 1.0]$) and falls with the gap in the gapped regime, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\text{eff}} > 0$, v76/v337). Keybox extended in `tfpt_research_contracts`; numerical module, Python-only by nature (Wolfram count unchanged, 376/376; the block identity flagged in the Wolfram README); ledger row GRAV.ENTROPIC.COMPRESS.01; AP2 (the continuum seam statement) stays **[O]**, no marker moves.

2026-07-13 (v475 — the R^2 kill test of the entropic-action bridge, *executed*: exact tensorial factors on the maximally symmetric background, raw entropic scalaron $m^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$ ($\approx 51.7 \bar{M}_{\text{Pl}}$, trans-Planckian) $\Rightarrow$ the light-trace-mode reading is dead, KMS renormalisation the only surviving R^2 route; plus the explicit Lorentzian-positivity witness; *no status moves*)

- **v475_entropic_scalaron_gate.py** (GRAV.ENTROPIC.SCALARON.01). Work package 5 of the v473 bridge. **[E]** On a maximally symmetric background the relative curvature operator $\tilde{\mathcal{R}}\tilde{g}^{-1}$ has eigenvalues *exactly* $\{R; R/4 \times 4; R/6 \times 6\}$ (Bianconi's Appendix-B flattening verified symbolically, incl. the 6×6 two-form block), giving the exact vacuum action $\mathcal{L}_B(R) = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$ — the 3 re-derives her Eq. (45) (= the v473 pin), $\frac{17}{24} = \frac{1}{2}(1 + \frac{1}{4} + \frac{1}{6})$ is the exact tensorial R^2 factor. **[E]** With the pinned $\beta' = c_3/6$ both mass routes (Starobinsky matching and $f'/3f''$) agree: the raw entropic scalaron is $m_{\text{raw}}^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$, i.e. $m_{\text{raw}} \approx 51.7 \bar{M}_{\text{Pl}} \approx 1.3 \times 10^{20}$ GeV — TRANS-PLANCKIAN, not light; v473's interpretation 2 (the TFPT scalaron as the light $\mathcal{G}$ -trace mode) is *dead in naive form*; a viable mechanism must supply exactly $\frac{72}{17}(8\pi)^9 \approx 1.69 \times 10^{13}$ in mass² (the kill-test number now WITH tensorial factors). **[E]** Domain: first log pole at $R=1/\beta=384\pi^2\bar{M}_{\text{Pl}}^2$ (trans-Planckian) — the obstruction is the mass, not the domain. **[E]** Lorentzian-positivity witness: a timelike gradient gives the Gg^{-1} eigenvalue $1 - \alpha v^2 \leq 0$ for $v^2 \geq 1/\alpha$ (ln undefined), spacelike/Euclidean stay positive — the v473 **[C]** caveat exhibited, the OS route sturdier. **[C]** Verdict: interpretation 1 survives (Einstein + Λ level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving R^2 route. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors all three blocks (376/376); ledger row GRAV.ENTROPIC.SCALARON.01; no gate closes, no marker moves.

2026-07-13 (v474 — the operator level of the entropic-action bridge: the D_5 Clifford/spinor structure exhibited on the carrier Fock space, the Hodge fold identified as the $5 \rightarrow \bar{5}$ conjugation, and the v473 Q -target *decided* — supports exactly $\{\mathbb{Z}_2, \text{rank } E_8, 2^{g_{\text{car}}}\}$, the naive pair-block reading killed; *no status moves*)

- **v474_entropic_hodge_carrier.py** (GRAV.ENTROPIC.HODGE.01). Work packages 1 and 4-algebra of the v473 bridge, machine-checked on the finite carrier Fock space $\Lambda^\bullet \mathbb{C}^5$ ($\dim 2^{g_{\text{car}}}=32$). **[E]** CAR $\Rightarrow$ Clifford: the ten Γ 's from $(a_k^\dagger, a_k)$ satisfy $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$ exactly (all 100 pairs, integer matrices); the 45 bilinears $[\Gamma_a, \Gamma_b]/4$ commute with $\gamma = (-1)^N$, span exactly $45 = \dim \mathfrak{so}(10)$ and leak nothing from the 16-dim even subspace — the carrier half-spinor *is* the even exterior algebra *with* its spinor structure (the operator level of v2/v44/v197). **[E]** the Hodge–Dirac symbol: $c(\xi) = a^\dagger(\xi) - a(\xi)$ satisfies $c(\xi)^2 = -|\xi|^2 \text{Id}$ for symbolic ξ — Bianconi's $D = d + \delta$ is Clifford multiplication at symbol level. **[E]** the fold is the $5 \rightarrow \bar{5}$ conjugation: the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the negatives of those of Λ^1 , so her fiber $1+5+10$ becomes the GUT half-spinor $16 = 1+5+10$ (the SM carrier decomposition). **[E]** honest obstruction: $\Omega^{\leq 2}$ is *not* Clifford-invariant ($\Lambda^2 \rightarrow \Lambda^3$ leak) — the bridge must go through the fold, never through degree truncation (a recorded constraint on the AP2 compression). **[E]** the Q -target enumeration (the v473 **[C]** conjecture decided at the algebraic level): $\text{Tr } Q^2 = d k^2 c_3^4 = 32 c_3^4$ with integer k has exactly the supports

$(d, k) = (2, 4), (8, 2), (32, 1) = \{|\mathbb{Z}_2|, g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8, 2^{g_{\text{car}}} = \dim \Lambda^\bullet\}$ — minimal uniform solution $q = c_3^2 = 1/(64\pi^2)$ per Fock mode, bonus identity $\text{Tr } Q^2 / \delta_{\text{top}} = 32/48 = 2/3 = \dim \Lambda^\bullet / \Omega_{\text{adm}}$; the two-form block 10 (the problem-note “pair sector” conjecture) and the fiber 16 need irrational multipliers $\sqrt{16/5}, \sqrt{2}$ — *killed* in uniform form. [C] the physical choice among the three admissible supports; [O] unchanged (AP2 compression + the R^2 mechanism). Keybox extended in `tfpt_research_contracts` (Research Contract 2, the v473 keybox); Wolfram mirrors the enumeration and the weight-conjugation identity (373/373); ledger row GRAV.ENTROPIC.HODGE.01; no gate closes, no marker moves.

2026-07-13 (v473 — the entropic-action bridge: Bianconi’s *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); arXiv:2408.14391) quantified as an *external candidate* for the missing action level of the parameter-free Einstein equation; her free constant pinned $\beta'_B = c_3/6 = 1/(48\pi)$, the R^2 gap $3(8\pi)^9$ pre-registered as kill test; *no status moves*, the v359 equation-of-state typing stays [O])

- **v473_entropic_action_bridge.py** (GRAV.ENTROPIC.ACTION.01). TFPT’s classical field equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ is derived thermodynamically (v358/v359) and honestly typed “equation of state, not a from-action quantisation” [O]. Bianconi’s entropic action $S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})$ (the quantum relative entropy between the spacetime metric and the matter-induced metric, Araki-related via $\Delta^{1/2} = \tilde{G}\tilde{g}^{-1}$) is a local covariant action whose variation gives Einstein + an emergent nonnegative Λ at low coupling — an explicit *external candidate* for exactly that missing variational layer. Machine-checked: [E] the carrier Hodge count (her $\Omega^{0,1,2}$ fiber on the five-slot carrier $\mathbb{C}^5$ counts $1+5+10=16=2^{g_{\text{car}}-1}=\dim S_{D_5}^+$, while on 4d spacetime it is only 11 — the 16 *requires* the carrier); [E] the one quantitative pin — matching her weak-coupling $3\beta R$ to $1/(16\pi G) = c_3/(2G)$ fixes her free constant *exactly*, $\beta'_B = c_3/6 = 1/(48\pi)$; [E] her $\Lambda_{\mathcal{G}} = \frac{1}{2\beta} \text{Tr}(\mathcal{G} - I - \ln \mathcal{G})$ is nonnegative and quadratic at the fixed point, and with $\varepsilon_\star = e^{-\alpha^{-1}} Q$ it reproduces the closed v60 branch $(3/4\pi^2)e^{-2\alpha^{-1}}$ with the exact target $\text{Tr } Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\delta_{\text{top}}$ ([C] until Q is constructed); [E] the R^2 kill test — the raw log expansion sits *exactly* $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$ below the TFPT Starobinsky coefficient $1/(12c_3^7)$, so a direct identification is *false* without a mechanism (pre-registered); [C] the compression conjecture $P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$ (the bridge proper) stays open, with Lorentzian positivity flagged (TFPT’s OS route is the sturdier construction). New keybox in `tfpt_research_contracts` (Research Contract 2, after the v359 keybox); residual pointers updated in `introduction/tfpt_4_frontier/status` card. Wolfram mirrors the three exact blocks (371/371); ledger row GRAV.ENTROPIC.ACTION.01; *no* gate closes, *no* marker moves, zero TFPT knobs added.

2026-07-03 (surfacing round — three already-computed unique readouts promoted to the public prediction surfaces: the axion–photon coefficient $g_{a\gamma\gamma} = -4c_3$, the free-seam Higgs band $m_H \approx 129\text{--}134$ GeV, and the never-run BH HFQPO $\times 1.5$ ladder-tooth search; no new vN, no status moves)

- **Uniqueness scan follow-up (no new results — presentation only).** A structure scan for “what does TFPT predict that nobody else pins?” found three concrete, falsifiable readouts that were fully computed and ledgered but missing from the public prediction surfaces; they are now surfaced, with their existing typing unchanged. (i) **Axion–photon coefficient:** the determinant-line anomaly coefficient $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$, $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$ (v207, the [E] leg of GRAV.ASYMP.01; the [C] caveat — coefficient, not a parameter-free $g_{a\gamma\gamma}$ in GeV^{-1} — stated verbatim) joins the axion card in `website/lib/predictions.ts` and the frontier dark-matter mirror in `papers.ts`; the TeX source (`tfpt_4_frontier`) already carried it. (ii) **Higgs free-seam band:** the double-criticality prediction $\lambda(\bar{M}_{\text{Pl}}) = \beta_\lambda(\bar{M}_{\text{Pl}}) = 0 \Rightarrow m_H = 133.5$ GeV (band 129–134, m_t/α_s -sensitive; measured 125.25 sits a few GeV below — the known slight

metastability, Buttazzo NNLO $\lambda(M_{\text{Pl}}) = -0.0143 \pm 0.0057, 2.5\sigma$) becomes a prediction card ([C]; v166, HIGGS.FREESEAM.01, kill: settled- (m_t, α_s) RGE pull off the surface at $> 5\sigma$). (iii) **BH HFQPO ladder tooth**: the one discriminating HFQPO test — a third tooth at $\nu_3 = (3/2)\nu_u$ (661.5/414/252/363 Hz), integer harmonics forbidden — that *no published search ever targeted* although the teeth lie in the searched RXTE band (experiments/hfqpo-ladder, preregistered): added as an [O] search-target bullet to the tfpt_horizon_readouts “Search targets (not claims)” gapbox (firewall language verbatim: non-canonical mapping, selection null 18.5%, J1859+226 +9.2 σ , GR resonance favored; even a tooth hit would be [C]) and as an [O] prediction card. Website mirrors updated in the same change (predictions.ts 23 $\rightarrow$ 25 cards, papers.ts frontier + horizon sections incl. the new search-targets mirror section); scorecard, ledger and all markers untouched.

2026-07-03 (v472 — the determinant line over the $U(1)$ -twist moduli of the collar model carries curvature = the inflow level: the v470 bridge lemma exhibited at the finite/model level; ALPHA.QUILLEN.EXACT.01 stays [O]. Plus: bird’s-eye rounds — electron-sector/Cabibbo whiteness probes, Calderón UV-universality, parameter-free flat-budget closure (sandbox); the flat-budget closure enters the evidence scorecard)

- **v472_quillen_detline_moduli.py** (ALPHA.QUILLEN.DETLINE.01). ALPHA.QUILLEN.EXACT.01 names the α^3 level as the curvature of the *determinant line over the $U(1)$ seam moduli* (Quillen 1985 / Bismut–Freed CMP 106 (1986) curvature; Dai–Freed JMP 35 (1994) section); v470 computed the bulk Chern invariant only over the Bloch Brillouin zone (the translation-invariant shortcut). v472 computes the Quillen-shaped object itself: the same collar Hamiltonian ($h(k) = \sin k_x \sigma_x + \sin k_y \sigma_y + (M - \cos k_x - \cos k_y) \sigma_z$, v367/v470) in real space on an $L \times L$ torus with *twisted* boundary conditions — the twist torus (θ_x, θ_y) is the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the *determinant line of the occupied frame* over that torus: $\oint = 2\pi \cdot 1$ exactly (integer to 10^{-9}) at $M=1$ for $L=4$ AND $L=6$ (size-independent); controls: trivial collar $M=3 \rightarrow 0$, orientation flip $M=-1 \rightarrow -1$; the twist-moduli integer equals the Bloch-BZ integer for all three M (Niu–Thouless–Wu PRB 31 (1985), exhibited on the S3 collar model); the Fermi gap (= 2.0) stays open over the whole twist torus, so the Dai–Freed section has no zero. **Reading [C]**: det-line holonomy over the $U(1)$ moduli = inflow level $k_0 = |C| = 1$ — the v470 bridge lemma “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” holds *verbatim at the finite/model level*, where $\log \det$ of the occupied frame replaces $\log \det_\zeta$. **Not closed [O]**: the *continuum* ζ -determinant identification on the abstract seam (= the SEAM.EQUIV.01 face); ALPHA.QUILLEN.EXACT.01 stays [O]; α^{-1} stays [E]. Structurally the same manoeuvre as v471 for SEAM.THEOREM.01’s replica side. 7/7 checks; suite green through v472 (466 modules); numerical (dense eigh + FHS twist grid), Python-only by nature. Papers updated: tfpt_1 (sixth honest step in the EM closure), tfpt_research_contracts (residual certification), tfpt_safeguards (α -form lever), root README, zenodo_description.html. Exploration original: experiments/tfpt-discovery/quillen_detline_twist_moduli.py (same day).
- **Bird’s-eye whiteness rounds (sandbox, experiments/tfpt-discovery/, nothing else promoted)**. (i) **electron_sector_cabibbo_probe.py**: the α metrology triangle (TFPT $\alpha^{-1} = 137.0359992168$ vs Rb-2020 +0.99 σ / CODATA22 +1.90 σ / a_e -route +3.39 σ / Cs-2018 +6.33 σ — TFPT takes the Rb side of the 5.5 σ metrology split; new honest pressure point recorded); the v204 seam-vertex universality catch (a flavor-universal reading of $a_\mu^{\text{seam}} = 2.879 \times 10^{-9}$ is excluded by the electron at $\sim 2 \times 10^4 \sigma$; linear scaling at $\sim 10^2 \sigma$; quadratic survives — the [C] bridge must scale at least quadratically in m_ℓ); the Cabibbo-angle anomaly as a third dissolution watchdog ($\lambda_C = 0.2243762$ exact sits +0.08 σ on the PDG-2026 rescaled kaon average, between $K_{\ell 3}$ and $K_{\mu 2}$; exact unitarity puts $V_{ud} = 0.97450$ — superallowed is the outlier at +2.6 σ , all neutron/pion routes agree). (ii) **kernel_calderon_uv_universality.py**: the raw-collar

Calderón/DtN datum is UV-universal (mode-by-mode first-order convergence to the exact Bessel symbol $mI'_\nu(mR)/I_\nu(mR)$, Richardson residual 3×10^{-4} ; N_θ -independent; $|k|$ principal symbol; μ_4 block structure at machine precision; both spec- T masses; 3-fold mesh-defect negative control) — the operator equality demanded by QGEO.KERNEL.01 has a well-defined discretization-independent left-hand side (gate stays **[O]**). (iii) `flat_budget_closure.py` + scorecard row: see next bullet.

- **Flat-budget closure enters the evidence scorecard (experiments-only, pattern candidate).** The zero-dial closure $\{\Omega_b = \varphi_0^{\text{ret}}(1 - \frac{1}{4\pi})$ [frozen], $\Omega_c = \frac{2}{7}(1 - \frac{1}{4\pi})$ [the 2026-07-02 numerology-flagged Nebenbefund], $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ [frozen], $\Sigma m_\nu = 0.0588$ eV [v468 route]] + spatial flatness determines the whole flat Λ CDM budget: $H_0 = 67.15$ km/s/Mpc (-0.39σ Planck18, -5.67σ SH0ES22 — the parameter-free Planck side of the Hubble tension), $\Omega_m = 0.3133$ (-0.27σ), $t_0 = 13.86$ Gyr; honest stresses recorded: $\omega_b -2.02\sigma$ (BBN-only -0.73σ), $\omega_c -1.19\sigma$, DESI-DR2 $H_0 -2.16\sigma$ (one correlated low- h direction). Scorecard row `cosmo / flat budget closure` (consistent, composite: shares the `phi0_seed` group (and the ρ_Λ leg the α engine), Ω_c leg in `alternative_group=axion_branch` vs the axion-spine DM reading — never double-counted); nothing promoted to `verification/`.

2026-07-02 (v471 — the Seam–Horizon replica chain exercised numerically on the discretized collar: the kernel-identification premise of SEAM.THEOREM.01 discharged at the finite/model level; residual retypes to the continuum MMST leg + the anchor; gate stays **[O]. Plus: the GW ringdown-echo channel hardened — 10 loudest events incl. GW250114, redshift-corrected templates, stacked search and a 12-variant signature battery, all null)**

- **v471_seam_horizon_replica.py (SEAM.EHMODEL.04).** The residual isolated by v150/v151/v152 — “the RP seam kernel *is* this BFK Calderón datum” — is exercised numerically on a discretized seam collar with the seam’s *own* kernel and *real* replica sheets: the matched conical deficit has slope $2C(\gamma)$ on deficit cones (0.01–0.05%) and replica sheets $n=2$ (0.8%), $n=3$ (1.5%); the EH coefficient is cutoff-independent (0.5% drift) while the ζ -scale μ_{lat} drifts — the v152 anchor exhibited on data, with *no* canonical $\ln(m/\mu)=3/4$ emerging (recorded honestly); the BFK/Schur split is exact ($\sim 10^{-16}$) with the Calderón factor conically clean *measured on the kernel* (pure cut-edge law, tip term 0, power-checked) and the halves carrying $2C_{\text{cone}}$ via Kac doubling; the v302 transfer masses $6\ln(3/2)$, $6\ln 3$ sit on the calibrated EH line ($\leq 0.05\%$); and the Perron/attractor mode ($m=0$) is IR-divergent — **the recovery gap $\Delta=6\ln(3/2)$ is what makes the induced-gravity coefficient finite** (the same gap that makes the attractor unique makes $1/G$ finite). SEAM.THEOREM.01 stays **[O]**; its residual retypes to the *continuum* scaling limit (the MMST class — the same single residual as SEAM.EQUIV.01, v336) plus the one declared anchor. 16/16 checks; suite green through v471 (465 modules); numerical (sparse lattice determinants), Python-only by nature. Papers updated: `origin_theory` (new collar paragraph in the Seam–Horizon gapbox + residual retyping), `tfpt_research_contracts` (R3 premise), `tfpt_4_frontier` (3/4 audit note), root README, `verification/README`, `zenodo_description.html`. Exploration original: `experiments/theory-contracts/seam_horizon_replica.py` (17/17, same day).
- **GW ringdown-echo channel hardened (experiments/gw-ringdown-echo, scorecard row `data_limited`→`consistent`).** (i) Strain coverage extended to the 10 loudest ringdown events incl. GW250114 (network SNR 78.6, O4b; new `fetch_strain_4096.py` crops the 4096 s GWOSC archive segments). (ii) **Redshift correction:** the GWTC catalogue reports source-frame masses, the observed ringdown is at $f_0/(1+z)$ — all templates moved to detector-frame $M(1+z)$ (before the fix GW190521, $z=0.56$, was filtered $\sim 1.6\times$ too high); all stages re-run. (iii) Stage 1b *stacked* search over 23 detector streams: stacked $p=0.262$, kernel-consistent streams 0/23 — STACKED NULL. (iv) Stage 1c *signature battery* (Bonferroni $\times 12$): ratio semantics

(amplitude $(2/3)^6$ vs *energy reading* $(2/3)^3$ vs step $2/3 \times \mu_4$ per-bounce phases $\{0, \pi/2, \pi, 3\pi/2\}$ (the boundary-birefringence analogue — path birefringence from parity violation $c_-=8$ cancels in the inter-echo ratio; only a per-reflection phase does not) $\times$ lags 0.5–350 ms (ECO through Planckian window), with joint $(2, 2, 0) + (2, 2, 1)$ subtraction and a template-agnosticism control: `NO_VARIANT_ECHO` on all 10 events — the null is robust against alternative signature readings. Search-target firewall unchanged: upper bound, no detection, no tension claim.

2026-07-02 (v295 robustness patch — external-review follow-up: the numeric a_2 -kernel tolerance must dominate its own finite-difference floor)

- `v295_flataway_a2_exact.py` (FLATAWAY.A2.01). Alessandro’s Git-baseline audit (commit 8cb8ad0) hit a platform-marginal failure in check 3: the exact divided-difference a_2 kernel is compared against an $\varepsilon=10^{-3}$ second finite difference whose *own* truncation error is $O(\varepsilon^2) \sim 10^{-6}$, yet the tolerance was 10^{-6} — on his machine $k=3$ landed at 1.1×10^{-6} (ours: 2.9×10^{-7}). Fix: tolerance widened to 5×10^{-6} so it *dominates* the control’s error floor, with the reasoning recorded in the check text; the load-bearing anchor remains the *closed form* at machine precision (v296), the numeric witness stays as the independent cross-check. A numerical-robustness patch, no claim or status change; ledger row updated; suite green (ALL CHECKS PASSED through v470).

2026-07-02 (v469/v470 — the closure-route round on the two named targets: the SEAM.EQUIV.01 extension leg re-founded on 1995–2001 peer-reviewed crossed-product theorems + the realisation axiom reduced to invariant level; the α^3 level = the *computed* bulk Chern invariant + the seam F -normalisation = the embedding index $k_Y=5/3$; both targets stay [O])

- `v469_seam_crossedproduct_route.py` (SEAM.EQUIV.CROSSEDPRODUCT.01). The 128-spinor extension leg of SEAM.EQUIV.01 no longer rests on 2024/2025 preprints: [E] the $SO(16)_1$ spinor has $h_s=N/16=16/16=1 \in \mathbb{Z}$, so its statistics phase is $+1$ — *exactly* the Longo–Rehren locality criterion — and the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1 \rtimes \{1, s\}$ is a *local* index-2 extension by the peer-reviewed subfactor package (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives $\mu(B)=4/2^2=1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ by the v463 pin. The AGT/AMT lattice-VOA route (v459) is *demoted to an independent second witness*. [E] the index-4 μ_4 glue runs on the same integer: $h(J^k)=\{1, 1, 1\}$ for $k=1, 2, 3$ (the v125 isotropy $q(k(1, 1))=k^2$ at net level), $\mu=16/4^2=1$, same endpoint. [E] the realisation input is reduced from model fiat to *invariants* (R1’): quasi-free [C] (v155/v160) + gap [E] (v302) + class D + $c_-=8$ [E] (v456, from P1), with the invariants *computed* — per-copy FHS Chern $|C|=1$ vs 0 (control), $\nu=16 \times |C|=16$, $c_-=\nu/2=8$ — and $\nu \equiv 0 \pmod{16}$ is exactly Kitaev’s class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (Ann. Phys. 321 (2006), sixteen-fold way); the phase-invariance of c_- is cited (Kapustin–Spodyneiko, PRB 101 (2020); Kim et al., PRL 129 (2022)). [O] *not closed*: the cited theorems stay cited; SEAM.EQUIV.01 stays [O].
- **Lean hardening.** `TfptCarrier/SeamResidualAxiom.lean` gains the *parallel* peer-reviewed derivation `seamResidualClosed’`: the invariant-level axiom `collar_invariants` (R1’) + *one* combined 1995–2001 package `crossedproduct_route_theorem`; the new joints are kernel facts with *no* axioms — `lr_locality_integer` ($h_s=16/16 \in \mathbb{Z}$), `glue_locality_integers` ($\mu_4 h(J^k) \in \mathbb{Z}$), `klm_holomorphic` ($4/2^2=1, 16/4^2=1$), `sixteenfold_pin` ($\nu=16 \equiv 0 \pmod{16}, c_-=8$), bundled as `crossedproduct_arithmetic` ([propext] only); `#print axioms seamResidualClosed’` = exactly four; the original `seamResidualClosed` (AGT/AMT witness) is kept unchanged as the independent second route; `lake build` OK (3285 jobs).

- **v470_alpha_inflow_level.py** (ALPHA.QUILLEN.INFLOW.01). Both residuals left on ALPHA.QUILLEN.EXACT.01 after v433–v435 are upgraded. [E] target re-verified ($\alpha^{-1}=137.0359992168$; Quillen split $<10^{-35}$). [C] *the level is computed, not minimal*: the a^3 coefficient $k_0=1$ equals the FHS bulk Chern invariant $|C|=1$ of the same p+ip collar that realises S3 (v367/v461) — “why exactly 1” is answered by computation plus the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey, Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez–Gaumé–Della Pietra–Moore 1985 / Witten, Rev. Mod. Phys. 88 (2016) η =CS part of $\delta \log \det$; Quillen 1985 / Bismut–Freed 1986 determinant-line integrality), replacing v435’s minimality assumption. [E] $k_Y = \text{tr}(Y^2)/\text{tr}(T_3^2) = (5/6)/(1/2) = 5/3$ (Ginsparg, Phys. Lett. B197 (1987)): $(3/5) \times \frac{41}{6} = \frac{41}{10} = b_1$ — the “GUT 3/5” is the inverse embedding index; $\frac{41}{6} = \frac{20}{3} + \frac{1}{6}$ (carrier decomposition). [C] the seam F -normalisation (the one [C] of v434) is *retyped*: once the seam net is $(E_8)_1$ level 1, $\langle JJ \rangle = k/(z-w)^2$ is fixed by k_Y — level-1 current-algebra rigidity, zero independent content, a face of SEAM.EQUIV.01 (per the v382 typing). [E] *structural unification*: one invertible phase, two quantised responses — $c_- = 8$ (gravitational, feeds SEAM.EQUIV.01/S3) and $C=1$ ($U(1)$, feeds EM1) — so closing R1’ feeds *both* named targets. [O] *not closed*: the bridge lemma “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” stays the named cited step; ALPHA.QUILLEN.EXACT.01 stays [O]; α^{-1} stays [E].
- **Integration.** Both registered in run_all.py + script_registry.csv (cluster frontier); ledger rows SEAM.EQUIV.CROSSEDPRODUCT.01 + ALPHA.QUILLEN.INFLOW.01; FORM.SEAM.RESIDUAL.01 updated (the seamResidualClosed’ route); the exact joints Wolfram-mirrored (365 $\rightarrow$ 368: the LR/glue locality integers, the KLM μ -arithmetic, the sixteen-fold pin, the $k_Y = 5/3/b_1$ identities); cited in tfpt_1 (α fixed point + EM-Ward residual prose), tfpt_research_contracts (seam residual + Lean section), introduction (closure ledger), tfpt_5_redteam (Target A residual wording) and the README/website status surfaces; exploration provenance experiments/tfpt-discovery/closure_route_*.py (2026-07-02 note in experiments/next.txt). Suite ALL CHECKS PASSED through v470. Curated version bumped to v5.4 (Zenodo deposit updated).

2026-07-02 (v467/v468 — two named post-hoc [O] flavor candidates, record unchanged: the three $\sim 2\sigma$ mixing tensions as *one* third-generation $-\varphi_0$ pattern, and the splitting ratio $\Delta m_{21}^2/\Delta m_{31}^2 = |J_{\text{PMNS}}|$)

- **v467_thirdgen_mixing_pattern.py** (FLAV.THIRDGEN.PATTERN.01). The suite’s three $\sim 2\sigma$ mixing tensions ($\sin^2 \theta_{13} + 2.0\sigma$, $\sin^2 \theta_{23} + 1.76\sigma$, $|V_{cb}| + 1.79\sigma$; v307) are exactly the three *third-generation* channels and share *one* relative deviation $-5.36\% \pm 1.67\%$ — non-zero at 3.2σ , -0.03σ from $-\varphi_0$. [E] vocabulary identities: $\lambda_C^2 = \varphi_0(1-\varphi_0)$ (the frozen Cabibbo); $\sin^2 \theta_{23} = (1-\varphi_0)/2 \Leftrightarrow \cos 2\theta_{23} = \varphi_0$; the suppression size $\varphi_0 = 5.3\%$ sits inside the seam-gapped band $(2/3)^6 = 8.8\%$ (v393). [N] the ladder-base forms $\lambda_C^2 e^{-5/6} = 0.021880 / (1-\varphi_0)/2 = 0.473414 / \lambda_C^2/(1+\lambda_C) = 0.041119$ drop the joint four-observable χ^2 from 10.4 to 0.11 with *zero* new parameters; $|V_{ub}|$ (no bare φ_0) and the 1–2 channels stay untouched and correct (dressing λ_C would fail at -24σ — scope checked, not chosen); Daya Bay alone already prefers the pattern form ($\Delta\chi^2 = 4.3$). [C] **honest**: post-hoc (5/16 equally-simple dressings fix the two PMNS tensions; $(1-\varphi_0)/(1-c_3)/\text{resolvent degenerate} < 1.4\%$). [O] no seam-transport derivation (the Bernoulli-variance reading $\lambda_Y^2 = \text{Var}[\text{Bernoulli}(\varphi_0)]$ is an exact identity but only an interpretation) — the frozen record (v84/REG.FREEZE.01) is *unchanged*. [X] pre-registered, symmetric: θ_{13} stable at 0.0231 kills the pattern, at ~ 0.0219 (sub-1%) kills the record’s bare- φ_0 reading, exclusive $|V_{cb}| \sim 0.0395$ kills both — JUNO-era reactors + Belle II decide. Python-only.
- **v468_dm2_ratio_jarlskog.py** (FLAV.DM2RATIO.01). The splitting ratio $\Delta m_{21}^2/\Delta m_{31}^2$ was previously *unpredicted*; the frozen PMNS matrix (v9/v268/v270, $\delta = 4\pi/3$) has one canonical derived CP invariant, and [N] $|J_{\text{PMNS}}| = 0.029653$ meets the NuFIT 6.0 measured ratio

0.029805 ± 0.000792 at -0.19σ (equivalently $m_2/m_3 = \sqrt{|J|} = 0.17220$ vs $0.17264(229)$) — zero dials. [N] the informal `tfpt_2` table heuristic $m_2/m_3 \sim \pi\varphi_0 = 0.16704$ (a “ $\sim$ ” entry, never frozen) is now -2.44σ and is *superseded as the comparator*. [C] spectrum consistency: $\Sigma m_\nu = m_3(1 + \sqrt{|J|}) = 0.0588$ eV reproduces the documented 0.0586 eV (v272); no absolute number is added. [O] no mechanism (the $\mu\tau$ texture gives $\theta_{13} = 0$ at leading order). [X] JUNO shrinks the window $\sim 4\times$; adopting the v467 pattern would shift $|J| \rightarrow 0.028849$ (-1.21σ today) — the two candidates *cross-discriminate* at JUNO precision. Python-only.

- **Integration.** Both registered in `run_all.py` and `script_registry.csv` (cluster neutrinos); ledger rows `FLAV.THIRDGEN.PATTERN.01` and `FLAV.DM2RATIO.01` (both [O] candidates); cited in `tfpt_2` (§ Neutrinos: pressure-point block + new candidate paragraphs, the m_2/m_3 table row retyped to the $|J|$ comparator, the octant prose kept honest — record does not select, the candidate would), `introduction` (closure ledger), `tfpt_4_frontier` (live kill-test board) and `tfpt_5_redteam` (seed-hyperplane caption); v328’s “single outlier” wording scoped to the seed observables; exploration provenance `experiments/tfpt-discovery/pattern_hunt_pmns_dressing.py` (2026-07-02 note in `experiments/next.txt`); Python-only (flagged in the Wolfram README).

2026-07-01 (v466 — a sixth seed channel from a new measurement sector: the charged-lepton mass ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ back-solves the same axiom seed to -0.11% , extending the empirical over-determination to *five* sectors)

- **v466_seed_leptonmass_channel.py (SEED.LEPTONMASS.01).** The seed over-determination of v306/v465 used five observables spanning three measurement sectors (neutrino mixing, CMB, quark mixing). This module adds the *charged-lepton mass* sector: the ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ is a sixth independent readout of the *one* axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$. [E] plumbing: the axiom φ_0 reproduces the frozen $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$ to 10^{-9} . [N] **new sector (headline):** back-solving φ_0 from the measured *pole* ratio $m_e/m_\mu = 0.004836$ (PDG 2024) gives $\varphi_0 = 0.053113$, only -0.11% from the axiom — a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. [C] **transport caveat:** the formula is the *source* ratio and the datum the *pole* ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6 = 8.78\%$, v393) but largely cancels for m_e/m_μ , so the 0.11% residual sits far inside the band — the channel is added as a *consistent new sector*, not a sharper pin. [N] **extends:** the six-channel cluster (v306’s five + this) has $\chi^2/\text{dof} < 1$ and the error-weighted mean stays at the axiom to $< 0.1\%$; with the corroborating heavy-quark m_t/m_b (-0.23%) it is a seven-channel, five-sector agreement. [N] **power:** a data↔formula shuffle explodes the cluster χ^2 by $> 100\times$, and the scheme-ambiguous $m_e/m_s = (34/47)/\varphi_0$ (FLAG 11–14) is deliberately not counted. [C] **honest:** theoretically still the *one* seed (v305 multiplicity 1) — an *empirical* strengthening, not theoretical independence; φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster registry); ledger row `SEED.LEPTONMASS.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside v305/v306/v465); Python-only (flagged in the Wolfram README).

2026-07-01 (v465 — the cross-sector one-parameter seed test: a CMB birefringence angle and a quark Cabibbo angle point to the same seed within 0.01σ , the θ_{13} -independent externally-runnable slice of v306)

- **v465_seed_crosssector_joint.py (SEED.CROSSSECTOR.01).** The sharpest falsifiable signature of a *shared-origin* theory is a correlation no non-shared-origin model predicts. In the Standard Model the cosmic-birefringence rotation β (a CMB *EB/TB* polarisation observable) and the Cabibbo angle $\lambda_C = |V_{us}|$ (a quark-flavour observable) are unrelated; TFPT reads both as functions of the *one* axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$. [E] plumbing: the axiom φ_0

reproduces the frozen λ_C , $\sin^2 \theta_{12}$, β to 10^{-9} . [N] **cross-sector (headline)**: back-solving φ_0 *independently* from β (CMB; ACT DR6 + Planck PR4 combined) and from λ_C (quark; PDG 2024) gives the same seed within $0.01\sigma_{\text{comb}}$. [N] **one-parameter fit**: the error-weighted mean of the three back-solved φ_0 reproduces the axiom to 0.06%; the joint χ^2 at the axiom (zero free parameters, 3 dof) is 0.007, $\chi^2/\text{dof} < 1$, combined pull 0.08σ . [N] θ_{13} **exclusion declared**: adding $\sin^2 \theta_{13}$ raises χ^2 by ~ 4 (the documented $\sim 2\sigma$ pressure point, v328/v338), so this slice is the honest θ_{13} -independent statement, not a hidden dilution. [N] **power**: a data $\leftrightarrow$ formula shuffle explodes the cluster χ^2 by $> 5 \times 10^4$. A consistency / out-of-sample statistic (a subset of v306); φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).

- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster registry); ledger row SEED.CROSSECTOR.01; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside v305/v306) and in the `origin_theory` birefringence dependency chain; Python-only (flagged in the Wolfram README).

2026-06-30 (v463/v464 + Lean + Wolfram — the $c=8$ holomorphic-uniqueness pin and the one-particle realisation rigidity: the SEAM.EQUIV.01 residual reduced to *entirely certification* on both halves)

- **The bird’s-eye reframing.** The recent G-block modules all sharpened the *target* of the scaling limit; this round attacks the two halves that actually remain. The *continuum existence* is already citable (v458 MMST + v459 AGT/AMT); what was implicit was (B) *why* a $c=8$ holomorphic limit must be $(E_8)_1$, and (A) the *realisation* input R_1 that the abstract seam IS the concrete quasi-free collar. v463 closes B; v464 reduces A to its one-particle data.
- **v463_seam_c8_holomorphic_uniqueness.py (SEAM.EQUIV.UNIQ.01).** The $c=8$ holomorphic-uniqueness pin. [E] $c=8$ is NOT unique at level 1: $c = \dim \mathfrak{g}/(1+h^\vee) = 8$ has THREE simple solutions A_8 ($\dim 80$), $D_8 = \mathfrak{so}(16)$ ($\dim 120$), E_8 ($\dim 248$), so the $\det K=4$ $SO(16)$ rival is a genuine same- c competitor. [E] HOLOMORPHY FORCES $\dim V_1=248$: the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3} = E_4/\eta^8$, q^1 coefficient 248 (the 1-dim candidate space, vacuum $a_0=1$). [E] $\dim 248 + \text{level-1 } c=8 \Rightarrow E_8$ uniquely ($248/(1+30)=8$); tower $\{1, 248, 4124, 34752, 213126\} = E_4/\eta^8 = j^{1/3}$. [C] Dong–Mason/Schellekens: the holomorphic $c=8$ VOA is unique $= V_{E_8}$ (even unimodular rank-8 lattice unique $= E_8$, Mordell–Witt) — so the identification half is classification-forced.
- **v464_seam_oneparticle_rigidity.py (SEAM.EQUIV.ONEPARTICLE.01).** The one-particle rigidity attacking R_1 . The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the net is the second quantisation of its one-particle symbol P . [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with $P =$ projection onto $K < 0$ ($P^2 = P$ to $< 10^{-12}$), K fixed by gap + chirality + reflection (v426). [E] ONE-PARTICLE LIMIT EXISTS: the kernel is Cauchy on a fixed window ($\max |P_{2N} - P_N| \sim 1/N \rightarrow 0$). [E] the limit is $c=1$ Dirac: entanglement $S(L) = (c/3) \ln L$ gives $c \rightarrow 1$, 16 copies $\Rightarrow c_- = 8$ (ties v450). [C] Shale–Stinespring makes the Bogoliubov map implementable, so R_1 is upgraded from a bald axiom to “the unique quasi-free realisation, modulo the named CAR functor”. Python-only (numpy).
- **Lean (SeamResidualAxiom.lean).** Adds the v463 kernel facts `c8_three_candidates` ($A_8/D_8/E_8$ each $\dim = 8(1+h^\vee)$) and `holomorphy_selects_e8` (forced $\dim V_1=248=E_8 \neq 80, 120$) to `residual_arithmetic` (still [propext]-only); v464 is numerical. `#print axioms seamResidualClosed` still lists exactly FOUR axioms.
- **Wolfram + integration.** v463 mirrored (the three candidates, the forced 248, the E_4/η^8 tower); the extension count is now 365/365. Integrated across the contracts paper (attacks (xxx)/(xxxi)), README, zenodo, ledger and website. Net effect: across (v)–(xxxi) the

SEAM.EQUIV.01 residual is now *entirely certification* — a named, hypothesis-audited package (MMST, AGT/AMT, Dong–Mason, Araki, Shale–Stinespring) with no open internal mechanism — yet SEAM.EQUIV.01 stays [O] because it still rests on cited continuum-existence theorems we do not re-prove.

2026-06-30 (v461/v462 + Lean axiom merge — the strict-locality obstruction made explicit, the 128-spinor extension exhibited at character level, and the SEAM.EQUIV.01 residual reduced to ONE realisation axiom + ONE combined cited theorem)

- **v461_seam_strict_locality.py (SEAM.S3.LOCALITY.01).** Sharpens v460’s verdict from a *cited* statement to an *exhibited* topological obstruction on the same v367 $p+ip$ collar. [E] CHERN INTEGER: the FHS Chern is $C=+1$ (chiral $M=1$), 0 (trivial $M=3$), -1 (opposite mass). [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre $\theta(k_y)$ winds by $|C|=1$ (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding = $|C|=1 \neq 0$ and $c_- = 8 = g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8 \neq 0$ (the same 8 as c_3 ’s one-sided count, v456). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0, $C=0$; since $C=1 \neq 0$ NONE exists, so v424 sub-step (i)’s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise). SEAM.EQUIV.01 stays [O]. Python-only (numpy); the winding/Chern integers Wolfram- and Lean-mirrored.
- **v462_seam_spinor_continuum.py (SEAM.EQUIV.SPINOR.01).** The constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual: the 128-spinor extension $SO(16)_1 \rightarrow (E_8)_1$ exhibited three ways. [E] $(E_8)_1$ TOWER FROM $SO(16)_1$ SECTORS: the exact Jacobi/ E_8 identity $\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4$ gives $\chi_{(E_8)_1} = E_4/\eta^8 = \chi_o^{SO16} + \chi_s^{SO16}$ — the holomorphic $(E_8)_1$ character IS the $SO(16)_1$ vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL $120 \rightarrow 248$: χ_o gives 120, χ_s gives 128, summing to the exact tower $\{1, 248, 4124, 34752\}$ ($248=120+128$). [E] FINITE 16-MAJORANA FOCK: $C(16, 2)=120$ NS bilinear currents and $2^{16/2}=256$ Ramond ground states = 128 spinor + 128 cospinor. [E] FINITE- $L \rightarrow$ CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir gives $c_- \rightarrow 8$ and the lowest scaled NS gap $x_1 \rightarrow 1$ with $|x_1 - 1| \sim 1/L^2$ ($L=64..1024$). [C][O] AGT/AMT (v459) supply the rigorous extension; SEAM.EQUIV.01 existence (v336) stays [O]. Mixed exact (θ identity + tower + $120/128/256$, Wolfram-mirrored) + numerical (finite- L , Python-only).
- **Lean (TfptCarrier/SeamResidualAxiom.lean, FORM.SEAM.RESIDUAL.01).** The two former cited axioms `mmst_existence` $\circ$ `agt_lattice_extension` are MERGED into ONE combined external axiom `seam_realisation_theorem` (dropping the intermediate `ChiralCFT_c8`), so `#print axioms seamResidualClosed` now lists exactly FOUR axioms (down from six). Two more kernel facts are added axiom-free: `strict_locality_forbidden` (v461: winding = $|C|=1 \neq 0$, $c_- \neq 0$) and `fock_sectors` (v462: $C(16, 2)=120$, $2^8=256$, spinor 128, $120+128=248$); `residual_arithmetic` (now including both) depends only on `[propext]`. `lake build` OK. So the residual is reduced to ONE realisation axiom + ONE combined cited theorem.
- **Wolfram. tfpt_readouts_extension.wl** adds 3 exact mirrors (total 363/363): the v461 obstruction integer-logic and the v462 θ -identity + $120+128=248$ sector / Fock counts.

2026-06-30 (v460 — the explicit flat-band commuting-projector (LTO) realisation of the S3 input: advancing the SEAM.EQUIV.01 sub-step (i) from abstract existence to an explicit gap-protected quasi-local projector net carrying the $\mathbb{Z}_2$ reflection and the μ_4 clock)

- **v460_seam_s3_lto.py (SEAM.S3.LTO.01).** NPW26 (arXiv:2605.10693) prove the intrinsic Bisognano–Wichmann lemma for *commuting-projector* (local-topological-order) models with a $\mathbb{Z}_2$ reflection; v424 reduced the keystone BW residual to that theorem modulo two open sub-steps, of which (ii) was settled by v426 and the continuum lift of BW by the uniform-in- N tower v455. This module attacks the *commuting-projector* side of sub-step (i) on the same explicit v367 $p+ip$ collar. [E] FLAT-BAND PARENT: the flattened Bloch $Q=h/|h|$ has $Q^2=I$ (flat ± 1 bands) and $P=(I-Q)/2$ is an exact projector EQUAL to v367’s occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state (machine precision over a k -grid). [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel $|P(n,0)|$ decays rapidly ($\xi \approx 1.14$ at $M=1$, $\exp R^2=0.995$, $|P(20,0)| < 10^{-6}$) and ξ grows as the gap closes (2.7 at gap 0.02); at the gapless point it decays as a clean power law $|R|^{-2.1}$ ($R^2=0.999$) — locality needs the gap. [E] $\mathbb{Z}_2$ REFLECTION + μ_4 INHERITED: the flattened modular Hamiltonian $K_{\text{flat}} = \text{sign}(K)$ inherits $\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}$ ($u_\Theta = J$) and $[\rho, K_{\text{flat}}] = 0$ EXACTLY (sign is odd; ρ commutes with any $f(K)$), with side-even and off- μ_4 neg controls firing. [C][O] the chiral $c_- = 8 \neq 0$ (FHS Chern $|C|=1$) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756; $C \neq 0$ forbids exponentially-localised Wannier), so the realisation is the quasi-local LTO net NPW26 use — the form sub-step (i) asks for. Advances v424 sub-step (i); SEAM.EQUIV.01 stays [O] (the continuum theorem MMST v336 is the backbone). Python-only (numpy); registered in run_all.py, the registry, the ledger and cited in tfpt_research_contracts (route (xxvii) of the convergent attacks).

2026-06-30 (v454–v459 + Lean + Wolfram — the post-F “G-block”: the lattice Sugawara current algebra, a uniform-in- N Tomita–Takesaki tower, chirality forced by P_1 , an $(E_8)_1$ character kill test, the exact MMST citation audit and the complementary lattice-VOA route)

- **v454_seam_edge_virasoro.py (SEAM.EQUIV.EDGE.VIRASORO.01).** Tests the cited MMST “Koo–Saleur lattice generators $\rightarrow$ continuum Virasoro with the correct c ” fact on our own free-fermion edge (G2). [E] the Affleck–Cardy Casimir energy fits $c_{\text{Dirac}} = 1.000$ ($\frac{1}{2}$ per Majorana, $L=64..1024$); [E] the lattice $U(1)$ current closes its affine level exactly ($\langle 0|J_n J_{-n}|0\rangle = n$, $n=1..8$; a gapped control vanishes); [E] the level-1 Sugawara $c = \dim G / (1+h^\vee) = 8$ for *both* $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31). [C] $16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$ is the level-1 current-algebra c_- ; SEAM.EQUIV.01 stays [O]. Sugawara Wolfram-mirrored.
- **v455_seam_bw_uniform.py (SEAM.EQUIV.BW.UNIFORM.01).** A uniform-in- N Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition $u_\Theta = J$ — the inductive-limit companion to v426, advancing v424 sub-step (i) (G1). [E] $\|\Theta K \Theta + K\| = 0$ exactly at every N ($N=8..128$); [E] an N -independent gap $6 \log \frac{3}{2}$; [E] $[\rho, K] = 0$ exactly at every N ; [E] an N -independent neg-control margin $2\|K\|$. [C][O] the inductive limit inherits $u_\Theta = J$; SEAM.EQUIV.01 stays [O]. Python-only.
- **v456_seam_chirality_from_c3.py (SEAM.S3.FROM-P1.01).** The edge chirality ($c_- \neq 0$, the S3 phase) is *forced* by the one-sidedness that defines $c_3 = 1/(8\pi)$ (P1) (G6). [E] c_3 ’s “8” is the one-sided count $8 = |\mathbb{Z}_2| \cdot (\int K/\pi) = 2 \cdot 4$; [E] a reflection sends the Chern integer $C \rightarrow -C$ ($+1 \rightarrow -1$ on the $p+ip$ model); [E] a two-sided boundary forces $C = -C = 0$. [C] the one-sided seam (no reflection) forces $C \neq 0$, magnitude $c_- = 8 = \text{rank } E_8$ — S3 is a consequence of P_1 ; SEAM.EQUIV.01 existence stays [O]. The arithmetic is Wolfram-mirrored.

- **v457_seam_character_killtest.py** (SEAM.EQUIV.KILLTEST.01). An explicit character/-sector kill test of $(E_8)_1$ against $SO(16)_1$ (G7). [E] the $(E_8)_1$ character $\chi=E_4/\eta^8$ with tower degeneracies $\{1, 248, 4124, 34752, \dots\}$; [E] the weight-1 count $248=120+128$ (not the free 120); [E] the sector count $|\det \text{Cartan } E_8|=1$ (not $|\det \text{Cartan } D_8|=4$). [X] a measured 120, four sectors, or a wrong coefficient would *falsify* $(E_8)_1$; the data give $248/1/\{1, 248, 4124\}$ (passed). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.
- **v458_seam_mmst_citation_audit.py** (SEAM.EQUIV.MMST.AUDIT.01). The exact hypothesis-by-hypothesis citation audit of MMST (arXiv:2107.13834) against the collar (G3, the “exakter Zitat-Abgleich”). [C] H1 CAR class, H2 massless limit, [E] H3 per-copy $c=\frac{1}{2}$ (Thm 4.11), [C] H4 decoupled additivity to $c=8$, H5 the WZW range $8 \leq 8 \leq 16$ with the 120 bilinear currents all match; [O] the single residual H6 is the 128-spinor extension $SO(16)_1 \rightarrow (E_8)_1$ ($\det K 4 \rightarrow 1$), outside MMST’s bilinear method. SEAM.EQUIV.01 stays [O], its open part narrowed to this one holomorphic extension.
- **v459_seam_lattice_voa_route.py** (SEAM.EQUIV.LATTICEVOA.01). The second, independent route to $(E_8)_1$ supplying exactly the residual of v458 (G5). [E] E_8 even unimodular ($\det=1$); [E] the weight-1 space $248=8+240$; [E] the 240 roots split 112 integer + 128 half-integer (the spinor, $248=120+128$). [C] the lattice net A_{E_8} (Adamo–Giorgetti–Tanimoto, arXiv:2506.01008; OS arXiv:2407.18222) constructs the holomorphic extension; [O] the two routes leave one shared realisation input (collar = A_{E_8} , tied to P_1 by v456). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.
- **Lean** (FORM.SEAM.RESIDUAL.01, `TfptCarrier/SeamResidualAxiom.lean`). Kernel-checks the whole G-block arithmetic (the one-sided $8=|\mathbb{Z}_2| \int K/\pi$, the reflection $C \rightarrow -C$, $c_-=8$, the MMST range $8 \leq 8 \leq 16$, decoupled additivity, $248=8+240=120+128$, $240=112+128$, $\det K 1$ vs 4) by `decide`, reducing the SEAM.EQUIV.01 residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus two cited external theorems (MMST v458, AGT v459); `lake build` OK.
- **Wolfram + integration**. Six exact mirrors added to `tfpt_readouts_extension.wl` (Sugawara $c=8$ for 120/15 and 248/31; the one-sided $8=2 \cdot 4$ and reflection $C=-C \Rightarrow 0$; the $(E_8)_1$ tower $\{1, 248, 4124, 34752\}$ with $248=120+128$ and $|\det \text{Cartan}| 1$ vs 4; the root split $112+128$ and $248=8+240$) — extension total now 360/360. Registry, ledger, master index, website mirror and manifests updated in the same change.

2026-06-30 (v449–v453 + Lean + Wolfram — the edge CFT named end-to-end: uniform-in- N convergence, three independent reads of $c_-=8$, the Ising operator tower, the $(E_8)_1$ modular data, and μ_4 -symmetry derived from the four marks)

- **v449_seam_mmst_uniform.py** (SEAM.EQUIV.MMST.UNIFORM.01). Strengthens the cited MMST hypothesis from convergence at a single cross-ratio (v444/v448) to the *uniformity* the theorem actually asserts, on the same v367/v368 collar. [E] cross-ratio A $Q(1, 2, 4, 7)/12 \rightarrow 2+\sqrt{3}$, the deviation shrinking monotonically; [E] an *independent* cross-ratio B $Q(1, 3, 5, 9)/12 \rightarrow$ its own value 2, so convergence is not a tuned point; [E] a uniform $1/N$ rate $|Q(N)-Q_\infty| \cdot N \leq 4$ for *both* observables at every $N_x \in \{36, 48, 60\}$ (one N -independent constant); [E] the bulk control is off-scale. [C] the MMST uniform-convergence hypothesis is empirically satisfied; SEAM.EQUIV.01 stays [O].
- **v450_seam_edge_entanglement.py** (SEAM.EQUIV.EDGE.ENTANGLEMENT.01). A *third* independent reading of the edge central charge (after the correlator v444 and the topology v447). [E] the Calabrese–Cardy entanglement law gives $c_{\text{Dirac}}=1.000$ for the critical free-fermion chain; [E] so c per Majorana = $1/2$; [E] a gapped control saturates ($c=0$). [C] bulk–edge: $c_-=N_{\text{Maj}} \cdot \frac{1}{2}=16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$. SEAM.EQUIV.01 stays [O].

- **v451_seam_edge_cardy_tower.py** (SEAM.EQUIV.EDGE.CARDY.01). Names the edge CFT by its *operator content*. [E] the Cardy finite-size spectrum gives the spin dimension $h_\sigma=1/16$ (the periodic–antiperiodic ground-energy split) and [E] the energy dimension $h_\epsilon=1/2$ (the lowest NS pair excitation), both [E] L -independent (conformal $1/L$ scaling). [C] $\{c, h_\sigma, h_\epsilon\}=\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$ is the chiral Ising (free-Majorana) minimal model $M(3, 4)$; 16 copies give $c_-=8$. SEAM.EQUIV.01 stays [O].
- **v452_seam_e8_modular.py** (SEAM.EQUIV.E8.MODULAR.01). The $(E_8)_1$ torus modular signature, extending v156’s q -expansion to the transformation data. [E] the character $\chi=E_4/\eta^8$ is S -invariant $\chi(-1/\tau)=\chi(\tau)$ (one primary, $S=[1]$); [E] T -phase $\chi(\tau+1)/\chi(\tau)=e^{-2\pi i/3}=e^{-2\pi i c/24}$ ($c=8$); [E] leading power $\chi q^{1/3} \rightarrow 1$ so $q^{-c/24}=q^{-1/3} \Rightarrow c=8$; [E] holomorphic $c=8 \equiv 0 \pmod{8}$ with T -phase order $24/\gcd(8, 24)=3$. [C] the $(E_8)_1$ identity is now pinned on the torus *and* the edge. SEAM.EQUIV.01 stays [O].
- **v453_seam_mu4_from_marks.py** (SEAM.RIGIDITY.MU4FROMMARKS.01). Folds the last structural premise of the rigidity chain (v446’s residual QGEO.SYM.01) into the existing realisation premise. [E] the four Gauss–Bonnet marks *are* μ_4 = roots of z^4-1 ; [E] the order-4 clock $z \rightarrow iz$ is a 4-cycle permuting them (the v445 clock); [E] it preserves the cross-ratio $\lambda=2$ (in the D_4 stabiliser, v168); [E] the form basis $\omega_k=z^{k-1}dz/(z^4-1) \rightarrow i^k\omega_k$ is μ_4 -graded (weights $(1, 2, 3)=A_3$ exponents). [C] so QGEO.SYM.01 follows from marks= μ_4 + the existing QGEO.REALIZE.01 — no new premise; the chain (v453→v446→v445) closes modulo realisation. SEAM.EQUIV.01 stays [O].
- **Lean: TfptCarrier/SeamEdgeChern.lean** (FORM.SEAM.EDGE.CHERN.01). Kernel-hardens the integer backbone the four edge reads converge on: the FHS Chern integers $(+1/0/-1)$, the bulk–edge channel count $|C|=1$, $N_{\text{Maj}}=2^{g_{\text{car}}-1}=16$, the assembly $2c_-=N_{\text{Maj}} \cdot |C|$ (so $c_-=8$), $c_-=8=g_{\text{car}}+N_{\text{fam}}=\text{rank } E_8$, holomorphic $c \equiv 0 \pmod{8}$ and the order-3 modular T -phase — all by kernel *decide* (edge_kernel_facts, axioms propext only). lake build OK, #print axioms clean.
- **Wolfram mirror.** tfpt_readouts_extension.wl gains nine exact checks (v452 modular data $\times 4$, v450 entanglement assembly, v451 Ising Kac weights, v453 marks= $\mu_4 \times 3$): 345 $\rightarrow$ 354 checks, all passing (GATE.WOLFRAM.02). E_4 is evaluated via its q -series since EisensteinE stays symbolic for complex τ .
- **Papers.** The adversarial audit (tfpt_5_redteam, Target A) and the research-contracts note record the edge CFT as named end-to-end — central charge $c_-=8$ from three independent observables (correlator v444, topology v447, entanglement v450), the Ising operator tower (v451), the $(E_8)_1$ torus modular data (v452), and the μ_4 premise derived from the marks (v453) — while the only residual that stays [O] remains the cited continuum-existence theorem (v336).

2026-06-30 (v446/v447/v448 + Lean + Wolfram — deriving clock-invariance from μ_4 -symmetry, an independent topological reading of the edge $c_-=8$, and pinning the one external MMST fact at four-point order)

- **v446_seam_clock_invariance.py** (SEAM.RIGIDITY.CLOCKINV.01). Closes v445’s single residual one notch further: the operator condition $[\rho, \Lambda_\Sigma]=0$ (clock-invariance) is shown to be a *consequence* of a structural symmetry, not an independent assumption. [E] the OS canonical transfer $\Lambda=C(1)C(0)^{-1}=e^{-h}$ (fixed by RP+reflection, v54) of a μ_4 -invariant RP covariance satisfies $[\rho, \Lambda]=0$ (swept $N=4..32$); [E] a μ_4 -breaking covariance gives $[\rho, \Lambda] \neq 0$ (iff-sharp neg control); [E] Λ is a positive contraction (spectrum in $(0, 1]$); [E] the free modular Hamiltonian $K=\log((1-C)C^{-1})$ (v258) of the invariant state has $[\rho, K]=0$ (Tomita–Takesaki), broken control $\neq 0$. [C] so with v445 (commute $\Leftrightarrow$ block-diagonal), RP-definability *plus* the four marks *force* the block-diagonal Λ_Σ ; the residual reduces from an operator commutation to “the seam RP data is μ_4 -symmetric” (QGEO.SYM.01). SEAM.EQUIV.01 stays [O].

- **v447_seam_edge_chern.py** (SEAM.EQUIV.EDGE.CHERN.01). An *independent* (topological) reading of the edge central charge, corroborating v444’s correlator route from a different observable. [E] the $M=1$ collar has integer Chern number $C=+1$ (Fukui–Hatsugai–Suzuki link-variable method, no fit); [E] $C=0$ in the trivial phase and $C=-1$ at $M=-1$ (opposite chirality); [E] the open strip carries exactly $|C|=1$ chiral edge branch per boundary; [E] the Li–Haldane entanglement spectrum has in-gap modes at $1/2$ only in the topological phase. [C] bulk–edge: $c_- = N_{\text{Maj}} \cdot \frac{1}{2} \cdot |C| = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$ — the same $c_- = 8$ as v444. SEAM.EQUIV.01 stays [O].
- **v448_seam_edge_fourpoint.py** (SEAM.EQUIV.EDGE.FOURPOINT.01). Pins the one external MMST fact (continuum existence of the chiral scaling limit) empirically at the next order. [E] the ground-state correlation matrix is an exact projector $G^2=G$ ($\sim 10^{-13}$), so the state is quasi-free (the MMST CAR hypothesis holds); [E] the edge four-point cross-ratio $Q=[G_{01}G_{23}]/[G_{02}G_{13}]$ at commensurate positions is Cauchy-convergent under refinement; [E] Q converges to the free-fermion conformal cross-ratio $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3} \approx 3.732$; [E] the bulk control is off-scale (exponential). [C] so the one external fact is supported at *both* two-point (v444) and four-point order. SEAM.EQUIV.01 stays [O].
- **Lean: TftptCarrier/SeamRigidityForcing.lean** (FORM.SEAM.RIGIDITY.FORCING.01). Kernel-hardens v445 (and v442’s uniformity): the entrywise forcing iff $\text{clock4 } a = \text{clock4 } b \Leftrightarrow a=b$ (i.e. $i^a=i^b \Leftrightarrow a \equiv b \pmod{4}$, residue-only hence uniform in N), the four distinct eigenphases, the exact commutant dimension $\sum n_s^2=64<256$ and the order discriminator ($128>64$) are all proved by kernel **decide** (**forcing_kernel_facts**, no domain axioms); only the operator-level lift to the full L^2 is the named premise. **lake build OK, #print axioms clean**.
- **Wolfram mirror. tftpt_readouts_extension.wl** gains three exact v445 checks (entrywise forcing for all $a, b \in 0..31$; commutant dimension $64<256$; order-2 commutant $128>64$ swept $N=4..64$): $342 \rightarrow 345$ checks, all passing (GATE.WOLFRAM.02).
- **Papers.** The adversarial audit (**tftpt_5_redteam**, Target A) records the five-direction narrowing of the SEAM.EQUIV.01 keystone (edge scaling v444, forcing converse v445 + Lean, derived clock-invariance v446, independent topological $c_- = 8$ v447, four-point pinning v448) explicitly as “not promoted”: the only residual that stays [O] is still the cited continuum-existence theorem (v336). The research-contracts note lists the same as “eleven convergent attacks”.

2026-06-30 (v445 — the rigidity FORCING converse: commuting with the order-4 clock is *exactly* μ_4 -block-diagonality, upgrading the rigidity step from “permit” to “force”)

- **v445_seam_rigidity_forcing.py** (SEAM.RIGIDITY.FORCING.01). The rigidity route (v398/v442) showed a μ_4 -block-diagonal K *commutes* with the order-4 clock $\rho = \text{diag}(i^n)$. This module proves the *converse* and the exact structure, isolating the single remaining physical premise. [E] FORCING (entrywise iff): $[\rho, K]=0$ iff $K_{ij}=0$ whenever $i \not\equiv j \pmod{4}$, verified entrywise and swept $N=4..128$ — commuting *forces* μ_4 -block-diagonality, it does not merely permit it. [E] EXACT COMMUTANT DIMENSION: $\dim\{K : [\rho, K]=0\} = \sum_s n_s^2$ (the ad_ρ nullspace), a proper subspace of the full N^2 algebra ($N=16$: $64<256$) — a genuine quantitative restriction. [E] OFF-BLOCK $\Leftrightarrow$ NON-COMMUTING: every off-block matrix unit has $\|[\rho, E_{ij}]\| \geq \sqrt{2} > 0$ (zero slack). [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors; the order-2 clock leaves a strictly larger commutant ($N=16$: 64 vs 128) — the four Gauss–Bonnet marks (order $4=|\mu_4|=h(A_3)$) force strictly more structure than two, and only the index-4 extension is $(E_8)_1$ ($\det=1$; v281/v154/v351). [E] NONDEGENERACY: exactly 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all $N \geq 4$. [C] SYNTHESIS: commuting with the order-4 μ_4 clock is *equivalent* to μ_4 -block-diagonality, so RP-definability (v54/v398)

+ clock-invariance *force* the block-diagonal Λ_Σ — v442’s “permit” upgraded to “force”. [O] the single residual is now precisely that the physical transfer lies in that commutant (clock-invariance = QGEO.SYM.01, v323/v335); SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy; det facts Wolfram-mirrored via v89/v281/v422). Registered in `run_all.py`, `script_registry.csv`, the ledger, and cited in `tfpt_research_contracts` (route (xii) of the convergent attacks, and the sharpened residual in the Seam State Rigidity contract).

2026-06-30 (Lean kernel-hardening of two of the seven SEAM.EQUIV.01 routes — the Borchers/Wiesbrock standard-pair relations and the applicability-ledger count, machine-verified with clean axioms; rigor without new mathematics)

- **Context.** Two of the seven convergent attacks on the keystone residual are turned from numerical witnesses into kernel-checked facts, in the audit-contract style of `SeamScalingLimit.lean` / `BWKeystone.lean`: the load-bearing algebra/bookkeeping is proved by the Lean kernel, while the cited theorems and the one open continuum input remain named axioms. `lake build OK`; `#print axioms clean`. Tracked as two new FORM.* ledger rows; `\veri`-context in `tfpt_research_contracts`; Lean manifest refreshed.
- **FORM.SEAM.BW.HSMI.01** (`TfptCarrier/SeamStandardPair.lean`) — **standard-pair relations, kernel-checked (hardens v438)**. The four Borchers/Wiesbrock relations on the centred 5-mode ladder over $\mathbb{Z}$ are proved by kernel `decide` (axioms `propext` / `Classical.choice` / `Quot.sound` only, no domain axiom): $[K, P]=P$, $J^2=1$, $JKJ=-K$, $JPJ=P_+$ (and $\text{tr } K=0$). The cited Borchers/Wiesbrock theorem and the one continuum input (the continuum standard pair) are named axioms; the composition pins face (ii) of SEAM.EQUIV.01 to exactly those two — intrinsic Bisognano–Wichmann, no rotation-covariance presupposed.
- **FORM.SEAM.APPLICABILITY.01** (`TfptCarrier/SeamApplicabilityLedger.lean`) — **the audit count, kernel-checked (hardens v441)**. The headline bookkeeping is itself machine-verified: of the 12 cited-theorem hypotheses (MMST/NPW26/Adamo, mirrored in order) exactly 11 are internal and 1 external (`total_count=12`, `internal_count=11`, `external_count=1`), plus `mmst_range` ($8 \leq 8 \leq 16$) and `npw_outside_bucket` ($\det K=1 \neq 4$) — ALL with NO axioms (pure kernel `decide`). The single external fact (continuum scaling-limit existence) is the named axiom; the keystone reduction pins its external dependence to exactly that one fact.
- **Net.** Both modules are imported by the root `TfptCarrier.lean`, axiom-audited in `AxiomCheck.lean`, and signature-locked in `AuditContract.lean`; full `lake build` succeeds. No new mathematics — the same facts v438/v441 verify numerically are now kernel-proved; SEAM.EQUIV.01 stays [O].

2026-06-30 (the chiral-edge scaling motor v444 — the edge half of the one SEAM.EQUIV.01 continuum residual: the lattice edge already carries the conformal data of a chiral $c=1/2$ Majorana CFT and converges under refinement, grounding the EDGE the way v439 grounded the BULK; the keystone stays [O])

- **Context.** v439 grounded the *bulk* half of the cited MMST existence (uniform gap, exponential clustering, finite light cone) and explicitly narrowed the one open analytic step to the chiral massless *edge* scaling limit. v444 attacks exactly that residual on the SAME explicit v367/v368 $p+ip$ collar. vN-registered, runs in `run_all.py` (suite ends ALL CHECKS PASSED), `\veri`-cited in the body of `tfpt_research_contracts` (the G_{net} contract). Python-only (numpy; no new exact identity to Wolfram-mirror).
- **v444_seam_edge_scaling.py** (`SEAM.EQUIV.CONTINUUM.EDGE.01`) — **chiral-edge motor**. The lattice edge carries the kinematic and conformal data of a chiral $c=1/2$ Majorana CFT:

[E] a linear, chiral in-gap dispersion (velocity $v \approx 0.98$, $R^2 = 0.9999$, with *opposite* per-edge velocities — one chiral Majorana per edge); [E] an *algebraic* equal-time edge two-point $\sim r^{-\eta}$ with $\eta \approx 1$, versus the *exponential* bulk row and the exponential trivial-phase ($M=3$, no edge) control; [E] a Cauchy-convergent scaling-collapse under refinement ($\eta = 1.001 \pm 0.007$, successive profile spread $0.028 \rightarrow 0.013$ across $N_x = 36..60$) — the numerical hallmark of an existing scaling limit. [C] $\eta = 1 \Leftrightarrow$ one chiral Majorana ($c = 1/2$) per edge, so the 16-copy carrier has a chiral $c_- = 8$ $(E_8)_1$ edge CFT ($c_- = g_{\text{car}} + N_{\text{fam}}$). [O] the rigorous uniform-convergence theorem (cited MMST, v336) is not supplied.

- **Net.** With v439 (bulk) and v444 (edge), *both halves* of the one continuum residual are numerically grounded on our explicit model and pinned to a single cited existence theorem; SEAM.EQUIV.01 stays [O]. Ledger row, `script_registry.csv` index and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) updated in the same change; manifest refreshed last.

2026-06-30 (six convergent attacks on the one SEAM.EQUIV.01 continuum residual — intrinsic Bisognano–Wichmann, a Lieb–Robinson continuum motor, the β -homotopy, an applicability audit, uniform rigidity, and falsifiable kill tests; none closes the keystone, together they reduce it to a single analytic fact and make the claim falsifiable)

- **Context.** The keystone SEAM.EQUIV.01 is “closed modulo a cited theorem”: its one open input is the continuum existence of the seam’s chiral massless scaling limit. Six new modules attack that residual from independent directions; each is honest about *not* closing it. All six are vN-registered, run in `run_all.py` (suite ends ALL CHECKS PASSED), and are \veri-cited in the body of `tfpt_research_contracts` (the G_{net} contract). Python-only (no new exact identity to Wolfram-mirror).
- **v438_seam_hsmi_borchers.py (SEAM.EQUIV.BW.HSMI.01) — intrinsic BW.** The seam supplies a Longo *standard pair* / +half-sided modular inclusion (positive lightlike translation $P \geq 0$, boost K with $[K, P] = P$, reflection J with $JKJ = -K$, $JPJ = P_-$), so by Borchers’ theorem the modular flow is geometric *intrinsically* — *without* presupposing rotation-covariance of the vacuum (complementary to Adamo–Giorgetti–Tanimoto). [E] ladder algebra exact, $sl(2, \mathbb{R})$ positivity ($\min \text{eig} > 0$); [C] synthesis; [O] the continuum standard pair.
- **v439_seam_lieb_robinson.py (SEAM.EQUIV.CONTINUUM.LR.01) — continuum motor.** The explicit gapped $p+ip$ collar (v367) has a uniform-in- N gap, exponential clustering with N -independent ξ , and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral *edge* scaling limit. [E]/[C], [O] the edge CFT.
- **v440_seam_lto_rp_beta.py (SEAM.EQUIV.LTORP.BETA.01) — β -homotopy.** The (LTO-RP)/BW data ($\Theta K \Theta = -K$, $[\rho, K] = 0$) are β -independent, and along the whole path $C_\beta = 1/(1+e^{\beta K})$ from the NPW26-proven *trace* state ($\beta=0$) to the seam KMS state ($\beta=1$) every covariance is reflection-related, μ_4 -invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii) of v424.
- **v441_seam_applicability_ledger.py (SEAM.EQUIV.APPLICABILITY.01) — audit.** A hypothesis-by-hypothesis ledger of the three cited theorems (MMST / NPW26 / Adamo–Giorgetti–Tanimoto) finds 11 of 12 hypotheses are TFPT-internal computed facts and exactly one is external — the continuum existence of the chiral scaling limit; covariance is an Adamo consequence (defuses the v215 circularity), not an input. The keystone’s external dependence is pinned to that one analytic fact.

- **v442_seam_rigidity_uniform.py** (SEAM.RIGIDITY.UNIFORM.01) — **uniform rigidity.** The band-limited μ_4 -character rigidity of v398 is uniform in N : exact 4-sector commutation to $N=256$ (not merely the $N \leq 64$ sweep), a uniform gap, and an N -independent clock-breaking lower bound $\|[\rho, V]\|=2$ — so it is not a small- N artifact (the full- L^2 forcing stays [O]).
- **v443_seam_e8_discriminator.py** (SEAM.E8.DISCIMINATOR.01) — **kill tests.** Mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest $c=8$ rival $SO(16)_1$: torus ground-state degeneracy 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots; $c=8$ alone cannot decide (v277), only $\det K=1$ (the genus-1 statement, v90) selects E_8 . A seam with $GSD>1$ would falsify $(E_8)_1$. [X]/[C].
- **Status discipline.** SEAM.EQUIV.01 stays [O] throughout; the six modules reduce, audit and falsify the residual but do not supply the one cited continuum-existence theorem. Ledger rows, the `script_registry.csv` index, and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) are updated in the same change; manifest refreshed last.

2026-06-30 (shadow-export coherence — the tfpt5-shadow mirror is now reproducible *as shipped*: the audit is shadow-aware, the content maps are non-destructive on the website-free subset, and an export gate blocks any incoherent mirror)

- **Context (reviewer finding, Alessandro).** On the exported `tfpt5_shadow.zip` the independent surfaces reproduced (`build.sh` notes $\rightarrow$ 11 ok; Wolfram base/extension; Red Team; Lean transcript), but the *package* layer did not: `make_manifest.py -check` reported a stale `verification/website_map.csv` and `build.sh` audit crashed because it unconditionally required `website/lib/release.ts`, which the shadow subset deliberately excludes.
- **make_docs_map.py is now shadow-aware.** `website_map.csv` mirrors the `website/` sources; on a tree without `website/` the generator now *skips* it instead of regenerating it down to a single `README.md` row. This removes the silent corruption that made `build.sh` notes/gen leave `website_map.csv` stale against the manifest on the export (it now matches the established skips in `make_script_index.py` and `make_changelog_web.py`).
- **audit_sync.py is now shadow-aware.** When `website/` is absent it skips the website mirror (section D), the website version stamps (E) and the website Wolfram-count card (F), and the website generated-surface freshness in (A), printing exactly which sections were skipped. `build.sh` audit therefore runs to a clean AUDIT OK on the shadow subset instead of raising `FileNotFoundError`.
- **Export coherence gate.** `scripts/export-shadow.sh` now re-runs `make_manifest.py -check` and the shadow-aware `audit_sync.py` on the *filtered tree* before the mirror is published. The manifest check catches the earlier v235–v298 symptom at the source: a module referenced by `run_all.py/script_registry.csv/manifest.sha256` but not committed (and thus dropped by `git ls-files`) now fails the export loudly instead of shipping a package that says ALL CHECKS PASSED while `run_all.py` stops at the first missing script.
- **Manifest refreshed.** `manifest.sha256` was regenerated as the last release step so the shipped digests match the committed `changelog.tex` and `website_map.csv`; the full Python suite reproduces end-to-end on the regenerated export. No theory content, status marker or vN module changed — this is an export/infrastructure fix only.

2026-06-29 (discoverability — an SEO/GEO pass on the public website: a generated `/llms.txt`, explicit AI-crawler `robots.txt`, query-targeted metadata, and the first real backlinks from the README and the Zenodo deposit)

- **New `/llms.txt` context file** for AI / answer engines (ChatGPT, Perplexity, Claude, Gemini), generated at build time from the same data layer as the site (`papers.ts`, `predictions.ts`, `version.ts`) so it never drifts. It carries the one-line summary, the “claims / does not claim” split, the document set with PDF links, the headline predictions, and an explicit *disambiguation* from the unrelated Brouwer–Lefschetz “topological fixed point theory” of mathematics.
- **`robots.txt`** now lists the major search and AI crawlers (Googlebot, Google-Extended, Bingbot, GPTBot, ChatGPT-User, OAI-SearchBot, PerplexityBot, ClaudeBot, anthropic-ai, ...) explicitly as allowed; the site-wide meta description and keywords are reworded toward natural search queries (“derivation of the fine-structure constant”, “Standard Model from first principles”, “where does 137 come from”) and carry the same physics/mathematics disambiguation.
- **Backlinks (the missing piece).** The root `README.md` and the Zenodo deposit description (`zenodo_description.html`) previously contained *no* outbound links; both now link to `fixpoint-theory.com` and the source repository, and the README gains a citation block with the archived deposit DOI 10.5281/zenodo.20846087.
- Website / documentation infrastructure only — no theory, suite, ledger or status change.
- **New module `v437 (E8.DEGREE.JOINT.01)` — a consolidation of `v6/v66/v355/v431` that derives more from the E_8 Casimir degrees *without* crossing the `v354/v355` anti-numerology line.** It adds *no* new per-degree coincidence — only one joint lock and a reconstruction.
 - **[E] all of E_8 is fixed by its degree multiset.** From $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$ alone: $\text{rank} = \# \deg = 8$, $h = \max \deg = 30$, $\# \text{positive roots} = \sum \exp = 120$, $\# \text{roots} = 240$, $\dim E_8 = 248$, $|W(E_8)| = \prod \deg = 696,729,600$, $\sum \deg = 128 = \dim S^+$.
 - **[E] the joint quadratic (the new lock).** The two TFPT structural integers ($g_{\text{car}}, N_{\text{fam}}$) are the *unique* root pair of $x^2 - (\text{rank } E_8)x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$: sum of roots = $\text{rank } E_8 = 8$ (the rank-fill, `v6`), product = $h/2 = 15 = g_{\text{car}}N_{\text{fam}}$ ($2g_{\text{car}}N_{\text{fam}} = h$, `v431`); $\{3, 5\}$ is the unique positive-integer factor pair of 15 summing to 8. So $g_{\text{car}} = 5$ and $N_{\text{fam}} = 3$ are forced *together* by two E_8 degree-invariants, not chosen one at a time. The three primary-readout degrees $\{2, 8, 30\}$ are exactly $\{\min, \text{rank}, \max\}$.
 - **[E] discipline upheld:** no new per-degree mining beyond the joint quadratic; the five non-primary degrees keep their `v431` family homes; the `v354/v355` line is not crossed.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`E8.DEGREE.JOINT.01`); cited in the `tfpt_5` degree-audit winbox beside `v431`; mirrored in `wolfram/tfpt_readouts_extension.wl` (extension 339 $\rightarrow$ 342). Exact integer/lattice; no new physical number, no status change.

2026-06-29 (`v436` — hardening the unconditional “is it numerology?” floor to an assumption-minimal counting floor; plus a literature note grounding the α^3 Chern level)

- **New module `v436 (OVERDET.FLOOR.02)` — it *hardens* `OVERDET.FLOOR.01 (v432)` so the numerology verdict no longer rests on any subjective chance assignment.** `v432`’s nominal $\sim 10^{-10}$ multiplies the counting census by hand-chosen plausibilities (input-“8” independence 0.1^3 , seed 0.012, kaon 0.15) a skeptic can contest; `v436` shows the conclusion survives without them.

- **[E] assumption-minimal floor.** The α cubic-class census (v100 layer D) is *pure counting*: of $N = 94500$ complexity-matched $F_{U(1)}$ variants only *one* lands in the 4×10^{-8} CODATA window — the TFPT instance, reproducing $\alpha^{-1} = 137.0359992$ — so $p_\alpha \leq 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$ with *no* subjective probability, prior or multi-observable grammar.
- **[E] monotone concession ladder.** $-\log_{10} p$ from generous to adversarial: full scorecard+ $\alpha \sim 30.7 \rightarrow$ scorecard-only $\sim 25.8 \rightarrow \alpha$ -census-only ~ 4.98 ; each rung uses strictly less evidence and the “not chance ($> 4\sigma$)” verdict holds at the *most adversarial* rung, so it never rests on one contestable piece. Stripping *all* of v432’s subjective factors ($\sim 1.8 \times 10^{-6}$) leaves the floor at 4.40σ — they only sharpen it.
- **[C] honest 5σ gap.** The conventional 5σ line ($p = 5.7 \times 10^{-7}$) is crossed unconditionally by the census times exactly *one* further independent factor ≤ 0.054 (the input-“8” four-fold over-determination, or one foreign witness) — stated, not hidden. **[C]** Firewall unchanged: bounds only “is it numerology?”, not the physics bridge (v187).
- **Literature note (research log, not a status change).** A `next.txt` entry grounds the α^3 Chern-level reading of v435 in named theorems — Dai–Freed (η -invariants and determinant lines, the exponentiated η -invariant as a section of the inverse Quillen determinant line), Bismut–Freed (determinant-line curvature = the α^2 Calderón side) and Redlich/Stora–Zumino (the cubic $\text{Tr } F^3$ anomaly descends to a Chern–Simons level, quantised to $\mathbb{Z}$). It records that the EM1 proof obligation reduces to constructing the seam Dirac family — i.e. it is a *facet* of SEAM.EQUIV.01 — and remains **[O]** external math (no fabricated closure).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (OVERDET.FLOOR.02); cited in the `tfpt_safeguards` “unconditional floor” keybox beside v432; mirrored on the website. Python-only; no new physical number, no status change.

2026-06-29 (v435 — a fourth honest step on the α^{-1} Quillen target: a π -power test isolates the cubic α^3 as the unique metric-independent (topological) rung, and types its coefficient as a conditional integer level)

- **New module v435 (ALPHA.QUILLEN.PROGRESS.04) — it attacks the *single* remaining **[O]** left after v434 (residual (2), the cubic α^3 Chern/Maxwell moment) and *sharpens* it; it does *not* close it.** The target ALPHA.QUILLEN.EXACT.01 (v382) stays **[O]**; $\alpha^{-1} = 137.0359992$ stays **[E]**.
 - **[E] π -power / metric-independence test.** The three EM-closure coefficients carry π -powers exactly $\{0, 3, 6\}$, equal to their c_3 -orders: Maxwell $k_0 = 1 \rightarrow \pi^0$, Calderón $-2c_3^3 = -1/(256\pi^3) \rightarrow \pi^3$, transport $-8b_1c_3^6 = -41/(327680\pi^6) \rightarrow \pi^6$. So the Maxwell a^3 is the *unique* π^0/c_3^0 (metric-independent) term. A heat-kernel boundary coefficient always carries the $(4\pi)^{-d/2}$ measure; a metric-independent term carries none — the signature of a *topological* term. The test thus partitions the closure into *one* topological level (a^3) + *two* heat-kernel boundary terms, confirming v342/v434’s “ a^3 is a level, not a Seeley–DeWitt curvature integral” from a *test* rather than a claim.
 - **[C] conditional level quantisation.** If the a^3 term is the seam $U(1)$ Chern–Simons/ η boundary level (the EM1 reading, v48), large-gauge invariance quantises the level to $\mathbb{Z}$, so $k_0 = 1$ is the *unit* (minimal) level — an integer, not a tunable real. This converts “the coefficient happens to be 1” into “the coefficient is an integer level and 1 is the unit”.
 - **[O] not closed:** the EM1 lemma itself — the from-first-principles Chern–Simons boundary proof that $\partial_\tau \log Z_{\text{Maxwell}} = a^3$ — stays external math (type A, v384); the seam F -normalisation stays the one **[C]** (residuals 1 & 3, v434); α^{-1} stays **[E]**.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv`

(ALPHA.QUILLEN.PROGRESS.04); cited in the `tfpt_1` EM-closure subsection beside `v433/v434`; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-28 (v434 — a third honest step on the α^{-1} Quillen target: b_1 is the a_4 heat-kernel coefficient, and the three residuals collapse to one [C] + one [O])

- **New module v434 (ALPHA.QUILLEN.PROGRESS.03)** — it settles the status of the three residuals named after **v433** and shows they are *not* three independent open problems. The target ALPHA.QUILLEN.EXACT.01 (v382) stays [O]; $\alpha^{-1} = 137.0359992$ stays [E].
 - [E] b_1 *is* the $U(1)$ a_4 **heat-kernel coefficient (the $\beta = a_4$ theorem)**. The one-loop β function of a gauge coupling *is* the a_4 (Seeley–DeWitt a_2 in $d=4$) heat-kernel coefficient of the charged-field operator (Vassilevich). Computed from the carrier hypercharge content with the standard spin weights ($\frac{2}{3}$ per Weyl fermion, $\frac{1}{3}$ per complex scalar), $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM) and $(3/5) \cdot \text{that} = 41/10 = b_1$ (GUT). So the b_1 in the counterterm $8b_1c_3^6$ is the a_4 coefficient of the carrier $U(1)_Y$ content (the same number as the β , v159/v246), the *same kind* of heat-kernel object as the α^2 Calderón a_4 (v342) — not a fitted multiplicity.
 - [E] **the geometry factors off**: $8b_1c_3^6 = b_1 \cdot (\text{rank } E_8) \cdot c_3^6$, with $\text{rank } E_8 = 8$ the $\chi=2$ seam boundary count (v216) and c_3^6 the six boundary insertions (the π -power test, v342).
 - [C] **residual (1) $\equiv$ residual (3)**. With b_1 and the c_3 -powers heat-kernel-forced, the only freedom left in the exact seam a_4 value is how the gauge curvature couples to the boundary measure — the seam F -normalisation. So “the actual seam $a_4 = 8b_1c_3^6$ ” and “the seam F -normalisation” are *one* [C] obligation, not two.
 - [E] **residual (2) located (not closed)**: the determinant-line variation factors $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$; a^2 is the Gilkey gauge-curvature (Quillen/Chern) density (v342), so the leading $a^3 = a \cdot a^2$ is the *first moment* of the Chern density — a topological level at boundary order c_3^0 , not a Seeley–DeWitt curvature integral. Its from-first-principles (Chern–Simons-type) origin stays [O].
 - [E] **net reduction**: the EM-Ward residual is *one* [C] (residuals 1 & 3) + *one* [O] (residual 2), not three independent problems — a genuine narrowing of the open surface. [O] ALPHA.QUILLEN.EXACT.01 is *not* closed (the α^3 Chern level and the from-first-principles proof stay external, type A, v384).
- **Surfaces updated**. Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.03); cited in the `tfpt_1` EM-closure subsection beside `v433`; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-27 (v433 — a second honest step on the α^{-1} Quillen target: a solvable 4D model reaches the a_4 order, connecting v391 and v342)

- **New module v433 (ALPHA.QUILLEN.PROGRESS.02)** — a second honest step on the external target ALPHA.QUILLEN.EXACT.01 (**v382**); it *grounds and connects*, it does *not* close it. The target stays [O] (external math, type A, v384); the value $\alpha^{-1} = 137.0359992$ stays [E]. The earlier honest attempt v391 gave a solvable model on a 1D circle whose ζ -determinant variation reached only the low Seeley–DeWitt orders a_0, a_1 ; separately, v342 grounded the quadratic Calderón term in the Gilkey gauge-curvature coefficient a_4 ($30/360 = \frac{1}{12}, \Omega^2/12$). This module supplies the missing link between the two.
 - [E] **a solvable 4D model reaches the a_4 order**: the flat operator $\Delta = -\partial^2 + m^2$ on a 4-torus of volume V has heat trace $\text{Tr } e^{-t\Delta} = V/(4\pi)^2 t^{-2} e^{-tm^2}$, with Gilkey coefficients $a_0 = V/(4\pi)^2$, $a_2 = -Vm^2/(4\pi)^2$ and $a_4 = Vm^4/(2(4\pi)^2)$; the t^0 (conformal-anomaly) order a_4 is *non-zero* — exactly the order v391’s 1D circle toy could not reach.

- **[E] it is the order v342 uses:** so v391 (“a ζ -determinant variation *is* closed heat-kernel data”) and v342 (“the seam term *is* the Gilkey a_4 ”) are unified — the variation is heat-kernel data *at the a_4 order*, not only the low orders v391 reached.
- **[E] the measure is c_3 -built:** the universal 4D heat-kernel measure $(4\pi)^{-2} = 1/(16\pi^2) = (2c_3)^2 = 4c_3^2$ in the seam’s one-sided 2D-boundary normalisation $c_3 = \frac{1}{2}(4\pi)^{-1}$; and the v382 split is re-verified at $\alpha^{-1} = 137.0359992$.
- **[O] not closed:** computing the *actual* seam $U(1)$ ζ -determinant a_4 coefficient ($= 8b_1c_3^6$) on the μ_4 seam moduli, and the origin of the cubic α^3 Maxwell/Chern moment, stay open (external math, type A, v384); the exact seam F -normalisation fixing the coefficient *value* stays **[C]** (v342). A second honest step that grounds and connects, it does not close.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.02); cited in the `tfpt_1` EM-closure subsection beside v382/v391; mirrored on the website (the residual-gap lab). Python-only (like v391); no new physical number.

2026-06-27 (v431, v432 — the “unmapped” E_8 degrees are forced spine+flavor structure; the unconditional over-determination floor)

- **New module v431 (E8.DEGREE.LADDER.01) — the reverse-audit “unmapped” E_8 Casimir degrees are NOT diffuse overhead but a forced two-family decomposition.** The structural complement of the reverse audit (v354/v355) and its sheet/deck complement (v430): $\deg(E_8) = 6 \cdot \text{spine}\{2, 3, 4, 5\} \sqcup (\{2\} \cup \text{det-ladder}\{8, 14, 20\})$. **[E]** the two-family split is *forced* by $h(E_8) = 30 = 2 \cdot 3 \cdot 5$ — the exponents are the $\varphi(30) = 8$ totatives of 30, hence coprime to 6, hence $\equiv \pm 1 \pmod 6$, so the degrees occupy only the residue classes $\{0, 2\} \pmod 6$. **[E]** the $6k$ family $\{12, 18, 24, 30\}/6 = \{2, 3, 4, 5\}$ is the v91 spine; the $6k+2$ family $\{8, 14, 20\} = (\det R, \det C, \det L)$ is the winding line $\det M(s, 0) = 6s + 8$ of the flavor diamond (v135). **[E]** “ $18 = 6N_{\text{fam}}$ ” is not a holdout (spine family), and the clean split is special to E_8 among the simply-laced exceptionals ($E_6/E_7/D_5$ fail). **[C] honest v355 line:** the arithmetic decomposition is exact, but the *functorial* claim “ $6k+2$ Casimir stratum = flavor sheet operators” needs a representation-theoretic map and stays **[C]** (12, 24 admit two canonical readings each) — a structure theorem, not a forced functorial flavor map. It reclassifies the v354 overhead to zero orphan degrees and closes no gate.
- **New module v432 (OVERDET.FLOOR.01) — the unconditional improbability floor and the $L_1 \leftrightarrow L_2$ bridge.** The complement of the conditional null model (v100) and the multiply-vs-compress framework (v427/v428): the defensible over-determination number that survives even if a skeptic *rejects* the declared formula grammar. **[E]** compression removed (v428), the genuinely multiplicative pieces are the input “8” forced four independent ways plus the foreign witness $\alpha^{-1} \sim 137$. **[C]** multiplying only across disjoint pieces (input over-determination $\times$ the α census $1/94500 \times$ the φ_0^{ret} -seed cluster counted once $\times$ one downstream witness) gives a floor $\leq 10^{-6}$ across a grid of generous assignments, nominal $\sim 10^{-10}$. **[E]** that floor sits ~ 20 orders *above* the v100 conditional $10^{-30.7}$; the gap is exactly the evidential value of proving the forms forced, so the α form-derivation (ALPHA.QUILLEN.EXACT.01, v382) and this floor are the *same* lever. **[C] honest scope:** bounds only the “is it numerology?” question, not the physics bridge (firewall v187); a conservative floor, not a posterior.
- *Surfaces synced in the same change:* v431/v432 added to `run_all.py`, `script_registry.csv` (master index + website `ScriptIndex`), `status_ledger.csv`, `tfpt_5_redteam` (reverse-audit winbox + Casimir-degree figure) and `tfpt_safeguards` (the over-determination layer); v431 mirrored in `wolfram/tfpt_readouts_extension.wl` (v432 is a conservative bound, Python-only).

2026-06-27 (v430 — the seam’s “other side” is forced-disjoint from E_8 ’s unmapped Casimir degrees)

- **New module v430 (E8.OTHERSIDE.AUDIT.01) — the sheet/deck complement of the reverse audit (v354/v355).** The reverse audit found five E_8 Casimir degrees $\{12, 14, 18, 20, 24\}$ with no primary readout (“hull overhead”); the double cover has an *other side* (the one-sided $\mathbb{Z}_2$ collar / conjugate half-spinor S^-), so the sharp adversarial follow-up is “*is the leftover where the second sheet lives?*” The answer is a disciplined **negative**. [E] the deck is the degree-2 invariant: the one-sided cover contributes $|\mathbb{Z}_2| = 2$ (the $\frac{1}{2}$ in $c_3 = 1/(|\mathbb{Z}_2| 4\pi) = 1/(8\pi)$, v58/v216), and $2 = \min \deg(E_8)$ is the quadratic/metric — one of the *matched* primary readouts $\{2, 8, 30\}$, never one of the unmapped degrees. [E] the two sheets are the 128-spinor: $\dim S^+ = \dim S^- = 16$, the spinor block $128 = \text{rank } E_8 \cdot \dim S^+ = 2^{\text{rank}-1} = \sum \deg$, the spinor half of $248 = 120 + 128$; S^- enters 248 as the conjugate $(\overline{16}, \overline{4})$ ($128 = 2 \cdot 64$), not a spare singlet. [E] forced-disjoint: the sheet/deck set $\{2, 16, 32, 128\}$ has empty intersection with $\{12, 14, 18, 20, 24\}$, its only degree contact the matched 2. [O] each unmapped degree, “explained” from sheet atoms, needs an unforced coefficient and admits ≥ 2 readings (the v355 discriminator), so there is no forced sheet $\rightarrow$ unmapped-degree identity.
- **A structural negative, not a status change.** It consolidates the S^- -dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier E_8 scan (tfpt_4_frontier): the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live “other side” question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five. Closes no gate, upgrades nothing.
- **Surfaces synced.** run_all.py, script_registry.csv, status_ledger.csv; the reverse-audit Outcome winbox + degree-audit table caption and the Target D winbox of tfpt_5_redteam, the magnitude/phase keybox of tfpt_3_e8_audit_bootstrap, the WIMP-no-go passage of tfpt_4_frontier; the E8 node of the website verification DAG; the Wolfram extension/README/ledger counts and the website verification count; manifests refreshed.

2026-06-26 (v429 — the ‘unmapped’ golden E_8 structure is the geometry of the one external input θ_i)

- **New module v429 (DM.AXION.PENTAGON.01) — a geometric re-reading answering “why does so much idle E_8 structure exist?” for the one case where the idle structure has a job.** The reverse audit (v354) had argued the golden ratio φ is *unmapped* (it appears only internally — the affine- E_8 attractor angle v312, the icosian lattice v348 — and in no physical readout), and used exactly that to conclude φ cannot be numerology. [E] this sharpens that statement: because $N_{\text{fam}} = g_{\text{car}} - 2$, the axion misalignment *spine* angle (v211, DM.AXION.SPINE.01) is $\theta_i = \pi N_{\text{fam}} / g_{\text{car}} = (g_{\text{car}} - 2)\pi / g_{\text{car}}$, which is *exactly* the interior angle of the regular g_{car} -gon — for $g_{\text{car}} = 5$ the pentagon, $\theta_i = 3\pi/5 = 108^\circ$. [E] its cosine is golden, $\cos(3\pi/5) = (1 - \sqrt{5})/4 = -1/(2\varphi)$ (partner $2\cos(2\pi/5) = 1/\varphi = \varphi - 1$, $\varphi = 2\cos(\pi/5)$). [E] the golden character is *unique* to $g_{\text{car}} = 5$: among the small regular n -gons only $n = 5$ has an irrational interior-angle cosine ($n = 3, 4, 6$ give $1/2, 0, -1/2$), so θ_i is golden *because* the carrier is 5 (P2).
- **An honest [C] bridge, not a status change.** It connects two standing facts — φ is the unmapped $g_{\text{car}} = 5$ signature (v354/v313) and $\theta_i = 3\pi/5$ is the one external cosmological input (v211) — and *refines* v354 from “ φ in no physical readout” (it is the unmapped $g_{\text{car}} = 5$ signature, v313) to “ φ touches *only* the [C] spine misalignment angle” (conditional, branch-level, never a frozen prediction). It is a geometric motive for the spine over the hilltop $\theta_i \approx 170^\circ$ (whose cosine is not a clean pentagon value), consistent with — not replacing — the

finite- T solver that decides the branch (v185/v211). It does *not* derive θ_i , does *not* upgrade DM.AXION.SPINE.01 (stays [C]) and closes no gate.

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the dark-matter section of `tfpt_4_frontier`; the Wolfram extension/README/ledger counts (GATE.WOLFRAM.02) and the website verification count; manifests refreshed.

2026-06-25 (External-works references appendix — web-verified)

- **New shared appendix `tex-artefacts/tfpt_references.tex`** (“External works built upon”), \input into the six math-heavy documents (`tfpt_1`, `tfpt_2`, `tfpt_4`, `tfpt_horizon_readouts`, `origin_theory`, `tfpt_research_contracts`). The active papers named the foundational results they build on by author only, with no reference list; the appendix supplies web-verified locators (journal, volume, year / arXiv) grouped by topic: modular theory & AQFT (Tomita–Takesaki LNM 128 1970; Bisognano–Wichmann J. Math. Phys. 16/17; Osterwalder–Schrader CMP 31/42; Doplicher–Haag–Roberts CMP 23/35; Haag–Ruelle; Longo CMP 126/130; Kawahigashi–Longo–Müger CMP 219), conformal nets/VOAs/lattices (Frenkel–Kac Invent. Math. 62; Segal CMP 80; MMST; Adamo–Moriwaki–Tanimoto; Adamo–Giorgetti–Tanimoto 2506.01008/2508.07109; Naaijken–Penneys–Wallick), lattices/quadratic forms (Conway–Sloane; Siegel Ann. Math. 36 1935), singularities (Brieskorn Invent. Math. 2 1966; Milnor AMS 61 1968) and spectral geometry/gravity/thermal time (Chamseddine–Connes CMP 186; Connes–Rovelli CQG 11; Wald PRD 48; Casini–Huerta–Myers JHEP 05 2011). Added to the `make_manifest.py` TEX list. No claim or status change — a provenance/citation completeness pass.

2026-06-25 (External AQFT citations + the four-fold μ_4 identity sharpened)

- **Two recent Adamo–Giorgetti–Tanimoto theorems cited where the keystone builds on them.** The open SEAM.EQUIV.01 continuum/realisation legs (v336) now cite, alongside MMST and Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, already in use): **(a)** the construction and classification of two-dimensional conformal nets from even lattices (arXiv:2506.01008, 2025) — “the lattice is a conformal net” is a theorem of the same operator-algebraic lineage, strengthening the lattice-realisation leg; and **(b)** automatic positive-energy/diffeomorphism covariance of conformal-net representations (arXiv:2508.07109, 2025), which defuses the “Bisognano–Wichmann presupposes the covariance it should produce” circularity (v215) on the intrinsic-BW residual (v424). Added to v336 docstring/registry, the SEAM.EQUIV.CONTINUUM.01 ledger `external_data`, and `tfpt_research_contracts`.
- **The four-fold μ_4 identity made explicit (v422).** v422 already proved that the seam deck/divisor μ_4 , the carrier Galois group $\text{Gal}(\mathbb{Q}(\zeta_5))$, the discriminant $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}/4$ and the $(E_8)_1$ simple-current glue are ONE cyclic $\mathbb{Z}/4$ (with the cyclic-vs-Klein negative control). It is now named the *four-fold μ_4 identity* and typed, per v428, as a *compression* (one object read four ways), NOT four independent witnesses – in v422, the registry, the SEAM.GALOIS.NET.01 ledger row and `origin_theory`. (No new module: a separate v429 would have duplicated v422 and violated the very compression discipline.)

2026-06-25 (v428 — honest self-correction of the over-determination map)

- **v428 (OVERDET.WITNESS.RECLASS.01) — applying v427’s own multiply-vs-compress test to v427 itself.** A pattern search and the right objection exposed that calling the seven arithmetic witnesses “disjoint grammars that *multiply*” overstates independence. [E] by the Brieskorn classification (v236) the $(2, 3, 5)$ singularity is the ONE generator behind the order-30 clock (Milnor $\mu = (2-1)(3-1)(5-1) = 8 = \text{rank } E_8$, $30 = 2 \cdot 3 \cdot 5 = h(E_8)$, $\varphi(30) = 8$), so each witness is a facet of that same object (primes 2, 3, 5 of 30, the clock 30, or E_8): all

seven collapse to one generator $\Rightarrow$ *compression*, not seven multiplications. [E] what genuinely multiplies is the input forced four independent ways — the “8” in $c_3=1/(8\pi)$ from rank E_8 , $h(D_5)=2(5-1)$, $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}\approx 137$. [C] refines and partly walks back v427, consistent with v236/v305/v313.

- **Surfaces corrected (same change).** v427’s prose, the ledger row (OVERDET.WITNESS.MAP.01, now noting the refinement), tfpt_safeguards Layer 3 (the seven compress one object; a second keybox states what multiplies), the witness-map figure (redrawn), the website (papers.ts, Safeguards.tsx, release.ts, the intro-video description) and the Zenodo description. The over-determination claim is now the narrower, harder-to-attack “one richly over-determined input + a foreign readout”.

2026-06-25 (Safeguards visuals + paper expansion + the website safeguards band)

- **Five safeguards figures** (make_figures.py, in manifest.sha256): the layered defense, the α^{-1} uniqueness scan (one $F_{U(1)}$ variant of 94,500 in the CODATA window), the reverse audit (3/8 vs 5/8), the witness-independence map (seven disjoint grammars), and the look-elsewhere null model (13/13 vs $\leq 5/13$).
- **tfpt_safeguards expanded.** Layer 5 now states the null model explicitly ($\prod p_i=10^{-25.8}$, $2\times 200k$ pseudo-theories $\leq 5/13$, $\varphi_0^{\text{ret}} \pm 10\% \rightarrow 6/13$, shuffle 1.2) and the α uniqueness ($\leq 10^{-30.7}$, ~ 102 bits, v100), with the data watchdog (v307) and kill board (v321); the five figures are embedded.
- **Website: a prominent animated Safeguards band** (components/Safeguards.tsx) in the home flow (after the trust beat) — headline statistics, the seven layers, the figures, links to the paper and the suite — to address the coincidence/numerology suspicion proactively.
- **Koide flow-time investigation (honest negative).** Post-v425: the flow-time is *not* native and does *not* reduce to one scale — v99 already rejects a scale reading as look-elsewhere; the only theory-side content is the conditional integer reading $t=N_{\text{fam}}=3 \Rightarrow m_\tau=1776.94$ MeV, a Belle II kill test. No new module.

2026-06-25 (Three closure-strategy modules v425–v427 + the Safeguards paper)

- **v425 (DYN.TRANSFER.UNIVERSAL.01) — the single-flow reduction of F_{transfer} .** The four frontier transfers do not each need a separate external solver: their *dynamics* is the ONE native flow (the seam modular/recovery semigroup, v238/v240, gap $6\ln(3/2)$) restricted to each sector. [E] the v99 Koide generator linearised at $q=2$ has rate $-\Delta$, time-1 contraction $(2/3)^6$; Boltzmann washout is that contraction (CP phase $4\pi/3$ native, v320); QCD rate $b_3=-7$. [C] only the anchors (v_{geo} , C_p , M_R) stay external; F_{relic} ’s cosmological IC θ_i is the one wall. A reduction (extends v383 static $\rightarrow$ dynamic), not internalisation; the firewall holds.
- **v426 (SEAM.EQUIV.LTORP.KMS.01) — the invertible $\beta=1$ -KMS variant of (LTO-RP).** The corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + KMS, not a trace). [E] a gapped reflection-symmetric μ_4 -even collar at $\beta=1$ KMS satisfies $\Theta K \Theta^{-1} = -K$ ($u_\Theta = J$) and $[\rho, K]=0$, so (LTO-RP) holds by direct Tomita–Takesaki; $|\det \text{Cartan}(E_8)|=1$ (no anyons), state not a trace; neg controls have teeth. Advances v424 sub-step (ii); SEAM.EQUIV.01 stays [O] (continuum sub-step (i) remains).
- **v427 (OVERDET.WITNESS.MAP.01) — the honest over-determination accounting.** A reviewer correction made machine-explicit: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. [E] seven disjoint-grammar witnesses each reproduce a carrier-skeleton integer from their own structure — Gauss $N(3+2i)=13$, Eisenstein $N(3+2\omega)=7$, cyclotomy $N(3+2\zeta_5)=55$, Galois $|(\mathbb{Z}/5)^\times|=4$, $|\det \text{Cartan}(E_8)|=1$, Pascal $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}=11$ ($2^4=16$), Coxeter $h(E_8)=30=2\cdot 3\cdot 5$ with $\varphi(30)=8$ — and the anchor $a=(1, 1, 2)$ is the compression generator. Extends v305.

- **New paper `tfpt_safeguards` (companion S).** A dedicated methodology paper collecting, in one language, every mechanism by which TFPT defends a claim against coincidence and numerology: the status calculus + ledger + audit, no-free-pattern + reverse audit, the over-determination map (`v427`), the firewall + No-Unit theorem, frozen predictions + null model, the independent Wolfram and Lean paths, and the red team — with a final kill-test summary. Registered in `build.sh`, `make_docs_map.py`, `make_manifest.py` and the document-set box; the document set is now **eleven** deliverables (five numbered papers + four companions + introduction + changelog).
- **Surfaces synced.** `run_all.py` (421 modules), the registry, the ledger, `tfpt_research_contracts` (the F_{transfer} and keystone sections), `tfpt_safeguards`; manifests refreshed.

2026-06-25 (The (LTO-RP) reduction of the keystone `SEAM.EQUIV.01: v424`)

- **New module `v424` (`SEAM.EQUIV.LTORP.01`).** An honest MMST-style reduction of the one open keystone — `SEAM.EQUIV.01`'s intrinsic Bisognano–Wichmann lemma (the `v308/v309` residual) — to the recent commuting-projector theorem of Naaijken–Penneys–Wallick (`arXiv:2605.10693`, [NPW26]), with the central mapping *corrected*. [E] the μ_4 clock is a modular *symmetry* ($[\rho, K]=0$, `v309/v199`), not the flow — $\sigma^\psi=\rho$ is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26's axiom (LTO-RP) means $u_\Theta=J$ (reflection = modular conjugation) = the intrinsic BW condition $\theta K \theta = -K$, *distinct* from $\omega \circ \rho = \omega$: a positive character-block-diagonal K has $[\rho, K]=0$ yet never $\theta K \theta = -K$, so clock-invariance does *not* imply (LTO-RP) (refuting that identification). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code $|\det K|=4$ via a trace; Levin–Wen anyonic), while the seam is invertible ($|\det \text{Cartan}(E_8)|=1$, no anyons) and a $\beta=1$ KMS state (not a trace) — in neither proved bucket. [C] verdict: the BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$ -reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, *not* a closure; `SEAM.EQUIV.01` stays [O]. Cited in `tfpt_research_contracts`; Python-only (numpy; the det discriminators mirrored via `v89/v281`).
- **Validation provenance.** The three pivotal references of the closure-strategy note were independently verified real and correctly described: NPW26 (`arXiv:2605.10693`), the RP–Yang–Mills complete-monotonicity reconstruction (Int. J. Geom. Meth. Mod. Phys. 2026), and Marolf (`arXiv:2203.07421`).
- **Surfaces synced.** `run_all.py`, the script registry, the status ledger and the `tfpt_research_contracts` keystone section; manifests refreshed.

2026-06-25 (No-Ambient-Dependency: no frozen readout is computed from `QG.AMB.01: v423`)

- **New module `v423` (`QG.AMB.NODEP.01`).** Sharpens the `v369/v384` prose redundancy (“`QG.AMB.01` is a [C] redundancy, not dynamics”) into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin $\Delta_{\text{eff}} = 6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$ ($31 = 2^{g_{\text{car}}} - 1$), susceptibility $\chi = 729/665$ finite (negative control gap $\rightarrow 0 \Rightarrow \chi \rightarrow \infty$); [E] `QG.AMB.01` is a certification *sink* (every claim that lists it is a gravity-gate/QG row, never a frozen prediction) and *not a value source* (no producing script, while α^{-1} , the mixing angles and cosmology are each *computed* by a compiler script, `v3/v9/v268/v7`); [E] the one transitive link is architectural — of the frozen readouts only θ_{13} references it, via the 4d-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`), a coherence edge, not a numerical input. [C] verdict: `QG.AMB.01` is gap-decoupled certification — `v369` as a DAG/script fact; honest scope — this does *not* discharge `SEAM.EQUIV.01`, on which the readouts still depend. Cited in `tfpt_4`; Python-only (gap algebra mirrored via `v337`).

- **Paper sharpening (seam selector, up front).** The `tfpt_4` quantum-gravity keybox now states explicitly that the open `SEAM.EQUIV.01` premise is the *holomorphic* $|\det K|=1$ point that selects $(E_8)_1$ — the central charge $c=8$ alone does *not*, being shared by the non-holomorphic $SO(16)_1$ counterexample (v277/v281).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `tfpt_4`; manifests refreshed.

2026-06-25 (The Galois $\leftrightarrow$ Net bridge: v422)

- **New module v422 (SEAM.GALOIS.NET.01).** The seam $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$ is the SAME cyclic $\mathbb{Z}/4$ as the $(E_8)_1$ simple-current glue — not a mere order-4 coincidence — bridging the Galois gearbox of v419 to $G_{\text{net}}/\text{SEAM.EQUIV.01}$. [E] $\text{disc}(A_3) = \text{disc}(D_5) = \mathbb{Z}/4$ (one Smith invariant factor $4 = \text{cyclic, det } 4$); D_n disc is $\mathbb{Z}/4$ for n odd, $\mathbb{Z}_2 \times \mathbb{Z}_2$ for n even, so the carrier D_5 (rank $5 = g_{\text{car}}$, odd) is cyclic while D_8 (rank 8, even) is Klein; [E] $(\mathbb{Z}/5)^\times = \langle 2 \rangle$ is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in $\mathbb{Z}_4 \times \mathbb{Z}_4$ ($16 = \dim S^+$) the glue $(1, 1)$ is order 4 with Lagrangian quotient $16/4^2 = 1 = (E_8)_1$ (v89/v154); [E] negative control: the Klein / order-2 $\langle (2, 2) \rangle$ gives $16/2^2 = 4 = |\text{disc}(D_8)| = SO(16)$ (det 4), NOT E_8 (det 1) — cyclic $\mathbb{Z}/4 \rightarrow E_8$, Klein $\rightarrow D_8$. [C] the bridge: one cyclic $\mathbb{Z}/4$ threads the seam deck μ_4 (v419), $\text{Gal}(\mathbb{Q}(\zeta_5))$ and the $(E_8)_1$ glue — the Galois gearbox $(\mathbb{Z}/30)^\times$ IS the holomorphic net's glue, cyclicity forced by 5, Klein excluded. Cited in `origin_theory`; Wolfram-mirrored ($324 \rightarrow 327$).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `origin_theory`; the Wolfram extension/README/ledger counts (327/327) and the website verification count; manifests refreshed.

2026-06-25 (F_pole external: the Koide source $\rightarrow$ pole transfer is one-loop EM: v420)

- **New module v420 (FR.POLE.QED.01).** The honest external-physics follow-up to v93 and the clock No-Go: the Koide source $\rightarrow$ pole transfer is a one-loop *electromagnetic* effect, not an algebraic step. [E] Koide Q is rescaling-invariant ($Q(cm) = Q(m)$), so flavor-universal running leaves it fixed and the source $\rightarrow$ pole shift is purely flavor-split; [C] the tree deficit $\frac{2}{3} - Q_{\text{src}} = 0.00220$ is one-loop-EM-sized ($= \varphi_0/24$ to 0.6%, $= \alpha/\pi$ to $\sim 5\%$), the PDG pole sits at $2/3$ (the IR attractor); [C] a 1-loop QED pole $\leftrightarrow \overline{\text{MS}}$ per-lepton-log estimate shifts Q by EM-loop order toward $2/3$ (the v82 direction); [O] the exact flow time $t \approx 2.84$ (v93) is scheme/scale-dependent external QED, only sized. [C] verdict: F_{pole} is a one-loop EM effect — the lens forces the rate $(2/3)^6$ (v327), external EM supplies the flow; `F_pole` stays [C], firewalled (v187), so F_{transfer} is confirmed genuinely external — the open theme sharpened (QED), not closed. Cited in `tfpt_4`; numerical (mpmath), Python-only (no Wolfram check, by the suite convention like v93/v371).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Koide section of `tfpt_4`; manifests refreshed.

2026-06-25 (The seam μ_4 is the carrier's Galois group: v419 + figure)

- **New module v419 (SEAM.GALOIS.01).** A positive resolution of the cyclic/Galois asymmetry (v409, `RES.COXETER.SYMMETRY.01`): why the seam μ_4 sits on the Galois side, not on the cyclic ζ_{30} dial. [E] $h(E_8) = 30$ is squarefree, so $\mathbb{Z}/30$ has element orders $\{1, 2, 3, 5, 6, 10, 15, 30\}$ — *no* order 4, so the seam μ_4 cannot be a rotation hand and is forced onto the Galois side; [E] by CRT $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$ (order $\varphi(30) = 8 = \text{rank } E_8$), and the μ_4 $(\mathbb{Z}/4)$ factor IS $(\mathbb{Z}/5)^\times$ (the carrier prime 5), the $\mathbb{Z}_2$ IS $(\mathbb{Z}/3)^\times$ (family 3 = CP conjugation, v316); [E] so the seam $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$, realised on the carrier clock C_5

(v418) by the Frobenius $\zeta_5 \mapsto \zeta_5^2$ — the explicit order-4 operator G with $GC_5G^{-1} = C_5^2$, $G^4 = I$, $G^2C_5G^{-2} = C_5^{-1}$. [C] this is the positive content of v409: the $2^2=4$ is the order of $(\mathbb{Z}/5)^\times$, and “no contraction rate” is because the seam is an *automorphism*, not a hand; so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (cyclic $h=30$, Galois rank=8). Cited in `origin_theory`; Wolfram-mirrored (321 $\rightarrow$ 324).

- **New figure `coxeter_galois.pdf`.** The order-30 cyclic dial with the four flavor hands (sheet/family/carrier/CP) beside the Galois gears ($\mu_4 = \text{Aut}(\zeta_5)$ the seam, $\mathbb{Z}_2 = \text{Aut}(\zeta_3)$ the CP conjugation); added to `make_figures.py`, the manifest figure list and `origin_theory`.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Wolfram extension/README/ledger counts (324/324) and the website verification count; manifests refreshed.

2026-06-25 (The cyclotomic norm triple + the carrier-5 clock: v418)

- **New module v418 (ARITH.NORMTRIPLE.01).** The carrier split (3, 2) read in the three atom-rings gives (7, 13, 55) = (scalaron, Δ_Q , quark numerator), and the carrier-5 clock missing from v417 is found as the 4×4 Φ_5 -companion C_5 . [E] $C_5^5 = I$, $\text{tr } C_5 = -1$, $\det C_5 = 1$ (the four primitive 5th roots), with the golden ratio as its real-part data ($2 \cos 72^\circ = 1/\varphi$, $2 \cos 144^\circ = -\varphi$; output-side, v313/v349; dim 4, not 3 — this refines v417’s gap from “no operator” to “no 3×3 operator”); [E] $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5) = \det(3I+2C_5) = 55 = 5 \cdot 11 = g_{\text{car}} \| \text{Pl}(K) \|_1$, the c_u/c_d numerator (v410/v411); [E] the triple $\det(3I+2 \text{Comp}(\Phi_n)) = (7, 13, 55)$ for $n=3, 4, 5$ over $\mathbb{Z}[\omega]$, $\mathbb{Z}[i]$, $\mathbb{Z}[\zeta_5]$ (atoms 3, 2, 5, v416); [E] NEG control: the anchor (5, 4) $\rightarrow$ (21, 41, 461) reaches named values only in the ω - and i -rings ($21 = N_\omega(5, 4)$, $41 = 10b_1$, v222) but 461 (prime) in the carrier ring, so the carrier ring separates (3, 2) $\rightarrow$ 55 from (5, 4) $\rightarrow$ 461. [C] honest: (3, 2) is the *forced* carrier split (v14), not the unique clean one — a scan finds $\{(2, 1) \rightarrow (3, 5, 11), (3, 2) \rightarrow (7, 13, 55)\}$, a small ladder. The sharper discriminator is *cross-sector*: only (3, 2) spans three physics sectors (gravity 7, flavor 13, masses 55), the unique cross-sector seed, while (2, 1) is all compiler-core. Cited in `origin_theory`; Wolfram-mirrored (318 $\rightarrow$ 321).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, a triple paragraph in `origin_theory`, the Wolfram extension/README/ledger counts (321/321) and the website verification count; manifests refreshed.

2026-06-25 (The Eisenstein/CP operator + the order-30 clock: v417)

- **New module v417 (DIAMOND.EISEN.01).** The hexagonal dual of v415: the $\tau=\omega$ flavor side put on operators and unified with the seam into one order-30 Coxeter clock. [E] the family rotation $P = (1\ 2\ 3)$ is the order-3 Eisenstein deck ($P^2+P+I = \text{ONES}$, so $\omega^2+\omega+1=0$ on the plane $\perp (1, 1, 1)$), the order-3 dual of $J^2 = -I$; [E] it realises the hex CM norm as $(3I+2P)(3I+2P^2) = 7I+6 \text{ONES}$, $\text{Spec}\{7, 7, 25\}$ ($7 = N_{\mathbb{Z}[\omega]}(3+2\omega) = \text{scalaron}$, $25 = g_{\text{car}}^2$) — the dual of v415’s $\{13, 13, 9\}$; [E] the CP clock $W = -P^2$ (ρ of v233) has $W^2 = P$, $W^3 = -I$, $\text{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$, giving $\delta_{\text{CKM,lead}} = \arg \zeta_6 = \pi/3$ and $\delta_{\text{PMNS}} = \arg \zeta_6^4 = 4\pi/3$ (v316/v320); [E] all flavor clocks are powers of ζ_{30} ($\zeta_{30}^{15} = -1$, $\zeta_{30}^{10} = \omega$, $\zeta_{30}^6 = \zeta_5$, $\zeta_{30}^5 = \zeta_6$) while the seam $\mu_4 = i$ is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30) = 8 = \text{rank } E_8$ — so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants ($h=30$ and rank=8). [C] honest gap: the carrier μ_5 has no 3×3 operator (the golden φ is an output-side trace, v313/v349); negatives: no U, V -diamond operator carries ζ_6 , $[D, P] \neq 0$ (no μ_{12}), $N(3+2\zeta_6) = 19 \neq 7$. Cited in `tfpt_2`; Wolfram-mirrored (315 $\rightarrow$ 318).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Sheet-Diamond keybox of `tfpt_2`, the Wolfram extension/README/ledger counts (318/318) and the website verification count; manifests refreshed.

2026-06-25 (The Gaussian operator + the atom trichotomy: v415, v416)

- **New module v415 (DIAMOND.GAUSS.01).** The square CM-norm dictionary of v222/v230 is *operator-realised*, and the μ_4 deck becomes an intrinsic diamond object. [E] the integer commutator $J := \frac{1}{3}[U, V] = \begin{pmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix}$ has $\text{Spec}(J) = \{i, -i, 0\}$, $J^2 = -I$ on the rank-2 image and $J^3 + J = 0$ — a μ_4 quarter-turn born from the binary V and ternary U compilers (v410/v95); [E] $3I + 2J$ and $5I + 4J$ have spectra $\{3+2i, 3-2i, 3\}$, $\{5+4i, 5-4i, 5\}$, so the Gaussian integers $3+2i$ (norm $13=\Delta_Q$) and $5+4i$ (norm $41=10b_1$) are *literal eigenvalues*; [E] $(aI+bJ)(aI-bJ)$ reads (a^2+b^2) on the μ_4 -plane and the real part² on the kernel ($\{13, 13, 9\}$, $\{41, 41, 25\}$); [E] the intrinsic order-4 deck $D = -I + J - J^2$ ($D^4 = I$, $\chi_D = (x+1)(x^2+1)$) has χ_2 -line $\ker[U, V] = a - 1$. [C] the geometric-deck identification sharpens (does not close) the v140 $\mathbb{Z}_3$ residue. Cited in `tfpt_2`; Wolfram-mirrored.
- **New module v416 (ARITH.TRICHOTOMY.01).** The atoms $\{2, 3, 5\}$ are ramified/inert/split in the two exceptional CM rings. [E] in $\mathbb{Z}[i]$ (square/seam) $2=|\mathbb{Z}_2|$ ramifies, $3=N_{\text{fam}}$ is inert, $5=g_{\text{car}}$ splits as $(2+i)(2-i)$; [E] in $\mathbb{Z}[\omega]$ (hex/flavor) $3=N_{\text{fam}}$ ramifies, $2, 5$ are inert; [E] the ramified prime of each ring is its own atom; [E] the three quadratic facets $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{5})$ of $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ (v403) ramify over exactly one atom each, product $2 \cdot 3 \cdot 5 = 30 = h(E_8)$, two imaginary (CM/dynamic) + one real (RM/static). [C] the seam/flavor/golden reading. Cited in `origin_theory`; Wolfram-mirrored.
- **Wolfram extension 308 $\rightarrow$ 315 + a pre-existing fix.** Added the seven exact checks for v415/v416 and corrected a one-character sign bug in the v412 `CharacteristicPolynomial` check (Wolfram returns $-\chi$ for odd rank), so `tfpt_readouts_extension.wl` now runs genuinely clean at 315/315 (GATE.WOLFRAM.02); README and ledger counts updated.
- **Surfaces synced.** `run_all.py`, `script_registry.csv` and `status_ledger.csv` updated; a new keybox paragraph in the Sheet-Diamond section of `tfpt_2` and a trichotomy sentence in `origin_theory`; manifests refreshed.

2026-06-24 (The sheet generator V as a binary internal compiler: v410–v414)

- **New module v410 (SHEET.GEN.BINARY.01).** The rank-2 sheet axis $V = Q \text{diag}(0, 1, 1)$ of the centered cross (v95/v218) is a binary internal compiler: its powers print the carrier spine. [E] the closed form $V^n = \begin{pmatrix} 0 & 2^{n-1} & 0 \\ 0 & 2^n & 0 \\ 0 & 2^{n+1}-2 & 1 \end{pmatrix}$ (proved by the inductive step $F(n)V=F(n+1)$), so $V^n \mathbf{1} = (2^{n-1}, 2^n, 2^{n+1}-1) = (1, 2, 3), (2, 4, 7), (4, 8, 15), (8, 16, 31)$; [E] the four bilinear families $\mathbf{1}^T V^n \mathbf{1} = 7 \cdot 2^{n-1} - 1$, $\mathbf{1}^T V^n a = 7 \cdot 2^{n-1}$, $a^T V^n a = 11 \cdot 2^{n-1}$, $a^T V^n \mathbf{1} = 11 \cdot 2^{n-1} - 2$ give $(6, 7, 9, 11)$ at $n=1$ and $\mathbf{1}^T V^2 \mathbf{1}=13=\Delta_Q$, $\mathbf{1}^T V^3 \mathbf{1}=27=\mathbf{1}^T R a$, $\mathbf{1}^T V^4 \mathbf{1}=55$, $\mathbf{1}^T V^4 a=56=\dim \mathbf{56}_{E_7}$; [C] the binary spine is FORCED by $\text{Spec}(V)=\{0, 1, 2\}$, so the carrier/Lie matches (rank $E_8=8$, $\dim S^+=16$, $248/8=31$) and the theta cross-link ($\sigma_3(2)=9=a^T V \mathbf{1}$, $\sigma_3(3)=28=\mathbf{1}^T V^3 a=\det(I+R)$, $\sigma_3(5)=126$) are audit-typed (the v95 G_2 -center typing).
- **New module v411 (FLAV.UD.VPOWER.01).** The quark ratio is a pure V -power readout, $c_u/c_d = (\mathbf{1}^T V^4 \mathbf{1})/((a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})) = 55/(9 \cdot 13) = 55/117$, identical to the established $g_{\text{car}} \|\text{Pl}(K)\|_1/(N_{\text{fam}}^2 \Delta_Q) = 5 \cdot 11/(9 \cdot 13)$ (v94). [E] the identity; [C] an exact *re-encoding* (no new physical input) — the value and the u/d identification stay conditional, coupled to the (U_{wall}) readout rigidity.
- **New module v412 (DIAMOND.SHEET.SOURCE.J.01).** Names the operator on the $\mathbb{Z}_2$ wall ($t=-2$, $\det=2$, v135), $J=M(1, -2)=C-2V$, completing the sheet axis $J, K, C, F=M(1, -2..1)$. [E] $\chi_J = \lambda^3 - 6\lambda^2 + 3\lambda - 2 \Rightarrow (\text{tr}, e_2, \det)=(6, 3, 2)$; $a^T J a=30=h(E_8)$, $\det(I+J)=12=\dim \mathbf{g}_{\text{SM}}$ and $\det(2I+J)=40=|R(D_5)|$; [C] the Lie readings audit-typed like the G_2/F_4 shadows.
- **New module v413 (DIAMOND.SHEET.CALCULUS.01).** On $M_t=C+tV$ the elementary-symmetric coefficients encode the atoms as difference orders: $e_1=3t+12$, $e_2=2t^2+15t+25$,

$e_3=4t^2+14t+14$ give $\Delta e_1=3=N_{\text{fam}}$, $\Delta^2 e_2=4=|\mu_4|$, $\Delta^2 e_3=8=\text{rank } E_8$ (the e_3 curvature is the $\sqrt{218}/\sqrt{224}$ sheet quadratic; the full triple is new); [E] the anchor energy $a^T M_t a = 52 + 11t$ ($30 \rightarrow 41 \rightarrow 52 \rightarrow 63$, step $11 = a^T V a$); [C] atom/Lie readings audit-typed.

- **New module v414 (DIAMOND.CENTER.RESOLVENT.01).** The center $C=M(1,0)$ is a resolvent portal: [E] $\det C=14=\dim G_2$, $\det(I+C)=52=\dim F_4$, $\det(2I+C)=120=|R^+(E_8)|$, the exceptional shell ladder $G_2 \rightarrow F_4 \rightarrow E_8^+$ as $\det(C+kI)$, $k=0,1,2$ (with $\text{tr } C=12=\chi_{E_8}(i)$, $\sum C=31=2^{g_{\text{car}}}-1$, v95); [C] the portal reading.
- **Surfaces synced.** Integrated into the Sheet-Diamond section of `tfpt_2` (new keybox, five \veri citations) and the E_8 theta-raster of `tfpt_3`; `run_all.py`, `script_registry.csv` and the ledger updated; Wolfram extension mirror $303 \rightarrow 308$.

2026-06-24 (Two new tracked targets + axion-branch consistency: v408, v409)

- **New module v408 (HYP.PHI0.PUNCTURE.01).** Names the φ_0 puncture term $48c_3^4$ as the order- $|\mu_4|=4$ local boundary heat-kernel (Seeley–DeWitt) contact term of the μ_4 puncture divisor, summed over $\Omega_{\text{adm}}=N_{\text{fam}} \dim S^+=48$ admissible states — the analytic complement of the icosahedral tree term (v396). [E] the exact split $\varphi_0=1/(6\pi)+48c_3^4$, the puncture count, the closed form $3/(256\pi^4)$ and the hypergraph negative (reused from v396/ARCH.QUAD.01); [O] the from-first-principles heat-kernel proof; [C] a FACE of the seam boundary analysis (the c_3 Gauss–Bonnet datum), not a new gate. Cited in `origin_theory`; no new number.
- **New module v409 (RES.COXETER.SYMMETRY.01 closed as a structural lemma).** prime-2 (the seam sheet involution) is *necessary* but *not autonomous*: $\sigma=\text{diag}(1, -1, -1)$ has spectrum $\{\pm 1\}$ and projector $(I+\sigma)/2$ spectrum $\{0, 1\}$, none in $(0, 1)$, so prime-2 carries no contraction rate, whereas the genuine rates $2/3$ and $(1/3)^6$ lie in $(0, 1)$ and carry prime-2 only as a factor. [E] no prime-2-only attractor exists; [C] dynamics begins only after coupling to family-3 or carrier-5 ($|\mu_4|=2^2$ the Galois bridge). Resolves the v394 open question’s falsifiable corollary; cited in `tfpt_research_contracts`; the ledger row RES.COXETER.SYMMETRY.01 retyped [O] $\rightarrow$ [E]/[C].
- **Axion branch consistency (tfpt_4_frontier, website predictions.ts, v185).** The dark-matter section’s opening + keybox are realigned with the finite- T result already stated later in the same section: the robust dominant-DM branch is the *spine* angle $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=3\pi/5=108^\circ$ (untuned, in band), while the seam hilltop $\theta_i \approx 170^\circ$ over-produces ($\sim 5.4\times$) and is disfavoured; the misalignment angle is retyped from a closed [E] input to a [C] branch decision. The website prediction card now leads with the spine angle.
- **Lean closure of the prime-2 corollary (FORM.COXETER.PRIME2.01).** The v409 result is now also machine-proved in Lean 4 / Mathlib (new module `TfptCarrier/CoxeterPrime2.lean`): an involution eigenvalue ($r^2=1$) is ± 1 , a projector eigenvalue ($r^2=r$) is 0 or 1, neither lies in the open interval $(0, 1)$, while $2/3$ does — so *no prime-2-only attractor exists*. Built clean (lake build); the new theorems are axiom-pure (`#print axioms = propext`, `Classical.choice`, `Quot.sound` only) and pass the `AuditCheck/AuditContract` signature locks. This lifts RES.COXETER.SYMMETRY.01 to the machine-proof tier.
- **Verified, no change (Plücker 11).** The quark constant $11=\|Pl(K)\|_1=\det(1|a|\sigma)$ is already typed honestly as an [E] identity with “why 11” the lone [O] residue (v407, `tfpt_1`); the repo carried no overstatement, so no edit was required.

2026-06-24 (Cross-surface consistency sweep: papers, website, gates, figures realigned)

- **Ledger hygiene.** Resolved a duplicate active `claim_id`: GATE.UWALL.06-09 was carried by two distinct chains. It is an ID collision (not a supersede): the monodromy/residue chain

v114–v117 keeps .06–.09 (as asserted by v123_inventory_update.py and run_all.py); the older exterior-leg chain v42–v45 is renumbered to GATE.UWALL.15–.18 (internal depends_on carried along). No claim content changed.

- **Website.** papers.ts document count $8 \rightarrow 9$ (the Red-Team paper was missing from the tally); Red-Team subtitle/highlight *Targets A–E* $\rightarrow$ *A–F* with a new Target F section + contribution bullet (the abstract already carried F); StatusMatrix.tsx Doc 7 $\rightarrow$ Doc 8 for the research contracts and the residual interfaces re-typed v_{geo} [O], G_{net} [C], F_{transfer} [C]; objections Paper 1 $\rightarrow$ Doc 1.
- **Papers (marker/wording alignment, no canonical status change).** tfpt_1 Strong CP [E] $\rightarrow$ [E]/[C] (given RP) and θ_{12} “genuinely open” $\rightarrow$ “conditionally derived”; tfpt_2 Starobinsky A_s [E] $\rightarrow$ [E]/[C] (M exact, A_s a band over $N_\star \in [50, 60]$) and the Part-II abstract “opens the one remaining gap” $\rightarrow$ “closes the last structural gap”; tfpt_research_contracts the QFT4D “nonperturbative measure stays open” line and the “only U_{point} stays open” headline reconciled with the [C] redundancy / v_{geo} reduction stated alongside; origin_theory class C re-typed to the ambient-measure [C] redundancy; introduction (U_{wall}) $\rightarrow$ v_{geo} . Forbidden old-paper attributions (“Papers 1–7”, “Papers 2/6”, “Paper-7”, “Paper-1”) replaced by current document / neutral wording (the tfpt_3 cascade bridge stays exempt).
- **Suite README.** The residual-gates table (Alessandro-5.0 vintage) brought to the current model (v_{geo} [O]; G_{net} /SEAM.EQUIV.01, F_{transfer} , $N_\star$, QG.AMB.01 [C]; Wolfram 116/116 + 303/303).
- **Figures.** Regenerated status_heatmap (ledger-driven) and extended the narrative figures script_timeline/residual_chain/qft_skeleton through the v303–v407 arc (typed solvers, parameter-free local Einstein equation v359, QG.AMB.01 [C] redundancy, seam closed modulo cited theorems); the qft_skeleton “Stelle ghost” open box is retyped (resolved truncation artefact). Added qft_skeleton/qft_unification to the manifest figure list.

2026-06-24 (The $\tau=\omega$ dual keystone + the flavor residuals it homes: v405–v407)

- [E]/[C]/[O] v405_seam_equiv_omega.py (SEAM.EQUIV.02): the *dual keystone* at $\tau=\omega$ – the family/flavor sector as the order-3 Eisenstein/ A_2 face of E_8 , dual to the seam $= (E_8)_1$ at $\tau=i$ (v404). A named contract (genre of v286/v398), NOT a closure. [E] $\chi_{E_8}(\omega) = 0$ (the family CM point degenerates E_8) vs $\chi_{E_8}(i) = 12$; [E] A_2 is the $\tau=\omega$ Eisenstein lattice (Cartan $\det = 3 = N_{\text{fam}}$, $h = 3 = N_{\text{fam}}$); [E] R lives in $E_6 \times A_2$ ($248 = 78 + 8 + 2 \cdot 27 \cdot 3$, $\det R = 8 = \dim A_2 = \text{rank } E_8$); [E] $\delta_{\text{PMNS}} = 4\pi/3$ is the $\tau=\omega$ CM phase; [E] the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck $= N_{\text{fam}} = \det Q$. [C] the family sector is the order-3 Eisenstein/ A_2 deck; the two keystones sit at the only two elliptic points of $\text{PSL}(2, \mathbb{Z})$. [O] residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the v_{geo} -class scale. Cited in origin_theory; Python-only.
- [E]/[C]/[O] v406_pmns_phase_origin.py (FLAV.PMNS.CLOSE.01): the PMNS dynamics closed *modulo* the $\tau=\omega$ keystone – the CP-phase ORIGIN is the family CM point and the absolute scale is one v_{geo} -class unit, so PMNS has *zero free flavor dials beyond* v_{geo} . [E] the three angles (v9/v268) $\rightarrow$ unitary U_{PMNS} (v270); [C] $\delta_{\text{PMNS}} = 4\pi/3$ is the $\tau=\omega$ hexagonal CM phase (v405), upgrading v270’s [O] “assembled input” to a keystone consequence; [C] $J_{\text{PMNS}} = -0.0297$ derived; [E] the absolute scale = one seesaw ratio = v_{geo} -class (v272/v153). [O] residual = the seesaw operator realisation (M_R , v184) + the Σm_ν floor kill test. Cited in tfpt_2_standard_model; Python-only.
- [E]/[C]/[O] v407_dn_pairings_omega.py (FLAV.SELECTOR.CLOSE.01): the $R4'$ selector residue folds into the $\tau=\omega$ keystone – the three (d, n) pairings ARE the $\tau=\omega$ family-slice atoms. [E] the frame $(\mathbf{1}, a, \sigma)$ has $\det = 11 = \|Pl(K)\|_1$ and $n = (5, -9, 6)$ is the unique covector with $n \cdot \mathbf{1} = 2$, $n \cdot a = 8$, $n \cdot \sigma = 121 = 11^2$ ($\Rightarrow (d, n) \Rightarrow R$, v136/v139);

[E] $n \cdot a = 8 = \text{rank } E_8 = \dim A_2 = \det R$ (the A_2 block reading R in $E_6 \times A_2$, v405); [E] the pairings $\{2, 8, 121\} = \{|Z_2|, \dim A_2 = \det R, \|Pl(K)\|_1^2\}$. [C] deriving (d, n) folds into SEAM.EQUIV.02 (d anchor-derived, v139); [O] the lone residue “why $11 = \|Pl(K)\|_1$ ”. Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (The arithmetic hull + the E_8 -character CM duality: v403, v404)

- [E]/[C] `v403_facet_compositum.py` (ARITH.HULL.01): the three single-prime number-field facets of the order-30 clock (v390/v394, used so far only one at a time) *compose* to one field $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$. [E] the squarefree parts $-1, -3, 5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so $[K : \mathbb{Q}] = 2^3 = 8 = \text{rank } E_8 = g_{\text{car}} + N_{\text{fam}}$ with Galois group $(\mathbb{Z}/2)^3$ (order 8); [E] exactly 7 quadratic subfields $\{-1, -3, 5, 3, -5, -15, 15\}$, splitting 4 imaginary $= |\mu_4|$ (CM/dynamic) + 3 real $= N_{\text{fam}}$ (RM/static); [E] ramified *only* over $\{2, 3, 5\}$ = the atoms = the primes of $h(E_8) = 30$; [E] $2^{\omega(h)} = \text{rank}$ is E_8 -special (the E_7 control $h = 18$ gives $2^2 = 4 \neq 7$). [C] K is the abelian $(\mathbb{Z}/2)^3$ *arithmetic* hull dual to the E_8 *lattice* hull; the 7 quadratic subfields are a candidate for the 7 E_8 audit slices. A synthesis composing the established facets, no new number; cited in `origin_theory`; Python-only by the synthesis-module convention (cf. v390/v394).
- [E]/[C] `v404_e8_char_cm_duality.py` (E8.CMDUAL.01): the $(E_8)_1$ vacuum character $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$ read at *both* CM points, extending v282 ($\chi_{E_8}(i) = 12$) to the second elliptic point. [E] $\chi_{E_8}^3 = j$ (q-series); [E] SEAM $\tau=i$ ($\mathbb{Q}(i)$, order 4): cross-ratio $2 \Rightarrow j = 1728 \Rightarrow \chi_{E_8}(i) = 1728^{1/3} = 12$ (E_8 present, v214/v282); [E] FAMILY $\tau=\omega$ ($\mathbb{Q}(\sqrt{-3})$, order 6): the equianharmonic λ ($\lambda^2 - \lambda + 1 = 0$) $\Rightarrow j = 0 \Rightarrow \chi_{E_8}(\omega) = 0$ (E_4 vanishes, v220/v267); [E] $\tau=i, \tau=\omega$ are the *only* elliptic points of $\text{PSL}(2, \mathbb{Z})$, so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where E_8 is present ($\chi = 12$, the keystone SEAM.EQUIV.01 closes) while the flavor/CP sector is the soft residual where E_8 degenerates ($\chi = 0$: $U_{\text{wall}}, v_{\text{geo}}$, the CP texture), a candidate dual keystone at $\tau=\omega$. Cited in `origin_theory`; core values $j = 1728/j = 0$ already mirrored (v214/v220/v267/v282), Python-only.

2026-06-24 (The five TOE/QFT closure contracts v398–v402 + a ledger rescope)

- [E]/[O] `v398_seam_state_rigidity.py` (SEAM.RIGIDITY.01, Paper A): the *Seam State Rigidity Theorem* as ONE named target, consolidating the scattered state-invariance reductions (v177/v199/v308/v309/v335) and *hardening* the finite-block evidence to a multi-mode sweep. [E] RP-definability hinge (OS canonical transfer $T=e^{-H}$ from RP + reflection alone, v54, quasi-free v155, no μ_4 normal form); [E] band-limited character-block-diagonal core $[\rho, |k|]=0$ swept $N=4.64$ (beyond the 3-dim H^1); [E] holomorphy $\Rightarrow (E_8)_1$ ($c=8, |\det \text{Cartan}|=1$ vs 4, index $4=|\mu_4|$). [O] the one residual: $[\rho, \Lambda_\Sigma]=0$ on the full L^2 (intrinsic Bisognano–Wichmann). NOT a closure of SEAM.EQUIV.01 (continuum = cited MMST v336).
- [E]/[C]/[O] `v399_gravity_complete.py` (GRAVITY.COMPLETE.01, Paper B): the explicit verdict *Gravity-complete* = *Boundary-complete* + *Ambient-redundancy*, combining the parameter-free local equation (v358/v359) with the ambient redundancy (v369). [E] the five gravitational readout classes (Newton 8π , Λ , scalaron c_3^7 , horizon $1/|\mu_4|$, recovery $(2/3)^6$) are all boundary/seam/gap, so $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$; [O] residual = SEAM.EQUIV.01 + intrinsic BW (no new gate).
- [E]/[O] `v400_grav_nonlocal_action.py` (GRAV.NONLOCAL.01, Paper C): the nonlocal spectral-gravity action with the local $R+R^2$ as its IR projection (consolidating v304/v370/v380/v386, adding the IR-matching check). [E] entire $a=e^u \Rightarrow$ one pole (ghost-free), every truncation carries a Stelle zero (monotone, v380), and the IR order reproduces $R+R^2$ with the atom-fixed scale $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$ (v36/v253). [O] the entire- a assumption + perturbative-only.

- **[E]/[O] v401_metrology_closure.py** (METROLOGY.CLOSURE.01, Paper D): the final input set $\{a, \pi, v_{\text{geo}}\}$ certified — 0 dimensionless dials + 1 unit (consolidating v153/v274/v364/v384). **[E]** dimensional bookkeeping $\text{mass}/1G/\Lambda/U_{\text{point}}/(m/\mu) = \text{number} \times v_{\text{geo}}^{1,2,4,1,0}$; **[E]** the G -vs- Λ over-determination 0.11% (v274); **[O]** v_{geo} theorem-forbidden ($\text{TOE}_{\text{strict}} = \text{TOE}_{\text{dimensionless}} + [v_{\text{geo}}]$).
- **[E]/[C] v402_ftransfer_suite.py** (FTRANSFER.SUITE.01, Paper E): the four frontier sectors as ONE functor F_{transfer} , the composition $F_{\text{obs}} \circ F_{\text{thr}} \circ F_{\text{RG}}$ (consolidating the FR.*.SOLVE solvers), each with an exact atom source, a committed kill test, and the firewall (physical prediction stays **[C]/[O]**, only the algebraic source **[E]**; never re-pressed into the compiler, v187).
- **Ledger rescope (repo hygiene)**: GATE.METRIC.01 no longer reads “full covariant metric-sector field equation open” — the *local* covariant equation is closed parameter-free (GRAV.NONLINEAR.01/v359); the gate is rescoped so the open item is *only* the global nonperturbative ambient measure (QG.AMB.01, a **[C]** redundancy v369), removing the stale contradiction with the status card.

2026-06-24 (φ_0 leading term from icosahedral combinatorics: v396)

- **[E]/[C] v396_phi0_icosahedral.py** (HYP.PHI0.01): the tree-level seed $\varphi_0^{\text{tree}} = 1/(6\pi)$ is exactly $F/(4h\pi) = F/(|\text{Aut}|\pi) = (F/(g_{\text{car}}N_{\text{fam}}))c_3$ on the icosahedral hypergraph ($F = 20$ triangular hyperedges, $h = 30$, $|\text{Aut}| = 120$); $F/h = |Z_2|/N_{\text{fam}} = 2/3$ (the same survival ratio as v327). **[E]** Gauss–Bonnet consistent with $c_3 = 1/(|Z_2|4\pi)$ (v216/v342). **[E]** honest negative: the puncture $48c_3^4$ is still analytic, not a hypergraph fraction. **[C]** φ_0^{tree} is a combinatorial reinterpretation, not a new independent injection; the puncture stays **[O]**.

2026-06-24 (Coupled hypergraph rewrite: v395 unifies carrier $\times$ family in one local rule)

- **[E] v395_hypergraph_coupled.py** (HYP.COUPLED.01): one coupled local rewrite on a 9×3 labelled grid (affine- E_8 nodes $\times$ family channels) unifies v299 (carrier growth), v327 (branching rule M) and v324 (fiber product). Micro-step = network lazy diffusion on each cusp column + M on each node’s family vector ($T_{\text{micro}} = T_{\text{net}} \otimes M$); joint attractor marks $\otimes e_0$; one clock hand (6 family steps) gives recovery $(2/3)^6$; v324’s $T_{\text{net}} \otimes T_{\text{cusp}}$ emerges as the cusp-readout basis. **[O]** the 3-arm seed (P_2) and analytic φ_0 remain irreducible (v312).

2026-06-24 (A falsifiable clock probe of the frontier: v397 how external is the “external physics”?)

- **v397 (FR.CLOCK.PROBE.01)** — **do the “external” frontier scheme-numbers land on the $\{2, 3, 5\}$ clock?** A falsifiable, anti-numerology probe (strict No-Free-Pattern, v100: “on the clock” means an *exact* match, not a wide-band fit). **[E]** the η_B decuple $A_\Lambda = 10 = |Z_2|g_{\text{car}}$ is ON the clock (v212); **[E]** the Koide rate $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$ is ON the clock (v303); **[E]** the axion spine angle $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ is ON the clock (v211/v373); **[E]** the proton $C_p \sim 2.83 \pm 0.15$ is *not* a discriminating clock value — within its $\pm 5.3\%$ band several simple values fit $(2^{3/2}, 17/6, 14/5)$, so by No-Free-Pattern it is declared EXTERNAL (the firewall v187/v374 is correct there). **[C]** verdict: 3 of the 4 frontier scheme-numbers land *exactly* on the clock, so the irreducible “external physics” residue shrinks to essentially the proton’s non-perturbative QCD number C_p . Cited in `tfpt_4_frontier`; a falsifiable probe whose answer could have been “none on the clock”; Python-only.

2026-06-24 (Clockwork coherence + an open question: v394 primes = atoms = facets, and RES.COXETER.SYMMETRY.01)

- **v394 (COXETER.ATOMS.01) — the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets.** A bird's-eye [E] re-reading tying v390 (the prime facets) to the anchor $a = (1, 1, 2)$ and v319 (the 5×6 clock). [E] $a = (1, 1, 2)$ has $e_3=2=|Z_2|$, $p_0=3=N_{\text{fam}}$, $e_2=5=g_{\text{car}}$, product $2 \cdot 3 \cdot 5 = 30 = h(E_8)$ — the clock's primes *are* the anchor's atoms. [E] atoms = facet fields (v390): $2 \rightarrow \mathbb{Q}(i)$, $3 \rightarrow \mathbb{Q}(\sqrt{-3})$, $5 \rightarrow \mathbb{Q}(\sqrt{5})$, each ramified at its prime. [E] the QG susceptibility is pure $\{2, 3\}$: $\chi = 1/(1-(2/3)^6) = 729/665 = 3^6/(3^6-2^6)$, $(2/3)^6 = 2^6/3^6$, exponent $6 = 2N_{\text{fam}}$. [E] both dynamic rates carry prime-2: $2/3 = |Z_2|/N_{\text{fam}}$ and $(\varphi+2)/4 = (\varphi+2)/|\mu_4|$ with $|\mu_4|=4=2^2$. [E] $30 = 5 \times 6 = g_{\text{car}}(2N_{\text{fam}})$ (v319). [C] the prime-2 seam is the universal dynamical engine; prime-3 (family) and prime-5 (carrier golden) the two attractors.
- **RES.COXETER.SYMMETRY.01 — a new registered open research question.** Is the prime-2-as-common-factor reading *forced* (the seam involution provably in every sector's gap, no independent prime-2 attractor), or is there a residual asymmetry? Resolving it would either tighten v383's "every sector is the same gapped operator" to "the same *seam-driven* operator" (with the falsifiable corollary that no prime-2-only attractor exists), or reveal a missing dynamical slot. Tracked in `tfpt_research_contracts`; a question, not a claim. Cited in `origin_theory`; Python-only.

2026-06-24 (Correction error-bars: v393 the typed budget as actual first-correction magnitudes)

- **v393 (CORRECTIONS.NUMERIC.01) — the typed correction budget (v388) as actual numbers.** The short-term, testable harvest of v388: instead of merely typing each prediction's correction, this gives the magnitude, so a parameter-free central value can be quoted as "value $\pm$ computed first correction". [E] FIXED-POINT band (the φ_0 -derived $\theta_{12}, \theta_{13}, \lambda_C, \beta_{\text{rad}}$) = the explicit φ_0 puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; $= \theta_{12}$'s ε first-correction, $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$); [E] SEAM-GAPPED band (Koide F_{pole} , recovery) $= (2/3)^6 \approx 8.78\%$, the QG-decoupling bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337); [E] EXACT-IDENTITY band $= 0$ structurally ($\det R = 8$, $N_{\Phi} = 1$, $\theta_{\text{eff}} = 0$); [E] EXTERNAL-RATE band = external physics, quoted $(m_p/m_e \pm 7.5\%$ v374; n_s/r the $N_{\star} \in [50, 60]$ band; $\eta_B 1.07 \times$ v212), fenced by v187. [C] the assembled budget gives each prediction a computed theoretical-uncertainty band. Cited in `origin_theory`, mirrored on the website prediction surface; a harvest of v388, no new number; Python-only.

2026-06-24 (Attacking the last open lemma: v392 the scaling-limit central charge converges)

- **v392 (SEAM.S3.SCALINGLIMIT.01) — an honest attack on SEAM.EQUIV.01's one open lemma (continuum existence, v336).** It does *not* close v336 (the abstract existence is the cited MMST theorem); it strengthens the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. [E] the Calabrese–Cardy entanglement slope $c_1(L)$ at $L = 66, 130, 258, 386, 514$ is a monotone Cauchy-like sequence approaching 1 ($|c_1(L) - 1|$ strictly decreasing, $4.6 \times 10^{-4} \rightarrow 7 \times 10^{-6}$); [E] a $1/L$ Richardson fit extrapolates to $c_1(\infty) \approx 1.000$, so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\text{car}} + N_{\text{fam}}$ is reached *in* the limit; [E] the successive differences are themselves strictly decreasing (a numerical Cauchy sequence). [O] convergent data are evidence, not a proof of existence (the cited MMST theorem, v336), and do not pin $(E_8)_1$ over the same- c $SO(16)_1$ (the holomorphy discriminator, v377/v378). Strengthens the residual, does *not* close it. Cited in `tfpt_research_contracts`; Python-only.

2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 extended to the finite-dimensional Perron–Frobenius case)

- **FORM.SPECTRALGAP.01** — the multi-dimensional case, reduced to the scalar core. The Lean module `TfptCarrier/SpectralGapAttractor.lean` now also proves the *finite-dimensional* gapped-operator statement (lake build OK, no sorry): [\[E\]](#) `component_tendsto` (a non-dominant eigencomponent $|r| < 1$ decays, $r^n c \rightarrow 0$); [\[E\]](#) `deviation_tendsto` (with the dominant eigenvalue normalised to 1, the deviation vector — all non-dominant components — tends to 0, so the iterate converges to the dominant eigendirection, the unique attractor, for any start); [\[E\]](#) `deviation_unique` (two starts give deviation vectors both $\rightarrow 0$, so the selected eigendirection is independent of the start — parameter-freeness in finite dimensions); [\[E\]](#) `flavor_deviation_tendsto` (a concrete instance on the cusp-transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, [v56/v82/v383](#)). Honest scope: this is the finite-dimensional *diagonalisable* case (the genuine multi-dimensional content, reduced to the scalar core); the general non-diagonalisable / infinite-dimensional Perron–Frobenius theorem stays the cited statement. Referenced in `origin_theory`.

2026-06-24 (Website: an animated “Universal Gap Lab” visualising [v383–v391](#))

- **Website** — the `UniversalGapLab` animation on /verification. A new client component (Framer Motion, no new dependency) with three linked, module-grounded views: (A) every sector as the gap contraction $\text{iter}_n = x_\star + r^n(x_0 - x_\star)$ racing to its attractor at $r = (2/3)^6$ (seam-gapped, [v303/v387/v388](#)) or the golden $(\varphi + 2)/4$ ([v312](#)), with a gapless $r = 1$ track that never forgets its start (the Lean theorem `FORM.SPECTRALGAP.01`) — the rate *is* the first-correction size ([v388](#)); (B) the entire-form-factor graviton ratio e^{-u} falling faster than any power, ghost-free and UV-soft ([v386/v389](#)); (C) the residual matrix as certification, not construction — every open item external, zero open internal mechanisms ([v384](#)). Mirror-only: every number is grounded in a module; no theory change.

2026-06-24 (Honest attempt at an external target: [v391](#) reduces it, not a closure)

- [v391](#) (`ALPHA.QUILLEN.PROGRESS.01`) — an honest attempt at the external target `ALPHA.QUILLEN.EXACT.01` ([v382](#), door 5). It REDUCES the open step to a sharper named sub-target via a solvable model but does *not* close it: `ALPHA.QUILLEN.EXACT.01` stays [\[O\]](#) (external math, type A, [v384](#)). [\[E\]](#) a solvable model $\det_\zeta(-\partial_\tau^2 + m^2)$ on a circle of circumference β equals $(2 \sinh(\beta m/2))^2$, whose m -variation $\beta \coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$ is a closed heat-kernel/Laurent series (the m^3 coefficient $-1/360$ verified). [\[E\]](#) so a ζ -determinant’s coupling-variation is structured heat-kernel data, and [v382](#)’s counterterm $8b_1c_3^6$ (a Gauss–Bonnet heat-kernel power) is the right *kind* of object; the [v382](#) split is re-verified at $\alpha^{-1} = 137.0359992$. [\[E\]](#) the reduction: the open step now reduces to the named sub-target “the seam $U(1)$ operator’s Seeley–DeWitt coefficient $= 8b_1c_3^6$ ”. [\[O\]](#) *not closed*: the actual seam-operator ζ -determinant proof stays open, and `SEAM.EQUIV.01` continuum existence (MMST, [v336](#)) stays the other external [\[O\]](#); the value α^{-1} stays [\[E\]](#). An attempt that reduces, does not close. Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 formalises the scalar core of the universal spectral-gap principle)

- **FORM.SPECTRALGAP.01** — a Lean 4 proof of the gap $\Rightarrow$ unique-attractor $\Rightarrow$ parameter-free core (door 4). New module `TfptCarrier/SpectralGapAttractor.lean` (lake build OK, no sorry, only the three standard kernel axioms), formalising the meta-theorem of [v303/v383/v387](#) at the one-dimensional reduction — the affine gap contraction $x \mapsto x_\star + r(x - x_\star)$ with multiplier $r = \lambda_2/\lambda_1$. [\[E\]](#) `iter_sub`: the closed form $\text{iter}_n - x_\star = r^n(x_0 - x_\star)$ (general n); [\[E\]](#) `iter_tendsto`: $0 \leq r < 1$ (the gap) $\Rightarrow \text{iter}_n \rightarrow x_\star$ (the unique attractor); [\[E\]](#)

starts_agree: two starts differ by $r^n(a-b) \rightarrow 0$, so the attractor is independent of the initial state (parameter-freeness); **[E] no_gap_no_forget**: $r = 1$ (no gap) leaves $\text{iter}_n - x_\star = x_0 - x_\star$ constant — the initial condition is then a free parameter, so the gap is exactly what removes it; **[E]** the seam rate $(2/3)^6$ and the golden $(\varphi + 2)/4 = (5 + \sqrt{5})/8$ both lie in $[0, 1)$. Honest scope: the scalar reduction (the contraction core **v303/v383** iterate numerically); the full multi-dimensional Perron–Frobenius statement stays the cited theorem. Referenced in **origin_theory**; ledger row **FORM.SPECTRALGAP.01**.

2026-06-24 (Completing the Coxeter clock: **v390 the prime-2 (Gaussian / $\mathbb{Z}/4$ -seam)** facet of $30 = 2 \cdot 3 \cdot 5$)

- **v390 (COXETER.PRIME2.01) — the prime-2 facet of the order-30 Coxeter clock.** **v383/v387** named two number-field facets of the clock $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ — the golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (prime-5, carrier) and $(2/3)^6 \in \mathbb{Q}$ (prime-3, family). This names the *prime-2* facet and identifies it with structure already in the theory: the Gaussian field $\mathbb{Q}(i)$, i.e. the CM point $\tau = i$ and the order-4 $\mu_4 = \mathbb{Z}/4$ *square* seam deck (**v282/v288/v333**). **[E]** $h(E_8) = 30$ factors into exactly three primes; one quadratic facet per prime, each ramified at its prime: $\text{disc } \mathbb{Q}(i) = -4$ (2 ramifies), $\text{disc } \mathbb{Q}(\sqrt{-3}) = -3$ (3 ramifies), $\text{disc } \mathbb{Q}(\sqrt{5}) = 5$ (5 ramifies); the ramified primes are $\{2, 3, 5\}$ with product $30 = h(E_8)$. **[E]** the prime-2 (Gaussian) facet is the $\mathbb{Z}/4$ seam deck: min. poly $x^2 + 1$, $j(i) = 1728 = 12^3$, $\tau = i$ fixed by the modular S of order $4 = 2^2$, $|\mu_4| = 4 = 2^2$ — the static, even, seam-orientation facet. **[C]** so the three primes of the clock now each carry a facet (square/hexagon/pentagon = seam/CP/carrier); the prime set $\{2, 3, 5\}$ is forced by $h(E_8) = 30$, not chosen. Cited in **origin_theory**; a synthesis, no new number; Python-only.

2026-06-24 (Graviton loop finiteness: **v389 the entire form factor is UV-finite at the loop level by power counting**)

- **v389 (GRAV.LOOP.FINITE.01) — the entire-form-factor graviton is UV-finite at the loop level.** The honest next step past **v386** (which established the tree amplitude), extending **v304/v370/v380/v386** from the propagator to the superficial degree of divergence of loop graphs. This is the standard infinite-derivative / nonlocal-gravity finiteness statement (Tomboulis; Biswas–Mazumdar–Siegel; Modesto), *not* a full proof of perturbative quantum gravity. **[E]** super-polynomial falloff: $p^k e^{-p^2/M^2} \rightarrow 0$ for every degree k . **[E]** one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$, while the GR $\int p^3/p^2 dp$ diverges quadratically. **[E]** the superficial degree $\rightarrow -\infty$ at every loop order L (the form factor multiplies the integrand by $e^{-(\#\text{internal})p^2/M^2}$; dressed cutoff-ratio $\rightarrow 1$ vs GR $\rightarrow 4$). **[E]** the regulator is the seam scale $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ (**v253/v259**), not a dial. **[C][O]** loop *unitarity* of the nonlocal form factor (Lorentzian continuation / macro-causality, Pius–Sen / Briscese–Modesto) stays the honest open point — tree unitarity is **v386**; not claimed solved. Cited in **tfpt_4_frontier**; Python-only.

2026-06-24 (Typed correction budget: **v388 the gap correction is keyed to the mechanism class, not a uniform band**)

- **v388 (CORRECTIONS.BUDGET.01) — the gap-driven correction (**v387**) is a *typed* budget.** A machine-checked answer to “does the $(\lambda_2/\lambda_1)^n$ correction apply to *all* predictions?” — the answer is *no*, and for one whole class a band would be *wrong*. Because the size is the distance from a gapped operator’s leading eigenvector, it is defined only where a subleading λ_2 exists, so the prediction surface splits into four classes. **[E]** *seam-gapped* (Koide F_{pole} **v303**, recovery **v221**, the QG bound capped by $\chi = 729/665$ **v337**): rate $(2/3)^6$; the discrete compiler: golden $(\varphi + 2)/4$ (**v312**). **[E]** *exact identity* ($\det R = 8$, $N_\Phi = 1$, $\theta_{\text{eff}} = 0$, $\sin^2 \theta_{13} = \varphi_0 e^{-5/6}$): no λ_2 , so the correction is 0 structurally and a band would contradict the **[E]** status. **[E]** *external-rate*

(η_B washout, axion freeze, m_p/m_e RG): the gapped *shape* with an external rate — only F_{pole} has the seam rate (v303), firewalled (v187). [E] *fixed-point*: the texture is already explicit (e.g. $\sin^2 \theta_{12}$, $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$). [E] the Koide first correction (door 2): F_{pole} is the one seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), the source quotient 0.33% below $2/3$ — the rate is the suppression *scale*, not the residual. [C] so a uniform band on all predictions is wrong; the size is keyed to the mechanism class. A typing of v387 (as v384 is of v383); cited in `origin_theory`; Python-only.

2026-06-24 (Harvesting the QFT/gravity machinery: v385 the UV-branch kill-test surface, v386 the graviton-exchange amplitude, v387 the gap-driven correction calculus)

- **v385 (UVBRANCH.KILLTEST.01) — the carrier-Pati–Salam UV branch on the all-order footing.** A consolidation that ties the all-order EG/BRST closure (v381) to the optional UV branch and computes the one genuinely new thing: the *proton-decay safety hierarchy*. [E] the carrier content is the SM with no new states, so the SM-non-unification (v246) is now all-order-robust; [X] the plain-SM 4D-GUT stays killed ($\sin^2 \theta_W = 3/8$ vs 0.231); [C] the optional carrier-PS branch sits at $M_{PS} = \kappa M_{\text{scal}} \approx 3 \times 10^{13}$ GeV ($\kappa \in [1.0, 1.2]$, v253); [E] minimal PS gauge bosons are $SU(4)$ leptoquarks that mediate rare LFV ($K_L \rightarrow \mu e$), *not* $p \rightarrow e^+ \pi^0$, so the branch clears the binding bound by $M_{PS}/M_{\text{bound}} \sim 10^7$ – proton-safe, with *no* fake τ_p window (v265). Cited in `tfpt_5_redteam`; Python-only.
- **v386 (GRAV.AMPLITUDE.01) — the graviton-exchange amplitude is finite and UV-soft.** Extends v304/v370/v380 from the propagator to the actual tree amplitude. [E] the dressed spin-2 propagator $e^{-p^2/M^2}/p^2$ has its only pole at $p^2 = 0$ (residue $1 > 0$, ghost-free), is exponentially softened in the UV (the ratio e^{-u} falls faster than any power) and reduces to GR + a calculable $-p^2/M^2$ correction at low energy ($M = M_{\text{scal}}$, v253); [C] a single positive-residue pole gives tree unitarity, so perturbative graviton scattering $A \sim (T \cdot P_2 \cdot T') e^{-p^2/M^2}/p^2$ is an explicit finite computation (perturbative-only; QG.AMB.01 stays a [C] redundancy, v369). Cited in `tfpt_4_frontier`; Python-only.
- **v387 (CORRECTIONS.GAP.01) — the gap-driven correction calculus.** The practical harvest of v383: the same spectral gap that makes a sector parameter-free also sets the *size* of its first correction ($\text{correction}_n \sim (\lambda_2/\lambda_1)^n$). [E] flavor, horizon recovery and the QG bound all share the $(2/3)^6$ rate (the family-3 facet; QG susceptibility $\chi = 729/665$), while the discrete compiler transient decays at the golden $(\varphi + 2)/4$ (the carrier-5 facet) — the two number-field facets of v314/v383. [C] the gap does double duty (why no free parameter *and* how big the first correction), so sub-leading magnitudes are organised, not free. Cited in `origin_theory`; Python-only.

2026-06-23 (Bird’s-eye synthesis: two new structural patterns — v383 the universal spectral-gap principle and v384 the residual is certification, not construction)

- **v383 (DYNAMICS.UNIVERSAL.01) — the universal spectral-gap / Perron–Frobenius principle.** A fresh bird’s-eye pass found that *every* TFPT sector is the *same* structural object: a gapped operator with a *unique* leading attractor (the physics) and a spectral gap (the reason there is no free parameter). v303 named this only for F_{transfer} ; this module extends it to the sectors made parameter-free *after* it. [E] three gaps are computed — the flavor transfer $\{1, (2/3)^6, (1/3)^6\}$ with gap $6 \ln(3/2)$ (v56/v82); the affine- E_8 adjacency with Perron eigenvalue 2 (Kac marks) and subleading $2 \cos(\pi/5) = \varphi$, gap $2 - \varphi = 1/\varphi^2$ (v312/v313); the decoupling margin $\Delta_{\text{eff}} \approx 1.648$ (v76/v337). [E] the two number-field facets (v314): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2 \cdot 3 \cdot 5 = h(E_8)$. [C] the other sectors are the same structure — gravity ($\delta S=0$ equilibrium, v358/v359), the QG measure (Gaussian saddle, v365),

the boundary QFT (holomorphic $c=8$ RG fixed point, [v157/v158](#)), the all-order S-matrix (adiabatic limit, [v381](#)), recovery (Petz fixed point, [v221](#)), $\tau=i$ ([v333](#)). [\[C\]](#) the meta-theorem: gapped $\Rightarrow$ unique attractor $\Rightarrow$ parameter-free, theory-wide. Cited in `origin_theory`; a synthesis (no new number), Python-only.

- **v384 (RESIDUAL.CERTIFICATION.01) — the residual is certification, not construction.** After [v369/v381/v382](#) the *kind* of every open item has shifted: there is *no* open TFPT physics mechanism left. A ledger-CI audit classifies every residual into exactly three external kinds: [\[E\]](#) (A) external math proof (SEAM.EQUIV.01 continuum existence = the cited MMST/Adamo theorem [v336](#); ALPHA.QUILLEN.EXACT.01 [v382](#); QG.AMB.01 the constructive measure, a [\[C\]](#) redundancy [v369](#)), [\[E\]](#) (B) theorem-forbidden (v_{geo} , the No-Unit theorem [v153/v364](#)), [\[E\]](#) (C) external physics (F_{transfer} , [v371–v374](#), firewalled [v187](#)). The two load-bearing facts are verified (No-Unit dimensionlessness; the gap-decoupling margin $1.648 > 0$), and the count of open *internal* TFPT mechanisms is **zero**. [\[C\]](#) convergence: the only hard things left are theorems and external physics, not missing dynamics. Cited in `tfpt_research_contracts`; an audit (no new number), Python-only.

2026-06-23 (External-review closure: the two genuine residual targets named — v381 the all-order Epstein–Glaser/BRST contract for S_{pert} , and v382 the Quillen determinant-line variation as a tracked target)

- **v381 (QFT4D.EG.ALLORDER.01) — the all-order 4D perturbative-QFT closing statement.** A mathematical-physics review of the QFT/TOE layers identified exactly one genuinely new, buildable item: the *all-order* Epstein–Glaser/BRST contract for the perturbative 4D S-matrix S_{pert} (the previous documents had the S_{pert} skeleton and one-loop, [v269/v271/v273/v278](#), but no named all-order claim). This module types the standard causal-perturbation-theory closure for the TFPT finite interaction class — it does *not* build a path integral. [\[E\]](#) power-counting: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator, so the EG singular order $\omega \leq 4$ and the counterterm space is finite at every order; [\[E\]](#) BRST nilpotency $s^2=0$ verified exactly via the Jacobi identity for $su(2)$ and $su(3)$ (the carrier weak+colour content); [\[E\]](#) the seam gap $\Delta = 6 \log \frac{3}{2} > 0$ ([v302](#)) gives the IR-safe adiabatic limit. [\[C\]](#) the two heavy legs are imported rigorous theorems — all-order T_n existence by causal factorisation (Epstein–Glaser 1973) and all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) — applied to the TFPT class. Matter+gauge *only*: the R^2/Weyl^2 gravity sub-sector is the entire-form-factor resummation ([v304/v370/v380](#)), and S_{pert} is not the ambient measure QG.AMB.01 (a [\[C\]](#) redundancy, [v369](#)). NET [\[C\]](#), prerequisites [\[E\]](#); not [\[E\]](#) and not a path integral. Cited in `tfpt_5_redteam` and `tfpt_2`; Python-only.
- **v382 (ALPHA.QUILLEN.EXACT.01) — the Quillen determinant-line variation, named as a target.** Per the same review, the “why *this* $F_{U(1)}$ functional” obligation (previously tracked only as EM.WARD.01’s residual, [v341/v342](#)) is elevated to its own citable target: $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1c_3^6 \log \varphi_{\text{seam}}) = 0$ on the determinant line over the μ_4 seam moduli. [\[E\]](#) the Quillen split holds at the $\alpha^{-1}=137.0359992$ root with every coefficient a named atom (index $8b_1$, heat-kernel $c_3^{3,6}$, discriminant $q(D_5)+q(A_3)=2$); [\[E\]](#) the value stays closed ([v3](#)); [\[O\]](#) the from-first-principles exactness proof is the target. [\[C\]](#) it is a *face* of SEAM.EQUIV.01 ([v336](#)) — the seam $U(1)$ determinant line lives on the same holomorphic net — so it adds *no* second independent gate; it makes the existing residual trackable. No new number (reuses the Wolfram-mirrored [v341](#) identities). Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-23 (Track 2 of the forward plan v2 — the AMBIENT REDUNDANCY statement (v369): the holographic route to Gravity-complete that sidesteps building the general Euclidean QG measure)

- **v369 (QGAMB.REDUNDANCY.01) — Track 2 (forward plan v2).** Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators that prove the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the QG measure (gate C7/QG.AMB.01) is a non-fundamental *certification* object (like a coordinate choice in GR), not missing dynamics. [E] holomorphy $\Rightarrow$ DHR($(E_8)_1$) = Vec: #primaries = $|\det \text{Cartan}| = 1$ for E_8 vs 4 for $SO(16)$ $\Rightarrow$ no nontrivial superselection charge a bulk could carry invisibly; [E] no torus ground-state degeneracy ($|\det K| = 1$); [E] finite Petz recovery rate $(2/3)^6$ (v221); [E] the gap-decoupling margin $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337). [C] with the Bisognano–Wichmann reconstruction map (v258/v309/v323) the redundancy statement $\mathcal{O}_{\text{phys}}^{\text{bulk}}/\text{Diff} \cong \mathcal{A}_{\Sigma}^{\mu_4}$ is well-posed. [O] residual: the two consumed premises — SEAM.EQUIV.01 (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct QG.AMB.01 but provides the *alternative* (holographic redundancy) route: Gravity-complete = Boundary-complete + Ambient-redundancy. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Status propagation: the v365–v380 + FORM.SEAM.MMST.01 status carried into the remaining papers, the README and the Zenodo description)

- **Deep-sync of the post-S3 status.** The canonical surfaces (`tfpt_status`, introduction) were already current; this pass propagated the same honest wording into `origin_theory`, `tfpt_1–tfpt_5`, `tfpt_research_contracts`, the root README.md, the Zenodo description and `website/lib/papers.ts`: SEAM.EQUIV.01 is *closed modulo a cited theorem* (lattice v367/v368 + the S_3 stack v376–v379 + Lean FORM.SEAM.MMST.01; residual [O] = the cited MMST/OS existence only); QG.AMB.01 is a [C] *redundancy* (v369/v379), a certification object not missing dynamics; perturbative graviton unitarity is established (the Stelle ghost is a Seeley–DeWitt truncation artefact, v304/v370/v380); and the frontier transfers are typed runnable solvers with kill tests (v371–v374) + the observatory CI (v375). No new claim — a wording propagation so every surface matches the ledger; AUDIT OK.

2026-06-23 (v380 — the KMS Entire Hessian: the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity)

- **v380 (GRAV.KMS.HESSIAN.01).** Upgrades v304/v370 from the *assumption* “the resummed graviton form factor is entire” to a derived statement. [E] the seam KMS cutoff $f(u)=e^{-u}$ (v259) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u)=e^u$ entire and nowhere zero (only the $p^2=0$ pole). [E] each finite Seeley–DeWitt truncation is a Taylor partial sum of a and carries a Stelle-ghost zero ($T_1=1+u \rightarrow u=-1$; $T_2 \rightarrow -1 \pm i$); the entire a has none. [E] the modulus of the nearest truncation-zero grows monotonically with the order $(1, 1.41, \dots, 3.07 \text{ for } n=1..8) \rightarrow \infty$ — the ghost decouples, “entire” = “do not truncate”. [C] $\Rightarrow$ perturbative graviton unitarity; [O] the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation, perturbative-only. Cited in `tfpt_5_redteam` (Target F); Python.

2026-06-23 (Lean: the S3 continuum leg formalised as “closed modulo a cited theorem” — FORM.SEAM.MMST.01)

- **FORM.SEAM.MMST.01 — Lean 4 formalisation.** The S3 continuum leg is now machine-formalised in a new module (`SeamScalingLimit.lean`), lake build OK and `#print axioms` clean, in the audit-contract style of `SeamEquivChain.lean`. The MMST applicability hypotheses for the seam collar are *provable* by kernel `decide` (no axioms): $D = 16$, $\text{rank} = c = 8$,

the range $8 \leq c \leq 16$, $c_- = 8 \neq 0$, and $\det K = 1$ vs $SO(16)$'s 4. The MMST scaling-limit theorem (CMP 2022) and the Adamo–Moriwaki–Tanimoto OS reconstruction (2024) are *named cited axioms*; the conclusion $(E_8)_1$ follows by a well-typed composition whose residual (`#print axioms`) is exactly those two published theorems plus the carrier gap — no hidden assumption, no sorry. Mirrors the S3 closure stack `v376–v379`; so `SEAM.EQUIV.01` is now “closed modulo a cited theorem”, the honest formalisation ceiling. Cited in `tfpt_research_contracts`; `lean_manifest` refreshed.

2026-06-23 (S3 closure stack — the target of the seam scaling limit pinned at EVERY computable level: central charge $c=8$ from the lattice (`v376`), the $(E_8)_1$ character (`v377`), the genus-1 torus count = 1 (`v378`), and reflection positivity (`v379`); only the abstract continuum existence (cited MMST) stays external)

- **`v376 (SEAM.S3.CENTRALCHARGE.01)`**. The Calabrese–Cardy finite-size entanglement-entropy scaling of the critical free-fermion content gives $c = 1.0000$ per complex mode (Peschel correlation-matrix EE; the $L=4m+2$ sizes avoid the half-filling zero-mode), so the 16-Majorana (= 8 complex) collar has $c = 8 = g_{\text{car}} + N_{\text{fam}}$, chiral ($c_- = 8$, `v367/v368`). Numerical $c=8$ from the lattice. Python (numpy).
- **`v377 (SEAM.S3.E8CHARACTER.01)`**. The exact affine character $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ gives $c = 8$ and 248 level-1 currents (one primary); the μ_4/GSO promotion $248 = 240 + 8 = 120 + 128$ ($240 = 16 \cdot 5 \cdot 3$) pins $(E_8)_1$ over the same- c rival $SO(16)_1$ (120 currents, 4 primaries). Python (sympy q-series).
- **`v378 (SEAM.S3.MODULAR.01)`**. The genus-1 discriminator: KLM condensation $\mu(\text{carrier})=16 \rightarrow \mu(E_8)=16/4^2=1$, $|\det \text{Cartan}(E_8)|=1$ vs $D_8=4$, and the single character's modular T -eigenvalue $e^{-2\pi i/3}$ give torus ground-state degeneracy 1 (holomorphic = $(E_8)_1$) — the “hard part” `v344` isolated. Python (sympy).
- **`v379 (SEAM.S3.RP.01)`**. Reflection positivity (the OS axiom) of the explicit gapped collar: the positive spectral measure makes the Hankel/reflection Gram matrix PSD (a ghost mode breaks it). With $c=8 + (E_8)_1$ in hand, the OS inputs are complete and `C7/QG.AMB.01` is then discharged by the redundancy theorem `v369`. Python (numpy).
- **Net:** S3 is the master key — on the explicit lattice model the target net $(E_8)_1$ is pinned exactly (central charge, character, genus-1 count, RP); the only remaining residual is the abstract *continuum existence* of the scaling limit (the cited MMST theorem `v336`), which lives outside the computational suite. Cited in `tfpt_research_contracts`.

2026-06-23 (Track 4 — the prediction OBSERVATORY: a status-typed CI over the frozen prediction registry that makes the falsifiability surface machine-checkable, with a live data scorecard (`v375`))

- **`v375 (OBSERVATORY.REGISTRY.01)`**. A CI over `freeze_file.csv`: **[E]** every prediction carries a kill criterion (falsifiability complete, no orphan claims); **[E]** the headline values re-derive from $\{\varphi_0^{\text{ret}}\}$ ($\sin^2 \theta_{12}=1/3-\varphi_0^{\text{ret}}/2$, $\sin^2 \theta_{13}=\varphi_0^{\text{ret}}e^{-5/6}$, $\beta_{\text{rad}}=\varphi_0^{\text{ret}}/(4\pi)$); **[C]** a live scorecard records θ_{12} vs the JUNO first run (0.28σ , compatible), θ_{13} vs NuFIT 6.0 NO (2.0σ , the flagged most-tensioned core prediction), β_{rad} vs ACT DR6 (0.37σ , a hint), $r < 0.036$ (compatible). Tooling that makes the closed pieces usable and the open ones honestly fenced; no new physics. Cited in `tfpt_5_redteam`; Python (stdlib).

2026-06-23 (Track 3 — the F_{transfer} functor promoted from contracts to TYPED runnable solvers with kill tests: Koide source→pole (v371), η_B Boltzmann (v372), axion finite- T relic (v373), m_p/m_e uncertainty budget (v374))

- **v371 (FR.POLE.SOLVE.01).** A clean *negative*: standard QED/EW running cannot be the Koide source→pole transfer (it pushes Q above $2/3$; the required ϵ is negative while QED's is positive), so the transfer is the non-QED $53/54$ operator / $(2/3)^6$ Moebius generator (v183/v82). [C], plain running excluded as the kill. Python (numpy+scipy).
- **v372 (FR.BOLTZMANN.SOLVE.01).** The integrated Buchmueller–Di Bari–Pluemacher Boltzmann ODE at the TFPT-frozen $M_1=M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ (v212) gives η_B within a factor ~ 1.1 of 6.1×10^{-10} with no M_R dial; replaces the v326 toy washout. [C]/[X].
- **v373 (FR.RELIC.SOLVE.01).** The finite- T axion misalignment ODE decides the branch: the frozen spine angle $3\pi/5$ gives $\Omega_a h^2 \sim 0.12$ (in $[0.08, 0.16]$, lands on Ω_{DM} untuned) while the hilltop 170° over-produces ($\sim 5.4\times$); replaces the v326 toy. [C]/[X] (lattice $\chi(T)$ input).
- **v374 (FR.QCD.BUDGET.01).** The m_p/m_e uncertainty budget: with $b_3 = -7$ [E] and the two declared externals ($\alpha_s(M_Z)$, C_p , $\sim 5.3\%$ each), the central ~ 1824 (v262) carries a propagated $\pm 7.5\%$ band [$\sim 1690, \sim 1960$] CONTAINING 1836.15; lattice tightening is the kill. The falsifiability companion to v262; [C].

2026-06-23 (Track 1 — the S3 input made CONCRETE: an explicit gapped chiral-Majorana ($p+ip$) lattice model with numerical Chern $|C|=1$ realises the $c_-=8$ edge (v367), and a strip shows the chiral edge exists by anomaly inflow (v368))

- **v367 (SEAM.S3.LATTICE.01) — Track 1.** The abstract S3 residual of v356/v366 (“the collar realised as a genuine lattice chiral free-fermion invertible phase”) is turned into a RUNNABLE model. [E] the $p+ip$ Bloch model is gapped at $M=1$ with a numerically computed Fukui–Hatsugai–Suzuki Chern number $|C|=1$ ($C=0$ trivially at $M=3$); [E] $N_{\text{Maj}} = \dim S^+ = 16$ copies give $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$, with the edge K-matrix $SO(16)_1$ ($\det 4$) $\rightarrow (E_8)_1$ ($\det 1$) by the order-4 μ_4 clock (v351/v369). [C] the μ_4 condensation (v154/v351); [O] the continuum scaling-limit existence (MMST, v336) stays open. Cited in `tfpt_research_contracts`; Python (numpy + sympy).
- **v368 (SEAM.S3.INFLOW.01) — Track 1.** The bulk-edge / anomaly-inflow leg (S4) made concrete on the v367 model: [E] a finite strip has an edge-localised in-gap chiral branch ($\min_{k_x} |E| \sim 0$) in the topological phase and a clean gap in the trivial phase (neg control); [E] bulk-edge correspondence gives one chiral Majorana per edge ($c_-=8$). [C] so by anomaly inflow the chiral edge EXISTS as a consequence of the invertible bulk, discharging the v336/v351 “does the edge exist?” residual on the lattice. [O] the continuum scaling limit stays open. Cited in `tfpt_research_contracts`; Python (numpy).

2026-06-23 (Track 2 — perturbative graviton unitarity sector-by-sector: the Barnes–Rivers spin decomposition localises the Stelle ghost in the spin-2 sector, cured by the entire KMS form factor; the spin-0 mode by the GHP contour (v370))

- **v370 (GRAV.SPIN2.UNITARITY.01) — Track 2.** Extends v304 (the scalar $1/(p^2(p^2+M^2))$ pole algebra) to the actual tensor spin structure. [E] the Barnes–Rivers projectors are complete ($\text{tr } P_2 = (d-2)(d+1)/2$, $\text{tr } P_1 = d-1$, $\text{tr } P_{0s} = \text{tr } P_{0w} = 1$ sum to $d(d+1)/2$; $d=4$ split $5+3+1+1=10$), the EH propagator is $P_2/p^2 - \frac{1}{d-2}P_{0s}/p^2$, and the Stelle ghost is a *spin-2* state. [E] the entire KMS form factor e^{p^2/M^2} (v304/v259) makes the spin-2 sector ghost-free; [C] the wrong-sign spin-0 conformal mode (v332) is cured separately by the GHP contour

(v334). [C] so the full *perturbative* graviton is ghost-free sector-by-sector (conditional on v304’s entire-analyticity assumption). [O] residual: perturbative-only + the open analyticity assumption; not the non-perturbative measure. [O] DECLINE: $\text{tr } P_2=5$ is dimension counting, not g_{car} (v354/v355). Cited in `tfpt_5_redteam` (Target F); Python-only.

2026-06-23 (Directions 3 & 7 — the two hard functional-analysis residuals ADVANCED: the QG measure as one-loop fluctuations around the parameter-free saddle (v365), and the seam collar verified IN MMST’s free-fermion class so the scaling limit is $(E_8)_1$ (v366))

- **v365 (QGAMB.SADDLE.01) — Direction 3.** With the parameter-free saddle (v358/v359) the QG *measure* (gate C7/QG.AMB.01) organises as a one-loop Gaussian fluctuation determinant. [E] the fluctuation stiffness is $1/c_3 = 8\pi$ (no free dial, so the one-loop problem is now well-posed); [E] the fluctuation operator is gapped $M \geq \Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ mode-by-mode convergence, finite model $\log\text{-det } \text{Tr}' \log M = 6 \log \frac{9}{2}$; [C] the conformal mode converges on the GHP contour (v334); [E] the admissible projective limit exists ($\chi = 729/665$, v330) with 0 new dials (v364). [O] residual: the full non-perturbative projective limit on the ambient sector. Sharpens C7 from “diffuse full QG” to a one-loop Gaussian problem around the parameter-free saddle; does not close it. Cited in `tfpt_research_contracts`; Python-only.
- **v366 (SEAM.MMST.INCLASS.01) — Direction 7.** The seam collar is verified IN MMST’s free-lattice-fermion scaling-limit class hypothesis-by-hypothesis: [E] $D = \dim S^+ = 16$ Majoranas, $c = g_{\text{car}} + N_{\text{fam}} = 8$, $\text{rank } E_8 = 8$ so $\text{rank} \leq c \leq D$ is $8 \leq 8 \leq 16$ (in range); [E] gapped ($\Delta = 6 \log \frac{3}{2} > 0$, derived v302); [C] quasi-free CAR class (v155/v160); [E] chiral ($c_- = D/2 = 8 \neq 0$). [C] the MMST scaling-limit theorem then applies, and [E] the order-4 μ_4 clock pins the target to $(E_8)_1$ ($\det K=1$ vs $SO(16)$ $\det K=4$, v351). [O] residual: the single S3 input (the collar as a genuine lattice chiral invertible phase, the seam one-sidedness, v297/v356). Reduces the continuum side of SEAM.EQUIV.01 to one named input; ADVANCES, does not close. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Consistency pass: the parameter-free *full covariant* Einstein equation (v358/v359) propagated across ALL stale paper and website surfaces; new v359 keybox; homepage gravity animation)

- **Deep-sync of the parameter-free gravity status.** A cross-surface audit found several documents and website components still framing gravity as “exact only in the $R + R^2$ regime” or the metric-sector field equation as “open”. These were reconciled with the canonical post-v358/v359 status: the *local* covariant field equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ is PARAMETER-FREE (both coefficients fixed); what remains open is the Jacobson equation-of-state status, the global ambient measure (QG.AMB.01/C7), and the single unit v_{geo} . Touched: `introduction`, `tfpt_2_standard_model`, `tfpt_3_e8_audit_bootstrap`, `tfpt_4_frontier` (summary table), `tfpt_research_contracts`, `origin_theory`.
- **New v359 keybox** added to `tfpt_research_contracts` (full covariant equation: fixed-volume $\Rightarrow$ Einstein tensor, Lovelock $\Rightarrow$ conservation an output); the v358 residual corrected from “linear $\rightarrow$ nonlinear” (stale — v359 closed it) to the equation-of-state + C7 + v_{geo} residual, including the generated script index (`script_registry.csv`).
- **Website.** The GravityEmergence animation (three origins of c_3 converging to $1/(8\pi)$; the parameter-free Einstein equation) was updated to the full covariant result and surfaced as a dedicated homepage section (`/#gravity`); gravity-status wording aligned across `ReconstructionChain`, `VerificationDag`, `StatusMatrix`, `WhatTfptClaims`, `ThreeDecoderMap`, `StatusPyramid`, `papers.ts`, `glossary.ts` and the review page.

2026-06-23 (Direction 6, v_{geo} sharpened: after the parameter-free gravity, the gravity sector adds NO new dimensionful input either, so v_{geo} is the SINGLE dimensionful input of the complete theory — final tally $\{v_{\text{geo}}, \pi\}$, ZERO dimensionless dials: v364)

- **v364 (VGE0.SHARPEN.01) — Direction 6.** The No-Unit Theorem (v153) already made v_{geo} a forced metrology primitive. [E] The SHARPENING after the parameter-free gravity (v358/v359/v361): the Einstein coefficient $1/c_3 = 8\pi$ and the Λ prefactor $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\text{geo}}^2$ and $\Lambda \sim (\text{dimensionless}) v_{\text{geo}}^4$ reduce to v_{geo} times atoms — the gravity sector adds no new scale. [E] FINAL TALLY: every dimensionful quantity = (a dimensionless ratio) $\times$ (a power of v_{geo}); TFPT's free content is $\{v_{\text{geo}}, \pi\}$ with ZERO dimensionless dials, a $\sim 26 \rightarrow 1$ reduction vs the SM. [O] v_{geo} stays irreducibly primitive (No-Unit; $1/G$ UV-sensitive, Sakharov v68). Direction 6 is clarity — the last dimensionful anchor named and shown to be the only one. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Direction 5, the matter–gravity backreaction: the carrier's gravitational anomaly is forced $c_- = 8$, and the matter vacuum-energy backreaction is finite ($\rightarrow$ the forced Λ from α , not M_{Pl}^4) — the loop closes; no new SM-quantum-number read-out (declined): v361)

- **v361 (GRAV.BACKREACT.01) — Direction 5.** With the parameter-free equation (v359) and the explicit J3 flux (v358), the backreaction of TFPT's carrier matter is computed. [E] the carrier's gravitational anomaly is forced: $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ (the chiral central charge of the 16-Majorana content). [E] the backreaction is FINITE $\rightarrow \Lambda$ from α : the matter vacuum energy through $G_{ab} = c_3^{-1} T_{ab}$ is gap-suppressed (Decoupling Thm v337) to $\rho_{\Lambda}/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ (v60), the 123 orders = $119.03 + 3.92$ — *not* the M_{Pl}^4 catastrophe; the loop (v359+v60 + the carrier vacuum) closes. [E] J3 explicit: $\delta\langle K_B \rangle = (8\pi^2 R^4/15)\delta\langle T_{00} \rangle$, carrier T_{ab} = the SM's (v159). [C] the SM trace anomaly is reproduced (content = SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond $c_- = 8$ and Λ — per v354/v355 declined. The backreaction is finite and consistent, adding no new dial. Cited in `tfpt_4`; Python-only.

2026-06-23 (Direction 2, the disciplined result: the gap-induced GR correction is the known R^2 scalaron — already falsifiable via n_s/r ; a *new* forced near-term deviation does NOT survive the discriminator and is declined: v360)

- **v360 (GRAV.GAPCORR.01) — Direction 2, run honestly.** Does the seam transfer gap $\{1, (2/3)^6, (1/3)^6\}$ yield a NEW, calculable, FORCED deviation from GR (a near-term kill test) beyond the known R^2 scalaron? With the v354/v355 discriminator the answer is *no*, and this is recorded rather than manufacturing a coefficient. [E] the leading higher-curvature term IS the R^2 scalaron (spectral-action $a_4 = R^2/72$, v36; $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^7$, exponent $7 = g_{\text{car}} + N_{\text{fam}} - 1$) — the calculable beyond-Einstein correction, already in TFPT. [C] it is the entanglement-equilibrium NEXT order (the Wald first law $\rightarrow f(R) = R + R^2$, v28/v359). [E] it is ALREADY falsifiable via $n_s = 1 - 2/N_*$, $r = 12/N_*^2$ (live CMB-S4 kill test). [E] the gap $\Delta = 6 \ln \frac{3}{2}$ is a dimensionless rate, so a distinct gap-induced curvature² term sits at $\xi v_{\text{geo}} \sim$ the seam/Planck scale (not near-term observable). [O] DECLINE: a new dimensionless coefficient is not atom-forced (the scalaron exponent $7 = g + N - 1$ and the gap exponent $6 = 2N_{\text{fam}}$ are different; c_3^7 and $(2/3)^6$ unrelated) — per v354/v355 declined. Direction 2 yields no new near-term forced prediction; the answer is the existing scalaron. Cited in `tfpt_4`; Python-only.

2026-06-23 (the FULL covariant Einstein equation, parameter-free: the fixed-volume entanglement equilibrium gives $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with BOTH coefficients TFPT-fixed ($8\pi = 1/c_3$ from v358, Λ from α via v60); B6 closed at the local level: v359)

- **v359 (GRAV.NONLINEAR.01) — Direction 1, the full covariant equation.** Extends v358 from the linearised (fixed-radius, Ricci-level) result to the FULL covariant equation. [C][E] Jacobson 2015: demanding stationarity of the small-ball entanglement entropy at fixed *volume* brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}) — trace $g^{ab}G_{ab} = (1 - d/2)R = -R$ in $d=4$, the structure that carries Λ . [E] Lovelock: G_{ab} is the unique symmetric divergence-free second-order metric tensor, so $\nabla^a(G_{ab} + \Lambda g_{ab}) = 0$ forces $\nabla^a T_{ab} = 0$ — conservation is an output. [E] coefficient 1: $8\pi G \rightarrow 8\pi = 1/c_3$ fixed (v358). [E] coefficient 2: Jacobson leaves Λ free, TFPT fixes it — $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$, prefactor $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$ (v60); so BOTH coefficients are TFPT-fixed, not free integration constants. [C] all timelike directions $\rightarrow$ the covariant tensor equation; the matter side is the assembled CHM ball modular flux (v358/v323). NET: the original B6 “full covariant Einstein-side field equation” is now PARAMETER-FREE at the local level. [O] residual: the Jacobson equation-of-state status (not a from-action quantisation) + the global ambient measure (C7/QG.AMB.01, separate, gap-decoupled); v_{geo} the one unit. Cited in `tfpt_research_contracts`; mirrored in Wolfram (303/303). (next-plan.md Direction 4 folded in: the Λ here matches the α -readout.)

2026-06-23 (the classical gravity field equation, parameter-free: the entanglement first law $\delta S = \delta\langle K \rangle$ with TFPT’s atoms gives the LINEARISED Einstein equation with G from c_3 ; the thermodynamic and geometric origins of c_3 coincide via $|\mu_4| = |\mathbb{Z}_2|\chi = 4$; the two formerly-open pieces (matter flux J3, entropy-density coefficient) closed/atom-fixed: v358)

- **v358 (GRAV.ENTROPY.EQUILIBRIUM.01) — the bridge to the dynamic world, instantiated on the gravity side.** Carrying out the Jacobson-2015 / Faulkner-et-al. “first law of entanglement $\Rightarrow$ Einstein equation” with TFPT’s atoms. [E] PARAMETER-FREE COEFFICIENT: $\delta S = \delta\langle K \rangle$ gives $G_{ab} = c_3^{-1} T_{ab}$ with $c_3^{-1} = 8\pi$ fixed; all four c_3 -roles consistent (Unruh $1/(2\pi)=4c_3$, EH $1/(16\pi)=c_3/2$, Einstein $8\pi=1/c_3$, Bekenstein $1/4=1/|\mu_4|$) — no free dimensionless Newton dial. [E] THERMO = GEO COINCIDENCE (new): the first law’s coefficient $2\pi/\eta$ equals the geometric $|\mathbb{Z}_2|2\pi\chi$ iff $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ (v216) — the thermodynamic and geometric origins of c_3 are ONE. [C] J3 SOLVED: the CHM ball modular Hamiltonian (boost via BW, v323) gives the matter flux $\delta\langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta\langle T_{00} \rangle$. [E] ENTROPY DENSITY ATOM-FIXED: $1/4=1/|\mu_4|$ (v57) + central charge $c = g_{\text{car}} + N_{\text{fam}} = 8$ (form v59). [O] residual: linear $\rightarrow$ full non-linear (the original B6) + the v_{geo} unit. Closes the LINEARISED Einstein equation parameter-free; cited in `tfpt_research_contracts`; mirrored in Wolfram (302/302).

2026-06-23 (the two frontier walls attacked directly: Direction A assembles the continuum chain of SEAM.EQUIV.01 theorem-by-theorem and ties its residual to the seam one-sidedness; Direction B was found ALREADY closed by v40 (harmonic metric = finite linear algebra, not a PDE): v356)

- **v356 (SEAM.EQUIV.CONTINUUM.03) — Direction A, the continuum chain assembled.** After the reverse-audit/bandwidth work showed no hidden *discrete* lever remains, the only fundamental progress is to assemble the *continuum* chain. Link-by-link: **S1** the gapped ($\Delta=6\ln(3/2)>0$, v302) quasi-free 16-Majorana collar (v155/v160) $\rightarrow$ **S2** invertible (gapped free fermions have no intrinsic topological order, Kitaev, v301) $\rightarrow$ **S3** the *nontrivial* invertible phase ($c_- = 8 \neq 0$) $\rightarrow$ **S4** a *non-gappable* anomalous chiral edge (anomaly inflow / bulk-edge), so edge *existence* is a CONSEQUENCE of S2+S3, not a separate assumption — this directly

reduces the v336/v351 “does the collar have a chiral edge?” residual $\rightarrow$ **S5** its chiral-CFT scaling limit (MMST, v336; range rank $\leq c \leq D$, $8 \leq 8 \leq 16$) $\rightarrow$ **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (v351/v297). [E] the only non-theorem link is S3, and [C] S3 is the seam being *one-sided* (chiral, $c_- \neq 0$) — the *same* one-sidedness that makes $c_3=1/(8\pi)$ a one-sided Gauss–Bonnet $(|\mathbb{Z}_2| \nmid K, \text{v58})$. [O] the one remaining input is “the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase” (the v297 Flat-Away). NET: SEAM.EQUIV.01 reduced to ONE realization input tied to the constant that *defines* the theory; it ADVANCES, it does not close.

- **Direction B — found ALREADY closed by v40.** The U-wall “transcendental Hitchin solve” (v31) is the worst case for a *generic* Higgs bundle; but at the *polystable* point TFPT selects, Mehta–Seshadri makes the bundle *unitary*, so the harmonic metric is the constant invariant Hermitian form (*finite linear algebra*, Higgs $\Phi=0$) and $\det R=8$ is read off — not a PDE. Only the absolute amplitude U_{point} (an anchor) remains. A redundant new B module was built, found to duplicate/contradict v40, and *deleted* (no padding). Honest outcome: B needs no new work.

2026-06-22 (bandwidth search of the unmapped E_8 region, with a strict forced-identity discriminator: the genuine overlooked structure is COLLECTIVE ($\sum \deg = 128 = \dim S^+$, $\prod \deg = |W(E_8)|$, exponents = totatives of 30); the per-degree atom matches are unforced and DECLINED; no new physical hit – reconciling v66 and v354: v355)

- **v355 (E8.UNMAPPED.BANDWIDTH.01) — the next step after v354, run honestly.** The obvious follow-up (“search those five degrees for hits we overlooked”) is itself the numerology temptation v354 warns about: a rich object like E_8 returns a numerical match for almost anything. So the scan uses a STRICT discriminator – a candidate counts only if it is a FORCED identity (a theorem about E_8), never a coincidence. [E] FORCED, SET-LEVEL (the genuinely overlooked structure, collective not per-number): $\sum \deg = 128 = \dim S^+$, the spinor half of $248 = 120 + 128$ (theorem: $\sum \deg = \# \text{pos roots} + \text{rank} = 120 + 8$) – the invariant budget of E_8 equals the fermion content the carrier reads; $\sum \exp = 120 = \# \text{pos roots}$; $\prod \deg = |W(E_8)| = 696,729,600$; exponents = the $\varphi(30)=8$ totatives of 30 (v223), so the unmapped degrees’ exponents are part of the forced set; $\max \deg = 30 = h(E_8)$. [C] SUGGESTIVE, not a theorem: $12=h(E_6)$, $18=h(E_7)$, $30=h(E_8)$ are all degrees of E_8 (the $2T/2O/2I$ McKay ladder, v219), but “ $\deg(E_8)$ contains its subalgebras’ Coxeter numbers” is not a general theorem – flagged, not claimed. [O] DECLINED (the numerology temptation): the per-degree atom matches $14=\dim G_2$, $20=\det L$, $24=|W(A_3)|$, $12=|R(A_3)|$, $18=p_4(a)$ (v66 already flagged 12, 18 as fittable) each admit ≥ 2 readings, and the lone extra prime in $|W(E_8)| = 2^{14}3^55^27$ “=” the scalaron exponent is one of many readings of 7 – all declined. [E] RECONCILES v66 (the 8 degrees coincide with the load-bearing integers) and v354 (only 2, 8, 30 are primary readouts): the search finds NO new forced physical hit, and the disciplined decline IS the anti-numerology result. (This also sharpens v354: the five are “no PRIMARY readout / hull overhead”, not literally “unmapped” – they coincide with structural atoms, but unforced.) Cited in `tfpt_5_redteam` (with a full audited degree table); mirrored in Wolfram (301/301).

2026-06-22 (the REVERSE numerology audit, and what the golden ratio means: of E_8 ’s 8 Casimir invariants only 3 feed a primary readout and 5 are hull overhead; the golden ratio is UNMAPPED structure, so it cannot be numerology: v354)

- **v354 (E8.REVERSE.AUDIT.01) — the reverse complement of v305.** TFPT’s forward discipline (v305) is one-directional: every physical READOUT must map onto an E_8 structure (anti-fitting). The sharp adversarial question runs the OTHER way — how much E_8 structure

maps onto NOTHING, and is the recurring golden ratio numerology? [E] THE REVERSE AUDIT: of E_8 's eight Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ (exponents $+1$), exactly THREE feed a readout — 2 (metric), 8 (rank $\rightarrow c_3=1/(8\pi)$), 30 (Coxeter $\rightarrow g_{\text{car}}=5$ and the order-30 cycle) — and FIVE carry no primary readout $\{12, 14, 18, 20, 24\}$ (they coincide with structural atoms per v66, but unforced – reconciled in v355); so 3/8 are primary readouts, 5/8 hull overhead. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, $h=30$, $\det K=1$, $D_5 \oplus A_3$, the 16 half-spinor) generates the ENTIRE readout set, and the higher Casimirs are never reached into to fit — a numerological use would be PROMISCUOUS (mine the rich structure), TFPT is ECONOMICAL (few invariants, many readouts) and leaves most of E_8 untouched, the anti-numerology signature being precisely the large unused remainder. [E] THE GOLDEN RATIO IS UNMAPPED $\Rightarrow$ NOT NUMEROLOGY: $\varphi = 2 \cos(\pi/5)$ appears only in internal structure (the affine- E_8 attractor eigenvalue v312, the icosian lattice v348) and in NO physical readout; numerology means a number FITTED to an observation, φ is fitted to nothing — it is the algebraic signature of the 5-fold ($h=30=2 \cdot 3 \cdot 5$), the carrier announcing it is icosahedral. [O] the 5/8 unmapped is the HULL OVERHEAD (E_8 is the minimal even-unimodular hull, not the world group); where a future higher-invariant readout could live, or genuine over-capacity — stated plainly, not hidden. Cited in `tfpt_5_redteam`; mirrored in Wolfram (300/300).

2026-06-22 (the bird's-eye rethink: TFPT is a unique self-consistent LOOP with ZERO free dials, not a linear axiom system; π is a fixed geometric constant, not a dial; the one residual is the continuum closure: v353)

- **v353 (TFPT.SELFLOOP.01) — refining v352.** Stepping all the way back: TFPT is not a LINEAR theory (axioms $\rightarrow$ theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs. [E] it is a loop: $g_{\text{car}}=5 = \max$ prime of the output $h(E_8)=30$, and the 8 in $c_3 = \text{rank } E_8$ – the outputs fix the inputs (v6), with a unique fixed point (v56); “which is the axiom?” is the wrong question for a loop. [E] ZERO free adjustable dimensionless dials: every load-bearing number is forced/over-determined. [E] π is a FIXED geometric constant (the boundary 2-sphere, $\oint K=4\pi$), not a dial. [C] this refines v352's “framework + π ”: π is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom – the honest characterisation is “a unique self-consistent loop with zero free dials”. [O] the one residual is the CONTINUUM closure of the loop (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork – no free numbers either way. A capstone synthesis.

2026-06-22 (two consequences: the $c=8$ which-net ambiguity resolved by the order-4 clock, and BOTH axioms shown bootstrap-forced so the only irreducibles are the framework + π : v351, v352)

- **v351 (SEAM.EQUIV.CONTINUUM.02) — the which-net ambiguity falls.** The long-standing “ $c=8$ ambiguity” ($E_8 \det 1$ vs $SO(16)=D_8 \det 4$, both $c=8$, flagged open in v277/v344) is RESOLVED, non-circularly, by the seam's ORDER-4 μ_4 clock: the index-4 simple-current extension of $D_5 \oplus A_3$ is E_8 (v154), the index-2 would be $SO(16)$, so the order-4 clock (the four Gauss–Bonnet marks, v216) selects E_8 (full μ_4 condensation $16 \rightarrow 4 \rightarrow 1$) not $SO(16)$ (partial $\mathbb{Z}_2$ $16 \rightarrow 4$). The bootstrap agrees ($h(E_8)=30$ max prime $5=g_{\text{car}}$ vs $h(D_8)=14$ max prime 7). [E] so $\det K=1$ is a consequence, not an open input. [O] the residual of SEAM.EQUIV.01 is then PURELY the chiral-edge / scaling-limit existence (v336), not which-net.
- **v352 (TFPT.IRREDUCIBLE.01) — the inputs are not axioms.** The deepest answer to “the inputs are back-determined”: BOTH P1 and P2 are bootstrap-forced fixed points. [E] $g_{\text{car}}=5$ is forced three ways (v6); the 8 in c_3 four ways (rank $E_8 = h(D_5) = \phi(30) = \det R$, v54); the E_8 hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles

are the FRAMEWORK (a discrete reflection-positive boundary with a finite carrier) and the transcendental π . [O] the framework is the META-AXIOM — not math-reducible, the foundational stance (like “spacetime is a manifold”), minimal, tested by its consequences not proved. TFPT’s true input is ONE framework + π . Wolfram mirror +2 (297 → 299).

2026-06-22 (correction: the inputs are bootstrap-forced fixed points, not free axioms – $g_{\text{car}}=5$ IS the icosahedral 5 of $h(E_8)=30$, and the golden ratio is emergent from the μ_4 -glue, not external: v350)

- **v350 (SEAM.EQUIV.BOOTSTRAP.01) — correcting v349.** An honest correction prompted by the right objection that the inputs are not axioms but are back-determined by the theory. [E] the inputs ARE bootstrap-forced (v6): $g_{\text{car}}=5$ is over-determined three ways – rank-fill ($g_{\text{car}}+N_{\text{fam}}=8$), Coxeter-match ($g_{\text{car}}=\max$ prime of $h(E_8)=30$), Pascal ($2^4=C(5,0)+C(5,1)+C(5,2)$). [E] so $g_{\text{car}}=5$ IS the icosahedral 5-fold (the 5 in $h(E_8)=30=2\cdot3\cdot5$), not the D_5 rank; the atoms (2,3,5) ARE the primes of the icosahedral Coxeter number 30. [E] the golden ratio is EMERGENT from the glue: D_5 ($h=8$)/ A_3 ($h=4$) are non-golden alone, but $D_5\oplus A_3 \rightarrow E_8$ (μ_4 index-4, v154) gives $h(E_8)=30$ (golden) – the seam’s own μ_4 produces the 5-fold, so v349 checked the un-glued pieces, the wrong object, and its pentagon-vs-count dichotomy was false. [O] the residual relocates: the golden is bootstrap-forced, the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Honest caveat: self-consistency within the framework, not from-nothing. Wolfram mirror +1 (296 → 297).

2026-06-22 (the honest test – does the raw seam carry φ ? – answer NO: the bare carrier gives $\sqrt{2}$, golden is the icosahedral input; the keystone reduces to “is $g_{\text{car}}=5$ a pentagon or a count?”: v349)

- **v349 (SEAM.EQUIV.GOLDEN.01).** The honest, computed test of the v348 question, and a clarifying NEGATIVE: the raw seam does *not* carry the golden ratio. [E] the golden ratio $2\cos(\pi/5)$ is the signature of a 5-fold rotation ($5 \mid h$); the carrier D_5 has $h=8$ (eigenvalue $\sqrt{2}$) and A_3 has $h=4$ – no 5-fold; the dynamical data (cusp $\{0, 1/3, 2/3\}$, recovery $(2/3)^6$, seed φ_0) are rational/transcendental. [E] $5 \mid h$ holds only for $A_4=SU(5)$ ($h=5$), H_3 ($h=10$), E_8 ($h=30$) – the output/icosahedral side, so φ is the icosahedral INPUT. [E] φ alone would not suffice ($2I \neq C_5 \times \mu_4$, $20 \neq 120$). [O] the resolution: the keystone reduces to one question – is $g_{\text{car}}=5$ a PENTAGON (a 5-fold rotation $\Rightarrow$ golden $\Rightarrow$ icosahedral $2I = E_8$) or just a COUNT (the rank of $D_5 \Rightarrow \sqrt{2}$)? The “absurdly simple solution” is real but is a *reading* of axiom P2 (the golden ratio is the pentagon content of $g_{\text{car}}=5$, not derivable from $g_{\text{car}}=5$ -as-a-count); there is no hidden φ . L2 is foundational: P2-as-pentagon closes it (mode C), else a rigidity theorem must produce the 5-fold (mode B); not closed. Wolfram mirror +1 (295 → 296). A NEGATIVE/clarification; prevents a false closure.

2026-06-22 (Route B carried out: rigidity + the icosian-ring identity reduce the whole keystone to ONE question – does the raw seam carry the golden ratio? – the simplest form of L2: v348)

- **v348 (SEAM.EQUIV.RIGID.01).** Executes the rigidity route (mode B of v347) and lands on a strikingly simple statement. [E] rigidity closes the crystallographic part: the order-4 clock has the unique normal form $z \mapsto iz$ (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] the downstream is a LATTICE identity, not just a graph: by Conway–Sloane the ring of icosians (120 unit icosians = $2I$, golden-ratio quaternion norm) IS the rank-8 E_8 lattice. [E] the single bridge is the golden ratio: the crystallographic restriction allows only orders $\{1, 2, 3, 4, 6\}$ (the 5-fold absent), so $\varphi = 2\cos(\pi/5)$ is exactly what extends the crystallographic μ_4 to the non-crystallographic icosahedral $2I$ (the quasicrystal). [O] so

the entire keystone reduces to ONE question: *does the raw seam carry the golden ratio?* – if yes, $\mu_4 + \varphi \Rightarrow 2I \Rightarrow E_8$ and rigidity fixes the rest. [O] honest subtlety: TFPT’s φ is currently E_8 -side (v312/v313); the raw-seam data so far are not manifestly golden, so showing the RAW seam carries φ non-circularly is the residual (= L2, sharpest form). Wolfram mirror +1 (294 $\rightarrow$ 295). An analysis/reduction; closes nothing.

2026-06-22 (can the one open arrow be *fully* solved? the closure-mode classification: the flat $\rightarrow$ spherical base locus, four modes, no free theorem – only rigidity or axiom P_3 : v347)

- **v347 (SEAM.EQUIV.CLOSURE.01).** A direct, honest answer to “is there any other way to fully solve the one open arrow L2?”. [E] the precise locus: the seam gives the *flat* pillowcase base $S^2(2, 2, 2, 2)$ ($\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$, v216) while the $2I/E_8$ structure is the Seifert base over the *spherical* $S^2(2, 3, 5)$ ($\chi_{\text{orb}} = 2 - (\frac{1}{2} + \frac{2}{3} + \frac{4}{5}) = \frac{1}{30}$), so L2 bridges two geometric types – why it needs the carrier $(2, 3, 5)$ input. [E] as a “physics-realises-geometry” statement L2 has exactly four closure modes: A construct (the analytic wall), B rigidity (a unique-realisation theorem – the only theorem-route w/o new input, pursued v284/v300/v331), C axiom P_3 (then TFPT is complete relative to 3 axioms), D empirical. [E] the “no new state” discipline (v246) blocks deriving L2 from extra physics, so no route makes it a free theorem. [E] verdict: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D); only B closes it as a theorem. [O] NOT closed – classifies the modes, locates the difficulty. Wolfram mirror +1 (293 $\rightarrow$ 294). An analysis/roadmap; closes nothing.

2026-06-22 (the geometric bridge made explicit: the keystone is ONE arrow – raw-seam $\Rightarrow \det K=1$ is a six-link chain, five theorems + one open orbifold realisation, the group forced by the $(2, 3, 5)$ atoms: v346)

- **v346 (SEAM.EQUIV.GEOM.01).** The honest capstone of the SEAM.EQUIV.01 investigation: it lays out the full path raw-seam $\Rightarrow \det K=1$ as six links and shows exactly one is open. [E]/[C] L1 raw seam $\rightarrow \mu_4$ + four marks (Gauss–Bonnet, v216); [O] L2 μ_4 + the $(2, 3, 5)$ atoms $\rightarrow$ the $2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE ONE OPEN ARROW = the geometric content of SEAM.EQUIV.01); [E] L3 $2I \rightarrow$ affine E_8 McKay (v219); [E] L4 $2I \rightarrow E_8$ du Val (v232); [E] L5 $\mathbb{C}^2/2I \rightarrow$ Poincaré link $H_1=0$ (v345); [E] L6 $H_1=0 \rightarrow \det K=1$. [E] five of six links are theorems; [E] the group is FORCED by the seam’s own atoms (the exceptional $SU(2)$ axis signatures $(2, 3, 3)/(2, 3, 4)/(2, 3, 5) \rightarrow 2T/2O/2I$; $(2, 3, 5)$ unique to $2I$, and $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$ ARE those axes); [E] everything downstream of L2 is then forced. [O] the irreducible open content is L2 alone (“the raw seam normal bundle is an ADE orbifold point $\mathbb{C}^2/\Gamma$ ”) = SEAM.EQUIV.01; not closed, but the bridge is now demonstrably ONE arrow with the group pinned and all downstream proven.

2026-06-22 (executing route R3: the $(2, 3, 5)$ -rewrite attractor link computed to be the Poincaré homology sphere ($\det K=1$), the combinatorial core complete, only the geometric bridge open: v345)

- **v345 (SEAM.DETK.02).** Carries out the combinatorial core of the hypergraph-homotopy route (R3, the most promising fresh angle from v344) as far as it computably goes. By plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has $H_1 = \text{coker}(\text{Cartan})$. [E] the Smith normal form of the E_8 Cartan matrix is the identity, so $\text{coker}(E_8) = 0$ – the E_8 -graph plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of $\det E_8=1$); among ADE only E_8 does this ($A_n \rightarrow \mathbb{Z}/(n+1)$, $D_n \rightarrow \mathbb{Z}/4$, $E_6 \rightarrow \mathbb{Z}/3$, $E_7 \rightarrow \mathbb{Z}/2$). [E] $\pi_1 = 2I = SL(2, 5)$ is PERFECT (commutator subgroup = the whole order-120 group, computed directly), so $H_1 = 0$ and the link is THE Poincaré sphere, $|2I| = 120 = |\mu_4| h(E_8)$. So R3’s combinatorial half is

[E]-complete: the E_8 attractor link is forced to be the Poincaré sphere ($\det K=1$), uniquely among ADE. [O] what stays open is only the geometric realisation bridge (the raw rewrite seam attractor *is* this E_8 du Val plumbing) = the SEAM.EQUIV.01 content; not closed, but the keystone is now reduced to that single geometric identification. Wolfram mirror +1 (292 $\rightarrow$ 293).

2026-06-22 (the one keystone bit $\det K=1$ mapped: six equivalent faces, the genus-1 obstruction, three access routes, the hypergraph-homotopy route the most promising fresh angle: v344)

- **v344 (SEAM.DETK.01).** A bird's-eye, full-spectrum map of SEAM.EQUIV.01's only open residual, the holomorphy bit $\det K=1$. [E] it is ONE statement with six equivalent faces (holomorphy / homology-sphere link $H_1=0$ / one 1-dim irrep / $\det \text{Cartan} = 1$ / full μ_4 condensation / $\tau=i$ CM), unified by $|\det \text{Cartan}| = |H_1(\text{link})| = \#(1\text{-dim irreps})$ across ADE ($A_n \rightarrow n+1$, $D_n \rightarrow 4$, $E_6 \rightarrow 3$, $E_7 \rightarrow 2$, $E_8 \rightarrow 1$) – only E_8 gives 1. [E] the GENUS-1 OBSTRUCTION explains why it is hard: every easy argument (plane vacuum, gap, free-fermion) is genus-0, necessary but not sufficient (v87); $\det K$ is the torus degeneracy. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the (2,3,5) rewrite attractor is the affine- E_8 graph (v312) and the E_8 du Val link is the Poincaré sphere (v232), so $\det K=1$ becomes a finite combinatorial statement (no continuum limit). [O] none closes it; the gap is the combinatorial-to-geometric bridge = the SEAM.EQUIV.01 content. A synthesis/roadmap; closes nothing.

2026-06-22 (the four black-hole-birth completion routes investigated: none closes all or is 100% provable alone; all four converge on SEAM.EQUIV.01 + the decoupled QG contour: v343)

- **v343 (FOUR.ROUTES.01).** A direct, honest answer to “which black-hole-birth route closes ALL problems and is 100% provable”: *none individually* — but the four converge on the same two residuals as v335. [E] (A) the finite causal diamond reframes QG.AMB to a finite thermal trace (de Sitter $S_{\text{dS}} = 32\pi^4 e^{2\alpha^{-1}}$ finite, $\chi = 729/665$) but inherits the GHP contour (v334, [O]); does not touch the seam. [C] (B) Carlip–Cardy gives a second route to existence+ c ($c=8$ consistent, but the Carlip ambiguity and no holomorphy $\det K=1$) — inherits SEAM.EQUIV.01. [E] (C) the modular/thermal flow (v239) is well-defined but its geometric content is QGEO.SYM.01, a corollary of SEAM.EQUIV.01 (v335) — a reduction. [E]/[O] (D) the unique attractor (v56) gives parameter-freeness but the cyclic dynamics is [C] interpretation. [E] **Verdict:** independent closures = 0; irreducible residuals after all four = 2 (unchanged from v335); the horizon premise makes both more natural but eliminates neither — focus = SEAM.EQUIV.01. An analysis module; closes nothing, fabricates nothing.

2026-06-22 (the heat-kernel origin of the α terms: the Calderón quadratic is the Gilkey gauge-curvature coefficient; the cubic Maxwell moment stays the EM.WARD.01 residual: v342)

- **v342 (EM.WARD.02) — the heat-kernel sharpening of EM.WARD.01.** Derives the ORIGIN and STRUCTURE of the $F_{U(1)}$ determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] (conjectured) to [E] for the term structure; no new gate. [E] $c_3 = 1/(8\pi)$ *is* the boundary Gauss–Bonnet coefficient ($\oint_{S^2} K = 2\pi\chi = 4\pi$, $\chi=2$, v216). [E] the α^2 (Calderón) term *is* the Gilkey gauge-curvature coefficient ($30\Omega^2/360 = \frac{1}{12}\Omega^2$, $\Omega=i\alpha F \Rightarrow -(\alpha^2/12)F^2$), so it is forced, not an ansatz. [E] the c_3 -orders $\{0, 3, 6\}$ are an arithmetic ladder and $2c_3^3 = 1/(256\pi^3)$ carries π^3 (three boundary insertions). [C] the exact coefficient needs the seam F -normalisation. [O] the cubic α^3

Maxwell moment (a topological/Chern level) + the full “this is the seam $U(1)$ ζ -determinant” identification stay the EM.WARD.01 residual, tied to SEAM.EQUIV.01. α^{-1} stays [E] (v3). Wolfram mirror +1 (290 $\rightarrow$ 291).

2026-06-22 (α as the stationarity of a $U(1)$ determinant line: every ingredient an index / heat-kernel coefficient / discriminant form, the open EM-Ward obligation named: v341)

- **v341 (ALPHA.QUILLEN.01).** Answers the external review’s sharpest valid point — “derive the form of α , not more decimals”. The EM closure $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6M \log(1/\varphi_{\text{seam}}) = 0$ is rewritten as a *stationarity* of the boundary $U(1)$ determinant line: $[\alpha^3 - 2c_3^3\alpha^2]_{\delta \log \det \Delta_{U(1)}} = [8b_1c_3^6]_{\text{counterterm}} \cdot [\log(1/\varphi_{\text{seam}})]_{\text{seam response}}$. [E] every ingredient is a NAMED object, not a knob: $M=41$ is a $U(1)$ index ($\frac{4}{5}M = 8b_1$, $b_1=41/10$ the SM $U(1)_Y$ beta, EM Ward v48); $c_3=1/(8\pi)$ the Gauss–Bonnet boundary coefficient ($8=\text{rank } E_8$); $-5/4 = -q(D_5)$ a discriminant form and $\varphi_{\text{base}}=1/(6\pi)=\frac{4}{3}c_3$ carries the other, $c_3/\varphi_{\text{base}}=3/4=q(A_3)$; $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4$. The Quillen split is verified at the $\alpha^{-1}=137.0359992$ root. **No new gate:** α^{-1} stays closed [E]; v341 adds no open item — it *sharpens* the pre-existing [O]/physical-origin obligation EM.WARD.01 (“why *this* function”, already conjectural). [O] the one still-open part *is* that residual: the from-first-principles AQFT proof that $F_{U(1)}$ is the EXACT Quillen functional (the variational principle derived, not assembled), unchanged in scope. Wolfram mirror +1 (289 $\rightarrow$ 290).

2026-06-22 (A_s reconciled with the archived derivation: predicted & dimensionless (so it does not define a mass), the tension is the reheating channel: v340)

- **v340 (AS.RECONCILE.01).** Answers “why don’t we have A_s / the old versions derived it” and “can a ratio define mass itself”. [E] A_s *is* predicted and *dimensionless*: $A_s = N_\star^2(M_{\text{scal}}/\bar{M}_{\text{Pl}})^2/(24\pi^2) = N_\star^2c_3^7/(24\pi^2)$ with $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = 1.26 \times 10^{-5}$ a pure ratio, so A_s pins the e-fold count $N_\star$, NOT $\bar{M}_{\text{Pl}}$ — it cannot define an absolute mass (No-Unit, v153, stands). [C] the -11.4σ at the slow Higgs channel ($N_\star=51.4$) is the slow-vs-fast reheating dichotomy; the A_s -preferred $N_\star=56.2 \sim$ the instantaneous bound. [C] the archived TFPT v4.5/v4.6 reported a clean $A_s = 2.124 \times 10^{-9}$ using the algebraic $N_\star = 3/\varphi_0^{\text{ret}} - 1 = 55.42$; at that $N_\star$ the current normalisation gives 2.047×10^{-9} (-1.9σ) — the old clean A_s is near-instantaneous reheating, not the slow channel. [E] nothing lost: the current theory replaced the posited algebraic $N_\star$ with one derived from reheating physics and exposed honestly that A_s requires fast (pre)heating. Record band $N_\star \in [50, 60]$ (v84) untouched.

2026-06-22 (the first-principles boundary, mapped: can M_{Planck} be computed? can F_{transfer} be first-principles? both NO, with the exact reason: v339)

- **v339 (FIRSTPRINCIPLES.BOUNDARY.01).** An honest boundary audit answering two head-on questions and fabricating nothing. [E] *The Planck mass cannot be computed* — $v_{\text{geo}}/M_{\text{Planck}}$ is forbidden by theorem (No-Unit, v153): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, v274); every dimensionless *ratio* is derived, only the one unit is not. [E] F_{transfer} *cannot be made first-principles* either: for m_p/m_e the structure is in place (v262, carrier $b_3=-7$ + closed electron) but two inputs stay external. [X] $\alpha_s(M_Z)$ is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread ~ 8.8 at 2×10^{16} GeV, v246). The lattice $C_p \sim 2.83$ is parameter-free but lattice-only (v35); η_B /axion ride on cosmological initial conditions; Koide $Q=2/3$ is structural [E]. [E] The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.

2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the θ_{13} budget: v336, v337, v338, FORM.CARTAN.DET.01)

- **v336 (SEAM.EQUIV.CONTINUUM.01) — the one open lemma, literature-anchored.** SEAM.EQUIV.01’s only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. [C] continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8, CMP **398** (2023) 219) give, building on the MMST (Morinelli–Morsella–Stottmeister–Tanimoto) operator-algebraic (wavelet) renormalization framework (CMP 387 (2021) 299; PRL 127 (2021) 230601), the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur $\rightarrow$ Virasoro, correct c); WZW range $\text{rank} \leq c \leq D$, and $(E_8)_1$ ($c=8$, rank 8, $D=16$ Majoranas, $SO(16)_1 \rightarrow (E_8)_1$) is inside it. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. [E] target pinned by $c=g_{\text{car}}+N_{\text{fam}}=8$, $\det \text{Cartan}(E_8)=1$. [O] narrowed residual = “the raw collar’s massless scaling limit is the holomorphic $(E_8)_1$ net” ($\det K=1$), not a from-scratch construction.
- **v337 (DECOUPLING.THEOREM.01) — the Decoupling Theorem.** [E] every dimensionless readout factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$; finite susceptibility $\chi = 1/(1 - (2/3)^6) = 729/665$ gap-suppresses the ambient contribution (margin $6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$, with $2 \dim(E_8)c_3^2 = 31/(4\pi^2)$ exactly), so no readout depends on the ambient measure $\Rightarrow$ QG.AMB.01 decoupled; negative control (gap closed $\Rightarrow \chi$ diverges) makes it non-vacuous. This is the theorem that makes the QG.AMB.01 reclassification (v335) honest; consolidates v76/v311/v330/v332.
- **v338 (THETA13.BUDGET.01) — the θ_{13} budget, fenced.** The $+2\sigma$ pressure point (v328) quantified against *both* NuFIT 6.0 NO variants: $+2.0\sigma$ vs the no-SK central (0.02195) but $+1.7\sigma$ vs the with-SK best fit (0.02215 ± 0.00057), a $\sim 0.3\sigma$ systematic spread; [E] 0.023108 lies inside the with-SK 3σ band (upper 0.02388), an upper-edge pressure point not a falsification; [X] JUNO’s sub-1% precision turns it into a $\sim 6\sigma$ kill-or-confirm.
- **Lean (FORM.CARTAN.DET.01).** CartanDeterminants.lean extended with the explicit $D_8 = SO(16)$ Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator: D_8 is even and symmetric just like E_8 (same $c=8$) but $4 = \det \text{Cartan}(D_8)$ is genuinely not a unit while $\det \text{Cartan}(E_8)$ is — so unimodularity (one anyon) is the load-bearing selector. lake build green.

2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: v335, FORM.QGEO.BW.01 qgeoSymIsCorollary)

- **v335 (SEAM.EQUIV.UNIFY.01) — the keystone unification.** A bird’s-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the “two open structural items” narrative. [E] QGEO.SYM.01 is a **COROLLARY** of SEAM.EQUIV.01 (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations $U(1)$ are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the μ_4 clock = rotation by $\pi/2$ ($(e^{i\pi/2})^4=1$) fixes it $\Rightarrow \omega \circ \rho = \omega$. So SEAM.EQUIV.01 $\Rightarrow$ QGEO.SYM.01 with NO extra premise (the v323 “rotation-invariant vacuum” is a net axiom) — the two bedrock items collapse to ONE. [E] **the one keystone theorem:** SEAM.EQUIV.01 = a construction theorem ($c=g_{\text{car}}+N_{\text{fam}}=8$, $\det \text{Cartan}(E_8)=1$, gap $6 \ln \frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). [E] **QG.AMB.01 decoupled, not a gate:** margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v311/v330/v332) $\Rightarrow$ no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from “TFPT structural frontier” to “decoupled general QG problem”. [E] **collapsed status:** open structural items

$2 \rightarrow 1$; live residual = $v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$. [O] the one residual = SEAM.EQUIV.01's continuum OS reconstruction.

- **Lean** (FORM.QGEO.BW.01, `qgeoSymIsCorollary`). `BWKeystone.lean` extended with the cited conformal-net vacuum axiom `conformal_net_vacuum_rotation_invariant`; with it, `StateInvariance` (=QGEO.SYM.01) follows from `SeamIsE8` alone, so QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01 with no independent open premise (`#print axioms` confirms only the chain + cited BW/net axioms). `lake build green` (3362 jobs).
- **Status deep-sync**. The ledger (QG.AMB.01 reclassified to “decoupled general QG; non-TFPT”; QGEO.SYM.01 marked a corollary; new SEAM.EQUIV.UNIFY.01 row), the canonical status card (`tfpt_status.tex`), introduction (status core + gate table), `origin_theory`, `tfpt_4_frontier`, `tfpt_5_redteam`, `tfpt_research_contracts`, `README`, `next.txt` and the website (`OpenGates`, `FAQ`, `papers.ts`) all moved to “one solvable theorem (SEAM.EQUIV.01) + one decoupled foreign problem (QG.AMB.01)”. No falsifiable prediction changed; this is a status-framing sharpening backed by v335 + the Lean corollary.

2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, FORM.CARTAN.DET.01)

- **v333_cm_selection.py** (QGEO.CMSELECT.01, [E]/[O]). The CM-selection mechanism for the last keystone residual: $\tau=i$ (the square) is the *unique* order-4 modulus — $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has $S^4=I$ and fixes i , and an order-4 deck selects $j=1728$ (vs the order-3 hexagon $j=0$) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal $\pi/2$ spacing). So given the μ_4 deck, $\tau=i$ is forced; the residual sharpens from “why the square” to “the deck acts geometrically” (= QGEO.SYM.01). NOT a closure.
- **v334_conformal_resolution.py** (QGAMB.CONFORMAL.01, [E]/[C]/[O]). The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation $\phi \rightarrow i\phi$ flips the wrong-sign kinetic term, making the divergent conformal Gaussian $\int e^{+|c|\phi^2}$ into the convergent $\int e^{-|c|\phi^2} = \sqrt{\pi/|c|}$; the entire IDG form factor e^{p^2/M^2} dresses the propagator pole-free (the polynomial $R+R^2$ ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour’s nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The Conway–Sloane lattice side: the E_8 Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ($|\det|=1$) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel `decide` / the integer-inverse argument. The remaining content (Witt’s uniqueness: E_8 is the only even unimodular lattice in $\dim < 16$) is the cited classification residual.

2026-06-22 (endgame round: the necessity of H reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem, $|\det E_8|=1$ and the μ_4 commutation grounded in Lean: v331, v332, FORM.MU4COMM.01)

- **v331_necessity_of_H.py** (QGEO.NECESS.01, [E]/[O]). The necessity companion to v329: a concrete Steklov (DtN) computation shows the seam $\Lambda=|k|+M_f$ commutes with the μ_4 clock *iff* the four marks sit at the square ($\tau=i$) — moving a mark off the square breaks $\mathbb{Z}_4$ and $[\rho, \Lambda] \neq 0$. The square has cross-ratio 2 ($j=1728$), the unique 4-config with an order-4 automorphism (v214/v267), so $H \Leftrightarrow \text{square} \Leftrightarrow \mu_4$ and “necessity of H ” sharpens to: the raw

dynamics places the four Gauss–Bonnet marks (v216) at the order-4 CM point. Residual = that dynamical selection.

- **v332_qgamb_metric_sector.py** (QGAMB.METRIC.01, [E]/[C]/[O]). The open metric sector of QG.AMB.01 is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient $-(D-1)(D-2)/4$ ($= -\frac{3}{2}$ at $D=4$), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (v330/v76); the only route is the conditional IDG taming (v304), with the polynomial $R+R^2$ ghost as the control. Open, but pinned to one named obstruction.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The rank-8 holomorphy discriminator $|\det \text{Cartan}(E_8)|=1$ (E_8 has one anyon, vs D_8 ’s four) is now proved principally and kernel-only: `cartanE8 * cartanE8inv = 1` by `decide` on the 8×8 PRODUCT gives `IsUnit (det E8)` via $\det E_8 \cdot \det E_8^{-1} = \det 1 = 1$ — no 8! determinant, no `native_decide` (Mathlib itself leaves `E8_det = 1` a `proof_wanted`).
- **Mu4Commutation.lean** (FORM.MU4COMM.01, Lean 4, [E]). The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers $\mathbb{Z}[i]$: $i^4=1$, $i^2=-1$, and (on concrete 6×6 matrices) an offset-4 coupling commutes with the μ_4 clock while an offset-2 coupling does not — all by kernel `decide`. Built lake build on Lean v4.29.1 + Mathlib.

2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: v329, v330, FORM.CARTAN.DET.01)

- **v329_os_gap_reduction.py** (QGE0.OSGAP.01, [E]/[O]). The one analytic input the v308 keystone chain still cited — “the OS-reconstructed bulk gap = the transfer gap” — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization $d\Gamma(-\log T)$, whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy $= 6 \ln \frac{3}{2}$, *independent of system size*. Together with $H \Rightarrow \omega \circ \rho = \omega$ and $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$, this exhibits the rotation-invariant flat-pillowcase premise H as the *single sufficient premise* of both bedrock IDs; residual = the necessity of H on the raw collar.
- **v330_qgamb_admissible_measure.py** (QGAMB.MEASURE.01, [E]/[O]). The constructive step v275’s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ($C(n)/C(n-1) \rightarrow (2/3)^6$), with finite susceptibility $\chi = 729/665$ so $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi$ converges $\Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit exists; OS reconstruction gives $H_{\text{OS}} \geq 0$ with gap $6 \ln \frac{3}{2}$. The full QG_{amb} (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **CartanDeterminants.lean** (FORM.CARTAN.DET.01, Lean 4, [E]). The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom): $|\det \text{Cartan}(A_3)|=4=|\mu_4|$ (3×3 , `det_fin_three`) and $|\det \text{Cartan}(D_5)|=4$ (5×5 , `decide`), giving the carrier $|\det \text{Gram}|=4 \cdot 4=16$ (the v281 anyon count). The rank-8 E_8/D_8 case stays the Nat discriminator — Mathlib’s own `Data.Matrix.Cartan` leaves `E8_det = 1` as a `proof_wanted`. Built lake build on Lean v4.29.1 + Mathlib.

2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the F_{transfer} solver suite, the cusp weight derived from a minimal rewrite, and the θ_{13} pressure point: FORM.QGEO.BW.01, v325, v326, v327, v328)

- **BWKeystone.lean** (FORM.QGEO.BW.01, Lean 4, [E]/[O]). The v323 Bisognano–Wichmann linkage formalised: the nodes `RotationInvariantVacuum` $\rightarrow$ `GeometricModularFlow` $\rightarrow$ `Mu4ModularSymmetry` $\rightarrow$ `StateInvariance` (=QGEO.SYM.01) composed over the v308 chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock QGEO.SYM.01 reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a `decide`-proven discriminator $(1 \mid 4) \wedge \neg(4 \mid 1)$ (BW geometric flow strictly stronger than v309 clock invariance). Built lake build on Lean v4.29.1 + Mathlib.
- **v325_pillowcase_keystone.py** (QGEO.KEYSTONE.01, [E]/[O]). The keystone as ONE citable theorem: IF the raw seam state is the flat $\tau=i$ pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks, μ_4 deck, $\omega \circ \rho = \omega$, holomorphic $c=8 \Rightarrow (E_8)_1$, with the recovery gap as input); all pieces machine-checked (incl. the pillowcase $\chi_{\text{orb}}=0$), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.
- **v326_ftransfer_suite.py** (FR.SUITE.01, [E]/[C]). The four F_{transfer} readouts productised into one runnable harness, each a gapped relaxation to a unique attractor: F_{pole} (Koide, seam rate $(2/3)^6$), $F_{\text{Boltzmann}}$ (η_B , H-theorem), F_{relic} (axion adiabatic invariant), F_{QCD} (m_p/m_e , RG to the UV fixed point), plus the F_{RareKaon} bridge. Only F_{pole} carries the seam rate; the four external rates are honestly fenced (v187/v303).
- **v327_hypergraph_rewrite.py** (HYP.REWRITE.01, [E]/[O]). The cusp weight $2/3$ is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum $\{1, 2/3, 0\}$, so $2/3 = (N_{\text{fam}} - 1)/N_{\text{fam}} = |\mathbb{Z}_2|/N_{\text{fam}}$ emerges from the rule arity and $(2/3)^6 = 64/729$ is the recovery gap; with a PROOF that $2/3$ is not a root of the affine- E_8 charpoly ($p(2/3) = -3520/19683 \neq 0$, sharpening v312). v324's injected datum is reduced to the anchor arity $\{2, 3\}$.
- **v328_theta13_pressure.py** (FLAV.TH13.PRESSURE.01, [C]/[X]) + **v307 hardening**. The honest weak spot named and quantified: $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.023108$ is $+2.0\sigma$ above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (v268), not relieved by any freeze-respecting correction (RG < 1%, exponent $5/6$ exact), pre-registered as the kill. The v307 watchdog gains a data provenance/date check and the rare-kaon F_{transfer} bridge row.
- **Wolfram mirror** 285 $\rightarrow$ 288. v325 ($\chi_{\text{orb}}=0$) and v327 (rewrite spectrum $\{0, 2/3, 1\}$, $(2/3)^6 = 64/729$, and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; GATE.WOLFRAM.02 and the Wolfram README updated to 288/288.

2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)

- **Keystone naming reconciled to SEAM.EQUIV.01**. Following v323/QGEO.BW.01 (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now SEAM.EQUIV.01 *everywhere*, with QGEO.SYM.01 stated as its *downstream* conformal-deck face. The ledger row QGEO.SYM.01 was updated (it no longer calls itself “the single structural residual of the whole theory”; it now depends on SEAM.EQUIV.01/QGEO.BW.01), and the superseded “GATE.QGEO open gate” / “two research gates (U_{wall}) and G_{metric} ” framing was replaced by the current $\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ with the two structural

items SEAM.EQUIV.01 + QG.AMB.01 (intro status card, tfpt_4_frontier, origin_theory, tfpt_5_redteam, tfpt_research_contracts, and the website).

- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs.** Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger (“honest bottom line”) and hard-reduction subsection; the `origin_theory` “residual inventory” (the five “Update/.../Terminal” passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ “What is still open?”, the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected.** Version label 5.0→v5.2 (intro, single-sourced via `\TFPTversion`); Lean build 1886→3357 jobs (`FORM.SEAMEQUIV.01`); README Python suite v213/212→v324/322 modules; Wolfram extension 269/269 / 249/249→285/285 (README, website replication page, `zenodo_description.html`); residual “Paper-3” old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: v202, v203, v305, v307, v321)

- **Bug fix — δ_{CKM} display value.** The predictions table (`tex-artefacts/predictions.tex`) showed $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$, but that formula evaluates to 68.65° (the canonical frozen `DELTA_CKM_RAD` = 1.198232 rad, used by v88/v202/FLAV.CP.01 everywhere else). Corrected to 68.65° . An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **v202_rare_kaon.py — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ ($\sim 1.2\sigma$ above) to the full 2016–2024 combination $\text{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$ (La Thuile 2026), on which the prediction 9.45×10^{-11} lands at $+0.08\sigma$. Framed honestly: a strong *consistency* hit for the closed CKM point, but an F_{transfer} bridge — *not* a unique TFPT-vs-SM discriminator (the SM $(8.6 \pm 0.4) \times 10^{-11}$ is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (v307, a new F_{transfer} bridge row), the forward board (v321, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from “a relation to the *measured* quark phase” to “a relation to the *leading* ($\pi/3$) component of the quark phase”: the lock is to the structural $\pi/3$, not the full measured $\gamma = \delta_{\text{CKM}}^{\text{lead}} + 3\lambda_C^2 \approx 68.7^\circ$, so $\delta_{\text{PMNS}} = 240^\circ$ (not 248.7°). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **v305_witness_independence.py — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule: $\phi = 2 \cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$ is the *static* carrier ($\mathbb{Z}/5$) field, every *dynamic* rate is rational ($\mathbb{Q}$, the $\mathbb{Z}/6$ hand), so a dynamic rate explained by ϕ is a cross-field false friend (v314/v315).
- **v203_eht_achromatic.py — the μ_4 Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at $m \equiv 0 \pmod{4}$ (the μ_4 fingerprint of the four marks, v201); a μ_4 -orbit profile has ~ 0 off-mode power while a $\mathbb{Z}_2$ control leaks at $m = 2$.

- **Seed-hyperplane figure (Fig. ??).** A new figure visualises v306: every scorecard observable inverts *independently* to the one latent seed φ_0 , the error-weighted mean matching the axiom to 0.04%, with $\sin^2 \theta_{13}$ the $\sim 2\sigma$ pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: v322, v323, v324)

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks, $275 \rightarrow 285$, *all passing*): v313 (the affine- E_8 characteristic polynomial with the golden factor x^2-x-1 ; the $(2, 3, 5)$ atoms as $2 \cos(\pi/k)$; the 31/30 selection), v314 (the rational kernel vs $\mathbb{Q}(\sqrt{5})$ dynamic-rate split), v315 ($30=5 \cdot 6$, $\varphi(30)=8$, the Galois group $\mu_4 \times \mathbb{Z}_2$, the $\sqrt{5}$ Gauss sum), v316 ($\rho=\zeta_6=\zeta_{30}^5$, $\rho^4=-\rho$, $\text{Gal}=\mathbb{Z}_2=\text{CP conjugation}$), v317 ($N_{\text{fam}}=3$ as the μ_3 orbit, the $1+2$ Galois split), v318 ($\varphi_0^{\text{ret}}=(4/3)c_3+48c_3^4$ a pure function of π) and v320 (the CP lock $\delta_{\text{PMNS}}=240^\circ$). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.
- **v322_cp_lock_sharpened.py (GALOIS.CP.SHARPEN.01, [E]/[C]/[X]).** The v320 CP-lock sharpened into a quantitative, pre-registered kill test. [E] the leading phase sits on the six hexagonal nodes $\arg \rho^k = k \cdot 60^\circ$, and *which* one is selected is the deck order: $\delta_{\text{PMNS}}^{\text{lead}} = |\mu_4| \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3)$ (node 4); [C] the sub-leading correction is bounded by the quark analogue $3\lambda^2 \approx 8.7^\circ$, so the prediction is the band $240^\circ \pm \sim 9^\circ$; [C] against NuFIT 6.0 (NO, $\delta_{\text{CP}}=212^{+26}_{-41}^\circ$) the leading value is $+1.08\sigma$, the band edge $+0.74\sigma$ – compatible today; [X] the nearest wrong node is 60° away, far beyond the budget, so DUNE/Hyper-K ($\sim 5\text{--}15^\circ$) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.
- **v323_bw_geometric_modular.py (QGEO.BW.01, [E]/[C]/[O]).** The keystone pushed one honest step via Bisognano–Wichmann. [E] the μ_4 clock is a geometric rotation $\rho=\text{diag}(i^n)=\exp(i\frac{\pi}{2}L)$, $\mu_4 \subset U(1)_{\text{rot}}$; [E] for a rotation-covariant covariance $C=f(L)$ the modular Hamiltonian $K=\log((1-C)/C)=g(L)$, so $[K, L]=0$ (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric, $[K, L] \neq 0$); [C] given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGEO.SYM.01` ($\omega \circ \rho = \omega$) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308). Cited in `origin_theory`; ledger `QGEO.BW.01`.
- **v324_hypergraph_fiber.py (HYP.FIBER.01, [E]/[C]).** The constructive v312 follow-up: the *minimal* substrate that carries both the E_8 skeleton and the recovery gap is a *product*. [E] the carrier network $T_{\text{net}}=(A+2I)/4$ (attractor = Kac marks = skeleton) fibred by the 3-node family cusp $T_{\text{cusp}}=\text{diag}((1-w)^{2N_{\text{fam}}})$, $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$ (spectrum $\{1, (2/3)^6, (1/3)^6\}$); the fibred $T_{\text{net}} \otimes T_{\text{cusp}}$ has top eigenvalue 1 (eigenvector marks $\otimes (w=0)$) and $(2/3)^6$ as a genuine eigenvalue (marks $\otimes (w=\frac{1}{3})$) – the $N_{\text{fam}}=3$ cusp is the minimal fibre and the substrate = carrier $\times$ family = the $30=g_{\text{car}}(2N_{\text{fam}})=5 \times 6$ of v315; [C] honestly (v312) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of $2/3$ from a pure rewrite. Cited in `origin_theory`; ledger `HYP.FIBER.01`.

2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: v321)

- **v321_killtest_board.py (KILL.BOARD.01, [E]).** The forward companion to v307 (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO $\sin^2 \theta_{12}=0.306747$ (fastest), #2 CMB-S4/LiteBIRD r

(kill $r > 0.01$), #3 DESI/Euclid w (kill $w \neq -1$), #4 DUNE/Hyper-K $\delta_{\text{PMNS}} = 240^\circ$ (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering + $m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2 \theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β . A forward decision ledger (publicly-stated experimental reach), complementing v307; no new physics. Cited in `origin_theory`; ledger row KILL.BOARD.01.

2026-06-22 (Lean: the SEAM.EQUIV.01 composition chain formalised — the keystone residual machine-pinned to the named cited steps: FORM.SEAMEQUIV.01)

- **SeamEquivChain.lean** (FORM.SEAMEQUIV.01, Lean 4, [E]/[O]). A Lean 4 mirror of the v308 closing chain: the four nodes $\text{GapPositive} \rightarrow \text{InvertibleSRE} \rightarrow \text{HolomorphicC8} \rightarrow \text{SeamIsE8}$ assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Mger / Kawahigashi–Longo–Mger holomorphy; Conway–Sloane E_8 uniqueness) as named `axioms` and the carrier recovery gap (the OS step) as the single open input. [E] `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); [E] a decide-proven discriminator `primariesE8 = 1  $\neq$  primariesD8 = 4` (the $|\det \text{Cartan}|$ that separates the holomorphic E_8 from the same- c rival $D_8 = SO(16)$). Built end-to-end (`lake build`, 3357 jobs) on Lean v4.29.1 + Mathlib. [O] The value is the machine-checked composition + axiom audit (pinning the keystone residual), *not* a from-scratch proof — it does NOT close SEAM.EQUIV.01. Ledger row FORM.SEAMEQUIV.01; mirrors v308.

2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$: v320)

- **v320_galois_cp_relation.py** (GALOIS.CP.PREDICT.01, [E]/[C]/[X]). The v316 follow-up turned into a testable cross-prediction. [E] both leading CP phases are powers of the *one* hexagonal unit $\rho = \zeta_6$ of the family Galois factor: $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$ (60°), $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$ (240°), and $\rho^4 = -\rho$ locks them as $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ (the $\mathbb{Z}_2$ sheet flip); [C] this upgrades the previously assigned $\delta_{\text{PMNS}} = 240^\circ$ to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); [X] kill test: a δ_{PMNS} robustly incompatible with 240° ($> 3\sigma$ at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation; 240° is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the v313–v319 arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row GALOIS.CP.PREDICT.01.

2026-06-22 (the translation clock: the discrete $\leftrightarrow$ dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands 5×6 , read law-inclusive (0..5) or live-only (1..5): v319)

- **v319_translation_clock.py** (TRANSLATE.CLOCK.01, [E]/[C]). A synthesis of the v313–v318 arc with v124/v55, answering the question “is the static $\leftrightarrow$ dynamic translation a clock running 0, 1, 2, 3, 4, 5 or 1, 2, 3, 4, 5?”. [E] the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ and $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (two coprime hands, v315). [E] the *dynamic* hand is $\mathbb{Z}/6 = 2N_{\text{fam}}$ (ticks 0, ..., 5): the recovery exponent is exactly $6 = 2N_{\text{fam}}$, rate $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$, order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$, v316). [E] position 0 is the *conserved law* — the resummed clock rate(n) = $-p_2 \ln(1 - n/N_{\text{fam}})$ (v124) has rate(0) = 0 (the attractor, eigenvalue 1, v56), the live decaying modes from $n=1$, and the E_8 live phases (totatives of 30) start at 1 ($\{1, 7, \dots, 29\}$), the 0 phase not a totative — so “0, 1, 2, 3, 4, 5” is law-inclusive, “1, 2, 3, 4, 5” live-only. [E] the *static* hand is $\mathbb{Z}/5 = g_{\text{car}}$ (ticks 1, ..., 5): the golden field $\mathbb{Q}(\sqrt{5})$ ($\varphi = 2 \cos(\pi/5)$), field-obstructed (no rational dynamic image, v314) — static-only; $\zeta_5 = \zeta_{30}^6$ carries no phase. [E] one full turn

$= \text{lcm}(5, 6) = 30 = h(E_8)$, rank $E_8 = 8 = \varphi(30)$ live phases pairing into $|\mu_4| = 4$ invariant planes (v55). [C] verdict: the discrete→dynamic translation *is* the order-30 clock = (static 5-ring) $\times$ (dynamic 6-ring), the dynamic hand carrying the rate (exponent $6 = 2N_{\text{fam}}$), the static hand the golden carrier (no rate), and “0 vs 1” = law-inclusive vs live-only; the arithmetic is [E], the “it is one clock” reading [C]. Python-only (sympy), like the rest of the arc. New figure [figures/translation_clock.pdf](#) (concentric 5- and 6-hands). Cited in [origin_theory](#); ledger row TRANSLATE.CLOCK.01.

2026-06-21 (capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30})$ + Galois $\mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\text{geo}}\}$ — zero dimensionless free parameters: v318)

- **v318_arithmetic_capstone.py** (ARITH.CAPSTONE.01, [E]/[O]). The synthesis of the v313–v317 arc. [E] the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field $\mathbb{Q}(\zeta_{30})$ with Galois group $\mu_4 \times \mathbb{Z}_2$ (degree $\varphi(30) = 8 = \text{rank } E_8$, $30 = |\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}} = h(E_8)$), forced by the anchor atoms $\{2, 3, 5\} = e(a)$; [E] the magnitude seed $\varphi_0 = (|\mu_4|/N_{\text{fam}})c_3 + \Omega_{\text{adm}}c_3^4 = \frac{4}{3}c_3 + 48c_3^4$ ($c_3 = 1/(8\pi)$) is a pure function of π (no free parameters, anchor-derived coefficients), so the magnitudes ($\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$) are $\{a, \pi\}$ -determined, not free; [E] free inventory: *zero* dimensionless free parameters, π the unique transcendental primitive (ARCH.CORE.01), v_{geo} the unique dimensionful input (No-Unit Theorem) — complete input $\{a, \pi, v_{\text{geo}}\}$. [O] honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (QGEO.SYM.01/SEAM.EQUIV.01 stay [O]). Cited in [origin_theory](#); ledger row ARITH.CAPSTONE.01.

2026-06-21 (are the 3 generations a μ_3 /Galois orbit? Yes (μ_3), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)

- **v317_galois_family.py** (GALOIS.FAMILY.01, [E]/[C]). The v316 follow-up to the generation structure. [E] the three cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ (with $\text{Spec}(Q_+) = \{1, 2, 3\}$, the A_3 exponents) are the cube roots of unity $\{1, \zeta_3, \zeta_3^2\}$ under $w \mapsto e^{2\pi i w}$, and the cyclic family symmetry $w \mapsto w + \frac{1}{3}$ is a *transitive* μ_3 orbit (the family IS the cube-root orbit); [E] the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at $w=0$); [E] the family Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$, = the v316 CP conjugation) refines the orbit to 1+2 — fixing $w=0$ (rational) and swapping $w=\frac{1}{3} \leftrightarrow \frac{2}{3}$; [E] so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. [C] verdict: the family STRUCTURE is the same arithmetic as the CP phase (a μ_3 orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in [origin_theory](#); ledger row GALOIS.FAMILY.01.

2026-06-21 (does the Galois $\mu_4 \times \mathbb{Z}_2$ organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois $\mathbb{Z}_2$ = CP conjugation: v316)

- **v316_galois_readout.py** (GALOIS.READOUT.01, [E]/[C]). The v315 follow-up to phenomenology: does the Galois coupling $\mu_4 \times \mathbb{Z}_2$ reach the physics? [E] it does, through the *family* factor — the hexagonal CP unit $\rho = \zeta_6 = \zeta_{30}^5$ is exactly the order-6 = $2N_{\text{fam}}$ factor c^5 of v315 (the carrier being $\zeta_5 = \zeta_{30}^6$); [C] the leading CP phases are its cyclotomic arguments $\delta_{\text{CKM}}^{\text{lead}} = \pi/3 = \arg \zeta_6$ and $\delta_{\text{PMNS}} = 4\pi/3 = \arg \zeta_6^4$ (power 4 = $|\mu_4|$), with $\zeta_6^4 = -\zeta_6$ the $\mathbb{Z}_2$ sheet flip and $\mu_6 = \mu_3 \times \mu_2$ (inherits v231/v233); [E] the family Galois $\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$ ($\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$) *is* CP conjugation $\delta \mapsto -\delta$; [E] the carrier factor ζ_5 (golden $\sqrt{5}$) carries no phase (μ_4 static), and the magnitudes ($\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$, transcendental φ_0) are not cyclotomic. [C] verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the v314 field split. Cited in [origin_theory](#); ledger row GALOIS.READOUT.01.

2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois = $\mu_4 \times \mathbb{Z}_2$: v315)

- **v315_coxeter_coupling.py** (COX.COUPLE.01, [E]/[C]). Answers the v314 question — does $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ couple the carrier (5-fold, golden, $\mathbb{Q}(\sqrt{5})$) and family ($6 = 2 \cdot 3$, rational) sectors, or just factorize? [E] the cyclic group *factorizes*: $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled, in line with the v314 field obstruction); [E] the E_8 exponents = totatives(30) split as totatives(5) $\times$ totatives(6), $\varphi(30) = 4 \cdot 2 = 8 = \text{rank } E_8$; [E] but the *field* couples: $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$ and the family $\mathbb{Q}(\zeta_3)$, degree $8 = \text{rank } E_8$; [E] the Galois group is exactly $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$ (order 8, max element order 4), so the μ_4 seam clock *is* the Galois group of the carrier's golden field and $\mathbb{Z}_2$ that of the family field (refining v223); [E] $\sqrt{5} = \phi + 1/\phi$ is the quadratic Gauss sum in $\mathbb{Q}(\zeta_5)$. [C] verdict: $30 = 5 \cdot 6$ does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static $\leftrightarrow$ dynamic bridge being *Galois*, not a dynamical map. Cited in *origin_theory*; ledger row COX.COUPLE.01.

2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split $\mathbb{Q}$ vs $\mathbb{Q}(\sqrt{5})$: v314)

- **v314_rate_translation.py** (RATE.TRANSLATE.01, [E]/[C]). Answers, exactly, whether the discrete kernel (the affine- E_8 network, carrying the golden ratio) and the dynamic part (the seam transfer, carrying the recovery rate $(2/3)^6$) translate. [E] FIELD SPLIT: the adjacency spectrum is the rational part $\{0, \pm 1, \pm 2\} \subset \mathbb{Q}$ plus the golden part $\{\pm \phi, \pm 1/\phi\} \subset \mathbb{Q}(\sqrt{5})$, while the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is *entirely* rational; [E] the family/sheet rates DO translate (the rational angles $2 \cos(\pi/2)=0 = |\mathbb{Z}_2|$ and $2 \cos(\pi/3)=1 = N_{\text{fam}}$ build $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}} \in \mathbb{Q}$, same field, same atoms), but [E] the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has NO rational dynamic image – a field obstruction ($2/3$ is neither an adjacency nor a lazy-update eigenvalue; heat-kernel $t = \Delta/(2 - \phi) \approx 6.37$ not clean), so the carrier is *static-only*; [E] the only object holding both is the order-30 Coxeter element $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$, and the ϕ -quasicrystal $E_8 = H_4 + \phi H_4$ ($240 = 2 \cdot 120$) carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in *origin_theory*; ledger row RATE.TRANSLATE.01.

2026-06-21 (the golden ratio made precise: the $(2, 3, 5)$ atoms are the affine- E_8 network spectral angles, $\phi = 2 \cos(\pi/5)$ the $g_{\text{car}}=5$ signature: v313)

- **v313_golden_atoms_spectral.py** (GOLD.ATOMS.01, [E]/[C]). A follow-up to v312's observation, made exact. [E] the affine- E_8 adjacency characteristic polynomial factors as $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$, so the golden minimal polynomial x^2-x-1 divides it and $\phi = 2 \cos(\pi/5) = (1+\sqrt{5})/2$ is an *exact* eigenvalue; [E] the non-trivial eigenvalues are $2 \cos(\pi/k)$ for exactly $k \in \{2, 3, 5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$ ($2 \cos(\pi/2)=0$, $2 \cos(\pi/3)=1$, $2 \cos(\pi/5)=\phi$, $2 \cos(2\pi/5)=1/\phi$), so the golden ratio is *specifically* the $g_{\text{car}}=5$ (5-fold) signature and $g_{\text{car}}=5$ gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the v305 multiplicity); [E] the $(2, 3, 5)$ spherical excess $1/2+1/3+1/5 = 31/30 > 1$ (the condition selecting the finite $2I \Rightarrow E_8$) ties the v63/v76 gap-margin numerator $31 = 2^{g_{\text{car}}} - 1 = 248/8 = 1+h(E_8)$ to the Coxeter $30 = h(E_8)$ — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, v305); [E] honest null: ϕ is a structural lattice/network quantity, not a phenomenological readout. [C] direction: ϕ is the E_8 -quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the v312 question. Cited in *origin_theory*; ledger row GOLD.ATOMS.01.

2026-06-21 (TOE-roadmap solver round: the SEAM.EQUIV.01 keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: v308–v312)

- **v308_seam_equiv_chain.py** (SEAM.EQUIV.CHAIN.01, [E]/[O]). TOE attack point 1: the keystone “the raw seam-Calderón net is the holomorphic $c=8$ $(E_8)_1$ net” assembled into ONE well-typed logical chain $\text{gap}>0 \rightarrow \text{invertible/SRE (Kitaev)} \rightarrow \text{holomorphic } c=8 \text{ (Müger/KLM)} \rightarrow (E_8)_1 \text{ (Conway–Sloane)}$, with the exact discriminators $|\det \text{Cartan}(E_8)|=1$ vs $|\det \text{Cartan}(D_8=SO(16))|=4$, the carrier tower $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$ and $c=g_{\text{car}}+N_{\text{fam}}=8$ all machine-checked; the input gap is the derived recovery gap $6 \ln(3/2)>0$. The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row SEAM.EQUIV.CHAIN.01.
- **v309_modular_bedrock.py** (QGEO.MODHAM.01, [E]/[O]). TOE attack point 2: the bedrock $\omega \circ \rho = \omega$ via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), so μ_4 is derived; the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ round-trips to the seam operator and $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$ (a $\mathbb{Z}_2$ profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicity of the raw collar; not a closure of QGEO.SYM.01. Cited in `tfpt_research_contracts`; ledger row QGEO.MODHAM.01.
- **v310_carrier_sm_anomaly.py** (CAR.SM.ANOM.01, [E]). TOE attack point 3: the carrier half-spinor $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$ is exactly one anomaly-free Standard-Model generation ($SU(5) \subset SO(10)$: $10+\bar{5}+1$), with $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ (exact) and the one-loop $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1 , so $N_{\text{fam}}=3$ pins the RG. Closes the spectrum/anomaly/ β sub-question; the interacting Epstein–Glaser S -matrix and the Yukawa sector stay [C]/[O]. Cited in `tfpt_research_contracts`; ledger row CAR.SM.ANOM.01.
- **v311_gap_decoupled_measure.py** (QGAMB.CLUSTER.01, [E]/[C]/[O]). TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$ (correlation length $1/(6 \ln(3/2))$), finite susceptibility $729/665$, and the v76 margin $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ decouples the physical sector from the un-built ambient. QG.AMB.01 stays [O] (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row QGAMB.CLUSTER.01.
- **v312_hypergraph_substrate.py** (HYP.INJECT.01, [E]/[O]). TOE attack point 5: the hypergraph substrate sharpened — the affine- E_8 Kac marks are the Perron eigenvector ($A m=2m$) and the subleading eigenvalue is exactly $2 \cos(\pi/5) =$ the golden ratio, but the recovery rate $(2/3)^6$ is *not* any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$ and the analytic seed φ_0 . The open question is precisely delimited, not closed. Cited in introduction; ledger row HYP.INJECT.01.

2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: v305–v307)

- **v305_witness_independence.py** (WITNESS.ECON.01, [E]). The *structural* anti-numerology firewall complementing the v100 phenomenological null test, against the sharpest self-deception worry (“the same integer counted many times as if independent”). Every headline integer ($|\mu_4|=4$, $g_{\text{car}}=5$, $|\mathbb{Z}_2|=2$, $N_{\text{fam}}=3$, $\dim S^+=16$, $\text{rank } E_8=8$, 240, 248, $|2I|=120$, $h(E_8)=30$, $|W(D_5)|=1920$, $\Omega_{\text{adm}}=48$, the gap exponent 6) is a documented image of the single anchor $a=(1, 1, 2)$ with π the only transcendental primitive (13 atoms from 2 free primitives, $6.5 \times$

compression made explicit), so the recurring $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$ and $(2/3)^6$ are *one* ratio; the E_8 spine is overdetermined (≥ 3 structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by **v100**, never multiplied); the negative control shows α^{-1} 's “137” is foreign to the anchor family (an independent witness via the $F_{U(1)}$ cubic, **v3**). Cited in the **tfpt_5_redteam** body; ledger row **WITNESS.ECON.01**. Python-only.

- **v306_seed_crossval.py** (**SEED.CROSSVAL.01**, **[E]**). A leave-one-out cross-validation turning the “many observables, one seed φ_0 ” worry into a falsifiable out-of-sample test: back-solving φ_0 from each of the five seed-grammar observables ($\sin^2 \theta_{12}$, $\sin^2 \theta_{13}$, β , Ω_b , λ_C ; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$ to 0.04%; leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned = $\sin^2 \theta_{13}$, $\sim 2\sigma$); a data $\leftrightarrow$ formula shuffle explodes the χ^2 by $> 100\times$. Cited in **tfpt_5_redteam**; ledger row **SEED.CROSSVAL.01**. Python-only.
- **v307_data_watchdog.py** (**DATA.WATCHDOG.01**, **[E]**). The operational decision pipeline over the frozen registry (**v84**): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live n_σ against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core prediction $\sin^2 \theta_{13}$ (2.0σ , the honest current pressure point); $\sin^2 \theta_{12} = 0.306747$ is compatible (0.02σ , the JUNO falsifier), the $r \in [0.0033, 0.0048]$ band is below the BK18 limit and the kill, and $w=-1$ is flagged as the DESI dark-energy watchdog. Cited in **origin_theory**; ledger row **DATA.WATCHDOG.01**. Python-only.
- **Target-A holomorphy kill test (freeze_file.csv)**. A new **seam_holomorphy** freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ($|\det K| > 1$, the same- c rival $SO(16)_1$) would break holomorphy and kill “seam net = $(E_8)_1$ ” — since holomorphic $c=8 \Leftrightarrow E_8$ is machine-proven (**v83/v143**) and $\det K=1 \Leftrightarrow$ invertible/SRE (**v237/v301/v302**).

2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the “constancy of c ” framing of **SEAM.EQUIV.01**: **v304**)

- **v304_idg_nonlocal_ghost.py** (**QGAMB.IDG.01**, **[C]/[O]**). A conditional, parameter-free narrowing of the red-team **rt_F/v278** Stelle-ghost attack on the perturbative gravity sector. **[E]** the local $R+R^2/\text{Weyl}^2$ four-derivative propagator $1/(p^2(p^2+M^2))$ carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; **[E]** for an *entire* graviton form factor $a(p^2) = e^{p^2/M^2}$ (nowhere zero) the dressed propagator $1/(p^2 a)$ has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; **[C]** the spectral-action cutoff is the seam KMS weight $f(u) = e^{-u}$ (**v259**), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. **[O]** It does *not* close **QG.AMB.01**: the trace regulator is not yet shown to resum to an entire $a(\square)$, and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row **QGAMB.IDG.01**.
- **Double-pushout figure (tfpt_1_architecture_e8)**. A new shared TikZ macro (**TFPTdoublepushout** in **tex-artefacts/tfpt_figures.tex**) draws the commuting square showing that the order- $|\mu_4|=4$ glue is *one* operation in two categories: the lattice pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ (**v1**) and the conformal-net / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$ (**v92/v281/v234/v277/v260**). Expository only; no new claim.
- **Framing (introduction, tfpt_research_contracts)**. **SEAM.EQUIV.01** is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of c plays in relativity (cf. **v267**). Prose sharpening; no status change.

2026-06-19 (the discrete→dynamic lens on **F_transfer**: one shape, one seam-rate, v303)

- **v303_ftransfer_dynamics.py** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update → the E_8 marks, gap $6 \ln(3/2)$) — is also the shape of the four **F_transfer** instances, or an unrelated bolt-on. It is the *same* shape: **[E]** **F_pole** (Koide) *is* the seam dynamics (Möbius multiplier $F'=(2/3)^6=\lambda_2$, the seam-clock eigenvalue $\Rightarrow$ a unique attractor at the seam rate, **v82/v54**); **[C]** **F_Boltzmann** (η_B) is a gapped positive contraction (washout $\kappa_f \in (0,1)$, H-theorem); **[C]** **F_relic** (axion) freezes at an adiabatic fixed point; **[E]** **F_QCD** (m_p/m_e) is the 1-loop RG flow with the carrier $b_3 = -7$ (asymptotic freedom, Gaussian UV attractor). **[O]** Verdict: one shape underlies all four; **F_pole** has the seam rate $(2/3)^6$ *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So **F_transfer** is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (**v187**), not reduced to the seam. **[E]** suite v303 5/5, AUDIT OK.

2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap $6 \ln(3/2)$, v302)

- **v302_seam_gap.py** closes the chain on the single statement **v301** had left: “the quasi-free seam bulk is gapped”. That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. **[E]** the frozen transfer spectrum is $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (**v160/v162**), a simple Perron eigenvalue 1 separated from the rest; **[E]** the gap is $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1 - (2/3)^6 = 665/729$; $(2/3)^6$ the μ_4 -deck transfer eigenvalue forced by the carrier); **[E]** with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (**v76**) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; **[C]** by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian. **[O]** **Bottom line:** with **v300/v301**, **SEAM.EQUIV.01**'s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the **v297** AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap $6 \ln(3/2)$) — *no undischarged TFPT-internal assumption remains*. It stays **[O]** (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. **[E]** suite v302 5/5, AUDIT OK.

2026-06-19 (Route A's last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, v301)

- **v301_route_a_invertible.py** attacks the single remaining *unconditional* gap of **SEAM.EQUIV.01**. Since **v300** collapsed Route B into Route A's rationality, that gap is Route A's open hypothesis (**v297** LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev's periodic table, class D). **[E]** the carrier is 16 Majoranas, $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$ (**v148**); **[E]** $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents + 128 spinor = 248, **v154**); **[C]** so a gapped quasi-free bulk is invertible automatically; **[E]** the invariant $|\det K| = \#\text{anyons}$ is 1 for E_8 (invertible) vs 4 for the gauged $D_8=SO(16)$ (anyons), which free fermions cannot produce. **[O]** LIT-A's ‘invertible bulk’ now *follows* from ‘gapped 16-Majorana quasi-free bulk’ (**v148/v160**+the mass gap), so the residual changes from a topological-order statement to a spectral one. **[O]** **Bottom line:** with **v300**, the whole open content of **SEAM.EQUIV.01** reduces to the *single spectral statement* “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed; the gap is not

manufactured. [E] suite v301 6/6, AUDIT OK.

2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$, v300)

- **v300_flataway_rigid.py** attacks the single external fact the three Flat-Away routes (v291/v293/v295) left open, on two fronts. *Harden*: the flat seam was only a *soft* minimum (“the spectral mismatch grows with ε ”); here the obstruction becomes a *discrete* on/off invariant. [E] the flat fingerprint — $\text{spec}(|D_\theta|)$ is the integer ladder with every level $n \geq 1$ of multiplicity exactly 2; [E] the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy 2×2 $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, + an $O(g^2)$ common shift), validated against the full Toeplitz operator; [E] any $g_k \neq 0$ lifts the resonant degeneracy ($2 \rightarrow 1+1$) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — strictly stronger than the “deviation grows” scan; [E] with the v295 convexity, flat is isolated analytically *and* discretely. *Derive the pin*: [C] the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q -coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), so by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, i.e. the external pin *is* the $(E_8)_1$ rationality (Route A). [O] verdict: Flat-Away carries *zero* new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact. [E] suite v300 6/6, AUDIT OK.

2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2, 3, r)$ seed reaches E_8 as the last finite point, v299)

- **v299_hypergraph_0d_autonomous.py** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): [E] the icosahedron is a 3-uniform hypergraph ($|\text{Aut}(\text{graph})| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$; the McKay group is the $SU(2)$ binary lift $2I$, a *different* order-120 group whose McKay graph is affine E_8 ; 30 edges $= h(E_8)$); [E] the $1 \rightarrow 4$ subdivision rewrite is equivariant (all 120 symmetries at every scale); [E] the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; [E] Smith ($\rho \leq 2 \Leftrightarrow$ affine ADE) + Collatz–Wielandt ($\rho \leq 2 \Leftrightarrow$ a local witness $Am \leq 2m$); [E]* the autonomous rule (diffusion-generated witness + *local* gate $Am \leq 2m$, Perron only as audit) runs $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$; [E] the negative controls $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$ and $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$ are < 2 for *all* n (closed form), so neither family ever reaches E_8 . Honest type: [O] the $(2, 3, \cdot)$ seed is the single irreducible input = P2 — *the rule reaches E_8 as the last finite point on this seed; it is not a derivation of P2*. [E] build green, AUDIT OK.

2026-06-18 (E_8 as a network attractor — the Wolfram-program reading, v298)

- **v298_e8_network_attractor.py** is an honest audit of the “TFPT as a hypergraph/rewrite system” idea, on the v219 McKay bedrock. [E] the affine- E_8 Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4m$ on the $(2, 3, 5)$ McKay graph (converges from any positive start; $Am = 2m$, top eigenvalue 2); the spherical tower terminates at E_8 and the cascade is nested-subgraph coarse-graining. [C] the Wolfram reading “ E_8 is the universal attractor of a minimal $(2, 3, 5)$ network” holds at the Coxeter-skeleton level. [O] a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives: φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. [E] build green, AUDIT OK.

2026-06-18 (a_2 closed form + Route A literature stack, v296–v297)

- **v296_flataway_a2_closed.py** (FLATAWAY.A2.CLOSED.01) evaluates the exact a_2 coefficient in *closed form*: the operator tails telescope geometrically ($C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$) to $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$, leaving a finite $(4k-1)$ -term middle; e.g. $W_1(t) =$

$t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$. Manifestly positive, small- t law $W_k = t/2 + O(t^3)$ (the t^2 term cancels). Reproduces v295 to machine precision.

- **v297_route_a_literature_stack.py** (SEAM.EQUIV.A.LIT.01) writes Route A’s one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins) $\rightarrow$ LIT-B (Müger; Kawahigashi–Longo–Müger) $\rightarrow$ LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk $\Rightarrow (E_8)_1$ ”, all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B’s Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). [E] build green, AUDIT OK.

2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **v295_flataway_a2_exact.py** (FLATAWAY.A2.01) upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of $\text{Tr } e^{-t\Lambda}$ vanishes (zero-mean off-mark curvature, Gauss–Bonnet), and since $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$ is convex (Klein/Peierls), $\varepsilon=0$ is the *global* minimum — so $\Delta \text{Tr} \geq 0$ for all deformations, $= 0$ iff $f = 0$. The exact second-order coefficient is the divided-difference kernel of e^{-tx} (Duhamel), validated to rel. err $< 10^{-6}$.
- **Lean FORM.FLATAWAY.01** (SeamDeckClosure.lean Part 5): the formal $\Delta=0 \Leftrightarrow$ flat step — a positive-weighted sum of squares is positive-definite (heatDeviation_eq_zero_iff); built and axiom-checked (only propext, Classical.choice, Quot.sound). So the heat route’s positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes a_2) remains. [E] build green, AUDIT OK.

2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **v292_flataway_heat_reduction.py** (FLATAWAY.HEAT.01): the heat route, made precise. The heat-trace deviation $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a *positive-definite* quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode, positive on all combinations), leading invariant $\|f\|_{L^2}$ — so Flat-Away reduces to the single fact “the seam fixes the a_2 heat coefficient”.
- **v293_flataway_spectral_hessian.py** (FLATAWAY.SPEC.01): the spectral route, upgraded from one direction to the *full* smooth- $\mathbb{Z}_4$ space. Mode $4k$ splits the degenerate Steklov level $|2k|$ at first order; the Hessian of the spectral mismatch over modes $\{4, 8, 12\}$ is positive-definite (eigenvalues $\{0.497, 0.500, 0.503\}$), so the flat seam is a strict isolated minimum — the “only one direction” objection is answered.
- **v294_flataway_rp_energy.py** (FLATAWAY.ENERGY.01): the variational route. With the rigid mark divisor (4 cone angles π , Gauss–Bonnet 4π), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy $\int f^2$ is strictly convex (Hessian $2\pi I$) with a unique minimiser — Flat-Away reduces to “RP+gap realises the energy-minimiser”. The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [E] build green, AUDIT OK.

2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **v290_z4_smooth_curvature_adversary.py** (SEAM.ADVERSARY.01) is the honest red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality: a smooth $\mathbb{Z}_4$ off-mark curvature $f = \varepsilon \cos(4\theta)$ [E] passes the v288 commutator ($\|[\rho, \Lambda]\| = 0$, same as mark-local) yet [E] shifts the DtN spectrum ($\max |\Delta\lambda| = 0.155$) and the heat trace ($\Delta = 0.019$). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not $[\rho, \Lambda]=0$ — which is exactly why Flat-Away cannot be read off the commutator.

- **v291_flataway_contract.py** (FLATAWAY.RP.01) states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family — deviation zero only at $\varepsilon=0$) and [C] RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, AUDIT OK.

2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **v289_marklocal_raw.py** (MARKLOCAL.RAW.01) decomposes the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks $\Rightarrow$ curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on $4\mathbb{Z}$ (v264/v288), L5 full-operator $[\rho, \Lambda]=0$ (v288). 4/5 discharged — Route B reduces to exactly L2.
- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as ‘full- L^2 QGEO proved’.
- **v287 literature matrix**: L3 (‘invertible bulk $\Rightarrow$ single-sector’) is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Muger/Kawahigashi–Longo–Muger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, AUDIT OK.

2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **v286_seam_equivalence_contract.py** (SEAM.EQUIV.01) names the one open theorem — *the raw RP seam state is the holomorphic $(E_8)_1$ net at $\tau=i$* — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no ‘ E_8 into the geometry and back’ circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **v287_free_rp_bulk_to_holomorphic_boundary.py** (SEAM.EQUIV.A01): Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible $\Rightarrow \mu=1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk $\Rightarrow$ single-sector boundary* (the AQFT form of $\det K=1$). 4/5 discharged.
- **v288_full_l2_subprincipal_z4.py** (SEAM.EQUIV.B01): Route B (DtN) — *proves* the full- L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \pmod{4}$, so $\Lambda = |n| + M_f$ commutes with $\rho = \text{diag}(i^n)$ on the full operator, $\|[\rho, \Lambda]\| \sim 2 \times 10^{-15}$; generic curvature $\rightarrow 4.44$), lifting v201/v284 to L^2 . The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, AUDIT OK.

2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **v284_route_i_rp_uniqueness.py** (QGEO.ROUTEI.01) turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1 RP $\rightarrow$ DtN (v194), L2 marks $\rightarrow$ pillowcase (v195/v216/v214), L3 flat Steklov = $\sqrt{-\Delta_{\text{flat}}}$ (verified here), L4 covariance from DtN (v258), L5 μ_4 -invariance (v276/Lean/v280); [C] Troyanov ($\chi_{\text{orb}}=0 \Rightarrow$ unique flat

$\tau=i$) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state*.

- **v285_route_ii_seam_condensation.py** (QGAMB.ROUTEII.01) turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E] $3/4$ discharged (no abelian sector $\Leftrightarrow \det K=1 \Leftrightarrow$ holomorphic $\Leftrightarrow E_8$; the tower $16 \rightarrow 4 \rightarrow 1$), and **its open lemma coincides with Route (i)’s** — both are “the raw seam is the $(E_8)_1$ -at- $\tau=i$ object” (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau=i$. Build green, AUDIT OK.

2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at- $\tau=i$ unification, live scorecard, v280–v283)

- **v280_pillowcase_steklov.py** (QGEO.STEKLOV.01) runs the QGEO chain on the actual flat $\tau=i$ pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$ is numerically exact — route-(i) evidence for QGEO.OBLIG.01; the premise stays [O].
- **v281_holomorphy_modular_data.py** (QGAMB.MODULAR.01): $\# \text{anyons} = |\det \text{Gram}|$ ($E_8 \rightarrow 1$, $SO(16) \rightarrow 4$), the anyon-condensation tower $16 \rightarrow 4 \rightarrow 1$, Gauss–Milgram $\rightarrow c=8$; holomorphy ($\det K=1$) is the single Tier-B discriminator.
- **v282_e8_tau_i_unification.py** (QGAMB.UNIFY.01): the candidate *simplification* — $\chi_{E_8}=j^{1/3}$ gives $\chi_{E_8}(i)=1728^{1/3}=12$, so $\tau=i$ is both the order-4 CM point (QGEO) and the $(E_8)_1$ modulus (QG.AMB): the two obligations are two faces of one object (the seam is $(E_8)_1$ at $\tau=i$), **obligation count** $2 \rightarrow 1$.
- **v283_live_killtest_scorecard.py** (PRED.LIVE.01): the recent round’s computable predictions vs current data — all consistent $< 2\sigma$, with two sharp near-term kill tests (Σm_ν floor 0.0586 eV; proton decay 1.55×10^{35} yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, AUDIT OK.

2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/FORM.QGEO.03)

- **v279_qgeo_obligation_lemma.py** (QGEO.OBLIG.01) writes the bedrock as **ONE precise constructive-QFT lemma**. Given the premise *the raw seam carries the flat $\tau=i$ pillowcase metric* (Trojanov-unique, $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$), then $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow \text{mark-locality} \rightarrow E_8$. [E] the reduction DAG from ‘free RP seam’ has exactly *one* open leaf — ‘the raw seam state is the flat state’; [E] the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Trojanov + OS).
- **FORM.QGEO.03: the all-orders step is now machine-proved in Lean**. `SeamDeckClosure.lean` gains `diagonal_commutatesClock` and `flat_all_orders_clock`: any spectral function $g(\Delta_{\text{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho = \text{diag}(i^n)$ — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full lake build green, 3356 jobs). The premise (the seam *is* flat) stays the [O] QGEO.SYM.01 input. Build green, AUDIT OK.

2026-06-18 (4D QFT: the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **v278_lszy_bridge_unitarity.py** (QFT4D.SPERT.04) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity. [E] LSZ bridge: the EG T -products of S_{pert} (v269) are the OS Wick-rotated correlators

(v240), so amputate+on-shell-residue gives $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$; [C] the gap $\Delta=6\log(3/2)>0$ gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2\text{Im } M = \sum \int d\Pi |M|^2$ (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector’s Stelle ghost (v269/rt_F) is non-unitary $\rightarrow$ QG.AMB.01. Exact core mirrored in Wolfram (273/273). Build green, AUDIT OK.

2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\rightarrow (E_8)_1$ match, v277)

- **v277_seam_calderon_e8_match.py** (QGAMB.TIERB.01) **reduces Tier-B to ONE bit**. [E] the full $(E_8)_1$ matching target is computed and matches: $c=8$, $\det \text{Cartan}(E_8)=1$, the holomorphic character $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$ with a single primary and exactly $248=\dim E_8$ weight-1 currents (240 roots). [E] the discriminator is holomorphy — the $c=8$ counterexample $(D_8)=SO(16)_1$ has $\det \text{Cartan}=4$ (non-holomorphic), so the one condition is $|\det K|=1$. [C] the seam side matches via v276 (flat $\tau=i$) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, AUDIT OK.

2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **v276_qgeo_flat_closes_commutator.py** (QGEO.SYM.03) **collapses the bedrock to ONE geometric postulate**. The state-level residual the whole bedrock rests on is the operator equation $[\rho, H]=0$. [E] if $H=f(\Delta_{\text{flat}})$ and the order-4 deck $\rho(z \mapsto iz)$ is a flat- $\tau=i$ isometry, then $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$ *exactly*, to all orders (verified on the discretised square torus: $\rho^4=I$, $[\rho, \Delta]=0$, $[\rho, \sqrt{-\Delta}]=0$; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full H . [C] $\tau=i$ is forced by order-4 ($j=1728$; hexagonal $j=0$ is order-6, excluded). [O] QGEO.SYM.01 therefore collapses to the single statement “the raw seam IS the flat $\tau=i$ pillowcase quasi-free state” — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, AUDIT OK.

2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt_F)

- **redteam/rt_F_qft4d.py** (REDTEAM.F.01) **stress-tests the just-published round** (the perturbative S_{pert} layer, the PMNS Jarlskog, the ν -scale, the over-determination, the QG.AMB.01 roadmap). SURVIVES (NARROWED): two attacks land and are folded in. (1) the gravity R^2/Weyl^2 sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator $1/(p^2(p^2+M^2))$ has a negative-residue massive spin-2 mode), so S_{pert} is unitary for the matter+gauge sector only — which is exactly why the nonperturbative QG.AMB.01 is the real frontier (caveat added to v269). (2) the 0.11% anchor over-determination is conditional on the Λ -branch readout (LAMBDA.BRANCH.01, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo**. Version 5.2 (rev 191) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI 10.5281/zenodo.20743347. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser S_{pert} skeleton v269, the

concrete one-loop quartic **v271** and gauge β -coefficients **v273**), the complete complex PMNS matrix and leptonic CP strength **v270**, the honest absolute ν -mass-scale typing **v272**, the over-determination of the single mass anchor **v274**, and the **QG.AMB.01** obligation roadmap **v275**. Suite 273 modules (highest ID **v275**; **v186/v226** skipped), Wolfram 271/271, **run_all.py** green, **AUDIT OK**, **npm** build clean. The Zenodo description (**zenodo_description.html**) and the public website were synced in the same release.

2026-06-18 (scale + frontier round: ν -mass scale, EG gauge running, over-determination, QG roadmap, **v272–v275**)

- **v272_nu_mass_scale.py** (**FLAV.NUSCALE.01**) **types the absolute neutrino-mass scale honestly** (no fabricated Σm_ν — would contradict the **v184** firewall). **[E]** the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{EW})^2 / M_R$ (one UV input, like v_{geo} , **v153**); **[C]** the observed $m_3 = 0.050$ eV with the TFPT PS→GUT window $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ GeV (**v249**) needs $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; **[O]** M_R is not a clean $\bar{M}_{Pl}$ power, so the absolute value stays open; **[X]** the NO floor $\Sigma m_\nu = 0.0586$ eV is the cosmological kill test. Closes the PMNS bookkeeping (the last **[O]** is one seesaw ratio).
- **v273_eg_gauge_running.py** (**QFT4D.SPERT.03**) **derives the SM one-loop b_i via the EG gauge self-energy**. **[E]** the gauge 2-point is the same scaling-degree-4 extension as the quartic (**v271**), giving $(b_1, b_2, b_3) = (41/10, -19/6, -7)$ from the carrier/SM content (the same 41 as the carrier algebra, **v159**); **[O]** two-loop + all-order stay open. Exact b_i mirrored in Wolfram.
- **v274_scale_overdetermination.py** (**SCALE.OVERDET.01**) **shows the single mass anchor is over-determined**. **[E]/[C]** two independent routes to $\bar{M}_{Pl}$ agree — gravity $(8\pi G)^{-1/2} = 2.4353 \times 10^{18}$ GeV and cosmology (the compiler prediction $\rho_\Lambda / \bar{M}_{Pl}^4 = (3/4\pi^2) e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4} = 2.24$ meV) = 2.438×10^{18} GeV, to 0.11%; the seam being a conformal $c=8$ CFT makes “no derivable absolute scale” a theorem, not a gap. Sharpens **v153/v78**.
- **v275_qgamb_roadmap.py** (**QGAMB.ROADMAP.01**) **gives the open gate QG.AMB.01 a module**. **[E]** Tier A gap decoupling ($\Delta_{eff} = 1.648 > 0$, **v36/v76**); **[E]/[C]** Tier B reduction to the holomorphic $(E_8)_1$ boundary net (**v77**); **[C]** the perturbative layer in hand (**v269/v271/v273**); **[O]** the nonperturbative ambient measure stays the one structural frontier.
- Suite 271/271 Wolfram, **run_all** green, **AUDIT OK**, build green.

2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, **v271**)

- **v271_eg_oneloop_quartic.py** (**QFT4D.SPERT.02**) **takes the **v269** S_{pert} skeleton from “exists” to an actual EG number** (no path integral). **[E]** the first renormalization is the order-2 product T_2 ; the massless propagator has scaling degree $sd=2$ in $d=4$, the one-loop bubble has $sd=4=d$, so $\omega=sd-d=0 \Rightarrow$ exactly one logarithmic local counterterm ($C(\omega+d, d)=1$, finite, no infinite tower); the loop factor $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly — the *same* factor that normalises the spectral-action a_4 geometric quartic. **[C]** the ϕ^4 one-loop $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry $3 = s, t, u$ channels), EG initial condition $\lambda_\Phi(UV) = 1/(16\pi^2)$. **[O]** order-2/one-loop only; the all-order EG construction and the nonperturbative measure **QG.AMB.01** stay open. Exact core mirrored in Wolfram (270/270). **[E]** build green, **AUDIT OK**.

2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, **v270**)

- **v270_pmns_jarlskog_assembly.py** (**FLAV.PMNS.03**) **assembles the four TFPT channels** ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{ret}}{2}$, $\sin^2 \theta_{23} = \frac{1}{2}$, $\sin^2 \theta_{13} = \varphi_0^{ret} e^{-5/6}$, $\delta = 4\pi/3$) **into one exactly-unitary**

complex PMNS matrix and derives the leptonic CP strength. [E] $U = R_{23}U_{13}(\delta)R_{12}$ is unitary; [C] the Jarlskog $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ ($J_{\text{max}} = 0.03424$) is a *derived* number, the leptonic analogue of the CKM J (v88); [C] data: J_{max} vs NuFIT 5.2 0.0332 ($\sim 3\%$), J vs best fit -0.026 , the negative CP-violating sign reproduced. [O] δ is the μ_6 /triality phase (assembled, not from M_ν/D_F) and the absolute ν -mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, AUDIT OK.

2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **v269_spert_paqft_skeleton.py (QFT4D.SPERT.01) builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers: S_{top} (2d DHR braiding $|M|=1$, statistics, v243), S_{pert} (4d Epstein–Glaser/BV S-matrix of the spectral action), S_{phys} (LSZ on the OS Wightman functions, v240). [E] the spectral-action a_4 interaction is power-counting renormalizable (all operators $\dim \leq 4$; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap $\Delta = 6 \log(3/2)$ gives an IR-safe adiabatic limit ($S_{\text{pert}} \rightarrow S_{\text{phys}}$ via LSZ). [O] perturbative only — the ambient measure QG.AMB.01 stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, AUDIT OK.

2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **v268_theta13_carrier_trace.py (FLAV.TH13.01) closes the θ_{13} exponent origin.** The value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ was already checked (v16); now the exponent is exact: [E] $\gamma = 5/6 = \text{tr}_E Y^2$, the carrier hypercharge-squared trace ($Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, roots of $6Y^2 - Y - 1 = 0$; complement $\frac{1}{6}$ the neutrino ratio). [E] own channel: the $\mu\tau$ -breaking $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel — correcting the tempting (and wrong) “ θ_{13} from M_ν ” route flagged by the new script atlas. [O] the CP magnitude and the absolute ν -mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, AUDIT OK.

2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **verification/make_script_atlas.py generates verification/script_atlas.md** — a single grouped, dependency-aware view fusing the existing single sources (status_ledger.csv for claim→status→dependencies→supersedes, script_registry.csv/script_clusters.csv for the theme grouping, docs_map.csv for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into bash build.sh gen. [E] build green, AUDIT OK.

2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for QGEO.SYM.01, v267)

- **v267_qgeo_rigidity_minimal_axiom.py (QGEO.SYM.02) sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks”

($= \omega \circ \rho = \omega$) forces *everything*: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 independent of a ; cross-ratio $2 \Leftrightarrow j=1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism on $y^2=x^3-x$; and the unique flat pillowcase metric, mark-locality, $[\rho, \Lambda]=0$ and $\omega \circ \rho = \omega$ all follow (v214/v201/v264/v198). [E] neg controls: generic config $j \neq 1728$ ($\mathbb{Z}_2$ only), hexagonal $j=0$ ($\mathbb{Z}_6$, order 6) — only the square (order-4) realises the μ_4 clock. [C] second, independent reason: $\{1, i, -1, -i\}$ over $\mathbb{Q}$ with $j=1728$ (CM by $\mathbb{Z}[i]$), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like $c=\text{const}$) — the *reduction* is closed (axiom $\Rightarrow$ everything), the axiom is not. [E] build green, AUDIT OK.

2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, $+(15, 1, 1)$ survives, Hyper-K decisive, v266)

- **v266_ps_threshold_proton.py** (PS.PROTON.02/PS.THRESHOLD.02) decides branch B of the **v265 fork**. The explicit two-step Pati–Salam thresholds give $d=6$ $\tau_p(p \rightarrow e^+ \pi^0)$ (writes `ps_threshold_proton.json`): [X] the minimal 16-Higgs content ($M_{GUT} \sim 2.4 \times 10^{15}$) gives $\tau_p \sim 4.4 \times 10^{33}$ yr $<$ Super-K and is *killed* even at the high hadronic band; [C] the E_8 -allowed $(15, 1, 1)$ [the single 45] raises $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$ yr, proton-safe *now*; [X] but that sits at the Hyper-K reach ($\sim 1.4 \times 10^{35}$ yr) — a *dated* kill test decisive within the decade. So branch B is promoted from “well-mannered candidate” to *content-selected* (*must be* $+(15, 1, 1)$), *Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an $O(3)$ τ_p band. Branch A (boundary-only) is unchanged and canonical. [E] build green, AUDIT OK.

2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- **v265_qft4d_fork_freeze.py** (QFT4D.FORK.01) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree (`qft4d_fork_freeze.json`), no new physics. **A (canonical)**: boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* (E_8 is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the $\alpha_1=\alpha_2$ scale α_3^{-1} off by ~ 5.6 , spread ~ 8.8 at 2×10^{16} GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional)**: carrier-native Pati–Salam is a falsifiable UV branch only with the E_8 -allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2 \theta_W = 3/8$ (v248), the $O(1)$ $\kappa = M_{PS}/M_s$ band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake τ_p window; carrier gauging stays the contract). [E] a prose guard (QFT4D.NO_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to `tfpt_research_contracts`; [E] build green, AUDIT OK.

2026-06-18 (Three open-residual advances: the F_{QCD} solver, M_ν from D_F , and the raw-seam $\Rightarrow$ mark-locality reduction, v262–v264)

- **v262_fqcd_mp_me.py** (FR.MPME.02) implements the one contract-only F_{transfer} pipeline. A real two-loop α_s solver runs $\alpha_s(M_Z)=0.1179$ with n_f thresholds (carrier $b_3 = -7$, [E]) to $\alpha_s(2 \text{ GeV}) \sim 0.30$ and $\Lambda_{\overline{\text{MS}}}^{(3)} \sim 0.33 \text{ GeV}$; with the lattice bridge C_p (external $O(1)$) and the closed m_e it reproduces $m_p/m_e \sim 1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
- **v263_mnu_from_df.py** (FLAV.PMNS.02) makes M_ν a seesaw operator of D_F . [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ is symmetric, seesaw-suppressed and (for $\mu\tau$ -symmetric inputs) $\mu\tau$ -symmetric;

[C] it reproduces $\theta_{23}=45^\circ$ and $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ (v9) under the seam misalignment; [O] $\theta_{13}=0$ in the leading texture, the CP *magnitude* (240°) is the μ_6 /triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

- **v264_raw_seam_marklocal.py (QGEO.ENERGY.03) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is $\mathbb{Z}_4$ -invariant (Fourier on $4\mathbb{Z}$, FFT-verified), block-diagonal, and $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho=\omega$ on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=QGEO.SYM.01). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the μ_4 orbit breaks the chain). Paper citations added in tfpt_4_frontier, tfpt_2_standard_model and tfpt_research_contracts. [E] AUDIT OK, build green.

2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ closed modulo QGEO.SYM.01 (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales (m_p/m_e), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The tfpt_2 “ambient closure” section now states that the cutoff of its S_{rel} is the seam KMS weight ($f_2/f_0=1$, v259) and D_{rel} the covariance induction (v258); the tfpt_2 “(4) Constructive QFT closure” section adds the relative-object pointer; tfpt_4_frontier’s full-quantum-gravity section records that the cutoff moment f_0 is pinned by the seam state (v255/v259); and origin_theory’s honest-status keybox notes that the whole boundary QFT collapses onto the same QGEO.SYM.01 postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (10 ok, 0 failed), AUDIT OK.

2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)

- **The v258–v261 closure is now reflected on every “what is open” surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary G_{net} entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise QGEO.SYM.01 via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card (“the structural residual is one condition”) carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, AUDIT OK.

2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, v258–v261)

- **v258_dirac_covariance_induction.py (PS.DIRAC.03) closes the “posited Yukawa vs derived operator” gap of v250/v252.** The finite Dirac operator D_F is the modular/covariance *induction* of the boundary seam quasi-free (KMS, $\beta=1$) state: a Gaussian state is fixed by its covariance C (a positive contraction) with one-particle modular Hamiltonian

$H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$. [E] the inversion identity holds exactly; [E] C_F is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252 D_F (same mass spectrum); [E] the Majorana/seesaw m_D^2/M_R is an off-diagonal covariance entry. [C] $C_F = P_F C_\Sigma P_F$, so D_F is the Kasparov shadow $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$ of the boundary geometry and the v18 Yukawas are the readout of C_Σ .

- **v259_modular_cutoff_kappa.py (PS.SPECACT.02) fixes the last open spectral-action input.** The cutoff f is the seam’s own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at $\beta=1$ (v239) and $2\pi=1/(4c_3)$ (v58) fixes β , so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*. [E] its moments give $f_2/f_0 = f_4/f_2 = 1$ exactly; [E] a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$ (the choice is non-vacuous); [E] hence $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$ loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim H_F (the RG $\kappa \approx 1.3$ pins $c_{PS}/c_{\text{grav}} \approx 1.7$).
- **v260_k3_kummer_unification.py (ARCH.K3.01) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient $T_{\tau=i}^2/(z \mapsto -z)$ (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the E_8 Cartan is even/det 1/positive-definite; [E] $H^2(K3, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$ (rank 22, det -1 , even, signature $(3, 19)$); [E] the seam is $T^2/(z \mapsto -z)$ with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the $|A[2]|=2^4=16$ Kummer nodes = $\dim S^+$ (v197), $16=4 \times 4$; [E] honest note — the rank-16 Kummer lattice is *not* the $E_8(-1)^2$ summand. [C] so the v1 “glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single object; E_8 appears both locally (du Val $\mathbb{C}^2/2I$, v232/v236) and globally (H^2 summand). A unification, not a closure.
- **v261_modular_spectral_closure.py (QFT.MSC.01) is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ reduced to *one* open premise. [E] the cross-consistency invariants agree: $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap $6 \log(3/2)$ (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)

- **The canonical changelog now also drives a public /changelog website page.** A new generator `verification/make_changelog_web.py` parses this file (dated `\subsection*` entries, `itemize` items, and the inline markup `\textbf/\emph/\texttt/\vref`, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx + website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] `build (npm run build green, /changelog prerendered static)` and `bash build.sh audit (AUDIT OK)`.

2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- **v257_quasiorient_modpd.py** (PS.NCG.ORIENT.02) **corrects the slightly too-generous v256 typing**. The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit **v252** triple: **[E]** strict orientability fails ($\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$); **[E]** *quasi-orientability* holds (the left/right A -irrep boxes are disjoint by chirality: $L = H$ weak-doublet, $R = \mathbb{C}$ singlet, so $L^{\text{LR}}(\mathcal{H}^{\text{even}}, \mathcal{H}^{\text{odd}}) = \{0\}$); **[E]** strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal- L/R -neutrino case); **[C]** modified Poincaré duality holds (CCM), Stephan’s alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **v254_inner_fluctuations.py** (PS.NCG.FLUCT.01) **closes v250 #5 + the no-junk obligation**. The inner fluctuations $\Omega_D^1(A_F)$ of the finite Dirac operator are computed explicitly: **[E]** they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet $(1, 2)_{1/2}$ — no junk, no triplet, no $(10, 1, 3)$ /**126**. **[E]** the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and **[E]** σ is an SM singlet $(1, 1, 0)$ with $B-L = 2$. **[C]** σ = scalaron.
- **v255_spectral_action_expansion.py** (PS.SPECACT.01) **closes the spectral-action expansion (v252 #11)**. The Seeley–DeWitt expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$: **[E]** structure (a_0 cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl² + R^2 +Higgs-quartic, $N = 96$); **[E]** $b/a^2 = 1/3$ (top-dominated); **[E]** $\sin^2 \theta_W = 3/8$; **[E]** the R^2 spin-0 mode = the Starobinsky scalaron; **[C]** Higgs quartic $\rightarrow \sim 125$ GeV with σ ; **[C]** $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$ explicit; **[O]** exact decimal needs the cutoff moments.
- **v256_orientability_poincare.py** (PS.NCG.ORIENT.01) — **honest status, no faked proof**. KO-dimension 6 forces the intersection form $\text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$ to be *antisymmetric*: **[E]** skew form (verified), **[E]** rank 2, corank 1, null $\sim (1, 1, -1)$ on the SM K-theory $\mathbb{Z}^3$; **[E]** the orientability local condition $[\gamma_F, \pi(a)] = 0$. **[C]** the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); **[O]** a self-contained machine proof needs the complete canonical bimodule.

2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and κ pinned to $O(1)$, v252–v253)

- **v252_full_finite_triple.py** (PS.DIRAC.02) **advances the Dirac contract v250 from a 7-obligation contract to a concrete object discharging 5 to [E]**. The full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ — H_F the particle $\oplus$ antiparticle doubling of three $SO(10)$ **16**-plets — is built explicitly and the NCG axioms are checked by direct computation: **[E]** reality ($J^2 = +1$, $JD = DJ$), grading ($\gamma^2 = 1$, $[\gamma, a] = 0$, $J\gamma = -\gamma J$), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to $\{\nu_R, e_R\}$ — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\text{light}} = m_D^2/M_R$. **[C]** Poincaré duality; **[O]** orientability, no-junk, spectral-action expansion.

- **v253_kappa_scalaron_ps.py** (PS.KAPPA.01) discharges residual 6(b) of **v249**. κ in $M_{PS} = \kappa M_s$ ($M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13}$ GeV, exact) is pinned by an RG scan (PDG errors $\times$ the two E_8 -allowed contents) to a tight $O(1)$ interval (≈ 1.3). [C] heat-kernel origin: $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$ is a scale-free ratio of heat-kernel traces over the 96-dim H_F (Λ cancels). [X] the **126**-content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs f_2/f_0 fixed.

2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/ σ — the CCvS Pati–Salam mechanism, v251)

- **v251_first_order_sigma.py** (PS.DIRAC.01) advances the Dirac contract **v250** from a contract to an exhibited mechanism. On a faithful lepton-block model ($H = H_F \oplus H_F^c$, $H_F = \mathbb{C}^3$), the real even structure is checked ($J^2=+1$, $\gamma^2=1$, $D\gamma=-\gamma D$, KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ($\nu_R \nu_R^c = \sigma$) block — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ . Honest correction to the review: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C] $\sigma = \text{scalaron}$ (v249); [O] the full 96-dim triple + spectral action.

2026-06-17 (Carrier-native Pati–Salam program: E_8 forbids the 126, PS algebra, unification kill-test, Dirac contract, v247–v250)

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** **v247_e8_branching_no126.py** (PS.E8BRANCH.01): the E_8 hull branches under $SO(10) \times SU(4)$ as $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4})$, so the $SO(10)$ content is exactly $\{1, 10, 16, 45\}$ and the **126** is *absent* — the renormalisable 126_H is algebraically forbidden and $10+16+45$ is E_8 -supplied (only one 45 $\Rightarrow$ proton-marginal). **v248_ps_algebra.py** (PS.ALGEBRA.01): $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \rightarrow SU(2)_L \times SU(2)_R \times SU(4)_c$, exact hypercharges $Y = T_{3R} + (B-L)/2$, matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2 \theta_W = 3/8$ (colour $SU(4)_c \subset D_5$, distinct from the family A_3). **v249_ps_unification.py** (PS.RGTEST.01): the two-step $PS \rightarrow SO(10)$ unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content), $M_{PS}/M_s = 1.0\text{--}1.7$. **v250_dirac_triple_contract.py** (CONTRACT.QFT4D.DIRAC.01): the spectral-triple contract ($H_F = 3 \times 16$, the seven NCG axioms, $\sigma = \text{scalaron}$) that would turn “carrier is gauged” from a fork into a theorem. The root-level **qft.txt** carries the full analysis.

2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)

- **v246_unification_data_crosscheck.py** (QFT4D.RGTEST.01) — the first hard data confrontation of the spectral-action prediction. TFPT’s gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the E_8 cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread $\sim 10^{13}\text{--}10^{17}$ GeV (residual $\Delta\alpha^{-1} \approx 9$ at 2×10^{16} GeV), and at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at 1.7×10^{14} GeV — the three never meet. Typed [X]/[O]: with “no new state” there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (**figures/qft_skeleton.pdf**, **figures/qft_unification.pdf**) added via **make_figures.py** (PDF + website PNG) and embedded in **tfpt_research_contracts.tex**; a root-level **qft.txt** summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

2026-06-17 (Spectral-action matter content: the $SO(10)$ $\mathbf{16}$ = one anomaly-free SM generation, v245)

- **v245_spectral_matter_content.py** (QFT4D.MATTER.01) discharges the computable core of the 4d contract **v244**. The carrier half-spinor (the $SO(10)$ $\mathbf{16}$) is verified, in exact rational arithmetic, to be one Standard-Model generation: $\mathbf{16} = SU(5)(\mathbf{10} + \bar{\mathbf{5}} + \mathbf{1})$, the SM multiplet dimensions sum to $16 = \dim S^+$ (15 Weyl fermions + ν_R), the electric charges $Q = T_3 + Y$ are exactly the SM values, all four gauge anomalies cancel ($\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$), the spectral-action normalisation gives $\sin^2 \theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3} g_1$), and the Higgs is the $(1, 2)_{1/2}$ inner fluctuation. So the matter-content + tree-relation half of CONTRACT.QFT4D.01 is now **[E]**; only the seam $\rightarrow$ Dirac realisation and the run-down couplings stay **[O]**. Wired into `run_all.py`, the ledger, the registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238)**. `v239_kms_thermal_time.py` (DYN.KMS.01): the modular flow is KMS at $\beta = 1$, and $2\pi = 1/(4c_3)$ fixes the modular temperature as $T_H = c_3/M$ — thermal time = horizon time. `v240_gns_os_reconstruction.py` (DYN.GNS.01): GNS reconstructs the Hilbert space from $(\mathcal{M}, \omega)$ and the OS Hamiltonian $H_{OS} = -\log T = -L$ is positive with gap Δ (the **v64** mass gap). `v241_dhr_sectors.py` (DHR.SECTORS.01): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower $16 \rightarrow 4 \rightarrow 1$ (**v235/v237**). `v242_spin_statistics_modular.py` (DHR.MODULAR.01): spin-statistics, the modular (S, T) data, Verlinde fusion, and Gauss–Milgram $c = 8$. `v243_haag_ruelle_braiding.py` (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the **v238** mass gap supplying Haag–Ruelle asymptotic states. `v244_spectral_action_contract.py` (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor $\mathbf{16}$ is one generation, the SM gauge group sits in the carrier, gravity is **v36**; the matter Lagrangian is the open **[O]** contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into `run_all.py`, the ledger and registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The static $\rightarrow$ dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New v238_modular_lindblad_dynamics.py** (DYN.SEMIGROUP.01, **cluster registry**). A companion to the modular round (**v196/v198**) and the recovery code (**v221/v64**) that reads the *same* seam objects *dynamically* rather than statically. **[E]** the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the μ_4 clock, $\rho \sigma_t \rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$, IS the **v198** state-invariance — so the static commutator is the conservation law of modular time. **[E]** the recovery channel’s generator $L = \log T$ is a doubly-stochastic/CPTP-classical Markov generator ($e^L = T$, $e^{L\tau} \rightarrow$ the democratic projector = the arrow of time). **[E]** keystone: the dissipative gap $= -\log((2/3)^6) = 6 \log(3/2) =$ the OS mass gap Δ (**v64**) — the static recovery bound (**v221**) and the dynamical mass gap (**v64**) are one number, one exponentiated. **[E]** the GKSL split $G = i\Lambda_\Sigma + L$ (imaginary + real ≤ 0 spectrum). **[O]** the residual is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra = QGEO.SYM.01 (**v198**) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

2026-06-17 (Experiments: black-hole-cosmology signatures, the η_B Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (experiments/, all data_limited, not load-bearing).** From the problem_b black-hole-cosmology survey: (i) **ccbh-dark-energy** — the de Sitter seam interior ($w_{\text{in}}=-1$) forces the Croker–Weiner cosmological-coupling index $k = -3w_{\text{in}} = 3$, hence a population $w = -1$; vs. Farrah+2023 $k = 3.11 \pm 0.79$ (-0.14σ), a *contested* downstream bridge and an *alternative reading* of the same $w = -1$ the DESI watchdog tests (**alternative_group w_de_eos**, never double-counted). (ii) **gravastar-compactness** — the Nariai $Q_{\text{geom}}=3/8$ equals the Jampolski–Rezzolla (2026) maximum horizonless compactness $\mathcal{C}=3/8$ (shared de Sitter endpoint $1/2$); since $1/3 < 3/8 < 4/9 < 1/2$ this is a light-trapping ECO, yielding an echo template (round-trip delay ~ 0.7 ms at $62 M_\odot$, amplitude $\leq (2/3)^6$) — a **[C]** structural echo, no $\mathcal{C} \leftrightarrow Q_{\text{geom}}$ map proven. (iii) **cosmic-handedness** — a parity watchdog (Shamir JADES $\sim 3.3\sigma$ spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).
- **η_B — full BDP Boltzmann ODE solve confirms the route at the frozen scale (experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py; tfpt_4_frontier, [C]).** Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency $\kappa_f = 0.092$, validating the analytic fit) at the *frozen* heavy scale $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV with $\delta_{\text{CP}} = 4\pi/3$ gives $\eta_B = 6.5 \times 10^{-10}$ — a factor 1.07 from the observed 6.1×10^{-10} , with *no* free M_R dial. This moves the leptogenesis route from “solve pending” to a consistent **[C]** readout (the full *flavored* density-matrix solve is the next refinement); tfpt_4_frontier Route C synced.
- **The Petz recovery map and the rank-one baby universe, realised numerically (experiments/recovery-channel; tfpt_horizon_readouts, [C]; companion to v221).** The gapped transport contracts to a *rank-one* projector at exactly $\|T^n - P_\infty\| = (2/3)^{6n}$ — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected $\lambda=1$ (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification v221 deferred is now verified at the channel level (**internal_consistency**, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain (experiments/gw-ringdown-echo; tfpt_horizon_readouts search-targets).** The Stage-1 machinery (Kerr-ringdown subtraction $\rightarrow$ matched filter on the residual, ratio $(2/3)^6$ frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in ≥ 2 detectors**, consistent with the $(2/3)^6$ *upper* bound (a single loud H1 excess has $\hat{q} \approx 2$, far from $(2/3)^6$, so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.
- **Scorecard + website synced.** experiments/evidence_scorecard.json now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website EXPERIMENTS_AUDIT mirror and the η_B prediction entry are updated to match. These are standalone experiments/ surfaces (not in the verification suite or ledger); the tfpt_4_frontier/tfpt_horizon_readouts and papers.ts/predictions.ts edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A** re-synced to `tfpt_5_redteam` (`rt_A_e8net.py`, `redteam_table.txt`, `redteam/README.md`, `infrastructure`). The Python red-team surface still emitted the historical *three-residual* “A reduced, not closed” form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement “the seam–Calderón boundary net is holomorphic with $c = 8$ ” ($\Leftrightarrow$ the index-4 simple-current extension), with the index-4 $(D5)_1 \times (A3)_1 \rightarrow (E8)_1$ glue realised at Lie level ([v143](#)) and its odd sectors twisted/Ramond ($E_8 = 120_{\text{NS}} + 128_{\text{R}}$, [v148](#)), while the $q(A_3)$ normalisation collapses into the one declared anchor ([v152](#)). Two exact adversarial checks added ($120+128 = 248$; glue index $= |\mu_4| = N_{\text{fam}}+1 = 4$); status stays [\[C\]](#)/[\[O\]](#) (reduced, not closed).
- **Shadow export now mirrors figures/ but not website/** (`scripts/export-shadow.sh`, `verification/make_script_index.py`, `infrastructure`). `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh` notes runs on the subset without crashing.

2026-06-17 (CP phases as μ_6 powers — a reduction of red-team Target D)

- **Both CP phases are μ_6 powers of one hexagonal CM unit, split by the sheet** ([v231_cp_mu6_phases](#), `tfpt_5_redteam`, [\[E\]](#)/[\[C\]](#)). Sharper than [v220/v225](#) (which only *located* the CP residual). With $\rho = e^{i\pi/3}$ the primitive 6th root (the $j=0$ Eisenstein CM unit): δ_{CKM} leading $= \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the phase lattice), with $\rho^4 = -\rho$, so $\delta_{\text{PMNS}} = \delta_{\text{CKM,lead}} + \pi = (\text{CM unit}) \times (\mathbb{Z}_2 \text{ sheet}, \rho^3 = -1)$. The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ($\pm 21 \sin(\pi/3)$). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction $3\lambda_C^2$ stays the [v88](#) misalignment, the deck stays $\mathbb{Z}/4$, and the physical CP derivation stays [\[C\]](#) (CP.MU6.01).
- **The seam as the E_8 Kleinian singularity** ([v232_e8_kleinian_seam](#), `origin_theory`, [\[E\]](#)/[\[C\]](#)). The du Val side of the McKay correspondence ([v219](#)) gives the open seam-realisation premise `QGE0.REALIZE.01` a canonical algebraic-geometric model: dropping the trivial-rep node of the affine- E_8 McKay graph of $2I$ leaves the finite E_8 Dynkin = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$; the eight curves = rank $E_8 = g_{\text{car}} + N_{\text{fam}}$ (a fourth reading of the 8), their negated intersection form is the E_8 Cartan (det 1, even, (-2) -curves), and the link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ perfect $\Rightarrow H_1=0$, $\pi_1=2I$ order 120). The raw seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes E_8 ; `TOP0.E8.01`, bedrock stays [\[O\]](#)).
- **CP is the universal family/triality phase, only sheet-split** ([v233_cp_triality_phase](#), `tfpt_5_redteam`, [\[E\]](#)/[\[C\]](#)). One step past [v231](#) in the $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$ factorisation: $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the $2/3$ cusp weight). Since $\rho^4 = \rho^1 \rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (CP.TRIALITY.01).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate** ([v234_seam_holomorphy_selection](#), `tfpt_research_contracts`, [\[E\]](#)/[\[O\]](#)). The whole structural residual is *one* condition — “the seam carries no nontrivial abelian sector” — with three provably-equivalent faces, all forcing E_8 : (i) holomorphy (μ -index 1; Target A), (ii) the

seam link is a homology 3-sphere (Γ perfect $\Leftrightarrow 2I$; v232), (iii) exactly one 1-dim irrep (v219). All equal $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$, which is 1 only for E_8 ; and holomorphic $c=8=g_{\text{car}}+N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice E_8 . So P2, G_{net} , Target A and the Kleinian seam are one gate. The stated closing theorem ([O]): $\text{RP}+\text{gap}+\text{chirality} \Rightarrow \text{quasi-free bulk}$ (v160) $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$; the single residual analytic step is “free bulk $\Rightarrow$ holomorphic boundary” (GATE.HOLO.01).

- **The closing step in Chern–Simons language: holomorphic $\Leftrightarrow \det K=1$** (v235_seam_chern_simons, tfpt_research_contracts, [E]/[O]). A genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer K -matrix with $\#\text{anyons} = |\det K|$, edge $c = \text{signature}(K)$, and the edge net holomorphic $\Leftrightarrow |\det K|=1$. The TFPT extension tower (v92) is the condensation tower read by determinant: carrier $D_5 \oplus A_3$ (det 16, 16 anyons) $\rightarrow SO(16)_1 = D_8$ (det 4) $\rightarrow (E_8)_1$ (det 1, holomorphic) — all rank 8, $c=8=g_{\text{car}}+N_{\text{fam}}$ — so holomorphic = E_8 = the Kitaev E_8 quantum-Hall state. Holomorphy $\Leftrightarrow$ condensing the order- $|\mu_4|$ Lagrangian glue (det 16 $\rightarrow$ 1, vs. a partial order-2 isotropic $\rightarrow 4=D_8$). The residual narrows to one integer condition “the free RP seam condenses the Lagrangian glue (det=1)” = the sheet/QGEO selection, now with holomorphy $\Leftrightarrow \det 1$ a theorem (GATE.HOLO.02).
- **The (2,3,5) Brieskorn singularity is the one generator of the skeleton** (v236_brieskorn_capstone, origin_theory, [E]). Capstone of the icosahedral round: the singularity $x^2+y^3+z^5$, whose exponents are the atoms ($|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2,3,5)$, has Milnor number $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$ (a *fifth* origin of the “8”), Milnor monodromy = the order-30 E_8 Coxeter element (eigenvalues the primitive 30th roots = the E_8 exponents, charpoly Φ_{30}), Milnor lattice E_8 , link the Poincaré sphere; the two clocks are facets of this one monodromy — the μ_3 triality (CP, v233) is h^{10} in $\langle h \rangle = \mathbb{Z}/30$, the μ_4 clock (v223) is the order-4 Galois automorphism of $\mathbb{Q}(\zeta_{30})$ (TOPO.BRIESKORN.01).
- **The closing step as a physical condition: no topological degeneracy $\Leftrightarrow \det K=1 \Leftrightarrow$ the seam is SRE** (v237_seam_sre_closure, tfpt_research_contracts, [E]/[O]). Sharpens GATE.HOLO.02’s residual from definitional to falsifiable: the K -matrix ground-state degeneracy on a genus- g surface is $|\det K|^g$ (torus: carrier 16, D_8 4, E_8 1), so “no topological degeneracy on any closed surface” $\Leftrightarrow \det K=1$; among rank-8 even positive-definite K ($c=8=g_{\text{car}}+N_{\text{fam}}$) only E_8 is short-range-entangled (the Kitaev E_8 phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$); so the one open step is the physical statement “the free RP seam is SRE (no topological order)” (GATE.HOLO.03).

2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are $\{2,3,5\}$** (v219_icosahedral_mckay, origin_theory, [E]). E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T \rightarrow \hat{E}_6$, $2O \rightarrow \hat{E}_7$, $2I \rightarrow \hat{E}_8$). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group $2I$ (element orders $\{1,2,3,4,5,6,10\}$ — it contains the 2,3,5 rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are $\{1,2,2,3,3,4,4,5,6\}$, sum $30=h(E_8)$, sum of squares $120=|R^+(E_8)|$; the McKay adjacency from the natural 2-dim rep *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8 , not a P2 proof (MCKAY.E8.01).
- **CM-norm duality: 41 (square) and 7 (hex)** (v222_cm_norm_duality, origin_theory, [E]). The square seam (Gaussian $\mathbb{Z}[i]$, $j=1728$) gives $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$ (the EM index *is* the Gaussian norm of the carrier-glue vector) and $N(3+2i)=13=\Delta_Q$; the hexagonal partner (Eisenstein $\mathbb{Z}[\omega]$, $j=0$) gives $N(3+2\omega)=7=\text{scalaron}$. The (3,2) split generates (5,6,7,13) by sum/product/Eisenstein/Gauss norm; rings rigid (CMNORM.DUAL.01).

- **The μ_4 clock is the order-4 character of the E_8 Coxeter cycle (v223_coxeter_totative_clock, origin_theory, [E]).** $(\mathbb{Z}/30)^\times$ are the E_8 exponents; the order-4 generator 7 ($\langle 7 \rangle = \{1, 7, 13, 19\}$) is a literal μ_4 clock permuting the four conjugate planes; only E_8 has $\varphi(h) = \text{rank}$ and $h = 2 \cdot 3 \cdot 5$ (COX.CLOCK.01).
- **248=120+128 as a magnitude/phase channel typing (v227_degree_exponent_channel_split, tfpt_5_redteam, [E]).** Exponents sum $120 = |R^+(E_8)|$ (magnitude channel), degrees sum $128 = \text{rank} \cdot \dim S^+$ (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong S^- reading; E8.CHAN.01).
- **P2 as the mode space of a degree-4 seam divisor (v228_rr_index_gate, origin_theory, [E]/[C]/[O]).** A Riemann–Roch index gate: $\deg D = 4 = |\mu_4|$ on $\mathbb{P}^1$ forces $h^0 = 5 = g_{\text{car}}$, $\text{rank } H_1 = 3 = N_{\text{fam}}$, $h^0 + H_1 = 8 = \text{rank } E_8$, $\Lambda^{\text{even}}(\mathbb{C}^5) = 16 = \dim S^+$ — bedrock language for P2, conditional on the seam producing the degree-4 divisor (QGE0.RR.01).
- **The seam as a finite recoverability code (v221_seam_qecc, origin_theory, [E]/[C]).** Code dimension $\dim S^+ = 16$; the gapped transport is a CPTP doubly-stochastic contraction with spectrum $\{1, (2/3)^6, (1/3)^6\}$, recovery bound $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$ (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (QEC.SEAM.01).
- **CP lives in the hexagonal phase fiber (v220_cp_hexagonal_modulus, tfpt_5_redteam, [E]/[C]).** The seam deck is the square modulus $j = 1728$ ($\mathbb{Z}/4$); its partner is the hexagonal $j = 0$ ($\mathbb{Z}/6$, $\arg \rho = \pi/3$). The frozen $\delta = \pi/3 + 3\lambda_C^2 = 68.654^\circ$ has leading term $\pi/3 = \arg \rho$; the deck stays $\mathbb{Z}/4$, CP is the $\mathbb{Z}/6$ fiber (CP.MOD.01).
- **The dual normal frame (d, n) with CP as orientation (v225_dual_normal_frame, tfpt_5_redteam, [E]/[C]).** $d = a^\top R^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1) = -\frac{1}{2}$ Nariai; $n = (5, -9, 6)$ reads $\det$ on the first-generation column; $\det(1, d, n) = 21 = 3 \cdot 7$; $\text{Im } \det(1, d, \rho n) = 21 \sin(\pi/3)$ — CP as orientation (DUAL.FRAME.01).
- **F_{transfer} as one path on the Sheet Diamond (v224_diamond_ftransfer_path, tfpt_2_standard_model, [E]/[C]).** $M(s, t) = R + Q \text{diag}(s, t, t)$: K, C, F on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps $(1, 8, 10), (1, 8, 16)$; the Koide source $\rightarrow$ pole is the 1-D projection (FTR.PATH.01).
- **The charged leptons as an étale Frobenius algebra (v229_lepton_frobenius_algebra, tfpt_2_standard_model, [E]/[C]).** $(c_e, c_\mu, c_\tau) = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, ring closure $c_e c_\tau = \frac{8}{3} = |\mathbb{Z}_2| c_\mu$, product $\frac{32}{9}$; $A = \mathbb{Q}[t]/(m)$ is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the μ_6 resolvent (LEP.FROB.01).
- **The center budget $(7, 11, 13)$ as three local norms (v230_center_budget_norms, tfpt_2_standard_model, [E]).** $C = R + Q \text{diag}(1, 0, 0)$ row sums $(7, 11, 13)$: $7 = N_{\mathbb{Z}[\omega]}(3 + 2\omega)$ (hex), $13 = N_{\mathbb{Z}[i]}(3 + 2i)$ (square), $11 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2}$ (QBL Fock count); the two CM rings and the boundary count meet in one operator center (CENTER.NORM.01).

2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (v218_diamond_axis_geometry, tfpt_2_standard_model, [E]).** Sharpens v94/v95 with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis $\det(C + xU) = 14 + 6x$ (slope $6 = |R^+(A_3)|$) and quadratic on the sheet axis $\det(C + yV) = 14 + 14y + 4y^2$ (2nd difference $8 = \text{rank } E_8$); the anchor block has $\det B(C + xU) = 2 \det(C + xU)$ and sheet curvature $6 = |R^+(A_3)|$ — two curvatures $(8, 6) = (\text{rank } E_8, |R^+(A_3)|)$. **(2) Plücker transfer ladder:** $\text{Pl}(K) \rightarrow \text{Pl}(C) \rightarrow \text{Pl}(F)$ lifts in steps $(1, 8, 10) = (N_\Phi, \text{rank } E_8, A_\Lambda)$ then $(1, 8, 16) = (N_\Phi, \text{rank } E_8, \dim S^+)$ — the

decouple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for Q, K, C, F the cubic discriminant factors as $q(r)^2 \text{Disc}(q)$ with squares $(1, 3, 4, 6)$ and kernels $(13, 48, 65, 105)$, F carrying $105 = 3 \cdot 5 \cdot 7$. The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ($28 = 2 \dim G_2$, $52 = \dim F_4$ stay audit-only, not spine, as in v94/v95); F as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger DIAMOND.AXIS.01, DIAMOND.PLUCKER.01, DIAMOND.SPECTRAL.01; Wolfram-mirrored (249/249).

2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed** (v216_marks_gauss_bonnet, tfpt_research_contracts, [E]; ledger QGEO.MARKS.03). The K1 lever of the v215 kill-test (“exactly four marks”) is *executed*: if the marks are $\mathbb{Z}_2$ branch points (deficit π) on the flat seam sphere ($\chi = 2$, the same χ as the 8 in c_3), Gauss–Bonnet forces $n\pi = 2\pi\chi = 4\pi$, so $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$. The closed Euclidean sphere 2-orbifolds are exactly $(2, 3, 6)$, $(2, 4, 4)$, $(3, 3, 3)$, $(2, 2, 2, 2)$, and three independent criteria — all cone orders = 2 (the $|\mathbb{Z}_2|$ sheet branch), rank $H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase $(2, 2, 2, 2)$. The numerical sibling v217_marks_emergence_scan ([C], ledger QGEO.EMERGE.01) confirms it on the actual DtN/state (number n and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap $(2/3)^6$). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. v216 exact (Wolfram extension → 248/248); v217 numerical (Python-only). Suite → 217 modules.

2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **QGEO.SYM.01 as a kill-test, not a proof-promise** (v215_seam_deck_killtest, tfpt_research_contracts, [E]/[O]; ledger QGEO.KILL.01). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock $\omega \circ \rho = \omega$ is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ with $|k|$ commuting exactly, so the residual is the bounded sub-principal M_f (equivalently the raw Gaussian bulk form A is μ_4 -deck-invariant). Levers: **K1** exactly four marks ($b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$); **K2** the square config (cross-ratio 2 $\Rightarrow j = 1728$, Aut = $\mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j = 0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support ($[\rho, M_f] = 0 \Leftrightarrow f$ $\mathbb{Z}_4$ -invariant); **K4** transfer gap $(2/3)^6 = 64/729$. Freeze-bound via freeze_file.csv (seam_deck_symmetry), carrier-side [E] (regression guards), with the *decisive* kill deferred to QGEO.REALIZE.01 [O] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts $j = 1728$ and $(2/3)^6$ are Wolfram-mirrored via v214/v54/v56). Suite → 215 modules.

2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of QGEO.SYM.01** (v214_seam_pillowcase, tfpt_research_contracts, [E]/[O]; ledger QGEO.PILLOW.01). The two separately-tracked open QGEO residuals — QGEO.ISO.01 (v180: the carrier clock is an order-4 *isometry* of the seam metric) and QGEO.SUBPRIN.01 (v201: the DtN sub-principal symbol is *mark-local*, the seam flat away from the μ_4 marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold $S^2(2, 2, 2, 2)$

($\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet $4\pi = 2\pi\chi$) — this *is* the v201 “flat away from the marks”. **(2)** New link: cross-ratio(μ_4) = 2 (v168) $\Rightarrow j = 1728$ ($j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2$); all six harmonic cross-ratios give 1728) $\Rightarrow$ the *square* modulus $\tau = i$, whose CM by $\mathbb{Z}[i]$ *derives* the order-4 clock $z \mapsto iz$ (explicit $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$); generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$. **(3)** Unification: “seam = uniformising flat pillowcase metric” implies *both* v201 and v180, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close QGEO.SYM.01: the choice “RP seam metric = the constant-curvature representative” stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/ j -invariant/CM arithmetic); Klein- J values mpmath-numerical. Suite $\rightarrow$ 214 modules.

2026-06-15 (the F_{transfer} transfer functor contract CONTRACT.F.01)

- F_{transfer} formalised as a typed functor with four axioms (v213_ftransfer_functor, tfpt_research_contracts, **[C]**). The four frontier transfers are consolidated into ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$, each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)** μ_4 -deck equivariance (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ is the deck transfer eigenvalue); **(2)** Plücker preservation ($53 = a^T(R+Q)\mathbf{1}$, $\text{Pl}(F) = (1, 22, 30)$, $\|\text{Pl}(K)\|_1 = 11$); **(3)** positivity/stochasticity ($\text{spec } T$ positive, $\kappa_f \in (0, 1)$); **(4)** external modules explicit ($b_3 = -7$ a carrier output). The four instances are F_{pole} (v183), $F_{\text{Boltzmann}}$ (v212), F_{relic} (v211), F_{QCD} (v164). CONTRACT.F.01 is the third research contract alongside (U_{wall}) and (G_{metric}), the structural complement of the v187 typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger CONTRACT.F.01.

2026-06-15 (frontier: two sharper **[C]** scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch** $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (v211_axion_spine_angle, tfpt_4_frontier, **[C]**). A more *robust* misalignment angle than the determinant-line hilltop $\theta_i \approx 170^\circ$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$): the central spine quotient gives $\theta_i = 3\pi/5 = 108^\circ$ exactly (no fit), the finite- T solver reaches Ω_{DM} at $\approx 106^\circ$, and 108° sits in the *mild*-anharmonic regime (62° below the hill-top) — not exponentially sensitive. An alternative ansatz to $\theta_i = \pi(1 - \varphi_\Sigma)$ (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); **[X]** $\Omega_a h^2$ outside $\sim [0.08, 0.16]$ demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decuple branch** (v212_leptogenesis_decuple, tfpt_4_frontier, **[C]**). Both Boltzmann inputs share the decuple $A_\Lambda = 10 = |E(K_5)|$: $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV and $\tilde{m}_1 = m_3/A_\Lambda \approx 5$ meV. M_1 uses *only* $\{M_{\text{scal}}, \varphi_0^{\text{ret}}, A_\Lambda\}$ — no hidden seesaw scale, cleaner than v184’s $M_R \varphi_0^4$ — and lands in the thermal window where $\eta_B \approx 1.2 \times 10^{-9}$ brackets the observed 6.1×10^{-10} . The coupling mechanism is posited not derived, so η_B stays **[C]**; **[X]** a precise Boltzmann/RGE solve outside the band falls the route, not the theory (FR.ETAB.04; Route A independent).

2026-06-15 (Lean: QGEO.COHO.M.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (CohomologyGrading.lean, ledger FORM.QGEO.03, **[O]**). A new module machine-proves the COHO.M node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the three $H^1(\mathbb{P} \setminus \mu_4)$ eigenforms $\omega_k = z^{k-1}/(z^4 - 1)$ ($k=1, 2, 3$) have pullback character $\rho^* \omega_k = i^k \omega_k$ under the clock $\rho: z \mapsto iz$ (each an exact rational-function identity, the denominator clock-invariant since $i^4=1$), so the grading is $(i, -1, -i)$ with weights $(1, 2, 3) =$ the A_3 exponents $= \text{Spec}(Q_+)$, rank 3 $= N_{\text{fam}}$. The MODULE *parity* is also

proved: the reflection $\sigma : z \mapsto 1/z$ satisfies $\sigma^*\omega_1 = \omega_3$, $\sigma^*\omega_2 = \omega_2$, $\sigma^*\omega_3 = \omega_1$ (swap $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed). Signature-locked in `AuditContract.lean`. The geometric COHOM/MODULE-parity nodes are now [E]; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay [O].

2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (`Möbius Uniformisation.lean`, ledger `FORM.QGEO.02`, [O]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (v177; AUDIT: PASS, no `sorry/admit`, axioms `propext`, `Classical.choice`, `Quot.sound` only): the clock $\rho : z \mapsto iz$ has $\rho^4 = \text{id}$ and $\rho^2 \neq \text{id}$ (order exactly 4); the reflection $\sigma : z \mapsto 1/z$ is an involution with $\sigma\rho\sigma = \rho^{-1}$ (so $\langle \rho, \sigma \rangle$ is the dihedral D_4 of order 8); a non-fixed orbit $\{a, ia, -a, -ia\}$ scales to $\mu_4 = \{1, i, -1, -i\}$; σ permutes μ_4 (fixes 1, -1 , swaps $i, -i$); and the multiplier classification “ $z \mapsto \zeta z$ has order exactly 4 $\Leftrightarrow \zeta = \pm i$ ”. Signature-locked in `AuditContract.lean`. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation `QGEO.MARKS.01` stays [O], *not* closed.

2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local $\Rightarrow \omega \circ \rho = \omega$ implication is now Lean-formalised** (`SeamDeckClosure.lean`, ledger `FORM.QGEO.01`, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of v201/v210 (AUDIT: PASS, no `sorry/admit`, no domain axioms — `#print axioms = propext`, `Classical.choice`, `Quot.sound` only): (i) the 4th-root character orthogonality $\sum_{j<4} \zeta^j = (\text{if } \zeta=1 \text{ then } 4 \text{ else } 0)$ for $\zeta^4=1$ (so a μ_4 -mark sum is supported only on modes $\equiv 0 \pmod{4}$); (ii) the clock generator $-i$ has order 4; (iii) `markLocal_blockDiagonal`: a mark-local Toeplitz symbol connects only equal clock-characters, $[\rho, M_f] = 0$; (iv) the premise is a typed structure `SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence $\omega \circ \rho = \omega$. So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN $\Rightarrow \omega \circ \rho = \omega$, certified on realistic Steklov profiles and at the state level** (`v210_mark_local_dtn`, `tfpt_research_contracts`, [O]). A numerical sharpening of `QGEO.SUBPRIN.01/v201`: a real von-Mises curvature bump summed over the four μ_4 marks, $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$, builds the Steklov DtN $\Lambda = |D_\theta| + M_f$ whose off-(mod 4) Fourier support vanishes to $< 10^{-16}$, $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the quasi-free covariance $C = \frac{1}{2}(1 + \text{sgn } H_1)$ is clock-invariant ($\rho C \rho^\dagger = C$ to $< 10^{-13}$), i.e. the actual $\omega \circ \rho = \omega$. Negative controls (single off-centre bump, $\mathbb{Z}_3$ source, four generic marks) break both by $O(1)$; convergence stable for $N \in \{16, 32, 64\}$. This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close `QGEO.SYM.01`; net existence and full-cone RP remain the [E] of v175). Ledger `QGEO.MARKLOCAL.01`. Python-only (the exact mark-sum identity is Wolfram-mirrored via v201).

2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon $g-2$ as a seam vertex** (`v204_muon_seam_g2`, `tfpt_4_frontier`, [C]). The carrier’s second-order topological defect $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$ projected through the seam-loop phase gives $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) \approx 2.879 \times 10^{-9}$ (0.81σ vs the dispersive Δa_μ ; $\sim 1.5\sigma$ vs

lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger FR.MUONG2.01; Wolfram-mirrored.

- **An independent gravitational 3/4** (v205_xi_threequarter, tfpt_4_frontier, [E]/[C]). $\xi = c_3/\varphi_{\text{tree}} = 3/4$ exactly (Einstein limit of $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$) — the same $3/4 = q(A_3) = \ln(m/\mu)$ as the seam-determinant replica (v152), an over-determination. Ledger GRAV.XI.01; Wolfram-mirrored.
- **Quantitative H_0 from the Λ branch** (v206_h0_lambda_branch, tfpt_4_frontier, [C]). $\delta_\Sigma = 48 e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$, the $(8\pi)^2$ Planck-convention split, and $H_0 = 66.5\text{--}67.1$ km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger GRAV.H0.01.
- **Asymptotic Safety as a second QG route** (v207_asymptotic_safety, tfpt_4_frontier, [O]). A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms; $y^2 = 16c_3^2 = 1/(4\pi^2)$) — a plausibility cross-check, *not* load-bearing, does *not* close G_{metric} . Ledger GRAV.ASYMP.01.
- **Black-hole thermodynamics from the seam** (v208_bh_thermodynamics, tfpt_horizon_readouts, [E]/[C]). Modular $T_H = \kappa/(2\pi)$ (the $2\pi = 1/(4c_3)$ seam unit, ties to v198/v199) and the scalaron-corrected Wald entropy $S_W = (A/4G)(1 + R_h/3M_s^2)$ (exact from v28). Ledger HOR.BHTHERMO.01; Wolfram-mirrored.
- **The black hole as a seam fixed point** (v209_bh_regular_core, tfpt_horizon_readouts, [C]). No singularity ($\varphi \rightarrow \varphi_*$ at the gapped attractor $(2/3)^6$); the maximal BH = the anchor (Nariai roots $(1, 1, -2)$); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger HOR.BHCORE.01.

2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure** (tfpt_3, [O]). Added a paragraph to the “Möbius origin of π ” section: $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$ (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock $z \mapsto iz$ is an elliptic $\text{PSL}(2, \mathbb{C})$ rotation (v180) and the RP→OS causal cone is the same Lorentz structure — so “Möbius origin of π ” and “one Lorentzian cone” are two faces of one boundary $\text{PSL}(2, \mathbb{C})$. A foundation citation, not a new claim (π stays primitive).

2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3 +$ retired small-angle reading)

- **Haloscope coupling tied to c_3** (tfpt_4_frontier, [C]). Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ($y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$), so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and $\beta_{\text{rad}} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$). The earlier small-angle $\theta_i = \varphi_0$ reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$, recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **HOR.EHT.01** (v203, [E]/[C]/[X]). The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a

horizon-scale black hole, $\beta_{\text{BH}}(r) = 16c_3^4 Q_e^{\text{eff}} Q_m^{\text{eff}} / r^2$. The coupling $16c_3^4 = 1/(256\pi^4) = \delta_{\text{top}}/3$ is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient $\delta_{\text{top}} = 48c_3^4$ that fixes the α -kernel precision-zone correction, one row from the cosmic-birefringence seed $\beta_{\text{rad}} = \varphi_0/4\pi$. TFPT fixes the $1/r^2$ shape, achromaticity and sign-flip ([C]); the amplitude $Q_e Q_m$ is an MHD/GR weight, never a pixel prediction. **Dated kill test [X]**: the residual intercept must pass three nulls (frequency, spatial $1/r^2$, sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

2026-06-15 (archive integration #1: the rare kaon sector $K \rightarrow \pi \nu \bar{\nu}$ — v202)

- **FR.RAREKAON.01 (v202, [C])**. The closed TFPT CKM point ($s_{12}=\lambda_C$, $s_{23}=\varphi_0/(1+\lambda_C)=|V_{cb}|$, $s_{13}=\lambda_C^3/3$, $\delta=\pi/3+3\lambda_C^2=1.19823$ rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou $X_t, P_c, \kappa_{\pm}$) it gives $\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = 9.45 \times 10^{-11}$ and $\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = 3.33 \times 10^{-11}$ — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted $\text{BR}(K_L) = 3.47 \times 10^{-11}$ from an $\Im(\lambda_t)/\lambda^5$ slip, corrected here). Grossman–Nir ratio $0.352 \ll 4.3$; NA62 $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ is $\sim 1.2\sigma$ above. **Dated kill test [X]**: a stable NA62 $\text{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\text{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new FR.RAREKAON. family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **QGE0.SUBPRIN.01 (v201, [E]/[O])**. The v199 bedrock residual — “the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements” — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature $f(\theta)$ (standard Steklov structure), M_f is μ_4 -character-block-diagonal iff f is $\mathbb{Z}_4$ -invariant ([E]; principal $|k|$ already commutes, v198); and a μ_4 -mark sum $f = \sum_{j=0}^3 g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \pmod{4}]$ for any profile g , hence is automatically $\mathbb{Z}_4$ -invariant ([E]). So the μ_4 -mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it (negative controls). The residual collapses to “the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck, v177)”, structurally definitional — the narrowest form yet of QGE0.SYM.01, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

2026-06-15 (F_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (experiments/ftransfer/axion_relic/relic_solver.py, [C])**. The first real F_{transfer} pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor $F(\theta_i) \approx 4$ at the hilltop. Honest finding: at the *predicted* $\theta_i \approx 170^\circ$ the estimate *over*-produces dark matter, $\Omega_a h^2 \sim 0.6\text{--}2.3$ (robustly > 0.12), because the large $\theta_i^2 \approx 8.8$ times F overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer $\theta_i \sim 120\text{--}130^\circ$ or extra dilution) — this *refines* v185’s optimistic “can be all-DM” toward an over-production tension. The near-hilltop regime is $O(1)$ -uncertain, so a full finite- T solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents; $\theta_i = \pi(1 - \varphi_{\text{seam}})$ only.

- **Full finite- T solve confirms it (axion_relic/full_finiteT_solve.py).** A converged nonlinear misalignment solve $(\theta'' + (3 + d \ln H/dN)\theta' + (m_a(T)/H)^2 \sin \theta = 0$, lattice $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$; normalised so $\theta_i=1 \rightarrow \Omega_a h^2 \approx 0.03$, the standard value) gives $\Omega_a h^2 \approx 0.66$ at the predicted $\theta_i \approx 170^\circ$ — $\sim 5.5\times$ above $\Omega_{\text{DM}} = 0.12$, reached only at $\theta_i \approx 106^\circ$. The over-production is thus *confirmed* (not the optimistic “all-DM”); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **v199_seam_state_invariance.py [E]/[O] (claim QGEO.STATE.01, work package A).** Attacking $\omega \circ \rho = \omega$ directly on the *raw* seam. (1) [E] the setup is non-circular: ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$, v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$, so $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$. (3) [E] $[\rho, H_1] = 0 \Leftrightarrow H_1$ is block-diagonal in the four μ_4 -character classes $\{n=r \bmod 4\}$ (verified). (4) [O] the residual is sharpened to “the bounded sub-principal symbol has no off-character matrix elements” — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- **v200_seam_variational_scan.py [C]/[O] (claim QGEO.VARI.02, work package B).** The discretised variational test: with the clock angle free, $E_{\text{fail}}(\theta)$ has its unique minimum at $\theta = \pi/2$ (the μ_4 clock $z \mapsto iz$) — order-4 *and* the three distinct characters $i, -1, -i =$ the $(1, 2, 3)$ grading; decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. A numerical sign that the variational principle selects μ_4 ; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, experiments/ only, not the ledger): the F_{transfer} four-pipeline scaffold (experiments/ftransfer/) with one CONTRACT per interface, and confirmation that the FRB preregistration (frb_tfpt_v1.yaml) and its status semantics already exist. Suite now 199 modules, highest ID v200; Wolfram extension 241/241.*

2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **v198_modular_commutator_reduction.py [O] (claim QGEO.MODULAR.01).** The decisive crack on the last residual. (1) [E] the DtN principal symbol $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$ and the clock $\rho: z \mapsto iz = \text{diag}(i^n)$ are both diagonal in the Fourier basis, so $[\rho, |k|] = 0$ *exactly* on all of L^2 (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to (M, Ω) , so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with $\log \Delta \sim \Lambda_\Sigma$: hence $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$, using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite H^1 block is already μ_4 -invariant (v177). So the entire residual collapses to the single *state-invariance* “the raw quasi-free seam state is μ_4 -invariant” ($\omega \circ \rho = \omega \Leftarrow$ the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of QGEO.SYM.01, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like $c = \text{const}$), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 197 modules, highest ID v198; Wolfram extension 240/240.

2026-06-14 (three review levers: Marks-via-Lefschetz, E_fail functional, RR→D5 — v195–v197)

- **v195_marks_lefschetz_character.py** [E]/[O] (claim QGEO.MARKS.02). The four seam marks are *forced* (not posited): $\rho: z \mapsto iz$ fixes only $\{0, \infty\}$, $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents, $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$) has no trivial component $\Rightarrow$ no puncture at a fixed point, and $\text{rank } H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$ a single free μ_4 orbit (cross-ratio 2). So MARKS reduces from the geometric posit “ D_4 marks” to the milder premise “ H^1 carries the A_3 -exponent rep” — less circular, still [O]. Cited in `tfpt_research_contracts`.
- **v196_seam_energy_functional.py** [E]/[O] (claim QGEO.VARI.01). The failure functional $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, zero locus = the QGEO.SYM.01 conditions; on the finite H^1 block it vanishes at the μ_4 config [E] and is > 0 for any μ_4 -breaking perturbation, so μ_4 is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **v197_rr_carrier_clifford_d5.py** [E]/[C] (claim ARCH.RRCAR.02). Hardens the Riemann–Roch carrier from g_{car} to D_5 itself: $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor — so D_5 is the Clifford spinor of the 5-dim mode space, closing $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$. [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop “ $\pi - 3\phi_0$ ” proposal was *rejected* as a $\pi \approx 3$ coincidence, not built). Suite now 196 modules, highest ID v197; Wolfram extension 239/239.

2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **v194_raw_seam_dtn_rp.py** [O] (claim QGEO.DTN.01). The next hard lever — “is the raw RP-seam DtN definable from RP data alone?” (sub-claim 2 of QGEO.ENERGY.02) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator $[\rho, \Lambda_\Sigma] = 0$ closes on the finite cohomology H^1 ($\rho^* w_k = i^k w_k$, distinct μ_4 characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without naming $\mathbb{P}^1 \setminus \mu_4$. (3) [O] the residual relocates: the full- L^2 commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So QGEO.SYM.01 reduces from a definitional postulate to “intrinsic BW geometric modular covariance” — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 193 modules, highest ID v194.

2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **v193_qgeo_energy_commutator.py** [O] (claim QGEO.ENERGY.02). The non-circular sharpening of QGEO.ENERGY.01: the energy commutator $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims — (1) [E] ρ is carrier-defined (Coxeter of $W(A_3)=S_4$, v117), not imported from the seam; (2) [O] the non-circularity hinge: Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [E] the commutator forces the $(1, 2, 3)$ grading on H^1 (QGEO.COHO.01). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ iff $\ln m = q(A_3) = \frac{3}{4}$ (*family*), not $q(D_5) = \frac{5}{4}$ (carrier, off by $5/3$) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, QGEO.SYM.01 becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID v193; Wolfram extension 236/236.

- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to “Constructive QFT closure of the admissible physical sector”, and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed; G_{metric} frontier). The two superseded standalone phrasings were added to the v188 prose guard so a frozen release cannot re-introduce them.

2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **v189_riemann_roch_carrier.py** [E]/[C] (claim ARCH.RRCAR.01). The pair $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$ (already QGEO.COHO.01) and the function side $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$, matched by $\text{rank}(D_5) = 5$. The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined $g_{\text{car}} = 5$). Cited in `origin_theory`; Wolfram-mirrored.
- **v190_nariai_entropy_bound.py** [E] (claim HOR.NARIAI.02). A variation-free one-line proof of the SdS entropy floor: $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$, so $S_{\text{tot}} \geq \frac{2}{3}S_{dS}$ with equality iff $x = 1$ (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v191_universal_branch_line.py** [C] (claim HOR.BRANCHLINE.01). The affine map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$ carries the SdS branch points $\pm 1/N_{\text{fam}}$ onto the flavor labels $\{2, 5\}$. *Honestly typed* [C] — an exact *relabeling*, not a theorem (a decoy pair $\{3, 4\}$ admits an equally exact map; the shared content is the known $2/3$ ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v192_qgeo_energy_clock.py** [O] (claim QGEO.ENERGY.01). An operator-checkable restatement of the bedrock: $\rho^* \Lambda_{\Sigma} \rho = \Lambda_{\Sigma}$ (the clock preserves the DtN energy form). *Honestly typed*: it *relocates* QGEO.SYM.01 to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless ρ is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID v192 (v186 skipped).
- *All four are validated external-review proposals (problem_b); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

2026-06-14 (Zenodo release 5.1 (rev 120) — F_transfer firewall, v184–v188)

- **Published to Zenodo.** Version 5.1 (rev 120) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20687564, DOI 10.5281/zenodo.20687564. This version carries the F_transfer-firewall round (v184–v188): the honest η_B anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger v187 and prose v188 guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-QGEO.SYM.01 two-layer clarification. The uploader’s DEFAULT_RECORD_ID was advanced to the new record for the next release. Suite 187 modules (highest ID v188; v186 skipped). `run_all.py` green, audit clean, website built.

2026-06-14 (F_transfer round: eta_B honest test v184, axion relic v185, discipline guard v187)

- **v184_etaB_anchored_boltzmann.py** [C] (claim FR.ETAB.03). An honest test of the external-review η_B proposal ($M_1 = M_R \varphi_0^4$, $\tilde{m}_1 = m_3/A_{\Lambda}$). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ($\tilde{m}_1 = m_3/10 \approx 5$ meV, $A_{\Lambda} = 10$ atom, in the strong-washout window) [C]; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3$ and the nearest clean TFPT powers of $\bar{M}_{P1}$ both miss it (c_3/φ_0 -powers off $\sim 75\%/ \sim 40\%$), so

$M_1 = M_R \varphi_0^4$ merely relocates the free input. η_B stays [C], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.

- **v185_axion_relic_solver.py** [C] (claim FR.DM.03). Honest misalignment estimate: the determinant-line axion $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$ GeV ($m_a \approx 23.8 \mu\text{eV}$) CAN be all dark matter, but the harmonic misalignment under-produces ($\sim 21\%$ at $\theta \sim O(1)$), so the $\theta_i \approx 170^\circ$ hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.
- **v187_ftransfer_laws.py** [E] (claim FR.TRANSFER.01). A CI-style firewall guard: the four continuous-transfer frontier observables (Koide, η_B , axion, m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or $b_3 = -7$ may be [E], the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- **v188_frontier_wording_guard.py** [E] (claim AUDIT.WORDING.01). A prose sentinel (no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording $\rightarrow$ v184 [C]; the old “open relic scale” dark-matter line $\rightarrow$ v185; the old “near, not exact” Koide line $\rightarrow$ v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-QGEO.SYM.01 two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID v188 (v186 was intentionally skipped as redundant — m_p/m_e is already typed [O] as a QCD matching contract, QCD.LAMBDA.01).
- *Review framing points (P1/P2 as projections of the seam-deck postulate, v_{geo} as unit gauge, G_{net} a corollary of QGEO.SYM.01, grammar tuning = data) were validated as already implemented in the prior rounds; m_p/m_e is already typed [O] as a QCD matching contract (QCD.LAMBDA.01), so no redundant module was added.*

2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **v183_koide_f_corner_transfer.py** [E]/[C] (claim FR.KOIDE.07). An external-review proposal, validated exactly: the Koide source $\rightarrow$ pole factor $\frac{53}{54}$ is not a bare arithmetic guess but an *operator* identity on the carrier, $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a) = 53/(2 \cdot 27) = \frac{53}{54}$. Here $\mathbf{1}^T R a = 27$ is the $E_6 \times A_2$ cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T R a)N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3$) and $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$, where $F = R+Q$ is the *missing* Sheet-Diamond corner with $\det B_F = 52 = \dim F_4$ (v80). Hard negative control: $a^T M \mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. This upgrades FR.KOIDE.03 from “near miss + conjecture” to “operator-sourced source $\rightarrow$ pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source $\rightarrow$ pole transfer reads the F corner) stays [C], an F_{transfer} special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, exponent ratio $|\mathbb{Z}_2|$, “one clock, two geometries”) is exactly HOR.TRISECT.01/v103 — a rediscovery of an existing result, no new module.

2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **v182_reviewer_residual_map.py** [E]/[C] (claim REVIEW.MAP.01). A careful external review raised four concerns (A mapping $2D \rightarrow 4D$, B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$: A (Plücker-11 = QBL boundary count v174 + holographic reduction), B ($\equiv \text{QGEO.SYM.01}$), D (Möbius forced by $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ on the deck; the missing generator = F_{transfer}). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a “Rosetta” translation lexicon** (website RosettaLexicon): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor, F_{transfer} , v_{geo} , QGEO.SYM.01) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website TrustContract): the frozen pre-data predictions ($\sin^2\theta_{12} = 0.3067$, JUNO; $r \approx 0.004$, CMB-S4) and the null model ($P \leq 10^{-30.7}$) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate**. QGEO.SYM.01 is now framed confidently as TFPT’s *one fundamental postulate* (the role “ $c = \text{const}$ ” plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website “what is open” card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

2026-06-14 (external-review response: CSV fix, G_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded**. Three rows (AX.P1.01, AX.P2.01, QGEO.ISO.01) had *unquoted commas* in a status/external field (e.g. “a=(1,1,2)”, “PSL(2,C)”); so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- **G_{net} typed in three explicit levels** (website “What is open” card): (1) *algebra* [E] $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (QGEO.SYM.01). The target object’s mathematics is closed; only the physical coupling of the raw seam is open.
- **QGEO.SYM.01 now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than CONF/ISO, what follows automatically (the conditional theorem), and *what would falsify it* (genus > 0 double; non-deck transport; $N_{\text{fam}} \neq 3$); with the explicit conditional framing “*given* QGEO.SYM.01, the closure follows” rather than “the theory derives the seam geometry”.
- **Claim-language detox**. “no free fundamental number” softened to “no free *dimensionless* compiler dial beyond the anchor and π (dimensionful scale + transfer physics stay typed)” (`origin_theory`, `introduction`, website); the site title softened from “the Standard Model Derived” to “the Standard-Model *Skeleton* Derived”.

- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version 5.1 (rev 114) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20686284, DOI 10.5281/zenodo.20686284. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise QGEO.SYM.01), the two overview figures (residual_chain, script_timeline), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The README and the Zenodo description were refreshed (181 modules), and the uploader’s DEFAULT_RECORD_ID was advanced to the new record for the next release.

2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (make_figures.py; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from “build a quantum-gravity measure” to the single definitional bedrock QGEO.SYM.01; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ~181 scripts (foundations → SM readouts → seam= horizon → horizon/flux geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor $a = (1, 1, 2)$ plus π (with $g_{\text{car}} = 5$ an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, FORM.LADDER.01) was already established (ANCHOR.GEN.01/v23, ARCH.CORE.01/v53) and stated in the paper-s/glossary, but the *axiom surfaces* still read as bare “declared inputs”. Fixed: the ledger rows AX.P1.01/AX.P2.01 now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ($c_3 = 1/(2e_1(a)\pi)$; $g_{\text{car}} = e_2(a)$, bootstrap-forced) rather than “declared axiom”. No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the FIG list). Suite unchanged at 181.

2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- `v181_clock_is_conformal_symmetry.py` [E]/[O] (claim QGEO.SYM.01). The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification QGEO.SYM.01: “the seam’s conformal structure *is* the μ_4 -deck structure of the carrier clock”. This is bedrock — a physical/definitional statement tying P2 to the seam’s conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE(v175)→⋯→SYM.01(v181)` ends here; we stop honestly. Python-only. Suite now 181 modules.

- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the introduction “latest reductions” prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A $\times 2$, G_{net}), the `glossary` G_{net} entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- `v180_clock_is_mobius.py` [E]/[O] (claim `QGEO.ISO.01`). Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is* $\mathbb{P}$ — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)=4$) is topologically $z \mapsto iz$; (3) *isometry* $\Rightarrow$ *conformal* $\Rightarrow$ *Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$; (4) an order-4 Möbius map is $z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i , free orbit μ_4). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does not name $\mathbb{P}/\mu_4$ (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- `v179_conformal_realisation.py` [E]/[O] (claim `QGEO.CONF.01`). Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$ and $\chi = 2 - 2g \Rightarrow g = 0$, so “genus-0” is the same $\chi = 2$ that gives the “8” in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly $4 = e_1(a)$ points). KERNEL then *follows* from the same geometry (DtN symbol $|k|$, Lee–Uhlmann v156; $c = 8$ rigidity v158; the cusped-surface residual $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: “the raw RP seam normal double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius automorphism”. The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- `v178_marks_kernel_attempt.py` [E]/[O] (claim `QGEO.REDUCE.01`). An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ , the marks are forced to be a generic ρ -orbit $\{1, i, -1, -i\} = \mu_4$ (*not* the fixed points), distinct iff $b_1 = 4 - 1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$

giving a faithful D_4 of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4 -equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol $|k|$, Lee-Uhlmann/v156; no drift v158). Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- `v177_seam_marking_kernel.py` [E]/[O] (claims QGEO.TREE.01, QGEO.MARKS.01, QGEO.KERNEL.01). A second external residual-class audit pointed out that the v176 statement takes the marks/ D_4 as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as `REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET` and closes *four* nodes *constructively* (all validated symbolically before adoption): `UNIFORM` [E] — an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4 , and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 (order 8), so a genus-0 four-marked faithful- D_4 boundary *is* $(\mathbb{P}, \mu_4)$ (strengthening v168's cross-ratio to the full uniformisation); `COHOM` [E] — $\rho^*\omega_k = i^k\omega_k$, grading $(1, 2, 3) = A_3$ exponents; `MODULE` [E] — a *unique* μ_4 -equivariant iso (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); `NET` [E] (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: QGEO.MARKS.01 (the raw RP seam collar canonically produces the genus-0 four-marked D_4 boundary) and QGEO.KERNEL.01 (the raw seam Calderón operator *is* the μ_4 -equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- `v176_seam_collar_realisation.py` [E]/[O] (claim QGEO.REALIZE.02). Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem*: given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks, four parabolic marks with μ_4 decomposition and admissible $c=8$ data, its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1, 2, 3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol $|k|$ (v156), H^1 character = generation grading $(1, 2, 3) = A_3$ exponents (v137/v168), μ_4 -equivariant contraction with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1 + \text{full-cone RP all } m$ (v154/v175). The whole structural residual thus reduces to the *one* open premise QGEO.REALIZE.01: that the physical seam collar's boundary *is* this $\mathbb{P} \setminus \mu_4$ — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note*: the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ($v_{\text{geo}} \mapsto \lambda^{-1}v_{\text{geo}}$, $1/G \mapsto v_{\text{geo}}^2$, dimensionless data invariant), and the Koide $u \rightarrow \frac{53}{54}u$ source→pole transfer is already typed [C] (FR.KOIDE.03); neither is re-derived.

2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version 5.1 (rev 105) of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record 20681451, DOI 10.5281/zenodo.20681451. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The README and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader’s DEFAULT_RECORD_ID was advanced to the new record for the next release.

2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **v175_net_existence_full_cone.py [E]/[O] (claim QFT.NETRP.01).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items — A_2 net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all m* : the CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16} = 65536$); by the exterior-power spectrum theorem every eigenvalue is a subset product of $\text{spec}(t)$, so all 2^{16} products lie in $[0, 1]$ (a contraction = full-cone RP), with eigenvalue-1 multiplicity $2^8 = 256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ — so full-cone RP reduces *for every m* (not only $m \leq 6$, v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2) A_2 *assembled + verified*: the $(E_8)_1$ character $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$ (level-1 = 248, extends the order-4 v156 check by one order), the embedding $248 = 120 + 128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation QGEO.REALIZE.01 (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **v174_seam_fock_readings.py [E] (claim QFT.FOCK.01).** Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$ directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ($2^{15}/16 = 2^{11}$) — the explicit count behind v161. (2) *Witten index = the Plücker norm 11*: the four μ_4 -corner fermions span a $2^4 = 16 = \dim S^+$ Fock space; QBL ($\deg \leq 2$) keeps $\sum_{k \leq 2} \binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$, so $c_u/c_d = 55/117$ reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*: $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$, so $\Delta c = 0$ forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **v170_diamond_slice_compression.py** [E] (claim E8.SLICE.01). E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P = (3, 4, 6, 10)$ ($p_n = 2 + 2^n$) and Sheet-Diamond determinants $(3, 4, 8, 14, 20, 32)$ summing to $81 = N_{\text{fam}}^4$. All seven 248-slices, the K_4 edge graph, and the row-budget cross $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$ follow; $78 = \dim E_6 = p_2(p_0 + p_3)$. Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).
- **v171_os_moment_cluster.py** [E]/[C] (claim QFT.OSMOMENT.01). Atomic OS Moment Lemma: $\text{spec}(T) = \{1, 64/729, 1/729\}$ makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster $729/665$, $\xi = 0.4111$. Sugawara Gap Safety: $31 = 1 + h^\vee(E_8)$, $c((E_8)_1) = 248/31 = 8$, $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$. Cited in `tfpt_research_contracts` (G5).
- **v172_trace_anomaly_seed.py** [E] (claim SEED.ANOMALY.01). The seed prefactor $\frac{4}{3}$ is the halved $(E_8)_1$ Weyl anomaly: over the seam S^2 ($\chi = 2$) the $c = 8$ trace anomaly is $c\chi/6 = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ to $\frac{4}{3} = 16/12$. Plus QCD $b_3 = 7 = \text{scalaron exponent } 48 - 41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf $-98/45 = -2 \cdot 7^2 / \dim(D_5)$ factorisation. Cited in `tfpt_2` (seed).
- **v173_pfaffian_cp_car.py** [E] (claim FLAV.STRONGCP.02). Strong CP $\theta_{\text{eff}} = 0$ as Pfaffian reality in the self-dual 16-Majorana CAR model: $\{D, \Sigma\} = 0$ pairs the spectrum, γ_5 -Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor, $E_8 = K_0$, GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version 5.1 (rev 97) of the ten-PDF document set (introduction, `origin_theory`, `tfpt_1–5`, `horizon`, `research_contracts`, `changelog`) was published as a new version of the TFPT deposit — record 20678568, DOI 10.5281/zenodo.20678568. The uploader’s `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

2026-06-13 (QGEO rigidity theorem v168 + η_B leptogenesis interface v169)

- **v168_qgeo_rigidity.py** [E]/[C] (claims QGEO.RIGID.01, QGEO.REALIZE.01). Hardens the *finite* half of the one remaining Tier-1 hinge GATE.QGEO (the seam = flavour = horizon identification). Exact, machine-checked: $\mu_4 = \{1, i, -1, -i\}$ has cross-ratio 2 and a faithful Möbius stabiliser of order $8 = |D_4|$; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1} dz / (z^4 - 1)$ ($k = 1, 2, 3$) are μ_4 -eigenforms with characters χ_1, χ_2, χ_3 of weights $(1, 2, 3) = A_3$ exponents = $\text{Spec}(Q_+)$. So a genus-0 boundary with faithful D_4 , four parabolic marks and this μ_4 -decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ — the QGEO Rigidity Theorem (QGEO.RIGID.01, [E]). The seam-collar realisation stays the one premise (QGEO.REALIZE.01, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167’s completeness claim).
- **v169_etaB_boltzmann_interface.py** [C] (claim FR.ETAB.02). Operationalises the Tier-3 F_{transfer} leptogenesis path. Type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron

28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted $\delta_{CP} = 240^\circ$. Over natural $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and washout, η_B brackets the observed 6.1×10^{-10} (a canonical point gives 6.0×10^{-10} , untuned) — the route is *viable*. But M_1 /washout are scenario inputs, so η_B stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream Ω_b readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- `v166_higgs_free_seam.py` [E]/[C] (claim `HIGGS.FREESEAM.01`). A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral $c=8$ fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there: $\lambda(M_{\text{seam}}) = 0$ and $\beta_\lambda(M_{\text{seam}}) = 0$ — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from M_Z up with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$ and $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$: the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from ~ 0.022 at one loop to ~ 0.002). The double condition predicts $m_H \sim 129\text{--}134$ GeV; the same condition at the scalaron scale gives ~ 107 GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only $N_\Phi = 1$); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer's SM run supplies the two-loop β_λ , reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

2026-06-13 (PyR@TE F_{transfer} follow-ups: flavor scale-tagging v163, QCD scale v164)

- `v163_rg_stability_flavor.py` [E]/[C] (claim `FLAV.RGSTAB.01`). Scale-tagging of the flavor sector. The lepton-mixing seeds ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are *RG-stable*: the PMNS running is y_τ^2 -suppressed (the only mixing-rotating Yukawa term is $\frac{3}{2} Y_e Y_e^\dagger Y_e$, PyR@TE SM RGEs), so $\kappa_\tau = y_\tau^2 / (16\pi^2) \ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$ and $|\Delta \sin^2 \theta_{ij}| \lesssim 10^{-4}$ — two to three orders below precision. Seam value = measured value; the y_t^2 -driven quark mass ratios ($m_t/m_b \sim 41$) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- `v164_qcd_scale.py` [E]/[O] (claim `QCD.LAMBDA.01`). The QCD confinement scale is parameter-free from the carrier $b_3 = -7$ (v159): running $\alpha_s(M_Z) = 0.1179$ down (two-loop, n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and confinement at ~ 0.5 GeV by dimensional transmutation — so the QCD-scale *numerator* of m_p/m_e is independently sourced. The $O(1)$ proton/ Λ factor is lattice, so $m_p/m_e = 1836$ stays the honest “[O] — not forced” frontier ratio. Cited in `tfpt_4_frontier`.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa β 's, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

2026-06-13 (PyR@TE external cross-check of the F_{transfer} gauge inputs — v159)

- **New module** `v159_pyrate_gauge_crosscheck.py` [E]/[C] (claim `EM.BUDGET.02`). The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes $\beta_{g_1} = (\frac{41}{10})g_1^3$, $\beta_{g_2} = -\frac{19}{6}g_2^3$,

$\beta_{g_3} = -7g_3^3$ verbatim (plus the full standard 2-loop matrix). So $b_1 = 41/10$ is pinned *three* independent ways — carrier algebra $10b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1 = 41$, the $U(1)$ hypercharge index $\sum \frac{3}{5} Y^2$, and PyR@TE — and the split $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$ reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of F_{transfer} is thus not a free knob.

- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge β functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind η_B); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F_{transfer} + irreducible core, **v167**)

- **New module `v167_inventory_post_closure.py`** [E]/[C]. The terminal residual inventory (line `v105`→`v123`→`v167`), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16-19`) the `v123` *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the Q -realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification = A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary = $\mathbb{P} \setminus \mu_4$, cohomology $b_1 = N_{\text{fam}} = 3$); the EH step (R3) folds into v_{geo} (**v152**) and the ambient measure into G_{net} (**v76**). **Terminal structure:** Tier 0 = the irreducible core $\{\pi, v_{\text{geo}}\}$ (a theorem, No-Unit **v153**); Tier 1 = one hinge; Tier 2 = folded; Tier 3 = F_{transfer} , one functor with four typed conditional interfaces (Koide, η_B , axion, m_p/m_e), η_B carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

2026-06-13 (premise A maximal honest closure: A_2 by assembly, $R_1 = \text{GATE.QGEO}$, zero independent open gates, **v165**)

- **New module `v165_premise_a_closure.py`** [C]/[O]. Pins the two residual gates of the `v160-v162` (A)-chain to their floor. A_2 (**net existence**) **closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic $c=8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q -system of index 4 (**v125**), $\mu(B)=16/16=1$, $248=120+128$, character $E_4/\eta^8=j^{1/3}$ (**v156**); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via $\text{DtN} = |k|$ (**v156/v157**). R_1 (**seam clock**) **reduces to `GATE.QGEO`** [C]: a μ_4 -equivariant flat gapped generator is *unique* ($= A_0^*$, **v115/v116**), so the gap $(2/3)^6$ is forced and the residual is the one geometric identification “seam boundary = $\mathbb{P} \setminus \mu_4$ ” = `GATE.QGEO` (cohomology established, **v137**, $b_1=3=N_{\text{fam}}$). **Final accounting:** by the No-Unit Theorem (**v153**) the irreducibles stay $\{\pi, v_{\text{geo}}\}$; premise (A) = (A_2 assembly) + ($R_1=\text{GATE.QGEO}$) + the $\{\pi, v_{\text{geo}}\}$ core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, **v162**)

- **New module `v162_seam_transport_identification.py`** [E]/[O]. Pushes the **v161** residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value (β) and a fixed space (α). (β) **is forced**: a μ_4 -equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection, $\sum_k U^k X U^{-k} = |\mu_4| \text{diag } X$ for $U = \text{diag}(1, i, -i)$; the commutant of U is exactly the diagonal algebra), so its spectrum is the cusp weights $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (**v115**) and the transfer eigenvalue $(1-\alpha)^{p_2}$ with $p_2=6=|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — all carrier data; the lift to the 16 Majoranas is the A_3 subnet grading (**v137**). (α) is the rank-8 Calderón polarisation of $\omega_k=|k|$ (**v113/v156**). **Closure**: with **v160+v161**, (A) closes given only μ_4 -equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, **HOR.CLOCK**). So premise (A) factors exactly into $A_2 + R_1$ — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger **GATE.METRIC.18**). Python-only (numpy/sympy + stdlib).
- **Bookkeeping**. Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; **tftpt_research_contracts** (premise-A keybox) and **next.txt** moved together.

2026-06-13 (the cone reduces to one particle: premise A’s scope premise discharged to a 16-dim statement, **v161**)

- **New module `v161_one_particle_reduction.py`** [E]/[O]. *Discharges* the scope premise of **v160** (“the transport is gapped on the *full* Schwinger cone”) via Bogoliubov second quantisation. A quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dim Majorana space; $\Gamma(t)$ acts on the degree- m correlator sector as the exterior power $\Lambda^m(t)$. Machine-checked: (1) $\text{spec}(\Gamma(t)|_{\deg m}) =$ products of m one-particle eigenvalues (compound-matrix theorem, all $m=0..6$); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is Λ^* of the fixed subspace (disconnected Wick powers), with uniqueness closed by **v113** (one polarisation $\Leftrightarrow$ one vacuum); (4) quasi-freeness is *forced* — the $120 = \dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and **v158** makes every non-Gaussian deformation irrelevant. **Consequence**: **v160**’s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ , gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger **GATE.METRIC.17**). Python-only (numpy + stdlib).
- **Bookkeeping**. Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; **tftpt_research_contracts** (premise-A keybox) and **next.txt** moved together.

2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, **v160**)

- **New module `v160_seam_gaussianity_from_pf.py`** [E]/[O]. Formalises the proposed closure of premise (A) (“the seam bulk is a reflection-positive free field”) as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants $\kappa_{2n}=0$ — verified by the *signed* set-partition/Pfaffian moment $\leftrightarrow$ cumulant inversion on a concrete pure Majorana covariance (the **v113** “one kernel = net” property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a second fixed ray $\Rightarrow$ breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim

needs only $\text{gap} > 0$, the value $(2/3)^6$ only sets the rate. **The honest residual: v56's** gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, while the 16-Majorana seam already carries $\binom{16}{4}=1820$ and $\binom{16}{6}=8008$ higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement “the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of v56”, which the suite does *not* yet establish (ledger GATE.METRIC.16). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.

- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection × {YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as 2026-06-07 / 2026-06-08 sort by the later date). A manual prepend had left five 2026-06-12 entries above seven 2026-06-13 blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral $c=8$ fixed point is provably stable (v158, exact dimension count).** The finite core of (A)'s “reaches-the-fixed-point” residual: the chiral Majorana has $h=\frac{1}{2}$, the bilinear current $h=1$, the quartic $h=2$. The relevant window $0 < h < 1$ for bosonic deformations is *empty*, the $h=1$ currents are chiral (v157, not $(1,1)$ -marginal), and the lowest interaction (quartic, $h=2$) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger GATE.METRIC.15). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the F_{transfer} interfaces (η_B leptogenesis RGEs, gauge-coupling running, the $b_1=41/10$ cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally* $|k|$ (Lee-Uhlmann — order-1 Ψ DO, principal symbol $|\xi|_g$; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic $c=8$ theory is *rigid* — every field has $\bar{h}=0$, so no marginal $(1,1)$ field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same $|\mathbb{Z}_2|$ that halves 8π in $c_3 = \frac{1}{8\pi}$ is the Ramond/GSO projection giving $\mu=1$ ($248=120_{\text{NS}}+128_{\text{R}}$). With $(E_8)_1$ unique (v83), premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ($\Delta=6\log\frac{3}{2}$) boundary flow reaches the holomorphic fixed point” — freeness then forced by rigidity, not fine-tuned (ledger GATE.METRIC.14).
- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** “ $B \cong (E_8)_1$ ” is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The

Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion $\omega_k = |k|$ (harmonic extension $e^{ikx-|k|y}$); (b) 16 free Majoranas give $c = 8 = 5+3$; (c) the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ ($SO(16)$ bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$ (verified by q -series) — the 248 at level 1 is $\dim E_8$, so the net *is* $(E_8)_1$. *Residual*: exact given (i) the free-bulk premise ($\text{DtN} = |k|$, **v155**) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger **GATE.METRIC.13**).

- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the $\text{DtN} = |k|$ identity, the $(E_8)_1$ character $E_4/\eta^8 = j^{1/3}$ and the $248=120+128$ split mirrored); contracts / redteam (Target A) / **origin_theory** / **next.txt** / website moved together.

2026-06-13 (the seam-side identification reduced to one shared premise, **v155**)

- G_{net} ’s last premise reduces to a single shared physical premise (**v155**, Python-only). The seam-side identification of the Simple-Current Extension Theorem (**v154**) splits (**v110**) into (a) “the Calderón involution is sheet-odd” (the double-cover deck, structural) and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression PTP of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from “the seam bulk is a reflection-positive free field”, the *same* premise already load-bearing in the area law (**v59**) and the EH mechanism (**v150/v151**). **The entire seam-side residual is therefore one physical premise**, shared by G_{net} , the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger **GATE.METRIC.12**).
- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (**v155** is numerical/numpy, Python-only); contracts / **origin_theory** / **next.txt** / website moved together.

2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, **v153–v154**)

- v_{geo} **re-typed: the No-Unit Theorem (**v153**)**. Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling $L \mapsto \lambda L$, whereas a mass / $1/G$ is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale. v_{geo} is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ($U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, $m/\mu = e^{3/4}$ a pure ratio); the irreducibles stay {one unit, π } (ledger **ANCHOR.VGEO.02**).
- G_{net} **stated as the central closing theorem (**v154**)**. The Simple-Current Extension Theorem: $A=(D_5)_1 \otimes (A_3)_1$ extended by the isotropic glue $L=\langle(1,1)\rangle$ has $[B:A]=|L|=4$, $c=5+3=8$, $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$ ($SO(16)_1$ excluded). Every algebraic ingredient is machine-checked (**v89/v92/v125/v143/v148**); the only residue is the seam-side identification. Pulled to the front of **tfpt_research_contracts**, **tfpt_5_redteam** (Target A final form) and **origin_theory** (ledger **GATE.METRIC.11**).
- **Status surfaces re-typed** (review): the canonical status card now reads v_{geo} as a metrology *primitive*, G_{net} as the simple-current extension $\cong (E_8)_1$ (**[E]/[C]**), and F_{transfer} as one functor with four typed interfaces — “not structural gaps”. The final one-line framing (dimensionless boundary compiler; irreducibles $\{\pi, v_{\text{geo}}\}$; one seam-net theorem; frontier = F_{transfer}) is in **origin_theory**, the introduction, the README and **next.txt**.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/**origin_theory**/contracts/redteam/status-card/website moved together.

2026-06-13 (currency sweep: remaining old-series “Paper N” cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series “Paper N” attributions and removed them: four literal “Paper 6” references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old “Paper 4” (OS/admissibility) and “Paper 5” (APS/Hodge) references in `tfpt_2`, the old “Paper 3” transport-kernel attributions in `tfpt_2`, and the internal “Paper 1/2/4” cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the E_8 decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts $\leftrightarrow$ papers $\leftrightarrow$ ledger $\leftrightarrow$ website) was already current (152 scripts, AUDIT OK).

2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, **v152**)

- **R3: the $q(A_3)$ normalisation merges into the one anchor (**v152**).** The replica EH coefficient is a *log-ratio* $k = \ln(m/\mu)/(12\pi)$ ($\ln m$ alone is scale-ambiguous, only m/μ is physical); pinning $k = c_3/2$ fixes the dimensionless ratio $m/\mu = e^{3/4}$, and in $d=2$ that ratio *is* the induced Newton constant $1/G$ (**v68**). So v_{geo} (**v78**), $1/G$ (**v68**) and m/μ are *one* dimensionful anchor in three readings — the irreducibles stay {one scale, π }, and SEAM.THEOREM.01’s residual reduces to the kernel-identification premise alone. *Audit (not promoted)*: the log-ratio is numerically $\frac{3}{4} = q(A_3)$, the target value the anchor takes in seam units — not a derivation (ledger SEAM.EHMODEL.03).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/origin_theory/next.txt/website moved together.

2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion}` (`rev {git count}`) (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (ZENODO_TOKEN secret).

2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to main, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`). ~ 270 files / ~ 3 MB — within Overleaf GitHub-sync limits.

2026-06-12 (origin_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a **keyeq**. The exact result keyboxes (**v53–v56**, **v101**, **v105**, **v113**, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class **[O]**/**[E]**.

2026-06-12 (tfpt_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) and the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon “Scope/typing” marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document’s `.aux/.toc/.out` *on failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, **v151**)

- **R3: the Calderón/DtN kernel is conically clean (**v151**).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ($C_N = C_D$), so by the Burghilea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ($C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$): the seam-reduced effective action inherits the **v150** EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by **v150** needs no separate derivation. `SEAM.THEOREM.01` narrows to the $q(A_3)$ normalisation plus the standing kernel premise; the gate stays **[O]** (ledger `SEAM.EHMODEL.02`).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/origin_theory/next.txt moved together.

2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, **v149–v150**)

- **R4': both dual normals are anchor data (**v149**).** In the cusp frame the torsion normal pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line (the cusp matrix is unimodular, so this determines n uniquely); with $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ the dual normal pair is anchor data through and through. Equivalences to the $(2, 8, 121)$ frame data and the w_0 -lift exact. $R4'$ residue = *one* discrete assignment (ledger `GATE.UWALL.14`).

- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and t -independent (Cheeger image sum $(N^2-1)/(12N)$); for a *gapped* operator the ζ -regularised replica variation is *finite and cutoff-independent*, $\Delta \log \det' = 2C(\gamma) \ln m$, and its linearisation is the Einstein–Hilbert form $(\ln m/12\pi) \int \sqrt{g} R$. The v73 normalisation becomes one exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays **[O]** (ledger SEAM.EHMODEL.01).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4': the pairing values are atom identities (v145).** Reading the v139 lift structurally ($n = w_0(\sigma) + |\mu_4|e_3$, w_0 = the A_3 exponent duality), all three pairings become atom arithmetic — including the *new* closure $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2$ ($4+5=9$, $\sigma \perp w_0(a)$) and $\|\sigma\|^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$. R4' = derive *one* map (ledger GATE.UWALL.13).
- **R5: the Möbius D_4 realisation (v146).** The *full* dihedral symmetry $\langle z \mapsto iz, z \mapsto 1/z \rangle$ of the seam curve acts on H^1 exactly as the integer model: ι^* is the exponent duality with parity +1 on the self-conjugate line (= T_A in the cusp basis, glue-swap parity), and Σ is the $\delta\iota$ class — the premise reduces to the module identification itself (ledger FLAV.QGEO.07).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is $\det'$ -clean (v147).** Variance ratio $(1 - \alpha)$ (forced by norm = area) gives the Born-squared ratio $(1 - \alpha)^{p^2}$, the $\ln$ series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend $\log_{3/2} 3$ carries no determinant correction — the v144 obstruction is localised to the $\alpha = 0$ reference (ledger HOR.CLOCK.13).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd μ_4 simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ($248 = 120_{\text{NS}} + 128_{\text{R}}$); the one remaining net statement is irreducibly twisted-sector (ledger GATE.METRIC.10). *Correction note (same day)*: the first version of v148 graded the Ramond labels by the sum $q_D + q_A$ (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + GATE.WOLFRAM.02 + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

2026-06-12 (external review integrated: Λ typing, marker migration, website parity)

- **Λ normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct — $\bar{M}_{\text{P1}}$ vs M_{P1} — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ($3/(4\pi^2) \Rightarrow 120.147$ reduced; $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$ unreduced) plus a typing note.
- **Stale references cleaned.** “How each of the seven papers simplifies” $\rightarrow$ “How the legacy layers simplify (bridge map)” with the lead-in “(Paper 6)” reference replaced by the scalaron-reheating chain (v86); “(Paper~6/7)” in the `tfpt_3` bridge table reworded; the `tfpt_1` “single remaining geometric input (H2)” passage and the `tfpt_2` “single remaining input (U)” passage

carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).

- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit, E_8 glue, $2/3$ motif and K_5 figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes `[E]/[C]/[O]` instead of the legacy `[A]/[I]/[L]/[B]/[P]/[R]` badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy + $c=8 \Leftrightarrow$ index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to `[E]/[C]/[O]/[X]` (fine types named in prose; the script index keeps quoting the scripts’ internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived; G_{net} : Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review’s narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed Λ error was in fact a correct-but-treacherous notation switch (see above).

2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140 $\mathbb{Z}_3$ choice is *derived*: the established integer deck $G = T_A \Sigma$ (v97/v98) pairs the $Q_+=1$ cusp line with the self-conjugate character 2, so only the sheet-twisted assignment $(2, 3, 1) = \text{chars}(-U)$ survives; the cusp exponential i^{Q_+} is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under $k \rightarrow -k$ only the established sheet $\mathbb{Z}_2$ flips. Consequence: Euler grading = $Q_+ \circ \rho$. GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4’ narrowed: three pairings $\rightarrow$ two (v142, frame integrality).** Integer covectors in the $(1, a, \sigma)$ dual frame form an index-11 sublattice ($x - 3y + z \equiv 0 \pmod{11}$; the index is $\|\text{Pl}(K)\|_1$); with $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8)$ the σ -pairing is forced $\equiv 0 \pmod{11}$, and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger GATE.UWALL.12).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The $\mathbb{Z}_4$ -graded hull carries exact coset duality ($-C_1 = C_3$; Killing pairing nondegenerate per $g_k \times g_{-k}$), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion = $\mathbb{C}[\mathbb{Z}_4]$) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger GATE.METRIC.09).
- **R1’s det-ratio step derived in-family (v144).** e_2 -rigidity of the traceless SdS cubic gives $r_b r_c = 1 - \Delta^2/3$ exactly; with the v132 anomaly the non-zero-mode determinant ratio is $(1 - \Delta^2/3)^{4/3}$ — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$ (audit). Finite-weight absorption stays `[C]` with the $6 \rightarrow \frac{14}{3}$ obstruction stated sharply (ledger HOR.CLOCK.12).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant $\rightarrow$ EH step (SEAM.THEOREM.01); the gate keeps its `[O]` typing untouched.

- **Wolfram mirror extended to v141–v144:** extension file now 211/211 (GATE.WOLFRAM.02 and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: `suite` ↔ `run_all.py` ↔ `registry` ↔ `ledger`; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri/\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the vN scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: `v17` and `v85` are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. `v91` (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` (“Three views of one spine”: anchor closure, edge/face products, volume 120, K_6 negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the ARCH.SPINE.01 ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare vN ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

2026-06-12 (tfpt_3 / tfpt_4 style alignment + box reduction)

- **Shared header/footer on tfpt_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): `tfpt_3` from 39 to 27 boxes (e.g. “Purpose and discipline”, the five “(i)–(v)” audit-lemma boxes, “Net”, “The connection in one sentence”, “The unification”, “The question and the honest answer”, “Net: two axioms”); `tfpt_4` from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item “Honest status of ...” keyboxes are kept — `tfpt_4` is the status authority for the frontier.

- **Colour-marked module references.** `\vref` applied to the inline `vN` references in `tfpt_3` and `tfpt_4` (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in `tfpt_4` (Koide flow-time) replaced by the 4-class `[C]`; status tables checked against the ledger.

2026-06-12 (tfpt_1 / tfpt_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant `[E]`) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): `tfpt_1` from 21 to 14 loud boxes (+3 `keyeq`); `tfpt_2` from 52 to 43 (e.g. “In one sentence”, “The one-paragraph version”, the four “Sharpening” notes, “What the simplification means net”, “The disciplined main statement”, “What improves...”, “Scope and honesty”). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout `tfpt_1` and `tfpt_2` (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** `tfpt_2`’s inflation-pivot table gains the `v86` scalaron-reheating point ($N_\star=51.4 \Rightarrow n_s=0.9611, r=0.0045, A_s$ -disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed `[C]`. The other status tables were checked against the ledger and are current under the 4-class markers.

2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: `[E]` exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), `[C]` conditional (physical / bridge / readout under named hypotheses), `[O]` open / axiom, and `[X]` kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ... \tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L] \rightarrow [E]`, `[N]/[P] \rightarrow [E]/[C]`). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction’s inflation/ $N_\star$ rows are made consistent with `v86/COSMO.NSTAR.01`: r and n_s are shown as the frozen *bands* over $N_\star \in [50, 60]$ with the scalaron-reheating conditional point $N_\star=51.4$ ($n_s=0.9611, r=0.0045$), replacing the stale point values $N_\star=55$ (predictions table) and $N_\star \simeq 57$ (residual card); the closure-ledger inflation row is re-typed `[E] \rightarrow [C]` (conditional on $M_{\text{Star}}=M_{\text{scal}} + \text{reheating}$), matching the other two tables and the ledger `[P]` typing.

2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today + version + revision`). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document’s title page and page footer carry

author, date and v5.4 (rev N) automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.

- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic (K_5 on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. `v108–v113`, `v49/v71`) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction’s callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and E_8 as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the “Reduction in one line” and the fermion-spectrum callouts now carry inline `\veri{}` citations (`v1`, `v6/v14/v23`, `v18/v20/v46`).

2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh` notes deliverable. The `make_manifest.py` TEX list and the workflow rule were updated to the new paths.
- **Master script index outsourced (tex-artefacts/verification.tex).** The long `vN→check` table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction’s eleven `keyboxes` are reduced to the single headline “In one sentence” card; the explanatory `keyboxes` (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*, $E_6 \times A_2$, *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible/follows next* `winboxes` become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winboxes`, one `gapbox`) plus the two shared orientation cards.

- **Document-count consistency.** The introduction’s document map gained the missing `tfpt_5_redteam` row; “seven/eight companion documents” and the ledger “six documents” wording are corrected to the true count (nine active documents); `README.md` (“10 ok”, 140 scripts) and the workflow rule (“9 docs”) are reconciled.

2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across `tfpt_1`, `tfpt_2`, `tfpt_3` and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer’s prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit* $a = (1, 1, 2)$ (B4), the E_8 *glue* diagram (B5), the $2/3$ *motif map* (B6) and the K_5 *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document’s layer; the topical figures are placed in their natural homes (cockpit + K_5 in the introduction, cockpit + glue in `tfpt_1`, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types `[E]` (exact algebra), `[E]` (exact number), `[C]` (readout / standard physics in TFPT units) to the existing `[C]` (bridge) and `[X]` (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain `[E]` where they are genuine exact identities.)

2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction’s one-sentence claim is softened from “derives all constants” to “constructs a discrete compiler; physical readouts run through named, status-typed bridges”; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says “stays [P]” — ratios are `[E]` closed on the derived stratum, only the absolute scale stays `[O]`, with an explicit forbidden-wording note); the seam misalignment is written explicitly as $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (leading c_3 , fourth-order puncture correction exact); θ_{12} has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNE’s first 59.1-day result (0.3092 ± 0.0087 , compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNE baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. “Complete solutions of the five open questions” becomes “Closure status of the five former open questions”.
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ (historical $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$ kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings $\rightarrow v_{\text{geo}} \rightarrow G_{\text{net}}$); F_{transfer} is named as one missing functor with $\text{Koide}/\eta_B/\text{axion}/m_p/m_e$ as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document $N \equiv \text{tfpt}_N$). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tftpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four `tcolorbox` families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tftpt_status.tex` (“TFPT status at a glance”) gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest TEX list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors (D_4 annihilator $n = (5, -9, 6)$, $\det R = 8$, $\det K = 4$) pin R uniquely among hexagon matrices ($64 \rightarrow 12 \rightarrow 1$); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4' selector establishment — replacing the discharged H2 dictionary — and R5 Q realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q -system (v124/v125).** R1's bend in closed form: $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ (pole at N_{fam} forces the three-level spectrum); the index-4 μ_4 extension exists explicitly as a Longo Q -system on $\mathbb{C}[\mathbb{Z}_4]$ (all Frobenius axioms exact) — R2 sharpens from “find” to “identify”.
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ($\text{spec } A_0^*$); $\det(\mathbf{1} - A_0^*) = \frac{2}{9}$ = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).** E_8 is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue Q -system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is $p_2(a)$, not $2p_2$ — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law $\rightarrow$ zeta budget (v129–v133).** The clock is $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ (Gibbons–Hawking form); $p_2 = 2h$ with $h = N_{\text{fam}}$ from mode counting + the Born rule (v130); the zero-mode norm is the area, $\|Y_{1m}\|^2 = A/(4\pi)$ exactly (v131); the det-ratio anomaly is the Koide constant, $\zeta(0)|_{\det'} = -\frac{2}{3}$ (v132); both $\zeta(0)$ budget routes computed exactly — the reduced seam route carries pure anchor ratios ($-\frac{4}{3}$ = minus the seed gain), the naive 4d route ($-\frac{109}{45}$) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).** $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ with $(1, 1, -2) = -2d$ — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg); $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ with iso-volume walls at the $\mathbb{Z}_2/\mu_4$ atoms, winding line $(8, 14, 20)$, $\det F = 32$, row-budget axes with $\text{rowsum}(V) = \text{Spec}(Q_+)$; micro-identities $41 = 16 + 25$, $52 = 16 + 36$.
- **Dual-normal selector, Q_+ cohomology, Volkov–Wipf firewall (v136–v138).** (d, n) pins R *columnwise* (each column the unique lattice point of the address box; kernel $(6, 8, 7)$); the A_0^* zero mode carries the spectral normal $\sigma = (2, -9, 5)$ with a $W(A_3)$ orbit control excluding the naive lift; $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ with degrees $(1, 2, 3) = \text{Spec}(Q_+) =$ the A_3 exponents and cusp weights = $\text{spec } A_0^*$; the external full-4d graviton/ghost value $-\frac{98}{45}$ is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.

- **Selector triangle + canonical map (v139/v140).** $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data (first selector *derived*); the frame $(\mathbf{1}, a, \sigma)$ has volume 11 and n is the unique covector with atom pairings $(2, 8, 121)$ — the $R4'$ residue is three pairings; the $H^1 \rightarrow$ generation equivariant map exists and is Schur-rigid (only sign-twisted μ_4 actions match), so the `GATE.QGEO` residue collapses to one $\mathbb{Z}_3$ choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded “single remaining input” claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltable` (it had silently truncated ~ 110 rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins $(E_8)_1$ (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface $M(s, t)$; centered cross $R, L = C \mp U$, $K, F = C \mp V$ with the cofactor seam normal $n = (5, -9, 6)$; documentation pass (“The Sheet Diamond and the Winding Line” in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots $(1, 1, -2)$, entropy bound $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ($\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ($g = 2K + 1 \Leftrightarrow$ the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank = c) — the QBL input merges with the $R2$ /holomorphy premise; communicated across all status surfaces.
- **U_{wall} made exact (v114–v118).** Torsion normal form and $\delta = \frac{1}{2}$ theorem; exact anchor residue A_0^* (diag = $a/|\mu_4|$, off-weights $(8, 0, 5)/144$); resonance theorem ($M_\infty = \mathbf{1} \iff a_{13} = 0$, one gauge orbit); the holonomy is the exact 24-element $W(A_3) = S_4$; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$; the 121 audit lemma; the lepton words are the compiler atoms $(8, 5, 3)$ with divmod-hexagon addresses; R is the unique hexagon matrix with atom margins and $\det 8$ — the address table carries zero residual information.

2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.

- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability $\leq 10^{-30.7}$, conditional on the declared grammar).

2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one θ_{12} number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82: E_8 glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.